CHAPTER 5
ROTOR SYSTEM

CHAPTER OVERVIEW

SECTION 5-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

5-1-1 MAIN ROTOR SYSTEM DESCRIPTION AND DATA.

5-1-1.1 Main Rotor Blade.

Four main rotor blades are installed on the main rotor head (Figure 5-1-1-1). The main rotor blade consists of:

a. Fiberglass Skin

b. Nickel And Titanium Abrasion Strips

c. Nomex Honeycomb Core

d. Pressurized Titanium Spar

e. Swept-Back Tip Fairing

A wire mesh is bonded to the surface of the blade, to protect the blade from lightning.

The spar of the main rotor blade is pressurized with nitrogen through a servicing valve at the inboard end of the blade. A BIM® (Blade Inspection Method) pressure indicator visually indicates that the spar pressure has not dropped below minimum.

The nickel and titanium abrasion strips, bonded to the leading edge of the blade, prevent damage that could occur from erosion.

Each blade is statically and dynamically balanced to permit replacement of individual blades. Balance strips painted around the blade locate the hoisting point.

A titanium cuff and two main rotor blade pins (both main rotor blade pins must be expandable or solid) attach the blade to the rotor head (Figure 5-1-1-2).

With the use of a blade fold set, each blade can be folded manually.

5-1-1.2 Blade Inspection Method (BIM®) Pressure Indicator.

The BIM® indicator is installed in the back wall of the spar at the root of the blade (Figure 5-1-1-1, Detail A). A color indicates if the blade becomes unserviceable. The indicator compares a reference pressure built into the indicator with the pressure in the blade spar. When the pressure in the blade spar is within the required service limits, indicating the blade is serviceable, three yellow stripes show. If the pressure in the blade spar drops below the minimum permissible service pressure, the indicator will show three red stripes (Figure 5-1-1-1, Detail B).

5-1-1.3 Main Rotor Head.

The main rotor head transmits the movements of the flight controls to the four main rotor blades (Figures 5-1-1-3 and 5-1-1-4). The main rotor head turns in a counterclockwise direction. The head is supported by the main rotor shaft extension. The shaft extension is splined to the main transmission main shaft, which drives the head. The lower pressure plate and cones, in conjunction with the main shaft nut, secure the shaft extension to the main shaft. The lower pressure plate also provides attachment for the rotating scissors. The principal components of the main rotor head are the main rotor hub (including the spindle modules), the droop stops, the bifilar vibration absorber, pitch control rods, dampers, antiflap assemblies, swashplate, and lead stops. For air transportability the rotor head may be lowered by using special support equipment.

5-1-1.4 Main Rotor Hub.

The hub consists of titanium spindle modules, hydraulic dampers, pitch control rods, antiflapping assemblies, and a titanium housing (Figures 5-1-1-3 and 5-1-1-5). Each blade is hinged through elastomeric bearings (rubber and steel laminates) in the spindle modules. The elastomeric bearings allow the blades to flap, lead, and lag. The bearing also permits the blade to move about its axis for pitch changes. The spindle module titanium endplate contains the lugs for blade attachment.

5-1-1.5 Antiflapping Assemblies.

An antiflapping assembly is installed on each of the four main rotor spindle modules, next to the hub (Figures 5-1-1-3 and 5-1-1-5). These are spring-loaded locks that prevent the main rotor blades from flapping when the main rotor head is slowing down or stopped. When the main rotor is rotating at above 35%, centrifugal force pulls the antiflapping assemblies outward and holds them in their locked positions to permit flapping and coning of the blades.

5-1-1.6 Droop Stops.

The droop stops, on the spindle module next to the hub, limit droop of the blades when the main rotor head is slowing down or stopped (Figure 5-1-1-3). When the main rotor head is rotating between 70% to 75% rotor speed (Nr), centrifugal force throws the droop stops out and permits increased vertical movement of the blade.

5-1-1.7 Lead Stops.

A lead stop and lead stop bracket are installed on each of the four main rotor spindle modules (Figure 5-1-1-4). The lead stop prevents extreme blade lead motion that occurs when the rotor brake is applied to stop the main rotor head.

5-1-1.8 Pitch Control Rods.

Four pitch control rods extend from the rotating swashplate to the blade pitch horn on the spindle (Figure 5-1-1-4). The pitch control rods transmit all movement of the flight controls from the swashplate to the main rotor blades. Each rod is adjustable. Rotation of the rod changes main rotor blade angle two minutes per notch. This allows for tracking adjustment.

5-1-1.9 Damper Indicators.

Damper indicators (dampers) are installed between each of the main rotor hub and spindle modules to restrain hunting (lead and lag motions) of the blades during rotation and to absorb rotor head engagement loads (Figure 5-1-1-4). Each damper is filled with hydraulic fluid. An indicator mounted on the side of the damper monitors hydraulic fluid quantity. When the damper is fully serviced, the indicator will show full gold.

5-1-1.10 Bifilar Vibration Absorber.

The bifilar vibration absorber (absorber) absorbs vibrations and stresses. It not only contributes to longer life of all components but to a smoother ride for the crew and passengers (Figure 5-1-1-4). The absorber support is a cross-shaped aluminum forging. A tungsten weight pivots on two points at the end of each arm. The absorber is bolted to the main rotor hub.

5-1-1.11 Swashplate.

The swashplate has stationary and rotating discs joined by a bearing (Figure 5-1-1-4). It transmits flight control movement to the main rotor head through the four pitch control rods. The swashplate is permitted to slide on the main rotor shaft and tilt in any direction following the motion of the flight controls.

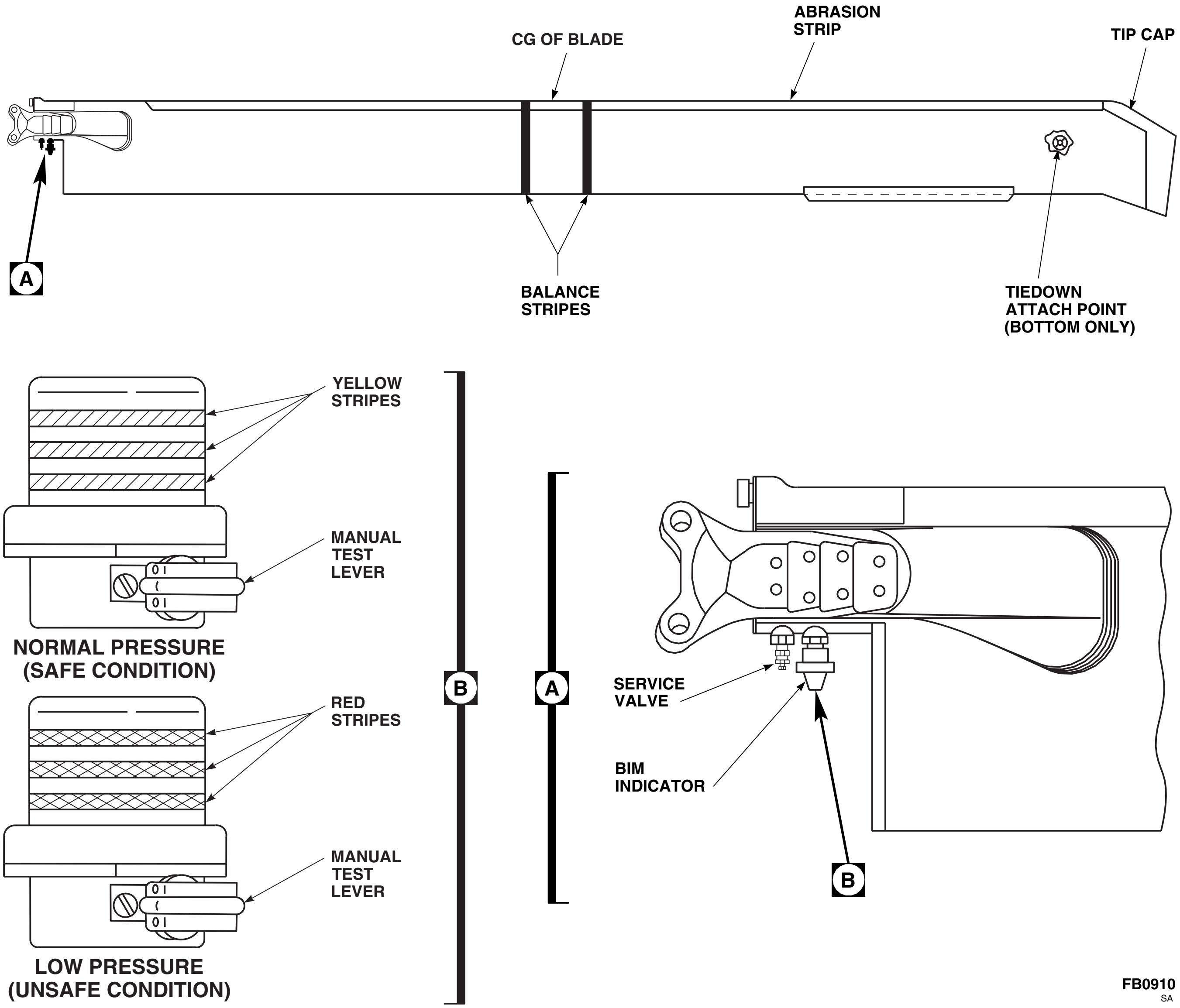
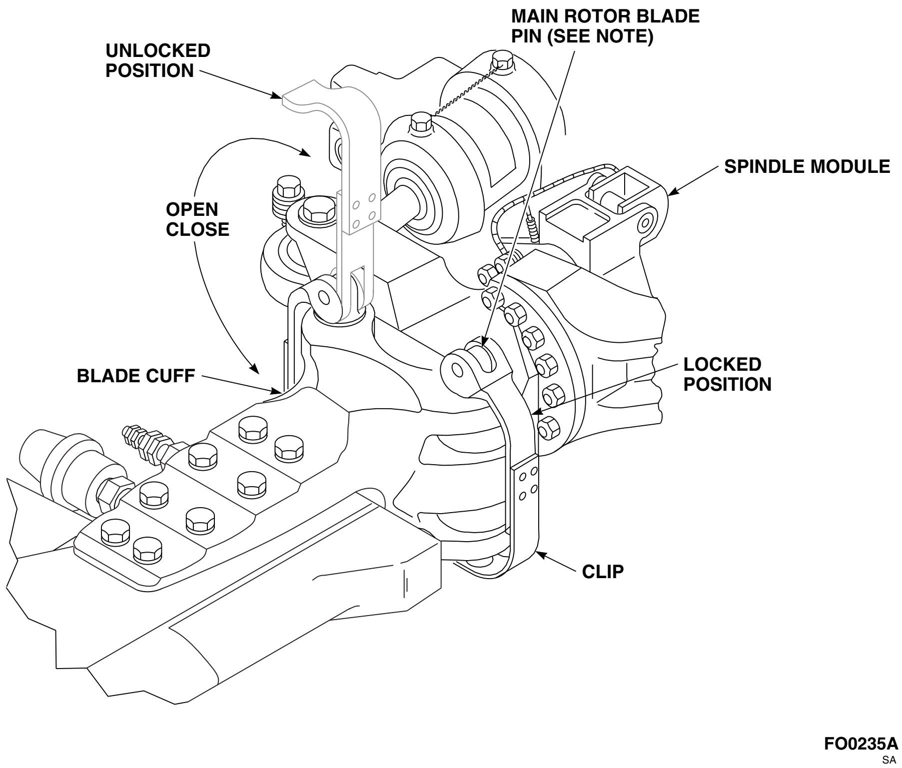


FIGURE 5-1-1-1. MAIN ROTOR BLADE AND PRESSURE INDICATOR



NOTE
MAIN ROTOR BLADE PIN CAN BE EXPANDABLE PIN OR SOLID PIN. EXPANDABLE PIN SHOWN.

FIGURE 5-1-1-2. MAIN ROTOR BLADE PINS

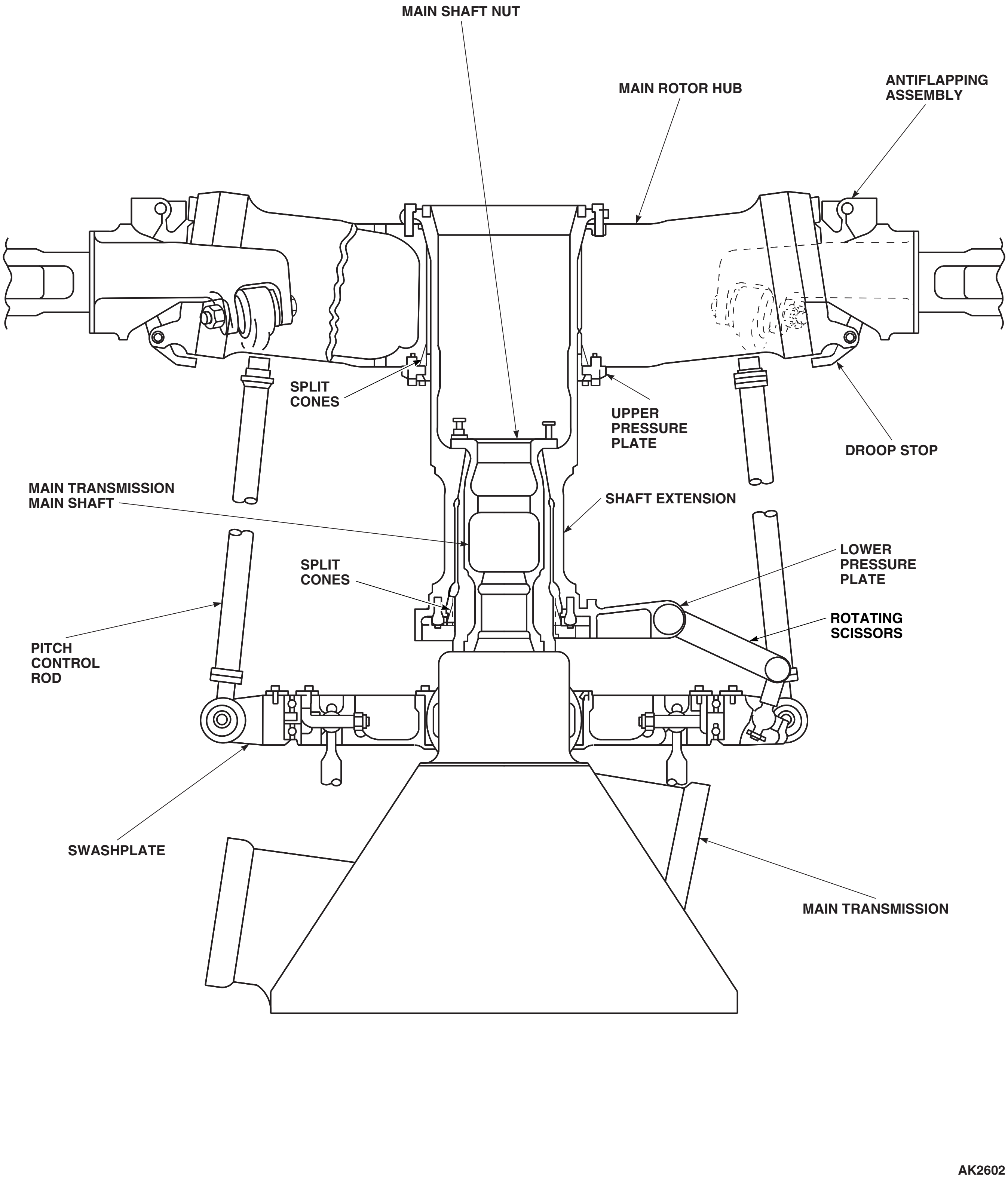


FIGURE 5-1-1-3. MAIN ROTOR HEAD AND MAIN TRANSMISSION

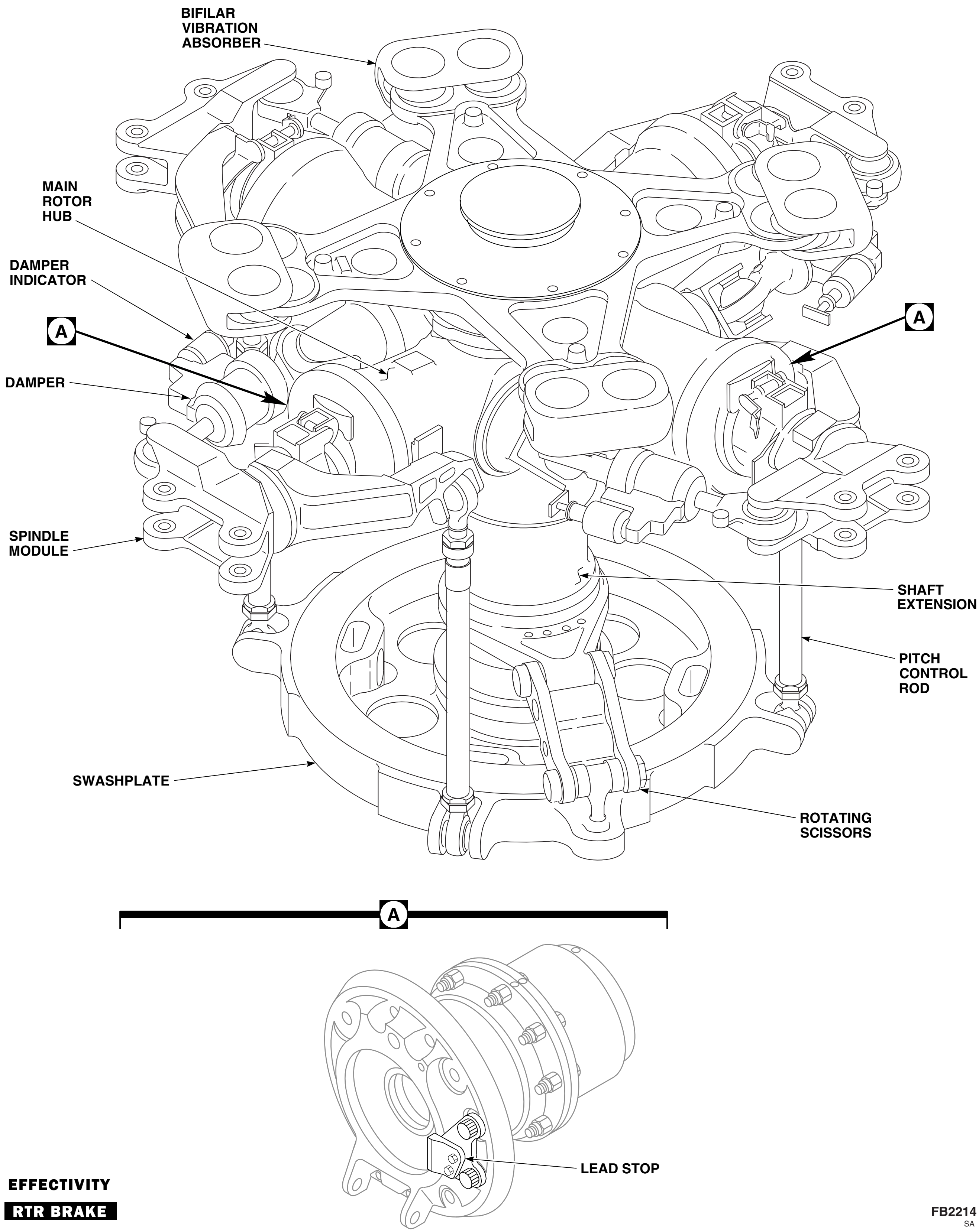
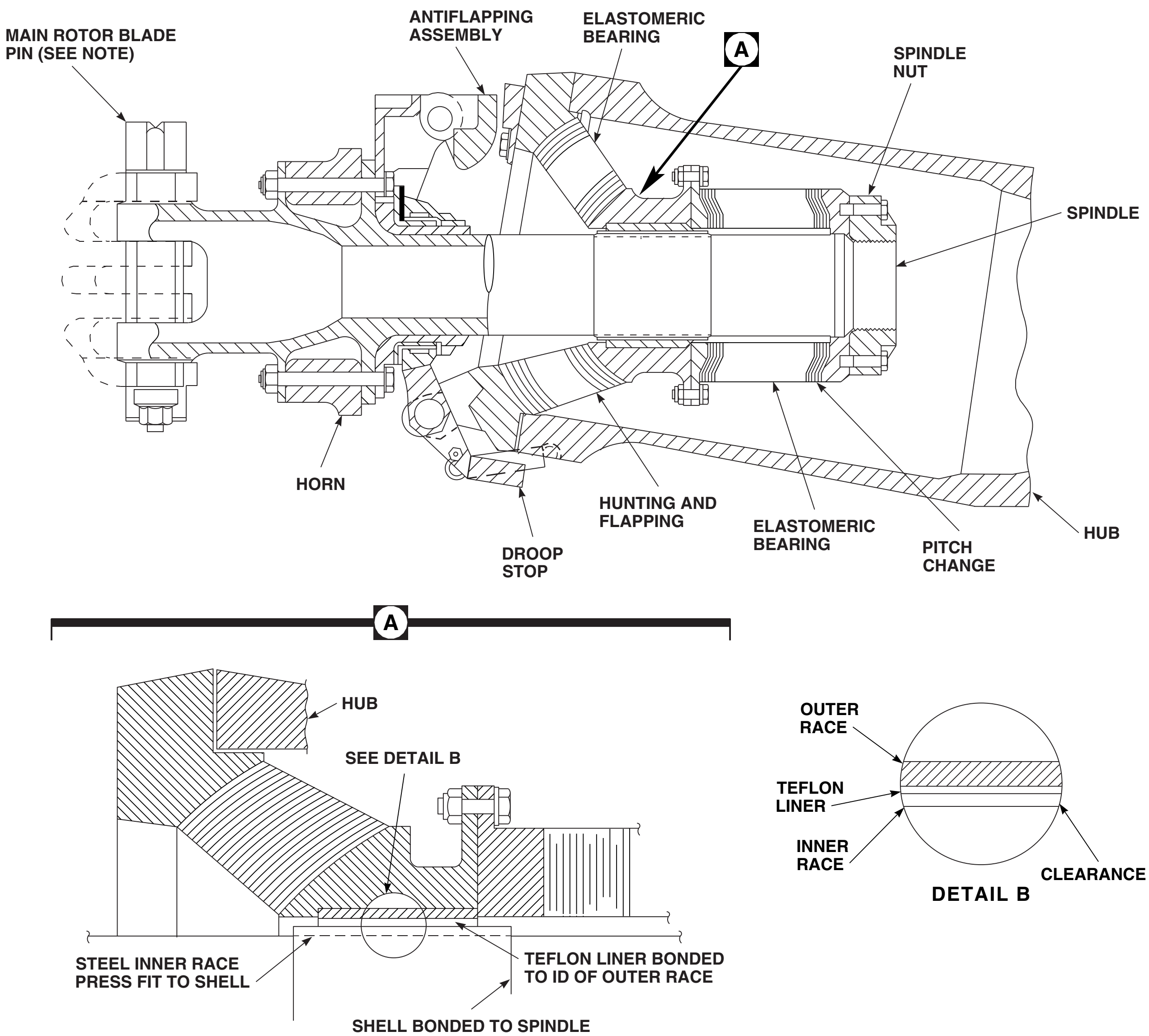


FIGURE 5-1-1-4. MAIN ROTOR HEAD



NOTE
MAIN ROTOR BLADE PIN CAN BE EXPANDABLE PIN OR SOLID PIN. EXPANDABLE PIN SHOWN.

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FIGURE 5-1-1-5. MAIN ROTOR HEAD SPINDLE MODULE - CUTAWAY

5-1-2 TAIL ROTOR SYSTEM DESCRIPTION AND DATA.

5-1-2.1 System Description.

The tail rotor is a bearingless, controllable pitch, cross-beam type system (Figure 5-1-2-1). The tail rotor blades are built around two interchangeable graphite composite spars which cross each other at their center. The two tail rotor blades are retained on the tail rotor hub by a set of retention plates. These plates bolt the tail rotor blades together to form four blades 90° apart. Counterweights are bolted to each tail rotor blade for balancing.

5-1-2.2 Principles Of Operation.

a. The tail rotor blades and rotor head are installed on the right side of the tail pylon, and are canted 20° upward.

b. Tail rotor blade flap and pitch change are accomplished by twisting the flexible graphite spars. This is done without any bearings or lubrication. In addition to providing directional control and antitorque reaction, the tail rotor provides 2.5% lifting force in a hover.

c. The tail rotor is controlled through the tail gear box. The quadrant transmits cable movements to the tail rotor servo. With a complete control cable failure, a spring-tension feature provides 7.5° (left pedal) positive pitch on the tail rotor. This is equivalent to antitorque requirements for a midposition collective power setting.

d. In case of a cable fracture, spring tension allows the quadrant to operate normally. This is accomplished by the quadrant controlling the opposite direction against spring tension. When a cable fails, a TAIL ROTOR QUADRANT caution capsule goes on to warn the pilot of the failure.

5-1-2.3 Component Description.

5-1-2.3.1 Tail Rotor Blades.

The tail rotor blades are built around two graphite composite spars running from tip-to-tip and crossing each other at the center to form the four blades (Figure 5-1-2-2). The two spars are interchangeable and may be replaced individually. The blade spars are covered with crossply fiber glass to form the airfoil shape. Polyurethane and nickel abrasion strips are bonded to the leading edge of the blades. Blade pitch changes are made by twisting the spar. Similar to the main rotor blades, the blade deice system ensures the removal of any ice that should accumulate on the tail rotor blades.

5-1-2.3.2 Pitch Beam.

A four-armed pitch beam is bolted to the end of the pitch change shaft. The pitch beam increases or decreases the pitch of all blades simultaneously through pitch links connected to the blades.

5-1-2.3.3 Pitch Control Links.

Four pitch control links are installed on the tail rotor head assembly. Each link connects an arm of the pitch beam to a pitch control horn on the blade. The links transmit movement necessary for blade pitch changes from the pitch beam. Each link consists of two rod ends with boots, locking devices, and a link. The rod end that is connected to the pitch beam is marked for proper installation.

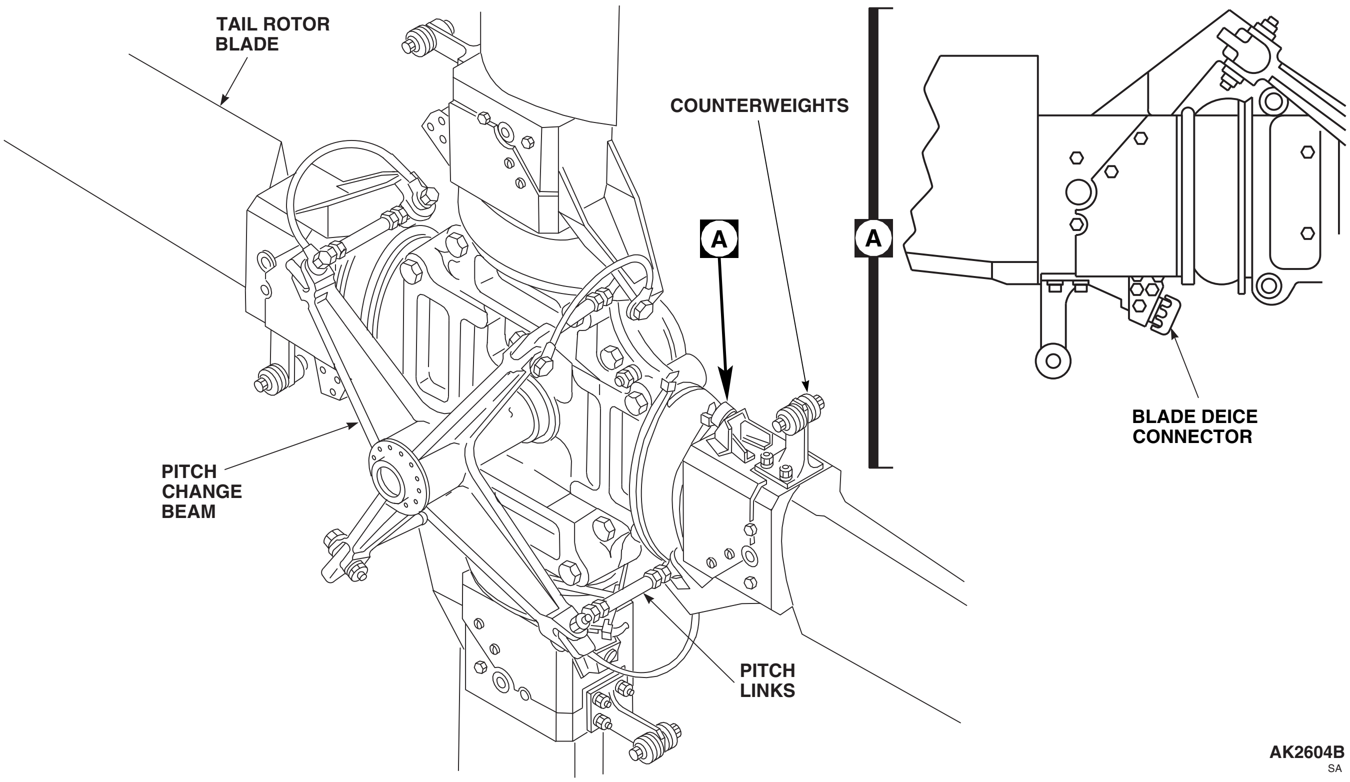


FIGURE 5-1-2-1. TAIL ROTOR SYSTEM

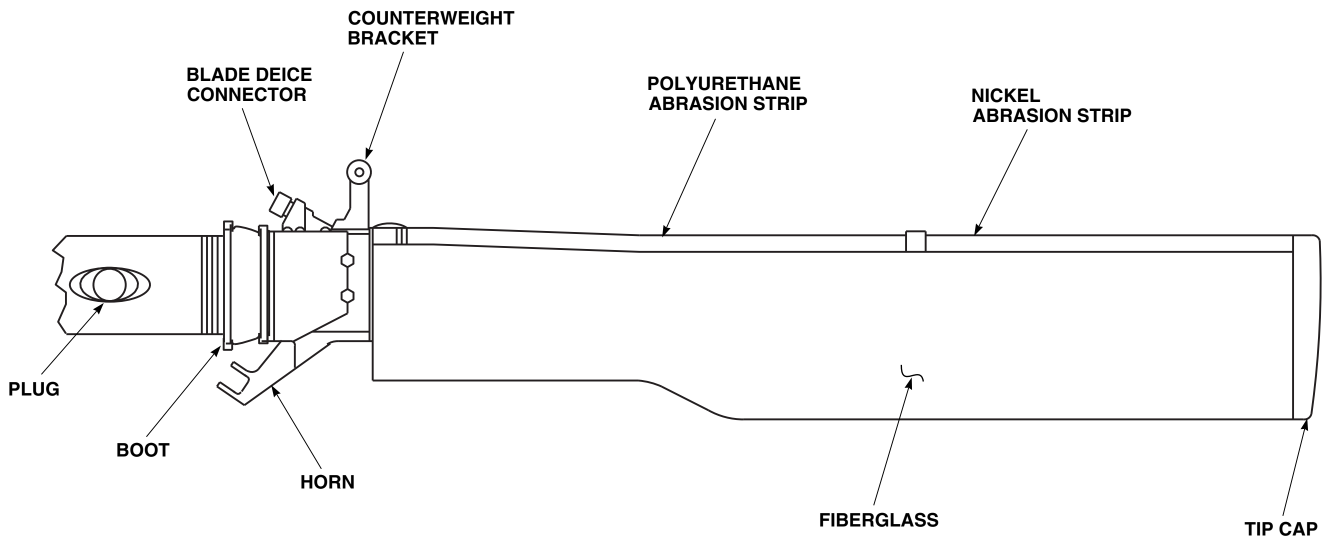


FIGURE 5-1-2-2. TAIL ROTOR BLADE

SECTION 5-2

TROUBLESHOOTING

SECTION OVERVIEW

5-2-1 BIM® INDICATORS.

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Leak Detecting Fluid, Item 189, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appropriate Flight Manual

WARNING

Visual inspection of the indicator must be performed while viewing the indicator perpendicular to its axis.

Upon inspection of each blade pressure indicator, if red is visible on all three inspection windows, maintenance must be performed prior to next flight. The cause of the unsafe condition must be positively identified and corrected. If the indicator is malfunctioning, it must be replaced, but only if the spar pressure is within permissible limits.

NOTE

Small slivers of red visible through one or two of the inspection windows is not cause for maintenance action.

NOTE

When a blade is forwarded to BENCH level or designated overhaul facility, it must be accompanied by a statement that accurately describes what maintenance action or actions have been taken, and what was originally wrong with the blade to require these actions. If applicable, blade shall be marked to identify it as having had an unsafe BIM® indication.

NOTE

Do the following operational/ troubleshooting procedure on all four main rotor blade BIM® indicators.

NOTE

When testing pressure indicator, press both thumbs on indicator's manual lever (grenade-type handle) to make sure that valve plunger under the lever is pushed all the way down to shut off all spar pressure to the indicator. Press on ridged part of handle, not on smooth tip. A partly depressed plunger may cause loss of spar pressure and a slow indication, or no indication at all.

NOTE

Do not hold indicator, as heat of hand may change internal reference pressure and result in erroneous indicator reading.

Special Instructions

The BIM® Indicator operational/troubleshooting procedure is subdivided as follows:

(1) BIM® Indicator Packing Test (to determine if the indicator is leaking at the base).

(2) BIM® Indicator Test (to determine if the indicator is functioning properly).

(3) One-Hour Whirl Test (to determine the final integrity of blade spars returned to organizational level from either BENCH level or designated overhaul facility).

5-2-1.1 BIM® Indicator Packing Test.

NOTE

Spar must be properly serviced before testing.

a. Apply leak detecting fluid or mild liquid detergent solution over and around base of indicator.

RESULT: No air bubbles shall form.

- If air bubbles form, replace BIM® indicator packing (PARA 5-4-34).

b. Rinse test fluid using clean fresh water before fluid dries. Dry blade using machinery towel.

5-2-1.2 BIM® Indicator Test.

a. Press in and hold manual test lever. Do not place hand on glass bulb.

RESULT: BIM® indicator stripes shall turn from yellow to full red (unsafe) within 10 to 30 seconds at - 6.7°C (20°F) or above, or within the following time limits and extended upper limit temperatures:

Temperature	Time
-7.2°C to -17.8°C (19°F to 0°F)	35 seconds
-18.3°C to -28.9°C (-1°F to -20°F)	40 seconds
-29.4°C to -40.0°C (-21°F to -40°F)	50 seconds
-40.5°C to -51.1°C (-41°F to -60°F)	60 seconds

- If indicator does not change color or completely change color, check main rotor blade spar for proper pressure (PARA 5-3-8).

  (a) If spar pressure is good, replace BIM® indicator (PARA 5-4-34).

  (b) If spar pressure is not good, replace main rotor blade (PARA 5-4-31).

- If indicator takes more than the maximum number of seconds given above or less than 10 seconds to change color completely, go to TABLE NO. 5-2-1-1.

b. Release indicator lever.

RESULT: Red indication must snap back to yellow immediately.

- If result is not as specified, go to TABLE NO. 5-2-1-1.

5-2-1.3 One-Hour Whirl Test.

NOTE

The whirl test is a final check on the integrity of a rotor blade spar that has had a loss of spar pressure. The test is conducted after all other corrective actions have been taken.

a. Measure and record spar temperature and pressure.

b. With qualified personnel at controls, ground run helicopter for two half-hour periods at 100% RPM, low collective stick, and neutral cyclic stick and directional control (Appropriate Flight Manual).

RESULT: At the end of first half-hour period, BIM® indicator stripes shall be yellow.

- If result is not as specified, send blade to BENCH level.

c. At the end of second half-hour period, measure and record spar pressure and temperature.

d. Correct for temperature variations as follows:

(1) For temperature increases, subtract 0.126 psi per degree Celsius (0.07 psi per degree Fahrenheit) of increase.

(2) For temperature decreases, add 0.126 psi per degree Celsius (0.07 psi per degree Fahrenheit) of decrease.

RESULT: Corrected spar pressure shall be no greater than 0.25 psi from pressure measured in step a.

EXAMPLE:

-INITIAL READING-

PRESSURE	**TEMPERATURE**
18.0 (PSI)	22.8 °C (73.0 °F)

-READING AFTER 1 HOUR-

PRESSURE	**TEMPERATURE**
17.0 (PSI)	17.2°C (63.0°F)

Temperature change is 5.6°C (10.0°F) decrease. 5.6 X 0.126 = 0.70 PSI (for °C) or 10.0 X 0.07 = 0.70 PSI (for °F). Correction to be added to reading. Therefore, the actual corrected pressure in the spar is 17.0 + 0.70 = 17.7 PSI.

- If result is not as specified, send blade to BENCH level.

NOTE

If blade has had an unsafe (red) indication at any time within the last 30 flight hours, send blade to designated overhaul facility.

TABLE NO. 5-2-1-1. BIM® Indicator Takes More Than The Maximum Number Of Seconds To Change Color Completely.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Check main rotor blade spar for proper pressure. Is pressure good?

Step 1. If pressure is good, go to **2.**
Step 2. If pressure is not good, replace main rotor blade (PARA 5-4-31). Go to **6.**

2. Wait for 6 hours. Go to 3.

3. Check if BIM® indicator stripes return to yellow.

Step 1. If stripes return to yellow, go to **4.**
Step 2. If stripes do not return to yellow, replace BIM® indicator (PARA 5-4-34). Go to **6.**

4. Perform whirl test. Go to 5.

5. Check if spar pressure is good.

TABLE NO. 5-2-1-1. BIM® Indicator Takes More Than The Maximum Number Of Seconds To Change Color Completely. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If spar pressure is good, continue blade in service. Go to **6.**
Step 2. If spar pressure is not good, send blade to bench repair facility. Go to **6.**

6. Procedure completed.

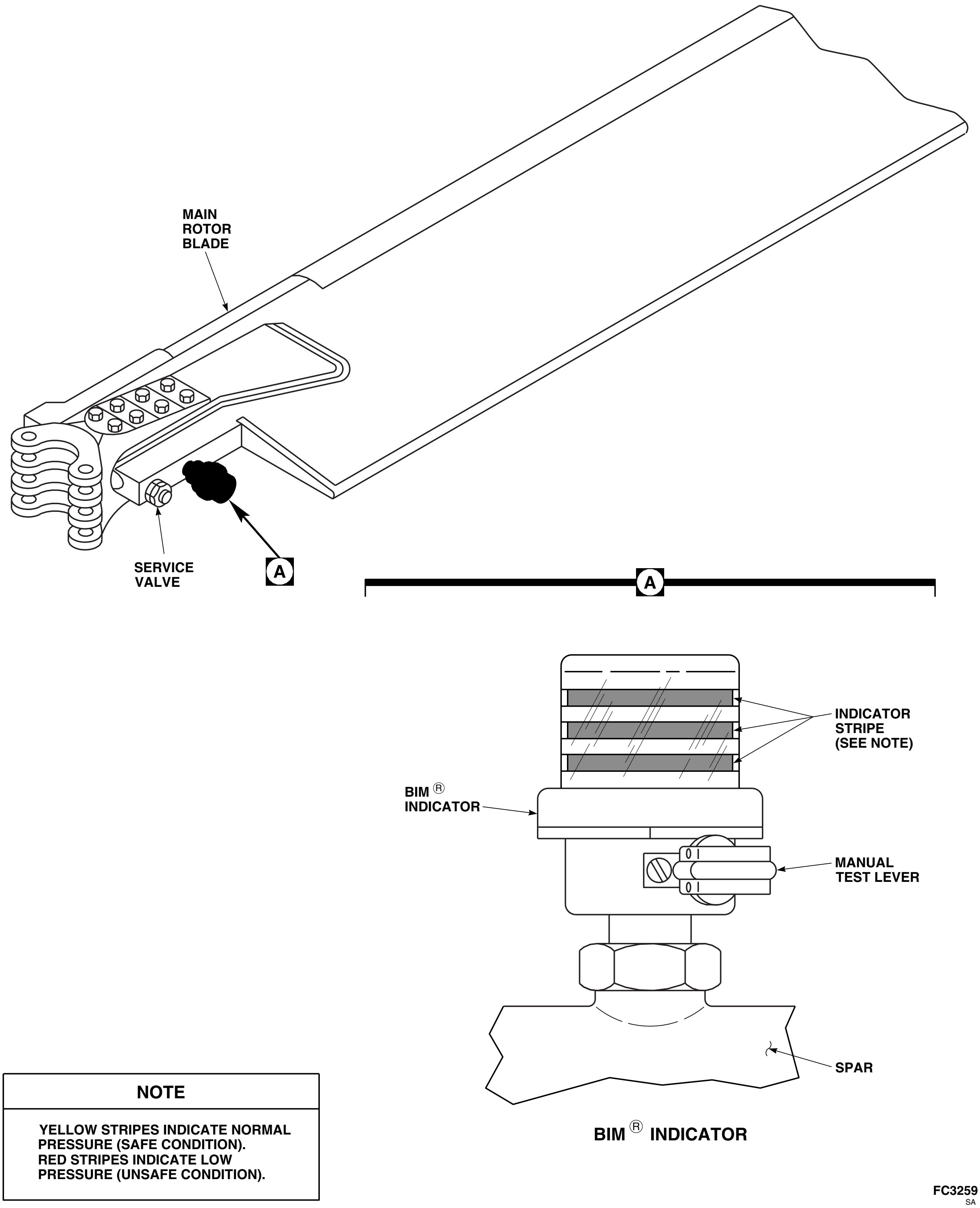


FIGURE 5-2-1-1. BIM® INDICATOR LOCATION DIAGRAM

5-2-2 MAIN ROTOR BLADES (BENCH).

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit, SC 5180-99-B01

References

- PARA 5-3-8

CAUTION

When BENCHlevel has repaired a blade that had an unsafe indication on its BIM® indicator, they must not repair that blade again if a second unsafe indication occurs within 30 flight-hours of the first. Should this happen, forward the blade to the designated overhaul facility.

NOTE

If a blade is sent to a designated overhaul facility, it must have with it a statement that accurately describes the specific maintenance action or actions that were done at BENCH level. The statement shall include spar pressure when blade was inducted into BENCH level and when it left, and specific reason for forwarding blade to designated overhaul facility. All data from organizational level, including identification of original problem, shall also be with blade. A blade component record card entry shall be made whenever a blade is removed from service stating the reason for removal.

5-2-2.1 Operational/Troubleshooting Procedure.

a. Check main rotor blade spar pressure (PARA 5-3-8).

RESULT: Spar pressure shall be correct, and BIM® indicator stripes shall be yellow.

- If spar pressure is correct but BIM® indicator stripes are red, perform a BIM® indicator test (PARA 5-2-1).
- If BIM® indicator fails the test, replace indicator (PARA 5-4-34) and return blade to organizational level for one-hour whirl test.
- If spar pressure is not correct, go to TABLE NO. 5-2-2-1.

TABLE NO. 5-2-2-1. Spar Pressure Is Incorrect.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Pressurize spar and test components for leakage (PARA 1-3-29).

Step 1. If spar leaks, go to **2.**
Step 2. If spar does not leak, go to **4.**

2. Is cause of leak clearly identified and is leak repairable?

Step 1. If spar leak is clearly identified and repairable, go to **3.**
Step 2. If spar leak is not clearly identified and repairable, send blade to DEPOT facility. Go to **5.**

3. Repair blade.

Step 1. Confirm repair(s). Go to **5.**
Step 2. Return blade to organizational level for one-hour whirl test (PARA 5-2-1). Go to **5.**

4. Has blade had previous BIM® indications within the last 30 hours?

Step 1. If blade had previous BIM® indications, send blade to designated DEPOT. Go to **5.**
Step 2. If blade has not had previous BIM® indications, return blade to organizational level for one hour whirl test (PARA 5-2-1). Go to **5.**

5. Procedure completed.

SECTION 5-3

ON AIRCRAFT INSPECTIONS

SECTION OVERVIEW

5-3-1 SPINDLE INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage, 0.005-inch
- Flashlight
- Inspection Mirror
- Powertrain Repairer's Toolkit

Materials/Parts

- Abrasive Wheel, Item 22, Appendix D

References

- Appendix D
- PARA 5-3-9
- PARA 5-4-5
- PARA 5-4-31
- PARA 5-5-4

5-3-1.1 Spindle Inspections.

NOTE

Spindle journal bearing inspection applies only to spindles with droop stop support ring, 70105-08056-101, with sleeve bearings.

a. Visually inspect exposed portion of spindle journal bearing (pressed on droop stop support) around entire circumference for cracks as follows (Figure 5-3-1-1, Sheet 1, Detail A):

(1) **IF HELICOPTER IS AT 500 HOUR PERIODIC INSPECTION AND A CRACK IS VISIBLE, REPLACE SPINDLE** (PARA 5-4-5).

(2) **IF HELICOPTER IS NOT AT 500 HOUR PERIODIC INSPECTION AND ONLY ONE CRACK IS VISIBLE, REPLACE SPINDLE AT NEXT 500 HOUR PERIODIC INSPECTION, OR UNTIL A SECOND CRACK IS FOUND** (PARA 5-4-5).

(3) **IF MORE THAN ONE CRACK IS VISIBLE, REPLACE SPINDLE** (PARA 5-4-5).

b. If spindle must be replaced, flight may be continued up to 10 hours under the following conditions:

(1) **FLIGHT IS A ONE-TIME FLIGHT TO SITE, CAPABLE OF REPLACING SPINDLE.**

(2) **CONTINUANCE OF FLIGHT IS ALLOWED TO ENABLE HELICOPTER**

TO RETURN TO SITE WHERE REPAIRS CAN BE MADE.

NOTE

- Inspection of spindle cuff lugs must be performed with handle of main rotor blade pin in open (unlocked) position.
- Main rotor blades may be secured by two different kinds of pins: main rotor blade expandable pins, 70103-08107-102, and main rotor blade solid pins, 70103-08108-041. In the following procedures, both types are referred to as main rotor blade pins, except where noted.

c. Open handle on main rotor blade pin as follows:

(1) On helicopters with main rotor blade expandable pins, perform the following (Figure 5-3-1-1, Sheet 1, Detail A):

(a) Release spring clip portion of handle from nut and open handle on expandable pin.

(b) Raise handle to fully unlocked (open) position.

(2) On helicopters with main rotor blade solid pins, perform the following (Figure 5-3-1-1, Sheet 1, Detail A and Detail B):

(a) Remove cotter pin securing solid pin.

(b) Raise handle to fully unlocked (open) position.

d. Visually inspect all exposed surfaces of spindle cuff lugs for cracks. **NONE ALLOWED.** If crack is found, perform the following:

NOTE

Main rotor blades may be secured by two different kinds of pins: main rotor blade expandable pins, 70103-08107-102, and main rotor blade solid pins, 70103-08108-041. In the following procedures, both types are referred to as main rotor blade pins, except where noted.

(1) Replace spindle (PARA 5-4-5), main rotor blade, and main rotor blade pins (PARA 5-4-31).

(2) Record part number, serial number, and flight hours of removed components.

(3) Ship recorded information, along with removed spindle and main rotor blade pin, to Sikorsky Aircraft Corporation for engineering analysis.

(4) Ship removed main rotor blade to approved Depot facility for removal of main rotor blade cuff.

e. Close handle on main rotor blade pin as follows:

(1) On helicopters with main rotor blade expandable pins, perform the following:

NOTE

Make sure blade tension is off main rotor blade expandable pins during tension check. Lower or raise blade as necessary to relieve tension.

(a) Perform spring clip/handle tension check (PARA 5-3-9).

(b) Close handle, making sure spring clip of pin snaps over and fully engages hex on nut. Visually check for expansion of lockring on end of pin. Make sure slots in spring clip line up with points of nut.

(2) On helicopters with main rotor blade solid pins, perform the following:

(a) Close handle, making sure retaining clip fork engages groove in pin and that bottom of pin protrudes through hole in spring clip.

(b) Install cotter pin from side of pin closest to main rotor hub.

f. Visually inspect white torque stripe on retaining rod and nut to ensure it is unbroken (Figure 5-3-1-1, Sheet 2, Detail C, View A-A). If torque stripe is broken, indicating potential torque loss, replace spindle (PARA 5-4-5).

g. Inspect spindle journal bearing for signs of movement on spindle shaft. Movement of bearing is identified by misalignment of slippage marks (white arrow and white stripe) (Figure 5-3-1-1, Sheet 2, Detail C, View B-B). If bearing displacement occurs, flight may be continued up to 10 hours after confirmation of slippage under the following conditions only:

(1) **BEARING HAS NOT DISENGAGED FROM OUTER RACE.**

(2) **BEARING HAS NOT MOVED BEYOND INBOARD CONTACT FACE OF TEFLON SLEEVE OR BROKEN INBOARD TIEDOWN STRAP.**

(3) **FLIGHT IS ONE-TIME FLIGHT TO SITE CAPABLE OF REPLACING SPINDLE.**

(4) **CONTINUANCE OF FLIGHT IS ALLOWED TO ENABLE HELICOPTER TO RETURN TO SITE WHERE REPAIRS CAN BE MADE.**

5-3-1.2 Elastomeric Bearing Assembly Inspection.

a. Inspect droop stop support for damage (Figure 5-3-1-1, Sheet 2, Detail D). Using abrasive wheel, Item 22, Appendix D, blend out damage in area around droop stop pin. **SURFACE DAMAGE MUST NOT EXCEED 0.030-INCH DEEP AFTER BLENDING. CORNER DAMAGE MUST NOT EXCEED 0.050-INCH DEEP AFTER BLENDING.** If damage exceeds limits, replace elastomeric bearing assembly (PARA 5-5-4).

b. Inspect droop stop support for cracks. Inspect upper and lower puller grooves of droop stop support for cracks (Figure 5-3-1-1, Sheet 3, Detail E). **NONE ALLOWED.** If droop stop support shows signs of cracking, replace elastomeric bearing assembly (PARA 5-5-4) and spindle (PARA 5-4-5). Send elastomeric bearing assembly and spindle to manufacturer.

NOTE

- Elastomeric bearing assembly consists of a spherical bearing and a thrust bearing. Elastomeric bearings consist of laminated elastomer and steel shells. Wear of elastomer in either bearing is a gradual process that will take several hundreds of hours from first detection of elastomer damage to point where bearing must be replaced. Visual comparison of bearings to Figure 5-3-1-2, Sheet 2, Sheet 3, Sheet 4, Sheet 5, and Sheet 6, is criterion for acceptance or rejection.
- Do not disassemble elastomeric bearing assembly to inspect.

c. Types of wear shown include eraser-type wear, elastomer extrusion, and shim cracks. Description of wear and general guidelines are as follows (Figure 5-3-1-2, Sheet 2, Sheet 3, Sheet 4, Sheet 5, and Sheet 6):

NOTE

White wax-like substance on rubber surface is normal and should not be removed.

(1) Eraser-type wear: pile-up of shredded rubber similar to result of using soft eraser.

(2) Elastomer extrusion: squeezing out of elastomer from between elastomeric bearing laminations, indicating disbonding of elastomer. Overlap of elastomer on elastomeric bearing assembly endplate outside diameter from manufacturing process is not cause for rejection. Peeling of elastomer mold lines, which run from inboard to outboard of elastomeric bearing, is acceptable.

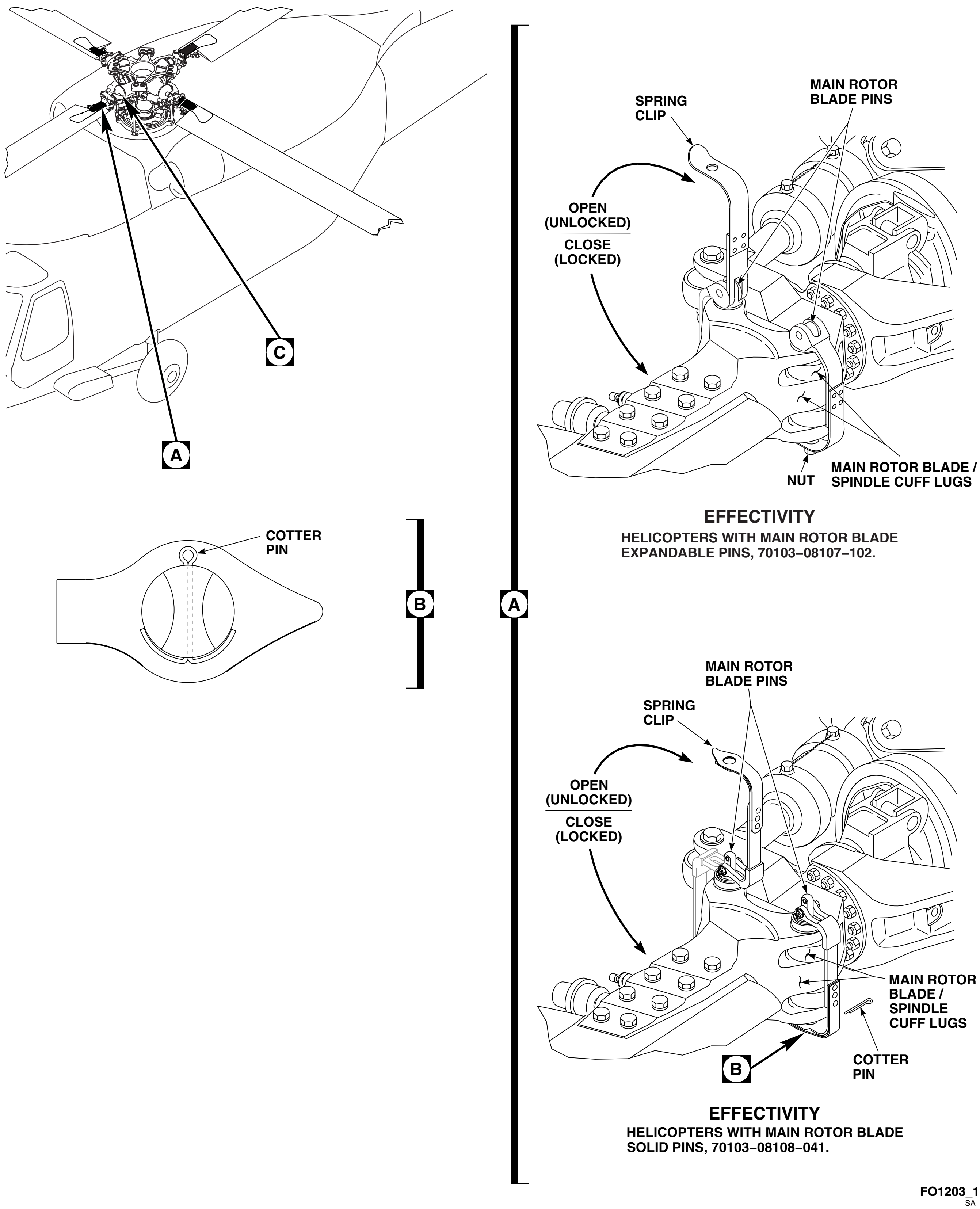


FIGURE 5-3-1-1. INSPECT SPINDLE (SHEET 1 OF 3)

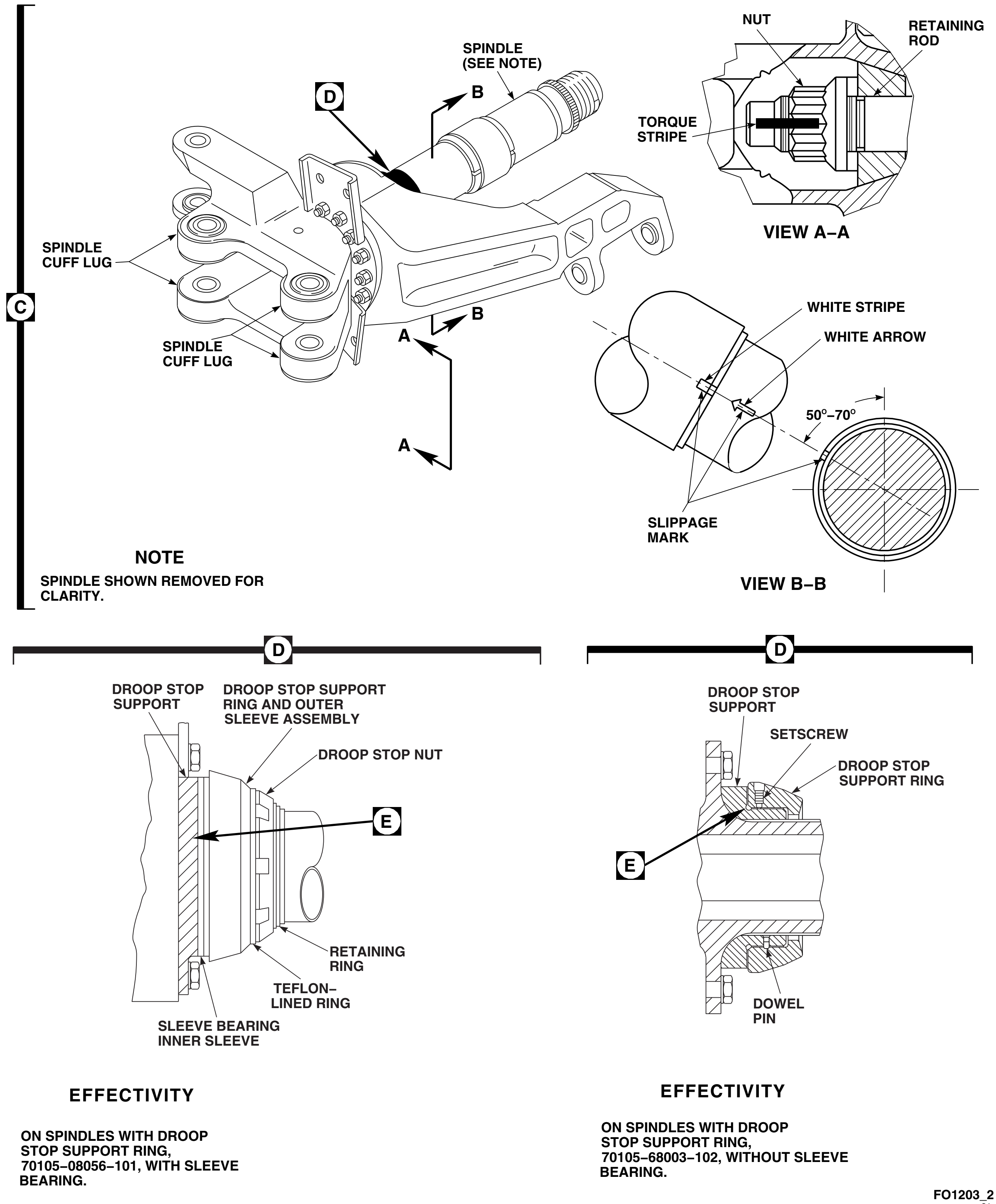


FIGURE 5-3-1-1. INSPECT SPINDLE (SHEET 2 OF 3)

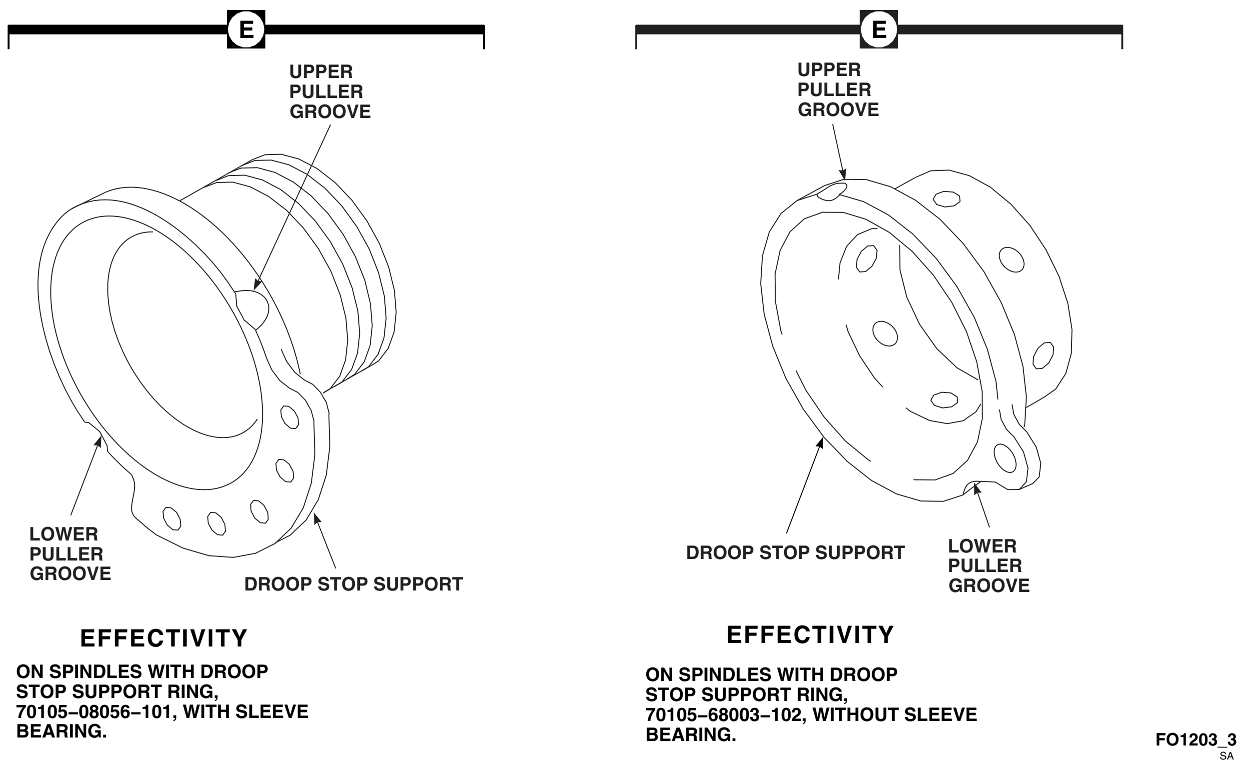


FIGURE 5-3-1-1. INSPECT SPINDLE (SHEET 3 OF 3)

d. Using flashlight and inspection mirror, visually inspect outer surface of spherical and thrust bearing for wear, laminate separations, and cracked shims through 2-inch diameter inspection holes as follows:

NOTE

- Do not remove spindle at 30 hour special inspection to inspect inner surface of thrust bearing.
- Elastomeric extrusion will normally be detectable on lower portion of thrust bearing.

(1) Inspect thrust bearing for elastomeric extrusion. **NONE ALLOWED.** If any elastomer extrusion is found, replace thrust bearing prior to next flight (PARA 5-5-4).

NOTE

Shim cracks will appear in edge of metal shims.

(2) Inspect shim for cracks. **NONE ALLOWED.** If crack is found, replace spherical or thrust bearing (PARA 5-5-4).

e. Using flashlight and inspection mirror, inspect spherical and thrust bearings for laminate separations through 2-inch diameter inspection holes as follows:

NOTE

Both inner and outer surfaces of spherical bearing must be inspected. Outer surface of thrust bearing must be inspected.

(1) Inspect between rubber layers and metal shims, and elastomeric bearing assembly endplate, and within rubber layers themselves.

CAUTION

Damage to spherical and thrust bearings will result if feeler gage is forced between rubber layers. Do not force feeler gage when inspecting for laminate separations.

(2) Using 0.005-inch feeler gage, determine amount of laminate separations by probing bearing. Carefully insert feeler gage in separation to measure depth and length of separation.

(3) Compare separations on spherical bearing with graph in Figure 5-3-1-3. If any separation extends into shaded area, replace spherical bearing (PARA 5-5-4). **MULTIPLE SEPARATIONS OF 0.061-INCH DEEP OR LESS ARE ALLOWED IN SPHERICAL BEARING IF THEY ARE AT LEAST 5-INCHES APART AROUND CIRCUMFERENCE, AND AT LEAST THREE METAL SHIMS APART.**

(4) Inspect thrust bearing for separations. **UP TO 3/8-INCH DEEP SEPARATION ALLOWED AROUND ENTIRE CIRCUMFERENCE.** If separation exceeds limits, replace thrust bearing (PARA 5-5-4).

(5) If spherical or thrust bearing has total delamination between rubber layer and metal shim endplate, replace spherical or thrust bearing (PARA 5-5-4).

f. Inspect elastomeric bearing assembly for surface cracking or crazing of rubber. **UP TO 0.060-INCH DEEP ALLOWED AROUND ENTIRE CIRCUMFERENCE, BETWEEN ALL LAMINATES.** Small hairline surface cracking or crazing of rubber surface is acceptable. If cracking or crazing exceeds limits, replace spherical or thrust bearing (PARA 5-5-4).

g. Inspect edges of shims for cracks. **NONE ALLOWED.** If crack is found, replace spherical or thrust bearing (PARA 5-5-4).

h. If droop stop pounding is experienced, inspect droop stop lugs on elastomeric bearing assembly endplate for cracks. **NONE ALLOWED.** If crack is found, replace elastomeric bearing assembly (PARA 5-4-5).

i. Inspect main rotor hub arm for signs of contact with spindle nut. **NONE ALLOWED.** If contact is found, replace spindle (PARA 5-4-5).

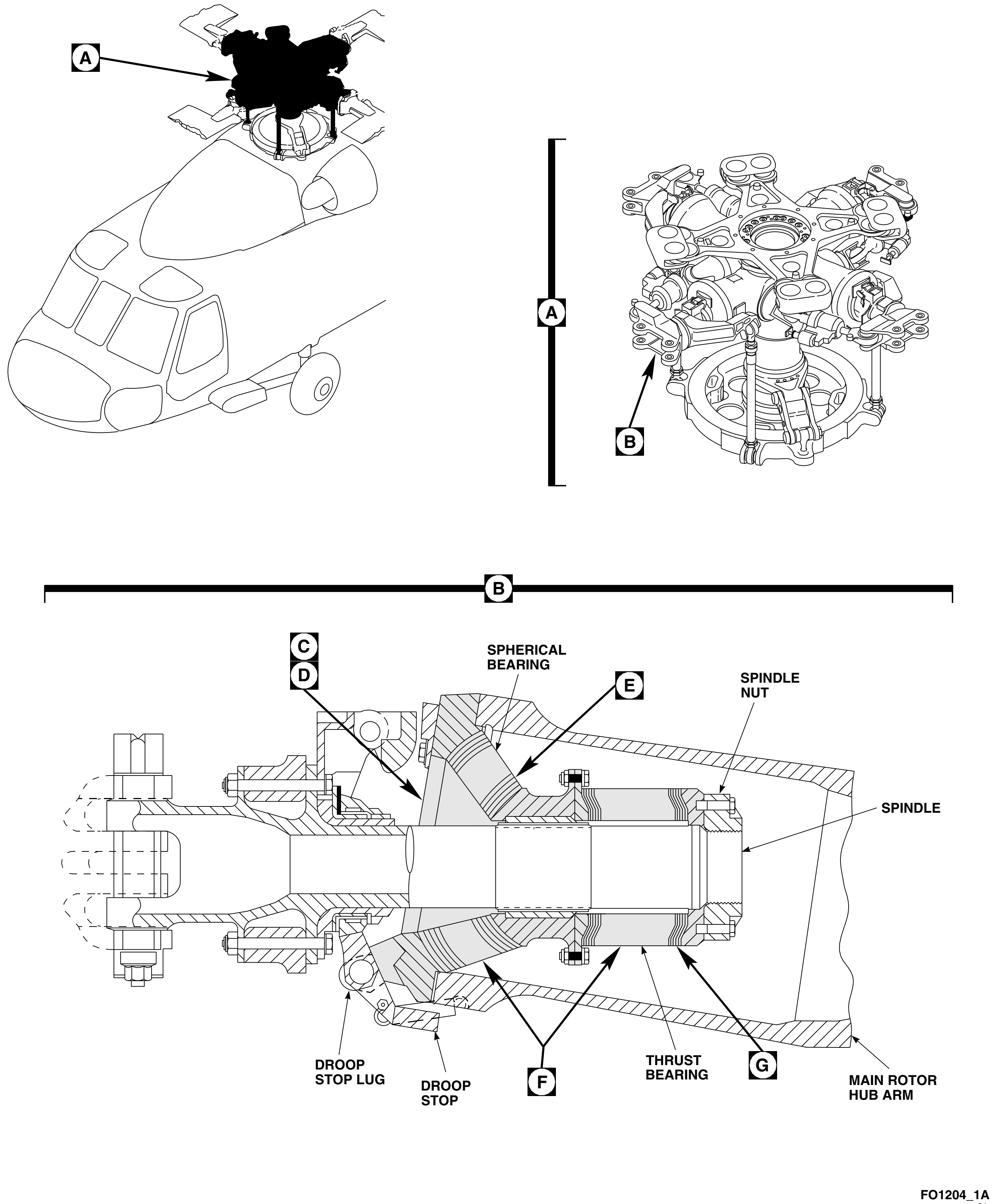


FIGURE 5-3-1-2. INSPECT ELASTOMERIC BEARING ASSEMBLY (SHEET 1 OF 6)

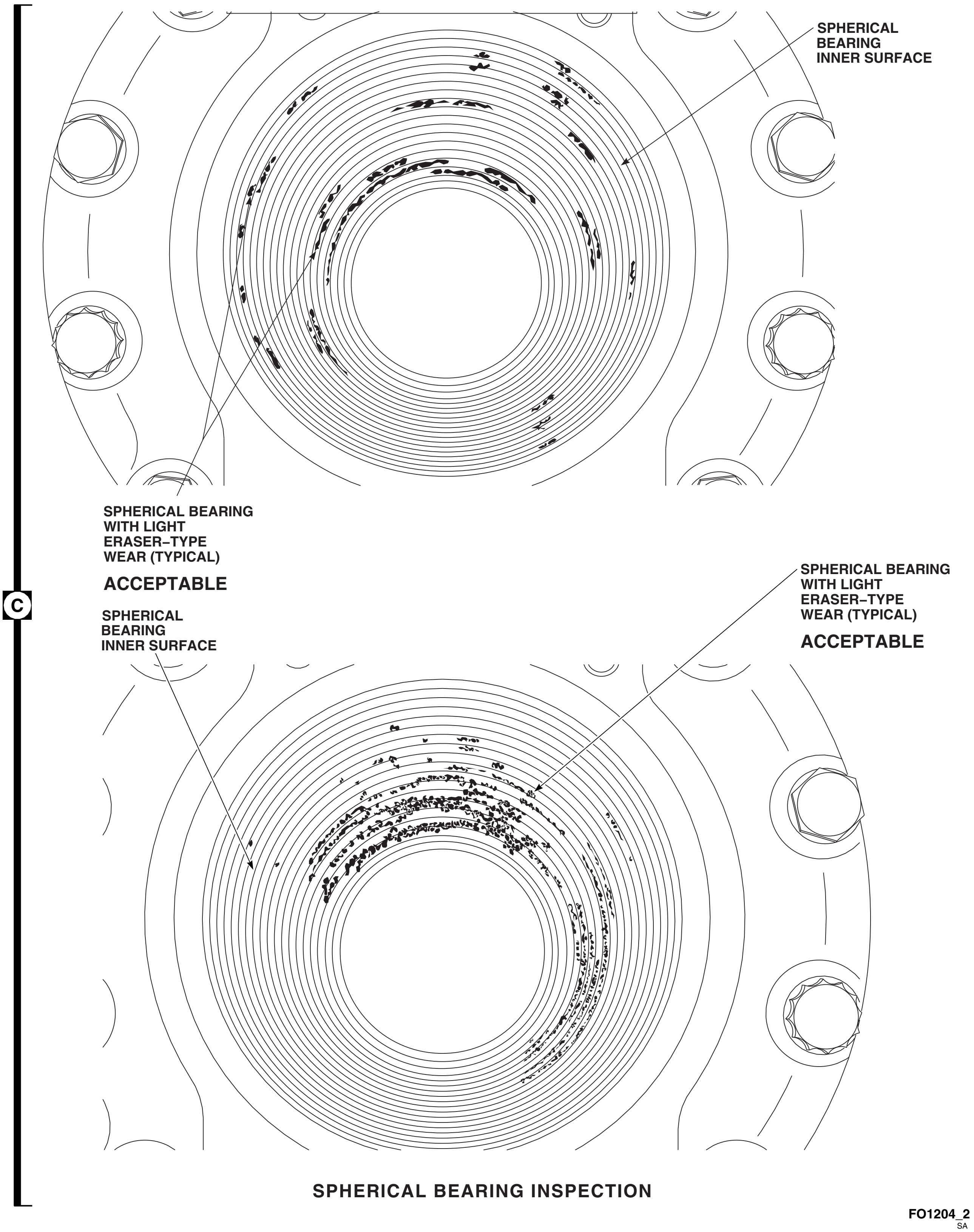


FIGURE 5-3-1-2. INSPECT ELASTOMERIC BEARING ASSEMBLY (SHEET 2 OF 6)

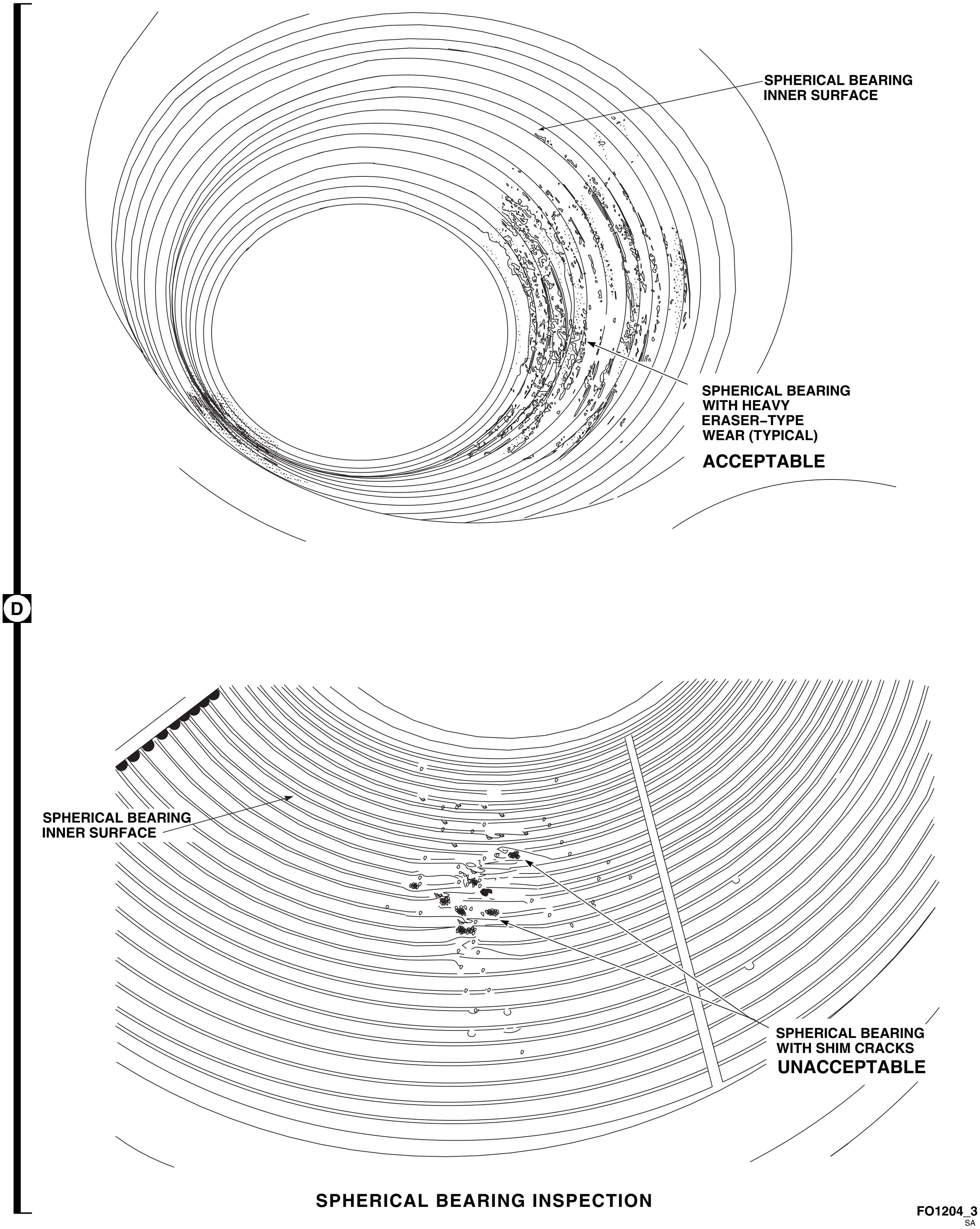
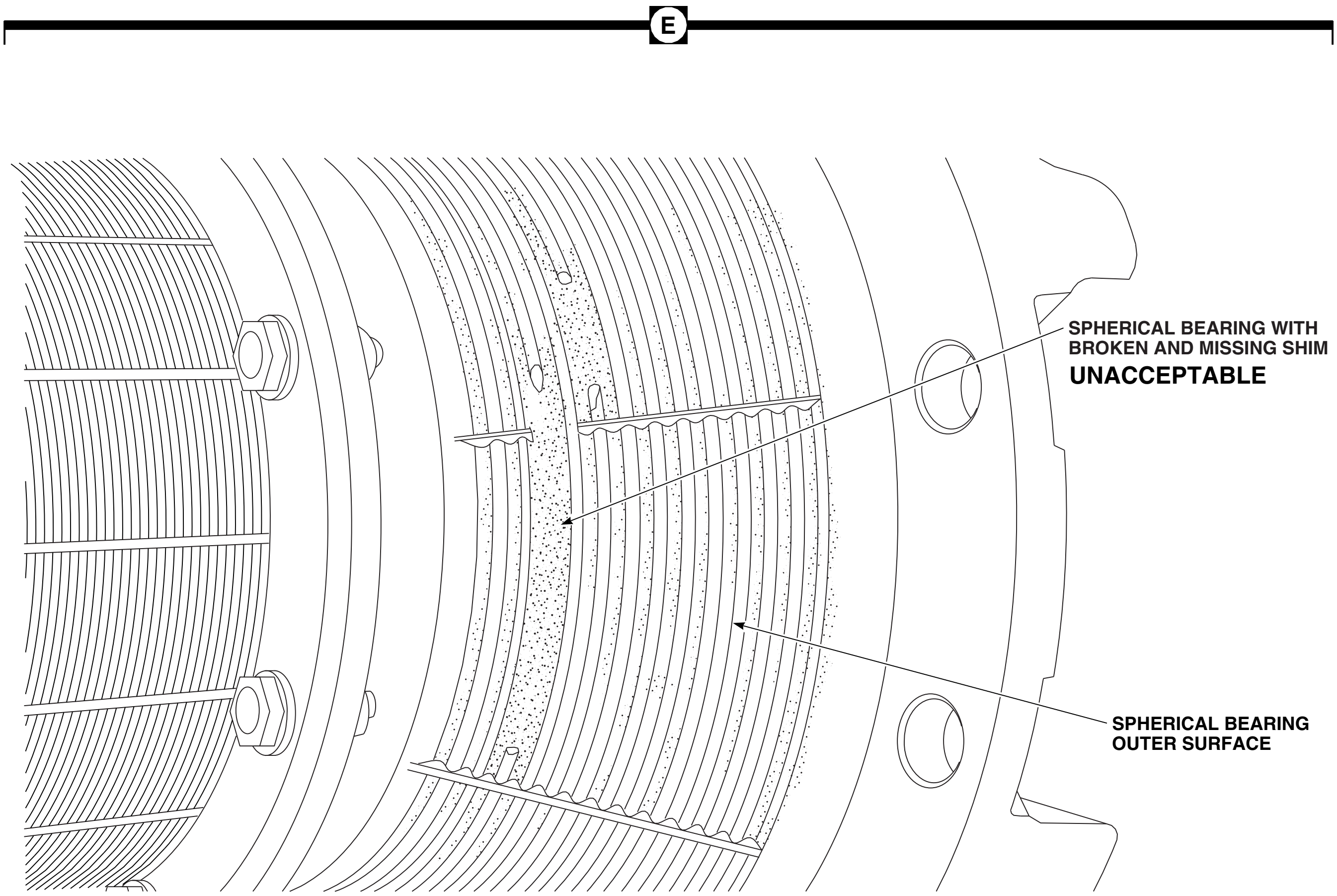


FIGURE 5-3-1-2. INSPECT ELASTOMERIC BEARING ASSEMBLY (SHEET 3 OF 6)



SPHERICAL BEARING INSPECTION

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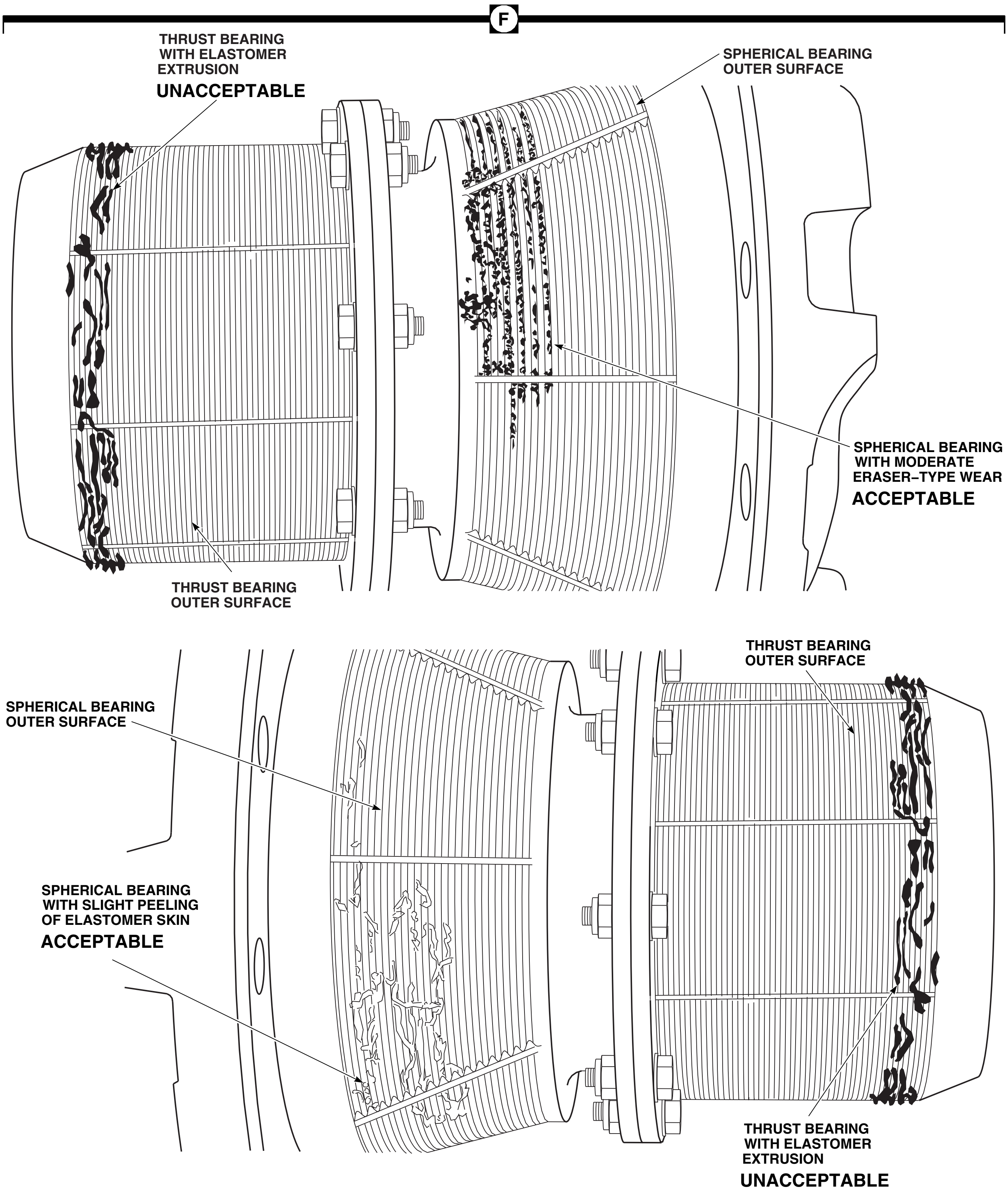
FIGURE 5-3-1-2. INSPECT ELASTOMERIC BEARING ASSEMBLY (SHEET 4 OF 6)



SPHERICAL AND THRUST BEARING INSPECTION

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FIGURE 5-3-1-2. INSPECT ELASTOMERIC BEARING ASSEMBLY (SHEET 5 OF 6)

G

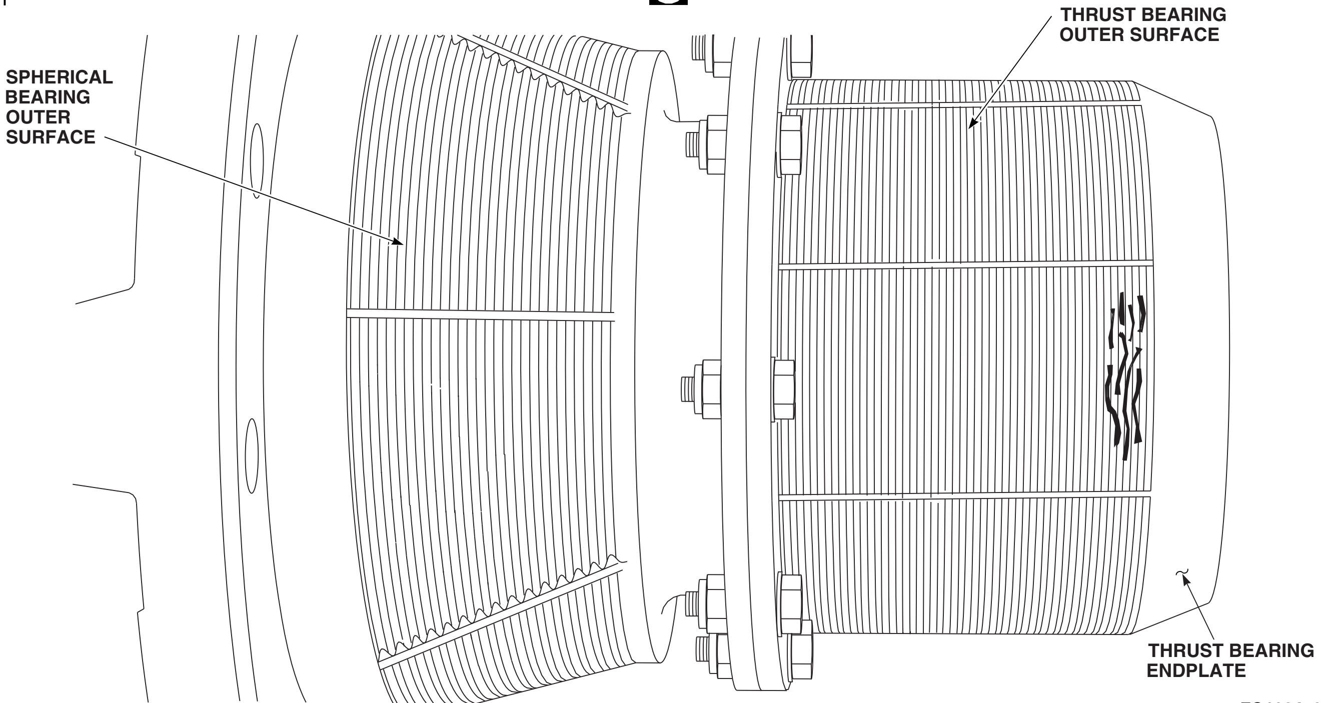
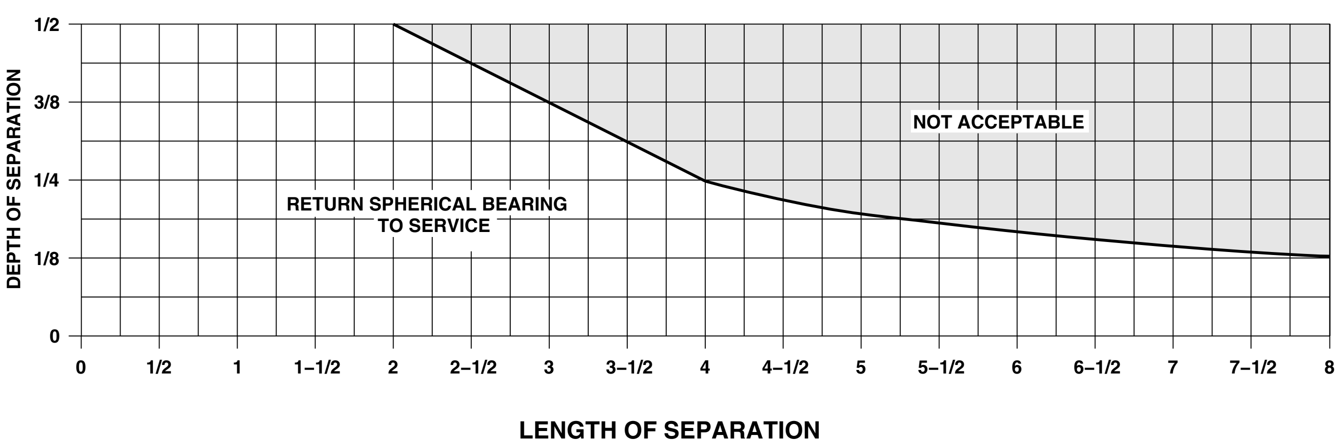


FIGURE 5-3-1-2. INSPECT ELASTOMERIC BEARING ASSEMBLY (SHEET 6 OF 6)



NOTE

ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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FIGURE 5-3-1-3. SPHERICAL BEARING SEPARATION GRAPH

5-3-2 SPINDLE HORN INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Powertrain Repairer's Toolkit

References

- PARA 2-6-5

5-3-2.1 Spindle Horn Inspection.

NOTE

The following procedure applies to spindle horn attachment boltheads only. Inspection is to be performed with bolts installed. Do not remove bolts to perform inspection.

a. Inspect spindle horn attachment boltheads for corner damage. **MAXIMUM CORNER DAMAGE SHALL NOT EXCEED 0.070 × 0.070-INCH.** If damage is beyond limits, replace bolts.

b. If damage is within limits, touch up finish on damaged areas (PARA 2-6-5).

5-3-3 DROOP STOP INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Medium Cut File
- Protection, Eye, Skin, and Breathing

References

- PARA 2-6-5
- PARA 5-4-5
- PARA 5-4-10

5-3-3.1 Droop Stop Inspection.

WARNING

The droop stop is made of beryllium copper, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds and eyes. To prevent injury, personnel must wear eye protection. Clean up visible dust piles and wash hands and arms after maintenance.

a. Visually inspect droop stop for cracks. **NONE ALLOWED, EXCEPT DROOP STOP CAM MAY HAVE SMALL CRACKS (UP TO ⅛-INCH) AT CORNER RADIUS OF IMPACT SURFACES** (Figure 5-3-3-1, Detail B). If cracks exceed limit, replace droop stop cam (PARA 5-4-10).

b. Inspect droop stop cam impact surfaces for deformation. **DEFORMATION UP TO 0.10-INCH IS ALLOWED.** If deformation exceeds limit, replace droop stop cam (PARA 5-4-10).

c. Inspect droop stop cam for corner damage. **CORNER DAMAGE UP TO ⅛-INCH IS ALLOWED AFTER REMOVING SHARP EDGES USING MEDIUM CUT FILE.** If corner damage exceeds limit, replace droop stop cam (PARA 5-4-10).

d. Touch up paint where required (PARA 2-6-5).

e. Visually inspect spring and attaching hardware for signs of corrosion. **NONE ALLOWED.** If corrosion is found, replace corroded parts (PARA 5-4-10).

f. If droop stop pounding is experienced, inspect droop stop lugs on elastomeric bearing endplate assembly for cracks. **NONE ALLOWED.** If crack is found, replace elastomeric bearing endplate assembly (PARA 5-4-5).

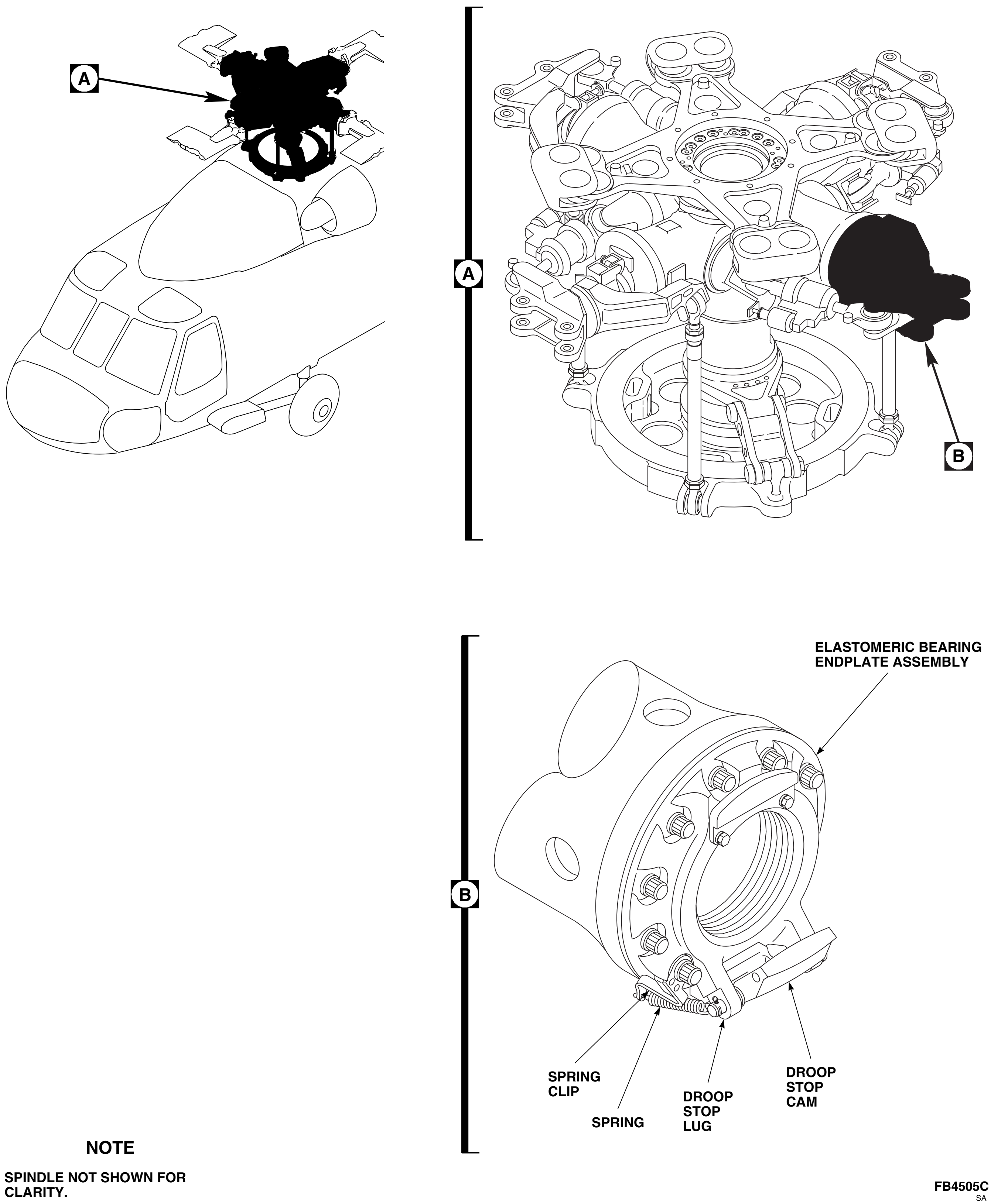


FIGURE 5-3-3-1. INSPECT DROOP STOP

5-3-4 DAMPER INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator (with Clamp)
- Teflon Lined Spherical Bearing Plastic Feeler Gage, Locally-Made, Figure H-209, Appendix H
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Scouring Pad, Item 270, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- PARA 1-6-18
- PARA 5-4-13
- PARA 5-4-14
- PARA 5-4-16
- PARA 5-5-5

5-3-4.1 Damper Indicator Inspection.

a. Inspect damper indicator gold band for wear or scoring. As long as level of servicing can be seen, damper indicator is usable.

b. Inspect for sticky damper indicator piston as follows:

NOTE

Too much bleeding will result in fluid loss and damper will require servicing.

(1) Slightly open each bleed plug on damper, one at a time.

(2) If any trapped air or small amount of fluid is expelled, damper indicator is usable.

(3) If no fluid or air comes out and damper indicator piston does not move, repair damper indicator (PARA 5-5-5).

c. Inspect damper indicator for leakage. **VISIBLE LEAKAGE OF ONE DROP OR MORE PER HOUR IS NOT ACCEPTABLE.** If leakage exceeds limits, repair (PARA 5-5-5) or replace (PARA 5-4-14) damper indicator.

5-3-4.2 Damper Leakage Inspection.

a. If damper indicator goes from full indication (full gold band visible) to refill indication (no gold visible) in less than 10 flight hours, inspect damper and damper indicator for leakage. Replace damper (PARA 5-4-13) or damper indicator (PARA 5-4-14) as required.

b. Inspect damper piston rod for continuous leakage. **VISIBLE LEAKAGE OF ONE DROP OR MORE PER HOUR IS NOT ACCEPTABLE.** If leakage exceeds limits, replace damper (PARA 5-4-13).

c. Inspect damper indicator for continuous leakage. **VISIBLE LEAKAGE OF ONE DROP OR MORE PER HOUR IS NOT ACCEPTABLE.** If leakage exceeds limits, replace damper indicator (PARA 5-4-14).

d. If there is leakage at damper cylinder head check valve, monitor damper indicator. If damper indicator goes from full indication (full gold band visible) to refill indication (no gold band visible) by next 10 hour/14 day inspection, replace damper (PARA 5-4-13).

5-3-4.3 Damper Bearing Inspection.

5-3-4.3.1 Bearing Surface Inspection.

NOTE

Accumulation of reddish-brown powder on spherical bearing is normal for Teflon bearing and is not cause for removal.

a. Visually inspect inboard and outboard damper bearings for gouges (Figure 5-3-4-1). **GOUGES MUST NOT EXCEED 0.040-INCH DEEP ON ½ OR MORE OF CIRCUMFERENCE AREA ON EDGE OF OUTER RACE.** If damage exceeds limits, replace damper bearing and nylon washer (PARA 5-4-16).

b. Inspect damper bearing surface for corrosion as follows:

CAUTION

Damage to Teflon liner will result if solvent is used to clean damper bearings. Do not use solvent to clean damper bearings.

(1) Using scouring pad, Item 270, Appendix D, remove minor surface corrosion.

(2) Inspect spherical bearing (ball) surface for corrosion pitting. Replace damper bearing (PARA 5-4-16) if:

(a) **FIVE OR MORE PITS EXCEED 0.030-INCH DIAMETER IN A ½-INCH DIAMETER CIRCLE.**

(b) **ANY PIT IS LARGER THAN 0.050-INCH.**

5-3-4.3.2 Damper Bearing Feeler Gage Wear Inspection.

a. Locally make Teflon lined spherical bearing plastic feeler gage (Figure H-209, Appendix H).

NOTE

Inspect at least half of each spherical bearing closest to damper body.

b. Inspect inboard and outboard damper bearings as follows:

(1) With assistant, position main rotor blade over nose.

(2) If not already done, engage gust lock (PARA 1-6-18).

(3) Grasp main rotor blade and apply load to blade in direction of full lead.

(4) Push damper toward side of damper bearings to be inspected (Figure 5-3-4-1).

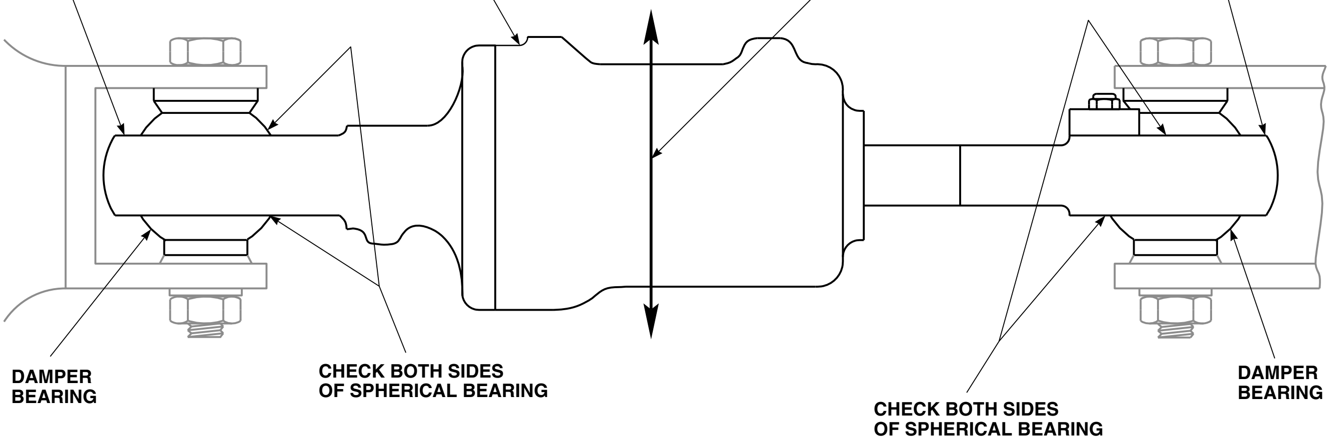


FIGURE 5-3-4-1. INSPECT DAMPER BEARINGS

CAUTION

- Damage to rod end spherical bearing will result if metallic feeler gage is used. Teflon race will be damaged. Use only Teflon lined spherical bearing plastic feeler gage when inspecting rod end spherical bearing.
- Damage to rod end spherical bearing will result if feeler gage is not inserted carefully. Teflon race will be damaged. Use care when inserting feeler gage.

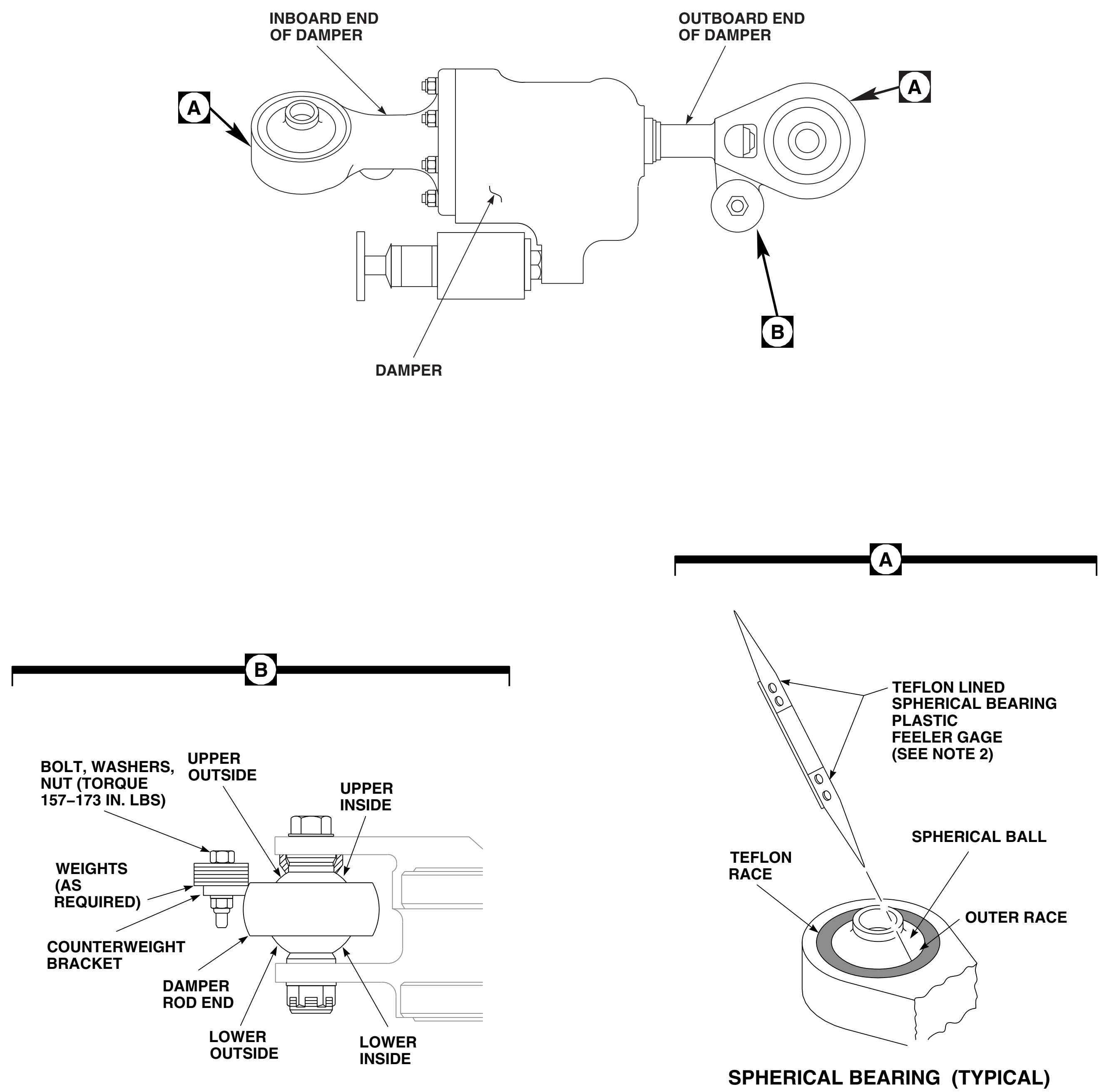
(5) Using feeler gage (0.010-inch end), probe both sides of damper bearing around entire circumference for clearance between spherical ball and outer race (Figure 5-3-4-2, Detail A).

(6) If feeler gage (0.010-inch end) cannot be inserted to 0.250-inch mark, damper bearing is acceptable.

(7) If feeler gage (0.010-inch end) can be inserted to 0.250-inch mark, or more, probe entire circumference again using 0.015-inch end of feeler gage.

(8) If feeler gage (0.015-inch end) cannot be inserted to 0.250-inch mark between spherical bearing and outer race, damper bearing is acceptable.

(9) If feeler gage (0.015-inch end) can be inserted between spherical ball and outer



NOTES

1. DAMPER SHOWN REMOVED FOR CLARITY.
2. PROBE BOTH SIDES OF SPHERICAL BEARING(S) AROUND ENTIRE CIRCUMFERENCE FOR CLEARANCE BETWEEN SPHERICAL BALL AND OUTER RACE.

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FIGURE 5-3-4-2. INSPECT DAMPER BEARINGS

race to 0.250-inch mark, inspect damper bearing for radial play using dial indicator method.

(10) Move damper in opposite direction and repeat inspection for opposite side of damper bearing.

NOTE

- Accumulation of reddish-brown powder on spherical bearing is normal for Teflon bearing and is not cause for removal.
- Metal-to-metal contact is present if worn or discolored spots are seen on load damper bearing surface of spherical bearing (ball), or if there is a clicking sound when spherical bearing (ball) is rotated in its outer race.

c. Inspect inboard and outboard damper bearings for evidence of metal-to-metal contact between spherical bearing (ball) and outer race. **NONE ALLOWED.** If either damper bearing shows any metal-to-metal contact between spherical bearing (ball) and outer race, replace damper bearing (PARA 5-4-16).

NOTE

When operating helicopter with high gross weights or external sling load operation, damper rod end bearing and nylon washer contact may occur.

d. If there is evidence of contact on lower outside or upper inside area of rod end bearing, remove one weight from counterweight bracket as follows (Figure 5-3-4-2, Detail B):

(1) Remove bolt, washers, and nut securing weights to counterweight bracket.

(2) Remove one weight from counterweight bracket.

(3) Install bolt, washers, and nut securing weights to counterweight bracket.

(4) **TORQUE NUT TO 157 - 173 INCH-POUNDS.**

(5) Apply bead of sealing compound, Item 292, Appendix D, around bolthead and nut.

NOTE

When operating helicopter with high gross weights or external sling load operation, damper rod end bearing and nylon washer contact may occur.

e. If there is evidence of contact on lower inside or upper outside area of rod end bearing, add one weight to counterweight bracket as follows:

(1) Remove bolt, washers, and nut securing weights to counterweight bracket.

(2) Add one weight to counterweight bracket.

(3) Install bolt, washers, and nut securing weights to counterweight bracket.

(4) **TORQUE NUT TO 157 - 173 INCH-POUNDS.**

(5) Apply bead of sealing compound, Item 292, Appendix D, around bolthead and nut.

f. If required, disengage gust lock (PARA 1-6-18).

5-3-4.3.3 Radial and Axial Play Inspection.

a. Clamp dial indicator to inboard end of damper (Figure 5-3-4-3).

b. Position main rotor blade over nose.

NOTE

Do not pull main rotor blade down more than 6-inches.

c. Grasp main rotor blade and apply load to blade in lead and lag direction. Do not pull main rotor blade down more than 6-inches. **MAXIMUM ALLOWABLE RADIAL PLAY MUST NOT BE MORE THAN 0.023-INCH.** If play exceeds limit, replace damper bearing (PARA 5-4-16).

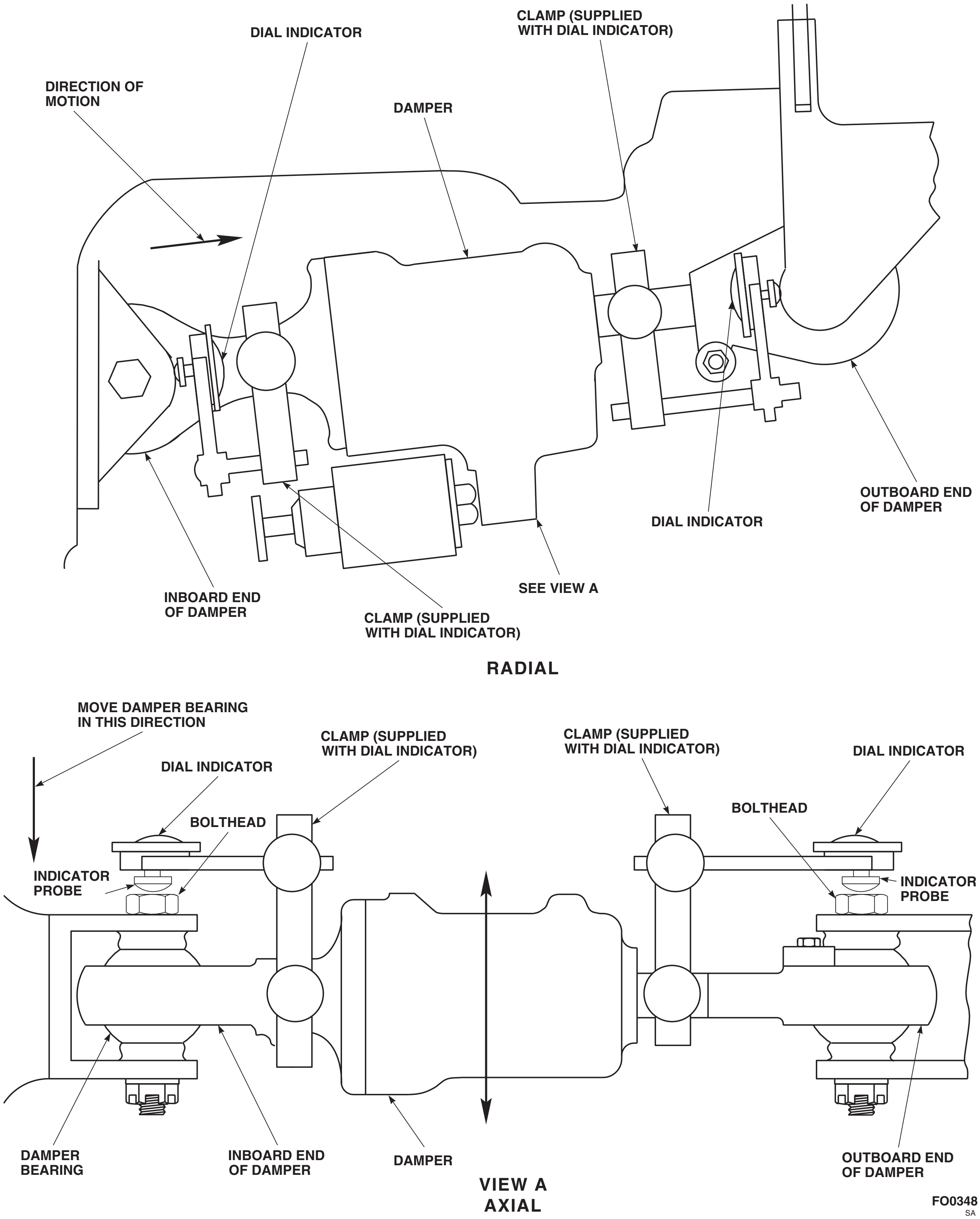


FIGURE 5-3-4-3. INSPECT DAMPER BEARINGS

d. Repeat inspection on outboard end of damper.

e. Clamp dial indicator to inboard end of damper, with indicator probe against bolthead.

f. Hold damper from turning and move damper bearing in direction shown. **MAXIMUM ALLOWABLE AXIAL PLAY AS SHOWN ON DIAL INDICATOR MUST NOT BE MORE THAN 0.042-INCH.** If play exceeds limit, replace damper bearing (PARA 5-4-16).

g. Repeat step a. through step f. on outboard end of damper.

5-3-5 PITCH CONTROL ROD INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Elastomeric Rod End Bearing Plastic Feeler Gage, Locally-Made, Figure H-232, Appendix H
- Powertrain Repairer's Toolkit
- Teflon Lined Spherical Bearing Plastic Feeler Gage, Locally-Made, Figure H-209, Appendix H

Materials/Parts

- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- Appendix H
- PARA 1-6-15
- PARA 1-6-16
- PARA 5-4-18
- PARA 5-4-19
- PARA 5-4-20

5-3-5.1 Rod End Spherical and Elastomeric-Bearing Surfaces Inspection.

a. Inspect upper and lower surfaces of pitch control rod end spherical bearing for corrosion by performing the following:

CAUTION

Damage to Teflon liner will result if solvent is used to clean bearings. Do not use solvent to clean bearings.

(1) Using scouring pad, Item 270, Appendix D, remove surface corrosion.

(2) Inspect spherical ball surface for corrosion pitting. **IF FIVE OR MORE PITS EXCEED 0.030-INCH DIAMETER IN A ½-INCH-DIAMETER CIRCLE, OR IF ANY PIT IS LARGER THAN 0.050-INCH, REPLACE PITCH CONTROL ROD END SPHERICAL BEARING** (PARA 5-4-19).

b. Inspect upper and lower pitch control rod end spherical bearings for gouges. **IF PITCH**

CONTROL ROD END SPHERICAL BEARING HAS GOUGES GREATER THAN 0.040-INCH DEEP ON ½ OR MORE OF THE CIRCUMFERENCE AREA ON EDGE OF OUTER RACE, REPLACE PITCH CONTROL ROD END SPHERICAL BEARING (PARA 5-4-19).

c. Inspect rod end elastomeric bearing surfaces as follows:

NOTE

- Elastomeric rod ends consist of laminated elastomer and steel shells. Wear of elastomer in bearings is a gradual process that will take several hundreds of hours from first detection of elastomer damage to point where rod end must be replaced.
- A wax-like substance on rubber surface is normal and should not be removed.
- Types of wear shown include eraser-type wear, elastomer extrusion, and shim cracks. Description of wear and general guidelines are as follows:

  Eraser-type wear: a pile-up of shredded rubber similar to the result of using a soft eraser.

  Elastomer extrusion: the squeezing out of the elastomer from between the elastomeric bearing laminations, and inner or outer members, indicating disbonding of elastomer. Peeling of elastomer mold lines which run radially across laminations is acceptable.

(1) Inspect for a concentration or pileup of eraser-type wear. A concentration or pileup of eraser-type wear may indicate an elastomeric crack or separation may exist. If a concentration or pileup of eraser-type wear is found, inspect rod end for elastomeric crack or separation (PARA 5-3-5.2).

(2) If an elastomer extrusion is detected, inspect rod end for elastomeric crack or separation (PARA 5-3-5.2).

(3) Visually inspect for cracking or disbonding of elastomer from shim inner or outer members. Cracking or disbonding of elastomer from shim inner or outer members will cause separation. If cracking or disbonding of elastomer from shim inner or outer members is suspected, inspect elastomeric rod end (PARA 5-3-5.2).

(4) Visually inspect all rod end elastomeric bearings for signs of inner member separation, using hand force, shaking pitch control rod. **NO INNER MEMBER LOOSENESS OR EXPOSED METAL OF INNER BALL SURFACE OR MISSING ELASTOMER ALLOWED.** If inner member separation is beyond limits, replace elastomeric rod end (PARA 5-4-20).

NOTE

Shims or steel shells alternate with elastomer layers of rod end elastomeric bearing and are composed of two 180° pieces (Figure 5-3-5-1). Within each shim layer, two pieces are separated by small gap measuring approximately 0.050-inch. These gaps are filled with elastomer during manufacture, and are evident on surfaces as a small lump of elastomer. Gaps becoming exposed over time are acceptable.

(5) Inspect for cracks or gaps in areas other than manufactured shim gaps. **NONE ALLOWED.** If cracks are found, or gaps exist, replace elastomeric rod end (PARA 5-4-20).

5-3-5.2 Rod End Elastomeric Bearing Feeler Gage Wear Inspection.

NOTE

Feeler gage inspection should only be performed if possible separation exists based on visual inspection of bearing surfaces.

a. Inspect rod end elastomeric bearing surfaces (PARA 5-3-5.1).

b. Locally make elastomeric rod end bearing plastic feeler gage (Figure H-232, Appendix H).

c. Inspect upper and lower rod end elastomeric bearings as follows (Figure 5-3-5-1):

CAUTION

- Damage to elastomeric rod ends will result if metallic feeler gage is used. Elastomer will be damaged. Use only plastic feeler gage when inspecting.
- Damage to elastomeric rod ends will result if feeler gage is forced between elastomeric layers. Do not force feeler gage when inspecting for laminate separations.

(1) On upper and lower rod end elastomeric bearings, using 0.005-inch end of feeler gage, inspect for elastomer/metal separations or disbonding of elastomer from shim inner or outer members in all areas carefully. If feeler gage can be inserted, separation has occurred. Using 0.005-inch end of feeler gage, measure depth of separations. **SMALL HAIRLINE SURFACE CRACKING, MINOR DISBONDING, AND CRAZING OF ELASTOMER UP TO 0.060-INCH ARE ACCEPTABLE IN ALL AREAS OF ROD END ELASTOMERIC BEARING.**

(2) On upper rod end elastomeric bearings, using 0.005-inch end of feeler gage, inspect for elastomer/metal separations or disbonding of elastomer from shim inner or outer members in zone I carefully. If feeler gage can be inserted, separation has occurred. Perform the following:

(a) Using 0.005-inch end of feeler gage, measure depth of separations. **SEPARATIONS UP TO 0.40-INCH IN ZONE I AND UP TO 0.060-INCH OUTSIDE ZONE I ARE ACCEPTABLE.**

(b) If separation is beyond limits, replace elastomeric rod end within next 10 flight hours (PARA 5-4-20).

(3) On lower rod end elastomeric bearings, using 0.005-inch end of feeler gage, inspect for elastomer/metal separations or disbonding of elastomer from shim inner or outer members in zone I carefully. If feeler gage can be inserted, separation has occurred. Perform the following:

(a) Using 0.005-inch end of feeler gage, measure depth of separations. **SEPARATIONS UP TO 0.20-INCH IN ZONE I AND UP TO 0.060-INCH OUTSIDE ZONE I ARE ACCEPTABLE.**

(b) If separation is beyond limits, replace elastomeric rod end (PARA 5-4-20).

5-3-5.3 Rod End Spherical Bearing Feeler Gage Wear Inspection.

a. Inspect bearing surfaces (PARA 5-3-5.1).

b. Locally make Teflon lined spherical bearing plastic feeler gage (Figure H-209, Appendix H).

c. Inspect upper and lower pitch control rod end spherical bearings as follows:

(1) Apply hand force to pitch control rod on opposite side of bearing that probing will be done (Figure 5-3-5-2, Sheet 1 or Sheet 2).

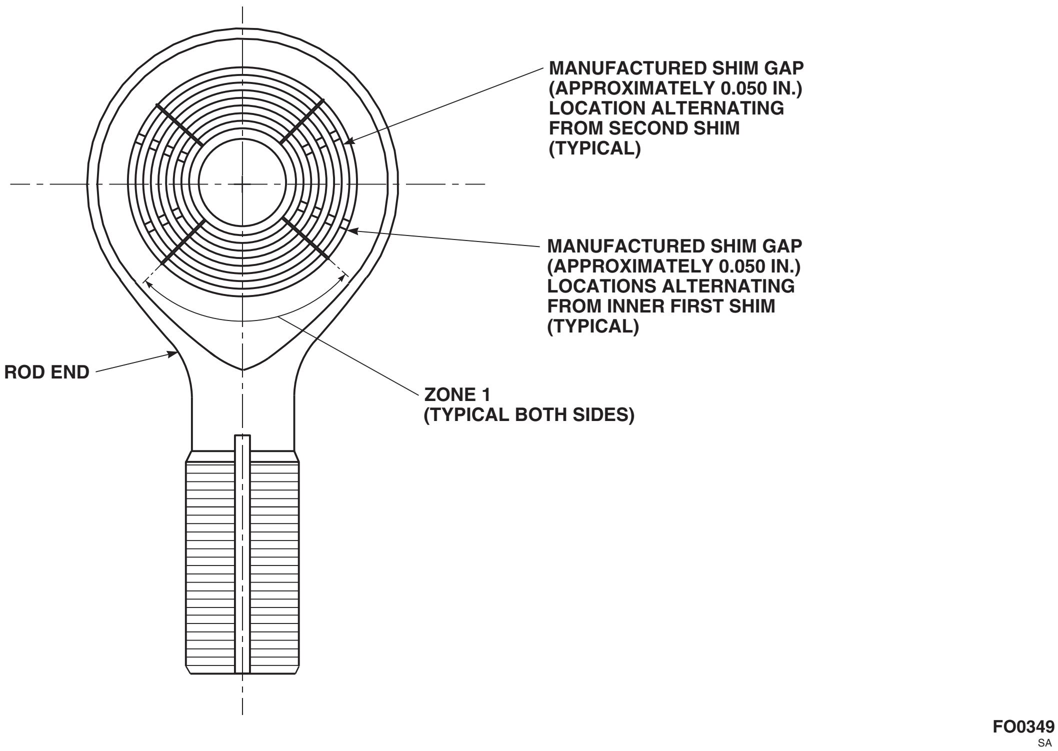


FIGURE 5-3-5-1. INSPECT ROD END ELASTOMERIC BEARINGS

CAUTION

- Damage to pitch control rod end spherical bearing will result if metallic feeler gage is used. Teflon race will be damaged. Use only plastic feeler gage when inspecting.
- Damage to pitch control rod end spherical bearing will result if feeler gage is not inserted carefully. Teflon race will be damaged. Use care when inserting feeler gage.

(2) Using 0.010-inch end of feeler gage, probe both sides of pitch control rod end spherical bearing around entire circumference for clearance between spherical ball and outer race.

(3) If feeler gage (0.010-inch end) cannot be inserted to 0.250-inch mark, pitch control rod end spherical bearing is acceptable.

(4) If feeler gage (0.010-inch end) can be inserted to 0.250-inch mark, or more, probe entire circumference again using 0.015-inch end of feeler gage.

(5) If feeler gage (0.015-inch end) cannot be inserted to 0.250-inch mark between spherical ball and outer race, pitch control rod end spherical bearing is acceptable. Repeat inspection every 15 flight hours.

(6) If 0.015-inch end of feeler gage can be inserted between spherical ball and outer race to 0.250-inch mark, perform the following:

(a) Disconnect upper or lower end of pitch control rod (PARA 5-4-18).

CAUTION

Damage to Teflon bearings will result if bearings are lubricated. Do not lubricate Teflon bearings.

NOTE

- An accumulation of reddish-brown powder on spherical bearing is normal for a Teflon bearing and is not cause for removal.
- Ratcheting indicates worn Teflon and metal-to-metal contact.

(b) Check for ratcheting by rotating spherical ball against outer race. **NONE ALLOWED.** If ratcheting is found, replace pitch control rod end spherical bearing (PARA 5-4-19) and pitch control rod end (PARA 5-4-20). If ratcheting does not exist, replace pitch control rod end spherical bearing only (PARA 5-4-19).

(c) Connect upper or lower end of pitch control rod (PARA 5-4-18).

d. Apply hand force to opposite side of pitch control rod and repeat step c.

NOTE

Metal-to-metal contact is present if worn or discolored spots are seen on load bearing surface of spherical ball.

e. Inspect upper and lower pitch control rod end spherical bearings for evidence of metal-to-metal contact between spherical ball and outer race. **NONE ALLOWED.** If metal-to-metal contact is found, replace pitch control rod end spherical bearings (PARA 5-4-19).

5-3-5.4 Pitch Control Rod Movement Inspection.

a. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

b. Inspect pitch control rod for movement (looseness) as follows:

(1) Firmly grasp control rod with two hands.

NOTE

- Make sure both ends of pitch control rod are inspected for movement.
- Aggressive repeated force must be used to detect movement.
- When inspecting for movement, pay particular attention to the area where locknut and link meet.

(2) Apply force to control rod link in all directions. While applying force, inspect for movement between link, locknut, and rod end. **NO MOVEMENT ALLOWED.**

c. If movement is detected, inspect upper and/or lower rod end, link, and locknut(s) (PARA 5-4-20).

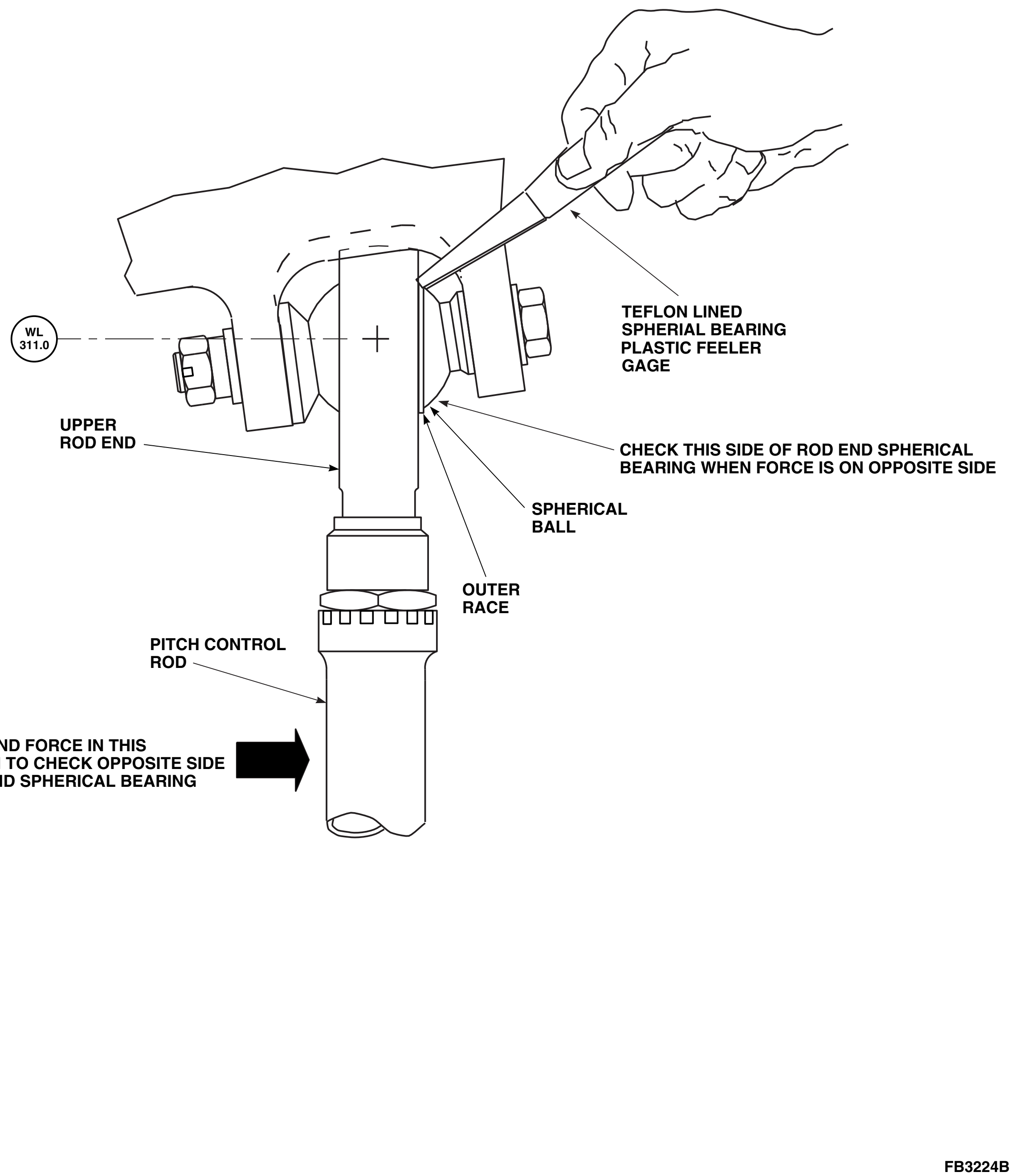


FIGURE 5-3-5-2. INSPECT PITCH CONTROL ROD END SPHERICAL BEARINGS (SHEET 1 OF 2)

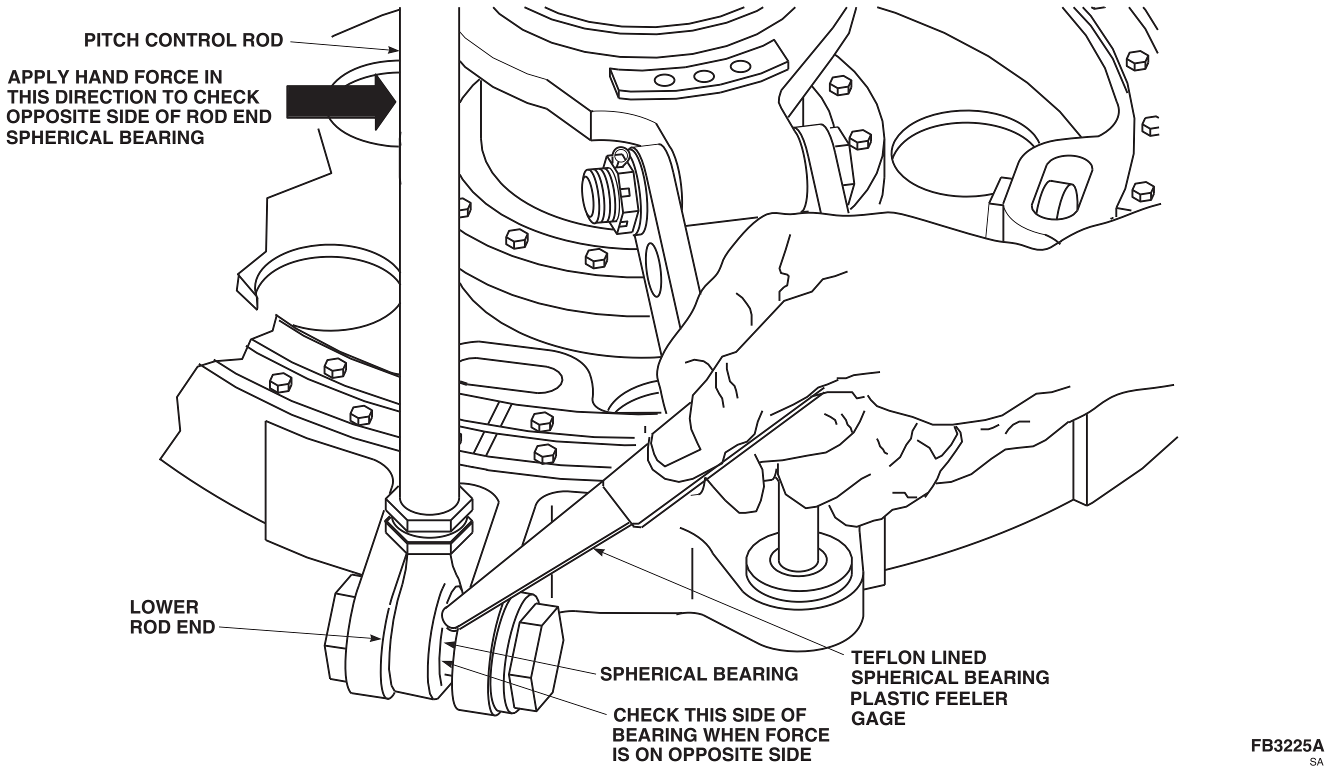


FIGURE 5-3-5-2. INSPECT PITCH CONTROL ROD END SPHERICAL BEARINGS (SHEET 2 OF 2)

5-3-6 SWASHPLATE INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator (with Clamp)
- Feeler Gage
- Flashlight
- Inspection Mirror
- Powertrain Repairer's Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry Lubricant, Item 123A, Appendix D
- Scouring, Pad, Item 270, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-15
- PARA 5-4-18
- PARA 5-4-22
- PARA 5-4-23
- PARA 5-4-27
- PARA 5-5-7
- PARA 11-2-2
- PARA 11-4-2
- PARA 11-4-95

5-3-6.1 Swashplate Large Spherical Bearing Inspection.

NOTE

- Swashplate large spherical bearing (uniball) is inspected by checking the condition of spherical surface guide and Teflon liners.
- Accumulation of reddish-brown powder on swashplate large spherical bearing is normal for Teflon bearing and is not cause for removal.
- Swashplate large spherical bearing may create squeaking, groaning, or feedback noises when moved.

a. If swashplate large spherical bearing creates noises when moved, lubricate as follows:

(1) Using clean cheesecloth, Item 90, Appendix D, wipe swashplate large spherical bearing surface, shaft surface, and shaft surfaces of swashplate guide.

(2) Using dry lubricant, Item 123A, Appendix D, apply liberal amount to swashplate large spherical bearing and shaft surface.

(3) Exercise controls to force penetration of dry lubricant.

(4) After exercising controls, reapply dry lubricant to swashplate large spherical bearing surface, shaft surface, and swashplate guide shaft surfaces.

b. Position swashplate to allow maximum exposure of swashplate large spherical bearing.

c. Inspect plated surfaces of swashplate large spherical bearing. If signs of metal-to-metal contact and/or peeling of plating greater than ¼-inch in diameter appear in more than two places, replace swashplate large spherical bearing (PARA 5-5-7).

d. Inspect Teflon surfaces for disbonding or separation from ball or outer race, or for pieces of Teflon coming out of swashplate large spherical bearing. **NONE ALLOWED.** If damage is found, replace swashplate large spherical bearing (PARA 5-5-7).

5-3-6.2 Swashplate Large Spherical Bearing Inner Race Inspection.

NOTE

Note location of swashplate expandable pins for reinstallation.

a. Disconnect swashplate links from swashplate (PARA 11-4-95).

b. Pull swashplate assembly toward self. **IF RATCHETING, CHATTERING, OR BINDING IS FELT IN FLIGHT CONTROLS, REPLACE SWASHPLATE** (PARA 5-4-22).

CAUTION

Damage to Teflon will result if feeler gage is forced between inner race of swashplate large spherical bearing and swashplate guide. Do not force feeler gage between inner race of swashplate large spherical bearing and swashplate guide.

c. Insert feeler gage between swashplate guide and inner race of swashplate large spherical bearing (Figure 5-3-6-1). **GAP MUST NOT EXCEED 0.028-INCH. NO METAL-TO-METAL CONTACT ALLOWED.** If gap exceeds limit, replace swashplate large spherical bearing (PARA 5-5-7).

d. Connect swashplate links to swashplate (PARA 11-4-95).

5-3-6.3 Swashplate Large Spherical Bearing Outer Race Inspection.

a. Set up dial indicator as follows (Figure 5-3-6-1):

(1) Clamp dial indicator to lower pressure plate.

(2) Position indicator probe on swashplate guide as near as possible to inner ring of bolts.

(3) Adjust indicator probe so that it is at 90° angle to main rotor shaft.

b. Disconnect swashplate links from swashplate (PARA 11-4-95).

c. Insert feeler gage between swashplate large spherical bearing and swashplate guide to lock out play from inner race. Feeler gage must be located directly in line with and above indicator probe.

d. Move swashplate assembly laterally by pushing and pulling on swashplate.

e. Measure gap between outer race and swashplate as follows:

(1) Read dial indicator when swashplate is pushed, then read dial indicator when swashplate is pulled.

(2) Gap is difference between two readings. **GAP MUST NOT EXCEED 0.012-INCH. NO METAL-TO-METAL CONTACT ALLOWED.** If gap exceeds limit, replace swashplate large spherical bearing (PARA 5-5-7).

f. Connect swashplate links to swashplate (PARA 11-4-95).

5-3-6.4 Ball Bearings between Stationary and Rotating Swashplate Inspection.

a. Inspect swashplate extruded grease for metal particles. **NONE ALLOWED.**

(1) If extruded grease is free of metal particles, proceed to step b.

(2) If extruded grease has metal particles, replace swashplate (PARA 5-4-22). Tag swashplate with reason for removal and send to approved maintenance facility.

NOTE

A newly lubricated swashplate will be more difficult to rotate. The following inspection should be performed prior to lubricating swashplate.

b. Disconnect pitch control rods from swashplate (PARA 5-4-18).

c. Disconnect rotating scissors from swashplate (PARA 5-4-27).

d. Check swashplate for smooth rotational motion as follows (Figure 5-3-6-2):

(1) Turn rotating swashplate by hand in direction of rotation (counterclockwise) through at least one complete rotation. **SWASHPLATE SHOULD ROTATE FREELY WITH NO SIGNS OF HESITATION, RATCHETING, OR UNEVEN MOVEMENT.**

(2) Turn rotating swashplate by hand in opposite direction of main rotor head rotation (clockwise) through at least one complete rotation. **SWASHPLATE SHOULD ROTATE FREELY WITH NO SIGNS OF HESITATION, RATCHETING, OR UNEVEN MOVEMENT.**

(3) If main rotor swashplate turns freely, proceed to step e.

(4) If swashplate does not turn freely or has signs of hesitation, ratcheting, or uneven movement, replace swashplate (PARA 5-4-22). Tag swashplate with reason for removal and send to approved maintenance facility.

e. Connect pitch control rods to swashplate (PARA 5-4-18).

f. Connect rotating scissors to swashplate (PARA 5-4-27).

g. Check ball bearings for wear as follows:

(1) Turn on all hydraulic power (PARA 1-6-15).

(2) Setup dial indicator as follows:

(a) Clamp dial indicator to rotating swashplate.

(b) Position indicator probe on stationary swashplate.

(c) Connect dial indicator as close to ball bearing as possible.

(3) Move collective stick through its normal range (PARA 11-2-2). Measure vertical movement between rotating and stationary swashplates. **MAXIMUM RELATIVE VERTICAL MOTION IS 0.010-INCH.** If relative vertical motion exceeds limit, replace swashplate (PARA 5-4-22).

(4) Remove dial indicator.

(5) Turn off all hydraulic power (PARA 1-6-15).

5-3-6.5 Swashplate Scissors Attachment Spherical Bearing Inspection.

a. Check surface of swashplate scissors attachment spherical bearing for corrosion as follows:

(1) Disconnect rotating scissors lower link from swashplate scissors attachment spherical bearing (PARA 5-4-27).

CAUTION

Damage to Teflon lining will result if solvent comes in contact with swashplate scissors attachment spherical bearings. Do not use solvent to clean swashplate scissors attachment spherical bearings.

(2) Using scouring pad, Item 270, Appendix D, remove all surface corrosion.

(3) Check spherical ball surface for corrosion pitting. **REPLACE SWASHPLATE SCISSORS ATTACHMENT SPHERICAL BEARING** (PARA 5-4-23) **IF**:

(a) Five or more pits are visible in a ½-inch diameter circle.

(b) Any pit is larger than 0.050-inch.

(4) Connect rotating scissors lower link to swashplate scissors attachment spherical bearing (PARA 5-4-27).

b. Check for radial play in bearing connecting rotating scissors lower link to swashplate as follows (Figure 5-3-6-3):

(1) Turn on all hydraulic power (PARA 1-6-15).

(2) Install rigging pin in cyclic stick (PARA 11-4-2).

(3) Place collective stick in full down position.

(4) Clamp dial indicator to rotating scissors lower link.

(5) Move rotating scissors lower link in direction shown (radial) and measure play. **MAXIMUM RADIAL PLAY IS 0.015-INCH.** If radial play exceeds limit, replace swashplate scissors attachment spherical bearing (PARA 5-4-23).

c. Check for axial play as follows:

(1) Clamp dial indicator to rotating scissors lower link.

(2) Move rotating scissors lower link in direction shown (axial) and measure play. **MAXIMUM AXIAL PLAY IS 0.038-INCH.** If axial play exceeds limit, replace swashplate scissors attachment spherical bearing (PARA 5-4-23).

(3) Remove rigging pin from cyclic stick (PARA 11-4-2).

(4) Turn off all hydraulic power (PARA 1-6-15).

5-3-6.6 Swashplate Expandable Pin Nuts Inspection.

Using flashlight and inspection mirror, visually inspect swashplate expandable pin nut for cracks or

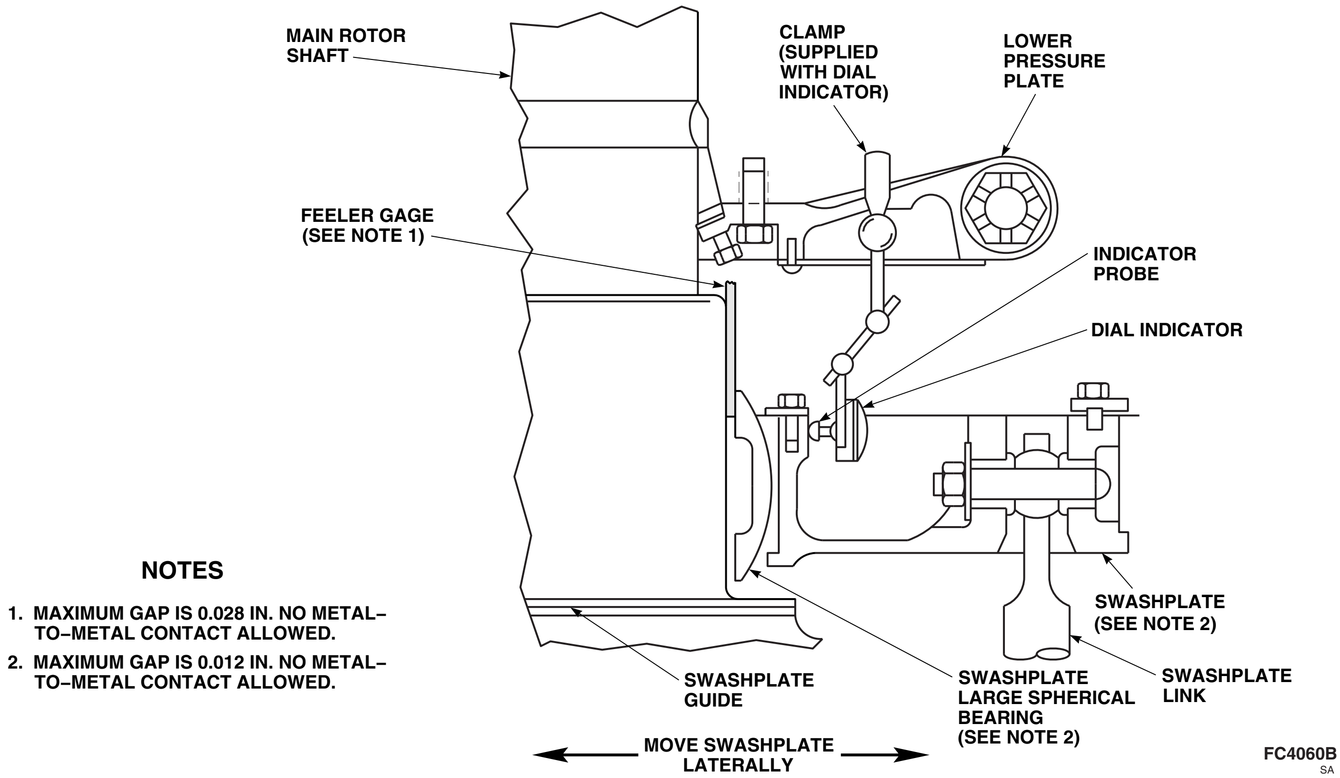


FIGURE 5-3-6-1. INSPECT SWASHPLATE LARGE SPHERICAL BEARING INNER AND OUTER RACE

fractures (Figure 5-3-6-4, Detail A). **NONE ALLOWED.** If crack or fracture is found, replace swashplate expandable pin and swashplate expandable pin nut (PARA 5-4-22).

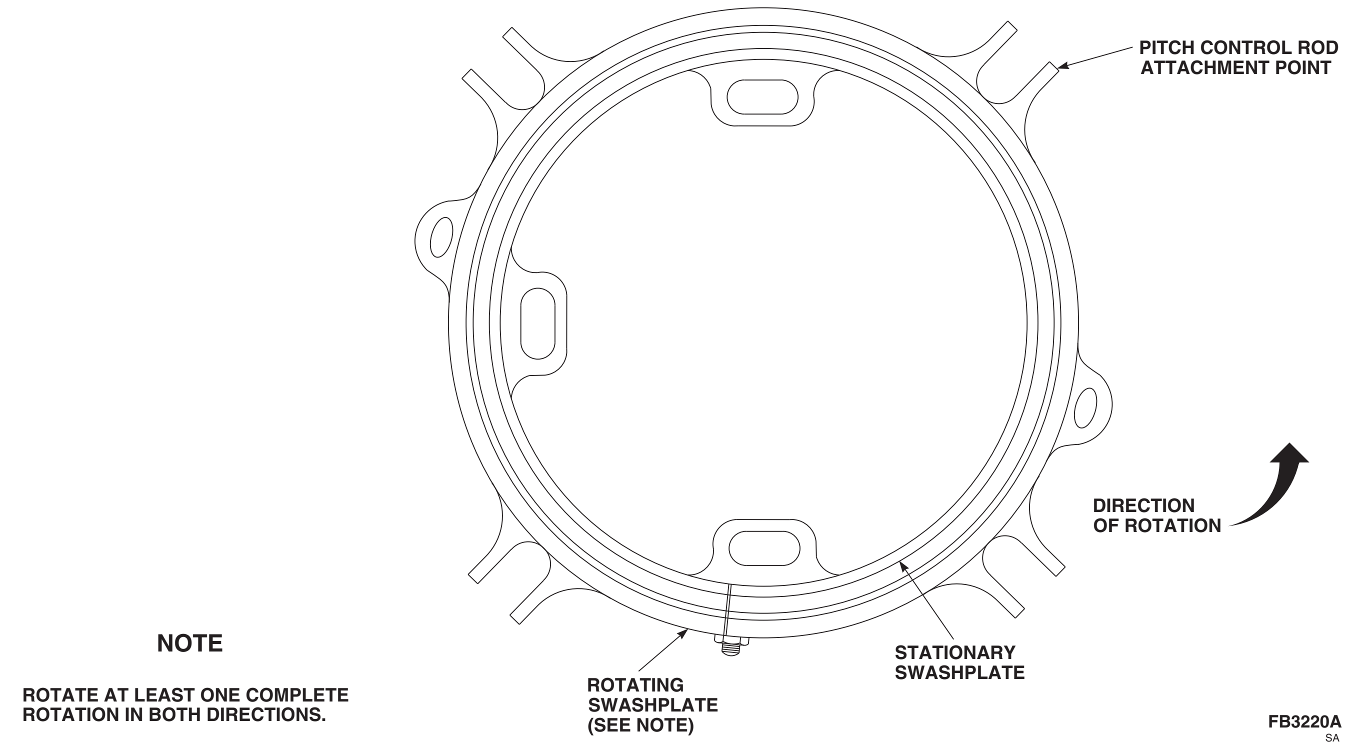


FIGURE 5-3-6-2. INSPECT SWASHPLATE BEARINGS

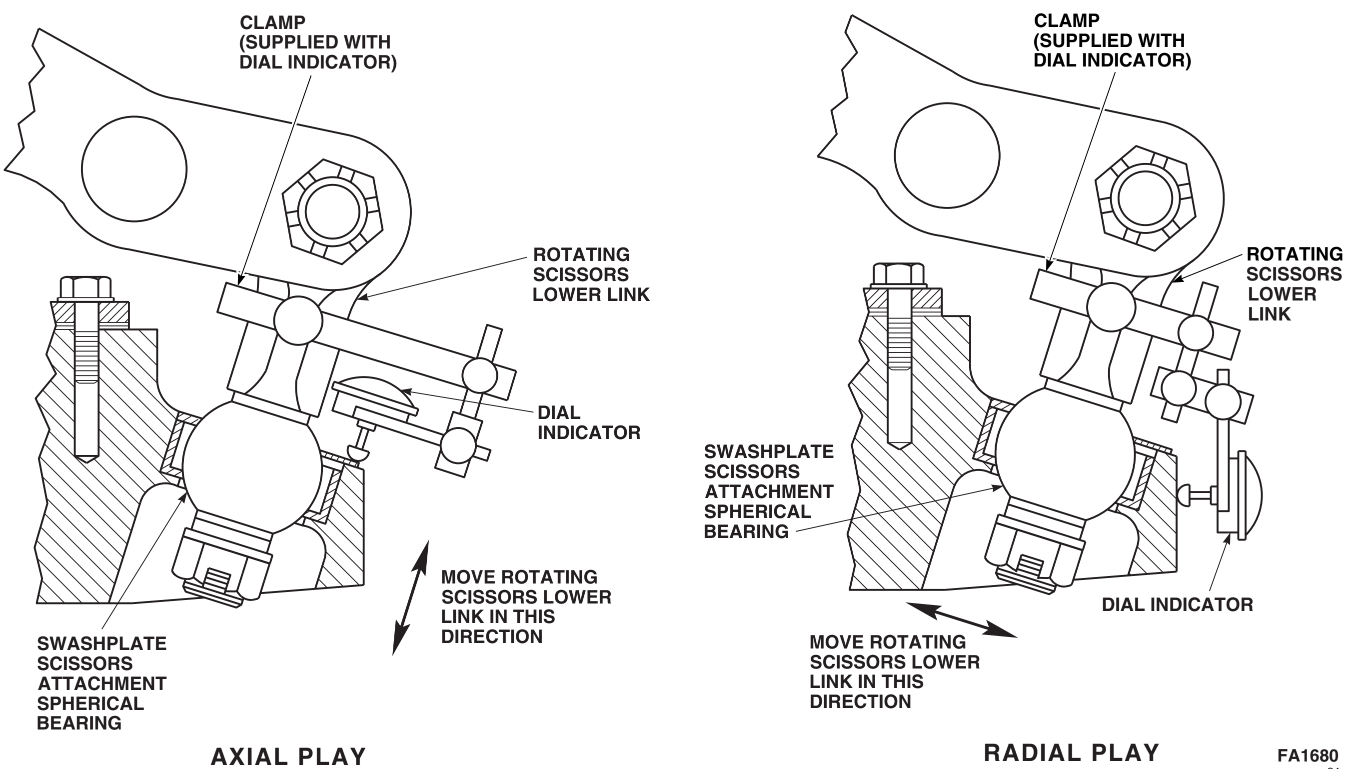


FIGURE 5-3-6-3. INSPECT SWASHPLATE SCISSORS ATTACHMENT SPHERICAL BEARINGS

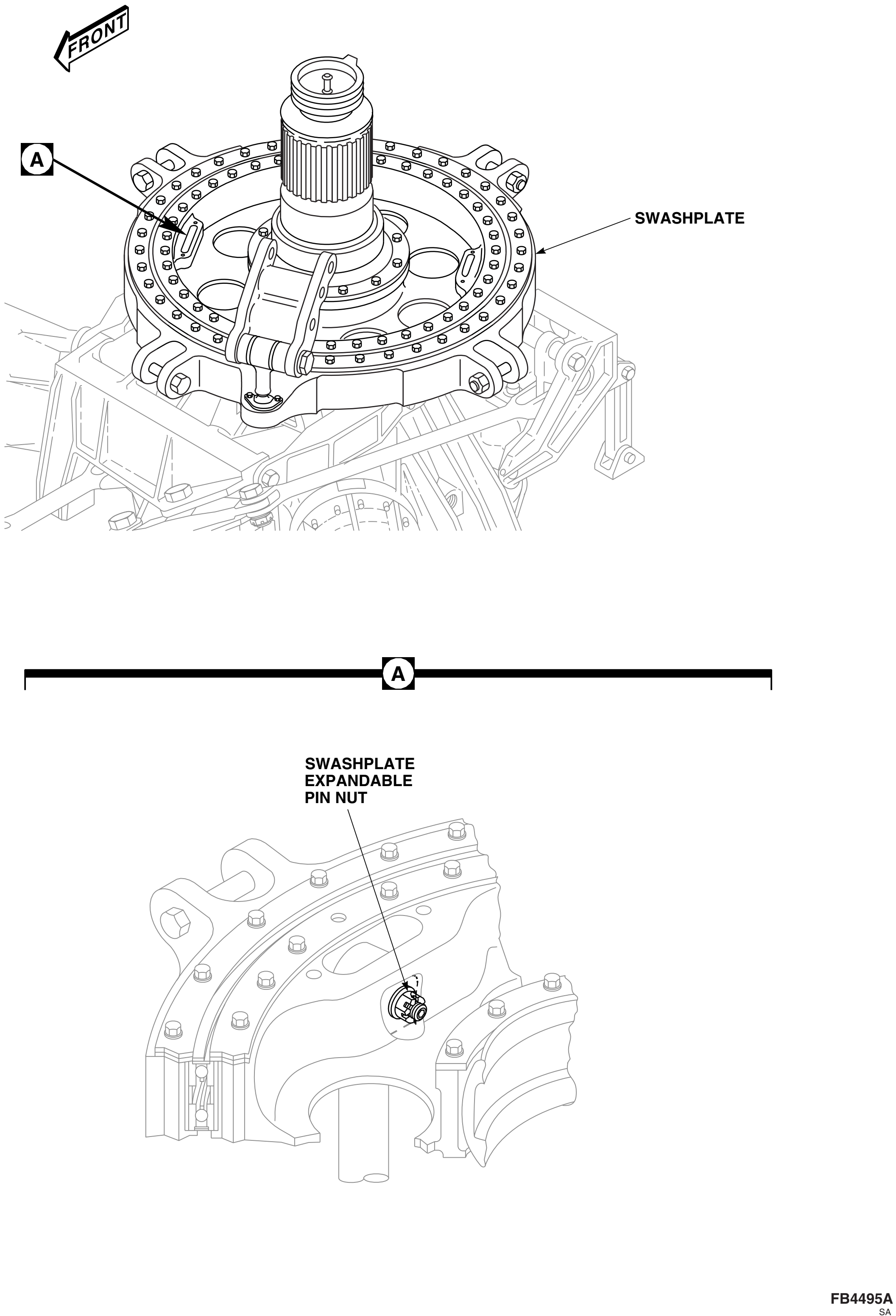


FIGURE 5-3-6-4. INSPECT SWASHPLATE EXPANDABLE NUTS

5-3-7 ROTATING SCISSORS UPPER LINK BEARING INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator (with Clamp)
- Feeler Gage
- Powertrain Repairer's Toolkit

References

- PARA 5-4-27
- PARA 5-5-8

5-3-7.1 Rotating Scissors Upper Link Bearings Inspection.

a. Rotate swashplate and rotating scissors in direction of rotation (counterclockwise). Make sure there is no load on rotating scissors upper link (Figure 5-3-7-1).

b. Inspect upper bearings as follows:

(1) Clamp dial indicator to lower pressure plate lug.

(2) Move rotating scissors upper link up and down at dial indicator. **RADIAL PLAY, AS SHOWN ON DIAL INDICATOR, MUST NOT EXCEED 0.012-INCH.** If play exceeds limit, remove rotating scissors (PARA 5-4-27) and replace rotating scissors upper link bearings (PARA 5-5-8).

(3) Using feeler gage, measure axial clearance between lower pressure plate lug and rotating scissors lug. **MAXIMUM ALLOWABLE AXIAL CLEARANCE IS 0.014-INCH.** If clearance exceeds limit, remove rotating scissors (PARA 5-4-27) and measure height of shoulder of outer rotating scissors upper link bearing (PARA 5-5-8).

c. Inspect lower bearings as follows:

(1) Clamp dial indicator to rotating scissors lower link (Figure 5-3-7-1, Detail A).

(2) Move rotating scissors upper link up and down at dial indicator. **RADIAL PLAY, AS SHOWN ON DIAL INDICATOR, MUST NOT EXCEED 0.012-INCH.** If play exceeds limit, remove rotating scissors (PARA 5-4-27) and replace rotating scissors upper link bearings (PARA 5-5-8).

(3) Using feeler gage, measure axial clearance between upper rotating scissors lug and rotating scissors lower link. **MAXIMUM ALLOWABLE AXIAL CLEARANCE IS 0.014-INCH.** If clearance exceeds limit, remove rotating scissors (PARA 5-4-27) and measure height of shoulder of outer rotating scissors lower link bearing (PARA 5-5-8).

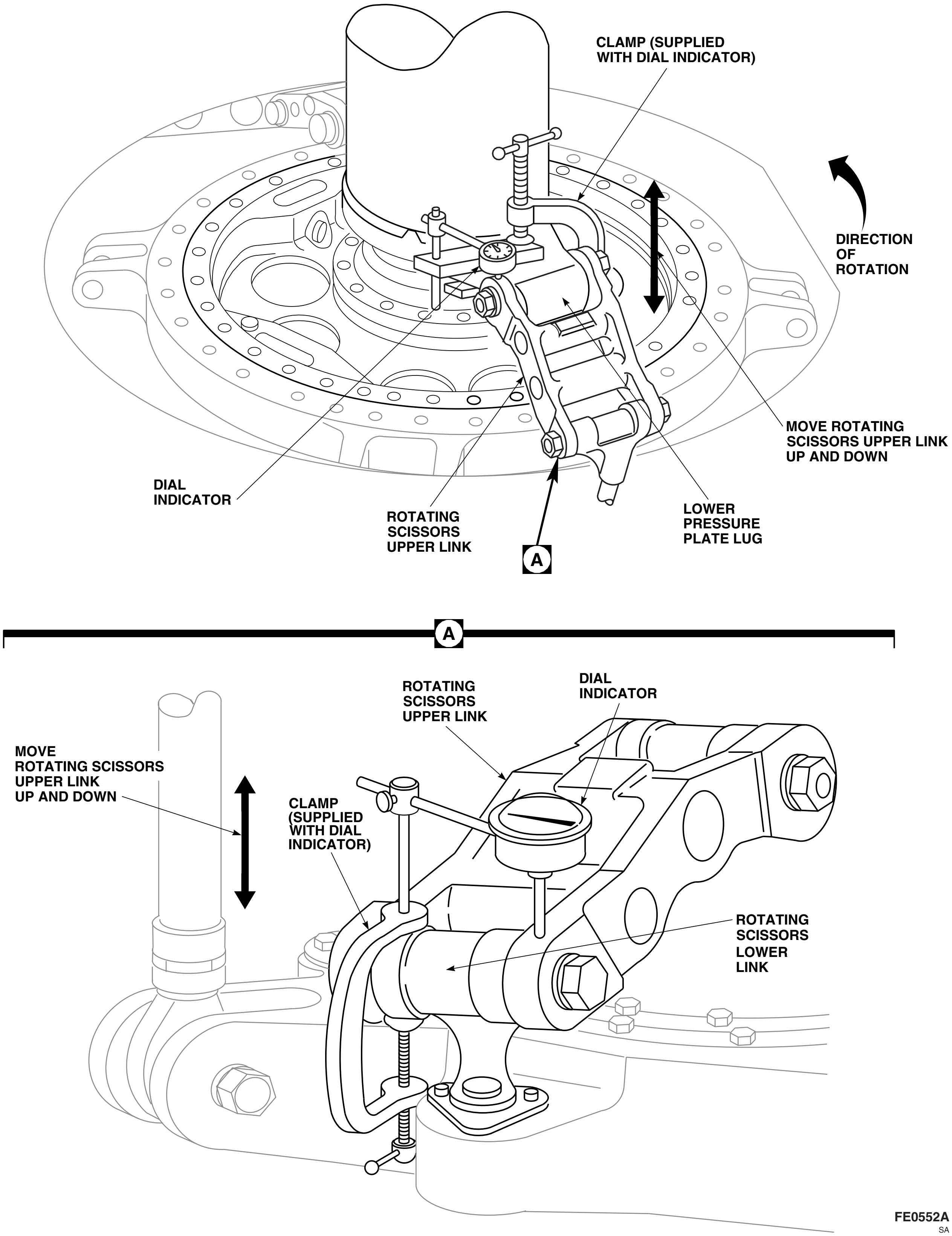


FIGURE 5-3-7-1. INSPECT ROTATING SCISSORS UPPER LINK BEARINGS

5-3-8 MAIN ROTOR BLADE SPAR INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- BIM® Check and Fill Unit, S1670-15000-25
- Checking Unit, S1670-15002-2
- Free-Air Thermometer
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dishwashing Compound, Item 122, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitrogen, Item 221, Appendix D

References

- Appendix D
- PARA 1-3-29
- PARA 5-4-31

NOTE

Check pressure in main rotor blade spar at intervals given in inspection requirements. Also, check main rotor blade spar pressure whenever new, overhauled, or stored main rotor blade is being placed in service.

5-3-8.1 Main Rotor Blade Spar Pressure Inspection (Using Checking Unit).

CAUTION

Do not attempt to pressurize main rotor blade spar if pressure is below limit indicated in ″Minimum When Checking Installed Blade″ column in PARA 1-3-29.

NOTE

- Do not attempt to measure main rotor blade spar pressure with main rotor blade exposed to sunlight. Solar heating of main rotor blade can raise main rotor blade spar pressure by over 2 psi above free-air-temperature pressure, resulting in inaccurate measurement. Allow minimum of 90 minutes after removing main rotor blade from sunlight, for main rotor blade spar temperature to stabilize, before measuring pressure in main rotor blade spar.

- If main rotor blade spar pressure is below limit indicated in ″Minimum When Servicing Blade″ column in PARA 1-3-29, but above ″Minimum When Checking Installed Blade″ pressure, increase main rotor blade spar pressure so it is between ″Minimum When Servicing Blade″ and ″Maximum″ pressures. Any main rotor blade with main rotor blade spar pressure below ″Minimum When Checking Installed Blade″ limit during normal checking must be treated as though BIM® pressure indicator had shown full red (unsafe) condition.

- Before measuring main rotor blade spar pressure, make sure calibration expiration date on checking unit has not expired.

- To make sure no pressure is lost from main rotor blade spar, connect coupling of checking unit to main rotor blade air valve before loosening outer hexnut.

a. Remove cap from main rotor blade air valve on main rotor blade (Figure 5-3-8-1, Detail A).

b. Connect checking unit coupling to main rotor blade air valve.

c. Check for tight connection.

d. Loosen outer hexnut 2 ½ turns while keeping inner hexnut and main rotor blade air valve adapter from turning (Figure 5-3-8-1, Detail A and Detail B).

e. Note gage reading.

NOTE

Free-air thermometer (FAT) should be in same lighting conditions as main rotor blade, and be within 10-feet of main rotor blade. Allow FAT to stabilize before taking final reading.

f. Using FAT, measure temperature in vicinity of main rotor blade being checked to nearest degree.

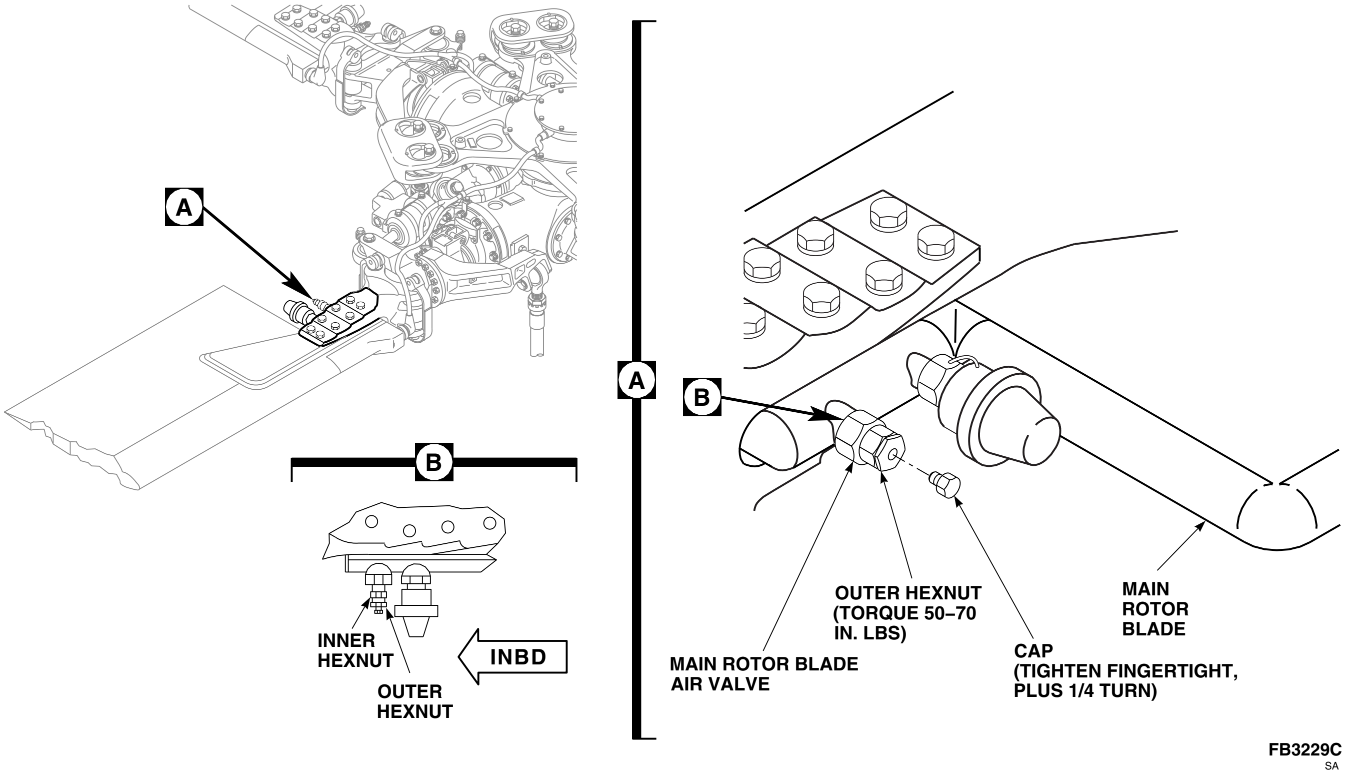


FIGURE 5-3-8-1. INSPECT MAIN ROTOR BLADE SPAR PRESSURE

g. Record temperature and pressure.

WARNING

Injury to personnel and damage to equipment will result if main rotor blade with main rotor blade spar pressure below acceptable limits is reserviced and released for flight. Cause of pressure loss must be determined and corrected before releasing main rotor blade for flight. Remove main rotor blade and send to Bench level maintenance or manufacturer.

NOTE

Make entry in helicopter logbook stating reason for removal whenever main rotor blade is removed from service.

h. If main rotor blade spar pressure is within acceptable limits for temperature recorded, per PARA 1-3-29, main rotor blade can be continued in service. If main rotor blade spar pressure exceeds acceptable limits for temperature recorded, per PARA 1-3-29, remove main rotor blade (PARA 5-4-31), and send to Bench level maintenance or manufacturer to determine cause of pressure loss.

i. **TORQUE OUTER HEXNUT TO 50 - 70 INCH-POUNDS** (Figure 5-3-8-1, Detail A).

j. Disconnect checking unit from main rotor blade air valve.

k. **INSTALL CAP ON MAIN ROTOR BLADE AIR VALVE AND TIGHTEN FINGER-TIGHT, PLUS ¼ TURN.**

l. Using lockwire, Item 197, Appendix D, lockwire outer hexnut to body of main rotor blade air valve.

m. Using test liquid, such as leak detection fluid, Item 189, Appendix D, or dishwashing compound, Item 122, Appendix D, check main rotor blade air valve for leakage by applying over and around entire main rotor blade air valve and watching for bubbles to form.

n. When check is completed and before test liquid dries, using clean, fresh water, clean area thoroughly to remove all traces of test liquid.

o. Using machinery towel, Item 211, Appendix D, dry main rotor blade.

p. Test BIM® pressure indicator for proper operation (PARA 1-3-29).

5-3-8.2 Main Rotor Blade Spar Pressure Inspection (Using BIM® Check and Fill Unit).

CAUTION

- Do not attempt to pressurize main rotor blade spar if pressure is below limit indicated in ″Minimum When Checking Installed Blade″ column in PARA 1-3-29.
- When using BIM® check and fill unit to check main rotor blade spar pressure, it is mandatory that 20-feet hose line be purged of air before opening valve core control nut on main rotor blade air valve.

NOTE

- Do not attempt to measure main rotor blade spar pressure with main rotor blade exposed to sunlight. Solar heating of main rotor blade can raise main rotor blade spar pressure by over 2 psi above free-air-temperature pressure, resulting in inaccurate main rotor blade spar pressure measurement. Allow minimum of 90 minutes after removing main rotor blade from sunlight, for main rotor blade spar temperature to stabilize, before measuring pressure in main rotor blade spar.
- If main rotor blade spar pressure is below limit indicated in ″Minimum When Servicing Blade″ column in PARA 1-3-29, but above ″Minimum When Checking Installed Blade″ pressure, increase main rotor blade spar pressure so it is between ″Minimum When Servicing Blade″ and ″Maximum″ pressures. Any main rotor blade with main rotor blade spar pressure below ″Minimum When Checking Installed Blade″ limit during normal checking must be treated as though BIM® pressure indicator had shown full red (unsafe) condition.

a. Open cover of BIM® check and fill unit.

b. Remove adjustment handle from cylinder regulator.

c. Remove cylinder regulator from bracket.

d. Reinstall handle on cylinder regulator.

e. Connect cylinder regulator of BIM® check and fill unit to source of nitrogen, Item 221, Appendix D, at fitting on cylinder regulator. If necessary, use one of the adapters supplied with BIM® check and fill unit to attach cylinder regulator to nitrogen source.

f. Make sure station regulator, cylinder regulator, and station valve, next to station regulator, are closed as follows:

(1) To close station regulator, turn shutoff knob counterclockwise to close.

(2) To close cylinder regulator, turn adjustment handle counterclockwise to close.

(3) To close station valve, turn shutoff knob clockwise to close.

g. Open valve on nitrogen source. Gage will now indicate pressure.

h. Using adjustment handle on cylinder regulator, turn on flow of nitrogen at cylinder regulator and adjust pressure to between 30 and 40 psi.

i. Connect BIM® check and fill unit coupling to main rotor blade air valve as follows:

(1) Remove cap from main rotor blade air valve on main rotor blade (Figure 5-3-8-1, Detail A).

(2) Open shutoff knob on station valve (counterclockwise) and shutoff knob on station regulator (clockwise) for a few seconds to purge air from lines.

(3) Close valves and connect BIM® check and fill unit coupling to main rotor blade air valve.

(4) Check for tight connection.

(5) Loosen outer hexnut 2 ½ turns while keeping inner hexnut and main rotor blade air valve adapter from turning (Figure 5-3-8-1, Detail A and Detail B).

(6) Note gage reading.

NOTE

Free-air thermometer (FAT) should be in same lighting condition as main rotor blade and should be within 10-feet of main rotor blade. Allow FAT to stabilize before taking final reading.

j. Using FAT, measure temperature in vicinity of main rotor blade being checked to nearest degree.

k. Record temperature and pressure.

WARNING

Injury to personnel and damage to equipment will result if main rotor blade with main rotor blade spar pressure below acceptable limits is reserviced and released for flight. Cause of pressure loss must be determined and corrected before releasing main rotor blade for flight. Remove main rotor blade and send to Bench level maintenance or manufacturer.

NOTE

Make entry in helicopter logbook stating reason for removal, whenever main rotor blade is removed from service.

l. If main rotor blade spar pressure is within acceptable limits for temperature recorded, per PARA 1-3-29, main rotor blade can be continued in service. If main rotor blade spar pressure exceeds acceptable limits for temperature recorded, per PARA 1-3-29, remove main rotor blade (PARA 5-4-31), and send to Bench level maintenance or manufacturer to determine cause of pressure loss.

m. If it is necessary to increase main rotor blade spar pressure, perform the following:

(1) Open shutoff knob on station valve (counterclockwise).

(2) Using station regulator shutoff knob, gradually increase pressure until gage reaches required pressure found in PARA 1-3-29, for measured temperature. Allow main rotor blade spar to fill to this pressure. Pressure in main rotor blade spar should be checked periodically by turning station valve shutoff knob fully clockwise and noting reading on gage at station regulator.

(3) When pressure in main rotor blade spar has reached specified limit, shut off nitrogen supply at station valve by turning station valve shutoff knob fully clockwise. Check gage at station regulator for reading within limits specified in PARA 1-3-29, for measured main rotor blade spar temperature.

n. **TORQUE OUTER HEXNUT TO 50 - 70 INCH-POUNDS** (Figure 5-3-8-1, Detail A).

o. Disconnect checking unit from main rotor blade air valve.

p. **INSTALL CAP ON MAIN ROTOR BLADE AIR VALVE AND TIGHTEN FINGER-TIGHT, PLUS ¼ TURN.**

q. Using lockwire, Item 197, Appendix D, lockwire outer hexnut to body of main rotor blade air valve.

r. Using test liquid, such as leak detection fluid, Item 189, Appendix D, or dishwashing compound, Item 122, Appendix D, check main rotor blade air valve for leakage by applying over and around entire main rotor blade air valve and watching for bubbles to form.

s. When check is completed and before test liquid dries, using clean, fresh water, clean area thoroughly to remove all traces of test liquid.

t. Using machinery towel, Item 211, Appendix D, dry main rotor blade.

u. Shut off nitrogen supply to cylinder regulator.

v. Open shutoff knob on station valve (counterclockwise) to release pressure in line.

w. Shut off station regulator and disconnect nitrogen supply from BIM® check and fill unit.

x. Test BIM® pressure indicator for proper operation (PARA 1-3-29).

5-3-9 MAIN ROTOR BLADE PIN INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Blade Clamp Assembly, 70700-20324-041
- Heat Gun
- Hoist, Capacity to Support Main Rotor Blade (Alternate for Cabin Maintenance Crane)
- Maintenance Platform
- Maintenance Sling, 70700-20323-043
- Safety Strap
- Spring Scale
- Spring Tester

Materials/Parts

- Abrasive Cloth, Item 12, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Acetone, Item 25, Appendix D
- Cotton Gloves
- Dry-Cleaning Solvent, Item 124, Appendix D
- Lubricant, Solid Film, Item 201, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 1-4-6
- PARA 5-4-8
- PARA 5-4-31
- PARA 18-4-1
- TM 1-70-VIB

5-3-9.1 Main Rotor Blade Expandable Pin Inspection.

CAUTION

Damage to main rotor blade will result if blade trim tab is bent or damaged. While working on main rotor blade, be careful not to bend or damage blade trim tab. Bending trim tab will cause blade out-of-track condition. Position maintenance platform under tip of blade for assistant to stand on to help guide blade clear of helicopter.

NOTE

- Remove one expandable pin at a time when performing the following inspections. Make sure pin is completely inspected and installed back in main rotor blade/spindle cuff lugs prior to inspecting subsequent pins.
- Expandable pin, 70103-08107-102, is manufactured by different suppliers and slight variations in design are evident; however, their function and operation are the same.

a. Prepare for inspection as follows:

(1) If cabin maintenance crane will be used, erect cabin maintenance crane (PARA 18-4-1).

WARNING

Blade clamp assembly may open during lifting, causing injury to personnel and damage to equipment. If maintenance sling is used, one leg of maintenance sling should be secured loosely around blade clamp assembly, in case blade clamp assembly opens. Install safety strap over blade clamp assembly before removing main rotor blade.

(2) Position blade clamp assembly between balance stripes and lock to main rotor blade.

(3) Install safety strap over blade clamp assembly and through shackle (Figure 5-3-9-1, Detail A).

(4) If using cabin maintenance crane, attach blade clamp assembly to crane.

(5) If hoist will be used, use maintenance sling to attach blade clamp assembly to hoist.

(6) Relieve load on expandable pins by slightly lifting main rotor blade.

b. Perform handle resistance inspection as follows:

(1) Release spring clip portion of handle from nut and open handle (Figure 5-3-9-1, Detail B).

(2) Raise handle to fully unlocked (open) position.

(3) Check lockring on each pin to see if it is fully compressed by rotating it around pin core (Figure 5-3-9-1, Detail B and Detail C). If lockring is not compressed, or shows evidence of dirt or contaminants, clean out lockring before removing pin (PARA 1-4-6).

NOTE

Resistance should be felt while lowering handle from unlocked (open) to locked (closed) position.

(4) Lower handle to fully locked (closed) position and evaluate as follows:

(a) If resistance is felt while lowering handle, secure spring clip portion of handle over nut. Proceed to step d.

(b) If resistance is not felt while lowering handle, perform severed pin core inspection. Proceed to step c.

c. Perform severed pin core inspection as follows:

(1) Remove expandable pin as follows:

WARNING

FSCAP

Expandable pins contain critical surfaces which must not be damaged during and after removal from main rotor blade. Use extreme caution when removing expandable pin.

(a) Release spring clip portion of handle from nut and open handle on expandable pin (Figure 5-3-9-1, Detail B).

(b) Raise handle to fully unlocked (open) position.

CAUTION

Damage to expandable pin and main rotor blade cuff will result if pin is removed before lockring is fully compressed. Pin will jam in main rotor blade cuff. Make sure lockring is fully compressed before attempting to remove expandable pin.

(c) Check lockring on each pin to see if it is fully compressed by rotating it around pin core (Figure 5-3-9-1, Detail B and Detail C). If lockring is not compressed, or shows evidence of dirt or contaminants, clean out lockring before removing pin (PARA 1-4-6).

(d) Remove expandable pin from main rotor blade/spindle cuff lugs. If expandable pin is stuck, remove stuck pin (PARA 5-4-31).

NOTE

A severed pin core will be in two pieces. Bottom portion of pin core may remain in main rotor blade/spindle cuff lugs.

(2) Inspect for severed pin core. **A SEVERED PIN CORE IS NOT ALLOWED.**

(3) If pin core is not severed, perform the following:

(a) Inspect spindle lugs for security of bonded washers. If bonded washers are loose, rebond washers (PARA 5-4-8).

(b) Install expandable pin (step j.).

(c) Proceed to step d.

(4) If severed pin core is found, perform the following:

(a) Remove any remaining pieces from main rotor blade/spindle cuff lugs.

(b) Inspect spindle lugs for security of bonded washers. If bonded washers are loose, rebond washers (PARA 5-4-8).

(c) Install replacement expandable pin (step j.).

(d) Contact Sikorsky Field Service Representative, attention: ASB 70-05-31 Disposition (Rotors Engineering Group), with the following information:

- Location of discrepant pin position (lead or lag).
- Part number, serial number, and component total time of main rotor spindle.
- Part number, serial number, and component total time of main

rotor blade.

- Part number, serial number, and component total time of main rotor blade cuff.

- Customer contact information, including fax number.

(e) Penalize main rotor blade cuff by adding 200 flight hours to total time on blade cuff. Annotate this on component log card. Inspection is complete.

d. Perform cam pivot pin inspection as follows:

(1) Release spring clip portion of handle from nut and open handle (Figure 5-3-9-1, Detail B).

(2) Raise handle to fully unlocked (open) position.

(3) Using spring tester, check cam pivot pin staking by applying 40 pounds of force to both ends. **TOTAL MOVEMENT MUST NOT EXCEED 0.025-INCH.**

(4) If movement is acceptable, proceed to step e. If movement exceeds limit, replace expandable pin as follows:

(a) Remove expandable pin (step c. (1)).

(b) Install replacement expandable pin (step j.). Inspection is complete.

e. Perform spring clip/handle tension check as follows:

(1) Using spring scale, place end of spring scale in nut hole in spring clip portion of handle and pull. **FORCE REQUIRED TO CLOSE HANDLE MUST BE 25 - 35 POUNDS.**

(2) If force is acceptable, proceed to step f. If force required to close and lock handle exceeds limits, perform the following:

NOTE

A minimum of two threads must extend beyond nut after adjusting tension.

(a) Adjust nut on bottom of expandable pin for proper tension making sure there is a minimum of two threads extending beyond nut (Figure 5-3-9-1, Detail C).

(b) Repeat tension check.

(c) If proper tension cannot be obtained after adjustment of nut, remove expandable pin (step c. (1)), then install replacement expandable pin (step j.). Inspection is complete.

f. Perform segments and wedges inspection as follows:

WARNING

FSCAP

Expandable pins contain critical surfaces which must not be damaged during and after removal. Use extreme caution when removing expandable pin.

(1) Remove expandable pin (step c. (1)).

(2) Using dry-cleaning solvent, Item 124, Appendix D, hand-clean or spray-wash expandable pin. Using Item 211, Appendix D, wipe up remaining solvent with clean machinery towel.

(3) Inspect expandable pin for missing or damaged segments or wedges (between segments and at top and bottom of pin). **NONE ALLOWED.** Evaluate as follows:

(a) If segments or wedges are acceptable, proceed to step g.

(b) If any segments or wedges are damaged or missing, install replacement expandable pin (step j.). Inspection is complete.

g. Perform cam inspection as follows:

NOTE

Denting or bruising of material can occur in clevis of handle. Dents occur when clevis of handle makes repeated contact with pin core handle attachment lug. This is a normal condition and not cause for replacement of expandable pin.

(1) Inspect cam for nicks, dents, scratches, and corrosion. If necessary, repair cam using abrasive cloth, Item 12, Appendix D, as follows:

(a) **BLEND SURFACE DAMAGE ON CAM SURFACE OF HANDLE UP TO 0.002-INCH DEEP. TOTAL REPAIR AREA MUST NOT EXCEED ¼ SQUARE-INCH.**

(b) **BLEND SURFACE DAMAGE TO HANDLE UP TO 0.020-INCH DEEP. TOTAL REPAIR AREA MUST NOT EXCEED 2 SQUARE-INCHES.**

(c) **BLEND EDGE DAMAGE TO HANDLE UP TO 0.040-INCH DEEP. TOTAL LENGTH OF REPAIR MUST NOT EXCEED 3-INCHES.**

(d) **BLEND SURFACE DAMAGE TO SPRING CLIP UP TO 0.005-INCH DEEP. NO LIMITS ON TOTAL REPAIR AREA.**

(2) If damage and/or repairs are acceptable, proceed to step h.

(3) If damage and/or repairs exceed limits, install replacement expandable pin (step j.). Inspection is complete.

h. Perform outer segment inspection as follows:

(1) Inspect outer segments' outside diameter (OD) for nicks, dents, and scratches. **DAMAGE MUST NOT EXCEED 0.005-INCH.** If necessary, remove metal pickup using abrasive wheel, Item 22, Appendix D.

(2) If damage and/or repairs are acceptable, proceed to step i.

(3) If damage and/or repairs exceed limit, install replacement expandable pin (step j.). Inspection is complete.

i. Perform outer segment solid film lubricant inspection as follows:

(1) Inspect outer segments' OD for condition of solid film lubricant. **UP TO 50% OF SOLID FILM LUBRICANT MISSING IS ACCEPTABLE.**

(2) If less than 50% of solid film lubricant is missing, install expandable pin (step j.).

(3) If more than 50% of solid film lubricant is missing, perform the following:

(a) Using machinery towel, Item 211, Appendix D, dampened with drycleaning solvent, Item 124, Appendix D, clean areas of bare metal. Allow to dry.

NOTE

Solid film lubricant will not adhere to contaminated surfaces. Make sure surface remains clean while handling parts.

(b) Using solid film lubricant, Item 201, Appendix D, apply two thin coats to OD of outer segment. Allow 15 - 20 minutes between coats.

(c) Allow expandable pin to cure for minimum of 12 hours (alternate accelerated cure time is 30 minutes air cure, plus using heat gun, 30 minutes at 150°F (66°C)).

(d) Install expandable pin (step j.).

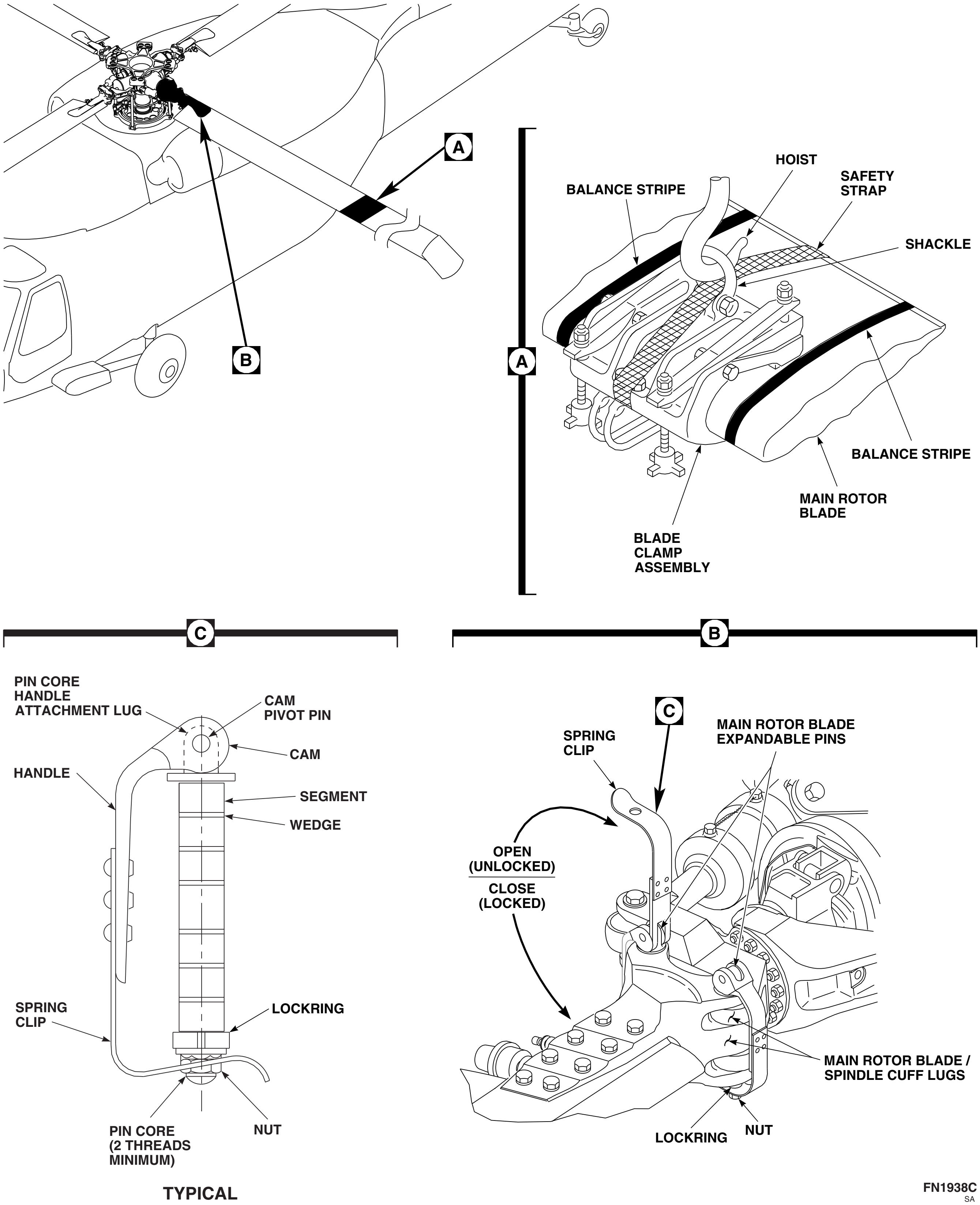


FIGURE 5-3-9-1. MAIN ROTOR BLADE EXPANDABLE PIN INSPECTION

j. Install expandable pin as follows (Figure 5-3-9-1, Detail B):

NOTE

- Expandable pin, 70103-08107-102, is manufactured by different suppliers and slight variations in design are evident; however, their function and operation are the same.
- Expandable pins manufactured by Shur-Lok Corporation weigh slightly less than those supplied by other suppliers. Expandable pins manufactured by Shur-Lok Corporation can be identified by CAGE code: 97393.
- Intermixing expandable pins manufactured by Shur-Lok Corporation with heavier expandable pins is permitted; however, this may cause helicopter main rotor vibrations to increase.
- Where possible, expandable pins of the same manufacturer must be used for all eight positions. If expandable pins of the same manufacturer are not available, install available expandable pins and check main rotor balance (TM 1-70-VIB).
- Unseating of expandable pins up to a maximum of 0.125-inch is acceptable, provided lockring is not contacting main rotor blade cuff.

(1) Insert expandable pin through main rotor blade/spindle cuff lugs.

NOTE

Do not hit or force expandable pin into position.

(2) Align handle with main rotor blade cuff decal. If necessary, gently rock main rotor blade to get proper alignment.

(3) Perform spring clip/handle tension check (step e.).

(4) Close handle, making sure spring clip of pin snaps over and fully engages hex on nut. Visually check for expansion of lockring on end of pin. Make sure slots in spring clip line up with points of nut.

k. If used, disassemble and remove cabin maintenance crane (PARA 18-4-1).

l. If required, check main rotor balance (TM 1-70-VIB).

5-3-9.2 Main Rotor Blade Solid Pin Inspection.

CAUTION

Damage to main rotor blade will result if blade trim tab is bent or damaged. While working on main rotor blade, be careful not to bend or damage blade trim tab. Bending trim tab will cause blade out-of-track condition. Position maintenance platform under tip of blade for assistant to stand on to help guide blade clear of helicopter.

NOTE

Remove one solid pin at a time when performing the following inspections. Make sure pin is completely inspected and installed back in main rotor blade/spindle cuff lugs prior to inspecting subsequent pins.

a. Prepare for inspection as follows:

(1) If cabin maintenance crane will be used, erect cabin maintenance crane (PARA 18-4-1).

WARNING

Blade clamp assembly may open during lifting, causing injury to personnel and damage to equipment. If maintenance sling is used, one leg of maintenance sling should be secured loosely around blade clamp assembly, in case blade clamp assembly opens. Install safety strap over blade clamp assembly before removing main rotor blade.

(2) Position blade clamp assembly between balance stripes and lock to main rotor blade.

(3) Install safety strap over blade clamp assembly and through shackle (Figure 5-3-9-1, Detail A).

(4) If using cabin maintenance crane, attach blade clamp assembly to crane.

(5) If hoist will be used, use maintenance sling to attach blade clamp assembly to hoist.

(6) Relieve load on expandable pins by slightly lifting main rotor blade.

b. Remove solid pin as follows:

WARNING

FSCAP

Solid pins contain critical surfaces which must not be damaged during and after removal from main rotor blade. Use extreme caution when removing solid pin.

(1) Remove cotter pin securing solid pin (Figure 5-3-9-2, Detail B).

(2) Raise handle to fully unlocked (open) position.

(3) Remove solid pin from main rotor blade/spindle cuff lugs. If solid pin is stuck, remove stuck pin (PARA 5-4-31).

NOTE

Solid pins should be handled with cotton gloves following cleaning and during inspection. Do not touch pin shank with bare hands.

c. Using machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, clean areas of bare metal on solid pin.

d. Perform handle inspection as follows:

(1) Inspect handle for cracks, nicks, dents, scratches, and corrosion (Figure 5-3-9-2, Detail C). **NO CRACKS ALLOWED. IF CRACKED, REPLACE IMMEDIATELY.** If corrosion, nicks, dents, and scratches are found, repair using abrasive cloth, Item 12, Appendix D, as follows:

(a) **BLEND SURFACE DAMAGE TO HANDLE UP TO 0.020-INCH DEEP. TOTAL REPAIR AREA MUST NOT EXCEED 2 SQUARE-INCHES.**

(b) **BLEND EDGE DAMAGE TO HANDLE UP TO 0.010 - 0.040-INCH DEEP. TOTAL LENGTH OF REPAIR MUST NOT EXCEED 3-INCHES. EDGE DAMAGE LESS THAN 0.010-INCH DEEP IS ACCEPTABLE.**

(c) **BLEND SURFACE AND EDGE DAMAGE UP TO 0.005-INCH DEEP. NO LIMITS ON TOTAL REPAIR AREA.**

(2) If damage and/or repairs are acceptable, proceed to step e.

(3) If damage or repairs exceed limits, install replacement solid pin (step i.). Inspection is complete.

e. Perform pin shank wear inspection as follows:

(1) Inspect pin shank for wear. Measure pin diameter in worn areas. **PIN DIAMETER MUST NOT BE LESS THAN 0.991-INCH.** Perform the following:

(2) If wear is acceptable, proceed to step f.

(3) If wear is less than limits, install replacement solid pin (step i.). Inspection is complete.

f. Perform pin shank scoring/groove inspection as follows:

(1) Inspect pin shank outside diameter for scoring or grooves. **BASE METAL AXIAL SCORING OR GROOVES UP TO 0.002-INCH DEEP ARE ACCEPTABLE. BLEND OUT BASE METAL AXIAL SCORING OR GROOVES 0.003 - 0.005-INCH DEEP. BLEND OUT BASE METAL CIRCUMFERENTIAL SCORING UP TO 0.005-INCH DEEP.**

(2) If damage and/or repairs are acceptable, proceed to step g.

(3) If damage and/or repairs exceed limits, install replacement solid pin (step i.). Inspection is complete.

g. Perform pin shank corrosion inspection as follows:

NOTE

Surface corrosion is not cause for rejection.

(1) Inspect pin shank for corrosion. **CORROSION PITTING UP TO 0.005-INCH DEEP IS ACCEPTABLE, PROVIDING PIN DIAMETER IS NOT LESS THAN 0.991-INCH AFTER BLENDING.**

(2) If damage or repairs are within limits, using abrasive cloth, Item 12, Appendix D, remove corrosion.

(3) If damage or repairs exceed limits, replace solid blade pin (step i.), and reapply solid film lubricant, Item 201, Appendix D, as necessary.

h. Perform pin shank solid film lubricant inspection as follows:

(1) Inspect pin shank for condition of solid film lubricant. **UP TO 50% OF SOLID FILM LUBRICANT MISSING IS ACCEPTABLE.**

(2) If less than 50% of solid film lubricant is missing, install solid pin (step i.).

(3) If more than 50% of solid film lubricant is missing, perform the following:

(a) Using machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, clean areas of bare metal. Allow to dry.

NOTE

- Solid film lubricant will not adhere to contaminated surfaces. Do not touch clean areas with bare hands. Use cotton gloves while handling parts.
- Apply solid film lubricant where ambient temperatures are between 65 - 85°F (18 - 30°C).

(b) Using solid film lubricant, Item 201, Appendix D, apply two thin coats to outside diameter of solid pin shank. Allow 15 - 20 minutes between coats. Total thickness of dry film lubricant to be 0.0003 - 0.0005-inches.

(c) Allow solid pins to air-dry for a minimum of 18 hours (alternate accelerated cure time is 6 - 9 hours at 120 - 150°F (49 - 66°C)).

(d) Install solid pin (step i.).

i. Install solid pin as follows:

(1) Insert solid pin through main rotor blade/spindle cuff lugs (Figure 5-3-9-2, Detail B).

NOTE

Do not hit or force solid pin into position.

(2) Align handle with main rotor blade cuff decal. If necessary, gently rock main rotor blade to get proper alignment.

(3) Close handle, making sure retaining clip fork engages groove in pin and that bottom of pin protrudes through hole in spring clip (Figure 5-3-9-2, Detail D).

(4) Install cotter pin from side of pin closest to main rotor hub (Figure 5-3-9-2, Detail B and Detail D).

j. If used, disassemble and remove cabin maintenance crane (PARA 18-4-1).

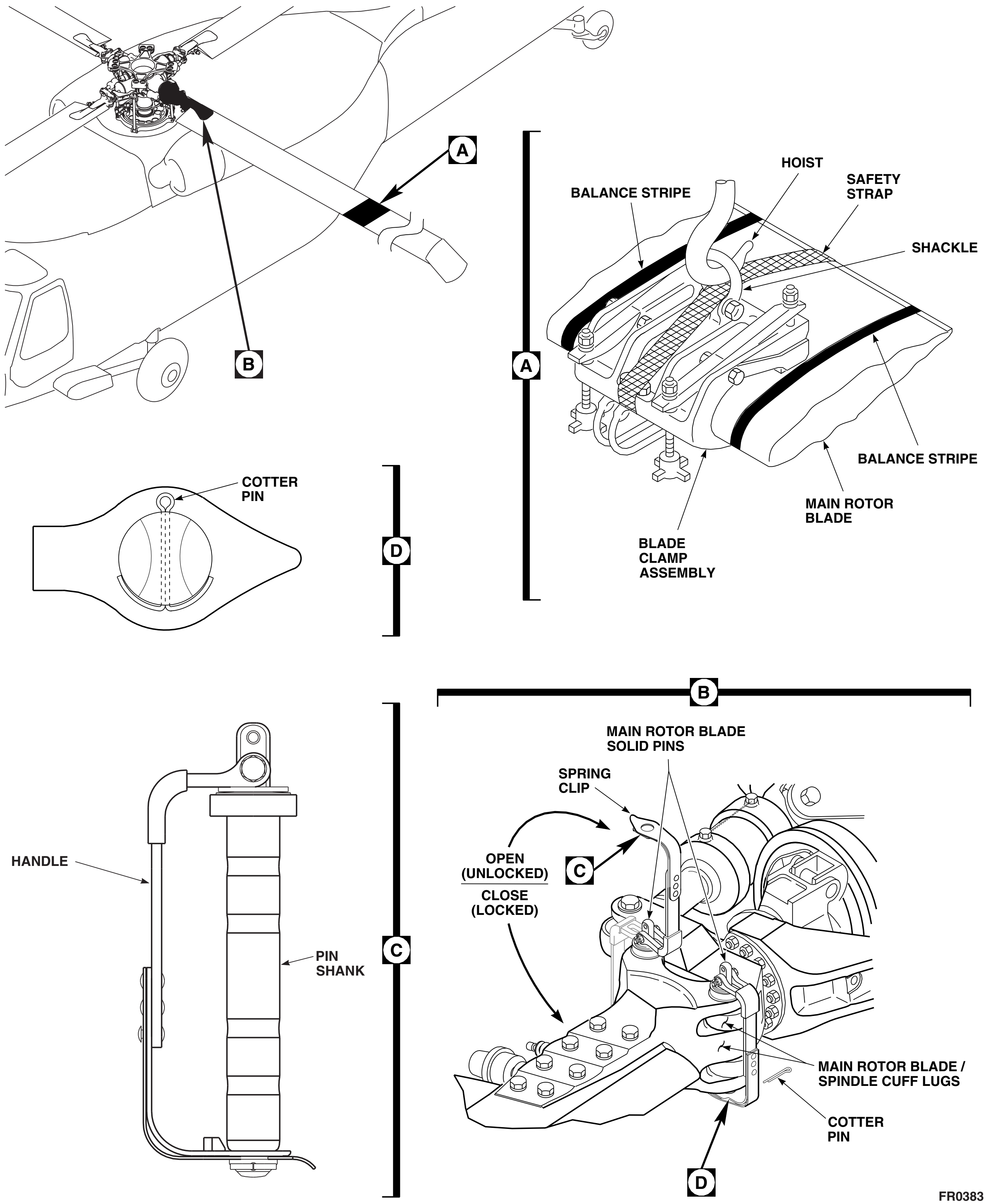


FIGURE 5-3-9-2. MAIN ROTOR BLADE SOLID PIN INSPECTION

5-3-10 BIFILAR INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 12, Appendix D
- Abrasive Cloth, Item 13, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Adhesive, Item 64, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 254, Appendix D
- Primer, Item 258, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5
- PARA 5-4-28
- PARA 5-4-29
- PARA 5-5-11

5-3-10.1 Bifilar Inspection.

NOTE

In the following steps, damage should be blended out using abrasive wheel, Item 22, Appendix D, and/or abrasive cloth, Item 12, Appendix D.

a. Inspect bifilar arm for corrosion (Figure 5-3-10-1). If required, repair as follows:

(1) **ON OPEN AREAS (NONMATING SURFACES), REMOVE CORROSION PITTING UP TO 0.050-INCH DEEP BY BLENDING OUT. NO TWO AREAS MAY BE CLOSER THAN 1-INCH APART.**

(2) **ON MATING SURFACES, REMOVE CORROSION PITTING UP TO 0.010-INCH DEEP BY BLENDING OUT OVER AN AREA UP TO 10% OF MATING SURFACE AREA.** If damage after

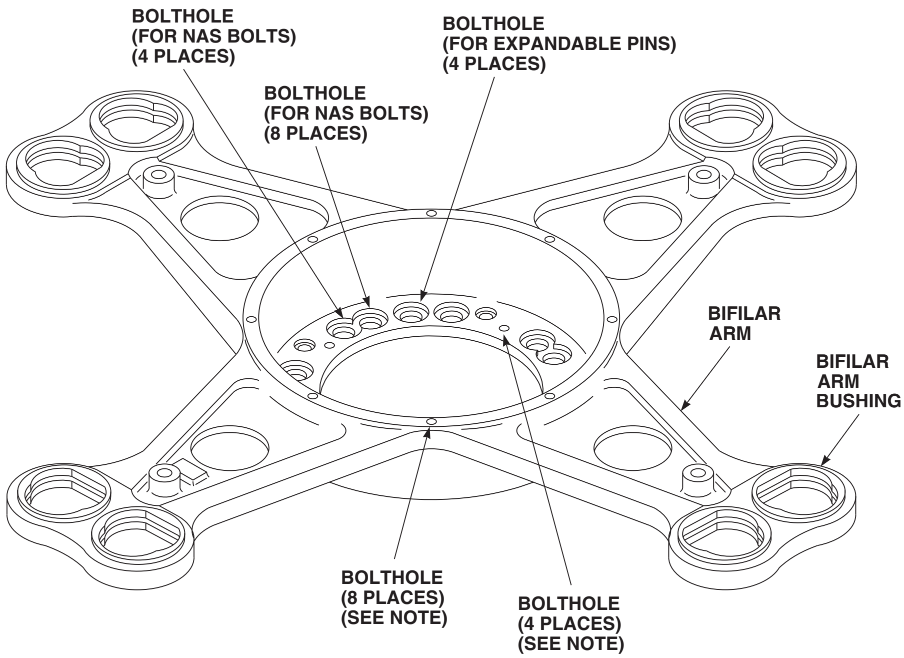


FIGURE 5-3-10-1. INSPECT BIFILAR

repair exceeds limits, replace bifilar (PARA 5-4-28).

(3) Using corrosion resistant coating, Item 118, Appendix D, touch up reworked areas.

(4) Apply primer, Item 258, Appendix D, to reworked areas.

(5) Touch up paint all reworked areas (PARA 2-6-5).

b. Inspect boltholes for corrosion. If required, repair as follows:

NOTE

The following inspection does not apply to all boltholes. Refer to Figure 5-3-10-1 for boltholes to which inspection applies.

(1) **IN BOLTHOLES, REMOVE CORROSION PITTING UP TO 0.005-INCH DIAMETER.** If damage after repair exceeds limits, replace bifilar (PARA 5-4-28).

(2) Using corrosion resistant coating, Item 118, Appendix D, touch up reworked areas.

(3) Apply primer, Item 258, Appendix D, to boltholes when installing bolts.

c. Inspect bifilar arm for nicks and gouges. If required, repair as follows:

(1) **CORNER DAMAGE UP TO 0.10-INCH MAY BE BLENDED OUT TO A ½-INCH RADIUS. MULTIPLE CORNER DAMAGE MUST BE AT LEAST 1-INCH APART.** If damage after repair exceeds limits, replace bifilar (PARA 5-4-28).

(2) Using corrosion resistant coating, Item 118, Appendix D, touch up reworked areas.

(3) Touch up paint all reworked areas (PARA 2-6-5).

d. Inspect mating surfaces and flanges for damage. If required, repair as follows:

(1) **ON MATING SURFACES AND FLANGES, REMOVE DAMAGE UP TO 0.020-INCH. DAMAGED AREA MUST NOT EXCEED 10% OF SURFACE AREA.** If damage after repair exceeds limits, replace bifilar (PARA 5-4-28).

(2) Remove raised material around damage and blend out to ½-inch radius.

(3) After repairing area, visually inspect for cracks. **NONE ALLOWED.**

(4) Using corrosion resistant coating, Item 118, Appendix D, touch up reworked areas.

(5) Touch up paint all reworked areas (PARA 2-6-5).

e. Visually inspect bifilar arm bushings for cracks. If crack is suspected, using fluorescent penetrant inspection kit, fluorescent penetrant inspect bifilar arm bushing inside diameter for stress cracks along its edges (ASTM E1417). **NONE ALLOWED.** If crack is found, replace bifilar arm bushings (PARA 5-5-11).

NOTE

- A knife edge on bifilar arm bushing edge is not cause for rejection.
- Original thickness of flange face on bifilar arm bushings, 70107-08202-106, 70107-08202-108, and 70107-08202-109, is 0.050-inch.
- Original thickness of flange face on bifilar arm bushings, 70107-08202-107, 70107-08202-110, and 70107-08202-111, is 0.0625-inch.

f. Inspect bifilar arm bushings for wear. **ON FLANGE FACES, UP TO 0.030-INCH WEAR IS ALLOWED.** If part number is not readable, or if wear exceeds limit, perform the following:

(1) If part number is not readable, use 0.050-inch as original thickness.

(2) If wear exceeds limit, replace bifilar arm bushings (PARA 5-5-11).

g. Inspect bifilar weight for nicks, gouges, and cracks. Visually inspect each bifilar weight for cracks between, and adjacent to, bifilar arm bushing bores on upper and lower surfaces of each flange. **NONE ALLOWED.** If crack is found, inspect/repair AREAS A, B, and C as follows (Figure 5-3-10-2, Detail A):

(1) **AREA A - NICKS AND GOUGES MUST NOT EXCEED 0.150-INCH DEEP.**

(2) **AREA B - CORNER NICKS AND GOUGES MUST NOT EXCEED 0.10-INCH DEEP. SURFACE NICKS AND GOUGES MUST NOT EXCEED 0.030-INCH DEEP. NO CRACKS ALLOWED.**

(3) **AREA C - NO NICKS OR GOUGES ALLOWED.**

(4) Using abrasive cloth, Item 12, Appendix D, or abrasive wheel, Item 22, Appendix D, blend nicks and gouges in AREAS A and B to ½-inch radius.

(5) If nicks, gouges, and cracks exceed limits, replace bifilar weight (PARA 5-4-29).

NOTE

Cracked paint in AREA C of bifilar weights with ″UT″ or ″X″ marked in area of part number is acceptable.

h. Inspect bifilar weights marked ″UT″ or ″X″ following part number. If letters ″UT″ or ″X″ are present, no further inspection is required. If letters ″PL″ or no part marking after part number, visually inspect each bifilar weight for cracked paint in bonded joint area (AREA C). **NONE ALLOWED.** If crack in paint is found, perform the following:

(1) Using methyl ethyl ketone, Item 217, Appendix D, remove paint.

(2) Inspect AREAS A and B for cracks. **NONE ALLOWED.** If crack is found, replace bifilar weight (PARA 5-4-29).

(3) Inspect AREA C for bond separation. **NONE ALLOWED.** If bond separation is found, replace bifilar weight (PARA 5-4-29).

(4) Touch up paint on bifilar weight (PARA 2-6-5).

i. Inspect bumper head for damage. **NONE ALLOWED.** If damage is found, replace bumper head as follows (Figure 5-3-10-2):

(1) Using nonmetallic scraper, remove damaged bumper head.

(2) Using abrasive cloth, Item 12, Appendix D, clean bumper head area on bifilar weight by rubbing until surface is clean and bright.

(3) Using methyl ethyl ketone, Item 217, Appendix D, wipe contact area.

(4) Using acid brush, Item 84, Appendix D, apply primer, Item 254, Appendix D, on new bumper head and bonding area on bifilar weight.

(5) Allow primer to cure at room temperature for 3 minutes.

(6) Apply bead of adhesive, Item 64, Appendix D, to bifilar weight. Immediately place bumper head on bumper head area on bifilar weight. Apply pressure for 15 minutes before handling bifilar weight.

(7) Allow adhesive to cure for 24 hours before installing bifilar weight on bifilar.

j. Inspect pins for damage. If required, repair as follows:

(1) **IN NONCONTACT AREAS, USING ABRASIVE CLOTH, ITEM 13, APPENDIX D, REMOVE DAMAGE UP TO 0.020-INCH DEEP BY BLENDING OUT.**

(2) **CORNER DAMAGE UP TO 0.040-INCH MAY BE BLENDED OUT.** If damage exceeds limit, replace pin(s) (PARA 5-4-29).

k. Inspect pins for out-of-round. **IF PINS EXCEED 0.002-INCH OUT-OF-ROUND, REPLACE PIN(S)** (PARA 5-4-29).

l. Inspect diameter of pins for wear. **IF PIN WEAR EXCEEDS 0.010-INCH, REPLACE PIN(S)** (PARA 5-4-29). Refer to Table 5-3-10-1 for original diameter of pins.

Table 5-3-10-1. Pin Diameters

Part Number	Original Diameter (Inches)
70107-08406-102	1.173
70107-08406-103	1.194
70107-08406-112	1.173
70107-08406-113	1.194
70107-08406-121	1.200
70107-08406-122	1.200

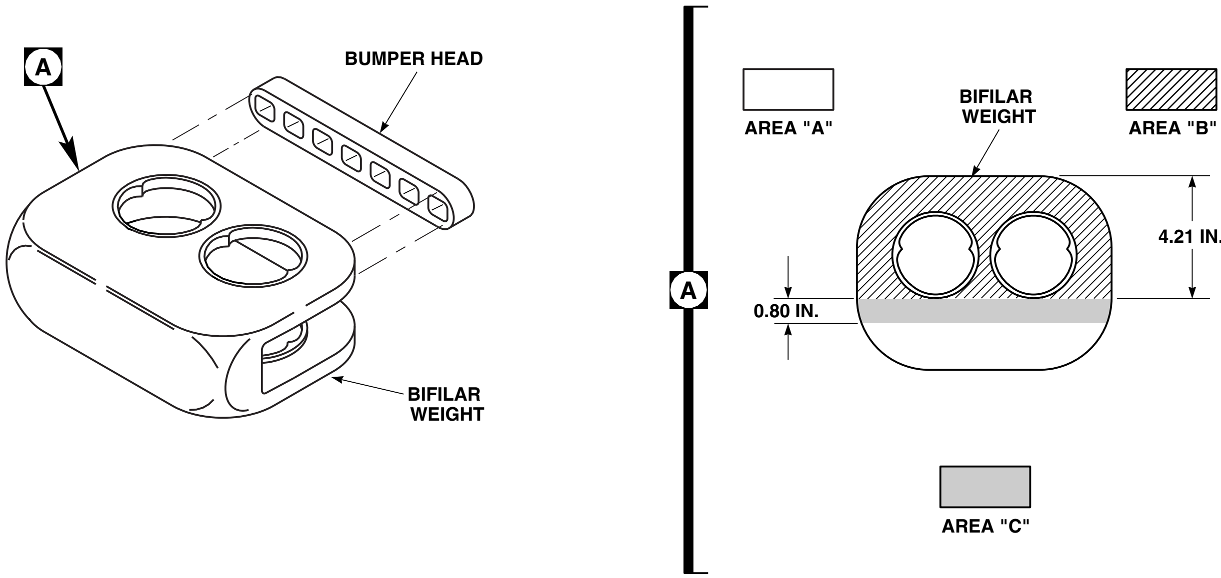
m. Inspect pins for cracks. **NONE ALLOWED.** If crack is suspected, perform the following:

(1) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean pin.

(2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace pin(s) (PARA 5-4-29).

n. Inspect tapered washers for damage. If required, repair as follows (Figure 5-3-10-3):

(1) **IN NONCONTACT AREAS, USING ABRASIVE CLOTH, ITEM 13, AP-**



NOTE

EFFECTIVE ON BIFILAR WEIGHT, 70107–08404–044.

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FIGURE 5-3-10-2. INSPECT BIFILAR WEIGHTS

PENDIX D, REMOVE DAMAGE UP TO 0.020-INCH DEEP BY BLENDING OUT.

(2) **CORNER DAMAGE UP TO 0.040-INCH MAY BE BLENDED OUT.** If damage exceeds limit, replace tapered washer(s) (PARA 5-4-29).

o. Inspect tapered washers for wear. If required, repair as follows:

(1) **IF MORE THAN 20% OF TAPERED WASHER CONTACT SURFACES ARE PITTED, WORN, OR HAVE METAL PICKUP ON SURFACE, REPLACE TAPERED WASHER(S)** (PARA 5-4-29).

(2) **CIRCULAR WEAR UP TO 0.020-INCH DEEP IS ACCEPTABLE IN CONTACT AREA BETWEEN BIFILAR ARM BUSHINGS AND TAPERED WASHERS.** If wear exceeds limits, replace tapered washer(s) (PARA 5-4-29).

p. Visually inspect tapered washers for cracks. If crack is suspected, using fluorescent penetrant inspection kit, fluorescent penetrant inspect for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace tapered washer(s) (PARA 5-4-29).

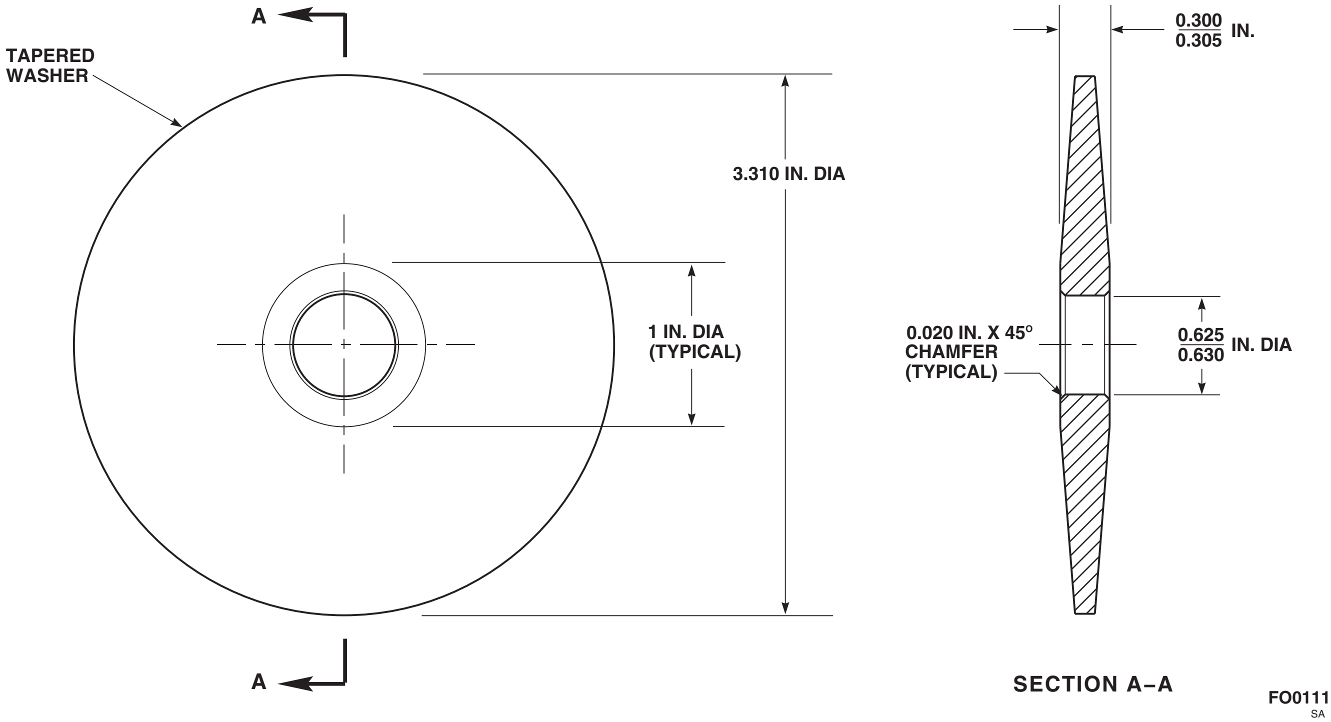


FIGURE 5-3-10-3. INSPECT TAPERED WASHERS

5-3-11 MAIN ROTOR BLADE INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Coin, Soft Copper (or equivalent metal), 1 ½ - 2-inch diameter × ⅛ - 3⁄16-inch thick, Procured or Locally-Made
- Drill
- Drill Bit, No. 40, 0.098-inch diameter
- Drill Bit, 0.190-inch diameter
- Grease Pencil
- Magnifying Glass, 10X
- Medium Cut File
- Nonmetallic Scraper, Locally-Made
- Punch, Soft Metal
- Scale
- Straight Edge
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Abrasive Cloth, Item 5, Appendix D
- Abrasive Cloth, Item 6, Appendix D
- Abrasive Cloth, Item 12, Appendix D
- Sealing Compound, Item 296, Appendix D

References

- Appendix D
- MIL-S-18729
- PARA 5-2-2
- PARA 5-4-31

- PARA 5-4-37
- PARA 12-2-5
- PARA 16-3-2
- PARA 16-6-9
- PARA 16-6-12
- QQ-A-250

5-3-11.1 General.

WARNING

The cause for appearance of red bands, or portions of red bands, in BIM® pressure indicator, must be determined and corrected before flight (PARA 5-2-2). Discovery of main rotor blade spar failure, or inability to determine cause for appearance of red bands, will require replacement of main rotor blade.

NOTE

Inspections of all main rotor blades are the same.

a. Refer to Table 5-3-11-1 and Figure 5-3-11-1, Sheet 1 and Sheet 2, for reference to damage limits and repair procedures on main rotor blade.

b. Damage to main rotor blade can be divided into two types, ″detectable″ and ″nondetectable″:

NOTE

If necessary, use scale to determine what four - five pounds of pressure feels like.

(1) Detectable damage can be seen and felt as blisters or dents as follows:

(a) Blisters can be seen as raised bumps above main rotor blade contour, indicating separation between main rotor blade skin and core. Four - five pounds of thumb pressure can move main rotor blade skin against core without ″crinkly″ sound.

(b) Dents indicate core crush, and can be felt and seen. Four - five pounds of thumb pressure against dent will result in a ″crinkly″ sound.

(2) Nondetectable damage can be found with coin-tapping, but cannot be seen or felt. It can be heard as follows:

(a) Four - five pounds of thumb pressure will not make sound with nondetectable blisters.

(b) Four - five pounds of thumb pressure will make ″crinkly″ sound with nondetectable dents.

c. If damage exceeds maximum skin, plug, or trailing edge patch limits, replace main rotor blade (PARA 5-4-31).

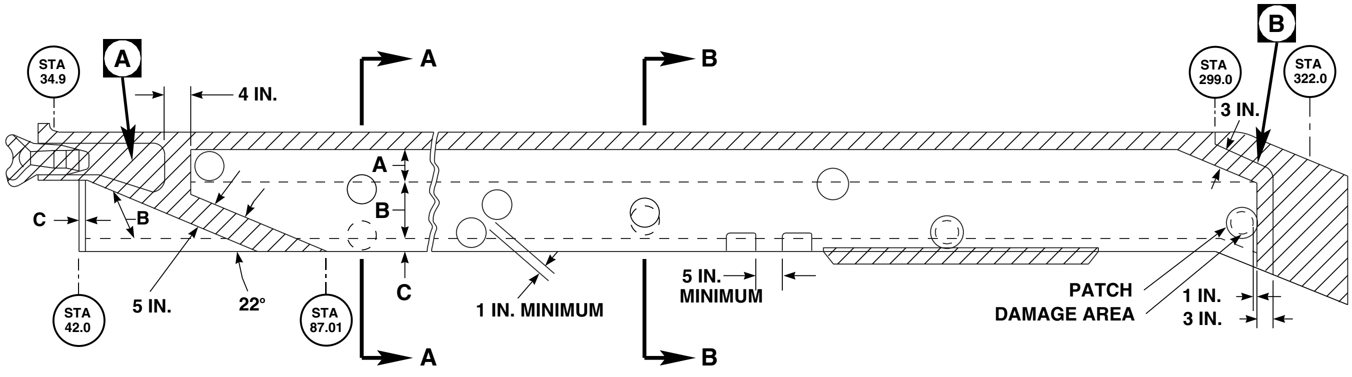
d. Skin patch portion of plug patch must overlap minimum of 1-inch beyond damage, and may extend 1-inch outside honeycomb core region (Figure 5-3-11-1, Sheet 1, Section A-A).

e. Damage not requiring skin, plug, or trailing edge patch repairs may extend into shaded area and main rotor blade spar (Figure 5-3-11-1, Sheet 1).

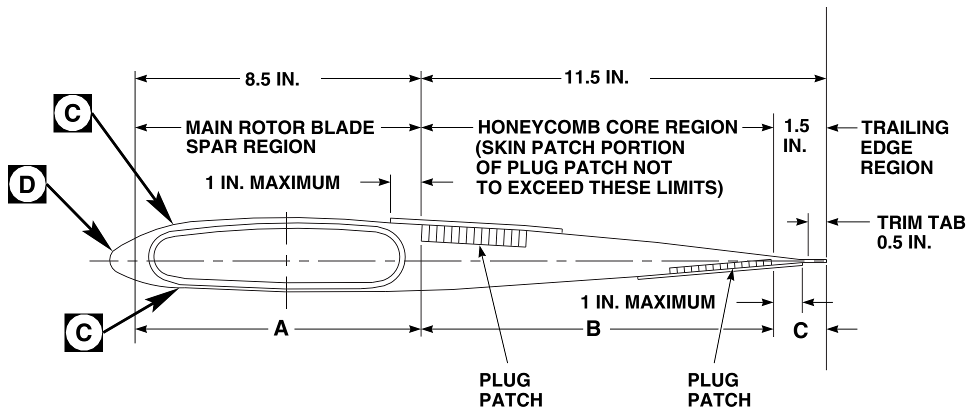
f. Edge of new skin and plug patch repairs are not to be closer than 1-inch from edge of previously installed skin and plug patch repairs on same side of main rotor blade. Trailing edge patch repairs are not to be closer than 5-inches from previously installed trailing edge patch repairs.

g. Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.

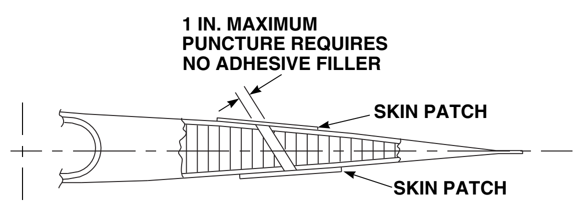
h. Damage to main rotor blade skin (reinforced plastic) will result if main rotor blade is sanded



MAIN ROTOR BLADE



TYPICAL REPAIR LIMITS
SECTION A–A



TYPICAL PUNCTURE REPAIR
SECTION B–B

LEGEND

A – MAIN ROTOR BLADE SPAR REGION – SURFACE REPAIR OF FIBERGLASS BY SKIN PATCH OR ADHESIVE FILLER.

B – HONEYCOMB CORE REGION–SURFACE, SUBSURFACE REPAIR BY SKIN AND PLUG PATCHES (ONE OR BOTH SIDES), AND ADHESIVE INJECTION.

C – TRAILING EDGE REGION–SURFACE, SUBSURFACE REPAIR BY SKIN PATCH, ADHESIVE FILLER, OR ADHESIVE INJECTION. CRACKS IN ADHESIVE, EA9309, ONLY FILL WITH SEALING COMPOUND, ITEM 296, APPENDIX D.

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FIGURE 5-3-11-1. MAIN ROTOR BLADE GENERAL REPAIR LIMITS (SHEET 1 OF 2)

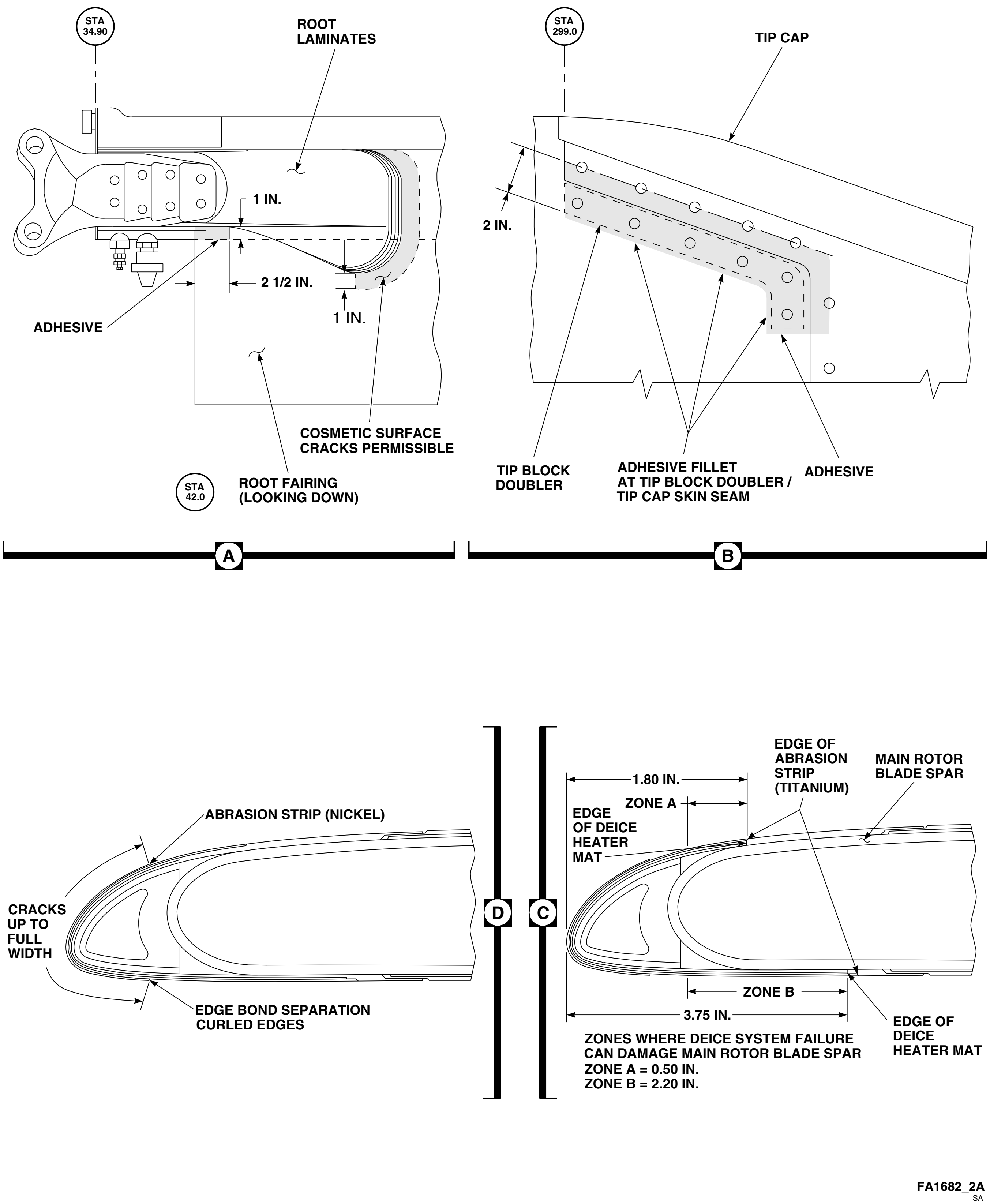


FIGURE 5-3-11-1. MAIN ROTOR BLADE GENERAL REPAIR LIMITS (SHEET 2 OF 2)

excessively. Heavy or careless sanding can produce enough damage to cause main rotor blade skin cracking. Sand main rotor blade skin lightly and carefully.

i. Send damaged tail rotor blades that exceed repairable limits to manufacturer.

j. Remove and install main rotor blades as required (PARA 5-4-31).

k. Record all damage and repairs to main rotor blade.

5-3-11.2 Coin-Tapping Inspection Method.

NOTE

- To verify suspected damage, use coin-tapping inspection method to determine size and shape of disbonds/delaminations between main rotor blade skin and main rotor blade spar, main rotor blade skin and honeycomb core, trim tab laminations, and tip cap and abrasion strip bond areas.
- Coin-tapping should be done by experienced personnel only.

a. Procure or locally make coin. Coin is smooth disc of soft copper (or equivalent metal), 1 ½ - 2-inches in diameter × ⅛ - 3⁄16-inch thick. Round and smooth edge of coin to prevent damage to main rotor blade skin.

b. Hold coin loosely between thumb and finger, tap lightly along bond line or areas suspected of having disbonds/delaminations, and listen for variations in tapping sound. A ″sharp-solid″ sound indicates a good bond and a ″dull-dead″ sound indicates a bond separation.

c. Tap across suspected area in horizontal, vertical, and diagonal lines, and using grease pencil, mark points on main rotor blade skin where sound varies. Continue tapping and marking until outline of bond separation is completed.

5-3-11.3 Main Rotor Blade Cuff Inspection.

NOTE

Liner disbond is repaired at manufacturer.

a. Visually inspect for security of attaching hardware (Figure 5-3-11-2, Sheet 2, Detail A).

b. Visually inspect for nicks, scratches, and corrosion.

c. Visually inspect for cracks in lugs of main rotor blade cuff. **NONE ALLOWED.** If crack is found, replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.

d. Visually inspect main rotor blade attach bolts and nuts for corrosion.

5-3-11.4 Root Laminates, Root Cap, and Root Fairing Inspection.

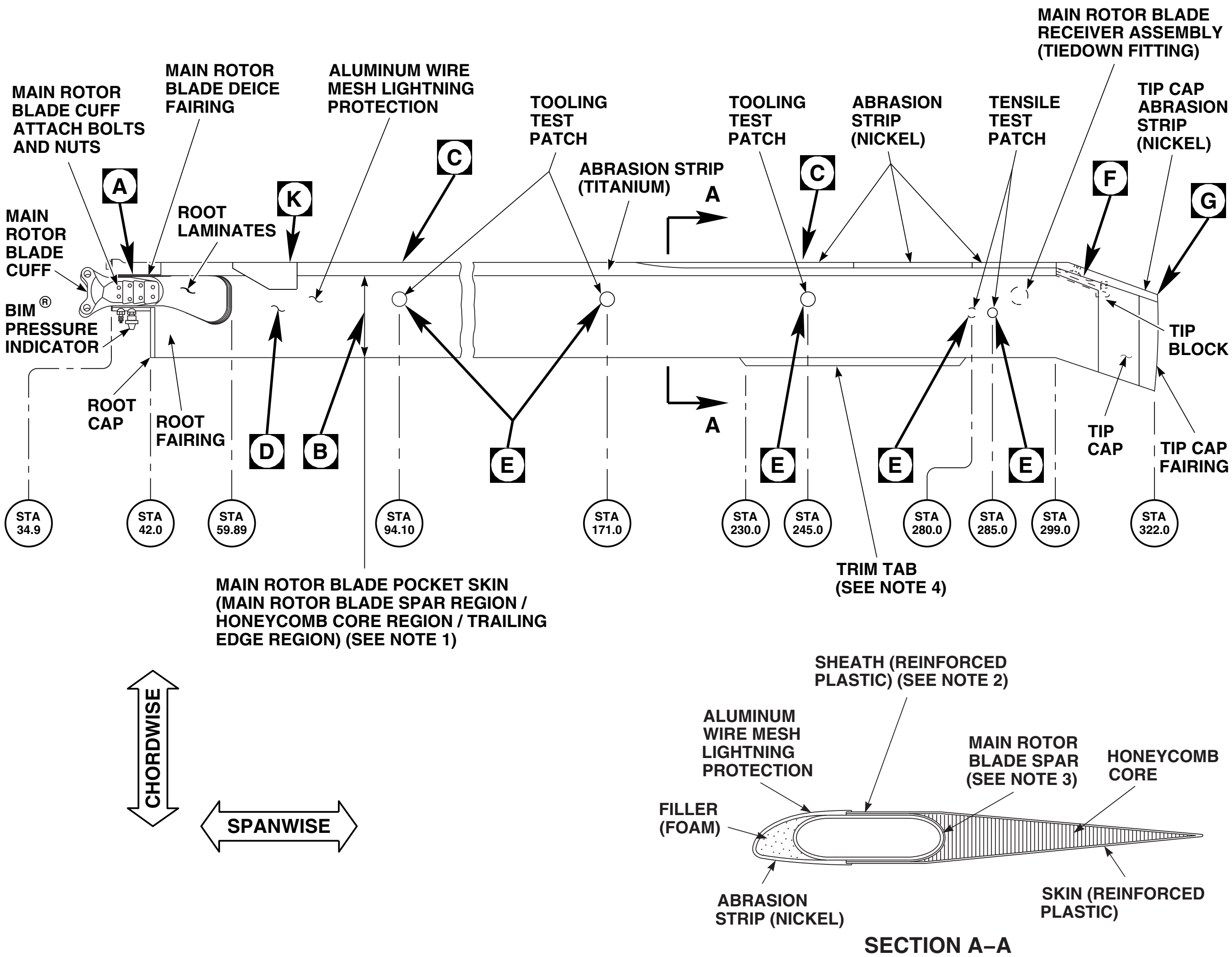
NOTE

Cracks found in paint will normally follow contour of root laminate and will be parallel to root laminate edge.

a. Visually inspect edge of root laminates for cracks in paint (Figure 5-3-11-2, Sheet 2, Detail A). If crack is found, proceed as follows:

(1) Using coin-tapping inspection method, inspect for disbonding/delamination (PARA 5-3-11.2). **NONE ALLOWED.** If inspection reveals disbonding/delamination, replace main rotor blade (PARA 5-4-31), and send blade to manufacturer.

(2) If no disbonding/delamination is noted, carefully blend out cracks and touch up paint (PARA 5-4-31).



MAIN ROTOR BLADE

NOTES

1. COMPOSITE MADE FROM GLASS FABRIC AND GRAPHITE PLIES, BONDED WITH ADHESIVE RESIN.
2. REINFORCED PLASTIC (0.090 IN.) INCORPORATES DEICE HEATER MAT.
3. INCORPORATES BIM® BLANKET.
4. HELICOPTERS WITH MAIN ROTOR BLADES SERIAL NOS A007–02550 TO A007–02900 WILL HAVE A NOTCH IN TRIM TAB.
5. ALL SHEET METAL CALLOUTS ARE 6061–T6 ALUMINUM ALLOY, QQ–A–250/11.
   A – BASIC SKIN (0.063 IN.) REDUCED TO 0.033–0.043 IN. THICKNESS.
   B – BASIC SKIN (0.063 IN.) REDUCED TO 0.018–0.023 IN. THICKNESS.
6. CRACKS ARE USUALLY FOUND AT HOLES 4 AND 5; 1/2 IN. OUTBOARD OF HOLE 6; BETWEEN HOLES 6 AND 7, 1 IN. AFT OF HOLE 6.
7. ON TIP CAP, 70150–09107–049 AND –055.

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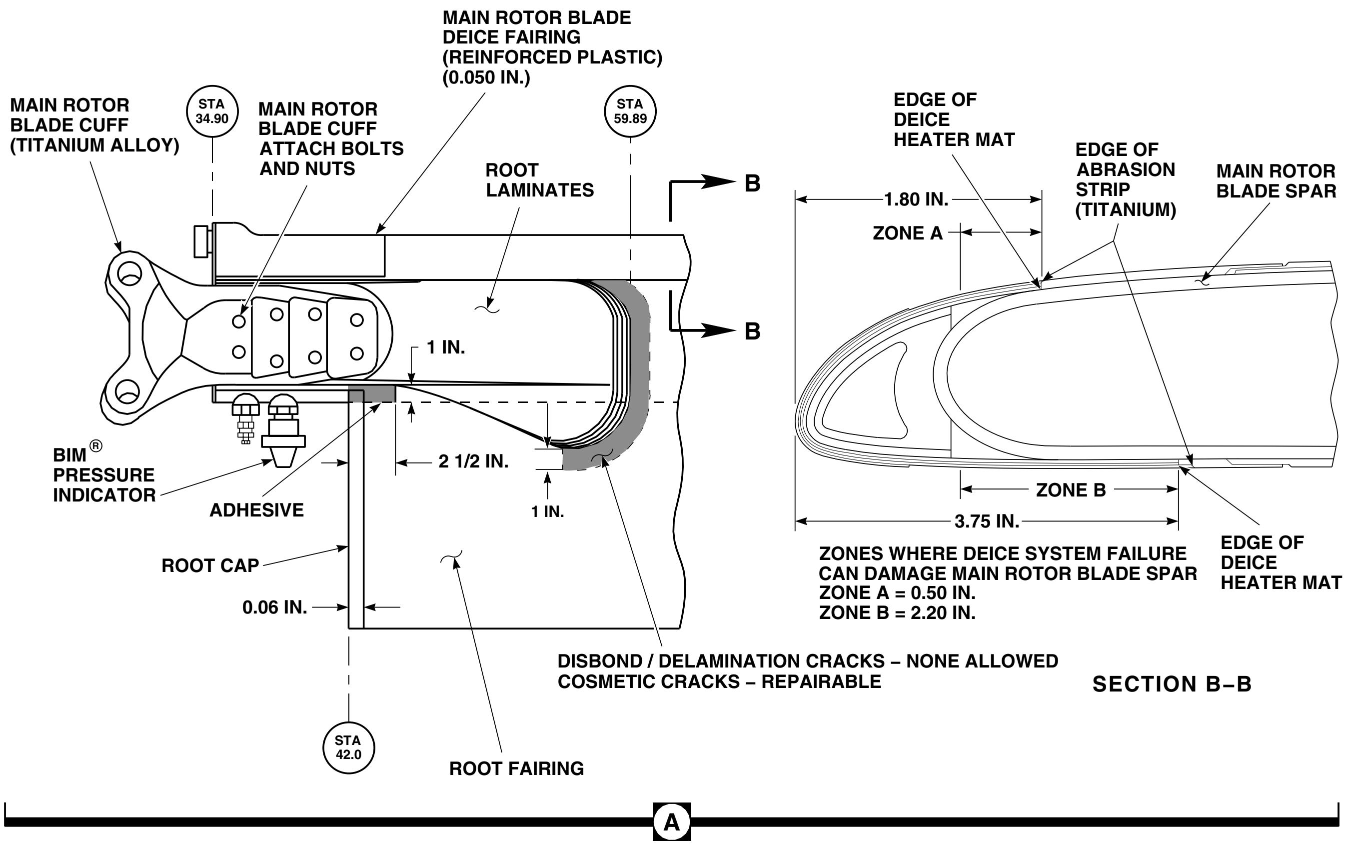
FIGURE 5-3-11-2. MAIN ROTOR BLADE INSPECTION AREAS (SHEET 1 OF 8)



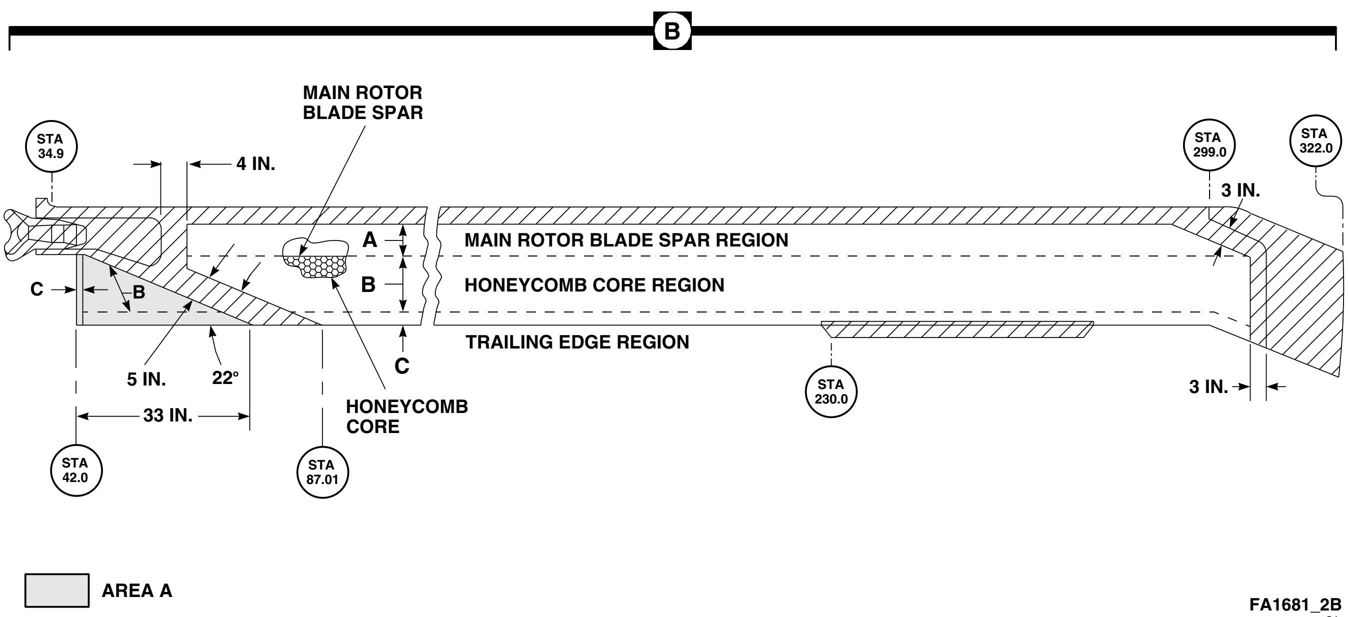
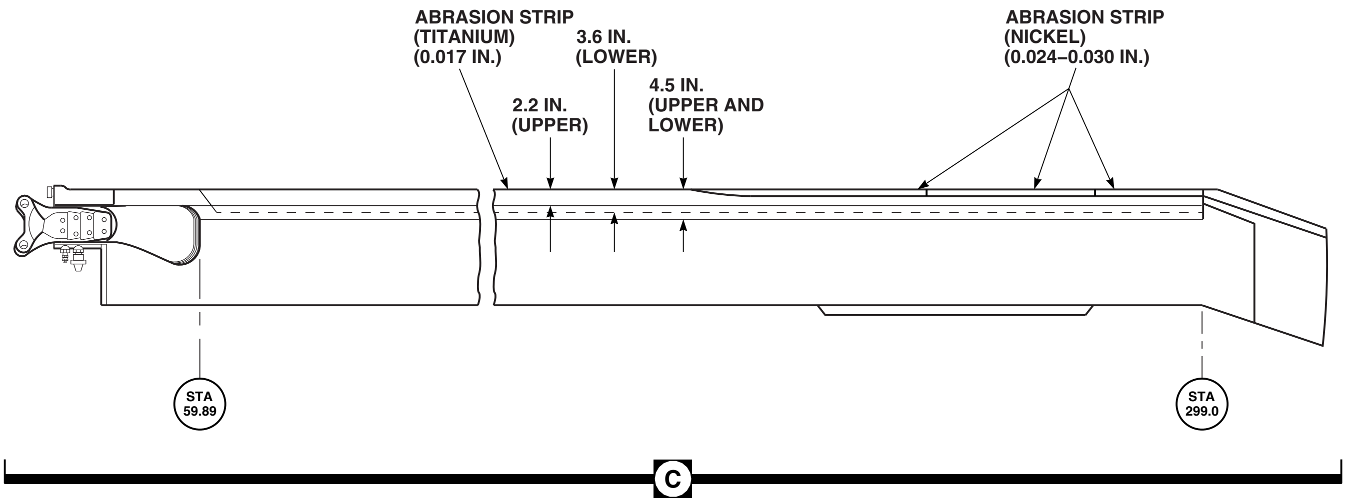


FIGURE 5-3-11-2. MAIN ROTOR BLADE INSPECTION AREAS (SHEET 2 OF 8)



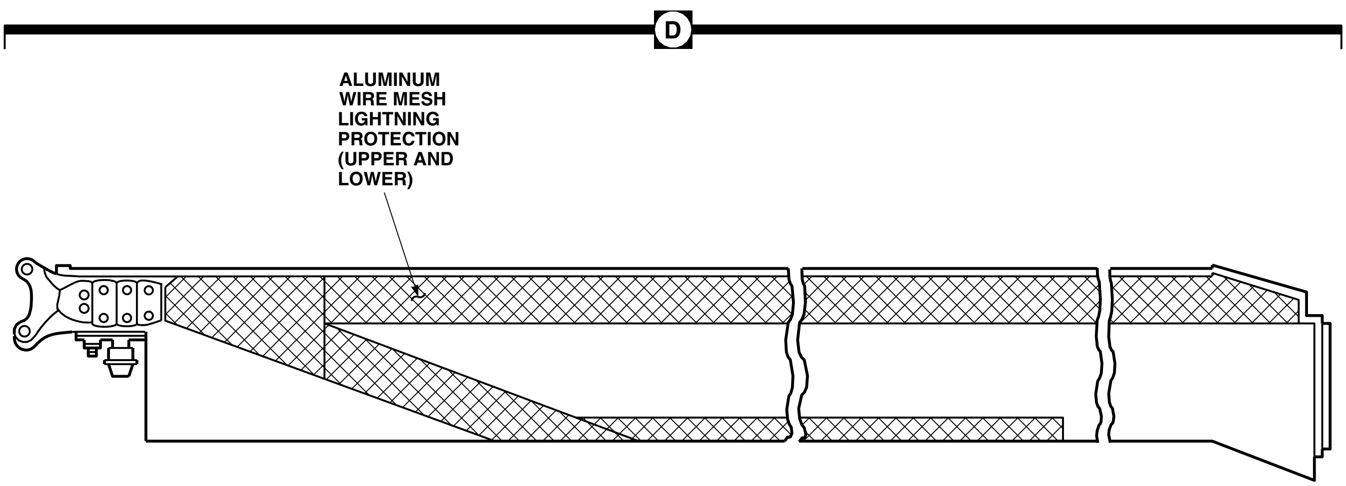


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FIGURE 5-3-11-2. MAIN ROTOR BLADE INSPECTION AREAS (SHEET 3 OF 8)

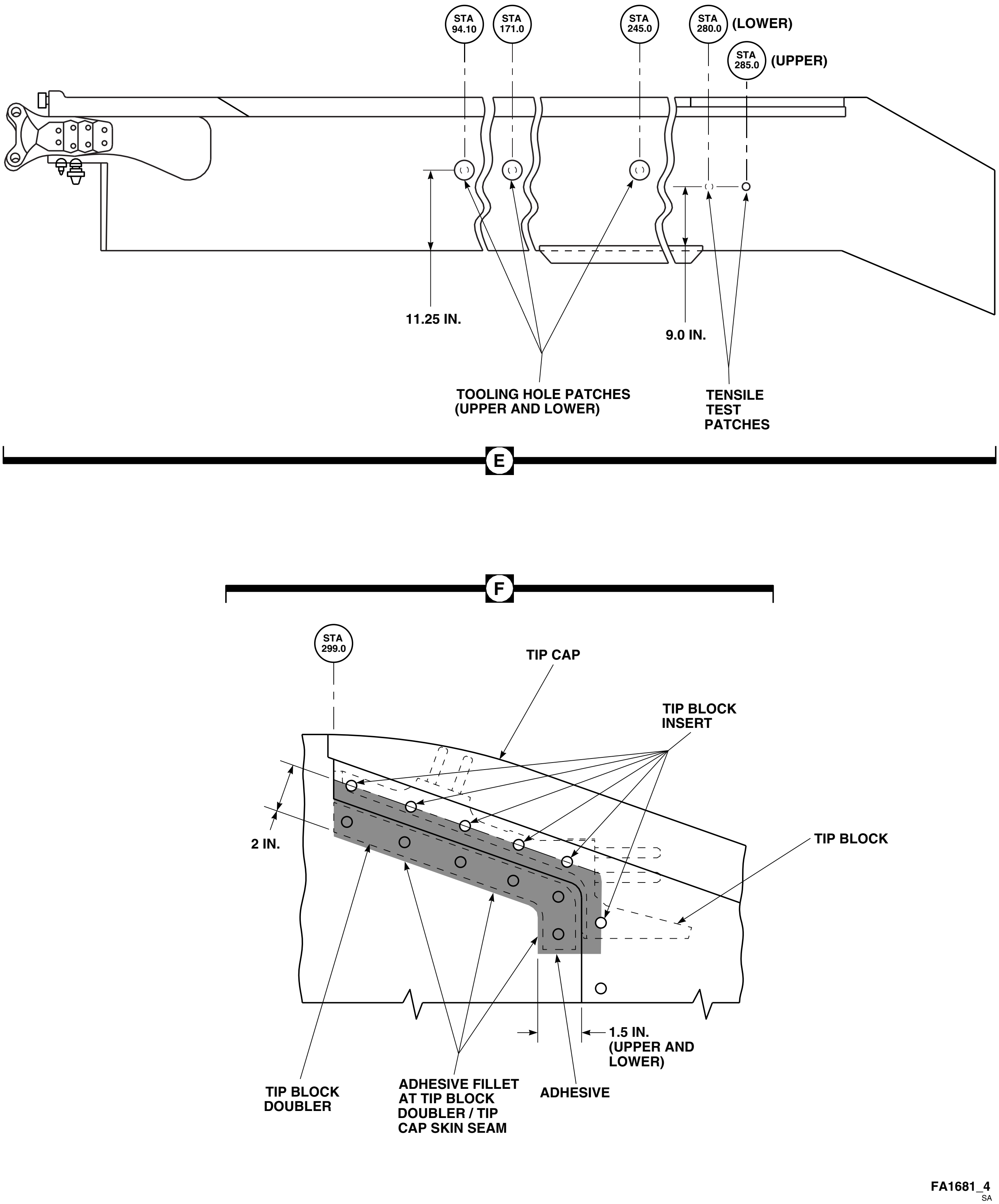


FIGURE 5-3-11-2. MAIN ROTOR BLADE INSPECTION AREAS (SHEET 4 OF 8)

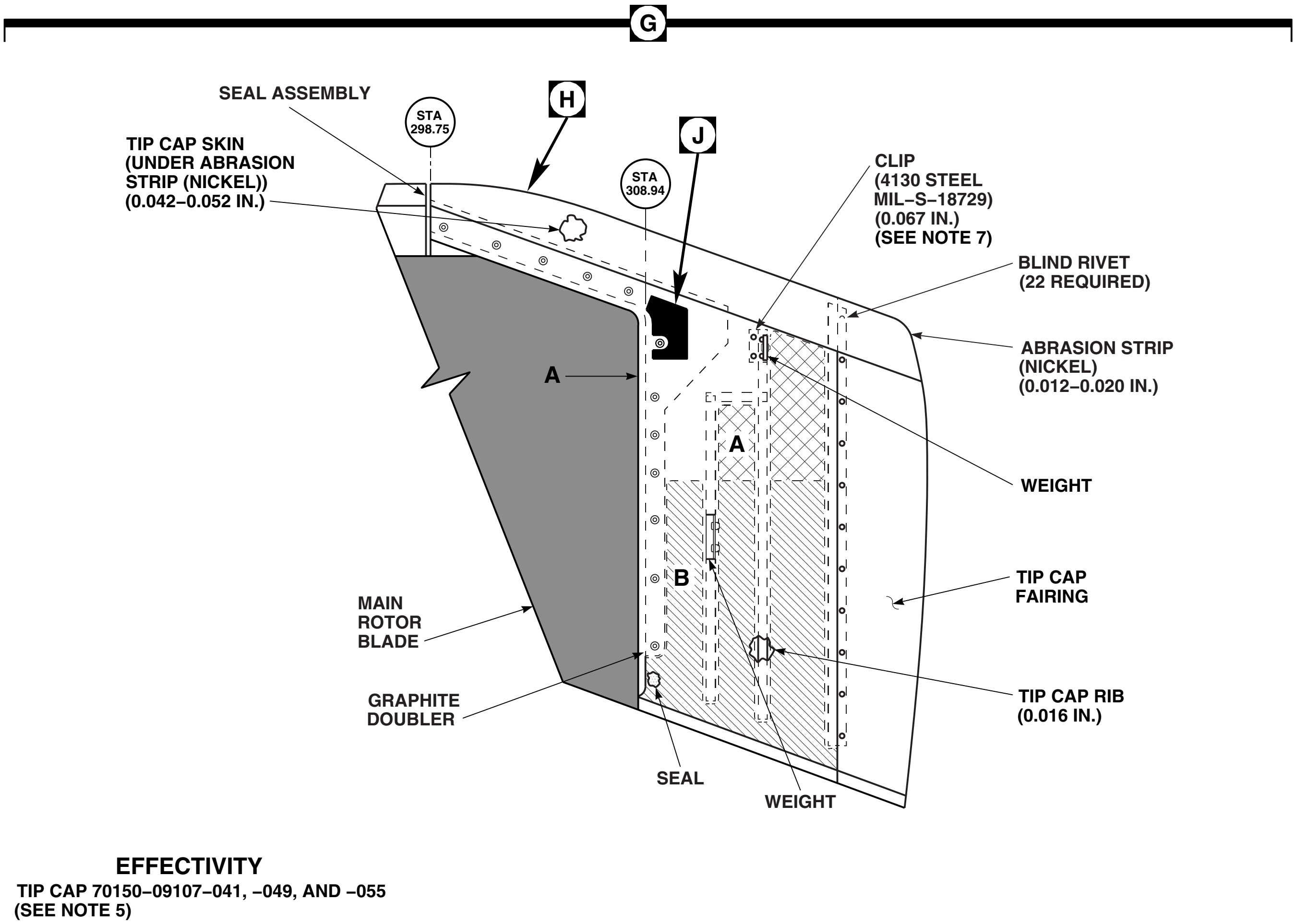


FIGURE 5-3-11-2. MAIN ROTOR BLADE INSPECTION AREAS (SHEET 5 OF 8)

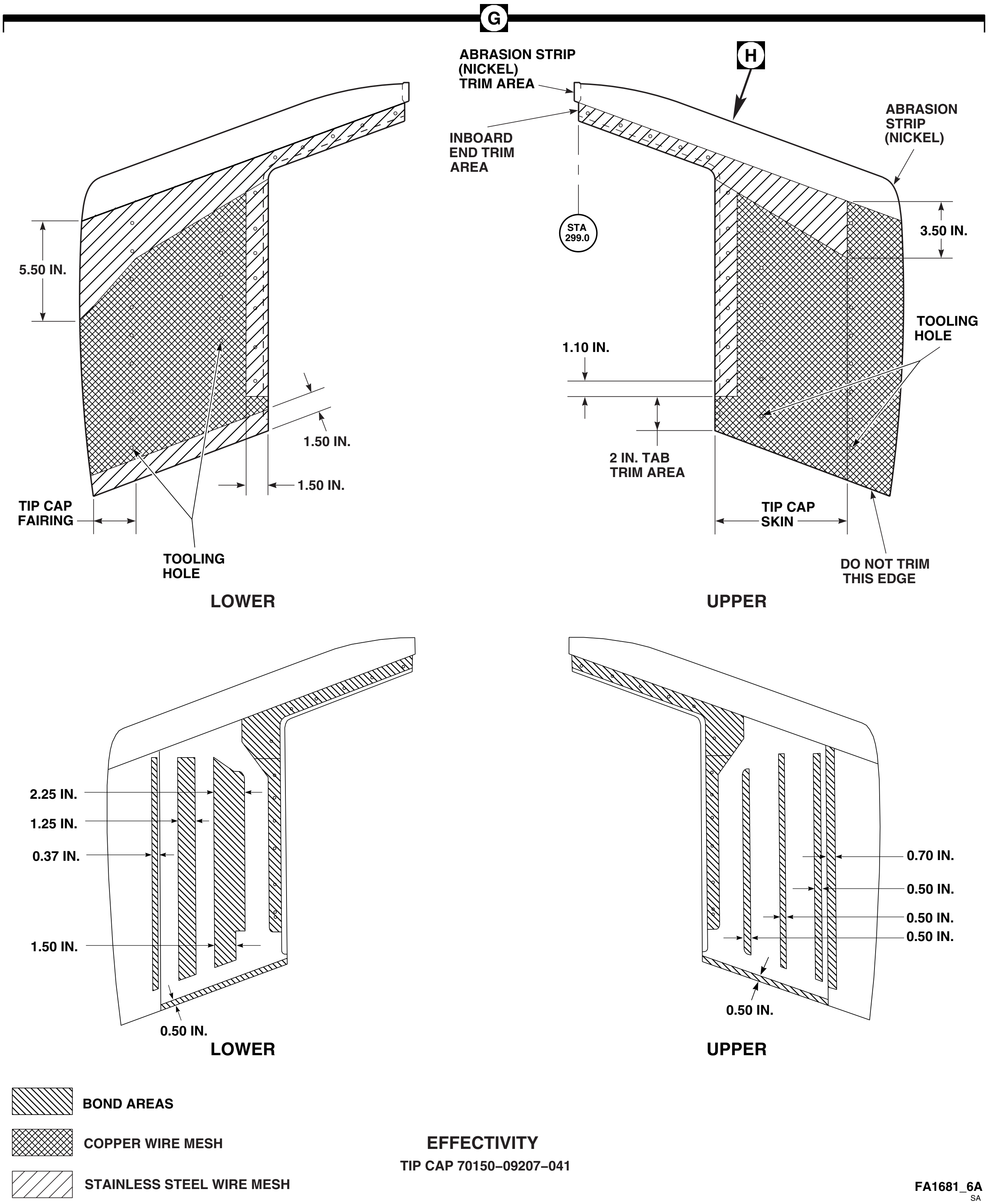


FIGURE 5-3-11-2. MAIN ROTOR BLADE INSPECTION AREAS (SHEET 6 OF 8)

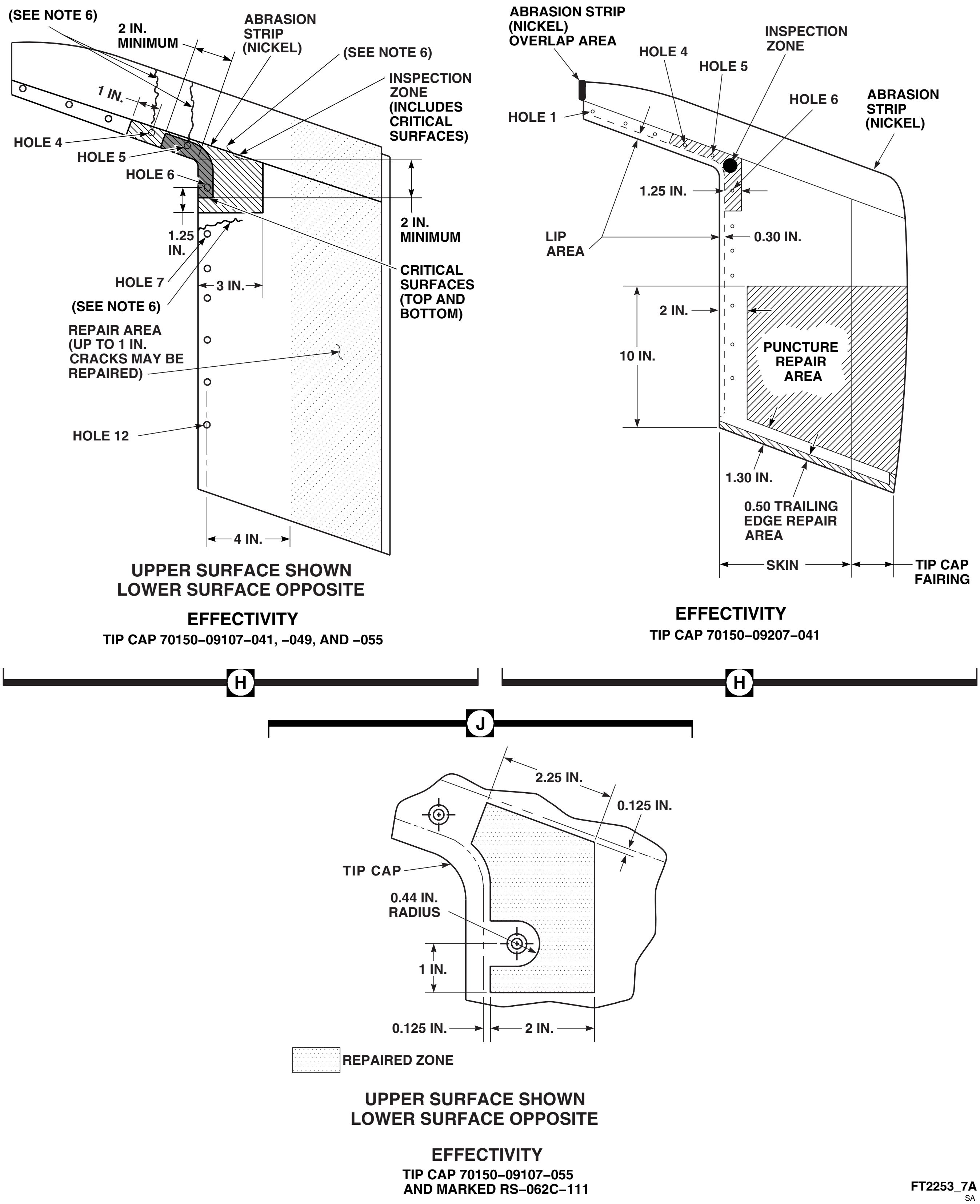


FIGURE 5-3-11-2. MAIN ROTOR BLADE INSPECTION AREAS (SHEET 7 OF 8)

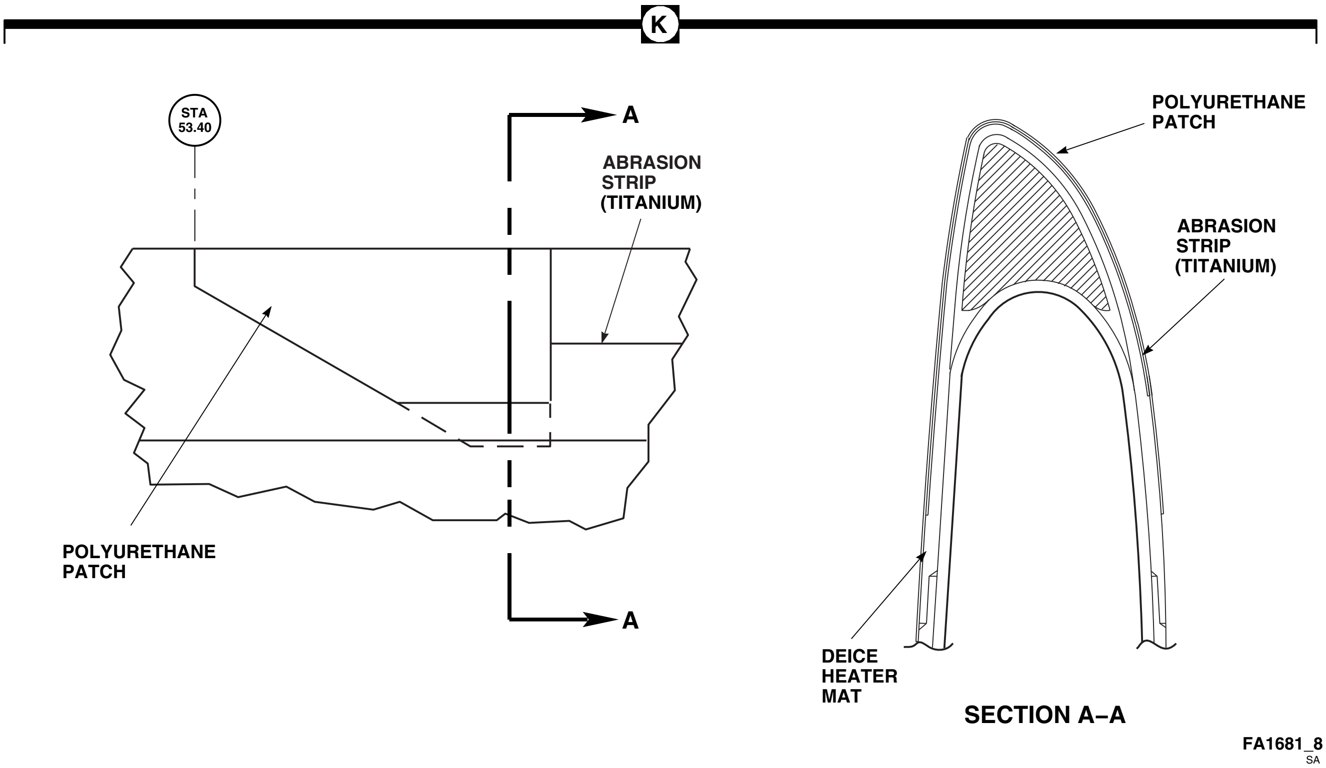


FIGURE 5-3-11-2. MAIN ROTOR BLADE INSPECTION AREAS (SHEET 8 OF 8)

NOTE

Cosmetic cracking is a frequent occurrence and is limited to cracking of adhesive and/or wire mesh only. Cosmetic cracking is repairable. Cracking of skin substrate is not cosmetic and is not an acceptable condition. Any cracking of skin substrate must be reported to Sikorsky Engineering.

b. Visually inspect for cosmetic cracks just outboard of root laminates. **NONE ALLOWED.** If crack is found, repair using cosmetic adhesive crack repair (PARA 5-4-31).

c. Visually inspect edge of root cap for cracks in paint. If crack is found, proceed as follows:

(1) Inspect for disbonding/delamination using coin-tapping inspection method (PARA 5-3-11.2). **NONE ALLOWED.** If inspection reveals disbonding/delamination, replace main rotor blade (PARA 5-4-31), and send blade to manufacturer.

(2) If no disbonding/delamination is noted, carefully blend out cracks and touch up paint (PARA 5-4-31).

d. Visually inspect deicing system for damage (Figure 5-3-11-2, Sheet 2, Detail A, Section B-B). **NONE ALLOWED.** If damage is found, replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.

NOTE

Sealing compound must be applied within first 50 flight-hours.

e. Seal cracks through main rotor blade skin and adhesive in shaded area (Figure 5-3-11-2, Sheet 2, Detail A) with layer of sealing compound, Item 296, Appendix D, using adhesive filler repair (PARA 5-4-31).

5-3-11.5 Main Rotor Blade Deice Fairing Inspection.

Main rotor blade deice fairing is repaired using adhesive filler repair or adhesive injection repair

(Figure 5-3-11-2, Sheet 2, Detail A) (PARA 5-4-31).

5-3-11.6 Main Rotor Blade Spar Region Inspection.

a. Visually inspect main rotor blade skin for dents, gouges, scratches, and surface imperfections (Figure 5-3-11-2, Sheet 2, Detail B).

b. Using coin-tapping inspection method, inspect main rotor blade skin over main rotor blade spar for disbonds/delaminations (PARA 5-3-11.2).

c. Inspect depth of dents and gouges to determine if damage has penetrated to main rotor blade spar.

d. Record location, dimensions, degree of damage, and surface area of disbonds/delaminations.

e. If necessary, using adhesive filler repair or skin patch repair (PARA 5-4-31), repair main rotor blade spar region. Adhere to the following guidelines:

(1) Area damage limited to 6-inch diameter (for 8-inch diameter patch) (Figure 5-3-11-1, Sheet 1).

(2) Two or more damaged areas falling within 6-inch diameter will be combined and patched as one repair.

(3) New damage occurring within previously repaired damage, with total diameter greater than 6-inches, will not be repaired. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.

5-3-11.7 Honeycomb Core Region Inspection.

a. Visually inspect main rotor blade skin/honeycomb core area for dents, crushes, breaks, blisters, gouges, and holes (Figure 5-3-11-2, Sheet 2, Detail B).

b. Using coin-tapping inspection method, inspect main rotor blade skin and area around main rotor blade receiver assembly (tiedown fitting) for disbonds/delaminations (PARA 5-3-11.2).

c. Record location, dimensions, degree of damage, and surface area of disbonds/delaminations.

d. If necessary, using skin patch repair or plug patch repair (PARA 5-4-31), repair honeycomb core region. Adhere to the following guidelines:

(1) Area damage limited to 6-inch diameter (Figure 5-3-11-1, Sheet 1).

(2) For plug patch, diameter of routed hole must be confined to honeycomb core region. Overlapping skin of plug patch may extend 1-inch beyond boundary of core.

(3) Two or more damaged areas falling within 6-inch diameter will be combined and patched as one repair.

(4) New damage occurring within previously repaired damage, with total diameter greater than 6-inches, will not be repaired. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.

5-3-11.8 Trailing Edge Region Inspection.

a. Visually inspect for dents, gouges, and scratches, at upper, lower, and thickness of trailing edge (Figure 5-3-11-2, Sheet 2, Detail B).

b. Using coin-tapping inspection method, inspect for disbonds/delaminations at trailing edge thickness. Inspect condition of bond between trailing edge skin and trim tab (PARA 5-3-11.2).

c. Record location, dimensions, degree of damage, and surface area of disbonds/delaminations.

d. Visually inspect edges of trim tab for cracks in paint or adhesive running spanwise at front edge or chordwise at inboard and outboard ends of trim tab. If crack is found, inspect for bonding voids, using coin-tapping inspection method (PARA 5-3-11.2).

NOTE

- Skin patches in honeycomb core region may extend into trailing edge region.
- Two trailing edge patches must be no closer than 5-inches along trailing edge.

e. Trailing edge region is repaired using trailing edge patch repair, adhesive filler repair, or adhesive injection repair (Figure 5-3-11-1, Sheet 1 and PARA 5-4-31).

5-3-11.9 Abrasion Strips Inspection.

a. Visually inspect abrasion strips for erosion, dents, holes, cracks, and scratches (Figure 5-3-11-2, Sheet 3, Detail C).

b. Using coin-tapping inspection method, inspect for bond voids/separations in leading and trailing edges, upper and lower (PARA 5-3-11.2).

c. Visually inspect for spanwise cracks in paint 2.2-inches from leading edge of abrasion strip on upper surface and 3.6-inches from leading edge of abrasion strip on lower surface. If crack is found, inspect for bonding voids, using coin-tapping inspection method (PARA 5-3-11.2).

d. Visually inspect for spanwise cracks, small pits, or voids in paint 4 ½-inches from leading edge on upper and lower surface. If crack is found, inspect for bonding voids, using coin-tapping inspection method (PARA 5-3-11.2).

CAUTION

Damage to main rotor blade spar will result if damage to abrasion strip (titanium) affects deice heater mat wires, if installed, under abrasion strip (titanium). Damaged deice heater mat wires may cause an arc burn to main rotor blade spar, resulting in main rotor blade spar crack.

e. If installed, check blade deice system for proper operation (PARA 12-2-5).

f. If necessary, using adhesive filler repair or adhesive injection repair (PARA 5-4-31), repair abrasion strip. Adhere to the following guideline:

(1) Erosion through abrasion strip will not be repaired (Figure 5-3-11-1, Sheet 2, Detail C and Detail D). Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.

5-3-11.10 Tooling and Tensile Test Patches Inspection.

Visually inspect tooling and tensile test patches for cracks in paint and slight contour changes (Figure 5-3-11-2, Sheet 4, Detail E). If crack or contour change is found, inspect for disbonding/delamination, using coin-tapping inspection method (PARA 5-3-11.2).

5-3-11.11 Tip Cap and Tip Cap Fairing Inspection.

5-3-11.11.1 General.

WARNING

Injury to personnel and damage to equipment will result if the following inspection is not carefully performed. Tip cap could separate from main rotor blade. If erosion protection coating or boots are applied, make sure they are completely removed from inspection zone to inspect for cracks. Do not reapply erosion protection coating or boots to inspection zone.

NOTE

This inspection applies to tip caps, 70150-09107-041, 70150-09107-049, 70150-09107-055, and 70150-09207-041 only.

a. If installed, remove polyurethane coating or boot in inspection zone of tip cap (Figure 5-3-11-2, Sheet 7, Detail H) (PARA 16-6-12).

b. If installed, inspect/repair polyurethane coating or boot (PARA 16-3-2).

c. Visually inspect for secure attachment.

WARNING

FSCAP

Tip caps, 70150-09107-041, 70150-09107-049, and 70150-09107-055, have critical surfaces which must not be damaged. Refer to Figure 5-3-11-2, Sheet 7, Detail H, for areas with critical characteristics.

d. Inspect main rotor blade skin for nicks, scratches, dents, scuffs, and edge damage in inspection zone (Figure 5-3-11-2, Sheet 7, Detail H). Nicks, scratches, dents, scuffs, and edge damage may be repaired, provided that damage does not exceed limits specified in Table 5-3-11-1.

e. Using at least 10X magnifying glass, inspect main rotor blade skin for cracks in inspection zone. **NONE ALLOWED.** If crack is found, replace tip cap (PARA 5-4-31).

f. Inspect main rotor blade skin for cracks in area a minimum of 4-inches outboard from center on attachment holes No. 6 through No. 12. Cracks up to 1-inch long may be repaired, provided that damage does not exceed limits specified in Table 5-3-11-1.

g. Inspect Kevlar surface for wear and fraying. Repair Kevlar surface using adhesive filler repair (PARA 5-4-31).

h. Inspect drain holes for elongation and wear. **OUTBOARD DRAIN HOLE ELONGATION MUST NOT EXCEED ¼-INCH VERTICALLY AND ½-INCH CHORDWISE. LOWER SURFACE DRAIN HOLE DIAMETER MUST NOT EXCEED ¼-INCH.**

i. Visually inspect tip cap attachment screws for corrosion. **NONE ALLOWED.** If corroded, replace screws (PARA 5-4-31).

NOTE

Tip caps marked, RS-062C-111, have been previously repaired at manufacturer. Refer to Figure 5-3-11-2, Sheet 7, Detail J, for repaired zone.

j. On tip caps marked, RS-062C-111, inspect for cracks progressing from previously stop-drilled hole (Figure 5-3-11-2, Sheet 7, Detail J).

k. Visually inspect for paint and adhesive cracks around tip block doubler 1 ½-inches inboard from joining edge of tip cap and main rotor blade and running parallel to edge of tip cap on upper and lower surface of main rotor blade (Figure 5-3-11-2, Sheet 4, Detail F). If crack is found, inspect for bonding voids, using coin-tapping inspection method (PARA 5-3-11.2). If bond is good, crack is acceptable and no repair is required.

l. Visually inspect for paint and adhesive cracks at end of tip block doubler running perpendicular to edge to tip cap approximately 1 ½-inches spanwise. If crack is found, inspect for bonding voids, using coin-tapping inspection method (PARA 5-3-11.2). If bond is good, crack is acceptable and no repair is required.

5-3-11.11.2 Tip Cap, 70150-09107-041, 70150-09107-049, and 70150-09107-055.

a. Visually inspect tip cap for erosion (Figure 5-3-11-2, Sheet 5, Detail G). If erosion is through tip cap abrasion strip, replace tip cap (PARA 5-4-31).

b. If necessary, patch holes in skin using reinforced plastic patches.

c. Inspect Kevlar surface for wear and fraying. If fraying is evident, repair Kevlar surface using adhesive filler repair (PARA 5-4-31).

d. Inspect main rotor blade skin for nicks, scratches, dents, and scuffs in inspection zone (Figure 5-3-11-2, Sheet 7, Detail H). Nicks, scratches, dents, and scuffs may be repaired, provided that damage does not exceed limits specified in Table 5-3-11-1.

e. Using at least 10X magnifying glass, inspect main rotor blade skin for cracks in inspection zone. **NONE ALLOWED.** If crack is found, replace tip cap (PARA 5-4-31).

f. Inspect main rotor blade skin for cracks in area a minimum of 4-inches outboard from center on attachment holes No. 6 through No. 12. Cracks up to 1-inch long may be repaired, provided that damage does not exceed limits specified in Table 5-3-11-1.

g. On tip cap, 70150-09107-055, scratches and gouges of any length in graphite doubler not exceeding 0.010-inch deep or penetrating into graphite core can be repaired using adhesive filler repair (PARA 5-4-31). If scratches and gouges exceed limits, replace tip cap (PARA 5-4-31).

h. Inspect drain holes for elongation and wear. Repair elongated outboard drain holes and erosion of surface surrounding outboard drain hole patched with fiberglass patches, using tip cap holes repair (PARA 5-4-31).

i. If lower surface drain holes are elongated, replace tip cap (PARA 5-4-31).

NOTE

Tip caps marked, RS-062C-111, have been previously repaired at manufacturer. Refer to Figure 5-3-11-2, Sheet 7, Detail J, for repaired zone.

j. On tip caps marked, RS-062C-111, if cracks progress from previously stop-drilled hole, replace tip cap (PARA 5-4-31).

k. On tip caps marked, RS-062C-111, if breaks or disbonding of internal graphite doubler are found, replace tip cap (PARA 5-4-31).

5-3-11.11.3 Tip Cap, 70150-09207-041.

a. Visually inspect tip cap for erosion (Figure 5-3-11-2, Sheet 6, Detail G). If erosion is through tip cap abrasion strip, replace tip cap (PARA 5-4-31).

b. If necessary, patch holes in skin using reinforced plastic patches.

c. Inspect Kevlar surface for wear and fraying. Repair Kevlar surface using adhesive filler repair (PARA 5-4-31).

d. Inspect main rotor blade skin for nicks, scratches, dents, and scuffs in inspection zone (Figure 5-3-11-2, Sheet 7, Detail H). Nicks, scratches, dents, and scuffs may be repaired, provided that damage does not exceed limits specified in Table 5-3-11-1.

e. Using at least 10X magnifying glass, inspect main rotor blade skin for cracks in inspection zone. **NONE ALLOWED.** If crack is found, replace tip cap (PARA 5-4-31).

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
Main Rotor Blade Cuff (Figure 5-3-11-2, Sheet 2, Detail A)	Cracks.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
	Nicks/Scratches/ Corrosion.			
	- - No deeper than 0.020-inch on lug surface and no deeper than 0.040-inch outside lug surface.	Using abrasive cloth, Item 12, Appendix D (or equivalent), blend out damage to ½-inch radius. Touch up paint (PARA 5-4-31).	x	
Main Rotor Blade Cuff Attach Bolts and Nuts (Figure 5-3-11-2, Sheet 2, Detail A)	Surface Corrosion/Pits.	Remove all sealing compound from area around corrosion.	x	
	- - No greater than 0.005-inch in base metal.	Using abrasive cloth, Item 5, Appendix D, blend out corrosion.	x	
	Bushing Movement.	Using soft metal punch, return to correct location by driving into position.	x	
Root Laminates (Figure 5-3-11-2, Sheet 2, Detail A)	Cracks (Disbond/ Delamination).	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
	Crack (Cosmetic).	Not acceptable. Repair, using cosmetic adhesive crack repair (PARA 5-4-31).		

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	Dents/Gouges.			
	- - No deeper than 0.050-inch × 4-inches long with no radiating cracks or bond separations.	Using adhesive, fill dent/gouge (PARA 5-4-31).		x
	- - More than 0.050-inch deep or 4-inches long. Also, if dent includes cracks or bond separation.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
	Bond Separation/ Delaminations.			
	- - Up to 3-inches long × 1-inch wide.	Repair, using tip cap-to-main rotor blade sealing repair (PARA 5-4-31).	x	
	- - More than 3-inches long or 1-inch wide.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
Root Fairing (Figure 5-3-11-2, Sheet 2, Detail A)	Cracks (If Deicing System Not Affected).	Patch, using bond repair (PARA 5-4-31).	x	
	Dents/Gouges (If Deicing System Not Affected).	Using adhesive, fill dent/gouge (PARA 5-4-31).	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	Damage (Affecting Deicing System).	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
	- - Bond separations up to 6-inches long.	Repair, using bond repair (PARA 5-4-31).	x	
	- - Loose or missing sealing compound at root fairing-to-main rotor blade spar joint.	Apply sealing compound, Item 296, Appendix D, to reseal joint.	x	
	Holes/Cracks.	If installed, check blade deice system for proper operation (PARA 12-2-5). Repair damage using corrosion removal repair (PARA 5-4-31).	x	
BIM® Pressure Indicator (Figure 5-3-11-2, Sheet 2, Detail A)	Indication of main rotor blade spar pressure loss or any type of damage to BIM® pressure indicator.	Refer to PARA 5-4-37.		
Pocket Skin/Main Rotor Blade Spar Region (Figure 5-3-11-2, Sheet 2, Detail B)	Scratches.			
	- - Less than 0.015-inch deep and any length.	No repair required. Touch up paint (PARA 5-4-31).	x	
	- - 0.015 - 0.030-inch deep and any length spanwise.	Using adhesive, fill scratch (PARA 5-4-31).	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Spanwise only, 0.015 - 0.050-inch deep (not through to main rotor blade spar) and not over 6-inches long.	Repair, using skin patch repair (PARA 5-4-31).		x
	- - Deeper than 0.050-inch, or more than 6-inches long.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
	Gouges/Digs.			
	- - 0.015 - 0.030-inch deep and 1 - 2-inch diameter.	Using adhesive, fill gouge/dig (PARA 5-4-31).		x
	- - 0.030 - 0.050-inch deep (not through to main rotor blade spar) and not over 6-inches long.	Repair, using skin patch repair (PARA 5-4-31).		x
	- - Deeper than 0.050-inch or longer than 6-inch diameter.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
	Tear/Rip (Skin Torn and Separated).			
	- - Not deeper than 0.040-inch and up to 2-inch diameter.	Repair, using skin patch repair (PARA 5-4-31).		x

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - More than 0.040-inch deep or longer than 2-inch diameter.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
	Bond Separations/ Delaminations.			
	- - Less than 2-inch diameter.	No repair required. Touch up paint (PARA 5-4-31). Daily inspection of area required.	x	
	- - More than 2-inch diameter, but not more than 6-inch diameter.	Repair, using skin patch repair (PARA 5-4-31).		x
	- - More than 6-inch diameter.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
Pocket Skin/ Honeycomb Core Region (Figure 5-3-11-2, Sheet 2, Detail B)	Dents (Smooth and Unbroken).			
	- - No deeper than 0.040-inch or longer than 3-inch diameter.	No repair required. Touch up paint (PARA 5-4-31).	x	
	- - No deeper than 0.040-inch and 3 - 6-inch diameter.	Trim out damage and patch, using skin patch repair (PARA 5-4-31).		x
	- - Deeper than 0.040-inch up to 6-inch diameter.	Repair, using plug patch repair (PARA 5-4-31).		x

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Deeper than 0.040-inch and greater than 6-inch diameter.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
	Gouges/Holes.			
	- - Up to 6-inch diameter, using damage to honeycomb core.	Repair, using plug patch repair (PARA 5-4-31).		x
	- - Damage through main rotor blade, but honeycomb core damage less than 1-inch diameter.	Repair both sides, using skin patch repair (PARA 5-4-31).		x
	- - Damage through main rotor blade and honeycomb core damage less than 6-inch diameter.	Repair both sides, using plug patch repair (PARA 5-4-31).		x
	- - Damage through main rotor blade and honeycomb core damage greater than 6-inch diameter.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
	Tear/Rip (Main Rotor Blade Skin Damage Only).			
	- - Up to 6-inch diameter.	Repair, using skin patch repair (PARA 5-4-31).		x

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Greater than 6-inch diameter.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
	Blisters/Bond Separations.			
	- - Less than 2-inch diameter and maximum blistering 0.020-inch high.	No repair required. Touch up paint (PARA 5-4-31).	x	

NOTE

Main rotor blades that have bond separation in Area A (Figure 5-3-11-2, Sheet 2, Detail B) greater than 2-inches, but with less than a 6-inch diameter, and show no further damage such as cracks or blisters, can remain in operational service. In this area, there is no minimum distance between damage found.

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - More than 2-inch diameter, but not more than 6-inch diameter.	Trim out damage and patch, using skin patch repair (PARA 5-4-31).		x
	- - More than 6-inch diameter.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
Pocket Skin/Trailing Edge Region (Figure 5-3-11-2, Sheet 2, Detail B)	Dents (Smooth and Unbroken).			
	- - No deeper than 0.040-inch.	No repair required. Touch up paint (PARA 5-4-31).	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - More than 0.040-inch deep and up to 1-inch diameter.	Using adhesive, fill dent (PARA 5-4-31).		x
	- - Deeper than 0.040-inch and up to 3-inches spanwise × 2-inches chordwise.	Repair, using trailing edge patch repair (PARA 5-4-31).		x
	- - Damage deeper than 0.040-inch and more than 3-inches spanwise × 2-inches chordwise.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
	Gouges/Scratches.			
	- - Up to 0.030-inch deep and 1-inch spanwise.	No repair required. Touch up paint (PARA 5-4-31).	x	
	- - Deeper than 0.030-inch and up to 3-inches spanwise.	Repair, using trailing edge patch repair (PARA 5-4-31).		x
	End Gashes/Slits.			
	- - No more than ½-inch chordwise and 1 ½-inches spanwise in trailing edge.	Repair, using trailing edge patch repair (PARA 5-4-31).		x
	- - Through damage more than ½-inch chordwise and 1 ½-inches spanwise.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
	Tear/Rip.			

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Up to 3-inches spanwise × 2-inches chordwise.	Repair, using trailing edge patch repair (PARA 5-4-31).		x
	Bond Separations/ Delaminations (in Trim Tab Edge).			
	- - Area up to ¼-inch × 3-inches spanwise.	No repair required. Touch up paint (PARA 5-4-31).	x	
	- - Area up to ½-inch × 6-inches spanwise.	Repair, using adhesive injection repair (PARA 5-4-31).	x	
Trim Tab	Scratch/Gouge.			
	- - Up to full thickness of tab, full chordwise width of tab, and less than 3-inches spanwise.	Using abrasive cloth, Item 6, Appendix D, blend out scratch/ gouge. Touch up chemical Alodine and touch up paint (PARA 5-4-31).	x	
	End Gashes.			
	- - ¼-inch or less through end of tab.	No repair required. Touch up paint (PARA 5-4-31).	x	
	Bond Separations/ Delaminations.			
	- - Up to ¼-inch deep chordwise × less than 3-inches spanwise (in trailing edge).	No repair required. Monitor disbond.	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Up to ½-inch deep chordwise × less than 3-inches spanwise between trim tab and blade skins.	Repair, using adhesive injection repair (PARA 5-4-31).		x
	- - Between ¼-inch and 1-inch deep chordwise, and less than 4-inches spanwise (in trailing edge).	Repair, using adhesive injection repair (PARA 5-4-31).		x

NOTE

Not all main rotor blades have slots in trim tab.

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	Cracks (Starting at Front End of Slot, Running Chordwise toward Leading Edge of Trim Tab).			
	- - Up to full width of tab.	No repair required.	x	
	Cracks (Not at Slot, Up to Full Chordwise Width).	No repair required.		
	- - Cracks in spanwise direction up to 2-inches.	Using drill, stop-drill if rear of skin. No repair required if over skin.	x	
	Bends.			
	- - Not greater than 6-inches spanwise.	Using straight edge for alignment, straighten with adjacent areas of trim tab.	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
Abrasion Strips (Nickel and Titanium) (Figure 5-3-11-2, Sheet 3, Detail C)	Cracks.			
	- - In abrasion strips (nickel) (outboard).	If bond separation is within limits, no repair required (refer to Bond Separation in this section). Seal crack using abrasion strip repair (PARA 5-4-31).	x	
	- - In abrasion strip (titanium) (up to full width of strip).	No repair required.If installed, check blade deice system for proper operation (PARA 12-2-5).	x	
	Dents.			
	- - In abrasion strip (titanium) up to ⅛-inch and not over main rotor blade spar.	No repair required.If installed, check blade deice system for proper operation (PARA 12-2-5).	x	
	- - Bond separation up to ½-inch beyond dent.	No repair required. Touch up paint (PARA 5-4-31).	x	
	- - Cracks in dent.	If crack does not extend into reinforced plastic, no repair required. Seal crack using abrasion strip repair (PARA 5-4-31).	x	
	Bond Separation.			

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Edge voids in abrasion strips (nickel) up to 1-inch deep × 2-inches long.	Seals voids using abrasion strip repair (PARA 5-4-31).	x	
	- - Edge voids in abrasion strips (nickel) up to ½-inch deep × 4-inches long.	Seals voids using abrasion strip repair (PARA 5-4-31).	x	
	- - Internal voids in abrasion strips (nickel) up to 3-inches long or wide.	No repair required.	x	
	- - Multiple voids in abrasion strips (nickel) totaling up to 3-inches long or wide.	No repair required.	x	
	- - Edge or internal voids in abrasion strips (nickel) within 2-inches of inboard or outboard end of strip.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
	- - Edge voids in abrasion strip (titanium) up to 1-inch deep × 6-inches long.	Repair, using abrasion strip repair (PARA 5-4-31).	x	
	- - Internal voids in abrasion strip (titanium) up to 3-inches long or wide.	No repair required.	x	
	- - Internal voids in abrasion strip (titanium) greater than 3-inches long or wide.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	Gouges.			
	- - Up to full depth of abrasion strip (nickel).	Using medium cut file, blend out damage. Using adhesive, fill gouge (PARA 5-4-31).	x	
	- - Up to full thickness of abrasion strip (titanium) (0.020-inch).	No repair required. If installed, check blade deice system for proper operation (PARA 12-2-5).	x	
	Wear.			
	- - Up to full depth of strip abrasion strip (nickel).	No repair until worn through to adhesive, then send damaged main rotor blade to manufacturer.		x
	- - Up to full depth of abrasion strip (titanium).	No repair until worn through to adhesive, then send damaged main rotor blade to manufacturer.		x

NOTE

For abrasion strip (nickel) inspections, if helicopter is not operated in high-erosion (desert sand) environment, ¼-inch dimension may be increased to ¾-inch.

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
Abrasion Strips (Nickel) (Figure 5-3-11-2, Sheet 3, Detail C)	- - Rear ¼-inch edge (both upper and lower).	No repair until worn or corroded through to adhesive, then send damaged main rotor blade to manufacturer.		x
	Loose/Curled Edges.			

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Rear ¼-inch edge (both upper and lower).	Using nonmetallic scraper, remove loose/ curled edges.	x	

NOTE

Aluminum wire mesh lightning protection is nonstructural. Corrosion is a cosmetic problem. Repair corrosion only if main rotor blade has already been removed from helicopter.

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
Aluminum Wire Mesh Lightning Protection (Figure 5-3-11-2, Sheet 3, Detail D)	Corrosion.			
	- - Less than 10 pinholes per square-foot.	No repair required.		
	- - More than 10 pinholes per square-foot.	Repair, using aluminum wire mesh lightning protection repair (PARA 5-4-31).		x
Main Rotor Blade Receiver Assembly (Tiedown Fitting)	Loose.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer.		x
Tip Block (Figure 5-3-11-2, Sheet 4, Detail F)	Stripped/Missing Inserts.	Repair, using tip block insert, RZA1287, repair (PARA 5-4-31).	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	Cracked Caulking.	Using coin-tapping inspection method (PARA 5-3-11.2), inspect caulking extending from upper to lower airfoil, between tip rib and tip block to ensure bond integrity. If bond is good, no repair required.	x	
Tip Block Doubler (Figure 5-3-11-2, Sheet 4, Detail F)	Cracks.			
	- - Visible in adhesive fillet.	Using coin-tapping inspection method (PARA 5-3-11.2), inspect tip block doubler to ensure bond integrity. If bond is good, no repair required.	x	
	Bond Separation.			
	- - Up to 2 square-inches.	Repair, using adhesive injection repair (PARA 5-4-31).		x
	- - Greater than 2 square-inches.	Not acceptable. Replace main rotor blade (PARA 5-4-31), and send damaged main rotor blade to manufacturer for repair.		x

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
Tip Cap Fairing (Kevlar-Reinforced Plastic) (Figure 5-3-11-2, Sheet 5, Detail G and Sheet 6, Detail H)	Cracks.			
	- - Cracks less than 2-inches long.	Repair, using tip cap skin and fairing crack repair (PARA 5-4-31).		x
	- - Crack in attachment holes.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	- - Cosmetic cracks in adhesive at joint between Kevlar fairing and aluminum skin. No limit if Kevlar fairing is securely attached.	No repair required.	x	
	- - Bond separation at edges or trailing edge.	Repair, using adhesive injection repair (PARA 5-4-31).		x
	Dents/Gouges.			
	- - Up to 0.020-inch deep.	Using adhesive, fill dent/gouge (PARA 5-4-31).		x
	- - Damage preventing secure attachment of tip cap fairing.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	Elongated/Worn Outboard Drain Hole.			

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Outboard drain hole worn more than ¼-inch vertically and ¼-inch chordwise.	Apply fiberglass patch up to 2-inches in diameter and 1 to 3 plies thick, using skin patch repair (PARA 5-4-31). After patch is cured, using drill with 0.190-inch diameter drill bit, reopen drain hole.		x
	Worn/Frayed Kevlar Surface.	Lightly sand frayed Kevlar. Using adhesive, fill Kevlar surface (PARA 5-4-31).	x	
	Missing/Broken Screws (Upper and Lower).			
	- - No missing or broken screws in positions 4, 5, and 6, both upper and lower locations.	Replace screws (PARA 5-4-31).	x	
	- - No more than one screw total in positions 1, 2, and 3, either upper of lower locations (one screw total).	Check that torque on remaining screws is no less than 25 inch-pounds (check in tightening direction only). Replace screws at next available opportunity (PARA 5-4-31).	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - No more than one screw per set in positions 7 through 12, both upper and lower locations (two screws total).	Check that torque on remaining screws is no less than 25 inch-pounds (check in tightening direction only). Replace screws at next available opportunity (PARA 5-4-31).	x	
	- - No more than two screws total per tip cap.	Check that torque on remaining screws is no less than 25 inch-pounds (check in tightening direction only). Replace screws at next available opportunity (PARA 5-4-31).	x	
Tip Cap, 70150-09107-041, 70150-09107-049, and 70150-09107-055 (Figure 5-3-11-2, Sheet 5, Detail G)	Dents (Smooth and Unbroken).			
	- - Less than ⅛-inch deep.	No repair required. Touch up paint (PARA 5-4-31).	x	
	- - More than ⅛-inch deep.	Remove tip cap and inspect for internal damage. If no damage is found, and dent can be straightened, reuse tip cap.	x	
	Nicks/Scratches/Dents/Scuffs.			

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Less than 0.020-inch deep and 1-inch long in inspection zone.	Repair, using adhesive filler repair (PARA 5-4-31).	x	
	- - More than 0.020-inch deep and 1-inch long in inspection zone.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	Edge Damage.			
	- - Less than 0.040-inch deep in inspection zone.	Repair, using adhesive filler repair (PARA 5-4-31).	x	
	- - More than 0.040-inch deep in inspection zone.	Not acceptable. Replace tip cap (PARA 5-4-31).	x	
	Holes.			
	- - Less than ½-inch diameter with no internal damage.	Patch, using tip cap holes repair (PARA 5-4-31).		x
	- - More than 1 ½-inch diameter or with internal damage.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	Cracks.			
	- - Skin crack less than 1-inch long in areas other than attachment holes, 4-inches outboard from center of attachment holes No. 6 through No. 12.	Using drill with No. 40 drill bit, stop-drill and patch, using tip cap skin and fairing crack repair (PARA 5-4-31).		x

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Skin crack more than 1-inch long in areas other than attachment holes.	Not acceptable. Replace tip cap (PARA 5-4-31). Visually inspect all tip cap attachment areas of main rotor blade for cracking or other damage.		x
	- - Skin crack progressing from previously stop-drilled hole.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	- - Attachment hole crack.	Not acceptable. Replace tip cap (PARA 5-4-31). If tip cap is replaced due to cracks, visually inspect all tip cap attachment areas of main rotor blade for cracking, or other damage.	x	
	- - Tip cap rib crack up to 1-inch long.	Using drill with No. 40 drill bit, stop-drill crack.	x	
	- - Tip cap rib crack greater than 1-inch long.	Repair, using tip cap rib cracks repair (PARA 5-4-31).	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Abrasion strip (nickel) cracks.	No repair required if disbond area is less than 1 square-inch. Seal crack, as required, using abrasion strip repair (PARA 5-4-31). Repair disbond area 1 square-inch to 3 square-inches, using bond repair (PARA 5-4-31).	x	
	- - Abrasive strip cracks in overlap region at inboard end.	Remove loose piece. Blend sharp edges. Apply sealing compound, Item 296, Appendix D, to seal crack.	x	
	- - Balance weight attachment blocks and hardware cracks.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	- - Edge cracks in balance shim weights.	No repair required if crack less than ⅛-inch long. If longer cracks are found, replace tip cap (PARA 5-4-31).		x
	Corrosion.			
	- - Corrosion pitting in thick portion of skin no more than 0.020-inch deep.	Remove pitting, using corrosion removal repair (PARA 5-4-31). Touch up paint (PARA 5-4-31).	x	
	- - Corrosion pitting in thin portion of skin no more than 0.010-inch deep.	Remove pitting, using corrosion removal repair (PARA 5-4-31). Touch up paint (PARA 5-4-31).	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Corrosion pitting deeper than 0.020-inch in thick portion of skin, or 0.010-inch in thin portion of skin.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	- - Localized erosion/ corrosion adjacent to No. 4 and No. 5 screws on lower side up to 0.030-inch deep.	Remove corrosion, using corrosion removal repair (PARA 5-4-31). Touch up paint (PARA 5-4-31). Apply sealing compound, Item 296, Appendix D, using tip cap-to-main rotor blade sealing repair (PARA 5-4-31).	x	
	- - Rib corrosion no deeper than 0.005-inch.	Remove corrosion, using corrosion removal repair (PARA 5-4-31). Touch up paint (PARA 5-4-31).	x	
	- - Corrosion of screws, studs, and mounting hardware.	Replace corroded screws, studs, and mounting hardware (PARA 5-4-31).	x	
	- - Corrosion of aluminum tip weight blocks no deeper than 0.005-inch.	Remove corrosion, using corrosion removal repair (PARA 5-4-31). Touch up paint (PARA 5-4-31).	x	
	Bond Separations.			
	- - Trailing edge separation up to ¼ × 1-inch.	No repair required. Touch up paint (PARA 5-4-31).	x	
	Bond Separation.			

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Trailing edge separation more than ¼ × 1-inch and up to 4-inches long.	Repair, using bond repair (PARA 5-4-31).	x	
	- - Rib separation less than 2-inches long.	No repair required.	x	
	- - Rib separation more than 2-inches long.	Repair, using bond repair (PARA 5-4-31).		x
	Erosion.			
	- - Through abrasion strip (nickel).	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	- - Local erosion (small isolated areas).	No repair required if erosion is less than ⅓ material thickness. Touch up paint (PARA 5-4-31).	x	
	Loose/Missing Sealing Compound (Tip Cap-to-Main Rotor Blade Joint).	Replace sealing compound, using tip cap-to-main rotor blade sealing repair (PARA 5-4-31).	x	
	Separations (in a 2-inch Wide Area behind Centerline of First Six Tip Cap Screws).			
	- - Separations less than 1/16-inch wide.	No repair required.	x	
	- - Separations parallel to tip cap/main rotor blade caulking line.	No repair required.	x	
	- - Separations not past tip cap leading edge.	No repair required.	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Separations beyond limits listed above.	Not acceptable. Replace main rotor blade (PARA 5-4-31).		x
	Elongated/Worn Lower Surface Drain Hole.			
	- - Lower surface drain hole worn more than ¼-inch in diameter.	Not acceptable. Replace tip cap (PARA 5-4-31).	x	
	Graphite Doubler.			
	- - Scratches and gouges less than 0.010-inch deep of any length not penetrating into graphite core.	Repair, using adhesive filler repair (PARA 5-4-31).		x
	- - Scratches and gouges deeper than 0.010-inch deep or penetrating into graphite core.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	- - Broken.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	- - Delaminated or disbonded less than ½-inch long.	Repair tip cap (PARA 5-4-31).		x
	- - Delaminated or disbonded greater than ½-inch long.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	- - Missing pieces less than ¼ square-inch.	Repair tip cap, using adhesive filler repair (PARA 5-4-31).		x

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Missing pieces greater than ¼ square-inch.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
Tip Cap Abrasion Strip (Nickel) (For Tip Cap, 70150-09107-041, 70150-09107-049, and 70150-09107-055) (Figure 5-3-11-2, Sheet 5, Detail G)	Erosion.			
	- - Through abrasion strip (nickel).	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	- - Local erosion (small isolated areas).	No repair required if erosion is less than ⅓ material thickness. Touch up paint (PARA 5-4-31).	x	
	Loose/Missing Sealing Compound (Tip Cap-to-Main Rotor Blade Joint).	Replace sealing compound, using tip cap-to-main rotor blade sealing repair (PARA 5-4-31).	x	
	Separations (in a 2-inch Wide Area behind Centerline of First Six Tip Cap Screws).			
	- - Separations less than 1⁄16-inch wide.	No repair required.	x	
	- - Separations parallel to tip cap/main rotor blade caulking line.	No repair required.	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Separations not past tip cap leading edge.	No repair required.	x	
	- - Separations beyond limits listed above.	Not acceptable. Replace main rotor blade (PARA 5-4-31).		x
	Elongated/Worn Lower Surface Drain Hole.			
	- - Lower surface drain hole worn more than ¼-inch in diameter.	Not acceptable. Replace tip cap (PARA 5-4-31).	x	
	Graphite Doubler.			
	- - Scratches and gouges less than 0.010-inch deep of any length not penetrating into graphite core.	Repair, using adhesive filler repair (PARA 5-4-31).		x
	- - Scratches and gouges deeper than 0.010-inch deep or penetrating into graphite core.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	- - Broken.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	- - Delaminated or disbonded less than ½-inch long.	Repair tip cap (PARA 5-4-31).		x
	- - Delaminated or disbonded greater than ½-inch long.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	- - Missing pieces less than ¼ square-inch.	Repair tip cap, using adhesive filler repair (PARA 5-4-31).		x

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Missing pieces greater than ¼ square-inch.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
Tip Cap, 70150-09207-041 (Figure 5-3-11-2, Sheet 6, Detail H)	Cracks.			
	- - Within inspection zone.	Not acceptable. Replace tip cap (PARA 5-4-31). If tip cap is replaced due to cracks, inspect tip cap attachment area of main rotor blade for cracks and damage.		x
	- - Within lip area.	No limit. Repair, using adhesive injection repair (PARA 5-4-31), or trim away loose material.	x	
	- - All other areas.	No through cracks allowed. Replace tip cap with through cracks (PARA 5-4-31).		x
	Delaminations/ Disbonds.			
	- - Within inspection zone.	Not acceptable. Replace tip cap (PARA 5-4-31). If tip cap is replaced due to delaminations or disbonds, inspect tip cap attachment area of main rotor blade for cracks and damage.		x

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Within lip area.	No limit. Repair, using adhesive injection repair (PARA 5-4-31), or trim away loose material.	x	
	- - Internal rib-to-skin up to 1-inch long.	Repair, using adhesive injection repair (PARA 5-4-31).		x
	- - Trailing edge up to 1-inch long.	Repair, using adhesive injection repair (PARA 5-4-31).	x	
	- - Trailing edge greater than 1-inch long up to 4-inches long.	Repair, using 4 ply wet lay up skin patch repair (PARA 5-4-31).		x
	- - Trailing edge greater than 4-inches long.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	- - All other areas up to ⅛-inch deep.	No repair required.	x	
	- - All other areas greater than ⅛-inch deep.	Repair, using adhesive injection repair (PARA 5-4-31).		x
	Surface Damage (Raised/Frayed Fibers, Tears, or Surface Delaminations).			
	- - Up to 0.015-inch deep.	Repair, using adhesive filler repair (PARA 5-4-31).	x	
	- - Greater than 0.015-inch deep, up to 0.030-inch deep.	Repair, using 2 ply wet lay up skin patch repair (PARA 5-4-31).		x

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Greater than 0.030-inch deep.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	Scratches/Abrasions (No Limit on Length or Number).			
	- - Up to 0.005-inch deep.	No repair required.	x	
	- - Greater than 0.005-inch deep, up to 0.010-inch deep.	Touch up paint (PARA 5-4-31).	x	
	- - Greater than 0.010-inch deep, up to 0.015-inch deep.	Repair, using adhesive filler repair (PARA 5-4-31).	x	
	Erosion.			
	- - Erosion which exposes stainless steel or copper wire mesh.	Touch up paint (PARA 5-4-31).	x	
	- - Erosion into stainless steel wire mesh, but not completely through.	Repair, using adhesive filler (PARA 5-4-31). Touch up paint (PARA 5-4-31).	x	

CAUTION

Sand only to remove loose or raised stainless steel wire mesh; do not remove excessive material from adjacent areas.

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Erosion completely through stainless steel wire mesh in local areas, including area adjacent to attachment holes (less than ¼ square-inch each).	Using abrasive cloth, Item 5, Appendix D, remove any loose or raised stainless steel wire mesh by lightly hand sanding. Then repair all exposed edges, using adhesive filler repair (PARA 5-4-31). Touch up paint (PARA 5-4-31). If flight will be continued in highly erosive environment, install main rotor blade tip cap erosion protection kit polyurethane tape, 70073-15000 (PARA 16-6-9).	x	
	- - Erosion completely through stainless steel wire mesh in large areas (greater than ¼ square-inch each).	Not acceptable. Replace tip cap (PARA 5-4-31).		x

CAUTION

Sand only to remove loose or raised copper wire mesh; do not remove excessive material from adjacent areas.

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Erosion completely through copper wire mesh, no limit on area.	Using abrasive cloth, Item 5, Appendix D, remove any loose or raised copper wire mesh by lightly hand sanding. Then repair all exposed edges, using adhesive filler repair (PARA 5-4-31). Touch up paint (PARA 5-4-31).	x	
	- - Erosion into composite material with no cracking, disbonding, or delamination.	Touch up paint (PARA 5-4-31).	x	
	- - Erosion with associated cracking, disbonding, or delamination.	Refer to specific limits for damage to composite material within this table.		
	Cracks/Torn Areas/ Missing Pieces (Trailing Edge Repair Area).			
	- - Up to ½-inch wide × 2-inches long.	Repair, using 4 ply wet lay up skin patch repair (PARA 5-4-31).		x
	- - Greater than ½-inch wide and 2-inches long.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	Holes.			
	- - Punctures through skin in puncture repair area only up to 1-inch diameter.	Repair, using 3-inch diameter patch or 4 ply wet lay up skin patch repair (PARA 5-4-31).		x

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Punctures through skin in puncture repair area only greater than 1-inch diameter.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	Attachment Hole Elongation at Attachment Holes 4, 5, and 6 (Upper and Lower).			
	- - Up to 0.030-inch.	No repair required.	x	
	- - Up to 0.040-inch.	No repair required, provided no more than 3 holes are elongated between 0.030 - 0.040-inch.	x	
	- - Greater than 0.040-inch.	Not acceptable. Replace tip cap (PARA 5-4-31). If tip cap is replaced due to attachment hole elongation, inspect tip cap attachment area of main rotor blade for cracks and damage.		x
	Delaminations/Disbonds/Other Damage around Attachment Holes 4, 5, and 6 (Upper and Lower).			
	- - Up to 0.015-inch deep, up to ⅝-inch diameter around hole.	Repair, using adhesive injection repair (PARA 5-4-31).	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Greater than 0.015-inch deep, or greater than ⅝-inch diameter around hole.	Not acceptable. Replace tip cap (PARA 5-4-31). If tip cap is replaced due to attachment hole delaminations, disbonds, and other damage, inspect tip cap attachment area of main rotor blade for cracks and damage.		x
	Attachment Hole Elongation at Attachment Holes 1 through 3, and 7 through 12 (Upper and Lower).			
	- - Up to 0.040-inch.	No repair required.	x	
	- - Greater than 0.040-inch.	Not acceptable. Replace tip cap (PARA 5-4-31). If tip cap is replaced due to attachment hole elongation, inspect tip cap attachment area of main rotor blade for cracks and damage.		x
	Delaminations/Disbonds/Other Damage around Attachment Holes 1 through 3, and 7 through 12 (Upper and Lower).			
	- - Up to 0.030-inch deep, up to 0.60-inch diameter around hole.	Repair, using adhesive injection repair (PARA 5-4-31).	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Greater than 0.030-inch deep, up to 0.60-inch diameter around hole.	Not acceptable. Replace tip cap (PARA 5-4-31). If tip cap is replaced due to attachment hole delaminations, disbonds, and other damage, inspect tip cap attachment area of main rotor blade for cracks and damage.		x
	Missing/Broken Screws (Upper and Lower).			
	- - No missing or broken screws in positions 4, 5, and 6, both upper and lower locations.	Replace screws (PARA 5-4-31).	x	
	- - No more than one screw total in positions 1, 2, and 3, either upper and lower locations (one screw total).	Check that torque on remaining screws is no less than 25 inch-pounds (check in tightening direction only). Replace screws at next available opportunity (PARA 5-4-31).	x	
	- - No more than one screw per set in positions 7 through 12, both upper and lower locations (two screws total).	Check that torque on remaining screws is no less than 25 inch-pounds (check in tightening direction only). Replace screws at next available opportunity (PARA 5-4-31).	x	

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - No more than two screws total per tip cap.	Check that torque on remaining screws is no less than 25 inch-pounds (check in tightening direction only). Replace screws at next available opportunity (PARA 5-4-31).	x	
Tip Cap Abrasion Strip (Nickel) (For Tip Cap, 70150-09207-041) (Figure 5-3-11-2, Sheet 6, Detail H)	Erosion.			
	- - Through abrasion strip (nickel).	Not acceptable. Replace tip cap (PARA 5-4-31).		x
	Cracks.			
	- - Overlap area at inboard end.	Remove loose pieces. Blend sharp edges. Apply sealing compound, Item 296, Appendix D, to seal crack.	x	
	- - All other areas.	Acceptable provided no pieces are missing.	x	
	Disbonds.			
	- - Edge disbonds up to ½-inch chordwise × 3-inches spanwise.	Repair, using adhesive injection repair (PARA 5-4-31).	x	
	- - Edge disbonds greater than ½-inch chordwise × 3-inches spanwise.	Not acceptable. Replace tip cap (PARA 5-4-31).		x

Table 5-3-11-1. Main Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Aircraft	Off Aircraft
	- - Internal disbonds more than ½-inch from any edge up to 6 square-inches.	No repair required.	x	
	- - Internal disbonds more than ½-inch from any edge greater than 6 square-inches.	Not acceptable. Replace tip cap (PARA 5-4-31).		x
Polyurethane Patch (Figure 5-3-11-2, Sheet 8, Detail K)	Any damage which would allow possible water entry.	Not acceptable. Replace polyurethane patch (PARA 5-4-31).		x

5-3-12 TAIL ROTOR INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Elastomeric Rod End Bearing Plastic Feeler Gage, Locally-Made, Figure H-232, Appendix H
- Teflon Lined Spherical Bearing Plastic Feeler Gage, Locally-Made, Figure H-209, Appendix H

References

- Appendix H
- PARA 5-3-13
- PARA 5-4-38
- PARA 5-4-39
- PARA 5-4-42
- PARA 5-5-15

5-3-12.1 Tail Rotor Blade Inspection.

a. Grasp tail rotor blade tip at midwidth. Apply light hand force in flatwise direction toward or away from tail pylon for 3 - 4-inches. Observe opposite tail rotor blade for equal deflection at tail rotor blade tip. Repeat for adjacent tail rotor blade.

b. If movement of an equal deflection is observed in opposite tail rotor blade, perform the following:

(1) Remove tail rotor blades (PARA 5-4-38).

(2) Inspect nylon surfaces of tail rotor blade spars and nylon shims on outboard retention plate for wear and heat damage (PARA 5-3-13). **NONE ALLOWED.** If wear and heat damage is found, replace tail rotor blade (PARA 5-4-38) and nylon shims on outboard retention plate (PARA 5-4-42).

(3) Inspect tail rotor blade spars for evidence of interlaminar cracking. Pay particular attention to leading and trailing edges of graphite spar in area from STA 8.0 through STA 0.0 to STA 8.0 on opposite end. **NONE ALLOWED.** If cracks are found, replace tail rotor blade.

(4) **CHECK SCREWS SECURING NYLON SHIMS TO OUTBOARD RETENTION PLATE TO MAKE SURE THEY ARE HANDTIGHT.**

(5) Install tail rotor blades (PARA 5-4-38).

5-3-12.2 Tail Rotor Pitch Control Rod Inspection.

5-3-12.2.1 Rod End Spherical Bearing Feeler Gage Wear Inspection.

a. Locally make Teflon lined spherical bearing plastic feeler gage (Figure H-209, Appendix H).

b. Inspect rod end spherical bearings on both rod ends as follows:

(1) Apply hand pressure to tail rotor blades, and twist each blade in direction of reduced pitch.

CAUTION

- Damage to rod end spherical bearing will result if metallic feeler gage is used. Teflon race will be damaged. Use only Teflon lined spherical bearing plastic feeler gage when inspecting rod end spherical bearing.
- Damage to rod end spherical bearing will result if Teflon lined spherical bearing plastic feeler gage is not inserted carefully. Teflon race will be damaged. Use care when inserting Teflon lined spherical bearing plastic feeler gage.

(2) Using 0.010-inch end of Teflon lined spherical bearing plastic feeler gage, probe both sides of rod end spherical bearing around entire circumference for clearance between spherical ball and outer race (Figure 5-3-12-1, Detail A).

(3) If Teflon lined spherical bearing plastic feeler gage (0.010-inch end) cannot be inserted to 0.125-inch mark, rod end spherical bearing is acceptable. Repeat inspection every 30 flight hours.

(4) If Teflon lined spherical bearing plastic feeler gage (0.010-inch end) can be inserted to 0.125-inch mark, or more, probe entire circumference again using 0.015-inch end of Teflon lined spherical bearing plastic feeler gage.

(5) If Teflon lined spherical bearing plastic feeler gage (0.015-inch end) cannot be inserted to 0.125-inch mark between spherical ball and outer race, rod end spherical bearing is acceptable. Repeat inspection every 15 flight hours.

(6) **IF TEFLON LINED SPHERICAL BEARING PLASTIC FEELER GAGE (0.015-INCH END) CAN BE INSERTED TO 0.125-INCH MARK BETWEEN SPHERICAL BALL AND OUTER RACE, REPLACE TAIL ROTOR PITCH CONTROL ROD END** (PARA 5-4-39).

NOTE

An accumulation of reddish-brown powder on spherical bearing is normal for a Teflon bearing and is not cause for removal.

c. Inspect rod end spherical bearings on both rod ends for evidence of metal-to-metal contact between spherical ball and outer race. Metal-to-metal contact is present if worn or discolored spots are seen on spherical ball, or if you hear a clicking sound when spherical ball is rotated in its outer race. **NONE ALLOWED.** If metal-to-metal contact is found, replace tail rotor pitch control rod end spherical bearings (PARA 5-5-15).

d. Inspect rod end spherical bearings for play. **IF RADIAL PLAY (MEASURE IN DIRECTION OF LOAD), IS GREATER THAN 0.012-INCH, OR IF AXIAL PLAY (MEASURE IN DIRECTION OF BOLT) IS GREATER THAN 0.023-INCH, REPLACE TAIL ROTOR PITCH CONTROL ROD END SPHERICAL BEARINGS** (PARA 5-5-15).

5-3-12.2.2 Rod End Elastomeric Bearing Surfaces Inspection.

a. Inspect rod end elastomeric bearing surfaces as follows:

NOTE

- Elastomeric rod ends consist of laminated elastomer and steel shells. Wear of elastomer in bearings is a gradual process that will take several hundreds of hours from first detection of elastomer damage to point where rod end must be replaced.
- A wax-like substance on rubber surface is normal and should not be removed.
- Types of wear shown include eraser-type wear, elastomer extrusion, and shim cracks. Description of wear and general guidelines are as follows:

  Eraser-type wear: a pile-up of shredded rubber similar to the result of using a soft eraser.

  Elastomer extrusion: the squeezing out of the elastomer from between the elastomeric bearing laminations, and inner or outer members, indicating disbonding of elastomer. Peeling of elastomer mold lines which run radially across laminations is acceptable.

(1) Inspect for a concentration or pileup of eraser-type wear. A concentration or pileup of eraser-type wear may indicate an elastomeric crack or separation may exist. If a concentration or pileup of eraser-type wear is found, inspect rod end for elastomeric crack or separation (PARA 5-3-12.2.3).

(2) If an elastomer extrusion is detected, inspect rod end for elastomeric crack or separation (PARA 5-3-12.2.3).

(3) Visually inspect for cracking or disbonding of elastomer from shim inner or outer members. Cracking or disbonding of elastomer from shim inner or outer members will cause separation. If cracking or disbonding of elastomer from shim inner or outer members is suspected, inspect elastomeric rod end (PARA 5-3-12.2.3).

(4) Visually inspect all rod end elastomeric bearings for signs of inner member separation, using hand force, shaking tail rotor pitch control rod. **NO INNER MEMBER LOOSENESS OR EXPOSED METAL OF INNER BALL SURFACE OR MISSING ELASTOMER ALLOWED.** If inner member separation is beyond limits, replace elastomeric rod end (PARA 5-4-39).

NOTE

Shims or steel shells alternate with elastomer layers of rod end elastomeric bearing and are composed of two 180° pieces (Figure 5-3-12-1, Detail B). Within each shim layer, two pieces are separated by small gap measuring approximately 0.050-inch. These gaps are filled with elastomer during manufacture, and are evident on surfaces as a small lump of elastomer. Gaps becoming exposed over time are acceptable.

(5) Inspect for gaps or cracks in areas other than manufactured shim gaps. **NONE ALLOWED.** If cracks are found, or gaps are found, replace elastomeric rod end (PARA 5-4-39).

5-3-12.2.3 Rod End Elastomeric Bearing Feeler Gage Wear Inspection.

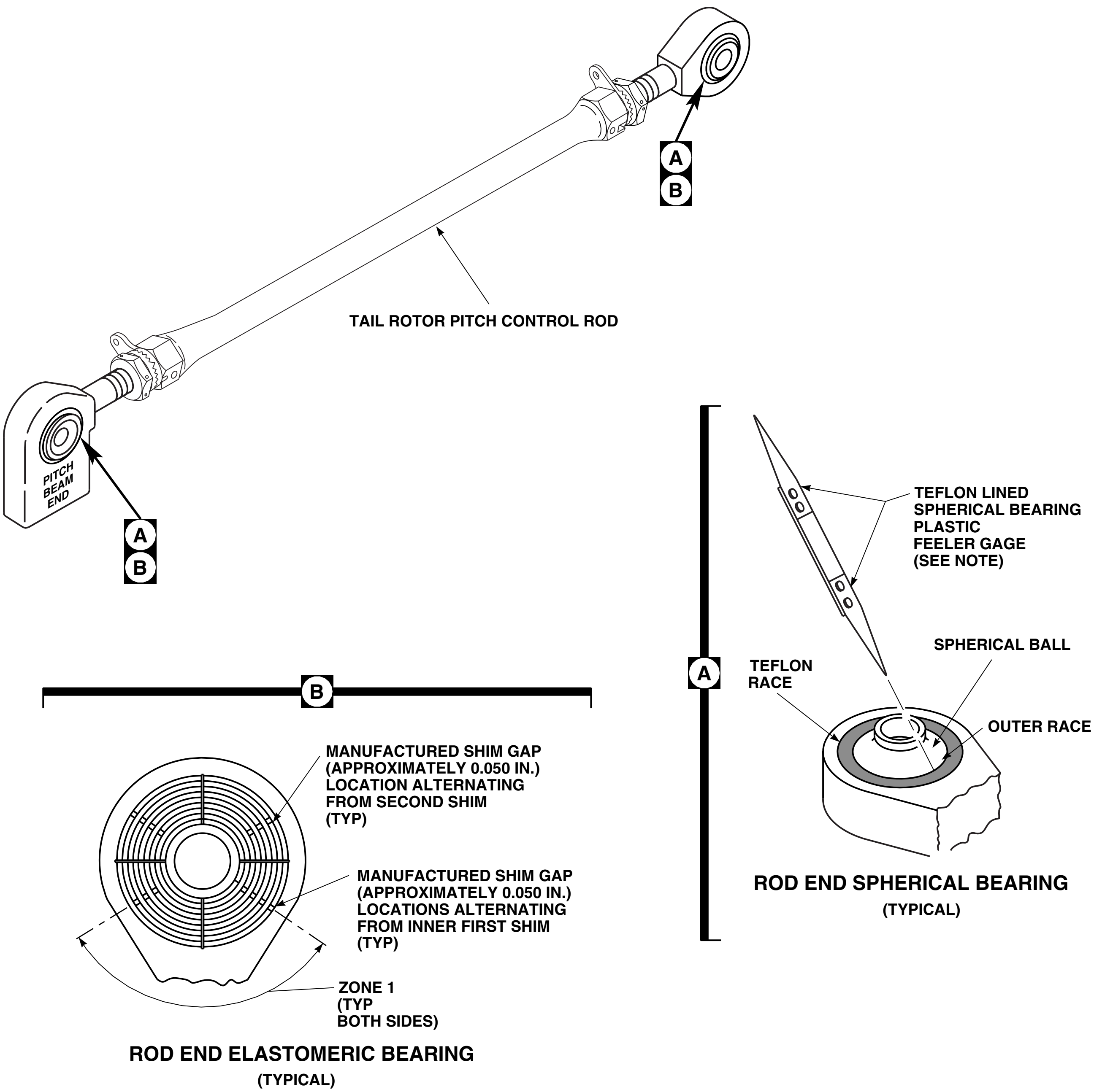
NOTE

Feeler gage inspection should only be performed if possible separation exists based on visual inspection of bearing surfaces.

a. Inspect rod end elastomeric bearing surfaces (PARA 5-3-12.2.2).

b. Locally make elastomeric rod end bearing plastic feeler gage (Figure H-232, Appendix H).

c. Inspect rod end elastomeric bearings as follows (Figure 5-3-12-1, Detail B):



NOTE

PROBE BOTH SIDES OF ROD END SPHERICAL BEARINGS AROUND ENTIRE CIRCUMFERENCE FOR CLEARANCE BETWEEN SPHERICAL BALL AND OUTER RACE.

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FIGURE 5-3-12-1. INSPECT TAIL ROTOR PITCH CONTROL ROD BEARINGS

CAUTION

- Damage to elastomeric rod end will result if metallic feeler gage is used. Elastomer will be damaged. Use only elastomeric rod end bearing plastic feeler gage when inspecting rod end elastomeric bearing.
- Damage to elastomeric rod ends will result if elastomeric rod end bearing plastic feeler gage is forced between elastomeric layers. Do not force elastomeric rod end bearing plastic feeler gage when inspecting for laminate separations.

(1) Using 0.005-inch end of elastomeric rod end bearing plastic feeler gage, inspect for elastomeric/metal separations or disbonding of elastomer from shim inner or outer members in all areas carefully. If elastomeric rod end bearing plastic feeler gage can be inserted, separation has occurred. Using 0.005-inch end of elastomeric rod end bearing plastic feeler gage, measure depth of separations. **SMALL HAIRLINE SURFACE CRACKING, MINOR DISBONDING, AND CRAZING OF ELASTOMER UP TO 0.060-INCH ARE ACCEPTABLE IN ALL AREAS OF ROD END ELASTOMERIC BEARING.**

(2) Using 0.005-inch end of elastomeric rod end bearing plastic feeler gage, inspect for elastomer/metal separations or disbonding of elastomer from shim inner or outer members in zone I carefully. If elastomeric rod end bearing plastic feeler gage can be inserted, separation has occurred. Perform the following:

(a) Using 0.005-inch end of elastomeric rod end bearing plastic feeler gage, measure depth of separations. **SEPARATIONS UP TO 0.20-INCH IN ZONE I AND UP TO 0.060-INCH OUTSIDE ZONE I ARE ACCEPTABLE.**

(b) If separation is beyond limits, replace elastomeric rod end (PARA 5-4-39).

5-3-13 TAIL ROTOR BLADE INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Coin, Soft Copper (or equivalent metal), 1 ½ - 2-inches diameter × ⅛ - 3⁄16-inch thick, Procured or Locally-Made
- Drill
- Drill Bit, No. 40, 0.098-inch diameter
- Fluorescent Penetrant Inspection Kit
- Grease Pencil
- Pencil, Soft
- Scale

Materials/Parts

- Abrasive Cloth, Item 5, Appendix D
- Abrasive Cloth, Item 7, Appendix D
- Abrasive Mat, Item 14, Appendix D
- Adhesive, Item 32, Appendix D
- Adhesive, Item 41, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- ASTM E1417
- MIL-P-15035
- MIL-T-9046
- PARA 5-4-38
- PARA 5-5-19
- PARA 16-3-3
- QQ-A-225
- QQ-A-250

5-3-13.1 General.

NOTE

Inspections of all tail rotor blades are the same.

a. Refer to Table 5-3-13-1 and Figure 5-3-13-1 for reference to damage limits and repair procedures on tail rotor blade.

b. Damage to tail rotor blade can be divided into two types, "detectable" and "nondetectable":

NOTE

If necessary, use scale to determine what four - five pounds of pressure feels like.

(1) Detectable damage can be seen and felt as blisters or dents as follows:

(a) Blisters can be seen as raised bumps above tail rotor blade contour, indicating separation between tail rotor blade skin and core. four - five pounds of thumb pressure can move tail rotor blade skin against core without "crinkly" sound.

(b) Dents indicate core crush and can be felt and seen. Four - five pounds of thumb pressure against dent will result in "crinkly" sound.

(2) Nondetectable damage can be found with coin-tapping, but cannot be seen or felt. It can be heard as follows:

(a) Four - five pounds of thumb pressure will not make sound with nondetectable blisters.

(b) Four - five pounds of thumb pressure will make "crinkly" sound with nondetectable dents.

c. Repairs to tail rotor blade can affect track and balance, causing tail rotor vibration. Due to nature of track and balance requirements, only minor repairs are made to tail rotor blade. Perform all repairs carefully and according to instructions. Total number of repairs will depend upon acceptable tracking and vibration levels.

d. Damage to tail rotor blade skin will result if tail rotor blade is sanded excessively. Heavy or careless sanding can produce enough damage to cause tail rotor blade skin cracking. Sand reinforced plastic skin lightly and carefully.

WARNING

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Bonding of skin along trailing edge of tail rotor blade is a critical characteristic. Damage must not exceed limits given in Table 5-3-13-1.

e. Send damaged tail rotor blades that exceed repairable limits to manufacturer.

f. Remove and install tail rotor blades as required (PARA 5-4-38).

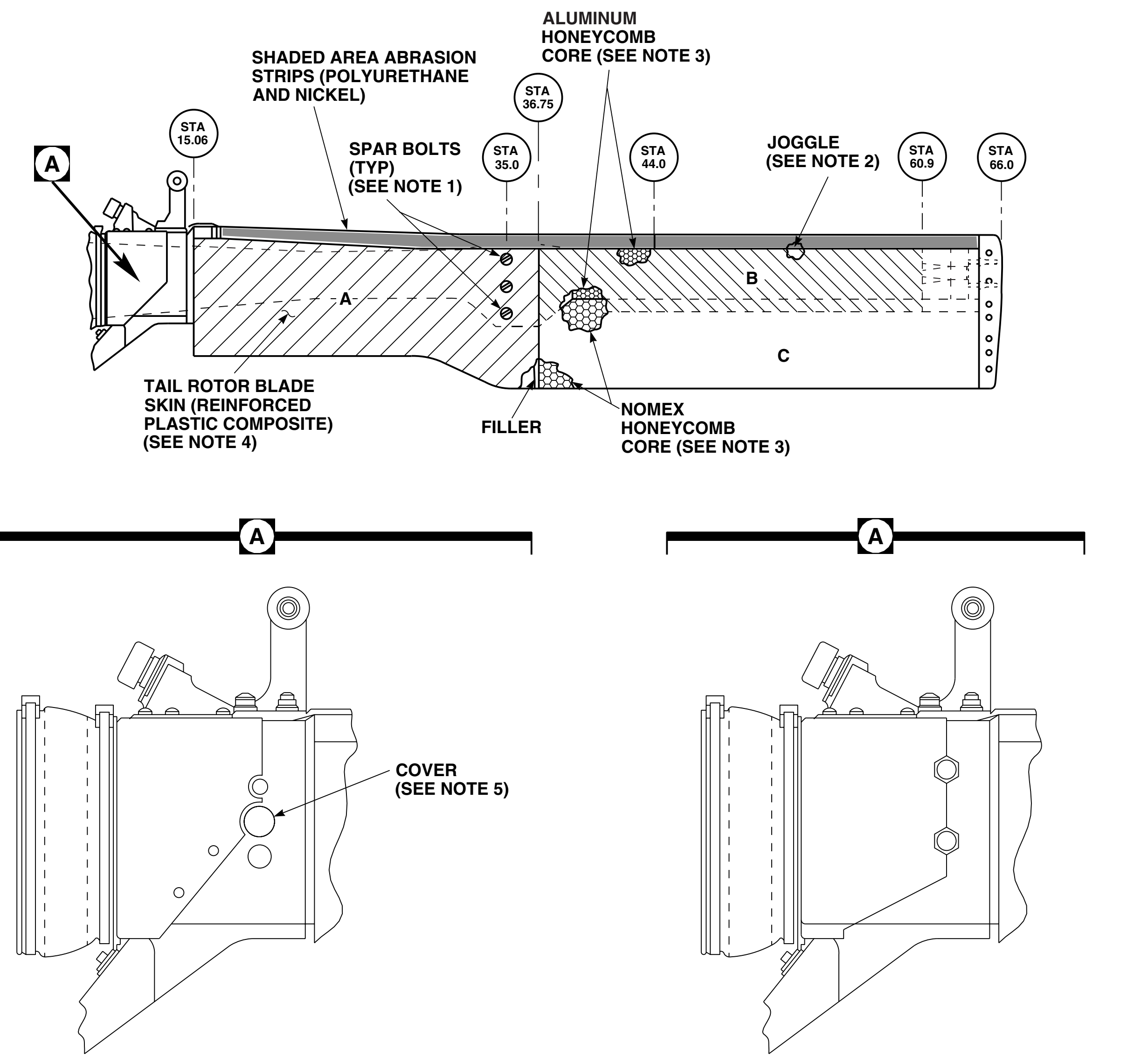
g. Record all damage and repairs to tail rotor blade.

5-3-13.2 Coin-Tapping Inspection Method.

NOTE

- To verify suspected damage, use coin-tapping inspection method to determine size and shape of disbonds/delaminations between tail rotor blade skin and honeycomb core, tail rotor blade skin and tail rotor pitch horn, and tail rotor blade skin and abrasion strip bond areas.
- Coin-tapping should be done by experienced personnel only.

a. Procure or locally make coin. Coin is smooth disc of soft copper (or equivalent metal), 1 ½ - 2-inches in diameter × ⅛ - 3/16-inch thick. Round and smooth edge of coin to prevent damage to tail rotor blade skin.



EFFECTIVITY

ON TAIL ROTOR BLADES WITH TAIL ROTOR PITCH HORN FAIRING, 70104–11104–043.

EFFECTIVITY

ON TAIL ROTOR BLADES WITH TAIL ROTOR PITCH HORN FAIRING, 70104–11104–044.

NOTES

1. SPAR BOLT THREADS AND TOP OF BOLTS ARE SEALED WITH ADHESIVE, ITEM 32, APPENDIX D. COVER BOLTHEADS EVEN WITH BLADE CONTOUR.
2. JOGGLE DEPRESSION FILLED TO BLADE CONTOUR WITH SEALING COMPOUND, ITEM 292, APPENDIX D.
3. CORE DAMAGE NOT REPAIRABLE.
4. SKIN DAMAGE (AREAS A, B, C) LIMITED TO ADHESIVE FILLER OR ADHESIVE INJECTION REPAIR.
5. BOND COVER TO TAIL ROTOR BLADE PIVOT BEARING HOLE WITH ADHESIVE, ITEM 41, APPENDIX D.

LEGEND

A – COMPOSITE SKIN REGION ONLY.
B – COMPOSITE SKIN/ALUMINUM HONEYCOMB CORE REGION.
C – COMPOSITE SKIN/NOMEX HONEYCOMB CORE REGION.

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FIGURE 5-3-13-1. TAIL ROTOR BLADE GENERAL REPAIR LIMITS

b. Hold coin loosely between thumb and finger, tap lightly along bond line or areas suspected of having disbonds/delaminations, and listen for variations in tapping sound. A ″sharp-solid″ sound indicates a good bond and a ″dull-dead″ sound indicates a bond separation.

c. Tap across suspected area in horizontal, vertical, and diagonal lines, and using grease pencil, mark points on tail rotor blade skin where sound varies. Continue tapping and marking until outline of bond separation is completed.

5-3-13.3 Tail Rotor Pitch Horn Fairing-to-Tail Rotor Blade Spar Clearance Inspection.

Inspect tail rotor blade spar for delamination or splintering. **NONE ALLOWED.** Corner damage up to maximum depth of 0.070-inch is allowable in area of inboard end of tail rotor pitch horn fairing (approximate RSTA 9.0).

5-3-13.4 Abrasion Strips Inspection.

a. Visually inspect tip cap for cracks, dents, gouges, and security of attachment (Figure 5-3-13-2, Sheet 4, Detail E).

b. Visually inspect abrasion strip (nickel) for cracks, dents, and gouges (Figure 5-3-13-2, Sheet 3, Detail C).

c. Visually inspect abrasion strip (polyurethane) for gouges, wear, and loose bonding.

d. If installed, inspect/repair polyurethane tape (Erosion Protection Kit) (PARA 16-3-3).

e. Abrasion strips bonded to leading edge of tail rotor blade pocket are repairable until complete wear through occurs.

f. Visually inspect tail rotor blade pocket skin for holes, dents, cracks, and gouges.

g. Using coin-tapping inspection method, inspect abrasion strip and tail rotor blade skin for bond voids/separations, particularly in area where tail rotor blade skin is bonded to tail rotor pitch horn, and around three spar bolts at center of tail rotor blade (PARA 5-3-13.2).

5-3-13.5 Erosion Protection Strips Inspection.

If installed, inspect erosion protection strips (PARA 16-3-3).

5-3-13.6 Tail Rotor Blade Pocket Skin.

NOTE

Region A, Figure 5-3-13-1, consists of composite skin with no filler. Region B consists of composite skin bonded to aluminum honeycomb core which is bonded to tail rotor blade spar. Region C consists of composite skin bonded to Nomex honeycomb core. Repairs to tail rotor blade pocket in all three regions is limited to adhesive filler and adhesive injection repairs.

a. Cracks or holes in tail rotor blade skin will not be repaired (Figure 5-3-13-2, Sheet 3, Detail C). Replace tail rotor blade (PARA 5-4-38).

b. Dents or gouges in tail rotor blade skin, within limits listed in Table 5-3-13-1, are repairable using adhesive filler repair procedure.

c. Core damage greater than dent limits will not be repaired. Replace tail rotor blade (PARA 5-4-38).

d. Bond separations between composite skin and aluminum honeycomb and Nomex honeycomb cores (Region B and Region C) are limited to ½ square-inch in area (Figure 5-3-13-1).

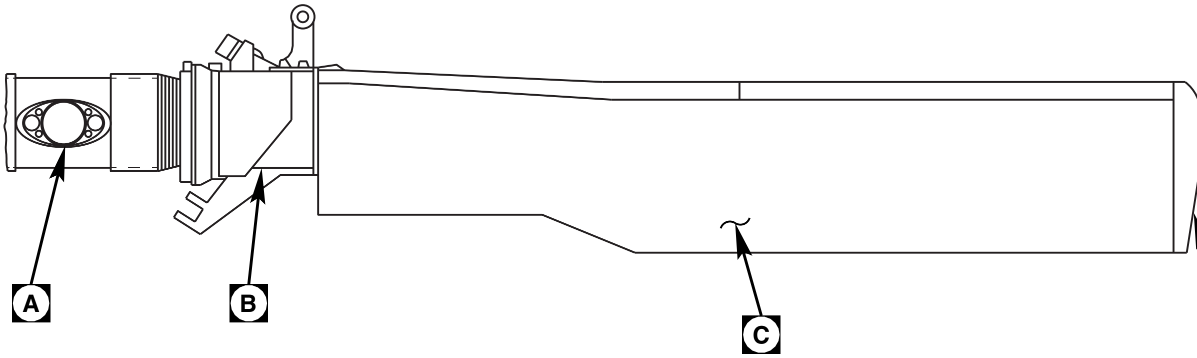
e. If installed, inspect/repair polyurethane tape (Erosion Protection Kit) (PARA 16-3-3).

f. Bond separation between pocket and tip rib is repairable using adhesive injection repair procedure.

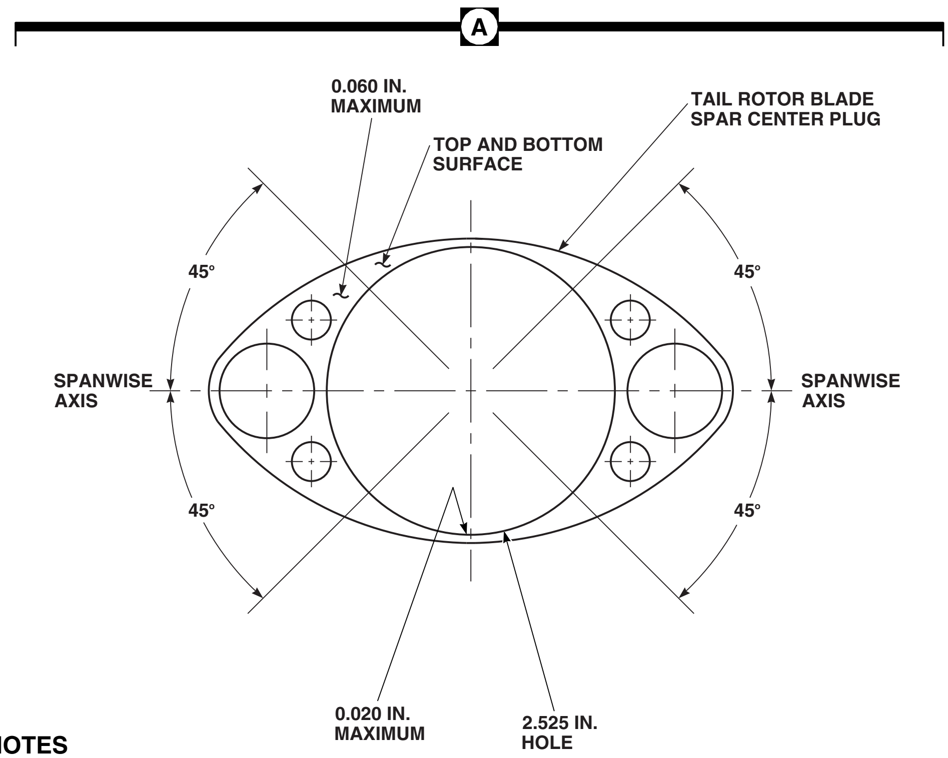
5-3-13.7 Tail Rotor Blade Horn-to-Torque Tube Inspection.

a. Remove tail rotor blades (PARA 5-4-38).

b. Measure 1-inch from face of tail rotor blade horn, and using soft pencil, draw light line on



TAIL ROTOR BLADE



NOTES

1. VARYING THICKNESS OF COMPOSITE SKIN MADE FROM GLASS FIBER AND GRAPHITE PLIES BONDED WITH ADHESIVE RESINS.
2. TIP CAP, 70101–31131–041, HAS ABRASION STRIP (NICKEL) BONDED AS SHOWN. TIP CAP, 70101–31131–042, HAS LEADING EDGE ELECTRO–PLATED NICKEL (0.050 IN. MAXIMUM).

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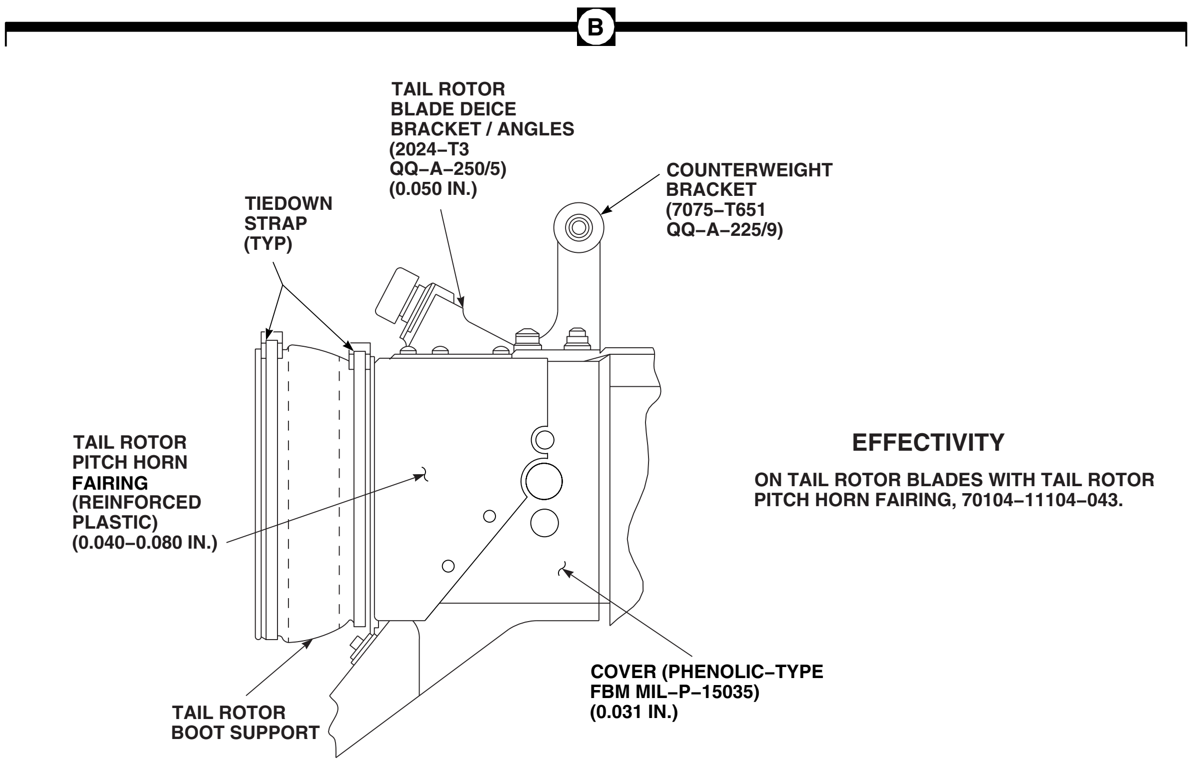
FIGURE 5-3-13-2. TAIL ROTOR BLADE INSPECTION AREAS (SHEET 1 OF 4)



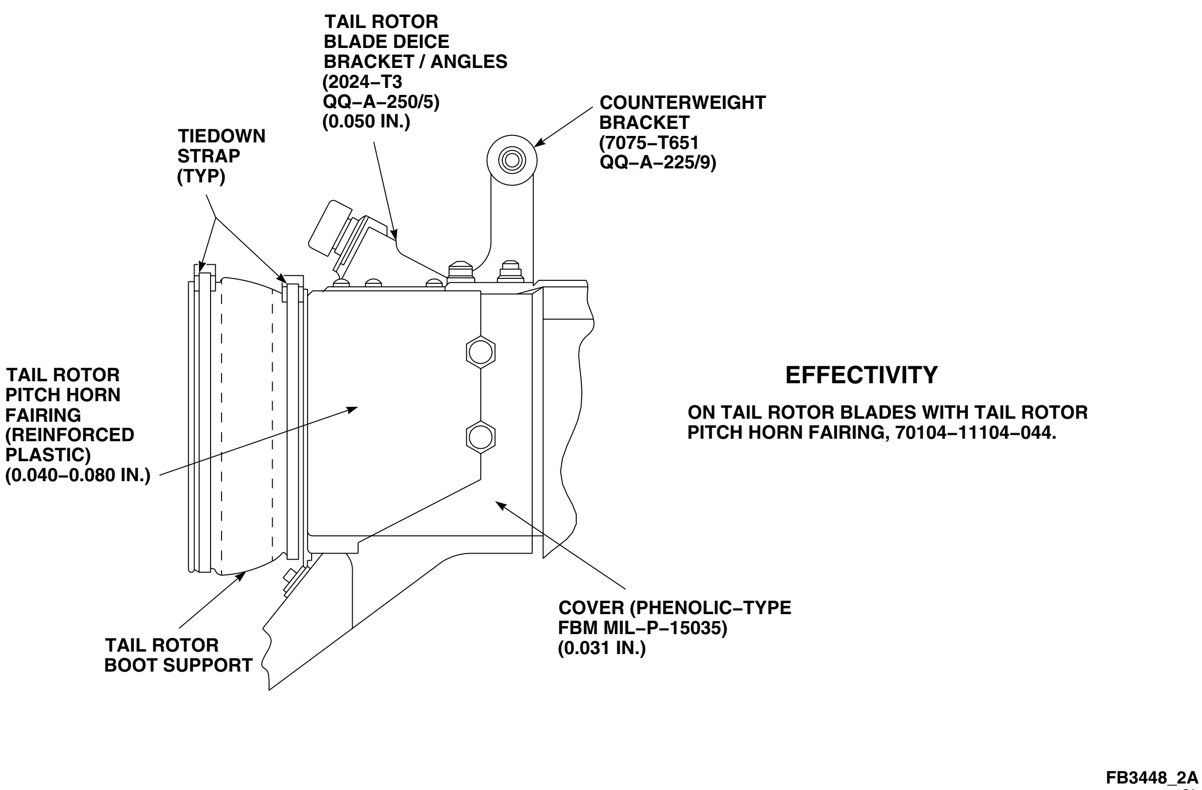
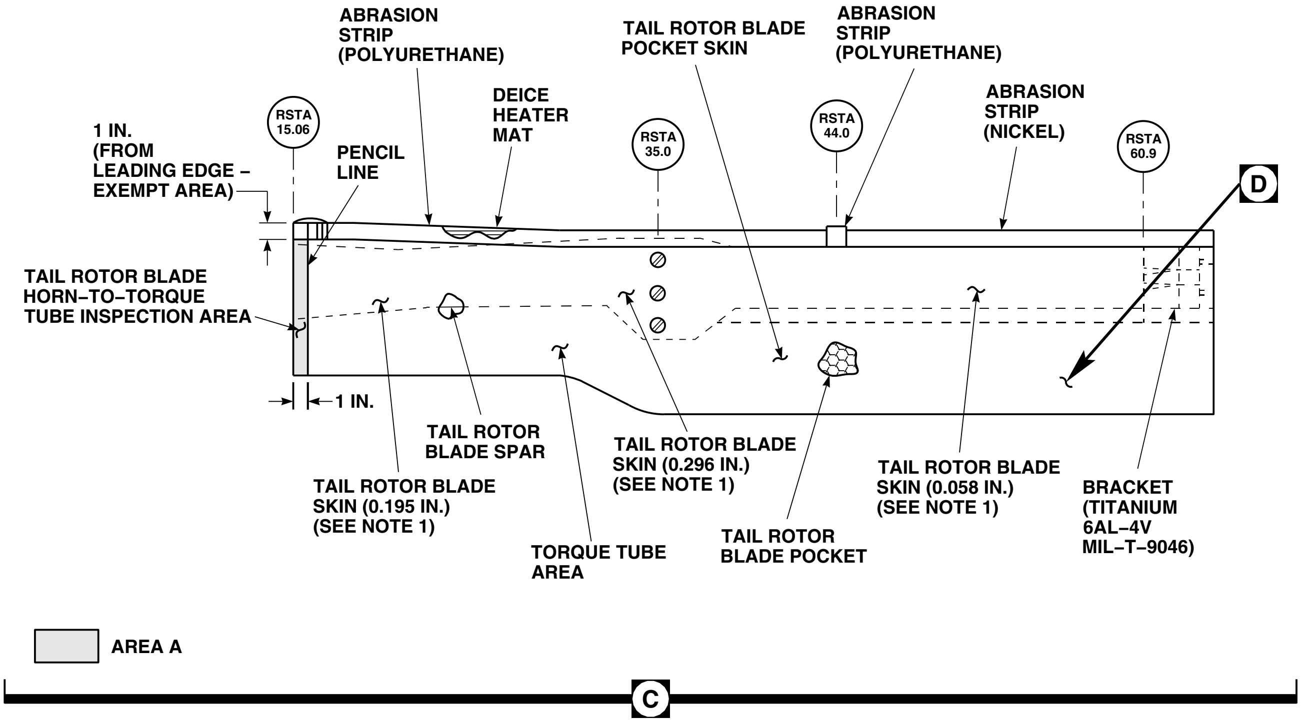


FIGURE 5-3-13-2. TAIL ROTOR BLADE INSPECTION AREAS (SHEET 2 OF 4)



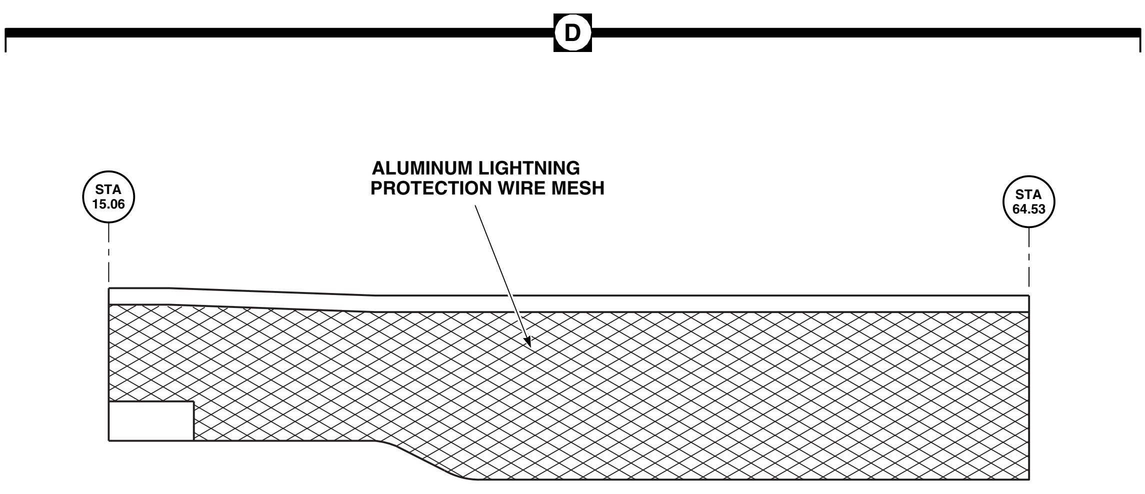


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FIGURE 5-3-13-2. TAIL ROTOR BLADE INSPECTION AREAS (SHEET 3 OF 4)

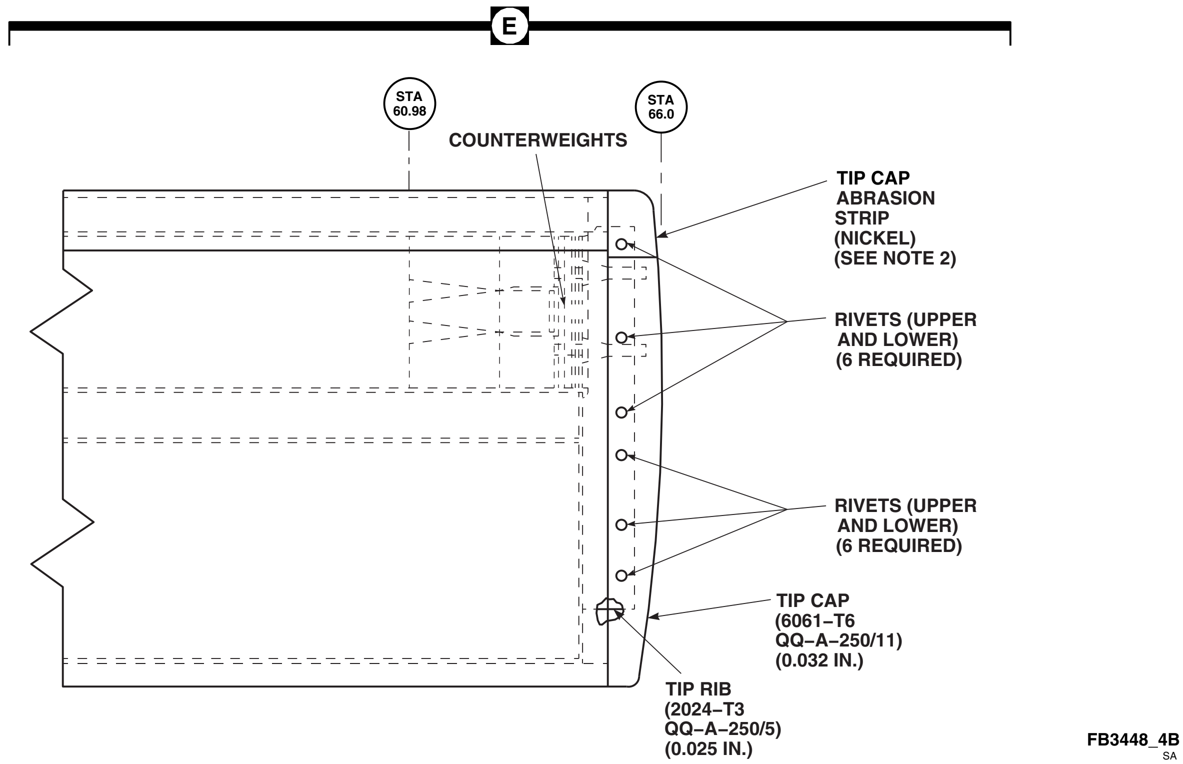


FIGURE 5-3-13-2. TAIL ROTOR BLADE INSPECTION AREAS (SHEET 4 OF 4)

tail rotor blade skin (Figure 5-3-13-2, Sheet 3, Detail C).

NOTE

- Coin tapping near trailing edge, over fiberglass doubler, may have different tone than skin-to-torque tube bond.
- Area 1-inch from leading edge, where heater mat is located, is exempt from this inspection.
- Multiple disbonds are allowed if each disbond is less than 1 square-inch. There is no limit to total number of disbonds less than 1 square-inch.

c. Using coin-tapping method, inspect horn-to-torque tube area (Area A) for disbonds (PARA 5-3-13.2). Inspect Area A on inboard and outboard sides of tail rotor blade on both ends of blade (4 locations per blade). **DISBOND(S) MUST BE LESS THAN 1 SQUARE-INCH.** Perform the following:

(1) If disbond(s) are less than 1 square-inch, perform the following:

(a) Touch up paint as required (PARA 5-4-38).

(b) Install tail rotor blades (PARA 5-4-38).

(2) If disbond(s) are 1 square-inch or greater, perform the following:

(a) Replace disbonded tail rotor blade (PARA 5-4-38).

(b) Send disbonded blade to manufacturer.

Table 5-3-13-1. Tail Rotor Blade Inspection/Repair Limits

Component	Type and Size	Repair Action	On Helicopter	Off Helicopter
Tail Rotor Blade Spar Center Plug (Figure 5-3-13-2, Sheet 1, Detail A)	Scratches/Gouges/ Corrosion/Fretting.			
	- - Surfaces up to 0.060-inch deep (upper and lower).	Using abrasive cloth, Item 5, Appendix D, blend out scratches/ gouges/corrosion/ fretting. Only remove rough and sharp edges. No need to entirely remove damage. Brush-anodize repaired area. If anodizing capabilities do not exist, using corrosion resistant coating, Item 118, Appendix D, apply coating to repaired areas.	x	
	- - On the 2.525-inch diameter hole surface up to 0.020-inch deep.	Using abrasive cloth, Item 5, Appendix D, blend out scratches/ gouges/corrosion/ fretting. No single blend area larger than 1.67-inches long over entire surface, measured along circumference. Total of all blends must not exceed 50% of circumference. Brush-anodize repaired area. If anodizing capabilities do not exist, using corrosion resistant coating, Item 118, Appendix D, apply coating to repaired areas.	x	

Table 5-3-13-1. Tail Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Helicopter	Off Helicopter
	- - Inside diameter of tail rotor blade spar center plug up to 2.533-inches diameter in an area ± 45° along spanwise axis. All other areas up to 2.536-inches diameter.			x

NOTE

Excessive wear in the spanwise area beyond limits may result in difficulty with dynamic balance. The additional wear allowance in the chordwise areas is allowable due to tail rotor blade positioning being controlled by the retention plate.

Component	Type and Size	Repair Action	On Helicopter	Off Helicopter
	Cracks.	Visually inspect for cracks. If cracks are suspected, using fluorescent penetrant inspection kit, fluorescent penetrant inspect for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace tail rotor blade spar center plug.		x
Tail Rotor Boot Supports (Figure 5-3-13-2, Sheet 2, Detail B)	- - Crack in adhesive fillet (any length, tail rotor boot support firmly attached).	Repair, using adhesive injection repair (PARA 5-4-38).		x
	- - Crack in tail rotor boot support.	Not acceptable. Replace tail rotor boot support (PARA 5-5-19).		x
	- - Disbond of tail rotor boot support to tail rotor blade spar.	Rebond tail rotor boot support (PARA 5-5-19).		x

Table 5-3-13-1. Tail Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Heli-copter	Off Heli-copter
Tail Rotor Blade Pocket Skin (Figure 5-3-13-2, Sheet 3, Detail C)	Cracks.	Not acceptable. Replace tail rotor blade (PARA 5-4-38).		x
	Dents.			
	- - Unbroken skin up to 0.040-inch deep × 2-inches diameter.	No repair required. Touch up paint (PARA 5-4-38).	x	
	- - Skin broken within dent or dent deeper than above limits.	Not acceptable. Replace tail rotor blade (PARA 5-4-38).		x
	Bond Separation.			
	- - Between skin and tip rib.	Repair, using adhesive injection repair (PARA 5-4-38).	x	
	- - Area A: Tail rotor blade horn-to-torque tube bond area less than 1 square-inch.	No repair required. Touch up paint (PARA 5-4-38).		x
	- - Area A: Tail rotor blade horn-to-torque tube bond area 1 square-inch or greater.	Not acceptable. Replace tail rotor blade (PARA 5-4-38).		x
	- - Skin to core up to ½ square-inch area.	No repair required. Touch up paint (PARA 5-4-38).	x	
	- - Larger than ½ square-inch.	Not acceptable. Replace tail rotor blade (PARA 5-4-38).		x

Table 5-3-13-1. Tail Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Heli-copter	Off Heli-copter

WARNING

FSCAP

Bonding of skin along trailing edge of tail rotor blade is a critical characteristic. Damage must not exceed the following limits.

Component	Type and Size	Repair Action	On Heli-copter	Off Heli-copter
	- - Between skin plies at trailing edge: inboard of RSTA 39.0, voids up to 0.10-inch chordwise × 0.050-inch spanwise; outboard of RSTA 39.0, voids up to ¼-inch chordwise × ½-inch spanwise.	Repair, using adhesive injection repair (PARA 5-4-38).		x
	- - Skin separation between RSTA 19.0 and RSTA 40.0, disbond up to 6-inches.	Repair, using trailing edge disbond repair (PARA 5-4-38).		x
	Core Damage (Nomex Honeycomb Core).			
	- - Greater than dent limits.	Not acceptable. Replace tail rotor blade (PARA 5-4-38).		x
	Core Damage (Aluminum Honeycomb Core).			
	- - Greater than dent limits.	Not acceptable. Replace tail rotor blade (PARA 5-4-38).		x
	Gouges.			

Table 5-3-13-1. Tail Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Helicopter	Off Helicopter
	- - Up to 0.030-inch deep × 3-inches long in Area A and Area B (Figure 5-3-13-1).	Repair, using adhesive filler repair (PARA 5-4-38).	x	
	- - Up to 0.005-inch deep × 2-inches long in Area C.	Repair, using adhesive filler repair (PARA 5-4-38).	x	
	- - Greater than above limits.	Not acceptable. Replace tail rotor blade (PARA 5-4-38).		x
	Holes (Skin/Core).	Not acceptable. Replace tail rotor blade (PARA 5-4-38).		x
	Erosion.			
	- - Up to one ply thickness (0.006-inch deep).	No repair required. Touch up paint (PARA 5-4-38).	x	
	- - Deeper than one ply thickness.	Not acceptable. Replace tail rotor blade (PARA 5-4-38).		x
Aluminum Lightning Protection Wire Mesh (Figure 5-3-13-2, Sheet 3, Detail D)	Corrosion (See Note).			
	- - Less than 10 pinholes per square-foot.	No repair required.		
	- - Greater than 10 pinholes per square-foot.	Repair, using aluminum wire mesh lightning protection repair (PARA 5-4-38).		x

Table 5-3-13-1. Tail Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Heli-copter	Off Heli-copter

NOTE

Aluminum wire mesh lightning protection is nonstructural. Corrosion is a cosmetic problem. Repair corrosion only if tail rotor blade has already been removed from helicopter.

Component	Type and Size	Repair Action	On Heli-copter	Off Heli-copter
Abrasion Strip (Polyurethane) (Figure 5-3-13-2, Sheet 3, Detail C)	Loose Bond.	Acceptable within limits if repaired using polyurethane abrasion strip repair (PARA 5-4-38).	x	
	Gouges.			
	- - Up to ⅛-inch wide.	Repair, using adhesive filler repair (PARA 5-4-38).	x	
	- - More than ⅛-inch wide.	Repair, using polyurethane abrasion strip repair (PARA 5-4-38).		x
	Erosion.	Acceptable until worn through, then repair, using polyurethane abrasion strip repair (PARA 5-4-38).		x
	Delaminations.			
	- - Less than 4 square-inches.	Repair, using polyurethane abrasion strip repair (PARA 5-4-38).		x
	- - Greater than 4 square-inches.	Not acceptable. Replace abrasion strip (polyurethane) (PARA 5-4-38).		x
Abrasion Strip (Nickel) (Figure 5-3-13-2, Sheet 3, Detail C)	Cracks.			

Table 5-3-13-1. Tail Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Helicopter	Off Helicopter
	- - 3 or less per abrasion strip (nickel).	If bond separation is within limits, no repair required (refer to Bond Separation in this section). Seal cracks using abrasion strip repair (PARA 5-4-38).	x	
	- - More than 3 per abrasion strip (nickel).	Not acceptable. Replace tail rotor blade (PARA 5-4-38).		x
	Dents.			
	- - Up to 1⁄16-inch deep × 1-inch across and bond separation less than ½-inch beyond edge of dent.	No repair required. Touch up paint (PARA 5-4-38).	x	
	- - Greater than above limits.	Not acceptable. Replace tail rotor blade (PARA 5-4-38).		x
	Bond Separations.			
	- - Up to ⅛ × ½-inch area.	Seal open voids, using abrasion strip repair (PARA 5-4-38).	x	
	- - Up to ½ × 2-inches area.	Repair, using adhesive injection repair (PARA 5-4-38).	x	
	- - Greater than above limits.	Not acceptable. Replace tail rotor blade (PARA 5-4-38).		x

Table 5-3-13-1. Tail Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Helicopter	Off Helicopter
	Gouges.	Acceptable within thickness of abrasion strip (nickel). Using abrasive cloth, Item 7, Appendix D, blend raised metal. Touch up paint (PARA 5-4-38).	x	
	Erosion (Wear/ Corrosion).	Acceptable until repair limits of 0.007-inch deep for large areas, or up to 0.012-inch deep, for no more than 1 square-inch, are exceeded. Using adhesive, Item 32, Appendix D, apply thin coat at point where abrasion strip (nickel) and composite skin/ aluminum honeycomb core mates. Touch up paint (PARA 5-4-38). If erosion exceeds limits, replace tail rotor blade (PARA 5-4-38).	x	
Tip Cap (Figure 5-3-13-2, Sheet 4, Detail E)	Cracks.			
	- - Up to ¼-inch long.	Using drill with No. 40 drill bit, stop-drill cracks (no more than two cracks allowed at attachment rivet holes). Touch up paint (PARA 5-4-38).	x	
	- - More than ¼-inch long, or more than two cracks at attachment holes.	Not acceptable. Replace tip cap (PARA 5-4-38).	x	

Table 5-3-13-1. Tail Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Heli-copter	Off Heli-copter
	Dents.			
	- - With no cracks or gouges.	No repair required. Touch up paint (PARA 5-4-38).	x	
	- - With cracks up ¼-inch long.	Using drill with No. 40 drill bit, stop-drill cracks. Touch up paint (PARA 5-4-38).	x	
	- - With cracks more than ¼-inch long.	Not acceptable. Replace tip cap (PARA 5-4-38).	x	
	Gouges/Erosion/ Corrosion.			
	- - Up to 0.005-inch deep (after cleanup).	Using abrasive mat, Item 14, Appendix D, blend out gouges/ erosion/corrosion. Touch up paint (PARA 5-4-38).	x	
	- - Deeper than 0.005-inch (after cleanup).	Not acceptable. Replace tip cap (PARA 5-4-38).	x	
Tip Cap Abrasion Strip (Nickel) (Figure 5-3-13-2, Sheet 4, Detail E)	Bond Separation.			
	- - Up to ⅛ × ½-inch area.	Seal voids using abrasion strip repair (PARA 5-4-38).	x	
	- - More than ⅛ × ½-inch area.	Repair, using adhesive injection repair (PARA 5-4-38).	x	

Table 5-3-13-1. Tail Rotor Blade Inspection/Repair Limits (Cont)

Component	Type and Size	Repair Action	On Heli-copter	Off Heli-copter
	- - Erosion (Wear/ Corrosion).	Acceptable until worn through, then replace tip cap (PARA 5-4-38).	x	
	Cracks.	Acceptable if bond separation within limits (refer to Bond Separation in this section).	x	

SECTION 5-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

5-4-1 MAIN ROTOR HEAD AND SHAFT EXTENSION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, 5⁄16-inch
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Grease Pencil
- Guide Rope
- Hoist, Capacity to Support Main Rotor Head (Alternate for Cabin Maintenance Crane)
- Hoisting Sling, 70700-20320-042
- Jacking Bolts, NAS566-41
- Jacking Bolts, NAS629-26
- Leather Mallet
- Main Rotor Hub Guide Assembly, 70700-20436-041
- Main Rotor Shaft Impact Tool, Locally-Made, Figure H-252, Appendix H
- Maintenance Trailer
- Nonmetallic Drift, Locally-Made
- Nonmetallic Scraper, Locally-Made
- Plastic Mallet
- Protective Covering
- Protective Padding
- Rawhide Mallet
- Rotary Wing Head Adapter, 70700-20421-041
- Shim, Aluminum, 0.032-inch × 1-inch × 4-inches, Locally-Made (4)
- Shim, Aluminum, 0.060-inch × 1-inch × 4-inches, Locally-Made (4)
- Special Wrench
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 500 - 2500 in. lbs
- Torque Wrench, 0 - 300 ft lbs

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Abrasive Cloth, Item 12, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Antiseize Compound, Item 74, Appendix D
- Cardboard (Used for Thread Protection)

- Corrosion Preventive Compound, Item 113, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Grease, Item 144, Appendix D
- Grease, Item 150, Appendix D
- Lubricant, Solid Film, Item 201, Appendix D
- Lubricating Oil, Item 209, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Pressure Sensitive Adhesive Tape, Item 249, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 279, Appendix D
- Sealing Compound, Item 299, Appendix D
- Sealing Compound, Item 301A, Appendix D
- Tags

References

- Appendix D
- Appendix H
- Appropriate MTF
- ASTM E1417
- PARA 1-6-18
- PARA 1-7-5
- PARA 2-6-5
- PARA 5-4-2
- PARA 5-4-3
- PARA 5-4-4
- PARA 5-4-18
- PARA 5-4-28
- PARA 5-4-30
- PARA 5-4-31
- PARA 5-5-8
- PARA 6-4-1
- PARA 6-4-7
- PARA 11-4-4
- PARA 12-4-26
- PARA 18-4-1
- TM 1-70-VIB

5-4-1.1 Replace Main Rotor Head and Shaft Extension.

NOTE

- Cabin maintenance crane may be used instead of hoist. In this procedure, hoist is illustrated.
- Main rotor head weighs approximately 875 pounds.

5-4-1.1.1 Prepare to Remove.

a. If cabin maintenance crane will be used, erect cabin maintenance crane (PARA 18-4-1).

b. If blade deice kit is installed, remove main rotor blade deice distributor and slipring/brush assembly (PARA 12-4-26).

c. If blade deice kit is not installed, remove bifilar cover (PARA 5-4-30).

d. Remove main rotor blades (PARA 5-4-31).

e. Disconnect pitch control rods from swashplate (PARA 5-4-18). Keep attaching hardware with swashplate.

f. If necessary, install rotary wing head adapter on maintenance trailer.

5-4-1.1.2 Remove Main Rotor Head and Shaft Extension.

WARNING

FSCAP

- Exposed surfaces of main rotor shaft and main rotor shaft extension are critical surfaces which must not be damaged during and after removal of main rotor head. Use extreme caution when removing and provide protective covering over critical surfaces after removal of main rotor head.
- Exposed surfaces of main rotor shaft and main rotor shaft extension are critical surfaces which must not be damaged while breaking torque. Use protective covering as required.

CAUTION

Damage to main rotor head will result if metallic scrapers are used to clean sealing compound from components. Do not use metallic items to remove sealing compound.

a. Using nonmetallic scraper, remove sealing compound from lower pressure plate and main rotor shaft extension.

b. Remove bolts and washers securing lower lockring to main rotor shaft nut (Figure 5-4-1-1, Sheet 1, Detail A and Section A-A). Remove lower lockring.

c. Gradually loosen bolt tension at main rotor shaft nut, in 1/4-turn increments in sequence numbered on main rotor shaft nut (Figure 5-4-1-1, Sheet 2, Detail B). Do not remove bolts. Repeat sequence as required until all bolts are loose.

d. Remove bolts securing lock to main rotor shaft nut and remove lock (Figure 5-4-1-1, Sheet 1, Detail A and Section A-A).

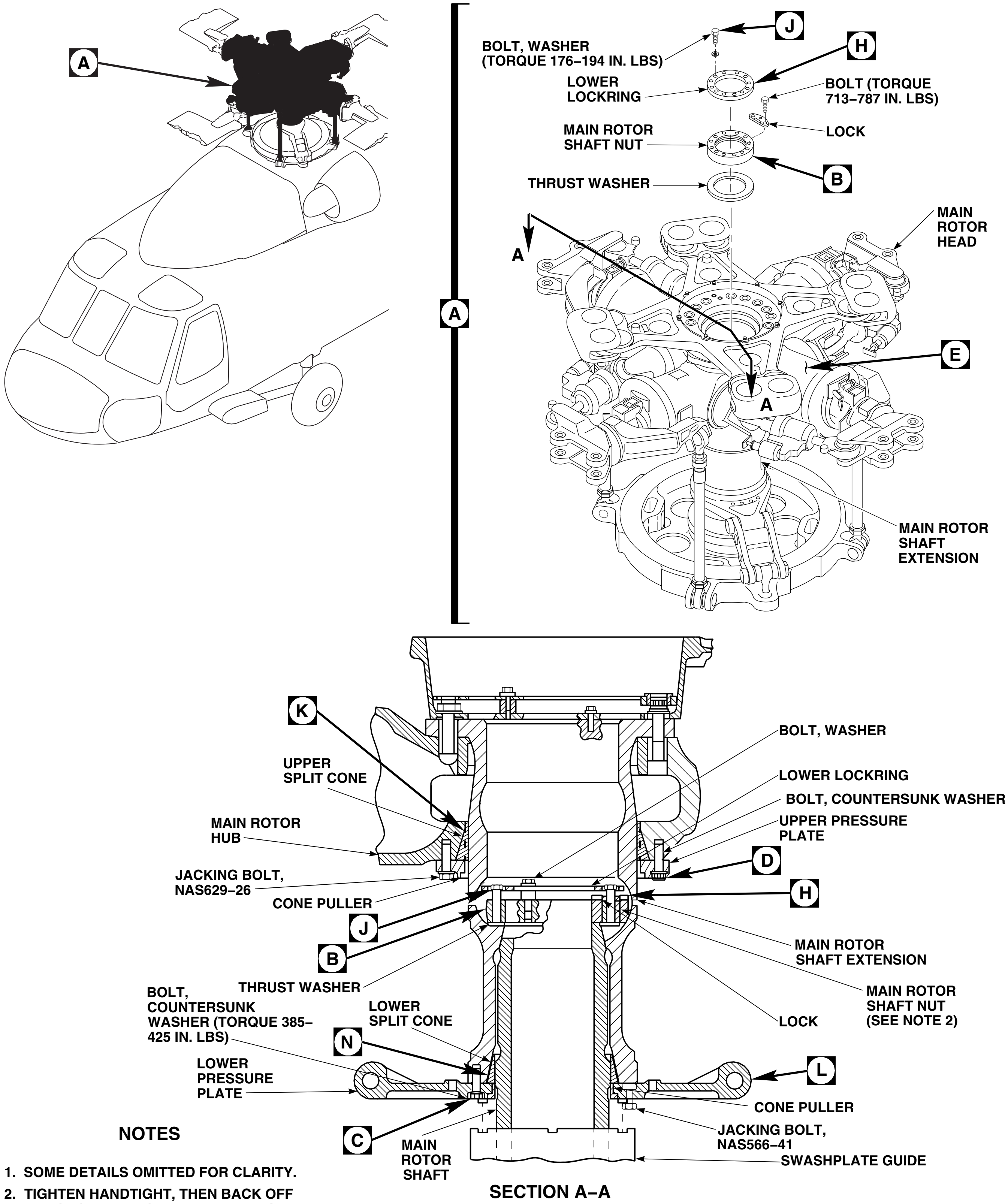
e. Remove bolts securing main rotor shaft nut (Figure 5-4-1-1, Sheet 1, Detail A). Remove main rotor shaft nut and thrust washer.

f. Remove lower pressure plate from shaft extension as follows:

NOTE

Do not completely remove bolts until split cones are free from main rotor shaft extension.

(1) Gradually loosen bolts at lower pressure plate, in 1/4-turn increments in sequence numbered on lower pressure plate (Figure 5-4-1-1, Sheet 2, Detail C). Do not remove bolts. Repeat sequence as required until all bolts are loose.



NOTES

1. SOME DETAILS OMITTED FOR CLARITY.
2. TIGHTEN HANDTIGHT, THEN BACK OFF UP TO 1/4 TURN TO LINE UP NEAREST WHITE MAIN ROTOR SHAFT TANG.

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FIGURE 5-4-1-1. REPLACE MAIN ROTOR HEAD AND SHAFT EXTENSION (SHEET 1 OF 5)

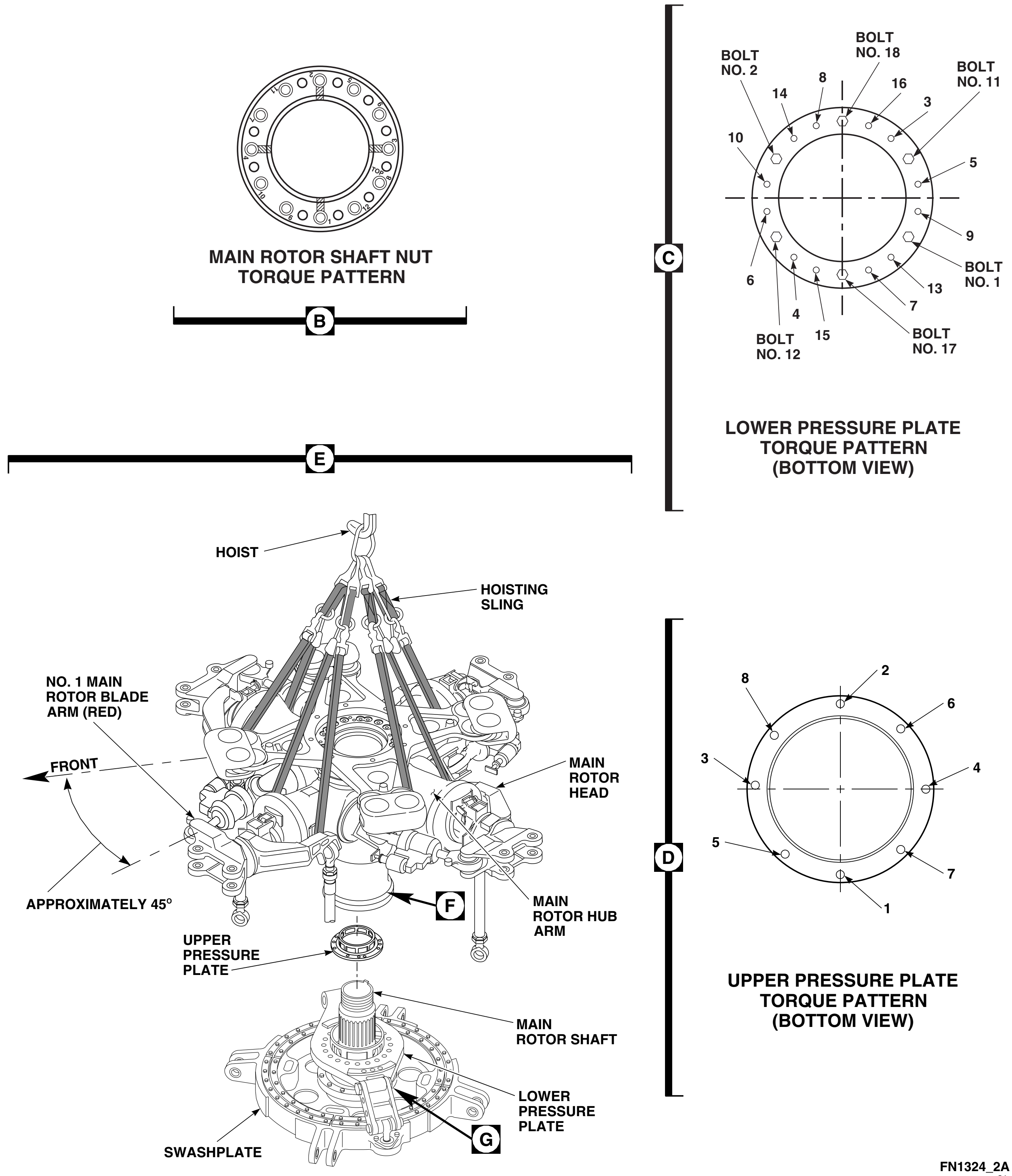


FIGURE 5-4-1-1. REPLACE MAIN ROTOR HEAD AND SHAFT EXTENSION (SHEET 2 OF 5)

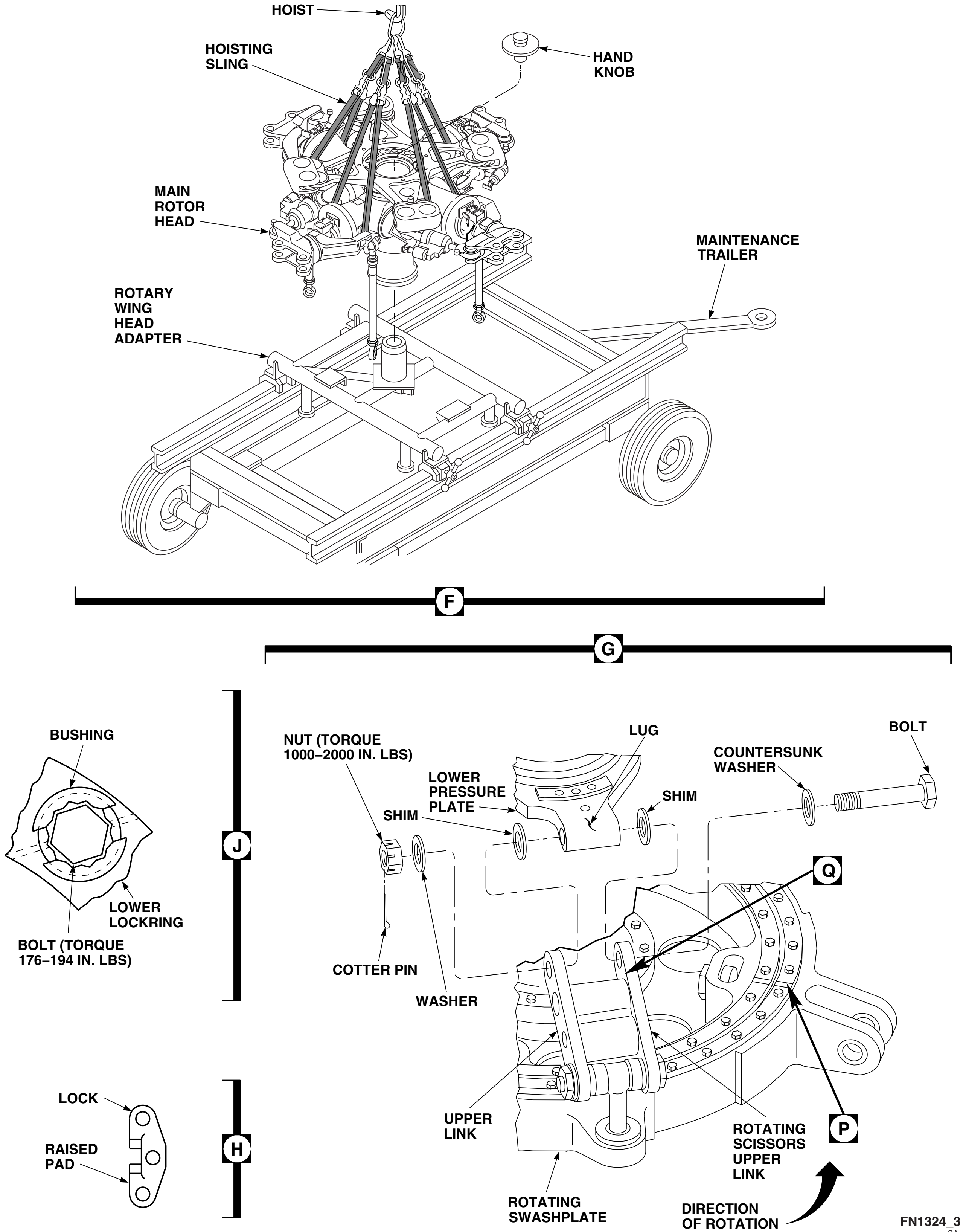


FIGURE 5-4-1-1. REPLACE MAIN ROTOR HEAD AND SHAFT EXTENSION (SHEET 3 OF 5)

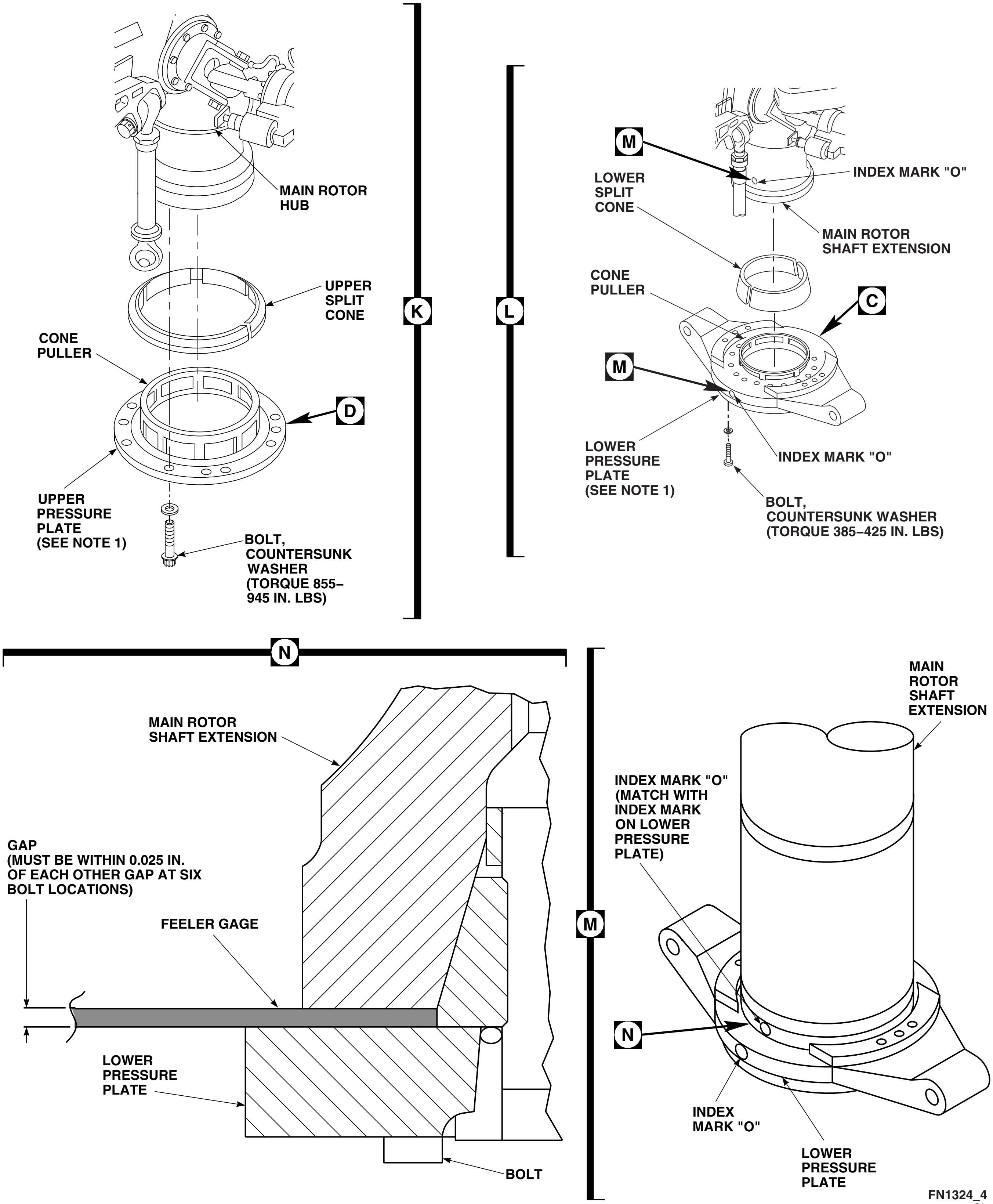
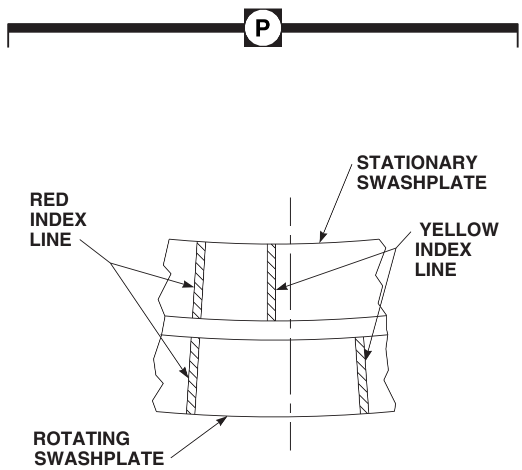


FIGURE 5-4-1-1. REPLACE MAIN ROTOR HEAD AND SHAFT EXTENSION (SHEET 4 OF 5)



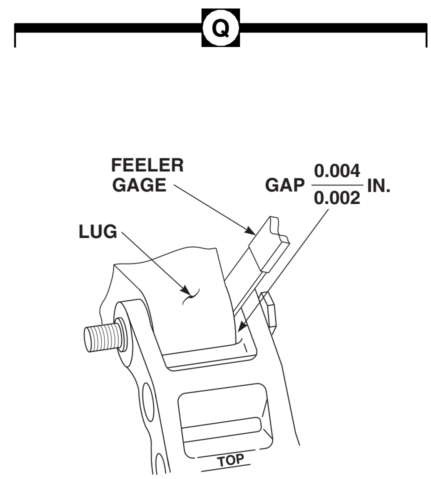


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FIGURE 5-4-1-1. REPLACE MAIN ROTOR HEAD AND SHAFT EXTENSION (SHEET 5 OF 5)

NOTE

When installing jacking bolts, make sure bolts securing lower pressure plate are loosened enough to prevent bottoming of plate on boltheads.

(2) If lower pressure plate does not drop freely from shaft extension, install six jacking bolts, NAS566-41, evenly spaced, in lower pressure plate (Figure 5-4-1-1, Sheet 1, Section A-A).

NOTE

Make sure jacking bolts are turned in evenly, and that lower split cones are free.

(3) Turn in jacking bolts until lower split cones are free from shaft extension.

(4) Remove all bolts, countersunk washers, and jacking bolts from lower pressure plate. Slide lower pressure plate down and rest it on swashplate guide.

(5) Remove lower split cones from shaft extension. Place in bag for reinstallation and tag.

(6) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean all bolts that have primer on them.

NOTE

If shaft extension or upper pressure plate do not require removal, disregard step g. and proceed to step h.

g. If shaft extension will be removed, or if upper pressure plate requires replacement, perform the following:

(1) Using nonmetallic scraper, remove sealing compound from upper pressure plate and shaft extension.

NOTE

Do not completely remove bolts until upper split cones are free from shaft extension.

(2) Gradually loosen bolts in upper pressure plate, in ¼-turn increments, in sequence numbered on upper pressure plate (Figure 5-4-1-1, Sheet 2, Detail D). Do not remove bolts. Repeat sequence as required until all bolts are loose.

NOTE

When installing jacking bolts, make sure bolts securing upper pressure plate are loosened enough to prevent bottoming of plate on boltheads.

(3) If upper pressure plate does not drop freely from main rotor head, install four jacking bolts, NAS629-26, evenly spaced, in upper pressure plate (Figure 5-4-1-1, Sheet 1, Section A-A).

NOTE

Make sure jacking bolts are turned in evenly, and upper split cones are free.

(4) Turn in jacking bolts until upper split cones are free from main rotor hub.

(5) Remove all bolts, countersunk washers, and jacking bolts from upper pressure plate. Slide upper pressure plate down so it rests on lower pressure plate.

(6) Remove upper split cones from main rotor hub. Place in bag for reinstallation and tag.

(7) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean all bolts that have primer on them.

h. Protect main rotor shaft threads with wrap such as cardboard or machinery towels, Item 211, Appendix D, and secure with pressure sensitive adhesive tape, Item 249, Appendix D.

CAUTION

Damage to main rotor head will result if hoisting sling is not properly positioned when hoisting main rotor hub. Hoisting sling must be put securely around main rotor hub arms.

i. Attach straps of hoisting sling around each main rotor hub arm (Figure 5-4-1-1, Sheet 2, Detail E).

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

j. Attach hoisting sling to hoist or cabin maintenance crane.

k. Adjust straps before lifting, for even distribution of weight and so main rotor head can be lifted straight up. If using cabin maintenance crane, make sure straps are adjusted so shaft extension will clear main rotor shaft when main rotor head is lifted.

l. Remove any slack from hoisting sling using hoist.

m. Attach guide rope to main rotor head.

n. Have assistant hold guide rope to help guide main rotor head.

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting main rotor head. Keep area clear when hoisting main rotor head.

o. Lift main rotor head clear of helicopter.

p. If main rotor head becomes seized to main rotor shaft at shaft extension, perform the following:

(1) Using lubricating oil, Item 209, Appendix D, apply light coat around split cone seat area where taper in shaft extension mates with taper in main rotor shaft.

(2) Apply tension to hoisting sling. If rotor head fails to become detached, relieve tension on hoisting sling.

(3) Locally make main rotor shaft impact tool (Figure H-252, Appendix H).

(4) Using main rotor shaft impact tool, position on main rotor shaft with steel rod of impact tool protruding out of shaft extension. Make sure slot in baseplate of impact tool lines up with tang on main rotor shaft.

WARNING

- Injury to personnel and damage to equipment will result if excessive tension is used and personnel and equipment is not clear of main rotor head when applying tension to hoisting sling and impacting main rotor shaft. Main rotor head may break free of main rotor shaft suddenly.
- Do not apply excessive tension to hoisting sling and make sure personnel and equipment are clear from main rotor head when performing this procedure.

(5) Apply tension to hoisting sling.

(6) Hold weight of impact tool 1-foot above baseplate of impact tool. Drop weight onto baseplate. Repeat until main rotor head is free from main rotor shaft.

q. Carefully lower main rotor head on rotary wing head adapter, making sure not to damage main rotor hub or pitch control rods (Figure 5-4-1-1, Sheet 3, Detail F).

r. Secure main rotor head to rotary wing head adapter with handknob.

s. Remove hoisting sling from main rotor head.

NOTE

When removed from main rotor head, upper pressure plate will be resting on lower pressure plate.

t. If not previously removed from rotor head in step g., remove upper pressure plate (Figure 5-4-1-1, Sheet 2, Detail E).

u. Remove bolts, countersunk washers, washers, shims, and nuts securing rotating scissors to lower pressure plate (Figure 5-4-1-1, Sheet 3, Detail G). Remove lower pressure plate.

v. If required, remove main rotor shaft extension (PARA 5-4-1.2).

w. Install protective covering to exposed surfaces of main rotor shaft and, if removed, main rotor shaft extension.

5-4-1.1.3 Inspect/Repair.

a. Inspect lower lockring for damaged bushings or missing spring pins from bushings. If damaged bushings or missing spring pins are found, replace as follows (Figure 5-4-1-2):

(1) To replace missing spring pin, perform the following:

(a) Apply thin coat of sealing compound, Item 279, Appendix D, on replacement spring pin.

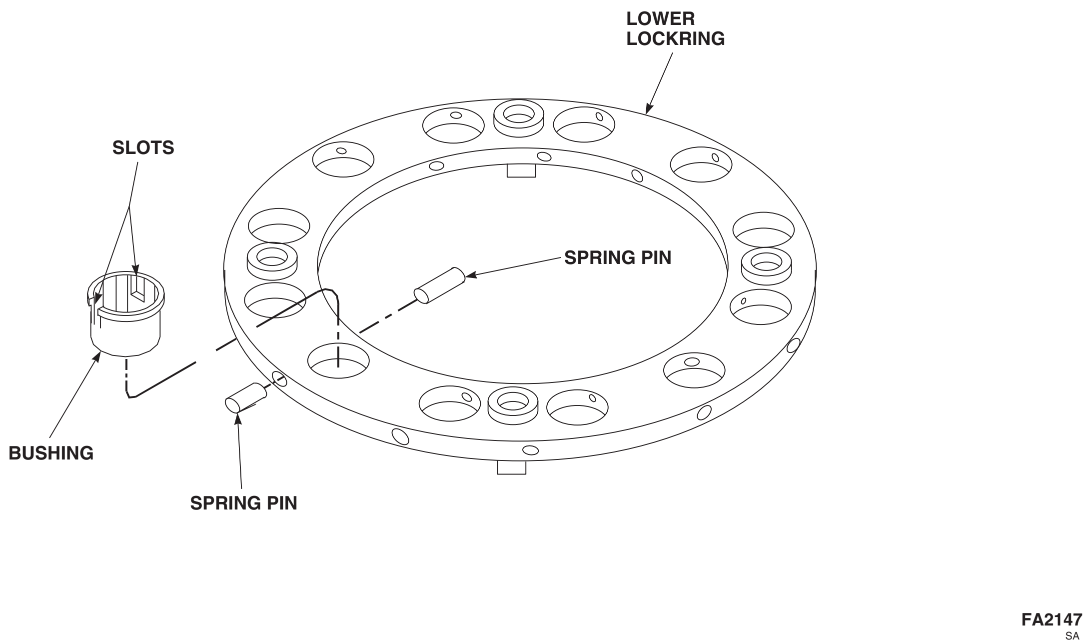


FIGURE 5-4-1-2. INSPECT/REPAIR LOWER LOCKRING

(b) Using rawhide mallet and nonmetallic drift, tap spring pin into lower lockring and bushing.

(2) To replace damaged bushing, perform the following:

(a) Using rawhide mallet and nonmetallic drift, tap spring pins from out of bushing and lower lockring.

(b) Remove bushing from lower lockring.

(c) Install replacement bushing in lower lockring. Position bushing so that spring pins will engage in slots.

(d) Apply thin coat of sealing compound, Item 279, Appendix D, on spring pins.

(e) Using rawhide mallet and nonmetallic drift, tap spring pins into lower lockring and bushing.

b. Inspect main rotor shaft nut for corrosion. If required, repair as follows:

(1) Remove light corrosion.

(2) Using abrasive cloth, Item 10, Appendix D, remove remaining corrosion on outside diameter (OD) and end of main rotor shaft nut by blending. Do not blend or chase threads. **MINIMUM SPHERICAL OD IS 7.290-INCHES. MINIMUM THICKNESS IS 1.240-INCHES.** If corrosion cannot be blended out within limits, replace main rotor shaft nut.

(3) Using epoxy primer coating kit, Item 135, Appendix D, apply two coats of epoxy primer.

(4) Touch up paint on all reworked areas (PARA 2-6-5).

c. Inspect thrust washer for corner damage, corrosion pitting, nicks, and scratches. If required, using abrasive cloth, Item 1A, Appendix D, repair thrust washer as follows:

(1) **CORNER DAMAGE MAY BE BLENDED OUT UP TO 0.003-INCH DEEP AROUND ENTIRE CIRCUMFERENCE OF THRUST WASHER. IF CORNER DAMAGE EXCEEDS 0.003-INCH, ONE SIDE OF THRUST WASHER MAY BE BLENDED TO MAXIMUM DEPTH OF 0.010-INCH OVER 10% OF THRUST WASHER CIRCUMFERENCE.** If repair exceeds limits, or if both sides of thrust washer were blended beyond initial limit of 0.003-inch, replace thrust washer.

(2) **CORROSION PITTING DAMAGE ON BOTH SIDES OF THRUST WASHER MAY BE BLENDED OUT TO MAXIMUM DEPTH OF 0.003-INCH OVER 10% OF THRUST WASHER SURFACE AREA. IF DAMAGE EXCEEDS 0.003-INCH, ONE SIDE OF THRUST WASHER MAY BE BLENDED TO MAXIMUM DEPTH OF 0.010-INCH OVER 10% OF THRUST WASHER SURFACE AREA.** If repair exceeds limits, or if both sides of thrust washer were blended beyond initial limit of 0.003-inch, replace thrust washer.

(3) **NICKS, GOUGES, AND SCRATCHES ON BOTH SIDES OF THRUST WASHER MAY BE BLENDED OUT TO MAXIMUM DEPTH OF 0.003-INCH OVER 10% OF THRUST WASHER SURFACE AREA. IF DAMAGE EXCEEDS 0.003-INCH, ONE SIDE OF THRUST WASHER MAY BE BLENDED TO MAXIMUM DEPTH OF 0.010-INCH OVER 10% OF THRUST WASHER SURFACE AREA.** If repair exceeds limits, or if both sides of thrust washer were blended beyond initial limit of 0.003-inch, replace thrust washer.

(4) Inspect thrust washer for warpage. **WARPAGE MUST NOT EXCEED 0.005-INCH.** If warpage exceeds limit, replace thrust washer.

d. Inspect main rotor shaft extension as follows:

(1) Where possible, visually inspect for voids in bonded liners and sleeves on main rotor shaft extension. **EDGE VOIDS UP TO 25% OF BONDED LINER OR SLEEVE WIDTH ARE ACCEPTABLE. VOIDS UP TO 5% OF TOTAL BONDED LINER OR SLEEVE AREA ARE ACCEPTABLE.** If voids exceed limits, replace main rotor shaft extension.

(2) Inspect shaft extension external surfaces for nicks, dents, scratches, and scuff marks (Figure 5-4-1-3). If damage is found, perform the following:

NOTE

- Areas shown in Figure 5-4-1-3 that are not labeled ZONE 2 are considered ZONE 1.

- Minimum length for any blended area must be 0.25-inch. For ZONE 1, this corresponds to a 500:1 blend ratio. For ZONE 2, this corresponds to a 50:1 blend ratio.

(a) ZONE 1 - Rework of surface discontinuities up to 0.0005-inch deep is permissible without re-shot peening. Rework is limited to 20% of area of any one surface.

(b) ZONE 2 - Rework of surface discontinuities up to 0.005-inch deep is permissible without re-shot peening. Rework is limited to 20% of area of any one surface, and no greater than 10% within any 90° quadrant.

(c) If damage is within limits, using abrasive cloth, Item 12, Appendix D, blend damage, making sure not to exceed indicated depth.

(d) If repaired areas exceed stated limits, replace shaft extension.

(e) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect reworked area for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace cracked hub.

(f) Touch up finish on all reworked areas.

e. Check lower pressure plate attachment bolt threaded inserts in main rotor shaft extension as follows:

(1) Using previously removed bolts, thread bolts in until boltheads are 0.580-inch away from bottom of main rotor shaft extension.

(2) Using torque wrench, back off each bolt individually. **MINIMUM BREAKAWAY TORQUE IS 9.5 INCH-POUNDS. MAXIMUM REMOVAL TORQUE IS 80 INCH-POUNDS.**

(3) If torque exceeds limits, inspect condition of bolt for corrosion and wear. If corrosion or wear is found, replace bolt and clean threaded insert.

(4) Thread bolts in until boltheads are 0.580-inch away from bottom of main rotor shaft extension.

(5) Using torque wrench, back off each bolt individually. **MINIMUM BREAKAWAY TORQUE IS 9.5 INCH-POUNDS. MAXIMUM REMOVAL TORQUE IS 80 INCH-POUNDS.**

(6) If breakaway or removal torque is still not acceptable, remove bolts from main rotor shaft extension, and replace threaded insert, 79698-624 (PARA 5-4-3).

f. Inspect upper split cones (if removed) and lower split cones as follows (Figure 5-4-1-4):

NOTE

Split cones are a matched set and cannot be replaced individually.

(1) Inspect split cones for matching serial numbers. If serial numbers do not match, replace split cones.

(2) Inspect for nicks and burrs. **NICKS AND BURRS MUST NOT EXCEED 0.010-INCH DEEP.** If nicks and burrs exceed limits, replace split cones.

(3) Inspect for corrosion pitting. **PITTING MUST NOT EXCEED 0.010-INCH DEEP.** If pitting exceeds limit, replace split cones.

(4) Inspect for fretting. **FRETTING UP TO 0.005-INCH DEEP OVER 50% OF CONTACTING SURFACE IS ACCEPTABLE AFTER REMOVAL OF METAL PICKUP USING ABRASIVE WHEEL, ITEM 22, APPENDIX D.** If fretting exceeds limits, replace split cones.

(5) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect split cones for cracks or fractures (ASTM E1417). **NONE ALLOWED.** If crack or fracture is found, replace split cones.

(6) Place split cones in bag for reinstallation and tag.

g. Inspect/repair upper and lower pressure plates (PARA 5-4-2).

h. Inspect upper pressure plate (if removed) and lower pressure plate cone puller for corrosion. **CORROSION UP TO 0.020-INCH DEEP CAN BE BLENDED OUT.** If corrosion is deeper than 0.020-inch deep, replace cone puller.

i. Inspect lower pressure plate cone puller for looseness. If cone puller is loose, apply coat of sealing compound, Item 299, Appendix D, to OD and flange face of cone puller. Position cone puller to allow split cones to be installed with gaps orientated approximately 90° from attachment lugs.

j. Inspect main rotor shaft (PARA 6-4-7).

k. Inspect main rotor shaft threads for crossed, stripped, or flattened threads. **NONE ALLOWED.** If threads are damaged, replace main module (PARA 6-4-1).

l. Inspect upper pressure plate bolts (if removed), lower pressure plate bolts, rotating scissors

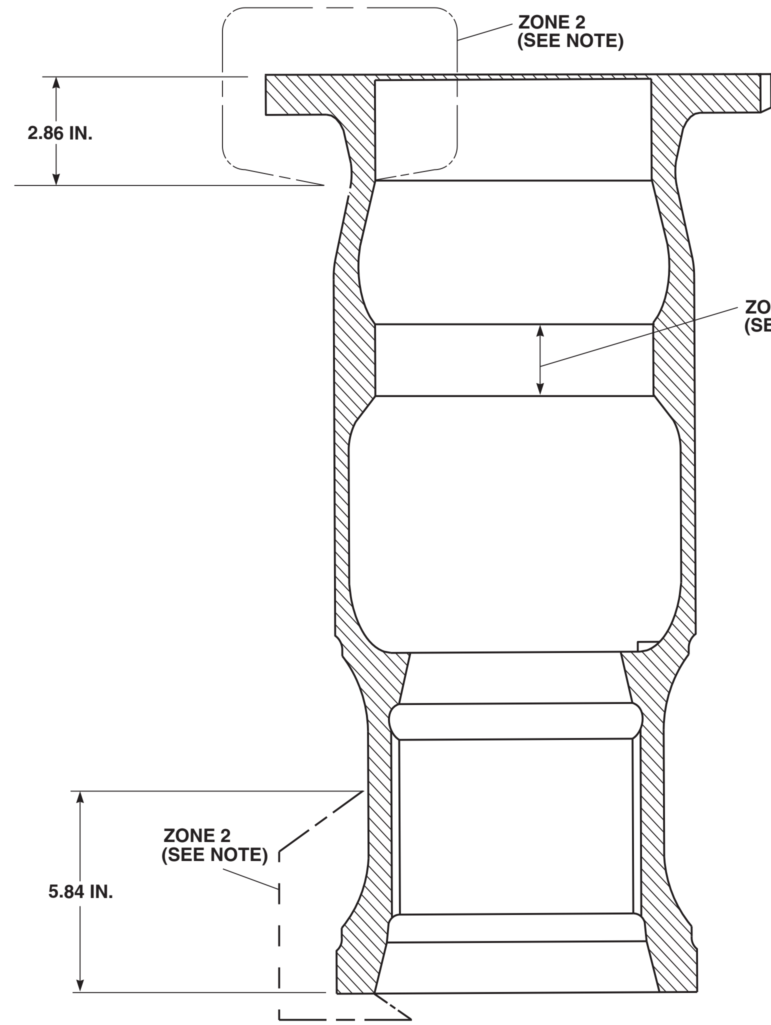


FIGURE 5-4-1-3. INSPECT/REPAIR MAIN ROTOR SHAFT EXTENSION

bolts, and lower pitch control rod bolts (PARA 5-4-4).

m. If upper pressure plate was removed, perform torque check of upper pressure plate threaded inserts as follows:

(1) Install upper pressure plate bolts with countersunk washers into hub until 0.700-inch clearance is obtained between bottom of hub and washer.

(2) Using torque wrench, back off each bolt separately. **MINIMUM BREAKAWAY TORQUE IS 18 INCH-POUNDS. MAXIMUM REMOVAL TORQUE IS 150 INCH-POUNDS.** If torque exceeds limits, replace threaded insert (PARA 5-4-4).

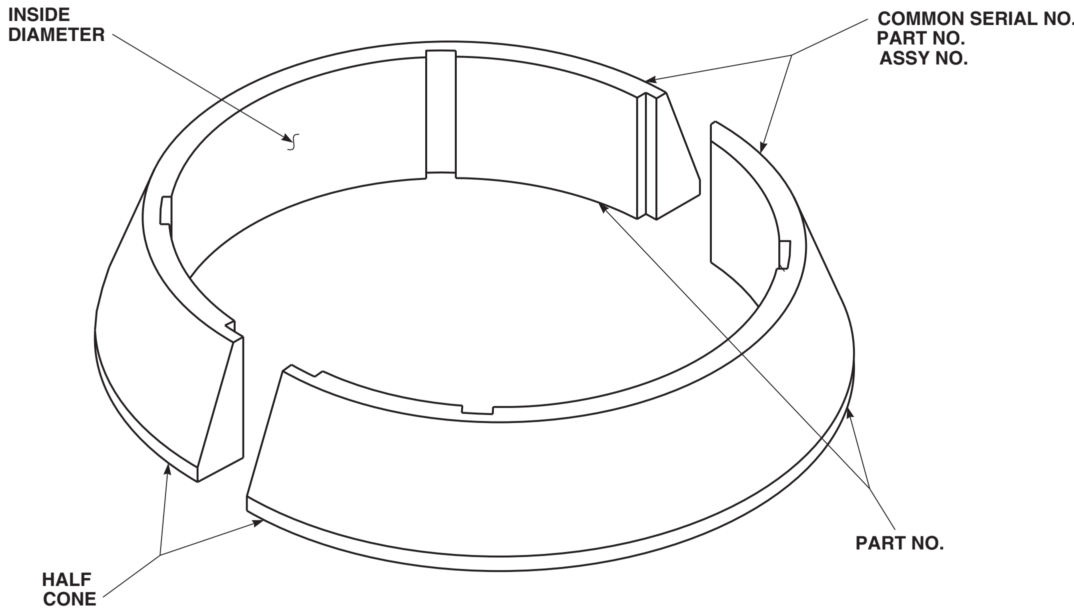


FIGURE 5-4-1-4. INSPECT/REPAIR UPPER/LOWER SPLIT CONES

5-4-1.1.4 Install Main Rotor Head and Shaft Extension.

a. If removed, install main rotor shaft extension (PARA 5-4-1.2.3).

CAUTION

Damage to main rotor head will result if hoisting sling is not properly positioned when hoisting main rotor hub. Hoisting sling must be put securely around main rotor hub arms.

b. Attach straps of hoisting sling around each main rotor hub arm (Figure 5-4-1-1, Sheet 2, Detail E).

c. Attach hoisting sling to hoist or cabin maintenance crane.

d. Attach guide rope to main rotor head.

e. Have assistant hold guide rope to help guide main rotor head.

f. Adjust straps before lifting for even distribution of weight.

g. Remove handknob securing main rotor head to rotary wing head adapter.

h. Using clean, dry machinery towel, Item 211, Appendix D, wipe main rotor shaft and rotor head mating surfaces clean.

i. Check main rotor shaft for thread protection. If none is present, tape cardboard or machinery towels, Item 211, Appendix D, around threads using pressure sensitive adhesive tape, Item 249, Appendix D.

j. Install lower pressure plate and let rest on swashplate guide.

k. If removed, install upper pressure plate over main rotor shaft and let rest on lower pressure plate.

l. Using grease, Item 144 or Item 150, Appendix D, lightly coat main rotor shaft splines.

NOTE

- Main rotor shaft nut, thrust washer, and bolts must be installed and torqued while epoxy primer is wet.
- Do not coat threads or spline on main rotor shaft. Do not coat entire surface.

m. Using epoxy primer coating kit, Item 135, Appendix D, apply light coat of epoxy primer to upper exposed portion (approximately 1/4-inch) in tapered cone area of main rotor shaft.

WARNING

- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.
- Injury to personnel and damage to equipment will result if caution is not taken when hoisting main rotor head. Keep area clear when hoisting main rotor head.

n. Hoist main rotor head to top of main rotor shaft.

o. Carefully lower main rotor head down main rotor shaft. Lower main rotor head so that No. 1 main rotor blade arm (red) is positioned approximately halfway between front of helicopter and left side of helicopter (approximately 45°).

WARNING

FSCAP

Exposed surfaces of main rotor shaft and main rotor shaft extension are critical surfaces which must not be damaged during installation of main rotor head. After removal of protective covering, use extreme caution when installing main rotor head.

p. Remove thread protection (or cardboard, or machinery towels) and protective covering from main rotor shaft.

q. Using antiseize compound, Item 74, Appendix D, lightly coat threads of main rotor shaft nut.

NOTE

- If one side of thrust washer was repaired per PARA 5-4-1.1.3, thrust washer must be installed with repaired side facing down.
- If both sides of thrust washer were repaired per PARA 5-4-1.1.3, thrust washer must be installed with side of deepest repair facing down.

r. Install thrust washer, making sure that it is properly seated and centered in main rotor shaft extension (Figure 5-4-1-1, Sheet 1, Detail A and Section A-A).

s. Install main rotor shaft nut. Make sure painted white stripes are facing up.

t. **TIGHTEN MAIN ROTOR SHAFT NUT HANDTIGHT, THEN BACK OFF NUT UP TO 1/4 TURN TO LINE UP NEAREST WHITE STRIPE ON NUT WITH MAIN ROTOR TANG.**

WARNING

FSCAP

Verification of proper hardware, lockring installation, and torque of bolts is a critical characteristic.

NOTE

Raised pads on lock must face up (Figure 5-4-1-1, Sheet 3, Detail H).

u. Place lock on top of main rotor shaft nut and line up notch in lock with tang of rotor shaft (Figure 5-4-1-1, Sheet 1, Detail A).

v. Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer or sealing compound to shank, threads, and area under boltheads securing main rotor shaft nut.

w. Install bolts in main rotor shaft nut while primer is still wet.

x. Torque bolts as follows (Figure 5-4-1-1, Sheet 1, Detail A and Sheet 2, Detail B):

(1) **TIGHTEN BOLTS UNTIL SLIGHT RESISTANCE IS FELT AS BOLTS CONTACT THRUST WASHER.**

(2) **TORQUE BOLTS TO 713 - 787 INCH-POUNDS IN 250 INCH-POUND INCREMENTS, FOLLOWING NUMBERED SEQUENCE ON MAIN ROTOR SHAFT NUT.**

(3) **STABILIZE TORQUE ON ALL BOLTS AT EACH 250 INCH-POUND INCREMENT BY TURNING EACH BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT.**

(4) **STABILIZE TORQUE AT 713 - 787 INCH-POUNDS BY TURNING EACH BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT.**

y. Make sure main rotor shaft tang is flush to above tang on lock.

z. Using machinery towel, Item 211, Appendix D, wipe off any primer that has seeped from under boltheads.

aa. Touch up paint on boltheads (PARA 2-6-5).

NOTE

Lockring must be installed over bolts on main rotor shaft nut before corrosion preventive compound dries.

ab. Apply corrosion preventive compound, Item 113, Appendix D, to main rotor shaft nut and to area inside main rotor shaft extension.

ac. Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer or sealing compound to shank, threads, and area under boltheads securing lower lockring.

CAUTION

Damage to lower lockring will result if it is not seated properly prior to torquing bolts. Lockring must sit flat on main rotor shaft nut. Turn bushings in lockring to allow lockring to seat over bolts (Figure 5-4-1-1, Sheet 3, Detail J).

ad. Install lockring over bolts on main rotor shaft nut. Make sure two shorter lockring spacers are resting on lock. Install bolts and washers while primer is still wet (Figure 5-4-1-1, Sheet 1, Detail A and Section A-A).

ae. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**

af. Using machinery towel, Item 211, Appendix D, wipe off any primer that has seeped out from under boltheads.

NOTE

Feeler gages may be substituted for aluminum shims.

ag. If required, locally make shims for upper pressure plate installation as follows:

(1) Using aluminum, make four shims measuring 0.032-inch × 1-inch × 4-inches.

(2) Using aluminum, make four shims measuring 0.060-inch × 1-inch × 4-inches.

ah. If upper pressure plate was removed or replaced, install as follows:

NOTE

- Upper split cones are a matched set and cannot be replaced individually.
- Serial number, part number, and assembly numbers may be found on either upper or lower surface of split cone half.

(1) Check serial number on upper split cones to make sure they are a matched set (Figure 5-4-1-4). If numbers do not match, replace split cones.

(2) Using machinery towel, Item 211, Appendix D, wipe inside diameter (ID) of upper split cones.

NOTE

Do not apply grease on outside tapered surface of upper split cones. Outside tapered surface of all split cones must be left dry.

(3) Using grease, Item 144 or Item 150, Appendix D, lightly coat ID of upper split cones and upper part of shaft extension where upper split cones are seated.

(4) Position upper split cones on upper pressure plate with equal gap between split cones (Figure 5-4-1-1, Sheet 4, Detail K).

(5) Lift upper pressure plate and upper split cones to seat cones in main rotor hub.

CAUTION

Damage to upper pressure plate and main rotor hub will result if wrong length attachment bolts are used. Make sure correct length bolts are used to install upper pressure plate.

(6) Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer or sealing compound to shank, threads, and area under boltheads securing upper pressure plate.

(7) While primer is still wet, install four bolts and countersunk washers in holes numbered 3, 4, 5, and 6 to hold upper pressure plate in place (Figure 5-4-1-1, Sheet 4, Detail K and Sheet 2, Detail D).

(8) Install remaining bolts and countersunk washers in pressure plate while primer is still wet (Figure 5-4-1-1, Sheet 4, Detail K).

NOTE

Shims are necessary to keep upper pressure plate level during installation. Drag on shims may or may not be reached prior to step ah. (11).

(9) Insert 0.092-inch thick shims in four places, equally spaced between upper pressure plate and bottom of main rotor hub.

(10) **TIGHTEN BOLTS IN SEQUENCE NUMBERED ON UPPER PRESSURE PLATE WHILE CHECKING FOR EQUAL DRAG ON SHIMS** (Figure 5-4-1-1, Sheet 2, Detail D). Shims should keep upper pressure plate and main rotor hub parallel.

(11) Remove 0.032-inch thick shims.

(12) **TIGHTEN BOLTS IN SEQUENCE WHILE CHECKING FOR EQUAL DRAG ON SHIMS.**

(13) Remove all shims.

(14) **TORQUE BOLTS TO 550 INCH-POUNDS, FOLLOWING NUMBERED SEQUENCE ON PRESSURE PLATE.**

(15) **TORQUE BOLTS TO 855 - 945 INCH-POUNDS IN 200 INCH-POUND INCREMENTS, FOLLOWING NUMBERED SEQUENCE ON UPPER**

PRESSURE PLATE. STABILIZE TORQUE AT 855 - 945 INCH-POUNDS BY TURNING EACH BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT (Figure 5-4-1-1, Sheet 4, Detail K and Sheet 2, Detail D).

(16) Measure gap between upper pressure plate and main rotor hub at four equally spaced places 90° from each other. **ALL FOUR GAPS MUST BE WITHIN 0.030-INCH OF EACH OTHER. IF NOT, REMOVE BOLTS AND COUNTERSUNK WASHERS AND REPEAT STEP AH. (6) THROUGH STEP AH. (16).**

(17) Check helicopter logbook to make sure upper pressure plate torque check is shown due 9 - 11 flight hours after installation (PARA 1-7-5).

ai. Install lower pressure plate as follows:

NOTE

- Lower split cones are a matched set and cannot be replaced individually.
- Serial number, part number, and assembly numbers may be found on either upper or lower surface of split cone half.

(1) Check serial number on lower split cones to make sure they are a matched set (Figure 5-4-1-4). If numbers do not match, replace split cones.

(2) Using machinery towel, Item 211, Appendix D, wipe split cones clean.

NOTE

Do not put grease on outside tapered surface of split cones. Outside tapered surface of all split cones must be left dry.

(3) Using grease, Item 144 or Item 150, Appendix D, lightly coat ID of lower split cones.

NOTE

Some lower pressure plates may have cone puller installed with cone puller ribs in line with attachment lugs on lower pressure plate. Cone pullers installed with ribs in line with attachment lugs will not allow lower split cones to be installed with gap orientated 90° from the lugs. In this case, split cones must be installed with gap located approximately 60° from attachment lugs.

(4) Install split cones on lower pressure plate so gap is located approximately 90° from attachment lugs (Figure 5-4-1-1, Sheet 4, Detail L).

(5) Line up ″O″ index marks on lower pressure plate and main rotor shaft extension (Figure 5-4-1-1, Sheet 4, Detail M).

(6) Torque bolts on lower pressure plate as follows:

(a) Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer or sealing compound to shank, threads, and area under boltheads securing lower pressure plate.

(b) Install bolts and countersunk washers in two holes while primer is still wet, 90° from split cone slots (Figure 5-4-1-1, Sheet 4, Detail L).

(c) **TIGHTEN BOLTS FINGER-TIGHT.**

(d) Install bolts and countersunk washers in third hole on each side of first two bolts.

(e) **TIGHTEN BOLTS FINGER-TIGHT.**

(f) **TIGHTEN THESE SIX BOLTS IN REVERSE SEQUENCE AS MARKED ON LOWER PRES-**

SURE PLATE TO 150 INCH-POUNDS. MAKE SURE TORQUE HAS STABILIZED.

NOTE

Number sequence on lower pressure plate may have to be shifted so that largest gap can be torqued first. For example, if largest gap is at bolthole No. 17, torque sequence is: 17, 18, 1, 2, etc.

(g) Using feeler gage, measure gap between lower pressure plate and main rotor shaft extension at each of six bolts (Figure 5-4-1-1, Sheet 4, Detail N). **GAPS MUST MEASURE WITHIN 0.025-INCH OF EACH OTHER.** If gaps are more than 0.025-inch of each other, loosen bolts and repeat step ai. (6) (e) and step ai. (6) (f).

(h) **STARTING AT BOLT WHERE GAP IS LARGEST, TORQUE THESE SIX BOLTS TO 285 - 315 INCH-POUNDS.**

(i) Using feeler gage, measure gap between lower pressure plate and main rotor shaft extension at each of six bolts.

(j) Install remaining bolts and countersunk washers. **STARTING AT BOLT WHERE GAP IS LARGEST, TORQUE ALL 18 BOLTS, IN SEQUENCE, TO 285 - 315 INCH-POUNDS IN 150 INCH-POUND INCREMENTS. STABILIZE TORQUE AT EACH 150 INCH-POUND INCREMENT** (Figure 5-4-1-1, Sheet 2, Detail C).

(k) Using feeler gage, measure gap between lower pressure plate and main rotor shaft extension at each of six bolts (Figure 5-4-1-1, Sheet 4, Detail N).

(l) **STARTING AT BOLT WHERE GAP IS LARGEST, TORQUE ALL 18 BOLTS, IN SEQUENCE, TO 385 - 425 INCH-POUNDS. MAKE SURE TORQUE HAS STABILIZED AT FINAL TORQUE** (Figure 5-4-1-1, Sheet 2, Detail C).

(m) Recheck torque on all 18 bolts. **TORQUE SHOULD BE EQUAL ON ALL BOLTS.**

(n) Using feeler gage, measure gap between lower pressure plate and main rotor shaft extension. **SMALLEST GAP ALLOWED IS 0.005-INCH. A MAXIMUM GAP DIFFERENCE OF 0.035-INCH IS ALLOWED BETWEEN SMALLEST AND LARGEST GAP AREAS.**

(o) If above gaps are not met, lower pressure plate and repeat procedure.

(p) Using machinery towel, Item 211, Appendix D, wipe off any primer that has seeped from under boltheads.

(q) Touch up paint on boltheads (PARA 2-6-5).

(r) Using corrosion preventive compound, Item 113, Appendix D, apply two coats between lower pressure plate and main rotor shaft extension and around locking ring boltheads.

(s) Check helicopter logbook to make sure lower pressure plate torque check is shown due 9 - 11 flight hours after installation (PARA 1-7-5).

aj. If not already done, disengage gust lock (PARA 1-6-18).

ak. Turn swashplate until red index lines on rotating and stationary swashplate line up (Figure 5-4-1-1, Sheet 5, Detail P).

al. Turn main rotor hub until rotating scissors upper link lines up with lugs on lower pressure plate (Figure 5-4-1-1, Sheet 3, Detail G).

am. Turn main rotor hub and swashplate together until No. 1 main rotor blade arm (red) is positioned approximately halfway between front of helicopter and left side of helicopter (approximately 45°) (Figure 5-4-1-1, Sheet 2, Detail E).

an. Engage gust lock (PARA 1-6-18).

ao. Make sure red index lines are lined up, scissors are lined up with lugs on lower pressure plate, and No. 1 main rotor blade arm is correctly positioned (Figure 5-4-1-1, Sheet 2, Detail E, Sheet 3, Detail G, and Sheet 5, Detail P).

ap. If previously removed rotating scissors will be installed, and lower pressure plate was not replaced, perform the following (Figure 5-4-1-1, Sheet 3, Detail G):

(1) Loosely connect rotating scissors upper links to lower pressure plate with bolts and previously installed shims.

(2) Using feeler gage, measure gaps (four locations) between lower pressure plate lugs and rotating scissors upper link (Figure 5-4-1-1, Sheet 5, Detail Q). **OVERALL GAP ON EACH SIDE MUST BE 0.002 - 0.004-INCH.** If gap is within limits, remove bolts and shims, and proceed to step ar. If gap is not within limits, perform the following:

(a) Remove bolts and shims.

(b) Inspect upper link bearing flange height (PARA 5-5-8).

(c) Repeat step ap. (1) and step ap. (2).

aq. If new rotating scissors will be installed, or if lower pressure plate has been replaced, perform the following:

(1) Loosely connect rotating scissors upper links to lower pressure plate with bolts (Figure 5-4-1-1, Sheet 3, Detail G). Do not install nuts at this time.

(2) Using feeler gage, measure gaps (four locations) between lower pressure plate lugs and rotating scissors upper link (Figure 5-4-1-1, Sheet 5, Detail Q). **OVERALL GAP ON EACH SIDE MUST BE 0.002 - 0.004-INCH.** If gap is within limits, remove bolts, and proceed to step ar. If gap is not within limits, perform the following:

(a) Remove bolts.

(b) Peel laminated shims equally on both sides of upper link to allow for 0.002 - 0.004-inch gap on each side of lugs.

(c) Repeat step aq. (1) and step aq. (2) until 0.002 - 0.004-inch gap on each side of lugs is obtained.

ar. Connect upper links to lower pressure plate as follows (Figure 5-4-1-1, Sheet 3, Detail G):

WARNING

FSCAP

Verification of proper hardware installation, torque, and safety installation are critical characteristics.

NOTE

Install bolts with boltheads in direction of rotation (counterclockwise).

(1) Position rotating scissors upper links on lower pressure plate lugs.

(2) Secure links to lower pressure plate with bolts, countersunk washers, shims, washers, and nuts. Make sure countersunk side of washers are toward boltheads.

(3) Using solid film lubricant, Item 201, Appendix D, lubricate bolt threads.

(4) **TORQUE NUTS TO 1000 - 2000 INCH-POUNDS.** Install cotter pins.

as. Disengage gust lock (PARA 1-6-18).

at. Connect pitch control rods to swashplate (PARA 5-4-18).

au. Install main rotor blades (PARA 5-4-31).

NOTE

If installed, main rotor blade deice kit may be left off until main rotor head torque check stabilizes. Bifilar cover must be installed.

av. If blade deice kit is installed, install main rotor blade deice distributor and slipring/brush assembly (PARA 12-4-26).

aw. If blade deice kit is not installed, install bifilar cover (PARA 5-4-30).

ax. Perform main rotor system rig check (PARA 11-4-4).

ay. Make sure area is clean and free of foreign material.

az. Perform main rotor blade track and balance (TM 1-70-VIB).

ba. Check helicopter logbook to make sure main rotor head torque check is shown due 9 - 11 flight hours after installation (PARA 1-7-5).

bb. Perform maintenance test flight (Appropriate MTF).

5-4-1.2 Replace Main Rotor Shaft Extension.

5-4-1.2.1 Remove.

a. Remove main rotor head (PARA 5-4-1.1.1).

b. Using grease pencil, place index marks on all main rotor head components, to make sure parts are positioned correctly during installation.

CAUTION

Damage to main rotor head will result if metallic items are used to clean off sealing compound. Do not use metallic items to remove sealing compound.

c. Using nonmetallic scraper, remove sealing compound from bifilar and main rotor shaft extension.

d. Install main rotor hub guide assembly on main rotor shaft extension below main rotor hub, and close latch (Figure 5-4-1-5, Sheet 1, Detail A).

e. Attach apex ring of hoisting sling to hoist.

CAUTION

Damage to main rotor head will result if hoisting sling is not properly positioned when hoisting main rotor hub. Hoisting sling must be put securely around main rotor hub arms.

NOTE

Straps must be adjusted to full length or main rotor hub will not completely lower and rest on rotary wing head adapter.

f. Attach straps of hoisting sling around each main rotor hub arm.

g. Remove handknob securing main rotor head to rotary wing head adapter.

h. Carefully raise hoist and remove any slack from hoisting sling until hoisting sling is supporting weight of main rotor head.

i. Remove bolts and washers securing upper lockring to bifilar. Remove upper lockring.

j. If necessary, using 5⁄16-inch Allen wrench, hold expandable pins from turning (Figure 5-4-1-5, Sheet 2, Section A-A and Detail B). Loosen four expandable pins by loosening nuts until top of nuts are flush with top of studs; then, using plastic or leather mallet, lightly tap. Use special wrench to loosen nuts.

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when removing hub-to-shaft extension bolts. After hub-to-shaft extension bolts are removed, entire main rotor hub and main rotor blades are supported only by straps and hoist. Always keep area underneath main rotor hub clear while lowering.

NOTE

Note location of washers and spacers for reinstallation on hub-to-shaft extension bolts.

k. Remove 12 hub-to-shaft extension bolts, washers, and spacers.

CAUTION

Damage to main rotor hub will result if not protected when being placed on rotary wing head adapter. Place protective padding on adapter prior to lowering main rotor hub.

l. Place protective padding on top of rotary wing head adapter. Slowly lower main rotor hub until it rests on top of rotary wing head adapter.

m. Lower hoisting sling to allow slack in straps, and remove hoisting sling from main rotor hub.

n. Attach straps of hoisting sling around each bifilar arm (Figure 5-4-1-5, Sheet 3, Detail C).

o. Shorten straps as much as possible after attaching to bifilar.

p. Attach guide rope to bifilar.

q. Have an assistant hold guide rope to help guide bifilar and main rotor shaft extension.

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting main rotor shaft extension and bifilar. Keep area clear when hoisting main rotor shaft extension and bifilar.

NOTE

Main rotor hub guide assembly will come up with bifilar and main rotor shaft extension.

r. Lift and remove bifilar and main rotor shaft extension from main rotor hub with hoist.

WARNING

FSCAP

Exposed surfaces of main rotor shaft extension are critical surfaces which must not be damaged during removal. Use extreme caution when removing, and provide protective covering after removal.

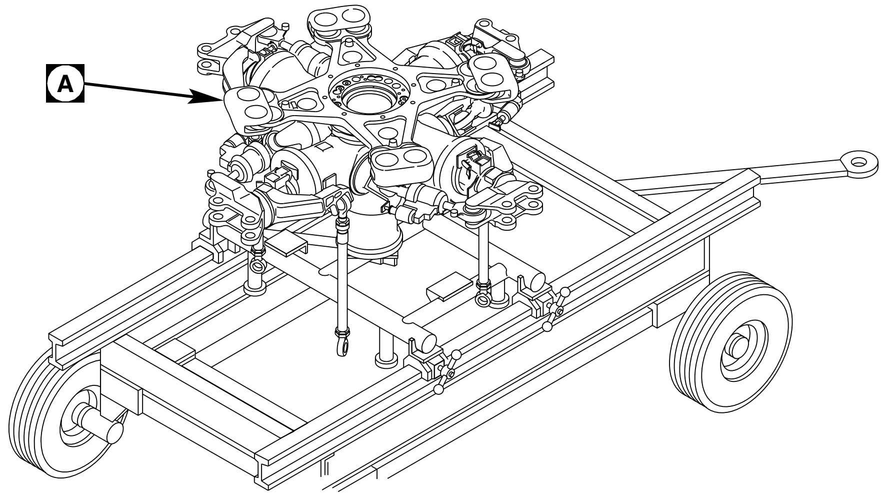
s. Install protective covering to exposed surfaces of main rotor shaft extension.

t. Remove bifilar from main rotor shaft extension (PARA 5-4-28).

5-4-1.2.2 Inspect.

a. Inspect hub-to-shaft extension bolts (PARA 5-4-4).

b. Visually inspect for voids in bonded liners and sleeves on main rotor shaft extension. **EDGE VOIDS UP TO 25% OF BONDED LINER OR SLEEVE WIDTH ARE ACCEPTABLE. VOIDS UP TO 5% OF TOTAL BONDED LINER OR SLEEVE AREA ARE ACCEPTABLE.** If voids exceed limits, replace main rotor shaft extension.



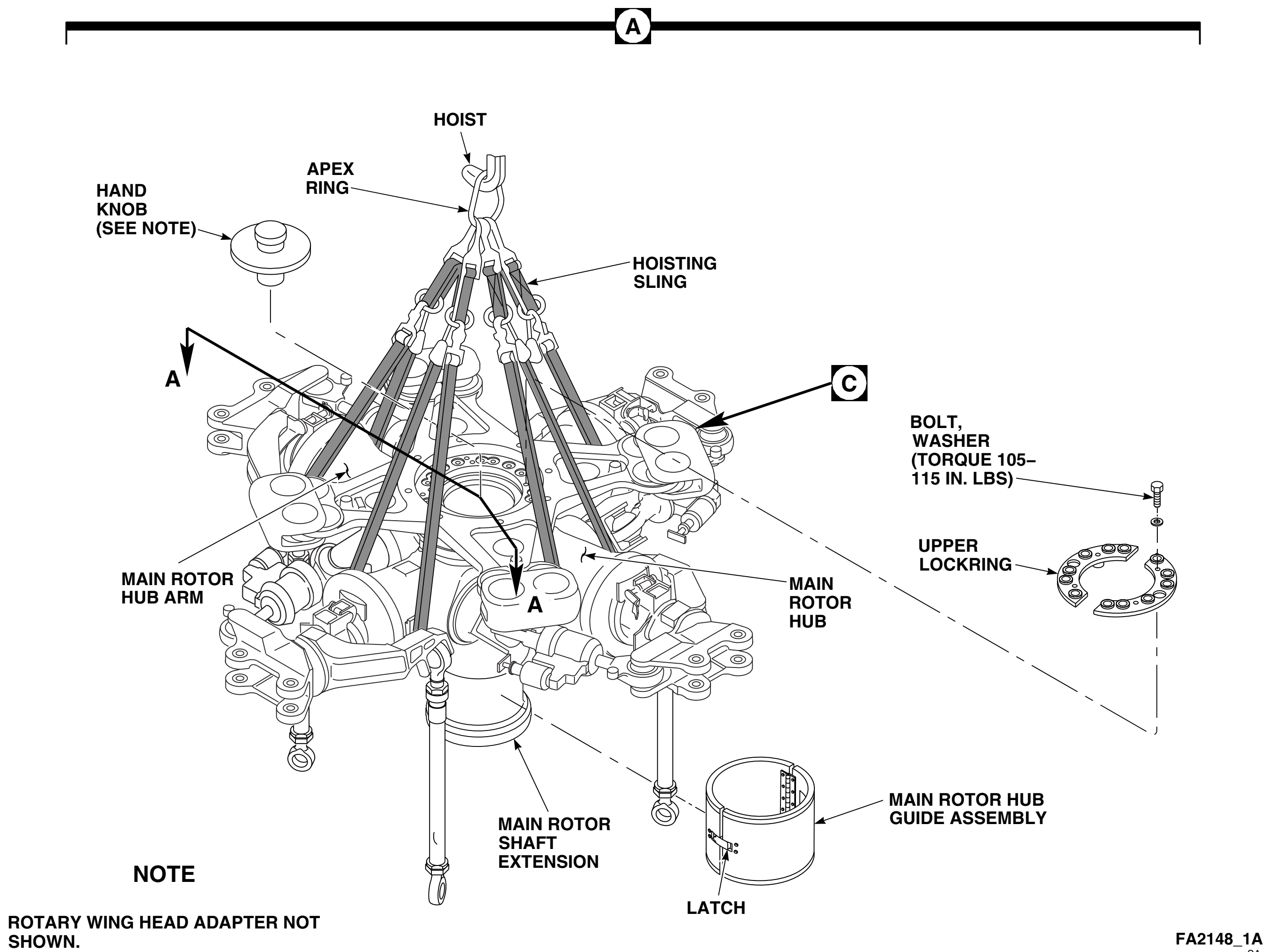
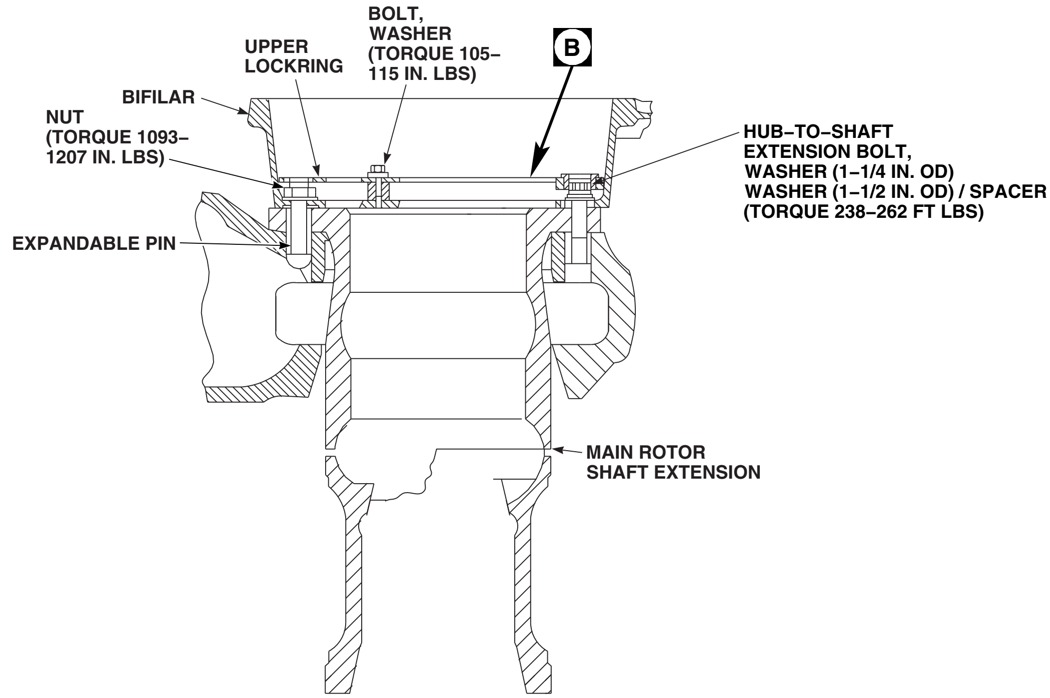


FIGURE 5-4-1-5. REMOVE MAIN ROTOR SHAFT EXTENSION (SHEET 1 OF 3)



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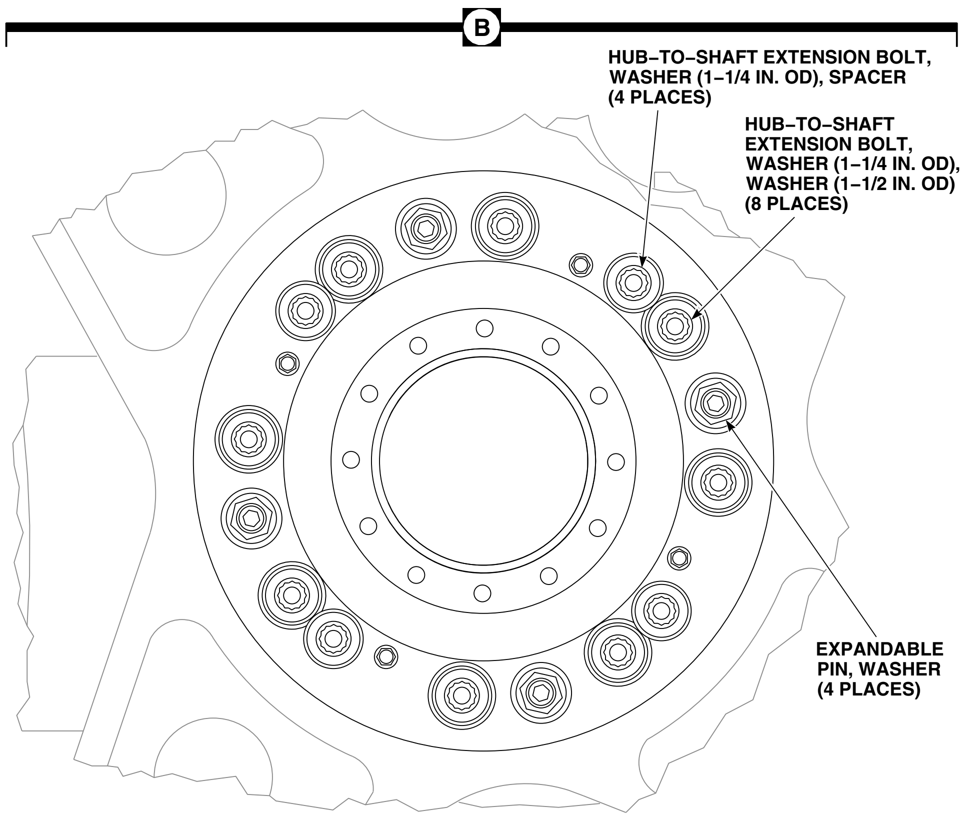

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FIGURE 5-4-1-5. REMOVE MAIN ROTOR SHAFT EXTENSION (SHEET 2 OF 3)

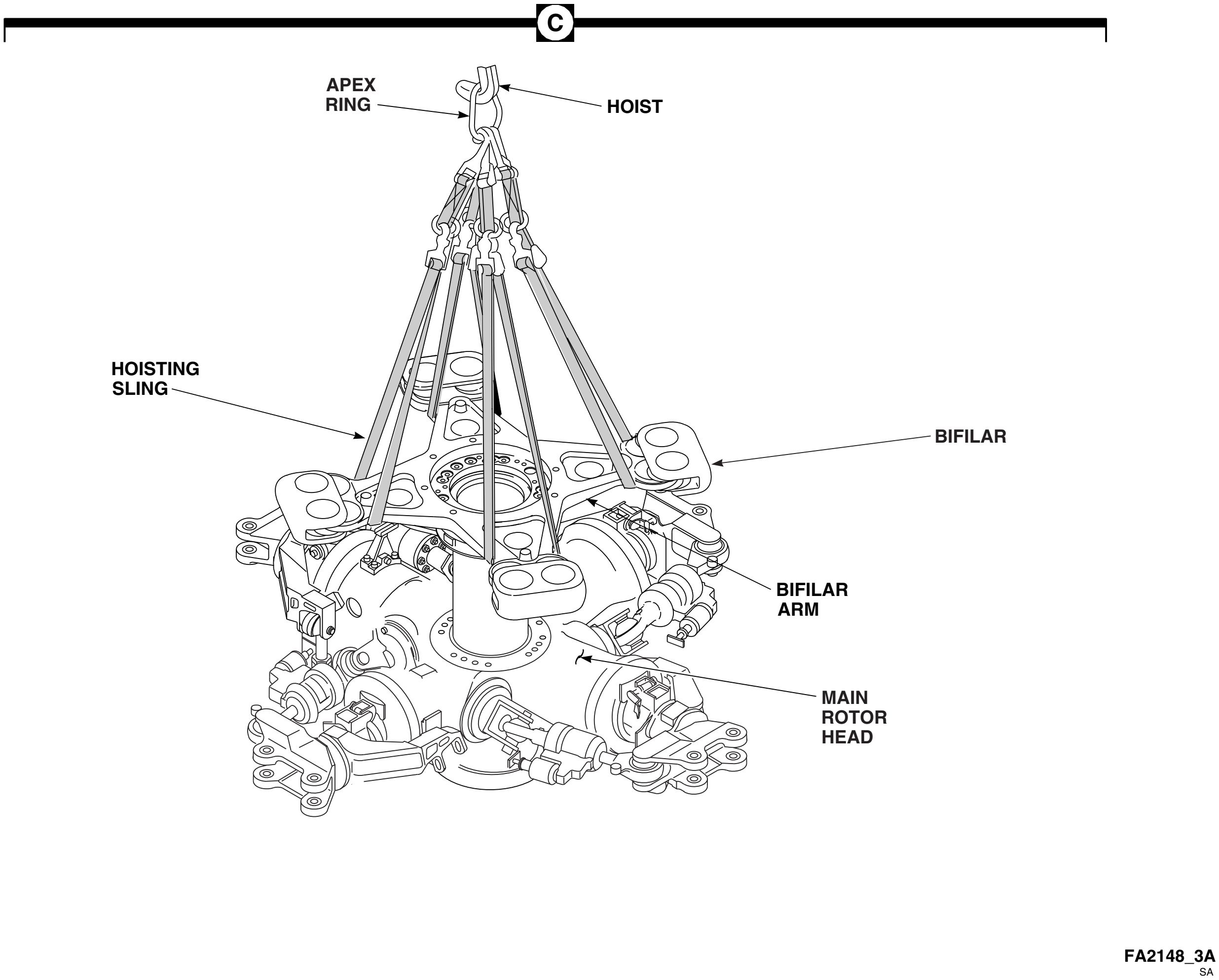


FIGURE 5-4-1-5. REMOVE MAIN ROTOR SHAFT EXTENSION (SHEET 3 OF 3)

WARNING

FSCAP

Main rotor shaft extension surfaces are critical surfaces which must not be damaged during installation. After removal of protective covering, use extreme caution when installing main rotor shaft extension.

c. Inspect shaft extension external surfaces for nicks, dents, scratches, and scuff marks (Figure 5-4-1-3). If damage is found, perform the following:

NOTE

- Areas shown in Figure 5-4-1-3 that are not labeled ZONE 2 are considered ZONE 1.
- Minimum length for any blended area must be 0.25-inch. For ZONE 1, this corresponds to a 500:1 blend ratio. For ZONE 2, this corresponds to a 50:1 blend ratio.

(1) ZONE 1 - Rework of surface discontinuities up to 0.0005-inch deep is permissible without re-shot peening. Rework is limited to 20% of area of any one surface.

(2) ZONE 2 - Rework of surface discontinuities up to 0.005-inch deep is permissible without re-shot peening. Rework is limited to 20% of area of any one surface, and no greater than 10% within any 90° quadrant.

(3) If damage is within limits, using abrasive cloth, Item 12, Appendix D, blend damage, making sure not to exceed indicated depth.

(4) If repaired areas exceed stated limits, replace shaft extension.

(5) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect reworked area for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace cracked hub.

(6) Touch up finish on all reworked areas.

5-4-1.2.3 Install.

a. Install bifilar on main rotor shaft extension (PARA 5-4-28).

WARNING

FSCAP

Exposed surfaces of main rotor shaft extension are critical surfaces which must not be damaged during installation. Use protective covering as required.

b. Attach apex ring of hoisting sling to hoist.

c. Attach straps of hoisting sling around each bifilar arm (Figure 5-4-1-5, Sheet 3, Detail C).

d. Shorten straps as much as possible after attaching to bifilar.

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting main rotor shaft extension and bifilar. Keep area clear when hoisting main rotor shaft extension and bifilar.

NOTE

Make sure main rotor hub guide assembly is installed on main rotor shaft extension (Figure 5-4-1-5, Sheet 1, Detail A).

e. Lift main rotor shaft extension and bifilar, and position above main rotor hub (Figure 5-4-1-5, Sheet 3, Detail C). Do not lower into hub yet.

NOTE

To keep ″O″ index marks lined up as accurately as possible, prevent shaft extension from turning during lowering.

f. Line up index mark on bifilar arm with No. 1 main rotor blade arm (red). Slowly lower main rotor shaft extension and bifilar into main rotor hub.

g. Lower hoisting sling to allow slack in straps, and remove hoisting sling from bifilar arms.

h. Attach straps of hoisting sling around each main rotor hub arm (Figure 5-4-1-5, Sheet 1, Detail A).

WARNING

- Always keep area under main rotor head clear while raising main rotor head, until hub-to-shaft extension bolts and expandable bolts are installed.
- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

i. Slowly lift main rotor head.

j. Verify hub-to-shaft extension bolts line up with holes in main rotor head. Stop hoisting when head is firmly against upper flange of main rotor shaft extension (Figure 5-4-1-5, Sheet 2, Section A-A).

NOTE

When installing expandable pins, hold each expandable pin from turning initially using 5⁄16-inch Allen wrench. Do not use Allen socket wrench when applying final torque requirement.

k. Install four expandable pins. Make sure expandable pins are fully seated (Figure 5-4-1-5, Sheet 2, Section A-A and Detail B).

l. **USING SPECIAL WRENCH, TORQUE NUTS ON EXPANDABLE PINS TO 238 - 262 INCH-POUNDS.**

m. Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer or sealing compound to shank, threads, and area under hub-to-shaft extension boltheads.

n. Install 12 hub-to-shaft extension bolts, washers, and four spacers while primer is still wet. Use 1 1⁄4-inch OD washers under all 12 bolts, and 1 1⁄2-inch OD washers under 8 bolts (bolts without spacers) (Figure 5-4-1-5, Sheet 2, Detail B).

o. **TORQUE HUB-TO-SHAFT EXTENSION BOLTS TO 950 - 1050 INCH-POUNDS. STABILIZE TORQUE AT 950 - 1050 INCH-POUNDS BY TURNING EACH BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT.**

p. Release torque on all four expandable pins.

q. **TORQUE 12 HUB-TO-SHAFT EXTENSION BOLTS TO 238 - 262 FOOT-POUNDS, IN SEQUENCE. STABILIZE TORQUE AT 238 - 262 FOOT-POUNDS BY TURNING EACH BOLT IN A TIGHTENING DIRECTION UNTIL NO MOTION IS FELT** (Figure 5-4-1-5, Sheet 2, Section A-A).

r. **USING SPECIAL WRENCH, TORQUE NUTS ON FOUR EXPANDABLE PINS TO 1093 - 1207 INCH-POUNDS.**

s. Remove hoisting sling from main rotor hub arms (Figure 5-4-1-5, Sheet 1 Detail A).

t. If used, disassemble and remove cabin maintenance crane (PARA 18-4-1).

u. Unlatch and remove main rotor hub guide assembly from main rotor shaft extension.

v. Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer or sealing compound to shank, threads, and area under upper lockring boltheads.

WARNING

FSCAP

Verification of proper installation and torque of bolt, and presence and proper installation of upper lockring is a critical characteristic.

w. Make sure upper lockring is properly seated over boltheads. Install upper lockring over hub-to-shaft extension bolts, and secure with bolts and washers while primer is still wet.

CAUTION

Damage to upper lockring will result if it is not seated properly prior to torquing bolts. Make sure upper lockring is properly seated over boltheads. If necessary, turn bushings to allow upper lockring to seat over boltheads. Upper lockring must sit flat before torquing bolts.

x. **TORQUE BOLTS SECURING UPPER LOCKRING TO 105 - 115 INCH-POUNDS.**

y. Secure main rotor head to rotary wing head adapter with handknob.

z. Install main rotor head and shaft extension (PARA 5-4-1.1.4).

5-4-2 MAIN ROTOR HUB.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Chisel, Plastic or Beryllium Copper, Not to Exceed Rockwell Hardness C25
- Fluorescent Penetrant Inspection Kit
- Nonmetallic Scraper, Locally-Made
- Pliers
- Powertrain Repairer's Toolkit
- Pressure Plate
- Syringe

Materials/Parts

- Abrasive Cloth, Item 11, Appendix D
- Abrasive Mat, Item 14, Appendix D
- Abrasive Paper, Item 20, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 32, Appendix D
- Adhesive Primer, Item 68, Appendix D
- Brush, Paint, Item 86, Appendix D
- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Liner, 70103-08002-106, Procured
- Liner, 70103-08002-115, Procured
- Liner, 70103-08002-116, Procured or Locally-Made, Figure H-265, Appendix H
- Liner, 70103-08002-120, Procured or Locally-Made, Figure H-263, Appendix H
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 230, Appendix D
- Parting Compound, Item 233, Appendix D
- Tape, Duct, Item 318A, Appendix D

References

- Appendix D
- Appendix H

- ASTM E1417
- PARA 2-6-5

5-4-2.1 Inspect/Repair Main Rotor Hub.

NOTE

- In the following steps, damage should be blended out using abrasive cloth, Item 11, Appendix D. Blend out all reworked areas to ½-inch radius.
- Record all repairs on component log card.

a. Check helicopter logbook to make sure area has not been previously reworked.

WARNING

FSCAP

Inspection holes at zone I of inspection area (Figure 5-4-2-1, Sheet 1, Detail C) are a critical characteristic area of main rotor hub.

b. Inspect zone I for nicks and gouges (Figure 5-4-2-1, Sheet 1, Detail C and Section A-A):

(1) Inspect inspection holes for damage penetrating through paint into base metal. **MAXIMUM ALLOWABLE BASE METAL DAMAGE IS 0.005-INCH.** If damage penetrating through paint into base metal exceeds limit, replace main rotor hub.

(2) Visually inspect zone I for cracks. **NONE ALLOWED.** If crack is found, replace main rotor hub.

(3) If cracks in zone I are suspected, using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, area for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace main rotor hub.

c. Inspect zone II for nicks and gouges:

(1) **DAMAGE UP TO 0.010-INCH DEEP CAN BE BLENDED OUT. DAMAGED AREA MUST BE LESS THAN 15% OF AREA. REPLACE REPAIRED MAIN ROTOR HUB WITHIN 500 FLIGHT HOURS OF REPAIR.**

(2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, immediately replace main rotor hub.

d. Inspect zone III for nicks and gouges:

(1) **DAMAGE UP TO 0.020-INCH DEEP, ½-INCH WIDE AND LONG CAN BE BLENDED OUT. DAMAGE MUST BE LESS THAN 15% OF CIRCUMFERENCE OF PART. REPLACE REPAIRED MAIN ROTOR HUB WITHIN 2000 FLIGHT HOURS OF REPAIR.**

(2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace main rotor hub.

e. Inspect zone IV for nicks and gouges:

(1) **CORNER DAMAGE UP TO 0.010-INCH DEEP CAN BE BLENDED OUT, EXCEPT FOR LUG AREAS. IN LUG AREAS, DAMAGE UP TO 0.050-INCH DEEP IS ALLOWED.**

(2) **ON OPEN AREAS (NONMATING SURFACES), SURFACE DAMAGE UP TO 0.040-INCH DEEP CAN BE BLENDED OUT. DAMAGED AREA MUST BE LESS THAN 20% OF CIRCUMFERENCE.**

(3) **ON MATING AREAS, SURFACE DAMAGE UP TO 0.020-INCH DEEP CAN BE BLENDED OUT. DAMAGED AREA MUST BE LESS THAN 10% OF SURFACE AREA AFTER BLENDING OUT AND REMOVING ANY METAL PICKUP.**

(4) **SURFACE DAMAGE ON BOTTOM OF MAIN ROTOR HUB DUE TO JACKING BOLTS BEARING ON MAIN ROTOR HUB SURFACE IS ACCEPTABLE.**

(5) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace main rotor hub.

f. Touch up finish on all reworked areas (PARA 2-6-5).

NOTE

Fretting on bonded liners is considered normal wear and is acceptable.

g. Inspect cone seat surfaces on shaft extension and zone IV for fretting. **FRETTING UP TO 0.002-INCH DEEP OVER ENTIRE CONTACTING SURFACE IS ACCEPTABLE, AS LONG AS THERE IS NO METAL PICKUP THAT CANNOT BE REMOVED USING ABRASIVE WHEEL, ITEM 22, APPENDIX D.**

h. Visually inspect bonded liners for edge voids as follows (Figure 5-4-2-1, Sheet 1, Detail A, Detail B, and Detail D, and Sheet 2, Detail E):

(1) **EDGE VOIDS UP TO 25% OF BONDED LINER WIDTH ARE ACCEPTABLE.**

(2) **FOR BONDED LINERS, 70103-08002-115, 70103-08002-116, AND 70103-08002-106, EDGE VOIDS UP TO 25% OF BONDED LINER AREA ARE ACCEPTABLE.**

(3) **FOR ALL OTHER BONDED LINERS, EDGE VOIDS UP TO 5% OF BONDED LINER AREA ARE ACCEPTABLE.**

(4) For bonded liners, 70103-08002-115 and 70103-08002-106, perform the following:

(a) If above void limits are exceeded, but are less than 50% of bonded liner width/area, repair bonded liner (PARA 5-4-2.2).

(b) If voids are greater than 50% of bonded liner width/area, replace bonded liner (PARA 5-4-2.3).

(5) For bonded liner, 70103-08002-116, if voids are greater than 25% of bonded liner width/area, replace bonded liner (PARA 5-4-2.3).

(6) For all other bonded liners, perform the following:

(a) If above void limits are exceeded, repair bonded liner (PARA 5-4-2.2).

(b) If liner is totally disbonded, replace main rotor hub.

i. Record time, type, and location of rework in helicopter logbook.

NOTE

- Fluorescent penetrant inspection of 11 o'clock damper bracket attachment holes is applicable for main rotor hub, 70103-08112-041, only.
- 11 o'clock damper attachment hole inspection should be performed every 200 flight hours.

j. Inspect 11 o'clock damper attachment holes as follows (Figure 5-4-2-1, Sheet 1, Detail D):

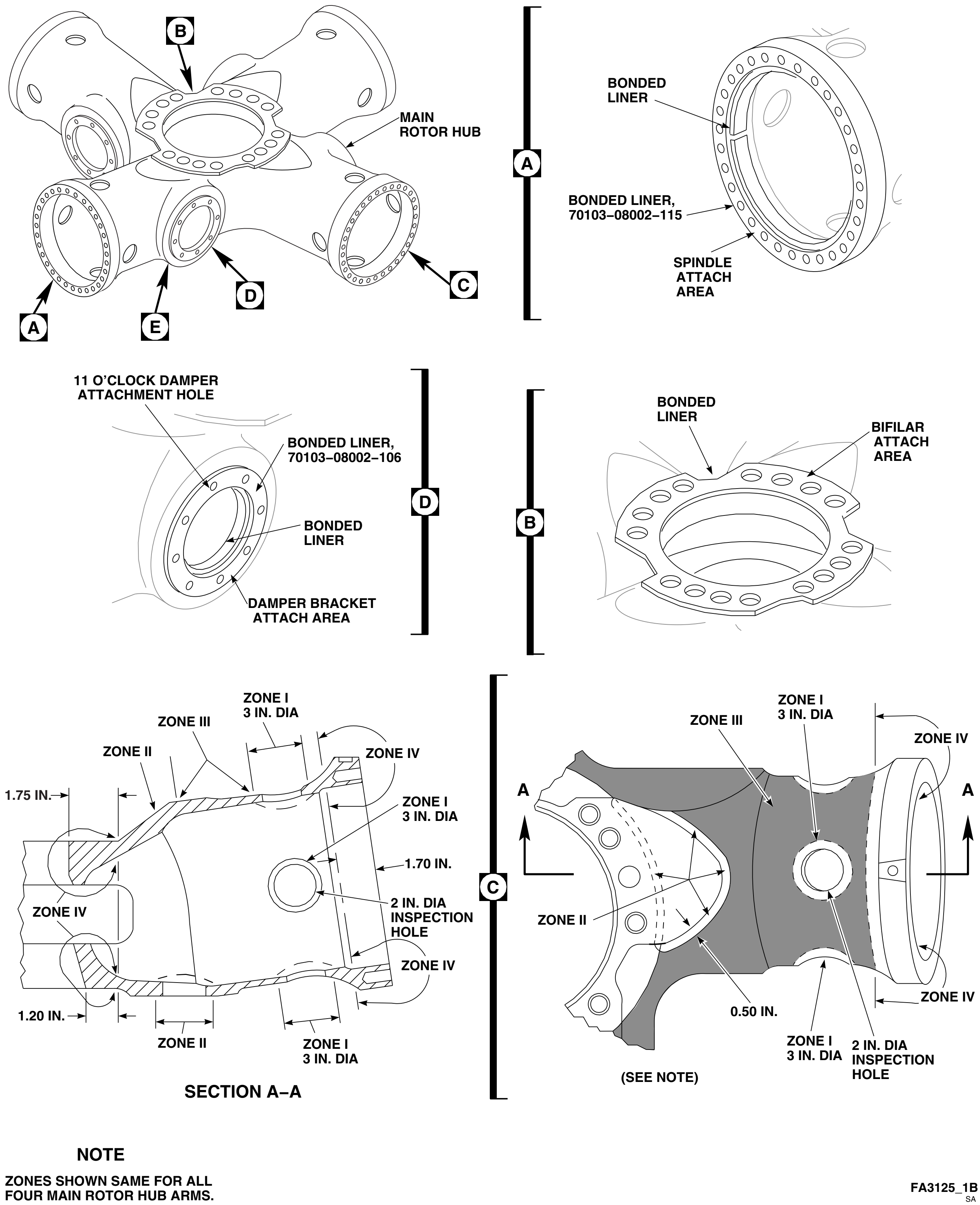


FIGURE 5-4-2-1. INSPECT/REPAIR MAIN ROTOR HUB (SHEET 1 OF 2)

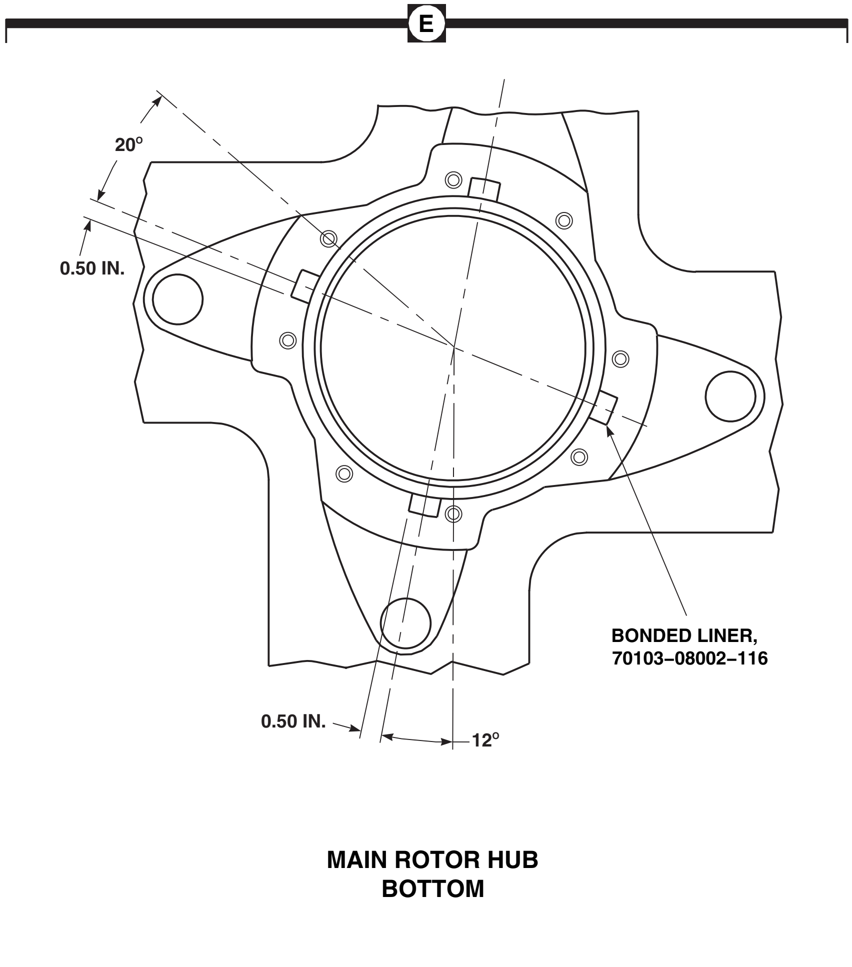


FIGURE 5-4-2-1. INSPECT/REPAIR MAIN ROTOR HUB (SHEET 2 OF 2)

NOTE

Main rotor hub must be cleaned with paint remover even if paint is not present.

(1) Clean a ½-inch area around inspection area on inside surface of 11 o'clock damper attachment holes as follows:

CAUTION

Damage to elastomeric bearing assembly will result if paint remover is allowed to seep into spherical and thrust bearings. Do not allow paint remover to come in contact with elastomeric bearings.

(a) Using duct tape, Item 318A, Appendix D, mask off areas adjacent to area being cleaned.

(b) Using machinery towels, Item 211, Appendix D, position around inboard end of spindle and elastomeric bearing and underneath area to be stripped to catch any paint remover during cleaning.

NOTE

If paint is present, allow paint remover to remain on surface until paint bubbles.

(c) Using paint brush, Item 86, Appendix D, apply paint remover, Item 230, Appendix D. Using machinery towel, Item 211, Appendix D, or nonmetallic scraper, remove paint remover.

(d) Using machinery towel, Item 211, Appendix D, saturated with water, thoroughly remove all remains of paint remover. Repeat if necessary.

(e) Using acetone, Item 25, Appendix D, clean area. Repeat.

(2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, Level 4 Sensitivity, 11 o'clock damper attachment holes (ASTM E1417). **NONE ALLOWED.**

(3) Record time of inspection in helicopter logbook.

5-4-2.2 Repair Bonded Liner.

a. Using acetone, Item 25, Appendix D (or equivalent), clean void areas. Using syringe, inject acetone into voids.

b. If possible, using abrasive paper, Item 20, Appendix D, sand voided surfaces. Using acetone, Item 25, Appendix D (or equivalent), clean again, and allow to dry.

c. Prepare adhesive, Item 32, Appendix D, per manufacturer's directions.

d. Using syringe, inject mixed adhesive into void areas.

e. Using parting compound, Item 233, Appendix D, apply thin coating to pressure plate (suitable, flat plate), on side to mate with bonded liner.

f. Position pressure plate on rebonded area and secure by bolting over entire bonded liner area.

g. Allow to cure for 24 hours at room temperature.

5-4-2.3 Replace Bonded Liners, 70103-08002-115 (70103-08002-120), 70103-08002-116, and 70103-08002-106.

a. Using pliers, peel bonded liner from main rotor hub.

CAUTION

Damage to main rotor hub will result if plastic or beryllium copper chisel harder than Rockwell hardness C25 is used. Do not use plastic or beryllium copper chisel exceeding Rockwell hardness C25.

b. Using cheesecloth, Item 90, dampened with acetone, Item 25, Appendix D (or equivalent), clean bonding surface of main rotor hub. Remove only loose debris. If necessary, use plastic or beryllium copper chisel to remove loose debris.

c. To replace liner, 70103-08002-115, prepare liner as follows:

NOTE

Liner, 70103-08002-115, must be replaced by liner, 70103-08002-120.

(1) Procure, or locally make liner, 70103-08002-120 (Figure H-263, Appendix H).

(2) Using cheesecloth, Item 90, dampened with acetone, Item 25, Appendix D (or equivalent), clean bonding surface of liner and main rotor hub.

d. To replace liner, 70103-08002-116, prepare liner as follows:

(1) Procure, or locally make liner, 70103-08002-116 (Figure H-265, Appendix H).

(2) Using cheesecloth, Item 90, dampened with acetone, Item 25, Appendix D (or

equivalent), clean bonding surface of liner and main rotor hub.

e. To replace liner, 70103-08002-106, prepare liner as follows:

(1) Procure liner, 70103-08002-106.

(2) Using cheesecloth, Item 90, dampened with acetone, Item 25, Appendix D (or equivalent), clean bonding surface of liner and main rotor hub.

NOTE

Procured liners should have adhesive primer already applied. If procured liners are received without adhesive primer, liner must be primed prior to installation.

f. If required, apply adhesive primer as follows:

(1) Using abrasive mat, Item 14, Appendix D, abrade side of liner to be bonded (side opposite part marking).

(2) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean liner.

(3) Using adhesive primer, Item 68, Appendix D, prime abraded side of liner, providing 100% coverage.

(4) Allow adhesive primer to cure at room temperature for a minimum of 30 minutes.

g. Prepare adhesive, Item 32, Appendix D, per manufacturer's directions.

CAUTION

Damage to main rotor hub will result if adhesive is allowed to enter threaded holes. Do not allow adhesive to enter threaded holes.

h. Apply adhesive, Item 32, Appendix D, to original bond area on main rotor hub and new liner.

i. Position liner on main rotor hub.

j. Apply pressure as follows:

(1) For bonded liner, 70103-08002-120, using elastomeric bearing assembly endplate or pressure plate (suitable, flat plate), apply pressure on rebonded area and secure by bolting over entire bonded liner area.

(2) For bonded liner, 70103-08002-106, using damper bracket or pressure plate (suitable, flat plate), apply pressure on rebonded area and secure by bolting over entire bonded liner area.

(3) For bonded liner, 70103-08002-116, using pressure plate (suitable, flat plate), apply pressure on rebonded area and secure by bolting over entire bonded liner area.

k. Allow to cure for 24 hours at room temperature.

5-4-2.4 Inspect/Repair Upper and Lower Pressure Plates.

NOTE

- Damage should be blended out using abrasive cloth, Item 11, Appendix D. Blend out all reworked areas to ½-inch radius.
- Record all repairs on component log card.

a. Check helicopter logbook to make sure area has not been previously reworked.

b. Inspect upper and lower pressure plates for nicks and gouges as follows:

(1) **CORNER DAMAGE UP TO 0.010-INCH DEEP CAN BE BLENDED OUT, EXCEPT FOR LUG AREAS. IN LUG AREAS, DAMAGE UP TO 0.050-INCH DEEP IS ALLOWED.**

(2) **ON OPEN AREAS (NONMATING SURFACES), SURFACE DAMAGE UP TO 0.040-INCH DEEP CAN BE BLENDED OUT. DAMAGED AREA MUST BE LESS THAN 20% OF CIRCUMFERENCE.**

(3) **ON MATING AREAS, SURFACE DAMAGE UP TO 0.020-INCH DEEP CAN BE BLENDED OUT. DAMAGED AREA MUST BE LESS THAN 10% OF SURFACE AREA AFTER BLENDING OUT AND REMOVING ANY METAL PICKUP.**

(4) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace main rotor hub.

c. Touch up finish on all reworked areas (PARA 2-6-5).

5-4-3 MAIN ROTOR HUB THREADED INSERTS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Hex Bit, 5⁄16-inch
- Hex Bit, 3⁄8-inch
- Powertrain Repairer's Toolkit
- Screwdriver Bit Set
- Soldering Iron
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 299, Appendix D
- Sealing Primer, Item 306, Appendix D

References

- Appendix D

5-4-3.1 Replace Main Rotor Hub Threaded Inserts.

NOTE

Replacement of all main rotor hub threaded inserts is the same, except as noted.

5-4-3.1.1. Remove.

CAUTION

Temperatures above 240°F (116°C) may degrade bond integrity of adjacent bonded liners. After removal of threaded inserts, inspect bonded liners per PARA 5-4-3.1.2 and repair per PARA 5-4-3.1.3.

a. Using hex bit from screwdriver bit set, remove damaged threaded insert (Figure 5-4-3-1).

NOTE

If it is difficult to remove threaded inserts, 79698-820, between spindle and main rotor hub, perform step b.

b. Using soldering iron inserted into threaded insert, bring temperature of threaded insert up to 300° - 450°F (149° - 232°C). Try to remove threaded inserts periodically during temperature increase.

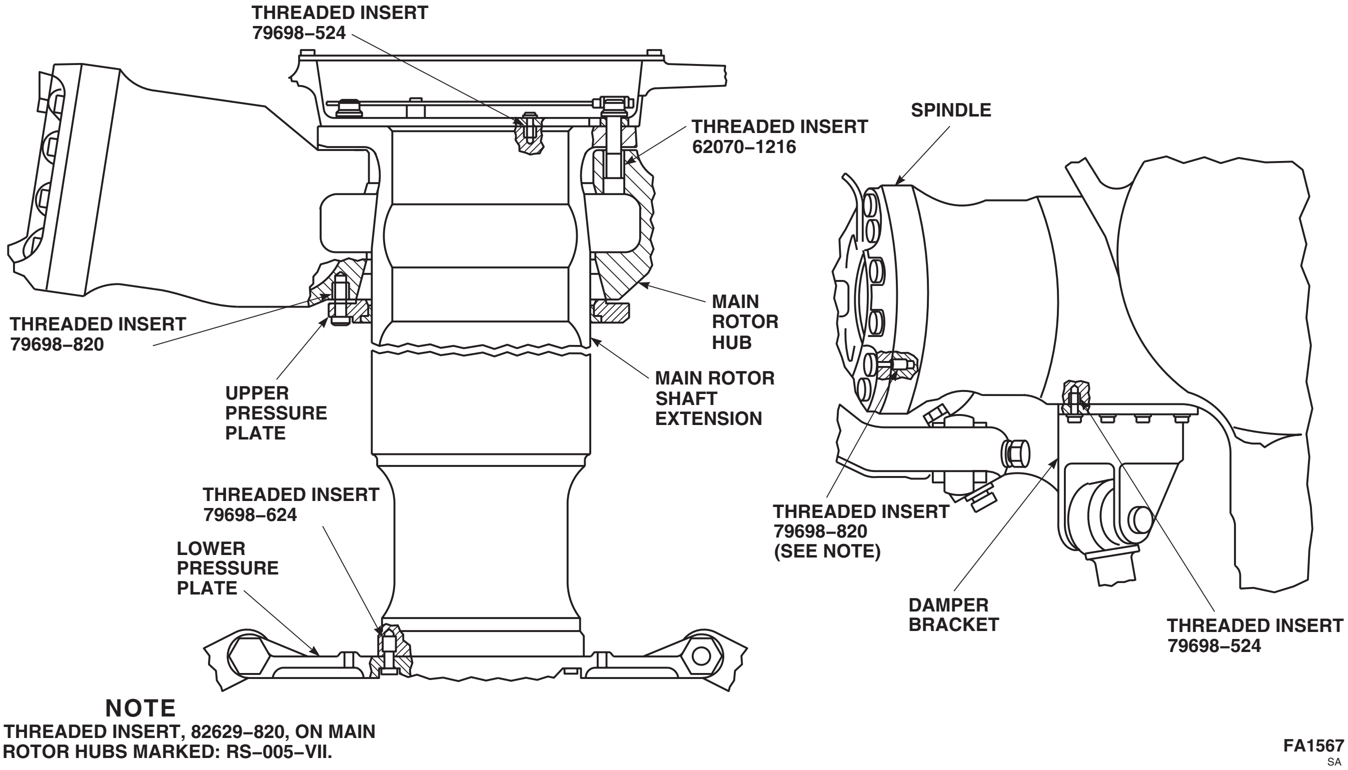


FIGURE 5-4-3-1. REPLACE MAIN ROTOR HUB THREADED INSERTS

5-4-3.1.2. Inspect.

a. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean any old sealing compound and sealing primer from hole. Dry area thoroughly.

b. Check threaded hole where new threaded insert will be installed as follows:

(1) Using hex bit, install threaded insert into threaded hole up to threaded insert locking feature. **THREADED INSERT MUST THREAD IN BY HAND WITHOUT USING TORQUE WRENCH.**

(2) If threaded insert will not thread in by hand, replace main rotor hub.

(3) Remove threaded insert from threaded hole.

5-4-3.1.3. Install.

a. Grind 5/16-inch hex bit to fit threaded insert, 79698-524, and 3/8-inch hex bit to fit threaded insert, 79698-624.

b. Using sealing primer, Item 306, Appendix D, treat outside threads of replacement threaded insert.

c. Apply a thin coat of sealing compound, Item 299, Appendix D, to only the first two threads of threaded insert.

d. Using hex bit from screwdriver bit set, install threaded insert in housing and torque wrench. Refer to Table 5-4-3-1 for correct torque limits and depth of threaded insert.

Table 5-4-3-1. Main Rotor Hub Threaded Inserts

Threaded Insert Part No.	Minimum Torque When Installing Threaded Insert (Inch-Pounds)	Maximum Torque When Installing Threaded Insert (Inch-Pounds)	Installation Depth (Inches)	Threaded Insert Location	Removal/ Installation Tool Key Size
79698-524	66	500	0.000 - 0.031 below face of main rotor hub (See Note 3)	Damper bracket to main rotor hub	0.264-inch
79698-524	66	500	0.000 - 0.031 below surface of main rotor shaft extension	Bifilar to main rotor shaft extension	0.264-inch
79698-820 (See Note 2)	165	790 (See Note 1)	0.000 - 0.031 below face of main rotor hub (See Note 3)	Spindle to main rotor hub	7/16-inch
82629-820	165	790 (See Note 1)	0.000 - 0.031 below face of main rotor hub (See Note 3)	Spindle to main rotor hub	7/16-inch
79698-820	165	790 (See Note 1)	0.000 - 0.031 below surface of main rotor hub	Upper pressure plate to main rotor hub	7/16-inch
62070-1216	432	1200	0.000 - 0.045 below face of main rotor hub (See Note 3)	Main rotor hub to main rotor shaft	5/8-inch
79698-624	88	500	0.000 - 0.031 below surface of main rotor shaft extension	Lower pressure plate to main rotor shaft extension	0.328-inch
79698-524	66	483	0.000 - 0.031 below face of bearing	Antiflap stop to elastomeric bearing assembly endplate	0.264-inch

NOTES

1. If proper torque cannot be achieved, up to two threaded inserts which are not next to each other, may be installed up to 0.030-inch below installation depth at each location.

2. On main rotor hubs marked RS-005-VII, replace threaded insert with threaded insert, 82629-820.

3. If threaded insert depth is measured from surface of bonded liner, add actual bonded liner thickness to installation depth.

5-4-4 MAIN ROTOR HEAD BOLTS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit
- Magnetic Particle Inspection Kit
- Powertrain Repairer's Toolkit

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

References

- Appendix D
- ASTM E1417
- ASTM E1444
- PARA 2-6-5

5-4-4.1 Inspect/Repair Main Rotor Head Bolts.

5-4-4.1.1 Inspect Titanium and Steel Bolts.

NOTE

- The following instructions apply to bolts used to attach damper, pitch control rods, and rotating scissors.
- Whenever one of these bolts is removed, it should be checked as follows before reinstallation.

a. Using dry-cleaning solvent, Item 124, Appendix D, clean any oil or grease from bolts.

b. Inspect bolts for any material displaced above normal surface of bolt. **NONE ALLOWED.** Using abrasive cloth, Item 10, Appendix D, remove any displaced material.

c. Inspect bolts for scoring or grooves. **LENGTHWISE SCORE MARKS OR GROOVES MAY BE SMOOTHED OUT, BUT CAN BE LEFT AS IS, IF WITHIN LIMITS. CIRCUMFERENTIAL SCORE MARKS MUST BE BLENDED OUT.** Refer to

Figure 5-4-4-1 for limits on depth of scoring. If necessary, blend out as follows:

WARNING

FSCAP

On bolts, 70103-08801-102/-103, the outside diameter (OD) of the shank is a critical characteristic. The OD dimension must not measure below the minimum allowable diameter after rework as reflected in Figure 5-4-4-1.

(1) On steel bolts, 70103-08114-101 or 70103-28702-101, using magnetic particle inspection kit, inspect for cracks (ASTM E1444), before rework. **NONE ALLOWED.** If crack is found, replace bolt.

(2) Before reworking bolt, using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, bolt for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace bolt.

(3) Using abrasive wheel, Item 22, Appendix D, blend out scratches. Using abrasive cloth, Item 10, Appendix D, blend area smooth. Inspect bolt as follows:

(a) On bolts, 70102-08117-102, corner damage may be permitted up to 0.070 × 0.070-inch. Prime and paint damaged areas (PARA 2-6-5). If corner damage exceeds limits, discard bolt.

(b) If bolt diameter is less than specified in Figure 5-4-4-1 after rework, discard bolt.

(c) If repaired area exceeds 30% of bolt shank area after rework, discard bolt.

(4) After reworking bolt, using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, bolt for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace bolt.

(5) After reworking steel bolts, 70103-08114-101 or 70103-28702-101, magnetic particle inspect bolt for cracks (ASTM E1444). **NONE ALLOWED.** If crack is found, replace bolt.

5-4-4.1.2 Inspect Plated Bolts.

NOTE

- The following instructions apply to both cadmium and silver-plated bolts. Silver-plated bolts are identified by an ″S″ after basic part number.
- Whenever one of these bolts is removed it should be checked as follows before installation.

a. Using dry-cleaning solvent, Item 124, Appendix D, clean any oil or grease from bolts.

b. Inspect bolts for corrosion pitting. **PITTING MUST NOT EXCEED 0.030-INCH ON EXPOSED BOLTHEAD AREA.** If limit is exceeded, replace bolt. Refer to Figure 5-4-4-2 for corrosion limits on other areas of bolt.

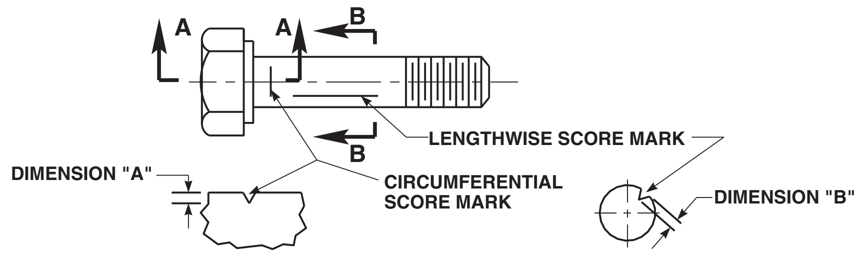
5-4-4.1.3 Inspect Non-Plated Bolts.

NOTE

Whenever one of these bolts is removed it should be checked as follows before reinstallation.

a. Using dry-cleaning solvent, Item 124, Appendix D, clean any oil or grease from bolts.

b. Inspect bolts for corrosion pitting. **PITTING MUST NOT EXCEED 0.030-INCH ON EXPOSED BOLTHEAD AREA.** If limit is exceeded, replace bolt. Refer to Figure 5-4-4-2 for corrosion limits on other areas of bolt.



BOLT PART NUMBER	LOCATION	DIMENSION "A" MAXIMUM	DIMENSION "B" MAXIMUM	MINIMUM ALLOWABLE DIAMETER AFTER REWORK
70103–28702–101	DAMPER (BOTH ENDS)	0.005 IN.	0.005 IN.	0.7435 IN.
70103–08801–103	PITCH CONTROL ROD (TOP)	0.002 IN.	0.003 IN.	0.7480 IN.
70103–08801–102	PITCH CONTROL ROD (BOTTOM)	0.001 IN.	0.003 IN.	0.6230 IN.
SS5011–16–80A	ROTATING SCISSORS (ROTATING SCISSORS TO LOWER PRESSURE PLATE)	0.002 IN.	0.003 IN.	0.9945 IN.
SS5011–12–81H	ROTATING SCISSORS (UPPER LINK TO LOWER LINK)	0.002 IN.	0.003 IN.	0.7445 IN.
70103–08114–101	DAMPER (BOTH ENDS)	0.005 IN.	0.005 IN.	0.7435 IN.
SS5032–12–66	PITCH CONTROL ROD (TOP)	0.002 IN.	0.003 IN.	0.7442 IN.
SS5032–10–59	PITCH CONTROL ROD (BOTTOM)	0.001 IN.	0.003 IN.	0.6202 IN.
NAS1956–9	SPINDLE HORN	0.001 IN.	0.001 IN.	0.3716 IN.
NAS1956–45	SPINDLE HORN	0.001 IN.	0.001 IN.	0.3716 IN.
70102–08117–102	SPINDLE HORN	0.001 IN.	0.001 IN.	0.3716 IN.
SS5021–10–34A	BIFILAR (BIFILAR WEIGHT TO SUPPORT)	0.001 IN.	0.003 IN.	0.6170 IN.

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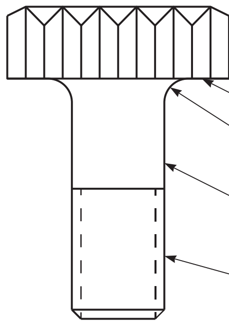
FIGURE 5-4-4-1. INSPECT/REPAIR MAIN ROTOR HEAD BOLTS



BOLT PART NO.	LOCATION	PIT LIMIT AREA A	PIT LIMIT AREA B	PIT LIMIT AREA C	PIT LIMIT AREA D
76683–5–7 OR 80489–5–7	BIFILAR SUPPORT	0.003 IN.	0.001 IN.	0.004 IN.	0.001 IN.
76683–5–8 OR 80489–5–8	DAMPER BRACKET	0.003 IN.	0.001 IN.	0.004 IN.	NONE
76683–6–10 OR 80489–6–10	LOWER PRESSURE PLATE	0.003 IN.	0.001 IN.	0.004 IN.	0.001 IN.
76683–8–10 OR 80489–8–10	SPINDLE ELASTOMERIC BEARING ENDPLATE	0.003 IN.	0.001 IN.	0.004 IN.	NONE
76683–8–12 OR 80489–8–12	UPPER PRESSURE PLATE	0.003 IN.	0.001 IN.	0.004 IN.	0.001 IN.
NAS1956–9	SPINDLE HORN	0.001 IN.	NONE	0.002 IN.	NONE
NAS1956–45	SPINDLE HORN	0.001 IN.	NONE	0.002 IN.	NONE
70102–08117–102	SPINDLE HORN	0.001 IN.	NONE	0.002 IN.	NONE

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FIGURE 5-4-4-2. INSPECT/REPAIR MAIN ROTOR HEAD BOLTS

5-4-5 SPINDLE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Cotton Gloves
- Dial Indicator (with Clamp)
- Droop Stop Nut Wrench, Locally-Made, Figure H-168, Appendix H
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Heat Gun
- Hoist, Capacity to Support Main Rotor Blade
- Hoisting Sling, 70700-20320-042
- Inside Micrometer Caliper
- Nonmetallic Scraper, Locally-Made
- Outside Micrometer Caliper
- Pliers
- Powertrain Repairer's Toolkit
- Protection, Eye, Skin, and Breathing
- Protective Covering
- Soft-Bristle Brush
- Soft-Faced Mallet
- Strap Wrench
- Torque Wrench, Dial, 0 - 600 in. lbs
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Wood, 2-inches × 4-inches × 3-feet

Materials/Parts

- Abrasive Cloth, Item 4, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Abrasive Cloth, Item 12, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Adhesive, Item 32, Appendix D
- Adhesive, Item 33A, Appendix D
- Adhesive, Item 64, Appendix D
- Adhesive, Item 66, Appendix D
- Cloth
- Corrosion Resistant Compound, Item 117, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dishwashing Compound, Item 122, Appendix D

- Dry Ice, Item 125, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Etching Solution, Item 136, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 179, Appendix D
- Lockwire, Item 197, Appendix D
- Lubricant, Solid Film, Item 201, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Mild Detergent
- Paint Remover, Item 230, Appendix D
- Primer, Item 254, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D
- Tags

References

- Appendix D
- Appendix H
- Appropriate Flight Manual
- ASTM E1417
- PARA 1-6-18
- PARA 1-7-5
- PARA 1-7-6
- PARA 1-7-20
- PARA 2-6-5
- PARA 5-3-1
- PARA 5-3-3
- PARA 5-4-2
- PARA 5-4-3
- PARA 5-4-4
- PARA 5-4-6
- PARA 5-4-7
- PARA 5-4-9
- PARA 5-4-10
- PARA 5-4-13
- PARA 5-4-18
- PARA 5-4-31
- PARA 5-5-2
- PARA 5-5-3
- PARA 5-5-4
- TM 1-70-VIB

5-4-5.1 Replace Spindle.

NOTE

Replacement of all spindles is the same.

5-4-5.1.1 Remove.

NOTE

One spindle assembly weighs about 65 pounds.

a. Turn main rotor head so spindle can be removed easily.

b. If not already done, engage gust lock (PARA 1-6-18).

c. Remove applicable main rotor blade (PARA 5-4-31).

WARNING

FSCAP

The pitch control rod attachment areas, and critical areas from spindle horn lugs to elbow are critical surfaces which must not be damaged during and/or after removal of spindle horn and/or pitch control rod. Use extreme caution when removing and provide protective covering after removal from spindle horn and attaching hardware. If damaged, inspect spindle horn (PARA 5-4-7).

CAUTION

Damage to spindle horn will result if allowed to come in contact with droop stop pin or cotter pin. Do not allow spindle horn to come in contact with droop stop pin or cotter pin.

NOTE

It may be necessary to move spindle horn slightly up and down, for easy removal of pitch control rod bolt.

d. Disconnect upper end of pitch control rod (PARA 5-4-18).

e. Disconnect damper from spindle (PARA 5-4-13).

f. Remove antiflap stop from spindle. Place bonding jumper aside (PARA 5-4-9).

g. Remove droop stop cam (PARA 5-4-10).

CAUTION

Damage to spindle will result if metallic items are used to remove sealing compound. Do not use metallic items to remove sealing compound.

h. Using nonmetallic scraper, remove sealing compound from spindle boltheads.

i. Prepare spindle for removal as follows:

(1) Remove bolts and washers securing spindle to main rotor hub (Figure 5-4-5-1, Detail B). Leave one upper bolt in until hoisting sling is attached.

NOTE

There may not be enough wrench clearance to remove bolt behind spindle horn and antiflap assembly.

(2) If necessary, insert piece of wood, 2-inches × 4-inches × about 3-feet long, between spindle cuff lugs, and turn spindle.

(3) Keep two clips that attach to droop stop springs with springs.

j. Place hoisting sling around spindle and adjust straps. Attach hoisting sling to hoist. Take up slack (Figure 5-4-5-1, Detail C).

k. Remove remaining bolt from spindle.

l. Using machinery towel, Item 211, Appendix D, with methyl ethyl ketone, Item 217, Appendix D, clean primer from bolts.

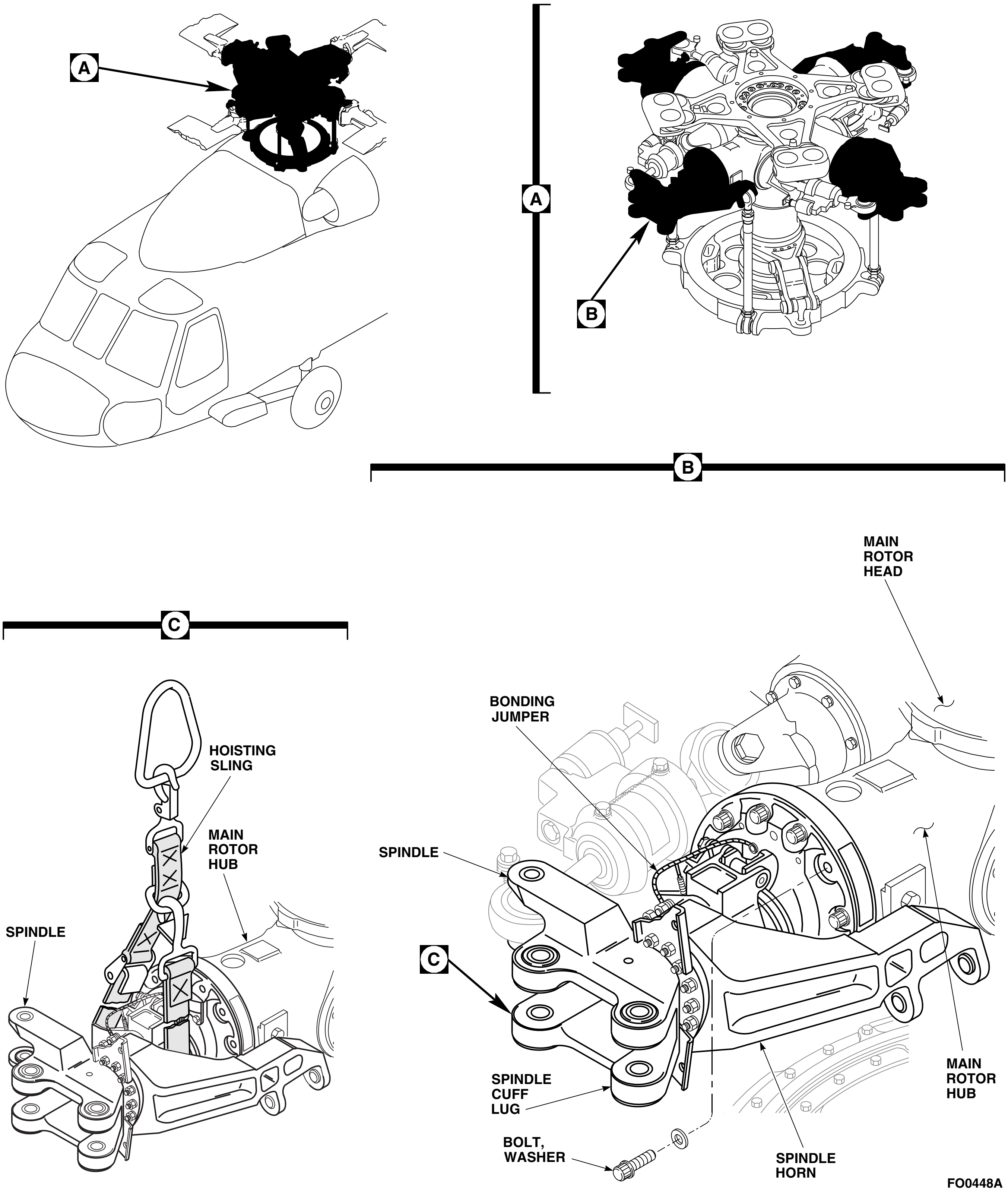


FIGURE 5-4-5-1. REMOVE SPINDLE

CAUTION

Damage to main rotor hub will result if metallic tools are used to pry spindle from main rotor hub. Do not pry spindle from main rotor hub with metal tools.

m. Slide spindle from main rotor hub.

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when handling spindle. Keep area clear when hoisting spindle.

n. Move spindle clear of helicopter onto workstand.

o. Remove hoisting sling.

p. Install protective covering from spindle horn lugs to elbow of spindle.

5-4-5.1.2 Disassemble.

a. If replacement spindle will be installed and previous spindle has balance weights installed, perform the following (Figure 5-4-5-2, Sheet 1, Detail A):

(1) Remove bolts, washers, balance weights, and nuts from balance bracket(s).

(2) Tag parts.

(3) Keep these parts for installation in new spindle.

b. If replacement spindle will be installed, remove bonding jumper from spindle as follows (Figure 5-4-5-2, Sheet 1, Detail B):

(1) Using nonmetallic scraper, remove sealing compound from bolthead securing bonding jumper to spindle.

(2) Remove short bolt, washer, and nut securing bonding jumper to spindle.

c. Remove bolts and washers securing spindle nut to thrust bearing (Figure 5-4-5-2, Sheet 2, Detail C). Remove spindle nut. If necessary, use strap wrench to loosen nut.

d. Slide elastomeric bearing assembly from spindle.

e. Install protective covering to threads of spindle, thread relief (undercut), and shank.

f. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean wet primer from bolts.

g. On spindles with droop stop support ring, 70105-08056-101, with sleeve bearing, disassemble spindle as follows (Figure 5-4-5-2, Sheet 2, Detail D):

(1) Using dial indicator, measure radial play between droop stop support ring and inner race of sleeve bearing. **MAXIMUM RADIAL PLAY IS 0.030-INCH.** If radial play exceeds limit, replace sleeve bearing outer sleeve (PARA 5-5-2).

(2) Using dial indicator, measure axial play between droop stop support ring and inner race of sleeve bearing. **MAXIMUM AXIAL PLAY IS 0.020-INCH.** If axial play exceeds limit, replace sleeve bearing outer sleeve and Teflon-lined ring (PARA 5-5-2).

(3) Visually inspect exposed sleeve bearing Teflon wear surfaces. **NO MORE THAN 30% OF CIRCUMFERENCE OF EITHER WEAR SURFACE MISSING ALLOWED.** If more than 30% of circumference of wear surface is missing, replace sleeve bearing outer sleeve and/or Teflon-lined ring (PARA 5-5-2).

(4) Visually inspect exposed sleeve bearing Teflon wear surfaces for metal-to-metal contact. **NONE ALLOWED.** If metal-to-metal contact is found, replace sleeve bearing outer sleeve and/or Teflon-lined ring (PARA 5-5-2).

(5) Remove retaining ring from droop stop nut.

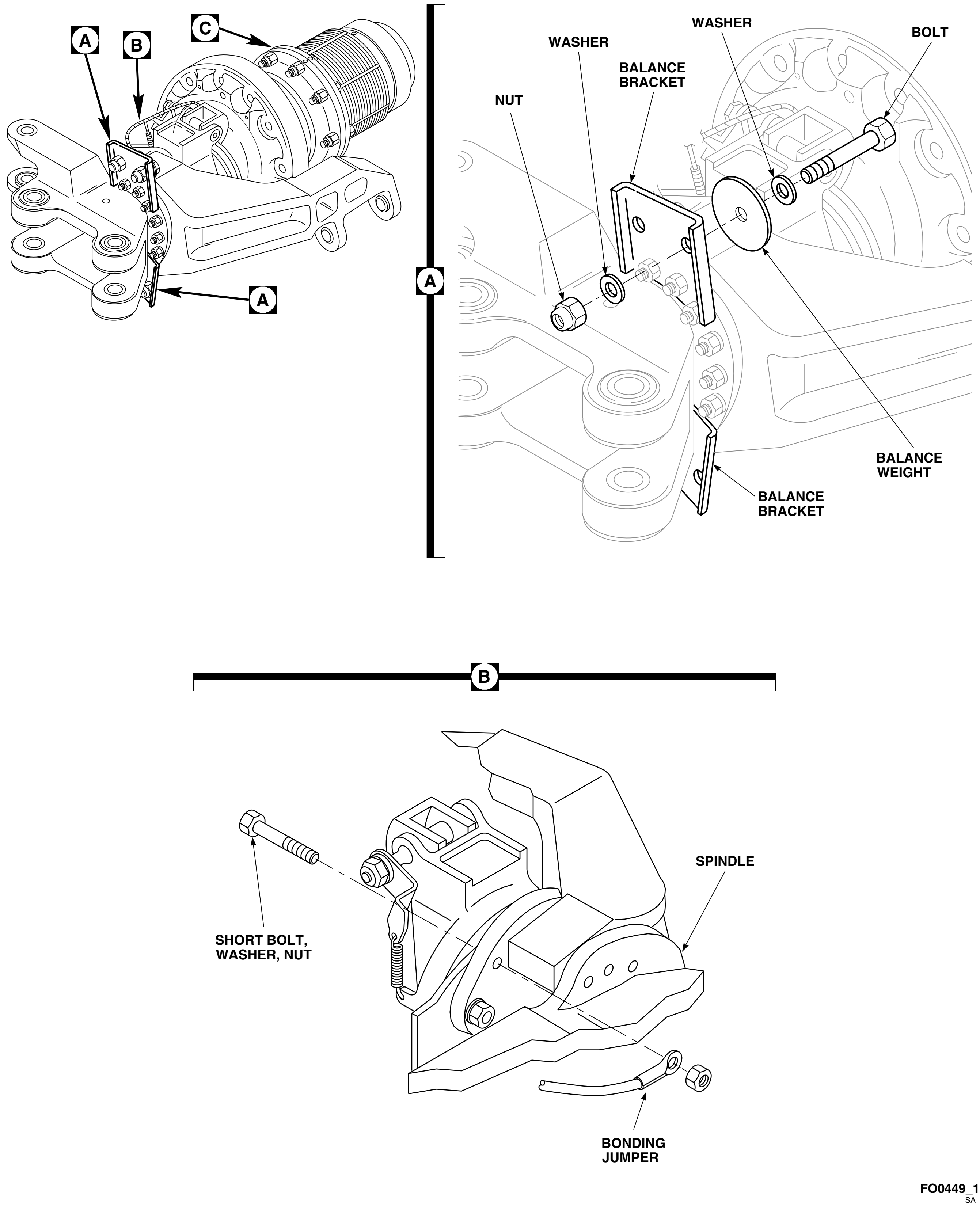


FIGURE 5-4-5-2. DISASSEMBLE SPINDLE (SHEET 1 OF 2)

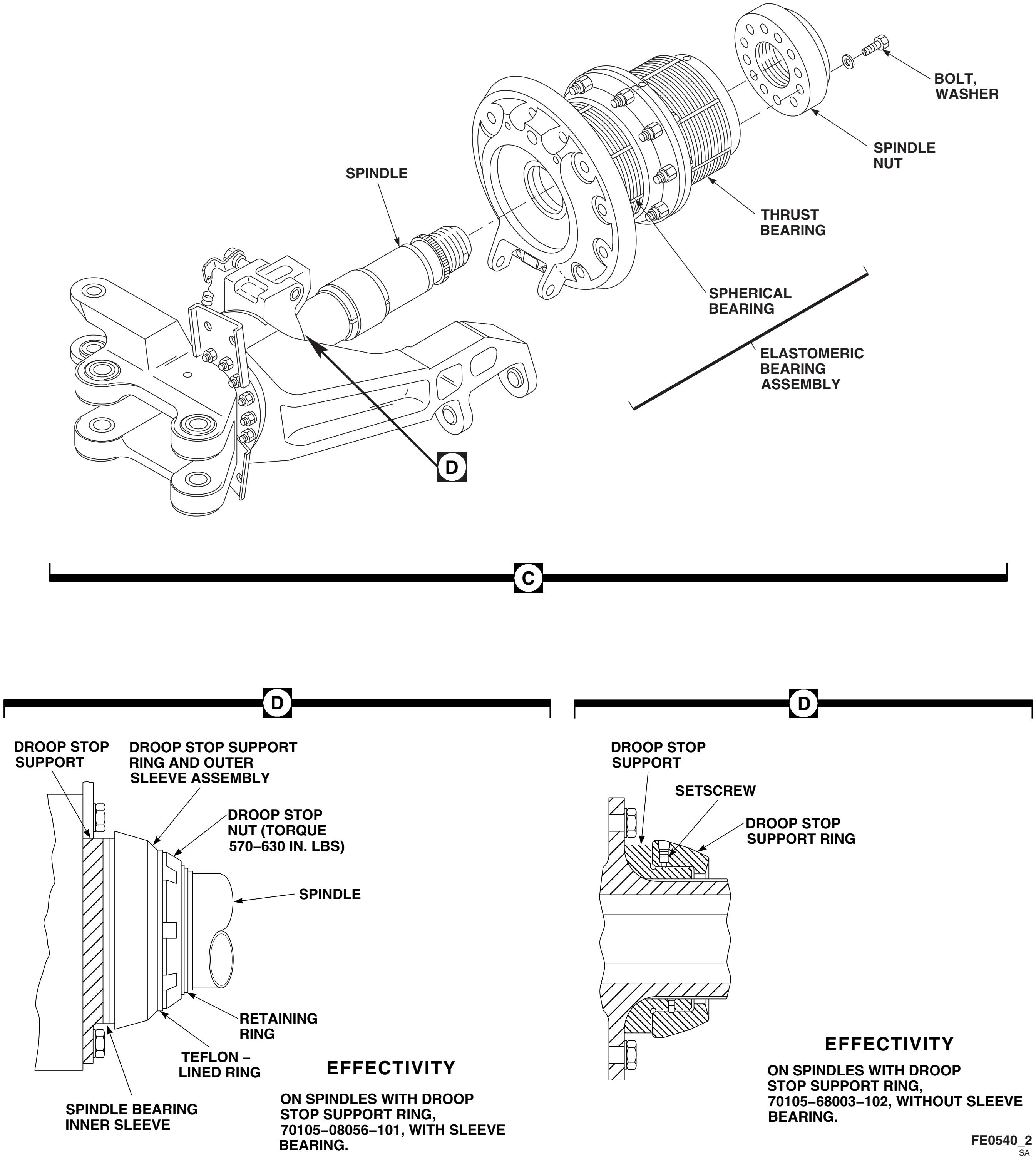


FIGURE 5-4-5-2. DISASSEMBLE SPINDLE (SHEET 2 OF 2)

(6) Locally make droop stop nut wrench (Figure H-168, Appendix H).

(7) Using droop stop nut wrench, back off droop stop nut.

(8) Slide Teflon-lined ring away from droop stop support ring.

(9) Slide droop stop support ring and outer sleeve assembly off sleeve bearing inner sleeve from spindle.

h. On spindles with droop stop support ring, 70105-68003-102, without sleeve bearing, disassemble spindle as follows (Figure 5-4-5-2, Sheet 2, Detail D):

(1) Using dial indicator, measure radial play in vertical direction between droop stop support ring and droop stop support. **MAXIMUM RADIAL PLAY IS 0.025-INCH.** If radial play exceeds limit, replace droop stop support ring.

(2) Using feeler gage, measure gap between droop stop support ring outboard Teflon surface and inboard surface of droop stop support. **MAXIMUM GAP MUST NOT EXCEED 0.090-INCH.** If gap exceeds limit, replace droop stop support ring.

(3) Remove droop stop support ring as follows:

(a) Using nonmetallic scraper, remove sealing compound to expose setscrew in droop stop support ring.

(b) Back out setscrew and rotate droop stop support ring while pulling slightly to remove droop stop support ring.

(4) Visually inspect droop stop support ring Teflon wear surface. **NO METAL-TO-METAL CONTACT ALLOWED.** If metal-to-metal contact is found, replace droop stop support ring.

5-4-5.1.3 Inspect Spindle Shaft.

WARNING

FSCAP

Thread relief and first seven full threads from thread relief on spindle shaft are critical characteristics of the part and must not contain any nicks, dents, scratches, or scuff marks. Perform detailed inspection of this area of spindle.

a. Perform detailed inspection of spindle critical areas. **REPLACE SPINDLE IF:**

(1) **DAMAGE SUCH AS NICKS, DENTS, SCRATCHES, SCUFF MARKS TO THREADS, OR THREAD RELIEF (UNDERCUT) IS FOUND** (Figure 5-4-5-3, Detail A).

(2) **SPINDLE NUT WILL NOT TURN FREELY WHEN IT IS THREADED ON** (Figure 5-4-5-3).

b. Inspect threads of spindle for condition of solid film lubricant. **REPLACE SPINDLE IF:**

(1) **THREADS OF SPINDLE ARE MISSING MORE THAN 50% SOLID FILM LUBRICANT.**

(2) **THREADS OF SPINDLE ARE MISSING SOLID FILM LUBRICANT NOT EXCEEDING 50% OF SURFACE AREA AND SPINDLE HAS PREVIOUSLY BEEN TOUCHED UP.**

c. If metal is showing on threads and spline of spindle, and spindle has not been touched up previously, using solid film lubricant, Item 201, Appendix D, reapply up to 50% of surface area as follows:

(1) Using dry machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean areas of bare metal. Allow to dry thoroughly.

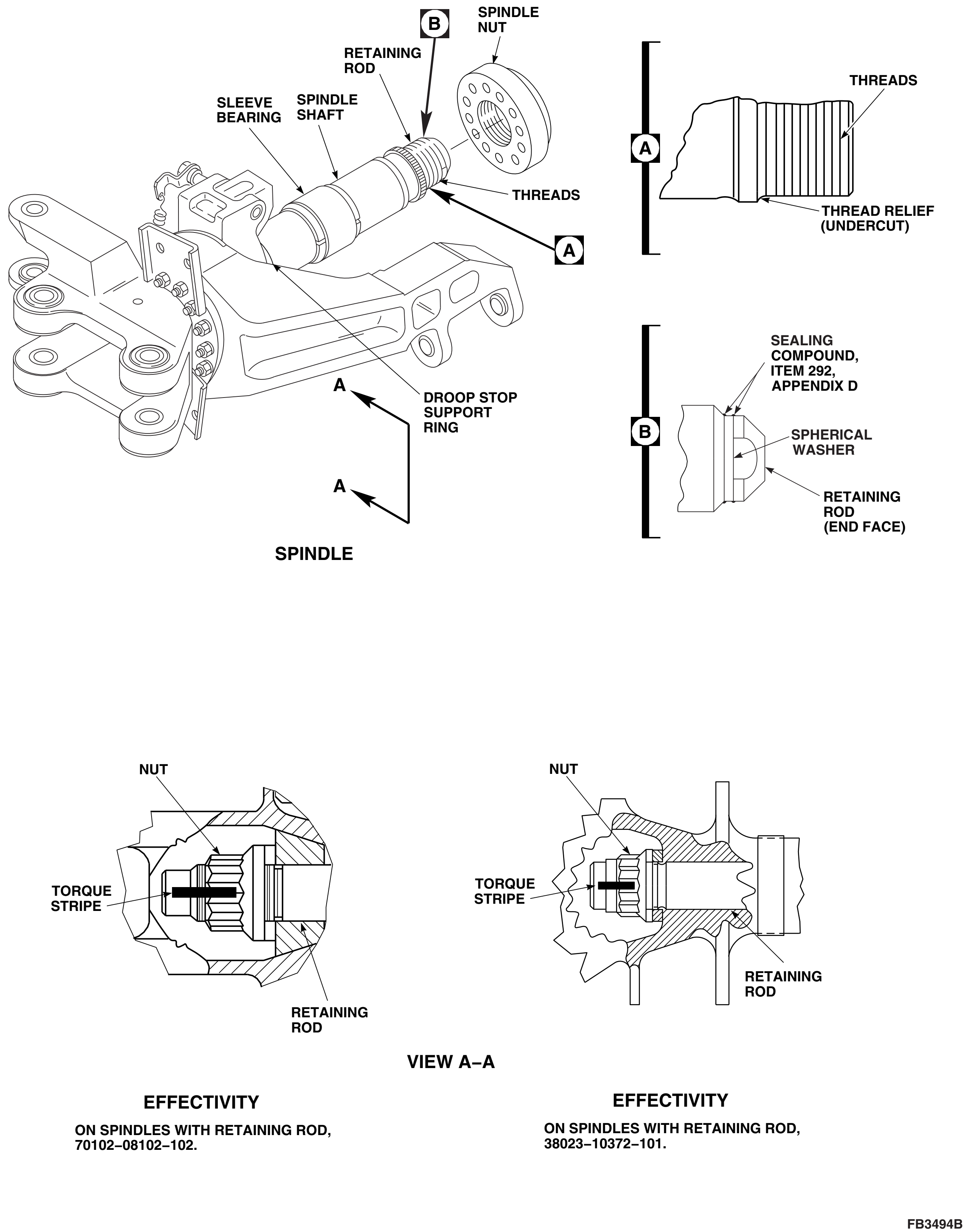


FIGURE 5-4-5-3. INSPECT SPINDLE SHAFT

NOTE

Solid film lubricant will not adhere to contaminated surfaces. Do not touch cleaned areas with bare hands. Use clean cotton gloves.

(2) At room temperature, using solid film lubricant, Item 201, Appendix D, spray exposed area. Apply multiple thin coats, allowing 15 - 20 minutes drying time between coats.

(3) Allow solid film lubricant to cure a minimum of 12 hours before using spindle.

d. Visually inspect threads and spline of spindle for condition of solid film lubricant. **REPLACE SPINDLE IF:**

(1) **COATING IS NOT ″SLATE-BLACK″ AND UNIFORM IN COLOR.**

(2) **COATING HAS ANY RUNS, BLISTERS, ROUGHNESS, OR SEPARATION OF INGREDIENTS.**

(3) **COATING HAS ANY OTHER SURFACE IMPERFECTIONS.**

e. Inspect solid film lubricant for excessive thickness by threading on spindle nut. **NUT MUST THREAD ON SMOOTHLY WITH LIGHT HAND FORCE.** If spindle nut will not thread on smoothly with light hand force, replace spindle.

f. If repair of solid film lubricant is successful, note component serial number, percentage of area touched up, and total time on part, in helicopter logbook.

g. Visually inspect condition of white torque stripe on retaining rod and nut for torque loss (Figure 5-4-5-3, View A-A). **NO BROKEN STRIPE ALLOWED.** If torque stripe is broken, indicating potential torque loss, replace spindle.

h. Inspect retaining rod for damage and corrosion as follows:

(1) Using nonmetallic scraper, remove sealing compound from around retaining rod head and spherical washer (Figure 5-4-5-3, Detail B).

(2) Using abrasive wheel, Item 22, Appendix D, blend any damage or corrosion found to retaining rod head. **REPLACE SPINDLE IF:**

(a) **DAMAGE OR CORROSION TO RETAINING ROD HEAD IS DEEPER THAN 0.010-INCH AFTER BLENDING.**

(b) **DAMAGE AREA EXCEEDS 0.050-INCH DIAMETER.**

(c) **DAMAGE COVERS THREE OR MORE 0.050-INCH DIAMETER LOCATIONS.**

(d) **DAMAGE COMES CLOSER THAN 0.150-INCH TO WASHER CONTACT FACE.**

(3) Using abrasive wheel, Item 22, Appendix D, blend out corrosion damage. **CORROSION DAMAGE MUST BE NO DEEPER THAN 0.015-INCH, COVERING NO MORE THAN 25% OF CIRCUMFERENCE AREA.** If corrosion damage to retaining rod washer exceeds limits, replace spindle.

(4) Apply coat of primer, Item 258, Appendix D, to blended area, retaining rod head, or retaining rod washer.

(5) Touch up paint on retaining rod (PARA 2-6-5).

(6) Apply sealing compound, Item 292, Appendix D, around retaining rod and washer.

i. Inspect spindle shaft for surface damage. Using abrasive wheel, Item 22, Appendix D, blend any damage found. **DAMAGE MUST BE NO DEEPER THAN 0.010-INCH AFTER BLENDING, AND NO GREATER THAN ¼ CIRCUMFERENCE OF SPINDLE.** If damage exceeds limits, replace spindle.

NOTE

- Crazing of adhesive in slits of spindle shank liners is acceptable.
- Teflon sleeve or thin metal retaining strap is located between inner race and the droop stop support ring.

j. On spindles, 70102-08200-043 and 70102-08200-044, inspect shrink tubing installed on bonded Teflon sleeve as follows:

(1) Inspect shrink tubing for signs of axial movement, displacement, or incorrect installation.

NOTE

Damage to shrink tubing may occur if care is not taken when applying axial force. Shrink tubing may tear. Use caution not to damage or tear shrink tubing.

(2) Grasp inboard end of shrink tubing, and apply axial force toward end of main rotor blade. Shrink tubing must not exceed limits of channel in sleeve bearing. Rotational movement is acceptable.

(3) **IF SHRINK TUBING HAS BEEN INSTALLED WITH EDGE OF TUBING BEYOND CHANNEL IN SLEEVE BEARING, HAS SHIFTED POSITION, OR EXCEEDS CHANNEL LIMITS IN SLEEVE BEARING, REPLACE SHRINK TUBING AT NEXT 500 HOUR INSPECTION.**

k. Any surface damage to Teflon sleeve and metal retaining strap is acceptable. Cracking of adhesive squeeze out from around Teflon sleeve is acceptable. Inspect journal bearings (PARA 5-4-5.1.7).

l. **REBOND METAL BAND IF IT BECOMES DISBONDED BY LIGHTLY SANDING (IF POSSIBLE), CLEAN WITH DRY CLOTH AND APPLY ADHESIVE, ITEM 32, APPENDIX D.**

m. **USING HEAT GUN, RESHRINK SHRINK TUBING IF TUBING BECOMES LOOSE (ONE-TIME ONLY). FLIGHT WITHOUT SHRINK TUBING IS ACCEPTABLE, PROVIDED TEFLON SLEEVE DOES NOT START PEELING FROM SPINDLE.**

5-4-5.1.4 Inspect Droop Stop Support.

a. Visually inspect upper portion (1 - 2 o'clock position, when looking inboard) and lower portion (6 - 9 o'clock position, when looking inboard) of droop stop support for cracks (Figure 5-4-5-4, Detail A). **IF CRACK IS VISIBLE IN ONE LOCATION ONLY, REPLACE SPINDLE WITHIN 500 FLIGHT HOURS.**

b. On spindles with droop stop support ring, 70105-08056-101, with sleeve bearing, perform the following:

(1) Visually inspect exposed portion of sleeve bearing inner race (pressed onto droop stop support) around entire circumference for cracks as follows:

(a) **IF HELICOPTER IS AT 500 HOUR PERIODIC INSPECTION AND CRACK IS VISIBLE, REPLACE SPINDLE.**

(b) **IF HELICOPTER IS NOT AT 500 HOUR PERIODIC INSPECTION AND ONLY ONE CRACK IS VISIBLE, REPLACE SPINDLE AT NEXT 500 HOUR PERIODIC INSPECTION, OR WHEN SECOND CRACK IS FOUND.**

(c) **IF MORE THAN ONE CRACK IS VISIBLE, REPLACE SPINDLE.**

(2) Inspect droop stop support ring sleeve bearing inner sleeve for nicks, gouges, and burrs. **DAMAGE MUST NOT EXCEED 0.005-INCH DEEP.** If nicks, gouges, and burrs exceed limits, replace spindle.

(3) Inspect droop stop support ring sleeve bearing inner sleeve for corrosion pitting. **PITTING MUST NOT EXCEED 0.005-**

INCH. If pitting exceeds limit, replace spindle.

c. On spindles with droop stop support, 70105-08059-101, inspect for corrosion. If corrosion is found, perform the following:

(1) Using abrasive cloth, Item 4, Appendix D, remove corrosion from droop stop support. **IF CORROSION IS DEEPER THAN 0.035-INCH ACROSS ENTIRE CIRCUMFERENCE OF FIRST TWO THREADS, REPLACE SPINDLE. REPAIR OF REMAINING THREADS LIMITED TO 30% AT THREAD CIRCUMFERENCE.**

(2) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean repaired area.

(3) Using corrosion resistant compound, Item 117, Appendix D, coat repaired area.

(4) Apply coat of primer, Item 258, Appendix D, to repaired area.

d. On spindles with droop stop support, 70105-68002-102, perform the following:

(1) Using nonmetallic scraper, remove sealing compound from setscrew counterbore.

(2) Remove setscrew from droop stop support ring.

(3) Turn droop stop support ring until slot in droop stop support ring aligns with dowel pin. Slide droop stop support ring off droop stop support.

(4) Inspect droop stop support for corrosion (Figure 5-4-5-4, Detail B). **REPLACE SPINDLE IF:**

(a) **CORROSION IS DEEPER THAN 0.020-INCH IN UNDERCUT, FLANGE, OR DROOP STOP SUPPORT RING SURFACE.**

(b) **CORROSION EXCEEDS 20% OF SURFACE AREA.**

(c) **CORROSION IS DEEPER THAN 0.050-INCH, OR EXCEEDS 20% OF SURFACE AREA ON ALL OTHER SURFACES.**

(5) If corrosion is 0.020-inch deep or less, or covers 20% surface area or less in undercut, flange, or droop stop support ring surface, using abrasive cloth, Item 4, Appendix D, remove corrosion from droop stop support.

(6) Inspect droop stop support for wear. Wear on bottom half (3 - 9 o'clock position) of droop stop support in droop stop support ring surface (shaded area) up to 0.002-inch deep is acceptable. Wear that is 0.003 - 0.050-inch deep is also acceptable, provided that droop stop support ring is replaced with a new droop stop support ring.

NOTE

Wear on droop stop support greater than 0.002-inch deep will accelerate wear of droop stop support ring.

(7) If new droop stop support ring was installed on a worn droop stop support, inspect droop stop support ring and droop stop support for wear within next 100 flight hours; repeat step d. (6). **NO METAL-TO-METAL CONTACT ALLOWED.**

(8) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect reworked area (ASTM E1417).

(9) After corrosion removal, using corrosion resistant coating, Item 118, Appendix D, touch up rework area.

5-4-5.1.5 Inspect Spindle Cuff Lug Bushings and Lug Area.

a. Inspect inside diameter (ID) of spindle cuff lug bushings for wear (Figure 5-4-5-5, View A-A). **MAXIMUM ID IS 1.0025-INCHES.** If wear exceeds limit, replace spindle cuff lug bushing (PARA 5-5-3).

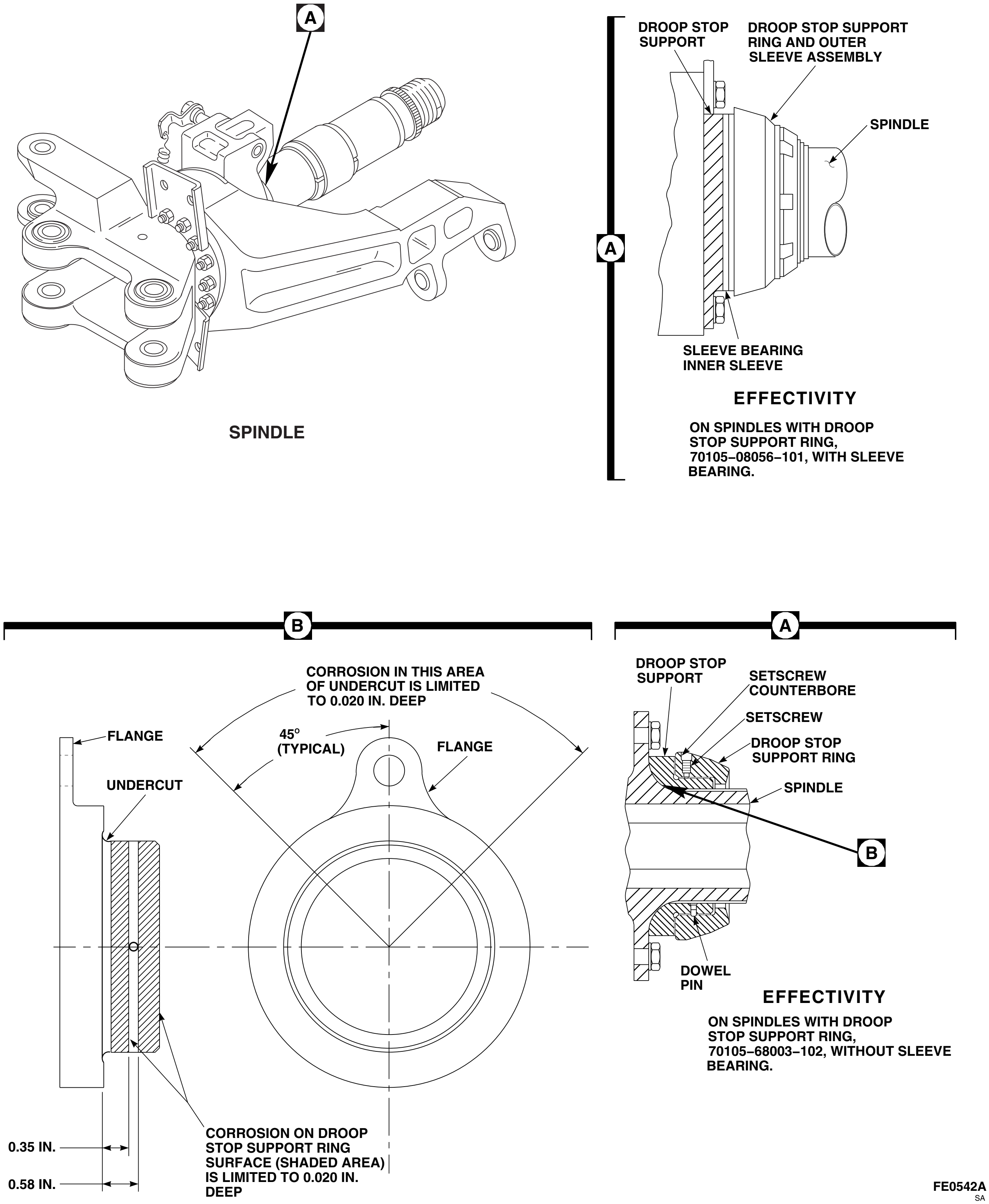


FIGURE 5-4-5-4. INSPECT DROOP STOP SUPPORT

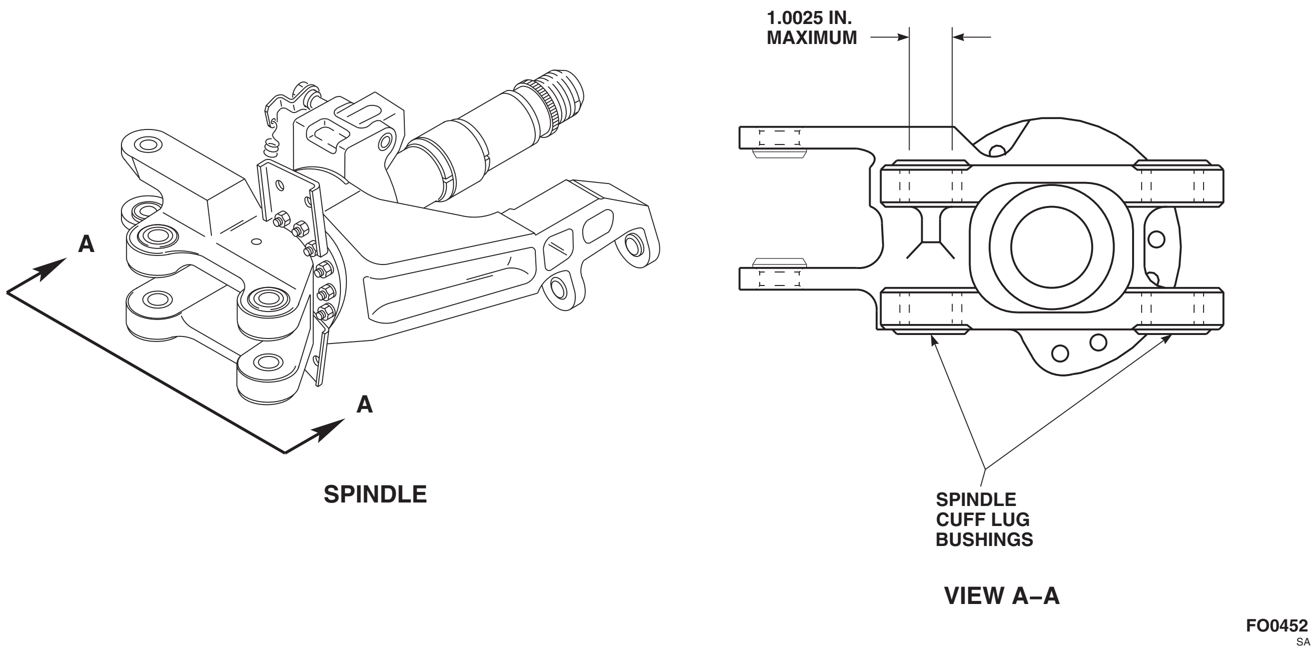


FIGURE 5-4-5-5. INSPECT SPINDLE CUFF LUG BUSHINGS AND LUG AREA

WARNING

Spindle cuff lug bushings are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds and eyes. To prevent injury, personnel must wear eye protection. Clean up visible dust piles and wash hands and arms after maintenance.

b. Using abrasive cloth, Item 10, Appendix D, repair spindle cuff lug bushings as follows:

(1) **BLEND NICKS, GOUGES, OR SCORING UP TO 0.005-INCH DEEP. TOTAL BLENDED AREA MUST NOT EXCEED 25% OF BORE AREA.**

(2) **BLEND CORROSION NO MORE THAN A MAXIMUM BUSHING ID OF 1.0025-INCHES.**

(3) If repairs exceed limits, replace spindle cuff lug bushings (PARA 5-5-3).

c. Inspect spindle cuff lug area for surface damage. Using abrasive wheel, Item 22, Appendix D, blend any damage found. **DAMAGE MUST NOT BE DEEPER THAN 0.020-INCH AFTER BLENDING.** If damage exceeds limit, replace spindle.

d. Inspect corners for surface damage. Using abrasive wheel, Item 22, Appendix D, blend any damage found. **DAMAGE MUST NOT BE DEEPER THAN 0.050-INCH AFTER BLENDING.** If damage exceeds limit, replace spindle.

5-4-5.1.6 Inspect Damper Shoulder Bushings.

a. Inspect wear on damper shoulder bushings where outboard end of damper attaches as follows (Figure 5-4-5-6, View A-A):

(1) Measure distance between shoulder of each damper shoulder bushing.

DISTANCE MUST NOT BE GREATER THAN 2.509-INCHES. If distance exceeds limit, replace spindle.

(2) Measure ID of damper shoulder bushing for wear. **ID OF EITHER DAMPER SHOULDER BUSHING MUST NOT BE GREATER THAN 0.7510-INCH.** If ID of either damper shoulder bushing exceeds limit, replace spindle.

b. Inspect for corrosion on damper shoulder bushings. **CORROSION ON CHAMFER OF DAMPER SHOULDER BUSHING MUST NOT BE GREATER THAN 0.040-INCH DEEP.** If corrosion on chamfer of damper shoulder bushing exceeds limit, replace spindle.

c. Using abrasive cloth, Item 12, Appendix D, blend out corrosion.

d. Apply coat of primer, Item 258, Appendix D, to blended area.

e. Touch up paint (PARA 2-6-5).

5-4-5.1.7 Inspect Journal Bearings.

NOTE

At least six intersections will be plotted. If any intersection falls into replace area of graph, replace component as applicable.

a. Inspect spindle journal bearing and Teflon journal bearing for wear as follows:

(1) Using inside micrometer caliper, measure ID of Teflon bearing, ½-inch in from both ends and at center at two places: one place at 12/6 o'clock position and one place at 3/9 o'clock position (Figure 5-4-5-7, Sheet 1).

(2) Using outside micrometer caliper, measure outside diameter (OD) of spindle bearing, ¾-inch in from both ends and at center at two places: one place at 12/6 o'clock position, and one place at 3/9 o'clock position.

(3) Enter each pair of measurements taken on graph using all clock positions, and both ends and center (Figure 5-4-5-7, Sheet 2).

(4) Find where intersection of readings is. Replace or keep sleeve bearing in elastomeric bearing assembly or spindle in use as determined by where intersection is on graph.

(5) Measure sleeve bearing in any other position where it seems to be worn. Measure mating sleeve bearings in same position. Enter these measurements on graph. Replace or keep sleeve bearing in elastomeric bearing assembly or spindle in use as determined by where intersection is on graph.

(6) If ID minus OD diameter is 0.017-inch or greater, remove spindle and spindle nut. Return as scrap.

b. Inspect spindle journal bearing for circumferential wear or score marks as follows:

NOTE

NO RAISED METAL ALLOWED. If raised metal is found, return spindle to Depot for replacement of bearing.

(1) Measure OD in wear area.

(2) Measure depth of score marks on inner race.

(3) Find intersection of measurements on graph (Figure 5-4-5-7, Sheet 3).

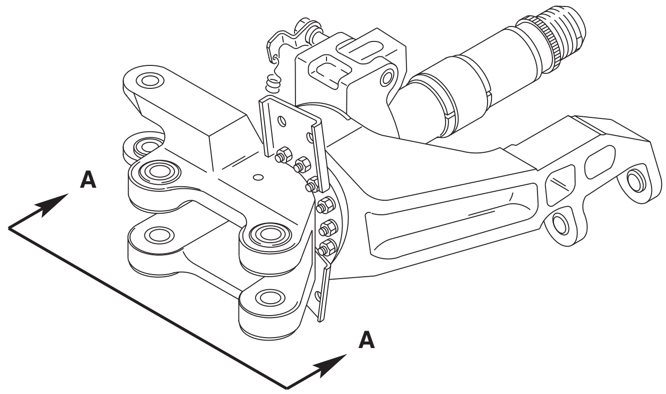
(4) Disposition spindle as determined by where intersection is on graph.

NOTE

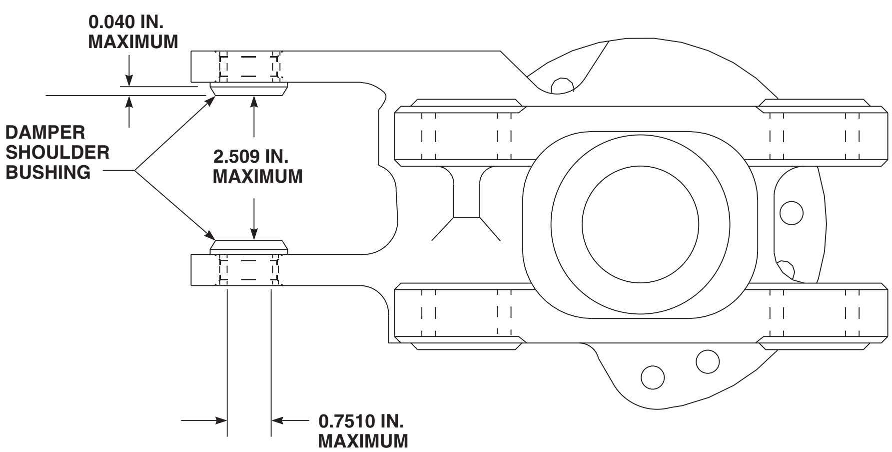
Some Teflon protrusion between races is not cause for rejection.

c. Inspect spindle journal bearings. **REPLACE SPINDLE AND/OR SLEEVE BEARING IF:**

(1) **BEARING IS CRACKED OR SHOWS SIGNS OF MOVEMENT ON SPINDLE**



SPINDLE



VIEW A–A

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FIGURE 5-4-5-6. INSPECT DAMPER SHOULDER BUSHINGS

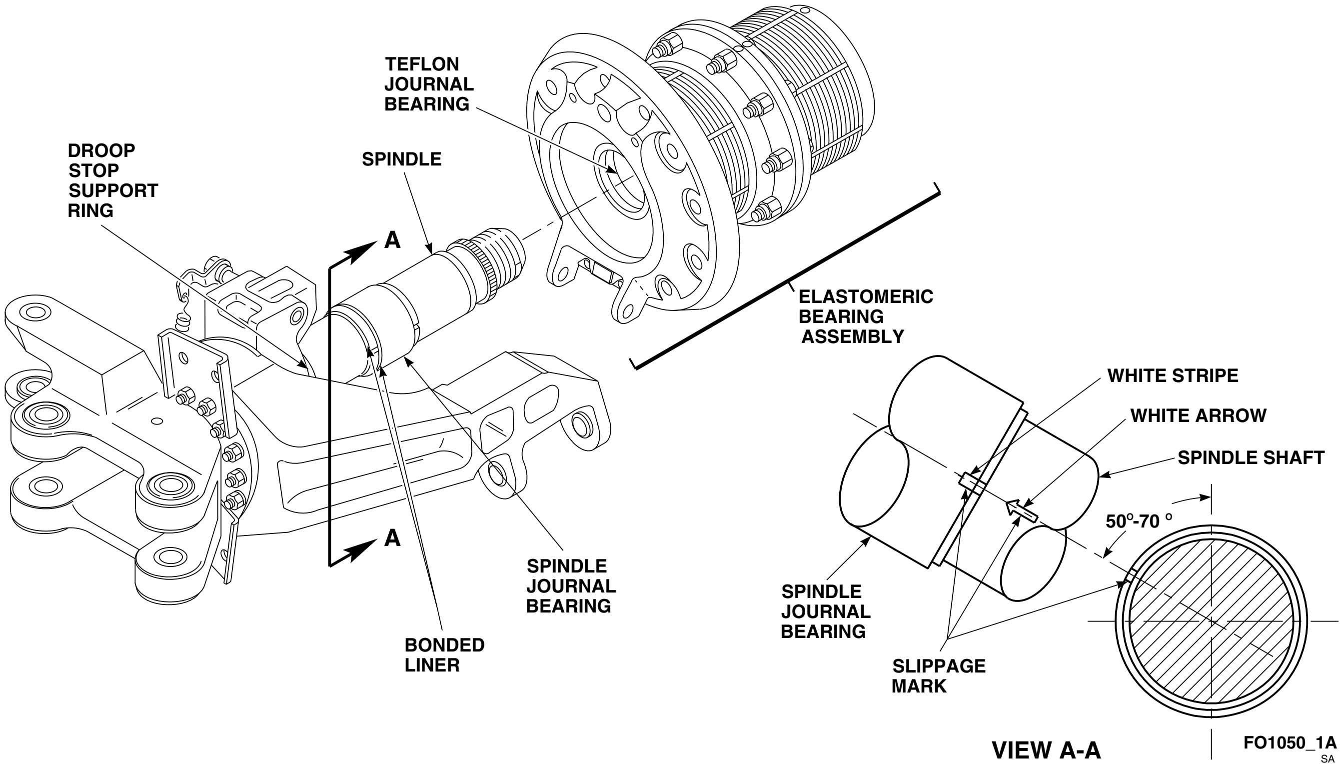


FIGURE 5-4-5-7. INSPECT JOURNAL BEARINGS (SHEET 1 OF 3)

SHAFT. Movement of bearing is identified by misalignment of slippage marks (white arrow and white stripe) (Figure 5-4-5-7, Sheet 1 and View A-A). If bearing displacement occurs, flight may be continued up to 10 hours after confirmation of slippage only under the following conditions:

(a) **BEARING HAS NOT DISENGAGED FROM OUTER RACE.**

(b) **BEARING HAS NOT MOVED BEYOND INBOARD CONTACT FACE OF TEFLON SLEEVE OR HAS BROKEN INBOARD TIEDOWN STRAP.**

(c) **FLIGHT IS ONE-TIME FLIGHT TO A SITE CAPABLE OF REPLACING SPINDLE.**

(d) **CONTINUANCE OF FLIGHT IS ALLOWED TO ENABLE HELICOPTER TO RETURN TO SITE WHERE REPAIRS CAN BE MADE.**

(2) **BEARING SHOWS SIGNS OF METAL-TO-METAL WEAR.**

(3) **DAMAGE ON ONE END OF BEARING EXCEEDS 3/8-INCH OR DAMAGE IS ON BOTH ENDS.**

d. Inspect spindle journal bearing for corrosion as follows:

(1) Using dishwashing compound, Item 122, Appendix D, and water, clean surface.

(2) Visually inspect bearing for corrosion pitting. **REPLACE SPINDLE IF:**

(a) **FIVE OR MORE PITS EXCEED 0.030-INCH OR GREATER IN DIAMETER IN 1/2-INCH DIAMETER CIRCLE.**

(b) **ANY PITS EXCEED 0.050-INCH IN DIAMETER.**

e. Inspect location of spindle journal bearing on spindle. **BEARING MUST NOT EXTEND BEYOND ENDS OF BONDED LINER** (Figure 5-4-5-7, Sheet 1). If bearing extends beyond ends of bonded liner, replace spindle.

f. If Teflon bearing becomes loose on spindle, rebond as follows:

(1) Remove shrink tubing and Teflon sleeve.

(2) Using paint remover, Item 230, Appendix D, and nonmetallic scraper, remove remaining adhesive from bonding areas of spindle and Teflon sleeve.

(3) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean bonding areas.

(4) Using adhesive, Item 66, Appendix D, apply to bonding area on spindle. Allow to cure for 1 hour at room temperature.

(5) Using etching solution, Item 136, Appendix D, apply to heat-shrinkable insulation sleeving, Item 179, Appendix D, and sleeve bearing bonding surfaces. Allow surfaces to turn brown in color. Using machinery towel, Item 211, Appendix D, remove etching solution.

(6) Apply adhesive, Item 33A, Appendix D, to spindle bonding area and ID of sleeve bearing.

(7) Install sleeve, butted against bonded liner.

CAUTION

Damage to droop stop support ring will result if adhesive is applied to droop stop support ring. Do not apply adhesive to droop stop support ring.

(8) Apply adhesive, Item 33A, Appendix D, to OD of sleeve bearing.

(9) Install insulation sleeving centered over sleeve bearing.

(10) Using heat gun, preshrink sleeving in place at 300° - 325°F (149° - 165°C).

(11) Allow adhesive to cure for 24 hours at 70° - 80°F (22° - 24°C), or using heat gun, for 1 hour at 140° - 160°F (60° - 71°C).

NOTE

Polishing of inner race (smooth finish) is acceptable if uniform. Reddish-brown powder on surface of either race is normal wear and not cause for replacement.

g. Inspect Teflon journal bearing in elastomeric bearing assembly. **BEARING MUST NOT SHOW SIGNS OF TEARS, RIPS, OR MISSING TEFLON MORE THAN ONE LAYER DEEP.** If bearing shows signs of tears, rips, or missing Teflon exceeding limit, replace bearing (PARA 5-5-4).

h. Inspect Teflon journal bearing assembly at 1:30/4:30 o'clock position and at 7:30/10:30 o'clock position for visible metal. **NONE ALLOWED.** If visible metal is found, replace bearing (PARA 5-5-4).

5-4-5.1.8 Inspect Spindle - General.

a. Inspect rest of spindle for damage (Figure 5-4-5-8). Using abrasive wheel, Item 22, Appendix D, blend out damage. **DAMAGE MUST NOT BE GREATER THAN 0.040-INCH AFTER REPAIR.** If damage exceeds limit, replace spindle.

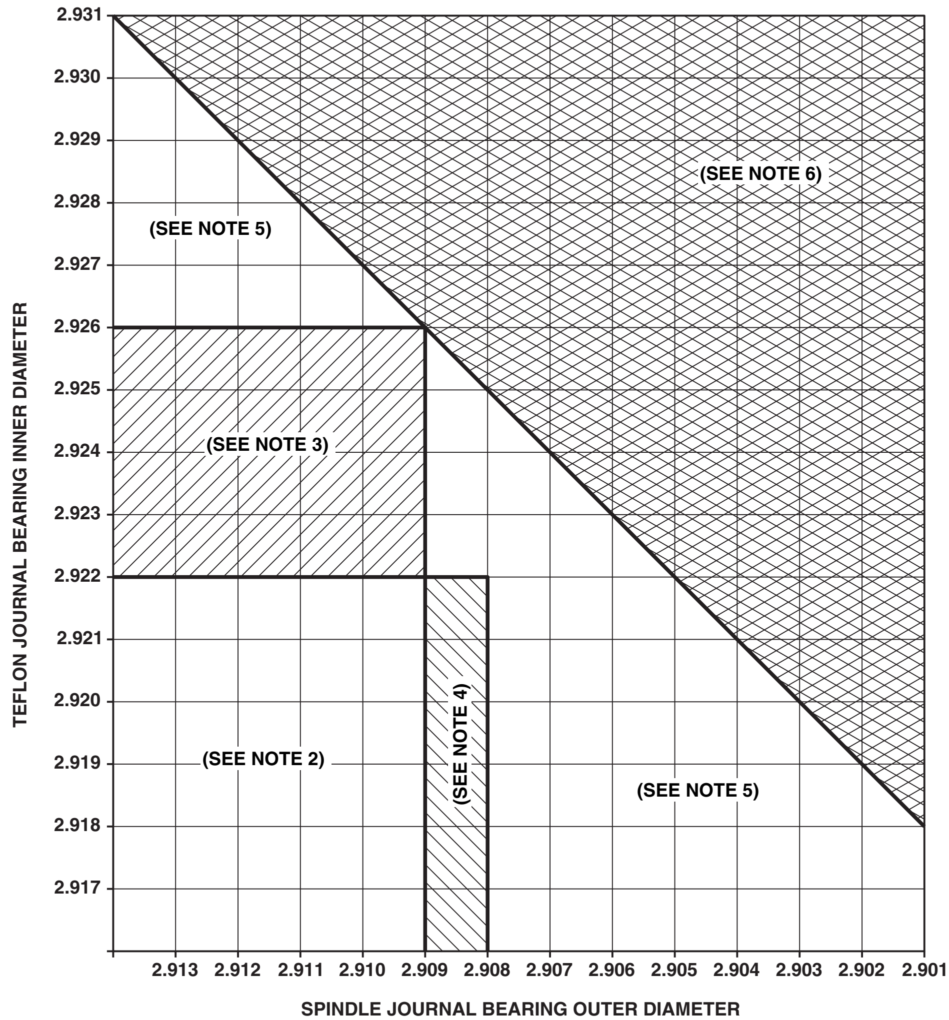
b. Inspect droop stop (PARA 5-3-3).

c. Inspect antiflap bracket (PARA 5-4-9).

d. **NO CRACKS ALLOWED IN OTHER AREAS OF SPINDLE.** If crack is found in other areas of spindle, replace spindle.

e. Inspect for corrosion. Using abrasive cloth, Item 12, Appendix D, blend any corrosion. **REPLACE SPINDLE IF:**

(1) **CORROSION ON MATING SURFACES COVERS MORE THAN**



(REPRODUCE LOCALLY)

NOTES

1. ALL MEASUREMENTS ARE IN INCHES.
2. REASSEMBLE SPINDLE AND ELASTOMERIC BEARING AND RETURN TO SERVICE.
3. RETURN ELASTOMERIC BEARING ASSEMBLY TO HIGHER LEVEL MAINTENANCE TO REPLACE TEFLON JOURNAL BEARING.
4. RETURN SPINDLE AND LINER ASSEMBLY TO HIGHER LEVEL MAINTENANCE TO REPLACE SPINDLE JOURNAL BEARING.
5. RETURN ELASTOMERIC BEARING ASSEMBLY AND SPINDLE AND LINER ASSEMBLY TO HIGHER LEVEL MAINTENANCE TO REPLACE SPINDLE AND TEFLON JOURNAL BEARINGS.
6. RETURN ELASTOMERIC BEARING ASSEMBLY TO HIGHER LEVEL MAINTENANCE TO REPLACE TEFLON JOURNAL BEARING. SCRAP SPINDLE AND LINER ASSEMBLY AND SPINDLE NUT.

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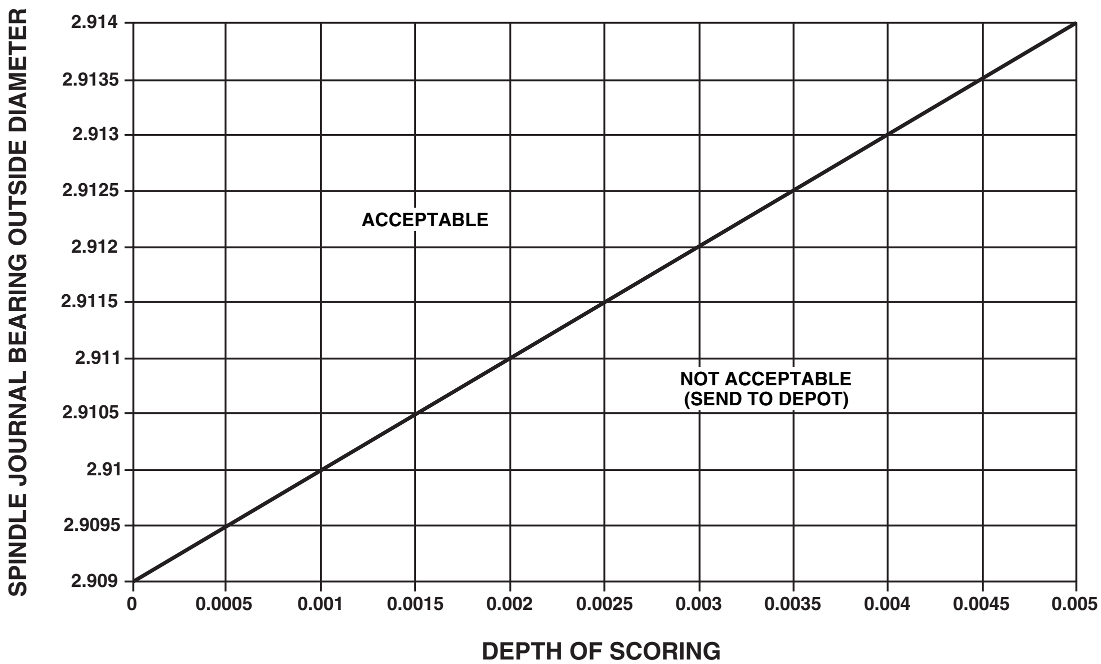
FIGURE 5-4-5-7. INSPECT JOURNAL BEARINGS (SHEET 2 OF 3)



NOTE
ALL MEASUREMENTS ARE IN INCHES, UNLESS NOTED.

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FIGURE 5-4-5-7. INSPECT JOURNAL BEARINGS (SHEET 3 OF 3)

10% OF SURFACE, OR IS DEEPER THAN 0.010-INCH AFTER REMOVAL.

(2) **CORROSION ON OPEN AREAS IS DEEPER THAN 0.030-INCH AFTER REMOVAL.**

5-4-5.1.9 Inspect Spindle Nut.

a. Inspect threads of spindle nut for condition of solid film lubricant (Figure 5-4-5-9). **REPLACE SPINDLE NUT IF:**

(1) **THREADS OF SPINDLE NUT ARE MISSING MORE THAN 50% SOLID FILM LUBRICANT.**

(2) **THREADS OF SPINDLE NUT ARE MISSING SOLID FILM LUBRICANT NOT EXCEEDING 50% OF SURFACE AREA AND SPINDLE NUT HAS PREVIOUSLY BEEN TOUCHED UP.**

b. If metal is showing on threads of spindle nut, and spindle nut was not touched up previously, using solid film lubricant, Item 201, Appendix D, apply up to 50% of surface area as follows:

(1) Using dry machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean areas of bare metal. Allow to dry thoroughly.

NOTE

Solid film lubricant will not adhere to contaminated surfaces. Do not touch cleaned areas with bare hands. Use clean cotton gloves.

(2) At room temperature, using solid film lubricant, Item 201, Appendix D, spray exposed area. Apply multiple thin coats allowing 15 - 20 minutes drying time between coats.

(3) Allow solid film lubricant to cure for a minimum of 12 hours before using spindle nut.

c. Visually inspect threads of spindle nut for condition of solid film lubricant. **REPLACE SPINDLE NUT IF:**

(1) **COATING IS NOT "SLATE-BLACK" AND UNIFORM IN COLOR.**

(2) **COATING HAS ANY RUNS, BLISTERS, ROUGHNESS, OR SEPARATION OF INGREDIENTS.**

(3) **COATING HAS ANY OTHER SURFACE IMPERFECTIONS.**

d. After solid film lubricant has cured, verify repair as follows:

(1) Firmly press a strip of 1-inch masking tape, Item 212, Appendix D, onto surface of touched up area.

(2) Remove masking tape with an abrupt motion. Inspect masking tape as follows:

(a) **A UNIFORM DEPOSIT OF POWDERY MATERIAL CLINGING TO TAPE IS ACCEPTABLE.**

(b) **SOLID FILM LUBRICANT THAT FLAKES OR LEAVES ANY BARE METAL EXPOSED IS UNACCEPTABLE.**

(3) If repair of solid film lubricant is successful, note component serial number, percentage of area touched up, and total time on part, in helicopter logbook. If repair does not meet criteria, replace spindle nut.

5-4-5.1.10 Inspect Droop Stop Support Ring and Nut.

a. On spindles with droop stop support ring, 70105-08056-101, with sleeve bearing, perform the following (Figure 5-4-5-10, Detail A):

(1) Inspect sleeve bearing outer sleeve for nicks, gouges, and burrs. **DAMAGE MUST NOT EXCEED 0.005-INCH DEEP.** If nicks, gouges, and burrs exceeds limit, replace sleeve bearing outer sleeve (PARA 5-5-2).

(2) Inspect sleeve bearing outer sleeve for corrosion pitting. **PITTING MUST NOT EXCEED 0.005-INCH.** If pitting exceeds limit, replace sleeve bearing outer sleeve (PARA 5-5-2).

(3) Inspect mating surface of droop stop support ring for nicks, gouges, and scratches. **DAMAGE MUST NOT EXCEED 0.002-INCH DEEP.** If damage exceeds limit, replace droop stop support ring.

(4) Inspect corners of droop stop support ring for nicks, gouges, and scratches. Using abrasive cloth, Item 4, Appendix D, blend out damage in not less than ½-inch radius. **DAMAGE AFTER REPAIR MUST NOT EXCEED 0.020-INCH DEEP.** If damage after repair exceeds limit, replace droop stop support ring.

(5) Inspect droop stop nut for cracks and corrosion.

(6) **IF RADIAL CRACK IS FOUND, REPLACE DROOP STOP NUT.**

(7) **IF CIRCUMFERENTIAL CRACK ON LUG EXCEEDS ¾ OF CIRCUMFERENTIAL WIDTH OF LUG, REPLACE DROOP STOP NUT.** If replacement droop stop nut is not available, perform the following:

(a) Using pliers, remove outer portion of damaged lug by breaking remaining section.

(b) Using abrasive cloth, Item 4, Appendix D, clean fractured area.

(c) Using corrosion resistant compound, Item 117, Appendix D, touch up reworked area.

(d) Apply coat of primer, Item 258, Appendix D, to reworked area.

(e) Replace droop stop nut within 500 flight hours.

(8) If corrosion is found on droop stop nut, perform the following:

(a) Using abrasive cloth, Item 4, Appendix D, remove corrosion. **IF CORROSION IS DEEPER THAN 0.035-INCH AFTER REMOVAL, REPLACE DROOP STOP NUT.**

(b) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean repaired area.

(9) Apply coat of primer, Item 258, Appendix D, to repaired area.

b. Inspect retaining ring on droop stop nut for damage. If damage is found, replace retaining ring.

5-4-5.1.11 Inspect Elastomeric Bearing Assembly.

a. Inspect elastomeric bearing assembly (PARA 5-3-1).

b. Inspect droop stop pad on elastomeric bearing assembly for damage or disbond (Figure 5-4-5-11). If damaged or disbonded, replace droop stop pad as follows:

(1) Using dry ice, Item 125, Appendix D, chill area of elastomeric bearing assembly with droop stop pad adhered to weaken bond.

(2) Remove elastomeric bearing assembly from dry ice.

(3) Using nonmetallic scraper and soft-faced mallet, carefully remove droop stop pad from elastomeric bearing assembly.

(4) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean area to be bonded.

(5) Using abrasive cloth, Item 4, Appendix D, lightly scuff bonding surface of droop stop pad.

(6) Apply light coat of primer, Item 254, Appendix D, to bonding area of droop stop pad and elastomeric bearing assembly. Let primer dry thoroughly at room temperature for at least 3 minutes.

(7) Apply bead of adhesive, Item 64, Appendix D, to droop stop pad and position 1 1/4-inches in from droop stop lugs on elastomeric bearing assembly (Figure 5-4-5-11, Detail A).

(8) Position and clamp droop stop pad to elastomeric bearing assembly, and let dry for at least 15 minutes.

5-4-5.1.12 Assemble.

WARNING

FSCAP

The spindle thrust bearing retaining nut, threads of spindle, thread relief (undercut) area next to threads, radius at antiflap mounting bracket and thread relief are critical surfaces which must not be damaged during assembly. After removal of protective covering from spindle, use extreme caution when assembling spindle.

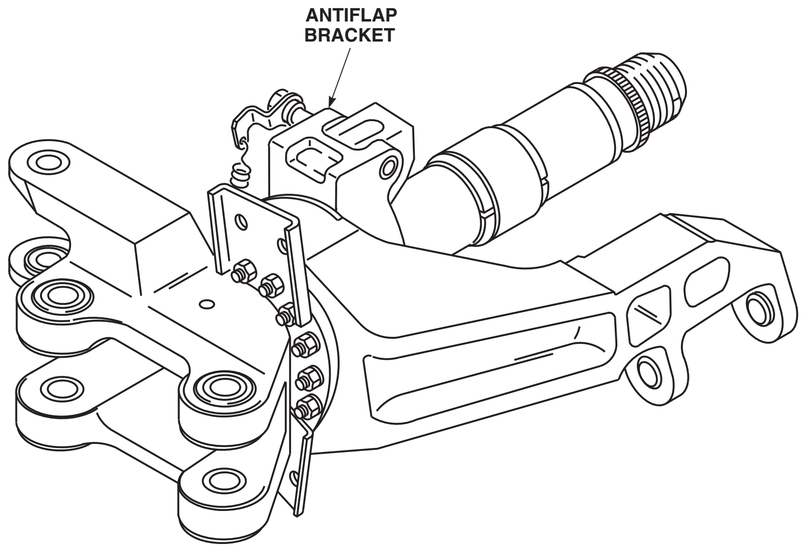
a. Remove protective covering from threads of spindle, thread relief (undercut), and shank.

b. On spindles with droop stop support ring, 70105-08056-101, with sleeve bearing, install droop stop support ring and outer sleeve assembly as follows (Figure 5-4-5-12, Detail A):

(1) Apply coat of primer, Item 258, Appendix D, to threads of droop stop support and face of droop stop nut.

(2) Slide droop stop support ring and outer sleeve assembly onto sleeve bearing inner sleeve.

(3) Slide Teflon-lined ring, with Teflon-lining facing droop stop support ring and outer sleeve assembly, against droop stop support ring.



NOTE

DROOP STOP CAM / ASSEMBLY NOT SHOWN.

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FIGURE 5-4-5-8. INSPECT SPINDLE - GENERAL

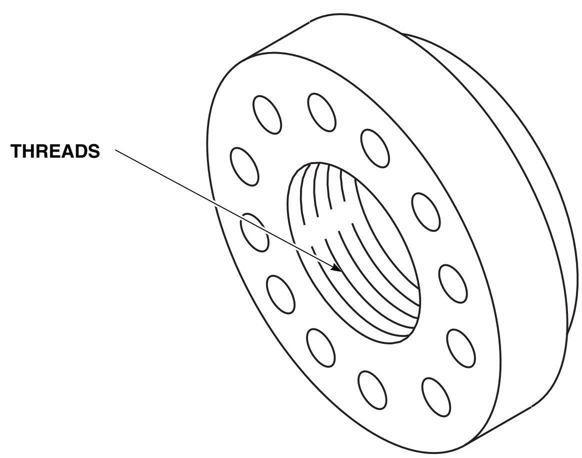

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FIGURE 5-4-5-9. INSPECT SPINDLE NUT

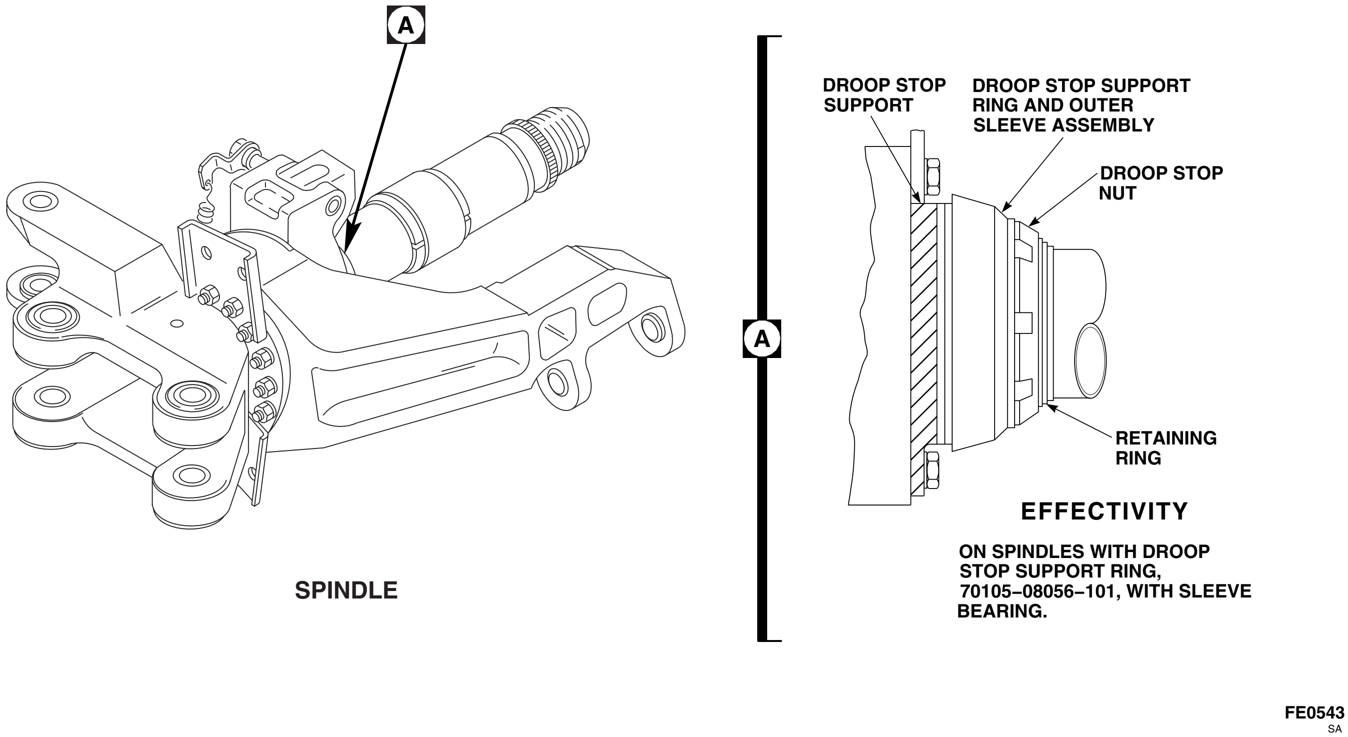


FIGURE 5-4-5-10. INSPECT DROOP STOP SUPPORT RING AND NUT

(4) Thread droop stop nut on droop stop support while primer is still wet.

(5) **USING DROOP STOP NUT WRENCH, TORQUE DROOP STOP NUT TO 570 - 630 INCH-POUNDS.**

(6) Using machinery towel, Item 211, Appendix D, wipe off excess primer from droop stop support and droop stop nut.

(7) Install retaining ring.

(8) Apply sealing compound, Item 292, Appendix D, to seal around droop stop nut.

c. On spindles with droop stop support ring, 70105-68003-102, without sleeve bearings, install droop stop support ring as follows (Figure 5-4-5-12, Detail A):

(1) Position droop stop support ring on spindle. Push support ring onto droop stop support, rotating slightly to line up slot and dowel pin, until fully seated.

(2) Rotate droop stop support ring until setscrew hole faces 180° away from dowel pin.

(3) Holding support ring against support, install setscrew until ring lifts off droop stop support.

(4) Back out setscrew one full turn.

(5) Apply sealing compound, Item 292, Appendix D, to fill setscrew counterbore. Using machinery towel, Item 211, Appendix D, wipe off excess sealing compound from droop stop support ring.

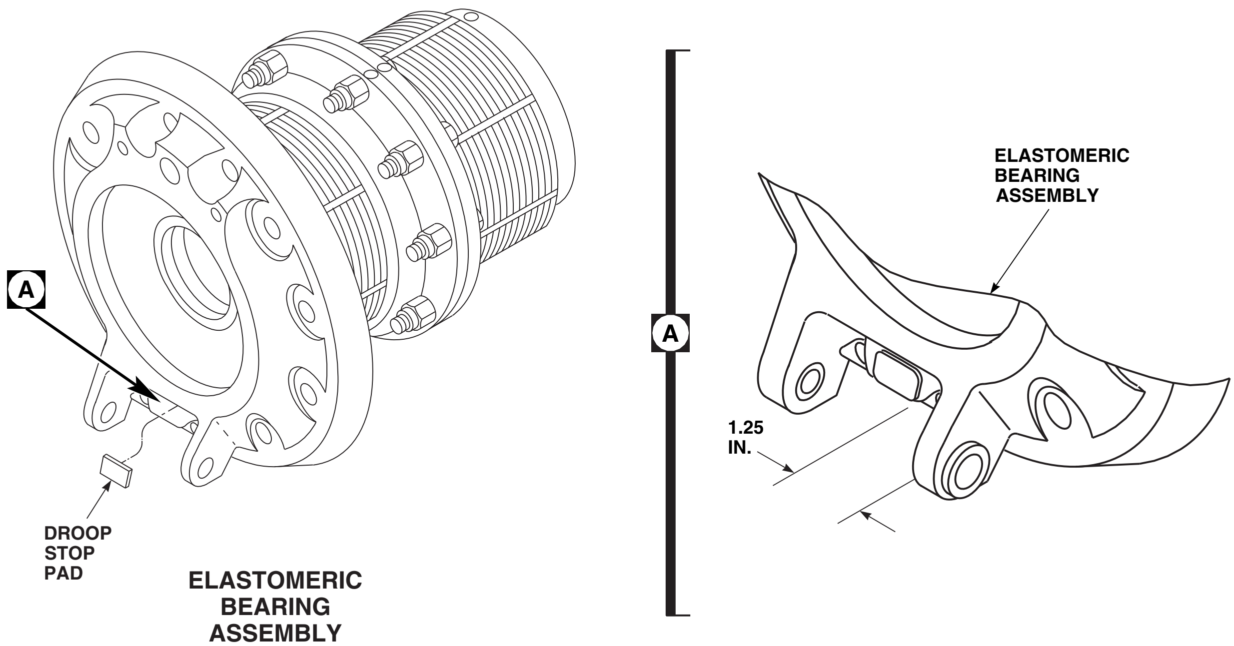


FIGURE 5-4-5-11. INSPECT ELASTOMERIC BEARING ASSEMBLY

WARNING

Injury to personnel and damage to equipment will result if elastomeric bearing assembly is installed without sleeve bearing outer race installed in ID of spherical bearing. Make sure sleeve bearing outer race is installed in spherical bearing prior to installing elastomeric bearing assembly.

NOTE

Teflon journal bearing is 2 ¼-inches long with Teflon fabric lining on ID.

d. Visually inspect elastomeric bearing assembly for presence of Teflon journal bearing in ID of spherical bearing (Figure 5-4-5-12). If bearing is not installed, install bearing (PARA 5-5-4).

e. Inspect journal bearings (PARA 5-4-5.1.7).

WARNING

FSCAP

The spindle thrust bearing retaining nut, threads of spindle, thread relief (undercut) area next to threads, radius at antiflap mounting bracket and thread relief are critical surfaces which must not be damaged during assembly. After removal of protective covering from spindle, use extreme caution when assembling spindle.

CAUTION

- If any oil, grease, or other fluids are spilled or accidentally get on elastomeric bearings, do not use solvent to remove them. Use only warm water, mild detergent, and a soft-bristle brush to clean elastomeric bearings. After cleaning, rinse with clear water, and then dry.

- To prevent damage to Teflon bearing outer race, use extreme care when sliding thrust bearing onto spindle.

NOTE

A white wax-like substance on the rubber surface is normal and should not be removed. If some of this white coating comes off during cleaning, it is not cause for rejection of elastomeric bearing assembly.

f. Line up wide tooth spline on thrust bearing with wide tooth on spindle, and slide bearing assembly on spindle.

g. Install spindle nut on spindle. Measure and adjust spindle nut. **DISTANCE FROM INBOARD END FACE OF SPINDLE NUT TO INBOARD END FACE OF RETAINING ROD MUST BE 1.160 - 1.180-INCHES** (Figure 5-4-5-12, Detail B).

h. Apply coat of primer to shanks, threads, and area under spindle nut attachment boltheads using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D.

i. Install bolts and washers securing spindle nut to thrust bearing while primer is still wet (Figure 5-4-5-12).

j. **TORQUE BOLTS TO 119 - 131 INCH-POUNDS.**

k. Using lockwire, Item 197, Appendix D, lockwire bolts.

l. Wipe off any primer that has seeped from under boltheads.

m. Using clamp supplied with dial indicator, secure dial indicator to spindle horn (Figure 5-4-5-12, Detail C and Detail D).

n. Measure movement between spindle spline and thrust bearing spline at spindle nut locking bolts. **MAXIMUM MOVEMENT IS 0.010-INCH.** If movement exceeds limit, replace spindle.

o. Apply sealing compound, Item 292, Appendix D, to seal boltheads and gap between thrust bearing and spindle nut (Figure 5-4-5-12, Detail B).

5-4-5.1.13 Install.

NOTE

Before installing new spindle onto main rotor hub, make sure preservative grease is removed.

a. If necessary, using warm, soapy water and clean cloth, remove preservation grease from mating flanges.

b. Inspect depth of main rotor hub threaded inserts. Threaded inserts installed to a maximum depth of 0.137-inch below face are acceptable. If any threaded insert is installed exceeding a depth of 0.137-inch, replace threaded insert (PARA 5-4-3).

c. Inspect main rotor hub bonded liner in spindle attach area (PARA 5-4-2).

d. Inspect bolts securing elastomeric bearing endplate to main rotor hub (PARA 5-4-4).

e. Inspect breakaway torque of spindle attachment bolts as follows:

NOTE

On spindles with droop stop support ring, 70105-68003-102, spindle attachment bolts securing lead stop bracket are longer than remaining spindle attachment bolts. Refer to PARA 5-4-6 for location of spindle attachment bolts securing lead stop bracket.

(1) On spindles with droop stop support ring, 70105-68003-102, thread in spindle attachment bolts securing lead stop bracket until boltheads are 0.800-inch away from surface of main rotor hub.

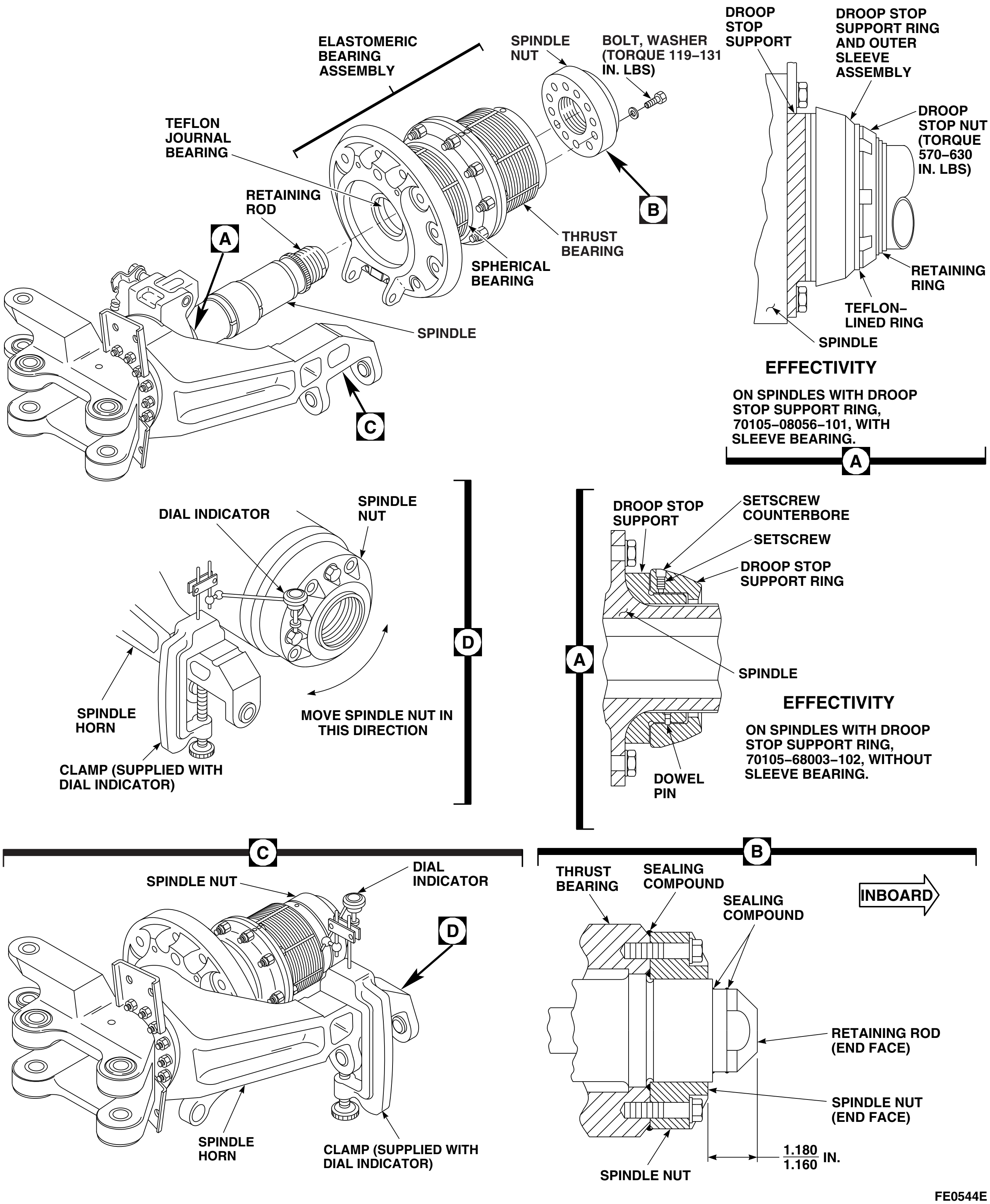


FIGURE 5-4-5-12. ASSEMBLE SPINDLE

(2) Thread in remaining bolts until boltheads are 0.610-inch away from surface of main rotor hub.

(3) Using dial torque wrench, measure torque required to breakaway and remove bolts. **MINIMUM BREAKAWAY TORQUE IS 14 INCH-POUNDS. MAXIMUM REMOVAL TORQUE IS 150 INCH-POUNDS.**

f. If breakaway or removal torque of spindle attachment bolts exceeds limits, perform the following:

(1) Inspect bolt threads for evidence of corrosion or wear. **NONE ALLOWED.** If corrosion or wear is found, replace bolts.

(2) Inspect threaded insert for cleanness. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean threaded insert.

(3) Repeat step e. If breakaway or removal torque still exceeds limits, replace threaded insert (PARA 5-4-3).

g. If not already done, if replacement spindle will be installed and previous spindle has balance weights installed, perform the following (Figure 5-4-5-13, Sheet 1, Detail B):

(1) Remove bolts, washers, balance weights, and nuts from balance bracket(s).

(2) Tag parts.

(3) Keep these parts for installation in new spindle.

h. Inspect and treat balance weights and bolts (PARA 1-7-20).

i. Install balance weights as follows:

NOTE

Make sure balance weights are installed in original location on replacement spindle.

(1) Remove tag.

(2) Slide attaching washer and balance weights on bolts.

(3) Install bolt in original location on balance bracket (with bolthead facing center of main rotor).

(4) Secure with washer and nut.

NOTE

Make sure sealing compound is applied to boltheads and nuts after main rotor track and balance is performed.

(5) **TORQUE NUT TO 261 - 289 INCH-POUNDS.**

j. If replacement spindle is being installed, install bonding jumper as follows (Figure 5-4-5-13, Sheet 1, Detail C and Sheet 2, Detail D):

(1) Apply coat of primer, Item 258, Appendix D, to shank, threads, and area under short bolthead.

(2) Install short bolt, washer, nut, and bonding jumper, while primer is still wet. Omit washer between bonding jumper and nut. Position bonding jumper to 45° angle facing up.

(3) **TORQUE NUT TO 304 - 336 INCH-POUNDS.**

CAUTION

To prevent damage to elastomeric bearings, wipe off grease from exterior surfaces of bearings, using warm, soapy water and clean cloth.

k. Remove protective covering from spindle.

l. Attach hoisting sling around spindle and adjust straps. Attach hoisting sling to hoist (Figure 5-4-5-13, Sheet 1, Detail A).

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when handling spindle. Keep area clear when hoisting spindle.

m. Lift and slide spindle on main rotor hub, with spindle horn facing direction of rotation. Line up offset hole in spindle with offset hole in main rotor hub.

n. Apply coat of primer to shank, threads, and area under spindle attachment boltheads using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D.

o. Secure spindle to main rotor hub as follows:

WARNING

FSCAP

Verification of proper hardware and spring clip installation and torque of bolts are critical characteristics.

(1) Install bolts, washers, and blade deice harness bracket around upper half of spindle (12 o'clock position) while primer is still wet (Figure 5-4-5-13, Sheet 2, Detail E and Detail F).

(2) Remove hoisting sling.

(3) On spindles with droop stop support ring, 70105-68003-102, install spindle attachment bolts securing lead stop bracket (PARA 5-4-6).

NOTE

Spring clips (no washers) will go on lower outer bolts.

(4) Install remaining bolts, washers, spring clips, and blade deice harness bracket on spindle (9 o'clock position) while primer is still wet (Figure 5-4-5-13, Sheet 2, Detail F and Detail G).

NOTE

There may not be enough wrench clearance to install bolt behind spindle horn and antiflap assembly.

(5) If necessary, insert piece of wood, 2-inches × 4-inches × about 3-feet long, between spindle cuff lugs and turn spindle.

(6) Make sure top blade deice harness bracket (12 o'clock position) is positioned vertically to prevent blade deice harness from chafing on damper (Figure 5-4-5-13, Sheet 2, Detail E).

(7) **TORQUE ALL BOLTS TO 910 - 1010 INCH-POUNDS IN CRISSCROSS SEQUENCE** (Figure 5-4-5-13, Sheet 2, Detail E, Detail F, and Detail G).

p. Using machinery towel, Item 211, Appendix D, wipe off excess primer.

q. Fill gap between spindle and hub with sealing compound, Item 292, Appendix D.

r. Install droop stop (PARA 5-4-10).

s. Connect spring from spring clip to droop stop.

t. Install antiflap stop (PARA 5-4-9).

u. Connect damper to spindle (PARA 5-4-13).

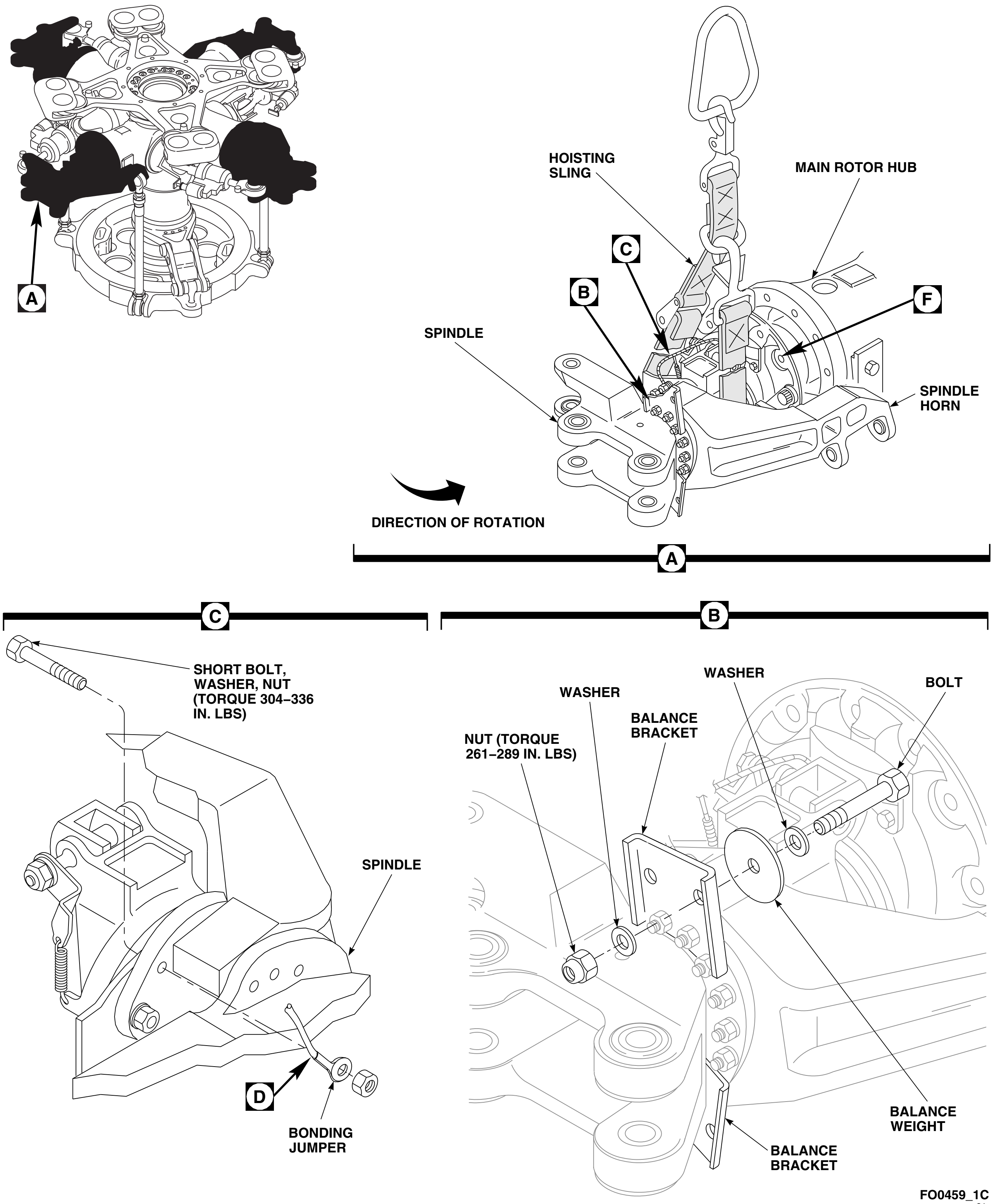


FIGURE 5-4-5-13. INSTALL SPINDLE (SHEET 1 OF 2)

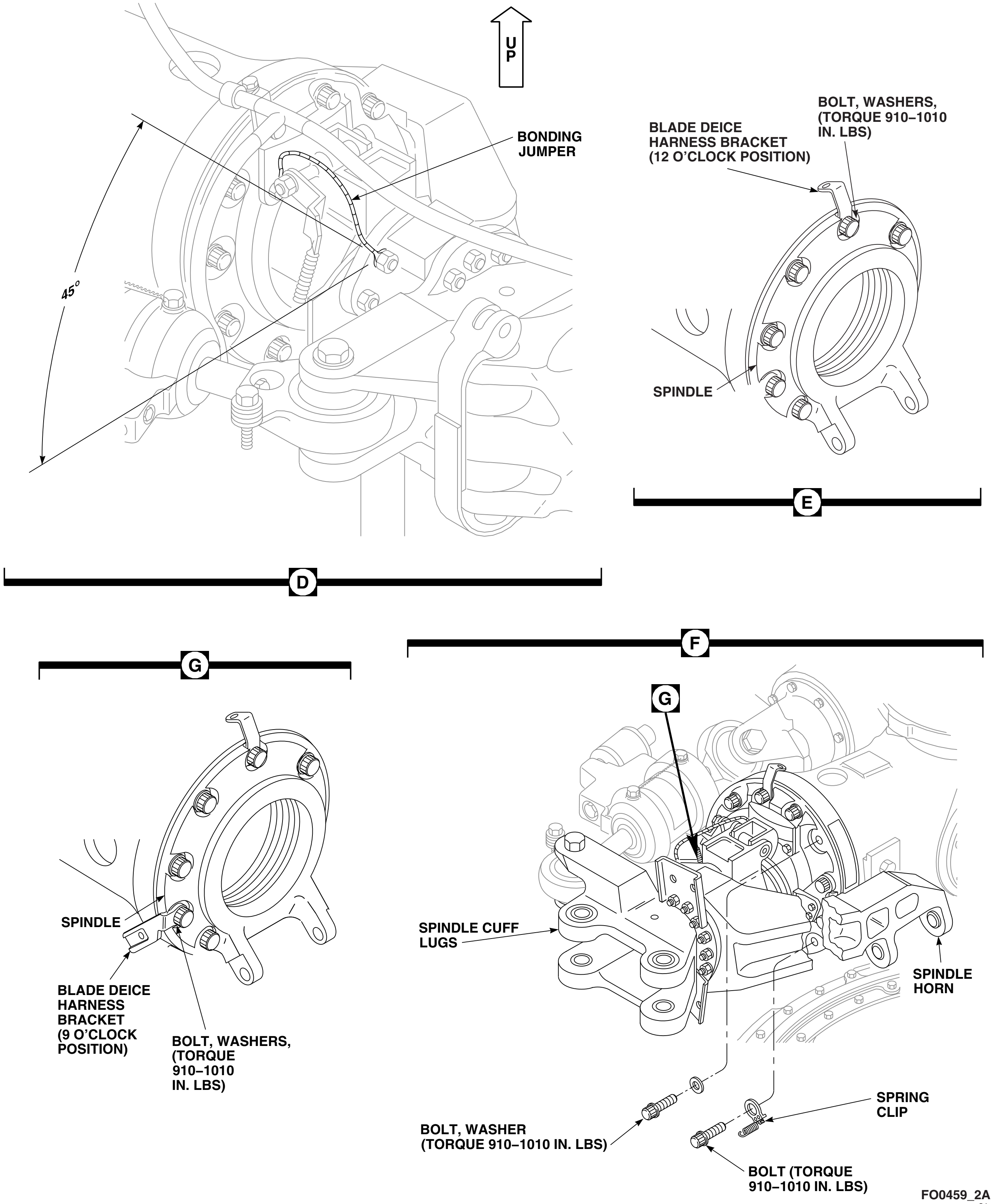


FIGURE 5-4-5-13. INSTALL SPINDLE (SHEET 2 OF 2)

WARNING

FSCAP

The pitch control rod attachment areas, and critical areas from spindle horn lugs to elbow are critical surfaces which must not be damaged during installation of spindle horn and/or pitch control rod. Use extreme caution when installing spindle horn and attaching hardware.

CAUTION

Damage to pitch control rod and spindle horn will result if nylon washer is not installed properly. Make sure nylon washer is on opposite side of bearing snapring.

NOTE

- It may be necessary to gently rock spindle horn to make it easy to line up pitch control rod bolthole.
- Make sure pitch control rod bolthead faces outboard.

v. Connect upper end of pitch control rod (PARA 5-4-18).

w. Touch up paint exposed boltheads, nuts, and threads (PARA 2-6-5).

x. Install main rotor blade on spindle (PARA 5-4-31).

y. Make sure area is clean and free of foreign material.

z. Check helicopter logbook to make sure spindle attachment bolts and damper attachment bolts retorque is shown due 9 - 11 flight hours after installation (PARA 1-7-5).

aa. Check helicopter logbook to make sure spindle sleeve bearing inspection is shown due 9 - 11 flight hours after installation (PARA 1-7-6).

NOTE

Spindle horn attachment nut torque check is only required if new or refurbished spindle is installed, or if it is unknown if the spindle horn was previously removed.

ab. Check helicopter logbook to make sure spindle horn attachment nuts torque check is shown due 9 - 11 flight hours (PARA 1-7-5).

ac. Check antiflap cam for free rotation. If antiflap cam contacts antiflap stop with main rotor blades installed and main rotor head in static position, perform the following:

(1) Run up helicopter to 100% rotor speed (Nr) in flat pitch for 15 minutes (Appropriate Flight Manual).

(2) Shut down helicopter and reinspect for antiflap cam contact. If antiflap cam contact continues, replace elastomeric bearing assembly (PARA 5-5-4).

ad. If required, disengage gust lock (PARA 1-6-18).

ae. Perform main rotor blade track and balance (TM 1-70-VIB).

af. Apply bead of sealing compound, Item 292, Appendix D, around boltheads and nuts securing balance weights to balance bracket.

5-4-6 LEAD STOP.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage
- Heat Gun
- Nonmetallic Scraper, Locally-Made
- Phenolic Wedge, Locally-Made
- Torque Wrench, Dial, 0 - 600 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Adhesive, Item 32, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D

References

- Appendix D
- PARA 1-6-18
- PARA 2-6-5
- PARA 5-4-4

5-4-6.1 Replace Lead Stop and Bracket.

NOTE

Replacement of all lead stops and brackets is the same.

5-4-6.1.1 Remove.

a. Turn main rotor head until lead stop is easy to reach.

b. If not already done, engage gust lock and rotor brake (PARA 1-6-18).

c. Remove bolts and washers securing lead stop to lead stop shim and lead stop bracket (Figure 5-4-6-1, Detail A).

NOTE

Do not discard lead stop shim. Retain for installation.

d. Remove lead stop and lead stop shim from lead stop bracket.

e. Using nonmetallic scraper, remove sealing compound from lead stop bracket bolts.

f. Remove bolts securing lead stop bracket and shims to elastomeric bearing assembly endplate.

g. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper, remove any remaining sealing compound from bolts.

h. Using phenolic wedge, remove lead stop bracket and shims from elastomeric bearing assembly endplate. Discard shims.

i. Using paint remover, Item 230, Appendix D, and nonmetallic scraper, remove any remaining adhesive from elastomeric bearing assembly endplate and lead stop bracket.

5-4-6.1.2 Inspect/Repair.

a. Inspect lead stop bracket for cracks. **NONE ALLOWED.** If crack is found, replace lead stop bracket.

b. Inspect lead stop shim for corrosion. **NONE ALLOWED.** If corrosion is found, remove as follows:

(1) Using abrasive cloth, Item 10, Appendix D, remove corrosion from lead stop shim.

(2) Apply primer, Item 258, Appendix D, to repaired area.

c. Inspect lead stop bracket/elastomeric bearing assembly endplate bolts (PARA 5-4-4).

5-4-6.1.3 Install.

a. Check breakaway and removal torque of bolts as follows:

(1) Thread in bolts until clearance of ¾-inch is obtained between underside of bolthead and elastomeric bearing assembly endplate.

(2) Using torque wrench, back off bolt and note breakaway and maximum removal torque. **MINIMUM BREAKAWAY TORQUE MUST BE GREATER THAN 14 INCH-POUNDS. MAXIMUM REMOVAL TORQUE MUST NOT EXCEED 150 INCH-POUNDS.**

(3) If torque exceeds limits, perform the following:

(a) Replace bolt.

(b) Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean threaded insert.

(c) Repeat step a. (1) and step a. (2).

(d) If torque still exceeds limits, replace threaded insert (PARA 5-4-4).

b. Install shims and lead stop bracket on elastomeric bearing assembly endplate and secure with bolts (Figure 5-4-6-1, Detail A).

c. Using feeler gage, measure gap between lead stop bracket and elastomeric bearing assembly endplate. Measure gap near bolthole. **GAP MUST BE 0.000 - 0.003-INCH** (Figure 5-4-6-1, Detail B). If gap exceeds limits, perform the following:

(1) Remove bolts, lead stop bracket, and shims (Figure 5-4-6-1, Detail A).

(2) Peel shim, SS52F0510Z0875, as necessary to allow for 0.000 - 0.003-inch gap.

(3) Repeat step b. and step c. until 0.000 - 0.003-inch gap is obtained.

d. After gap is obtained, remove bolts, lead stop bracket, and shims.

e. Apply adhesive, Item 32, Appendix D, between mating surface of lead stop bracket and elastomeric bearing assembly endplate.

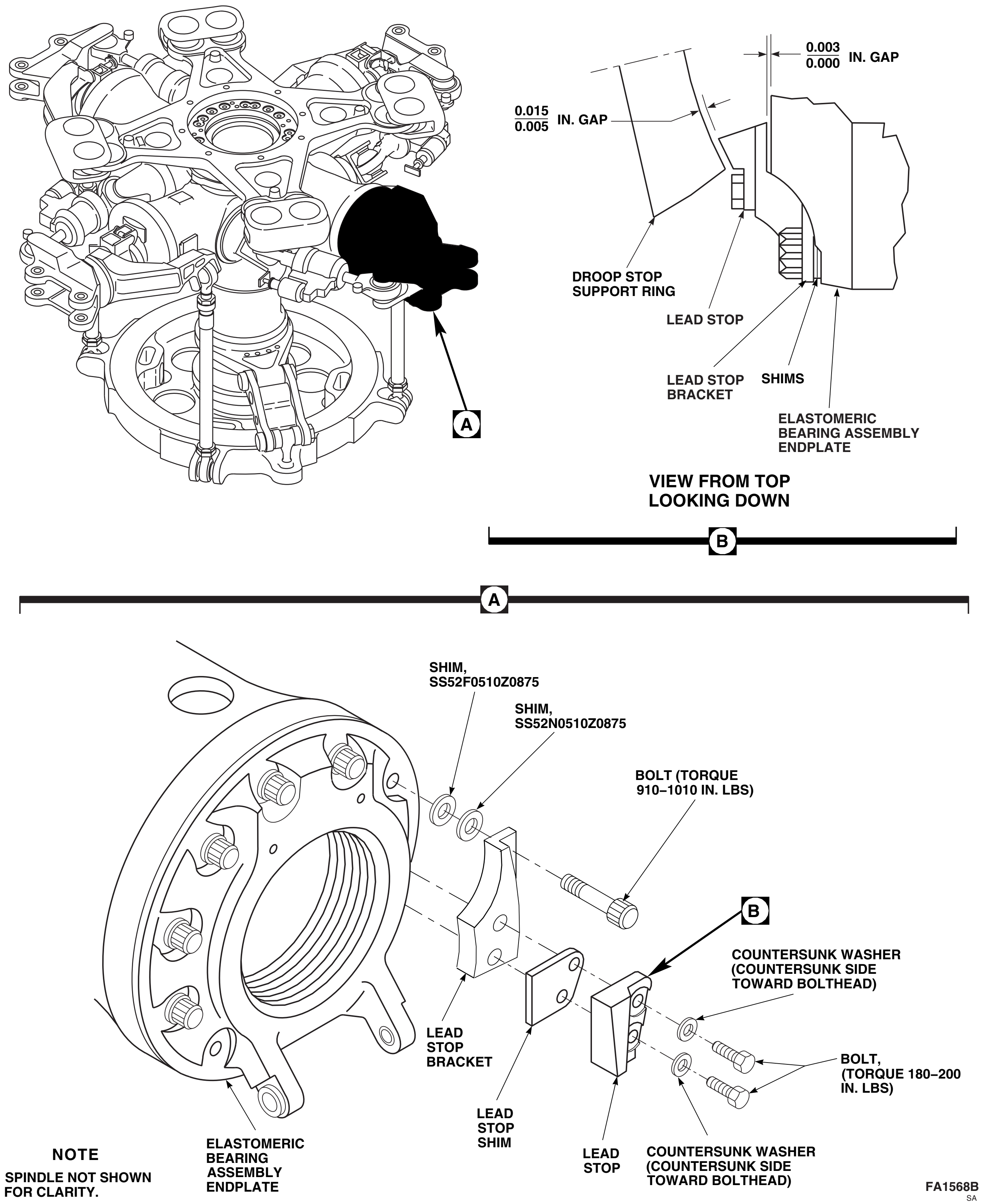


FIGURE 5-4-6-1. REPLACE LEAD STOP AND BRACKET

f. Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer to shanks, threads, and area under boltheads. Install bolts while primer is still wet.

WARNING

FSCAP

Verification of proper hardware installation and proper torque is a critical characteristic.

g. Install bolts, lead stop bracket, and shims on elastomeric bearing assembly endplate (Figure 5-4-6-1, Detail B).

h. **TORQUE BOLTS TO 910 - 1010 INCH-POUNDS.**

i. Using machinery towel, Item 211, Appendix D, wipe off excess primer.

j. Using adhesive, Item 32, Appendix D, fill area behind lead stop bracket.

k. Allow adhesive to cure at 70° - 80°F (22° - 24°C) for 24 hours, or using heat gun, cure at 140° - 160°F (60° - 71°C) for 1 hour.

l. Install lead stop shim and lead stop to lead stop bracket with bolts and countersunk washers. Make sure countersunk side of washer is toward bolthead.

m. **TORQUE BOLTS TO 180 - 200 INCH-POUNDS.**

n. Using lockwire, Item 197, Appendix D, lockwire bolts.

o. Apply force on blade tip in direction of rotation (counterclockwise), and using feeler gage, inspect for gap between droop stop support ring and lead stop. **GAP MUST BE 0.005 - 0.015-INCH.** If gap exceeds limits, replace shims and recheck gap (Figure 5-4-6-1, Detail A).

p. Apply sealing compound, Item 292, Appendix D, around boltheads.

q. Apply torque stripe to boltheads (PARA 2-6-5).

r. If required, disengage gust lock and rotor brake (PARA 1-6-18).

5-4-7 SPINDLE HORN.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Droop Stop Nut Wrench, Locally-Made, Figure H-168, Appendix H
- Fluorescent Penetrant Inspection Kit
- Inside Micrometer Caliper
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer's Toolkit
- Protective Covering
- Spindle Horn Nut Torque Adapter, Locally-Made, Figure H-205, Appendix H
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 4, Appendix D
- Abrasive Cloth, Item 11, Appendix D
- Abrasive Cloth, Item 12, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D

References

- Appendix D
- Appendix H
- ASTM E1417
- PARA 1-6-18
- PARA 1-7-5
- PARA 2-6-5
- PARA 5-4-5
- PARA 5-4-10
- PARA 5-4-18

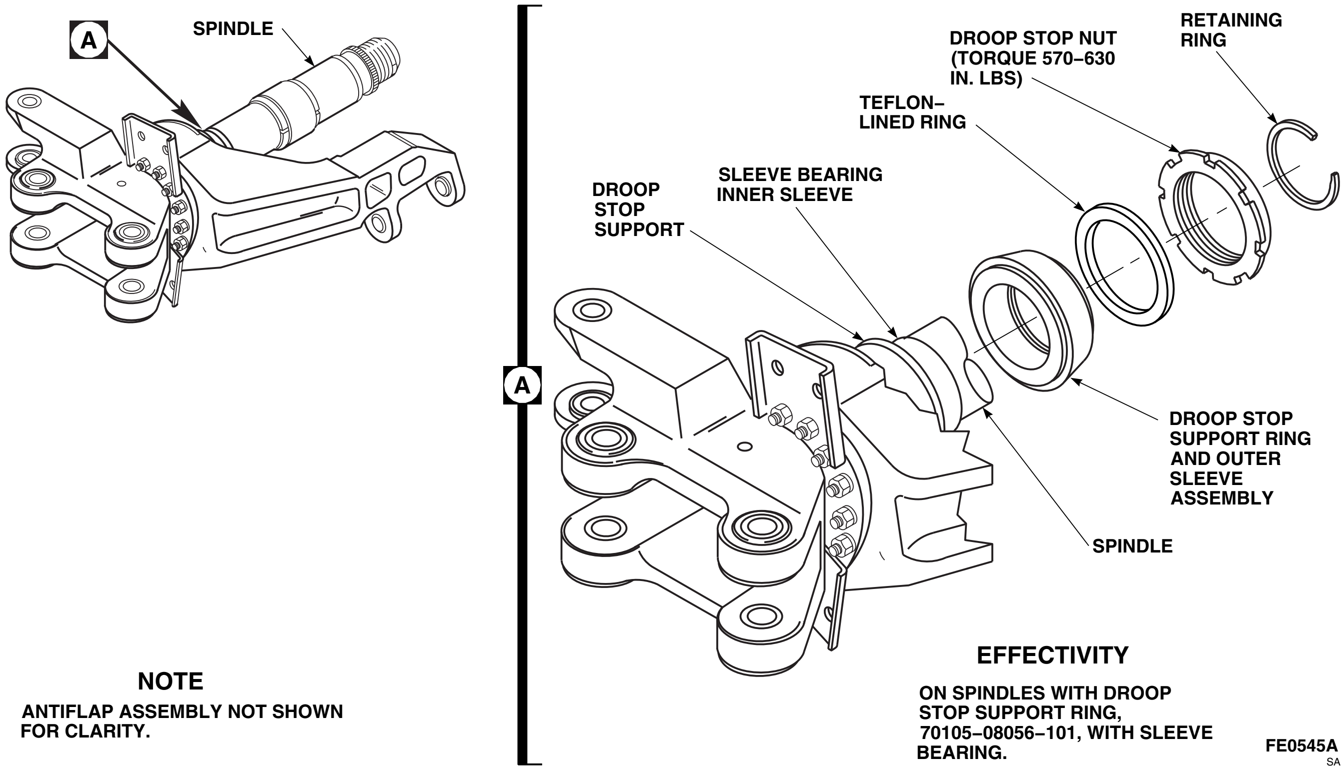


FIGURE 5-4-7-1. ACCESS SPINDLE HORN BOLTS

5-4-7.1 Replace Spindle Horn.

NOTE

Replacement of all spindle horns is the same.

5-4-7.1.1 Remove.

a. Turn main rotor head so spindle horn can be easily removed.

b. If not already done, engage gust lock (PARA 1-6-18).

WARNING

FSCAP

The pitch control rod attachment areas, and critical areas from spindle horn lugs to elbow are critical surfaces which must not be damaged during and after removal of spindle horn and/or pitch control rod. Use extreme caution when removing and provide protective covering for spindle horn and attaching hardware after removal.

c. Disconnect upper end of pitch control rod (PARA 5-4-18).

d. Remove droop stop cam (PARA 5-4-10).

CAUTION

Damage to spindle horn will result if metallic scraper is used to clean spindle horn mounting bolts. Do not use metallic items to remove sealing compound.

e. Using nonmetallic scraper, remove sealing compound from spindle horn bolts.

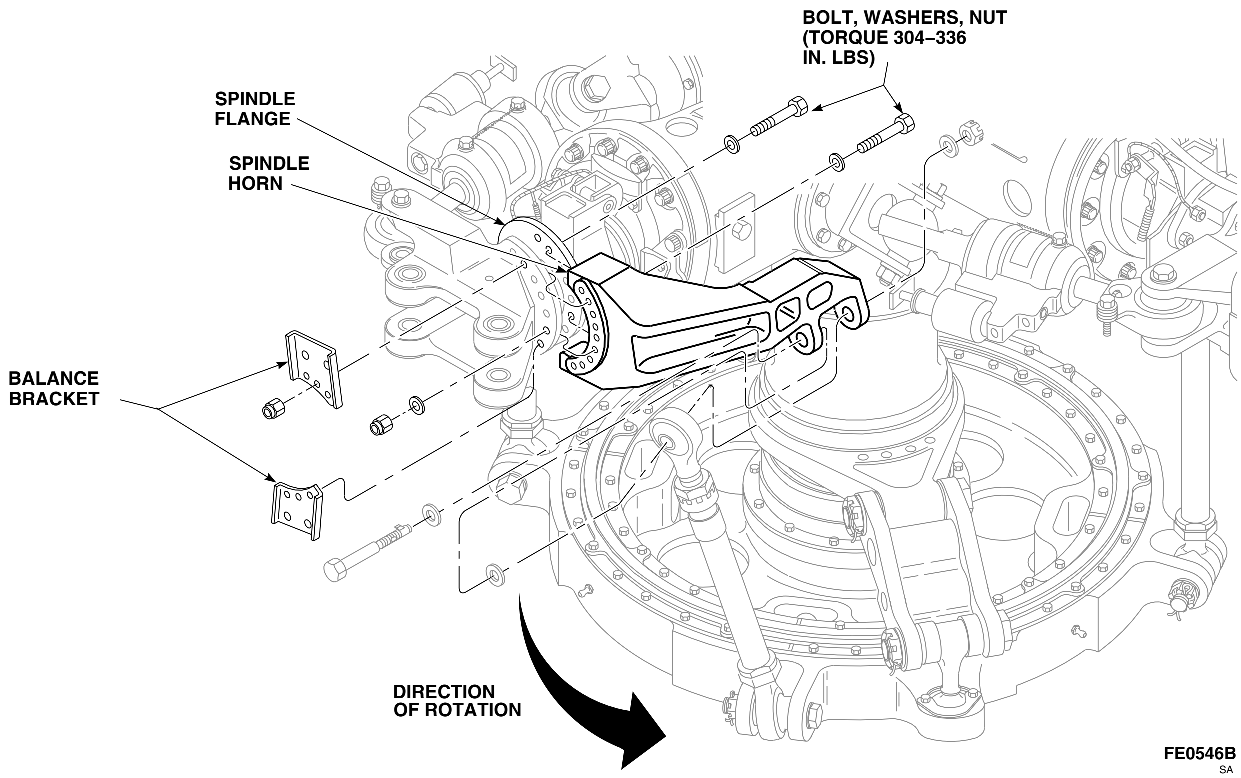


FIGURE 5-4-7-2. REPLACE SPINDLE HORN

NOTE

Positioning of droop stop support ring to access spindle horn bolts is not required on spindles with droop stop support ring, 70105-68003-102, without sleeve bearing.

f. On spindles with droop stop support ring, 70105-08056-101, with sleeve bearing, access spindle horn bolts as follows (Figure 5-4-7-1, Detail A):

(1) Remove retaining ring from droop stop nut.

(2) Locally make droop stop nut wrench (Figure H-168, Appendix H).

(3) Using droop stop nut wrench, back off droop stop nut.

(4) Slide Teflon-lined ring away from droop stop support ring.

(5) Slide droop stop support ring and outer sleeve assembly off sleeve bearing inner sleeve far enough to allow spindle horn bolts to be removed.

g. Remove bolts, washers, and nuts securing upper and lower balance brackets (Figure 5-4-7-2). Remove balance brackets.

h. Remove remaining bolts, washers, and nuts securing spindle horn to flange of spindle. Discard old bolts.

i. Remove spindle horn.

j. Using nonmetallic scraper, remove all sealing compound from spindle flange.

k. Using methyl ethyl ketone, Item 217, Appendix D, clean primer from bolts and balance brackets.

5-4-7.1.2 Inspect/Repair.

a. Inspect balance brackets for surface damage and corrosion. **NONE ALLOWED.** If corrosion is found, remove as follows:

(1) Using abrasive cloth, Item 4, Appendix D, blend damage. **IF DAMAGE OR CORROSION IS DEEPER THAN 0.004-INCH TOTAL THICKNESS AFTER BLENDING, REPLACE BALANCE BRACKET.**

(2) Apply coat of primer, Item 258, Appendix D, to blended area.

(3) Touch up paint (PARA 2-6-5).

b. Inspect balance brackets for cracks. **NONE ALLOWED.** If cracks are found, replace balance brackets.

WARNING

FSCAP

Shaded area of spindle horn (Figure 5-4-7-3 and View A-A) is a critical surface. No repairs allowed in this area, except for condition specified in step e.

c. Visually inspect shaded area of spindle horn. **DAMAGE TO PAINT OR PROTECTIVE FINISHES NOT PENETRATING INTO BASE METAL MAY BE REPAIRED USING TOUCH UP PAINT** (PARA 2-6-5). If damage penetrates into base metal, replace spindle horn.

d. Check helicopter logbook to see if any inspection areas on spindle horn have been previously reworked.

e. Inspect spindle horn AREA A for scratches, nicks, and gouges (Figure 5-4-7-3, View A-A). Using abrasive cloth, Item 11, Appendix D, blend out all reworked areas to 0.030-inch radius as follows:

(1) **DAMAGE TO LUG SURFACE UP TO 0.010-INCH DEEP CAN BE BLENDED OUT. TOTAL BLEND AREA MUST NOT EXCEED 1-SQUARE-INCH.** If damage after blending exceeds limits, replace spindle horn.

(2) **DAMAGE TO LUG EDGE UP TO 0.020-INCH DEEP CAN BE BLENDED OUT. TOTAL REPAIR LENGTH TO LUG EDGE MUST NOT EXCEED 0.075-INCH.**

(3) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace spindle horn.

f. Inspect spindle horn AREA B for scratches, nicks, and gouges (Figure 5-4-7-3, and View B-B). If necessary, using abrasive cloth, Item 11, Appendix D, blend out all reworked areas to 0.030-inch radius as follows:

(1) **DAMAGE TO ATTACHMENT FLANGE UP TO 0.010-INCH DEEP CAN BE BLENDED. TOTAL LENGTH OF REPAIR MUST NOT EXCEED 3-INCHES.** If damage after blending exceeds limits, replace spindle horn.

(2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace spindle horn.

g. Inspect spindle horn AREA C for scratches, nicks, and gouges (Figure 5-4-7-3, View A-A). If necessary, using abrasive cloth, Item 11, Appendix D, blend out all reworked areas to 0.030-inch radius as follows:

(1) **DAMAGE IN RADIUS BETWEEN SPINDLE HORN LUGS UP TO 0.010-INCH DEEP CAN BE BLENDED. TOTAL BLEND AREA MUST NOT EXCEED 1 ½-SQUARE-INCHES.** If damage after blending exceeds limits, replace spindle horn.

(2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace spindle horn.

h. Inspect spindle horn AREA D for scratches, nicks, and gouges (Figure 5-4-7-3, and View

A-A and View B-B). If necessary, using abrasive cloth, Item 11, Appendix D, blend out all reworked areas to 0.030-inch radius as follows:

(1) **DAMAGE TO "I" BEAM AND RECTANGULAR SECTION UP TO 0.050-INCH DEEP CAN BE BLENDED. TOTAL REPAIR WIDTH AT ANY CROSS SECTION MUST NOT EXCEED 0.600-INCH.** If damage after blending exceeds limits, replace spindle horn.

(2) **TOTAL BLEND AREA ALONG LOWER FLANGE MUST NOT EXCEED 2-SQUARE-INCHES.** If damage after blending exceeds limits, replace spindle horn.

(3) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace spindle horn.

i. Inspect for corrosion. Using abrasive cloth, Item 12, Appendix D, remove any corrosion. **IF CORROSION IS DEEPER THAN 0.030-INCH ON OPEN AREAS OR 0.010-INCH ON MATING AREAS AFTER REMOVAL, REPLACE SPINDLE HORN.**

j. Using inside micrometer caliper, measure distance between shoulders of bushing where pitch control rod attaches to spindle horn. **IF DISTANCE IS GREATER THAN 2.509-INCHES, REPLACE SPINDLE HORN.**

k. Inspect silver-plated bushings for wear. **IF DEPTH OF WEAR IS DEEPER THAN 0.005-INCH IN SHOULDER AREA OR 0.040-INCH IN CHAMFER AREA, REPLACE SPINDLE HORN.**

l. Inspect antiflap attachment lugs for damage. Using abrasive wheel, Item 22, Appendix D, blend damage in areas. **IF DAMAGE EXCEEDS 0.015-INCH DEEP AFTER BLENDING, REPLACE BEARING ASSEMBLY.**

m. Inspect antiflap attachment lugs for cracks. **NONE ALLOWED.** If cracks are found, replace spindle (PARA 5-4-5).

n. Touch up paint reworked areas (PARA 2-6-5).

5-4-7.1.3 Install.

a. Locally make spindle horn nut torque adapter (Figure H-205, Appendix H).

b. Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer to shanks, threads, and area under boltheads and exposed portion of each balance bracket at mounting holes.

c. Place spindle horn on flange while primer is still wet (Figure 5-4-7-2).

WARNING

FSCAP

Torque requirement for spindle to spindle horn attachment nuts is a critical characteristic of the assembly. Verification of the required torque is required.

d. Install bolts, washers, and nuts. Install upper balance bracket on three bolts at top of spindle horn/flange and lower balance bracket on three bolts at bottom of spindle.

e. **USING SPINDLE HORN NUT TORQUE ADAPTER, TORQUE ALL NUTS TO 304 - 336 INCH-POUNDS. BOLTS MUST EXTEND AT LEAST 0.040-INCH BEYOND NUT.**

f. Using machinery towel, Item 211, Appendix D, wipe off any primer that has seeped from under bolthead.

g. On spindles with droop stop support ring, 70105-08056-101, with sleeve bearing, assemble droop stop support ring and outer sleeve assembly as follows (Figure 5-4-7-1, Detail A):

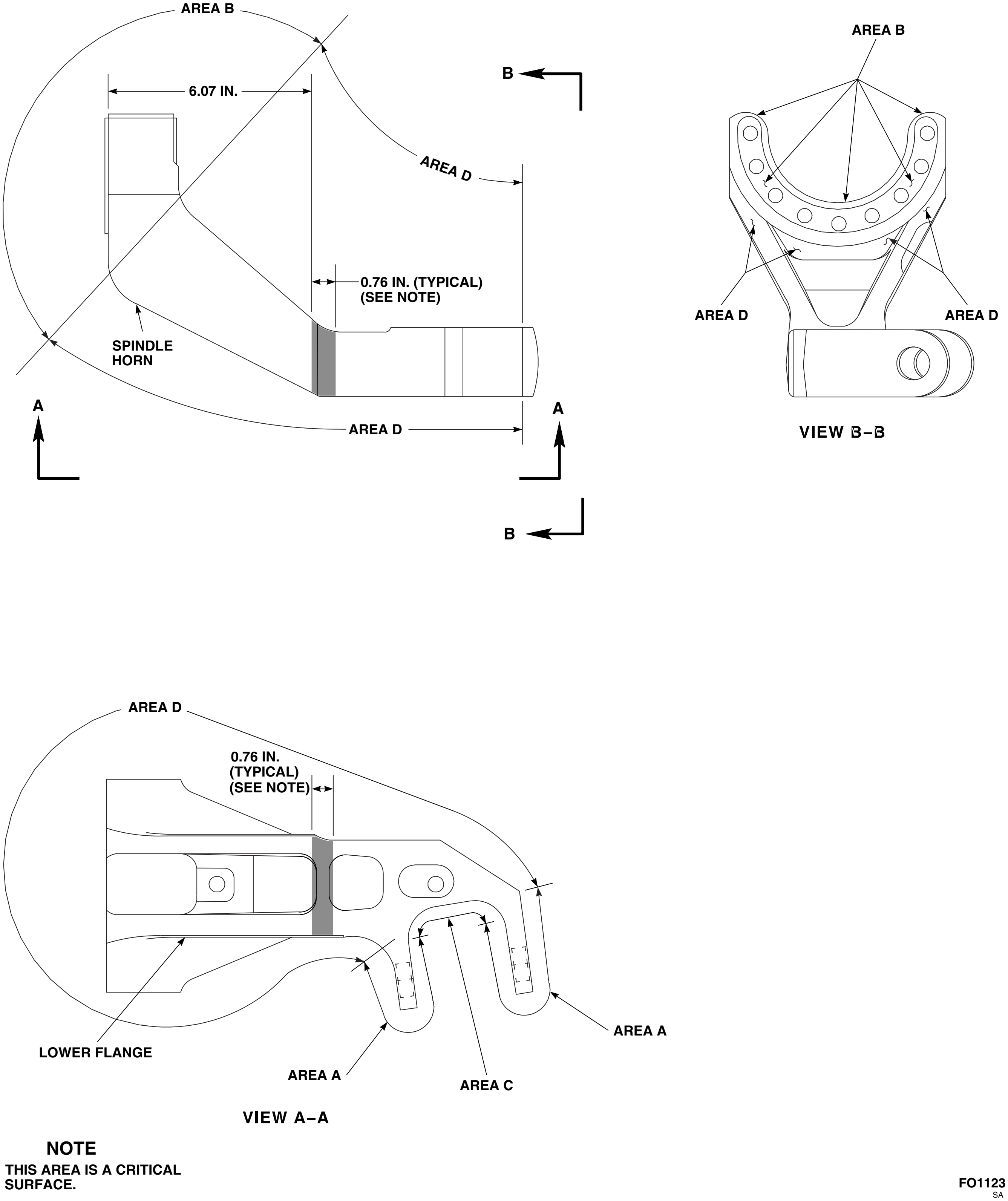


FIGURE 5-4-7-3. INSPECT/REPAIR SPINDLE HORN

(1) Apply coat of primer, Item 258, Appendix D, to threads of droop stop support and face of droop stop nut.

(2) Slide droop stop support ring and outer sleeve assembly onto sleeve bearing inner sleeve.

(3) Slide Teflon-lined ring against droop stop support ring.

(4) Thread droop stop nut on droop stop support while primer is still wet.

(5) **USING DROOP STOP NUT WRENCH, TORQUE DROOP STOP NUT TO 570 - 630 INCH-POUNDS.**

(6) Using machinery towel, Item 211, Appendix D, wipe off excess primer from droop stop support and droop stop nut.

(7) Install retaining ring.

(8) Apply sealing compound, Item 292, Appendix D, to seal area around droop stop nut.

WARNING

FSCAP

The pitch control rod attachment areas on spindle horn lugs are critical surfaces which must not be damaged during installation of spindle horn and/or pitch control rod. After removal of protective covering, use extreme caution when installing spindle horn and attaching hardware.

NOTE

- It may be necessary to gently rock spindle horn to make it easy to line up pitch control rod bolthole.

- Make sure pitch control rod bolthead faces outboard.

h. Connect upper end of pitch control rod (PARA 5-4-18).

i. Touch up paint on exposed boltheads, nuts, and threads (PARA 2-6-5).

j. Check helicopter logbook to make sure spindle horn attachment nuts torque check is shown due in 9 - 11 flight hours (PARA 1-7-5).

k. Make sure area is clean and free of foreign material.

l. Install droop stop cam (PARA 5-4-10).

m. If required, disengage gust lock (PARA 1-6-18).

5-4-8 SPINDLE BONDED WASHER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- C-Clamp
- Heat Gun
- Locating Tool, Locally-Made, Figure H-43, Appendix H
- Nonmetallic Scraper, Locally-Made
- Sizing Block, Locally-Made, Figure H-44, Appendix H
- Vernier Caliper

Materials/Parts

- Abrasive Cloth, Item 4, Appendix D
- Adhesive, Item 32, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Parting Conpound, Item 233, Appendix D

References

- Appendix D
- Appendix H
- PARA 5-4-5

5-4-8.1 Replace Spindle Bonded Washer.

NOTE

Replacement of all spindle bonded washers is the same.

a. Locally make two sizing blocks to be used as compression plates (Figure H-44, Appendix H).

b. Locally make locating tool (Figure H-43, Appendix H).

c. Using nonmetallic scraper and methyl ethyl ketone, Item 217, Appendix D, clean surface of spindle.

NOTE

Only one bonded washer from each main rotor blade pin hole can be missing or unserviceable. If both bonded washers are missing or unserviceable from one main rotor blade pin hole, spindle must be replaced.

d. If both bonded washers are missing or unserviceable from one main rotor blade pin hole, replace spindle (PARA 5-4-5).

e. If original washer is unserviceable or missing, perform the following (Figure 5-4-8-1, Detail A):

(1) Using vernier caliper, measure outside diameter (OD) of spindle cuff lug bushing. **IF SPINDLE CUFF LUG BUSHING OD IS GREATER THAN 1.14-INCHES, REPLACEMENT WASHERS CANNOT BE USED.** If spindle cuff lug bushing OD exceeds limits, replace spindle (PARA 5-4-5).

(2) Measure from existing bonded washer to surface of spindle that washer will be bonded to. Record as dimension Y.

(3) Subtract dimension Y from 4.181. The total will be replacement washer thickness required to obtain a distance of 4.187 - 4.189-inches between inner faces of sizing blocks (compression plates) (Figure 5-4-8-1). **REPLACEMENT WASHER THICKNESS SHALL NOT BE MORE THAN THE THICKEST WASHER AVAILABLE IN TABLE 5-4-8-1.** If thickness is beyond limits, replace spindle (PARA 5-4-5).

(4) Select washer thickness required to obtain a distance of 4.187 - 4.189-inches between inner faces of sizing blocks (compression plates).

f. Using abrasive cloth, Item 4, Appendix D, clean bottom surface of washer. Using clean machinery towel, Item 211, Appendix D, soaked with methyl ethyl ketone, Item 217, Appendix D, wipe clean.

g. Spray thin film of parting compound, Item 233, Appendix D, onto areas on sizing blocks (compression plates) and locating tool that might touch adhesive.

h. Prepare adhesive, Item 32, Appendix D, using manufacturer's instructions.

i. Apply a thin coat of adhesive, Item 32, Appendix D, to spindle on area where washer will be bonded. Make sure to cover entire area.

i. Apply a thin coat of adhesive, Item 32, Appendix D, to bottom of washer.

k. Place locating tool through bolthole in spindle. This will keep washer centered over hole until C-clamp can hold it in place.

l. Place washer over locating tool, bottom side towards spindle, and press onto spindle.

m. Place sizing blocks (compression plates) on washer.

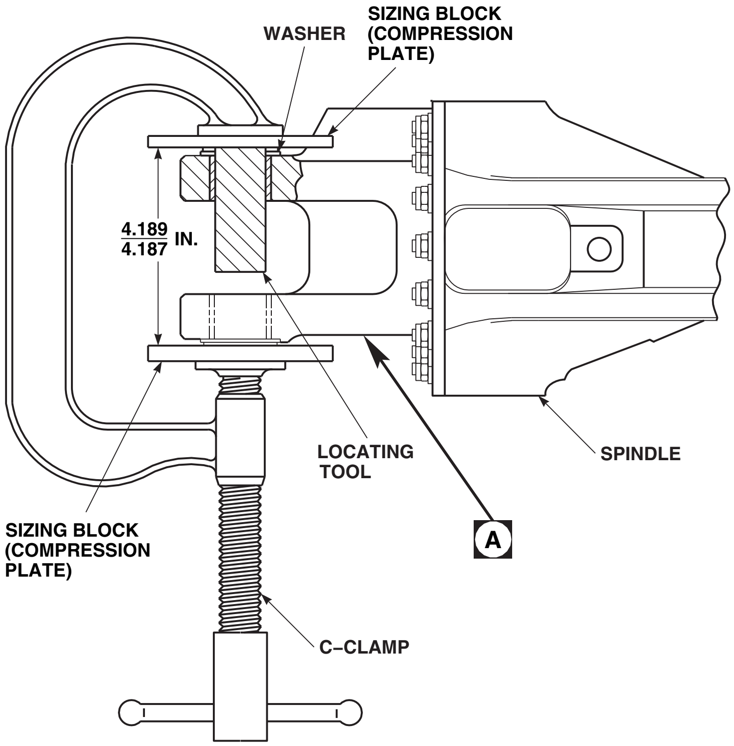
n. Clamp sizing blocks (compression plates) with C-clamp, applying pressure until distance between inner faces of sizing blocks (compression plates) is 4.187 - 4.189-inches.

CAUTION

Damage to elastomeric bearing assembly will result if it is exposed to direct heat from heat gun. Do not expose elastomeric bearing assembly to direct heat greater than 148°F (65°C).

o. Allow adhesive to cure for 12 hours at room temperature, or using heat gun, cure adhesive for 45 minutes at 190° - 210°F (87° - 99°C), or 2 hours at 140° - 160°F (60° - 71°C).

p. Remove locating tool, C-clamp, and sizing blocks (compression plates).



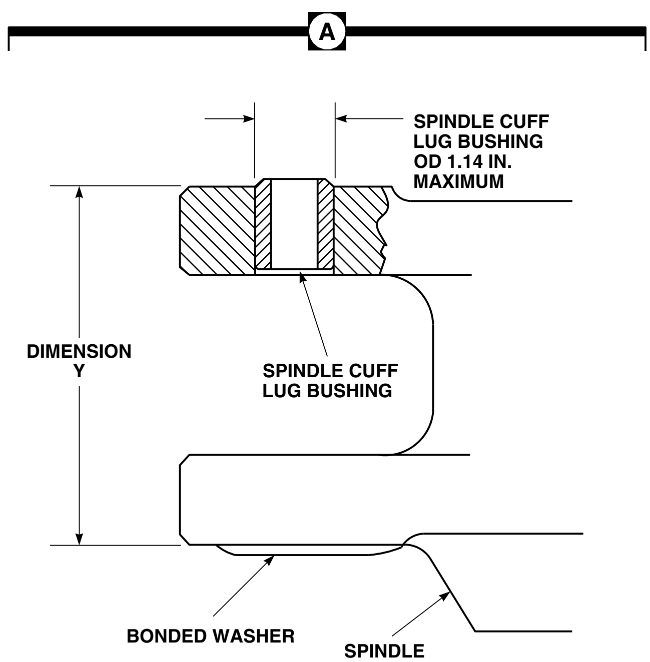


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FIGURE 5-4-8-1. REPLACE SPINDLE BONDED WASHER

Table 5-4-8-1. Washer, 70102-08010, Thickness

Washer, -105, is 0.079 - 0.085-inch thick.	Washer, -110, is 0.062 - 0.064-inch thick.
Washer, -106, is 0.050 - 0.052-inch thick.	Washer, -111, is 0.065 - 0.067-inch thick.
Washer, -107, is 0.053 - 0.055-inch thick.	Washer, -112, is 0.068 - 0.070-inch thick.
Washer, -108, is 0.056 - 0.058-inch thick.	Washer, -113, is 0.071 - 0.073-inch thick.
Washer, -109, is 0.059 - 0.061-inch thick.	Washer, -114, is 0.074 - 0.076-inch thick.

5-4-9 ANTIFLAP ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Blade Clamp Assembly, 70700-20324-041
- Droop Stop Nut Wrench, Locally-Made, Figure H-168, Appendix H
- Hoist, Capacity to Support Main Rotor Blade (Alternate for Cabin Maintenance Crane)
- Maintenance Sling, 70700-20323-043
- Medium Cut File
- Nonmetallic Drift, Locally-Made
- Nonmetallic Scraper, Locally-Made
- Pin Punch
- Rubber Mallet
- Safety Strap
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 171, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 299, Appendix D
- Sealing Compound, Item 301A, Appendix D

References

- Appendix D
- Appendix H
- Appropriate Flight Manual
- PARA 1-7-5
- PARA 1-7-43
- PARA 2-6-5
- PARA 5-4-3
- PARA 5-5-4
- PARA 18-4-1

5-4-9.1 Replace Antiflap Assembly.

NOTE

- Replacement of all antiflap assemblies is the same.
- It is not necessary to remove antiflap bracket from spindle to remove antiflap shaft from antiflap cam. If only antiflap shaft and/or antiflap cam will be removed, proceed to PARA 5-4-9.1.2.

5-4-9.1.1 Remove.

a. If cabin maintenance crane will be used, assemble and install cabin maintenance crane (PARA 18-4-1).

b. Turn main rotor head until antiflap assembly is easy to reach.

WARNING

Blade clamp assembly may open during lifting causing injury to personnel and damage to equipment. If maintenance sling is used, one leg of maintenance sling should be fastened loosely around blade clamp assembly in case blade clamp assembly opens. Install safety strap over blade clamp assembly before removing main rotor blade.

c. Position blade clamp assembly between balance stripes and lock to main rotor blade. Install safety strap over blade clamp assembly and through shackle (Figure 5-4-9-2, Sheet 1, Detail A.

d. Secure antiflap cam in flight position.

e. Attach blade clamp assembly to hoist or cabin maintenance crane. Use maintenance sling to attach blade clamp assembly to any hoist other than cabin maintenance crane.

f. Relieve load on droop stop cam by slightly lifting main rotor blade.

NOTE

Positioning of droop stop support ring to access antiflap assembly bolts is not required on spindles with droop stop support ring, 70105-68003-102, without sleeve bearing.

g. On spindles with droop stop support ring, 70105-08056-101, with sleeve bearing, access antiflap assembly bolts as follows (Figure 5-4-9-1, Detail A):

(1) Remove retaining ring from droop stop nut.

(2) Locally make droop stop nut wrench (Figure H-168, Appendix H).

(3) Using droop stop nut wrench, back off droop stop nut.

(4) Slide Teflon-lined ring away from droop stop support ring.

(5) Slide droop stop support ring and outer sleeve assembly off sleeve bearing inner sleeve far enough to allow antiflap assembly bolts to be removed.

h. Remove spring and allow antiflap cam to swing down (Figure 5-4-9-2, Sheet 1, Detail B).

i. If antiflap stop is damaged, remove bolts and washers securing antiflap stop and bonding jumper to elastomeric bearing assembly endplate. Remove antiflap stop, and move bonding jumper aside.

CAUTION

Damage to antiflap bracket will result if metallic scraper is used to clean mounting bolts. Do not use metallic items to remove sealing compound.

j. Using nonmetallic scraper, remove sealing compound from antiflap bracket boltheads.

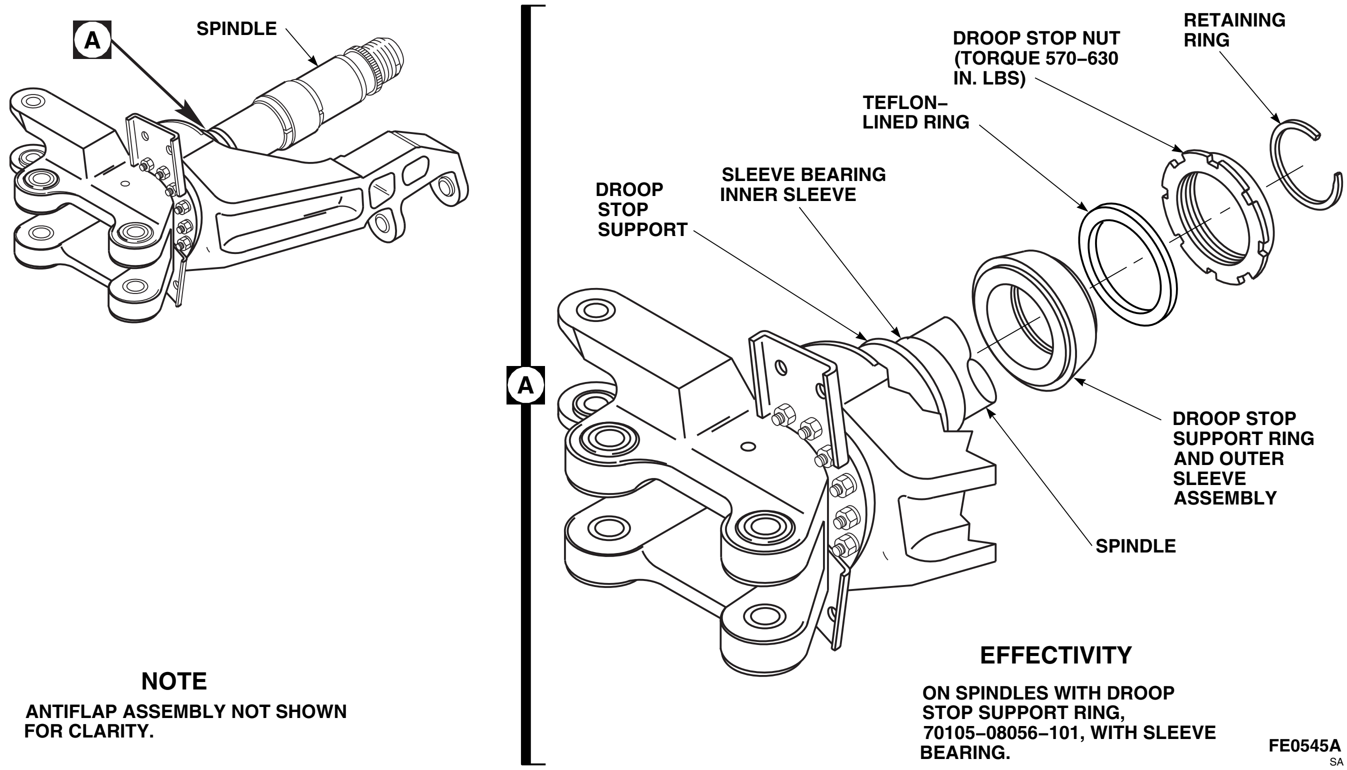


FIGURE 5-4-9-1. ACCESS ANTIFLAP ASSEMBLY BOLTS

NOTE

Bonding jumper and balance bracket will be removed when bolts are removed.

k. Remove bolts, washers, and nuts securing antiflap bracket to spindle.

l. Remove antiflap bracket. Discard long bolts.

5-4-9.1.2 Disassemble.

a. Remove self-locking nut, washer, and spring retainer from antiflap shaft (Figure 5-4-9-2, Sheet 2, Detail C).

b. Using pin punch, gently tap out spring pin until free of antiflap cam and antiflap shaft.

c. Slide antiflap shaft out of antiflap bracket and remove antiflap cam.

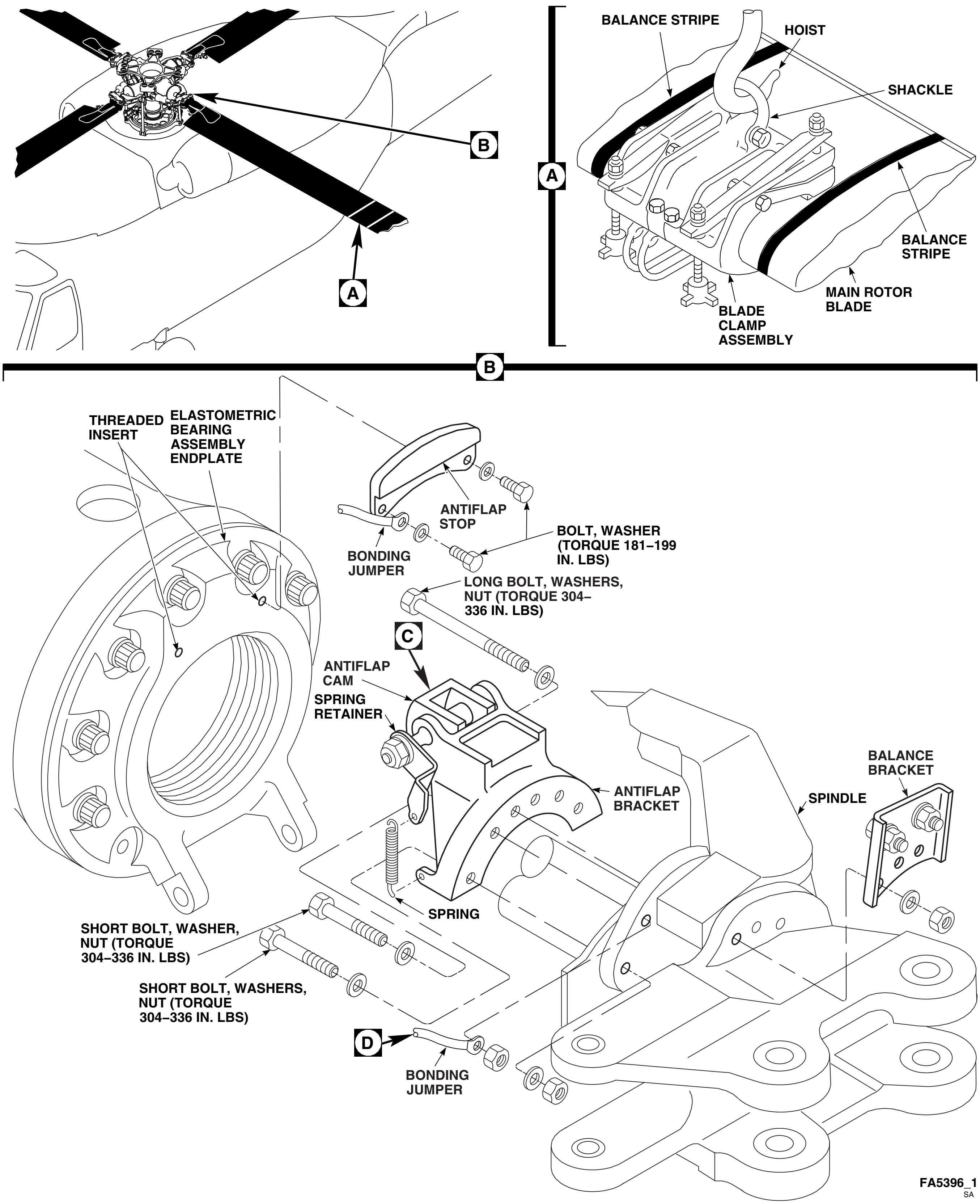
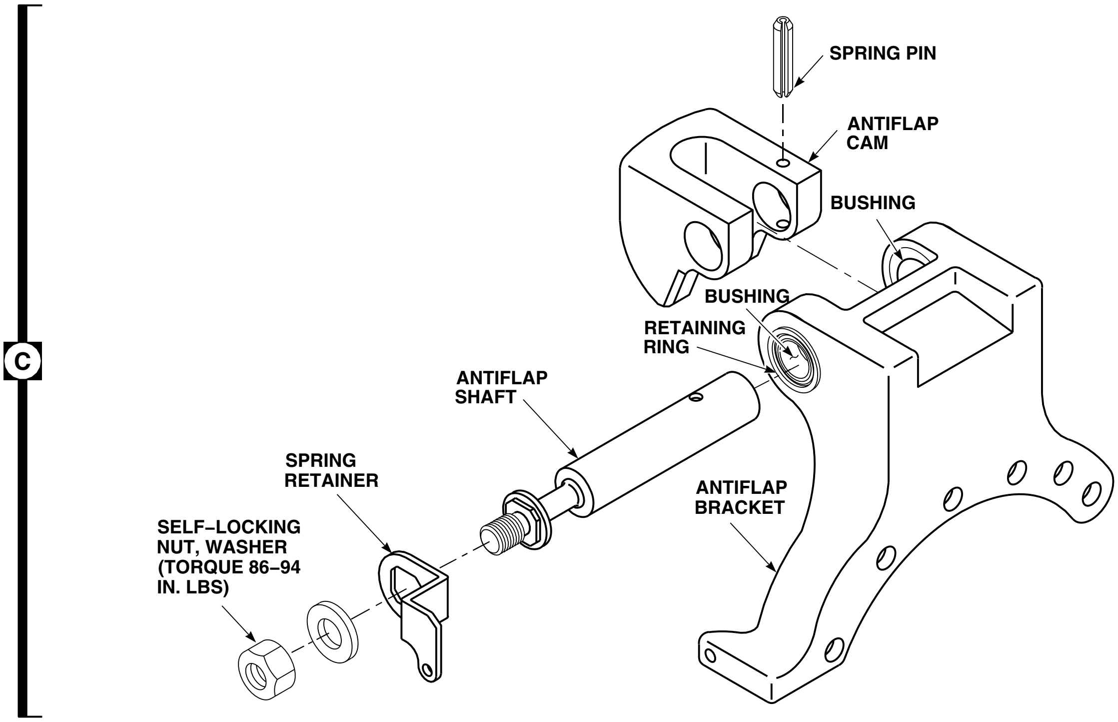


FIGURE 5-4-9-2. REPLACE ANTIFLAP ASSEMBLY (SHEET 1 OF 2)



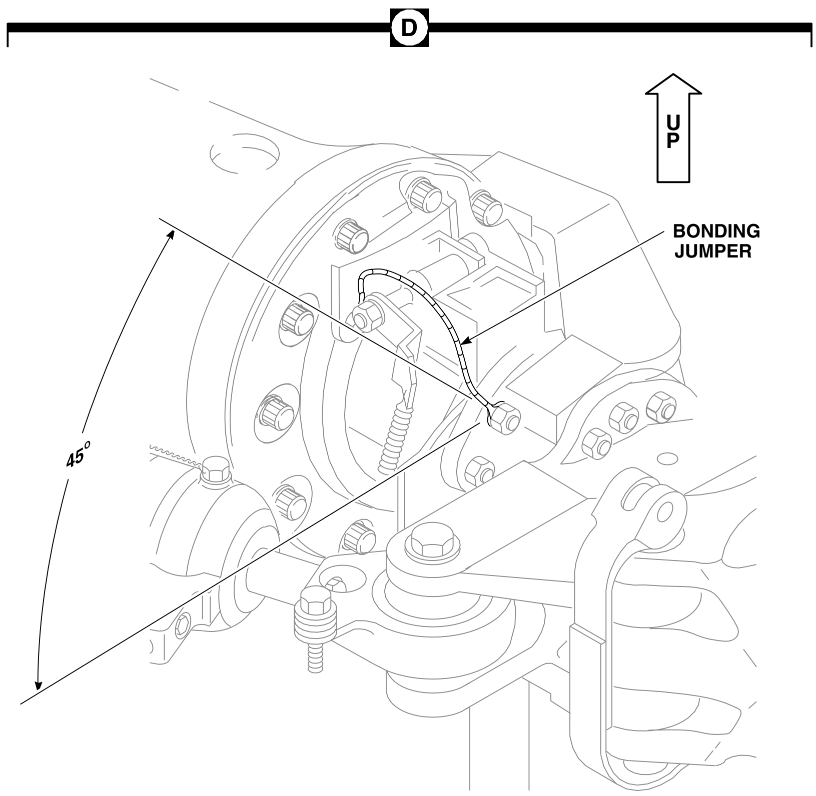


FIGURE 5-4-9-2. REPLACE ANTIFLAP ASSEMBLY (SHEET 2 OF 2)

5-4-9.1.3 Inspect/Repair.

a. If antiflap stop was removed, inspect threaded inserts in elastomeric bearing assembly endplate as follows (Figure 5-4-9-2, Sheet 1, Detail B):

(1) Thread new bolt into threaded insert until bolt reaches threaded insert locking feature. **BOLT SHOULD THREAD DOWN SMOOTHLY WITH NO EVIDENCE OF STRIPPED OR CROSSED THREADS. MAXIMUM TORQUE REQUIRED TO THREAD BOLTS IS 20 INCH-POUNDS.**

(2) If threaded insert exceeds limits, replace insert (PARA 5-4-3).

b. Visually inspect antiflap assembly for cracks. **NONE ALLOWED, EXCEPT ANTIFLAP CAM MAY HAVE SMALL CRACKS UP TO 1/8-INCH AT CORNER RADIUS OF IMPACT SURFACES.** If cracks exceed limits, replace antiflap assembly.

c. **IF ANTIFLAP BRACKETS ARE CRACKED DUE TO HIGH WINDS, INSPECT PER PARA 1-7-43. IF CRACKS ARE FOUND, REPLACE ANTIFLAP BRACKET.**

d. Inspect antiflap bracket for nicks, gouges, burrs, or corrosion. **BLENDING OUT OF NICKS, GOUGES, BURRS, OR CORROSION UP TO 0.020-INCH IS ACCEPTABLE. BLENDING OF CORNERS AND SURFACES UP TO 0.030-INCH IS ACCEPTABLE.** If nicks, gouges, burrs, or corrosion exceed limits, replace antiflap bracket.

e. Inspect antiflap cam impact surfaces for deformation. **DEFORMATION UP TO 0.10-INCH IS ACCEPTABLE.** If deformation exceeds limits, replace antiflap cam.

f. Inspect antiflap cam for corner damage (nicks, gouges, etc.). **CORNER DAMAGE UP TO 1/8-INCH IS ACCEPTABLE AFTER REMOVING SHARP EDGES WITH A MEDIUM CUT FILE.** If corner damage exceeds limits, replace antiflap cam.

g. Touch up reworked areas (PARA 2-6-5).

h. Visually inspect antiflap stop attachment boltheads and spring for signs of corrosion. **NONE ALLOWED.** If corrosion is found, replace corroded bolts and spring.

i. 701044–SUBQ On antiflap assemblies with retainer rings installed, verify presence of retaining rings, securing bushings, on lead side and lag side of antiflap assembly (Figure 5-4-9-2, Sheet 2, Detail C). If necessary, repair as follows:

(1) If bushing retaining ring is missing on lead side, perform the following:

(a) Using epoxy primer coating kit, Item 135, Appendix D, apply coating to new retaining ring and bushing to insulate dissimilar metals.

(b) Install new retaining ring, M2742610112D, on lead side antiflap bushing.

(c) Touch up paint as necessary (PARA 2-6-5).

(2) If bushing retaining ring is missing on lag side, perform the following:

(a) Remove nut, washer, and spring retainer securing antiflap shaft to antiflap assembly.

(b) Using epoxy primer coating kit, Item 135, Appendix D, apply coating to new retaining ring and bushing to insulate dissimilar metals.

(c) Install new retaining ring, M2742610112D, on lag side antiflap bushing.

(d) Install spring retainer, washer, and nut on antiflap shaft.

(e) Touch up paint as necessary (PARA 2-6-5).

j. 70583–701043 On antiflap assemblies without retainer rings installed, inspect antiflap bushings for displacement.

CAUTION

Damage to antiflap bracket will result if metallic tools are used to clean or remove bushings. Do not use metallic items to remove sealing compound or bushings.

k. 70583–701043 If bushings are displaced, perform the following:

(1) Using nonmetallic scraper, remove sealing compound from antiflap bushing.

(2) Using nonmetallic drift and rubber mallet, remove bushing (Figure 5-4-9-2, Sheet 2, Detail C).

(3) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean outside diameter (OD) of bushing and inside diameter of hole in antiflap bracket. Using abrasive cloth, Item 10, Appendix D, wipe OD of bushing. Using machinery towel, Item 211, Appendix D, wipe OD of bushing and antiflap bracket dry.

NOTE

Do not allow sealing compound to contact inside diameter of bushing or bushing face.

(4) Apply sealing compound, Item 299, Appendix D, to outside diameter of bushing.

(5) Place bushing in antiflap bracket.

(6) Apply sealing compound, Item 292, Appendix D, around outside diameter of bushing flange.

5-4-9.1.4 Assemble.

a. Position antiflap cam in antiflap bracket and install antiflap shaft through bushings in antiflap bracket and holes in antiflap cam (Figure 5-4-9-2, Sheet 2, Detail C).

b. Line up spring pin hole in antiflap cam with hole in antiflap shaft. Press or tap spring pin into hole until flush with antiflap cam.

NOTE

Do not reuse self-locking nut.

c. Install spring retainer, washer, and self-locking nut on antiflap shaft. Make sure spring retainer is correctly seated on square end of shaft with small hole for spring facing down.

d. **TORQUE SELF-LOCKING NUT TO 86 - 94 INCH-POUNDS.**

e. Using lockwire, Item 197, Appendix D, lockwire spring pin to antiflap cam.

5-4-9.1.5 Install.

a. Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer or sealing compound to shanks, threads, and area under boltheads used to secure antiflap cam.

WARNING

FSCAP

Torque requirement for spindle to spindle horn attachment nuts is a critical characteristic of antiflap assembly. Verification of required torque is required.

b. Place antiflap assembly on spindle. Install long bolts, washers, balance brackets, and nuts as shown while primer or sealing compound is still wet (Figure 5-4-9-2, Sheet 1, Detail B).

c. **TORQUE NUTS TO 304 - 336 INCH-POUNDS.**

d. Install short bolts, washers, bonding jumper, and nuts while primer or sealing compound is still wet. Omit washer between bonding jumper and nut. Orientate bonding jumper to 45° up position (Figure 5-4-9-2, Sheet 1, Detail B and Sheet 3, Detail D).

e. **TORQUE NUTS TO 304 - 336 INCH-POUNDS** (Figure 5-4-9-2, Sheet 1, Detail B).

f. Install spring between spring retainer and antiflap bracket.

g. Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer or sealing compound to shank, threads, and area under boltheads used to secure antiflap stop.

h. If removed, install antiflap stop and bonding jumper to elastomeric bearing assembly endplate with bolts and washers while primer or sealing compound is still wet. Orientate bonding jumper to 45° up position (Figure 5-4-9-2, Sheet 1, Detail B and Sheet 3, Detail D).

i. **TORQUE BOLTS TO 181 - 199 INCH-POUNDS** (Figure 5-4-9-2, Sheet 1, Detail B).

j. Using lockwire, Item 197, Appendix D, lockwire bolts.

k. Using heat-shrinkable insulation sleeving, Item 171, Appendix D, cover lockwire.

l. Using machinery towel, Item 211, Appendix D, wipe off any primer that has seeped from under boltheads.

m. Apply sealing compound, Item 292, Appendix D, to seal areas around boltheads and nuts.

n. On spindles with droop stop support ring, 70105-08056-101, with sleeve bearing, assemble droop stop support ring and outer sleeve assembly as follows (Figure 5-4-9-1, Detail A):

(1) Apply coat of primer, Item 258, Appendix D, to threads of droop stop support and face of droop stop nut.

(2) Slide droop stop support ring and outer sleeve assembly onto sleeve bearing inner sleeve.

(3) Slide Teflon-lined ring against droop stop support ring.

(4) Thread droop stop nut on droop stop support while primer is still wet.

(5) **USING DROOP STOP NUT WRENCH, TORQUE DROOP STOP NUT TO 570 - 630 INCH-POUNDS.**

(6) Using machinery towel, Item 211, Appendix D, wipe off excess primer from droop stop support and droop stop nut.

(7) Install retaining ring.

(8) Apply sealing compound, Item 292, Appendix D, to seal area around droop stop nut.

o. Lower main rotor blade until droop stop ring is resting on droop stop cam.

p. If used, remove maintenance sling from main rotor blade. Remove safety strap; unlock and remove blade clamp assembly from main rotor blade.

q. If used, disassemble and remove cabin maintenance crane (PARA 18-4-1).

r. Touch up paint on bolts and exposed threads (PARA 2-6-5).

s. Make sure area is clean and free of foreign material.

t. Check for free rotation of antiflap cam. If antiflap cam contacts antiflap stop with main rotor blades installed and main rotor head in static position, perform the following:

(1) Run up helicopter to 100% Nr in flat pitch for 15 minutes (Appropriate Flight Manual).

(2) Shut down helicopter and reinspect for antiflap cam contact. If antiflap cam contact continues, replace elastomeric bearing assembly (PARA 5-5-4).

u. Check helicopter logbook to make sure spindle-to-horn attachment bolts is shown due

9 - 11 flight hours after installation (PARA 1-7-5).

5-4-10 DROOP STOP CAM.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Blade Clamp Assembly, 70700-20324-041
- Hoist, Capacity to Support Main Rotor Blade (Alternate for Cabin Maintenance Crane)
- Maintenance Sling, 70700-20323-043
- Nonmetallic Scraper, Locally-Made
- Protection, Eye, Skin, and Breathing
- Safety Strap
- Torque Wrench, 0 - 75 in. lbs
- Torque Wrench, 700 - 1,600 in. lbs

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Primer, Item 258, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-5
- PARA 1-7-20
- PARA 5-3-3
- PARA 5-4-11
- PARA 18-4-1

5-4-10.1 Replace Droop Stop Cam.

NOTE

Replacement of all droop stop cams is the same.

5-4-10.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. If cabin maintenance crane will be used, assemble and install cabin maintenance crane (PARA 18-4-1).

WARNING

Droop stop cam is made of beryllium copper, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds and eyes. To prevent injury, personnel must wear eye protection. Clean up visible dust piles and wash hands and arms after maintenance.

c. Turn main rotor head so droop stop is easy to reach.

WARNING

Blade clamp assembly may open during lifting, causing injury to personnel and damage to equipment. If maintenance sling is used, one leg of maintenance sling should be fastened loosely around blade clamp assembly in case blade clamp assembly opens. Install safety strap over blade clamp assembly before removing main rotor blade.

d. Position blade clamp assembly between balance stripes and lock to main rotor blade (Figure 5-4-10-1, Sheet 1, Detail A). Install safety strap over blade clamp assembly and through shackle.

e. Attach blade clamp assembly to hoist or cabin maintenance crane. Use maintenance sling to attach blade clamp assembly to any hoist other than cabin maintenance crane.

f. Relieve load on droop stop cam by slightly lifting main rotor blade.

g. Disconnect springs from spring pins (Figure 5-4-10-1, Sheet 1, Detail B).

h. If blade deice kit is installed, perform the following (Figure 5-4-10-1, Sheet 2, Detail C):

(1) Disconnect electrical connector, P845, P846, P847, or P848, from heater pin.

(2) Install protective caps and plugs on electrical connectors.

i. Slide out droop stop pin or heater pin (if blade deice kit is installed) and remove droop stop cam (Figure 5-4-10-1, Sheet 1, Detail B or Sheet 2, Detail C).

j. If necessary, remove spring clips as follows:

(1) Using nonmetallic scraper, remove sealing compound from bolts securing spring clips to elastomeric bearing assembly endplate. Remove bolts.

(2) Remove spring clips and springs.

(3) Remove springs from spring clips.

k. Remove nuts, washers, and spring pins from droop stop cam.

5-4-10.1.2 Inspect.

a. Inspect droop stop (PARA 5-3-3).

b. Inspect droop stop bushings (PARA 5-4-11).

5-4-10.1.3 Install.

a. Install nuts, washers, and spring pins on sides of droop stop cam (Figure 5-4-10-1, Sheet 1, Detail B).

b. **TORQUE NUTS TO 29 - 31 INCH-POUNDS.**

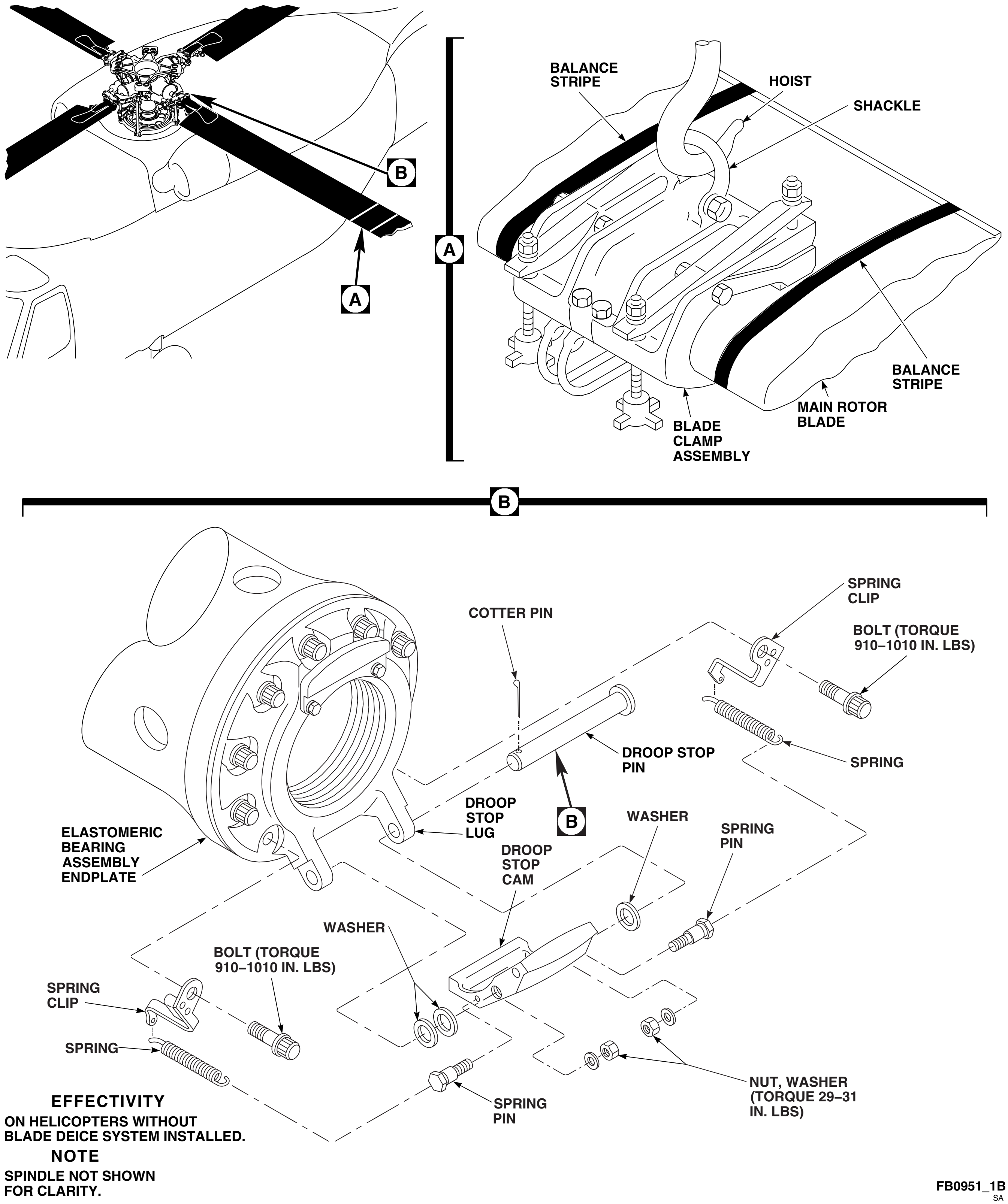


FIGURE 5-4-10-1. REPLACE DROOP STOP CAM (SHEET 1 OF 2)

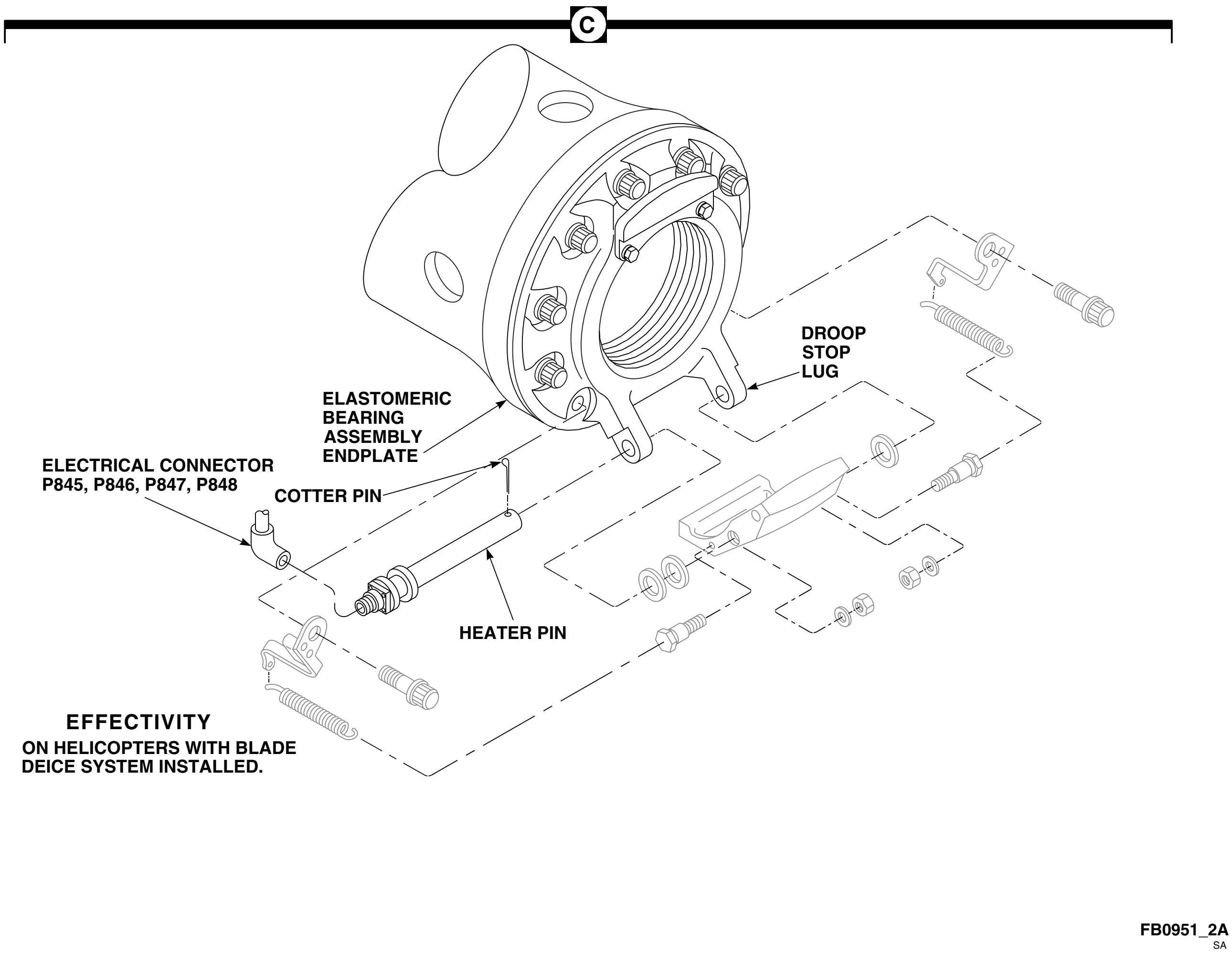


FIGURE 5-4-10-1. REPLACE DROOP STOP CAM (SHEET 2 OF 2)

c. If removed, install spring clips as follows:

(1) Apply coat of primer, Item 258, Appendix D, to shank, threads, and area under spindle attachment boltheads.

WARNING

FSCAP

Verification of proper hardware and spring clip installation and torque of bolts are critical characteristics.

(2) Position spring clips on elastomeric bearing assembly endplate and install bolts.

(3) **TORQUE BOLTS TO 910 - 1010 INCH-POUNDS.**

(4) Using machinery towel, Item 211, Appendix D, wipe off excess primer.

(5) Connect springs to spring pins.

(6) Apply bead of sealing compound, Item 292, Appendix D, around boltheads.

d. Apply bead of sealing compound, Item 292, Appendix D, around nuts.

NOTE

If blade deice kit is installed, install heater pin in the direction shown in Figure 5-4-10-1, Sheet 2, Detail C.

e. Install droop stop cam, washers, and droop stop pin or heater pin (if blade deice is installed) in droop stop lugs. Install cotter pin.

f. Check for gap between cotter pin and shoulder on droop stop bushing. **GAP MUST NOT BE LESS THAN 0.010-INCH.**

g. If blade deice kit is installed, perform the following (Figure 5-4-10-1, Sheet 2, Detail C):

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) Connect electrical connector, P845, P846, P847, or P848, to heater pin.

h. Connect springs to spring pins.

i. Lower main rotor blade until droop stop ring is resting on droop stop cam.

j. If used, remove maintenance sling from main rotor blade. Remove safety strap; unlock and remove blade clamp assembly from main rotor blade (Figure 5-4-10-1, Detail A).

k. If used, disassemble and remove cabin maintenance crane (PARA 18-4-1).

l. Make sure area is clean and free of foreign material.

m. Check helicopter logbook to make sure spindle attachment bolts retorque is shown due 9 - 11 flight hours after installation (PARA 1-7-5).

5-4-11 DROOP STOP BUSHINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Drift
- Inside Micrometer Caliper
- Nonmetallic Scraper, Locally-Made
- Outside Micrometer Caliper
- Protection, Eye, Skin, and Breathing

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Cheesecloth, Item 90, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 302, Appendix D

References

- Appendix D
- PARA 5-4-5
- PARA 5-4-10
- PARA 5-5-4

5-4-11.1 Replace Droop Stop Bushings.

NOTE

Replacement of all droop stop bushings is the same.

5-4-11.1.1 Inspect.

WARNING

Droop stop cam is made of beryllium copper, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds and eyes. To prevent injury, personnel must wear eye protection. Clean up visible dust piles and wash hands and arms after maintenance.

a. Inspect droop stop bushings for nicks and gouges. **USING ABRASIVE CLOTH, ITEM 10, APPENDIX D, NICKS AND GOUGES ON SHOULDER OF BUSHINGS MAY BE BLENDED UP TO 0.020-INCH DEEP AND UP TO 25% OVERALL DAMAGE.**

b. Using inside micrometer caliper, measure inside diameter (ID) of bushings. **ID OF EITHER BUSHING MUST NOT EXCEED 0.762-INCH.** If ID exceeds limits, replace droop stop bushing.

c. Inspect droop stop bushings for displacement or disbonding. **NONE ALLOWED.** If displacement or disbonding has occurred, perform the following:

(1) Remove displaced or disbonded droop stop bushing (PARA 5-4-11.1.2).

(2) Using outside micrometer caliper, measure droop stop bushing outside diameter (OD). **DROOP STOP BUSHING OD MUST NOT BE LESS THAN 0.8755-INCH.** If OD exceeds limits, replace droop stop bushing.

(3) Inspect spherical bearing lug bore (PARA 5-4-11.1.3).

(4) Install droop stop bushing (PARA 5-4-11.1.4).

5-4-11.1.2 Remove.

WARNING

Droop stop cam is made of beryllium copper, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds and eyes. To prevent injury, personnel must wear eye protection. Clean up visible dust piles and wash hands and arms after maintenance.

a. Remove spindle (PARA 5-4-5).

b. Remove droop stop cam (PARA 5-4-10).

c. Apply methyl ethyl ketone, Item 217, Appendix D, to droop stop bushing(s) to soften sealing compound.

d. Using drift and arbor press, press droop stop bushing from droop stop lug (Figure 5-4-11-1, Detail A).

e. Using nonmetallic scraper, and cheesecloth, Item 90, Appendix D, soaked in methyl ethyl ketone, Item 217, Appendix D, clean hole in droop stop lug.

5-4-11.1.3 Inspect.

a. Inspect droop stop lug bore for scoring. **NONE ALLOWED.** If scoring is found, using abrasive cloth, Item 10, Appendix D, remove scoring.

b. Using inside micrometer caliper, measure ID of droop stop lug bore in two places minimum. **AVERAGE DROOP STOP LUG BORE DIAMETER MUST NOT EXCEED 0.8760-INCH.** If ID exceeds limits, replace spherical bearing (PARA 5-5-4).

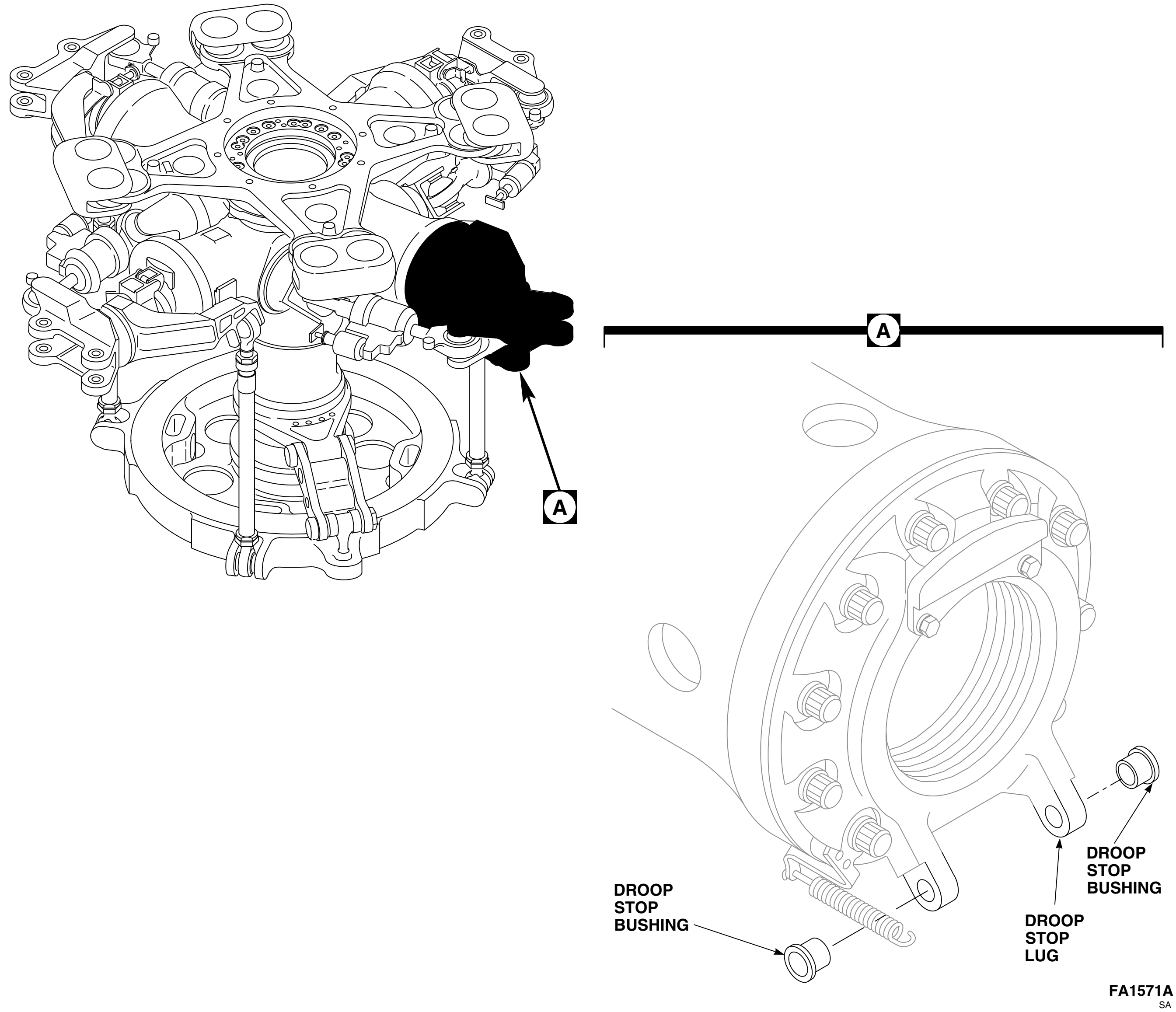


FIGURE 5-4-11-1. REPLACE DROOP STOP BUSHINGS

5-4-11.1.4 Install.

a. Apply coat of sealing compound, Item 302, Appendix D, to new droop stop bushing.

b. Using arbor press, press bushings into droop stop lugs (Figure 5-4-11-1, Detail A).

c. Using machinery towel, Item 211, Appendix D, wipe extra adhesive off droop stop bushing and droop stop lugs. Allow to cure for 24 hours.

d. Install droop stop cam (PARA 5-4-10).

e. Install spindle (PARA 5-4-5).

5-4-12 DROOP STOP PAD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Abrasive Cloth, Item 5, Appendix D
- Adhesive, Item 37, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 2-6-5
- PARA 5-4-10

5-4-12.1 Replace Droop Stop Pad.

NOTE

Replacement of all droop stop pads is the same.

5-4-12.1.1. Remove.

a. Remove droop stop cam (PARA 5-4-10).

b. Remove old droop stop pad (Figure 5-4-12-1).

5-4-12.1.2. Install.

a. Using abrasive cloth, Item 5, Appendix D, clean bond areas on droop stop cam and new droop stop pad.

b. Using cleaning cloth, Item 102, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, wipe roughed areas on droop stop cam and droop stop pad.

c. Apply adhesive, Item 37, Appendix D, on droop stop cam and droop stop pad.

d. Allow adhesive to dry for 10 minutes, until most of methyl ethyl ketone has evaporated and adhesive is tacky.

e. Put droop stop pad on droop stop cam so that surfaces to be bonded are joined together firmly, making sure entire mating surfaces are in direct contact (Figure 5-4-12-1). Do not move or disturb bonded surfaces after mating.

f. Allow bond to cure a minimum of 24 hours at room temperature. Greatest strength is obtained in about seven days.

g. Touch up paint (PARA 2-6-5).

h. Install droop stop cam (PARA 5-4-10).

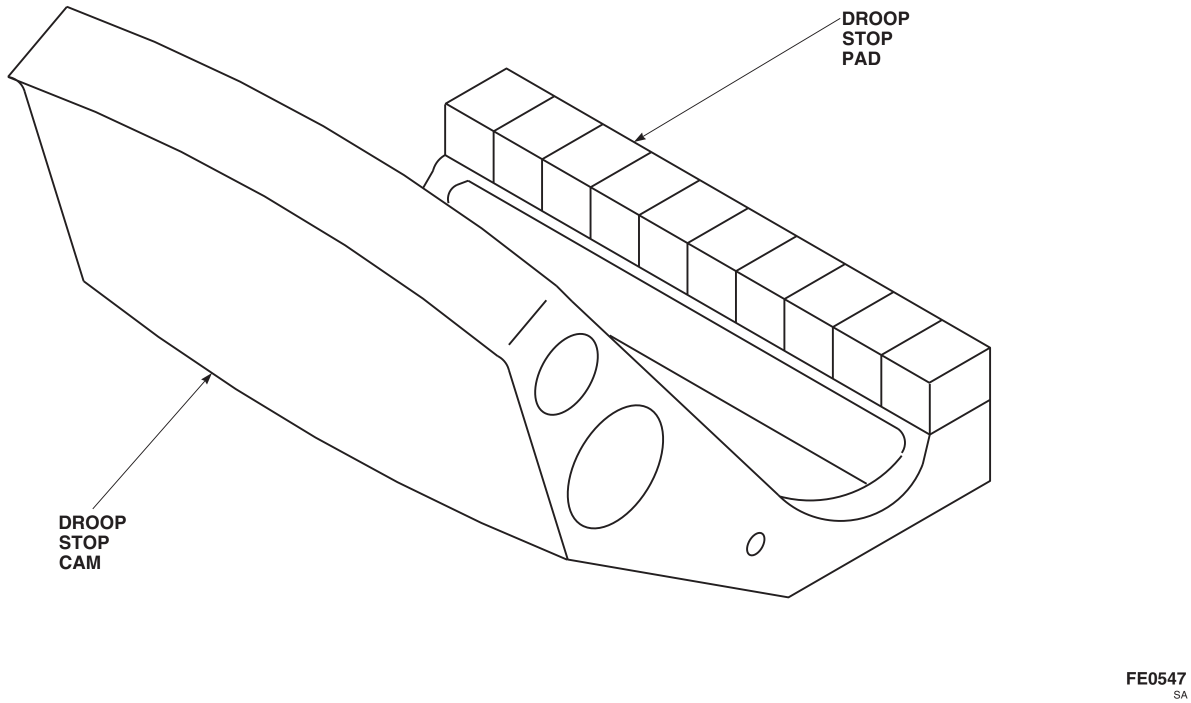


FIGURE 5-4-12-1. REPLACE DROOP STOP PAD

5-4-13 DAMPER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Medium Cut File
- Nonmetallic Drift, Locally-Made
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer's Toolkit
- Rawhide Mallet
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 500 - 2500 in. lbs

Materials/Parts

- Abrasive Cloth, Item 12, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Antiseize Compound, Item 74, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D

References

- Appendix D
- PARA 1-6-18
- PARA 1-7-5
- PARA 2-6-5
- PARA 5-3-4
- PARA 5-4-4
- PARA 5-4-5
- PARA 5-4-15
- PARA 5-4-16
- PARA 5-4-17

5-4-13.1 Replace Damper.

NOTE

Replacement of all dampers is the same.

5-4-13.1.1 Remove.

a. Turn main rotor head until damper is easy to reach.

b. If not already done, engage gust lock (PARA 1-6-18).

c. Using nonmetallic scraper, remove sealing compound from around damper attachment bolts and nuts.

NOTE

If required, when removing inboard and outboard bolts, lightly tap them with rawhide mallet and nonmetallic drift.

d. Remove bolt, washers, and nut securing damper to spindle (Figure 5-4-13-1, Detail A).

e. Remove bolt, washers, and nut securing damper to damper bracket.

f. Remove damper.

NOTE

- Keep restraining cable for possible installation on new damper.
- Restraining cables are not required on dampers, 70106-08100-044, 70106-08100-046, or marked, RS-001C-VI.

g. If necessary to remove restraining cable from damper cylinder head, perform the following:

(1) Note location and position of cable clip attached to stud on cylinder head.

(2) Remove nut securing cable clip to stud on cylinder head.

(3) Remove restraining cable.

5-4-13.1.2 Inspect.

a. Inspect shoulder bushings for wear (PARA 5-4-5). If wear is detected, perform the following (Figure 5-4-13-1, Detail D):

(1) Remove bolt, washers, and nut.

(2) Remove one weight from counterweight bracket if wear is on lower outside or upper inside area of damper spindle attachment lug washer.

(3) Add one weight to counterweight bracket if wear is on lower inside or upper outside area of damper spindle attachment lug washer.

(4) Install bolt, washers, and nut.

(5) **TORQUE NUT TO 157 - 173 INCH-POUNDS.**

(6) Apply bead of sealing compound, Item 292, Appendix D, around bolthead and nut.

b. Inspect damper bracket (PARA 5-4-15).

c. Inspect damper bearings as required (PARA 5-3-4).

NOTE

- Retaining ring installed on damper cylinder head should be located on same side as cylinder head vent.
- Retaining ring installed on rod end should be located on bottom side (side opposite of counterweight bracket).

d. Inspect damper bearing retaining rings for correct orientation. If orientation is incorrect, remove and install damper bearing (PARA 5-4-16).

e. Inspect/repair damper (PARA 5-4-13.2).

f. Inspect bolts securing damper to damper bracket and spindle (PARA 5-4-4).

5-4-13.1.3 Install.

CAUTION

At temperatures below -40°F (-40°C), only hydraulic fluid, Item 154, Appendix D, must be used.

a. If replacement damper is filled with hydraulic preservative fluid, drain damper, refill with hydraulic fluid, Item 154 or Item 155, Appendix D, and bleed (PARA 5-4-17).

NOTE

Bearing retaining rings will face down when damper is installed. Make sure nylon washers are installed on opposite side of retaining ring.

b. Install nylon washer on damper bracket and spindle. Position slot on washer toward center-line of damper body (Figure 5-4-13-1, Detail B and Detail C).

c. Apply bead of sealing compound, Item 292, Appendix D, around edge of nylon washer and lug of damper bracket or spindle.

NOTE

Restraining cables are not required on dampers, 70106-08100-044, 70106-08100-046, or marked, RS-001C-VI.

d. If required, install restraining cable on damper cylinder head as follows:

(1) Apply coat of primer, Item 258, Appendix D, to restraining cable clip. Install clip on stud of cylinder head while primer is still wet.

(2) Make sure bend in cable clip faces away from damper.

(3) Secure cable clip to stud on cylinder head with nut.

(4) **TORQUE NUT TO 167 - 183 INCH-POUNDS.**

(5) Using primer, Item 258, Appendix D, touch up paint on nut.

e. Apply coat of antiseize compound, Item 74, Appendix D, to bolt threads and shank.

NOTE

- Install nylon washer between damper bearing and bracket.
- If necessary when installing bolt, lightly tap with rawhide mallet.

f. Place damper between damper bracket and spindle (Figure 5-4-13-1, Detail A). Make sure damper is positioned with word ″TOP″ facing up and out. Install bolt, countersunk washer, and nylon washer at damper bracket end. Make sure countersunk side of washer is toward bolt-head.

NOTE

Restraining cables are not required on dampers, 70106-08100-044, 70106-08100-046, or marked, RS-001C-VI.

g. If restraining cable is required, position cable clip on bolt with bend toward damper assembly.

NOTE

To prevent excessive wear of bolts, install enough washers under nut so that at least half of cotter pin will be inside castellations of nut when cotter pin is installed.

h. Install thick and thin washers, as required, to align cotter pin hole and nut. Install nut.

i. **TORQUE NUT TO 1320 - 1635 INCH-POUNDS.** Install cotter pin.

j. Apply coat of sealing compound, Item 301A, Appendix D, to bolt threads and shank.

NOTE

- Install nylon washer between damper bearing and spindle.
- If necessary when installing bolt, lightly tap with rawhide mallet.

k. Install bolt, countersunk washer, and nylon washer at spindle end. Make sure countersunk side of washer is toward bolthead.

NOTE

To prevent excessive wear of bolts, install enough washers under nut so that at least half of cotter pin will be inside castellations of nut when cotter pin is installed.

l. Install thick and thin washers, as required, to align cotter pin hole and nut. Install nut.

m. **TORQUE NUT TO 1320 - 1635 INCH-POUNDS.** Install cotter pin.

n. Touch up paint on boltheads, nuts, and exposed threads (PARA 2-6-5).

o. Make sure area is clean and free of foreign material.

p. Check helicopter logbook to make sure damper attachment bolts torque check is shown due 9 - 11 flight hours after installation (PARA 1-7-5).

q. If damper was drained, record type of hydraulic fluid used to fill damper in helicopter logbook.

r. If required, disengage gust lock (PARA 1-6-18).

5-4-13.2 Inspect/Repair Damper.

NOTE

In the following steps, damage should be blended out using abrasive wheel, Item 22, Appendix D, abrasive cloth, Item 12, Appendix D, and/or medium cut file, as required.

a. Inspect damper for corrosion pitting as follows:

(1) **ON OPEN AREAS (NONMATING SURFACES), REMOVE PITTING UP TO 0.030-INCH DEEP ON NO MORE THAN 10% OF ANY CIRCUMFERENCE BY BLENDING OUT.** If pitting exceeds limits, replace damper.

(2) **ON MATING SURFACES, REMOVE PITTING UP TO 0.010-INCH DEEP ON NO MORE THAN 10% OF SURFACE BY BLENDING OUT.** If pitting exceeds limits, replace damper.

(3) On aluminum areas that were reworked, perform the following:

(a) Using corrosion resistant coating, Item 118, Appendix D, touch up areas that were reworked.

(b) Using primer and paint, touch up areas (PARA 2-6-5).

(4) On nonaluminum areas that were reworked, using primer and paint, touch up areas (PARA 2-6-5).

b. Inspect damper for nicks and gouges as follows:

(1) **CORNER DAMAGE UP TO 0.060-INCH DEEP MAY BY BLENDED OUT TO A ½-INCH RADIUS.** If nicks and gouges exceed limits, replace damper.

(2) **SURFACE DAMAGE UP TO 0.040-INCH DEEP AND LESS THAN 10% OF CIRCUMFERENCE MAY BE BLENDED OUT TO A ½-INCH RADIUS.** If surface damage exceeds limits, replace damper.

(3) Visually inspect reworked areas for cracks.

(4) On aluminum areas that were reworked, perform the following:

(a) Using corrosion resistant coating, Item 118, Appendix D, touch up areas that were reworked.

(b) Using primer and paint, touch up areas (PARA 2-6-5).

(5) On nonaluminum areas that were reworked, using primer and paint, touch up areas (PARA 2-6-5).

c. Inspect damper piston surface for damage. If any damage can be felt with fingernail, replace damper.

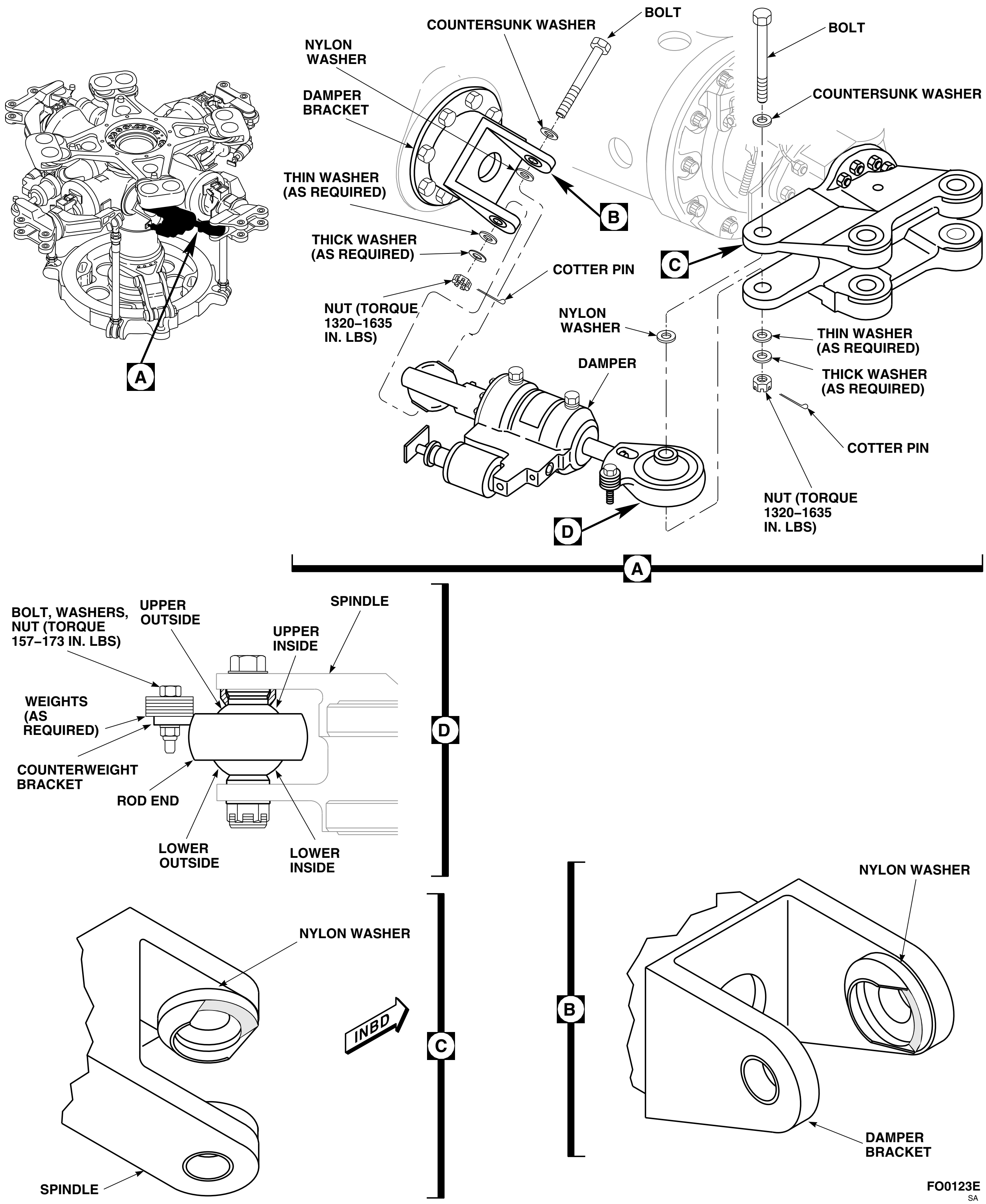


FIGURE 5-4-13-1. REPLACE DAMPER

5-4-14 DAMPER INDICATOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Insulation Sleeving, Item 160D, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 1-6-18
- PARA 5-4-13
- PARA 5-4-17

5-4-14.1 Replace Damper Indicator.

NOTE

Replacement of all damper indicators is the same.

5-4-14.1.1 Remove.

a. If not already done, engage gust lock (PARA 1-6-18).

b. Loosen bleed plugs on damper enough to relieve pressure, so damper indicator bottoms at empty position (handle in). Using machinery towel, Item 211, Appendix D, catch any overflow of fluid (Figure 5-4-14-1, Detail B).

c. **TIGHTEN BLEED PLUGS HANDTIGHT.**

NOTE

Have new damper indicator ready to install; otherwise, insert plug to stop fluid loss.

d. Remove damper indicator and packing from damper. If required, install plug in damper. Discard packing.

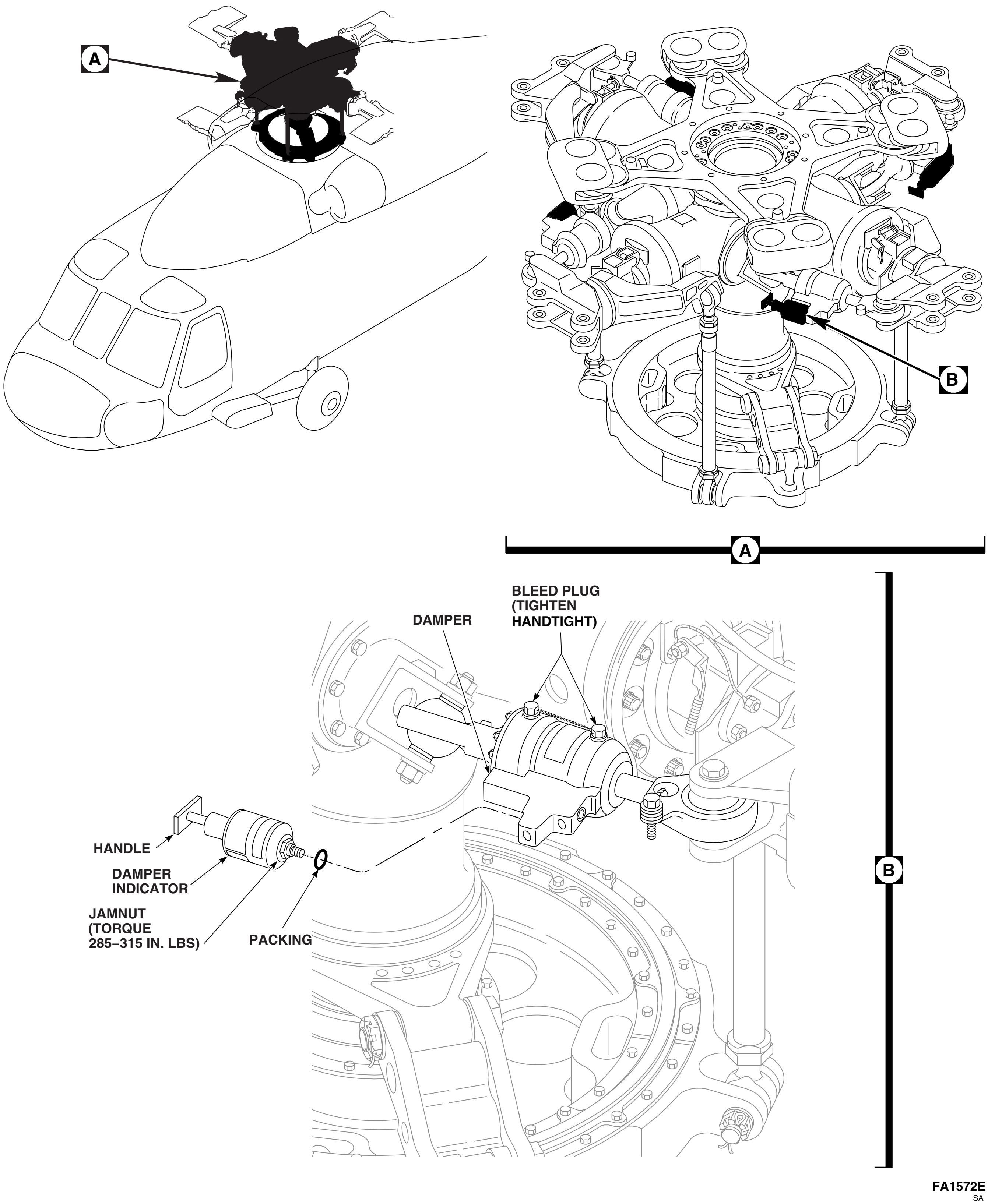


FIGURE 5-4-14-1. REPLACE DAMPER INDICATOR

5-4-14.1.2 Install.

a. Using hydraulic fluid, Item 154, Appendix D, coat packing. Install packing on damper indicator (Figure 5-4-14-1, Detail B).

b. Make sure handle is all the way in and install damper indicator in damper. **TORQUE DAMPER INDICATOR JAMNUT TO 285 - 315 INCH-POUNDS.**

c. Secure damper indicator to damper by performing the following:

(1) Using lockwire, Item 197, Appendix D, install lockwire on damper indicator.

(2) After twisting lockwire, but before securing loose end to bolthead on damper, install insulation sleeving, Item 160D, Appendix D, over lockwire.

(3) Secure loose end of lockwire to bolthead on damper.

d. Bleed damper (PARA 5-4-17).

e. If required, disengage gust lock (PARA 1-6-18).

5-4-15 DAMPER BRACKET.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit
- Nonmetallic Scraper, Locally-Made
- Rawhide Mallet
- Torque Wrench, Dial, 0 - 75 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 500 - 2500 in. lbs

Materials/Parts

- Abrasive Cloth, Item 12, Appendix D
- Acetone, Item 25, Appendix D
- Brush, Paint, Item 86, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D
- Tape, Duct, Item 318A, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 1-6-18
- PARA 1-7-5
- PARA 2-6-5
- PARA 5-4-3
- PARA 5-4-4

5-4-15.1 Replace Damper Bracket.

NOTE

Replacement of all damper brackets is the same.

5-4-15.1.1 Remove.

a. Turn main rotor head until damper bracket is easy to reach.

b. If not already done, engage gust lock (PARA 1-6-18).

NOTE

Restraining cables are not required on dampers, 70106-08100-044, 70106-08100-046, or marked, RS-001C-VI.

c. Using nonmetallic scraper, remove sealing compound from damper attachment bolt and nut and damper bracket attachment bolts and nuts.

d. Remove bolt, nylon washer, washers, restraining cable (if installed), and nut securing damper to damper bracket and swing damper out of way (Figure 5-4-15-1, Detail A).

e. Remove bolts and washers securing damper bracket to main rotor hub.

f. Remove damper bracket.

g. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean primer off bolts.

h. Using nonmetallic scraper, remove sealing compound from damper bracket.

5-4-15.1.2 Inspect.

a. Inspect bolts securing damper bracket to main rotor hub (PARA 5-4-4).

b. Measure distance between shoulder bushings (Figure 5-4-15-1, Detail B). **DISTANCE MUST NOT EXCEED 2.509-INCHES.** If distance exceeds limit, replace damper bracket.

c. Inspect bushing for wear. **WEAR MUST NOT EXCEED 0.7510-INCH.** If wear on bushing exceeds limit, replace damper bracket.

d. Inspect bushing for corrosion. **CORROSION ON CHAMFER OF BUSHING MUST NOT EXCEED 0.040-INCH DEEP.** If corrosion on chamfer of bushing exceeds limit, replace damper bracket. If corrosion on damper bracket is within limits, perform the following:

(1) Using abrasive cloth, Item 12, Appendix D, blend out corrosion.

(2) Apply coat of primer, Item 258, Appendix D, to blended area.

(3) Touch up paint (PARA 2-6-5).

e. Inspect damper bracket for cracks. **NONE ALLOWED.** If crack is found, replace damper bracket.

f. Inspect damper bracket for nicks and gouges. Using abrasive cloth, Item 12, Appendix D, blend out damage. **REPLACE DAMPER BRACKET IF:**

(1) **DAMAGE AFTER BLENDING EXCEEDS 0.050-INCH DEEP.**

(2) **DAMAGE ON MATING AREAS AFTER BLENDING EXCEEDS 0.020-INCH DEEP, OR COVERS MORE THAN 10% OF SURFACE AREA.**

NOTE

- Fluorescent penetrant inspection of 11 o'clock damper bracket attachment holes is applicable for main rotor hub, 70103-08112-041, only.
- 11 o'clock damper attachment hole inspection should be performed every 200 flight hours.

g. Inspect 11 o'clock damper attachment hole as follows:

NOTE

Main rotor hub must be cleaned with paint remover even if paint is not present.

(1) Clean ½-inch area around inspection area on inside surface of 11 o'clock damper attachment holes as follows:

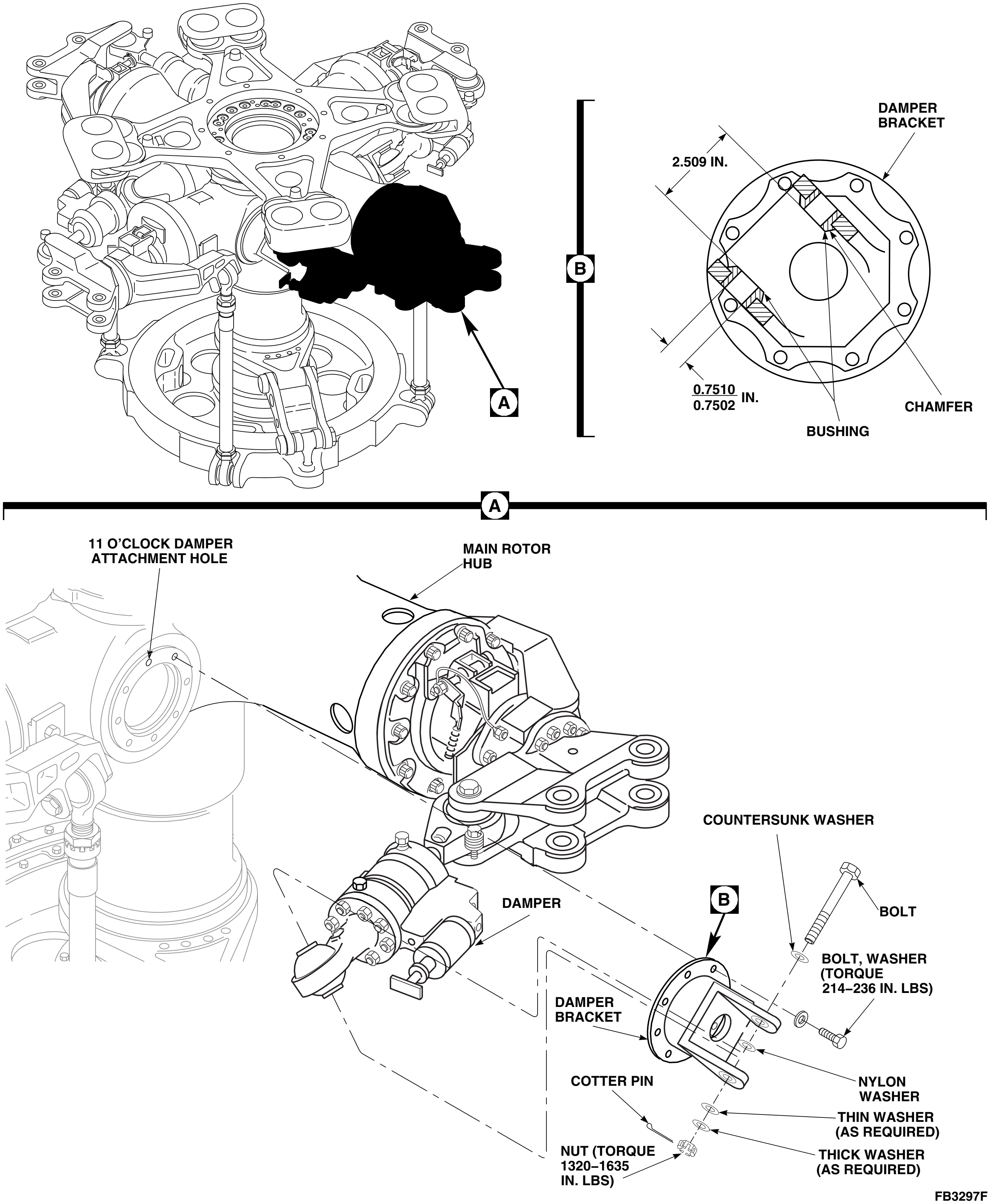


FIGURE 5-4-15-1. REPLACE DAMPER BRACKET

CAUTION

Damage to elastomeric bearing assembly will result if paint remover is allowed to seep into spherical and thrust bearings. Do not allow paint remover to come in contact with elastomeric bearings.

(a) Using duct tape, Item 318A, Appendix D, mask off areas adjacent to area being cleaned.

(b) Using machinery towels, Item 211, Appendix D, position around inboard end of spindle and elastomeric bearing assembly and underneath area to be stripped to catch any paint remover during cleaning.

NOTE

If paint is present, allow paint remover to remain on surface until paint bubbles.

(c) Using paint brush, Item 86, Appendix D, apply paint remover, Item 230, Appendix D. Using machinery towel, Item 211, Appendix D, or nonmetallic scraper, remove paint remover.

(d) Using machinery towel, Item 211, Appendix D, saturated with water, thoroughly remove all remains of paint remover. Repeat if necessary.

(e) Using acetone, Item 25, Appendix D, clean area. Repeat.

(2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, Level 4 Sensitivity, 11 o'clock damper attachment holes (ASTM E1417). **NO CRACKS ALLOWED.**

(3) Record time of inspection in helicopter logbook.

5-4-15.1.3 Install.

a. Before torquing damper bracket attachment bolts, check threaded inserts as follows:

NOTE

On main rotor hub, 70070-10046-055, 11 o'clock damper bracket attachment hole has been modified. Attaching hardware is not used at this position.

(1) Hold bracket up against hub and thread in bolts and washers until surface of washers are 0.030-inch away from surface of damper bracket.

(2) Using dial torque wrench, back off each bolt. **MINIMUM BREAKAWAY TORQUE IS 6.5 INCH-POUNDS. MAXIMUM REMOVAL TORQUE IS 60 INCH-POUNDS.**

(3) If torque exceeds limits, perform the following:

(a) Inspect bolt for corrosion and wear. **NONE ALLOWED.** If corrosion or wear is found, replace bolts.

(b) Repeat step a. If breakaway or removal torque still exceeds limits, replace threaded insert (PARA 5-4-3).

b. Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer or sealing compound to shank, threads, and area under damper bracket boltheads.

c. Place damper bracket on main rotor hub with word "TOP" facing up (Figure 5-4-15-1, Detail A).

NOTE

On main rotor hub, 70070-10046-055, 11 o'clock damper bracket attachment hole has been modified. Attaching hardware is not used at this position.

d. Install bolts and washers while primer is still wet.

e. **TORQUE BOLTS TO 214 - 236 INCH-POUNDS.**

f. Using machinery towel, Item 211, Appendix D, wipe off any primer that has seeped from under boltheads.

g. Apply coat of sealing compound, Item 301A, Appendix D, to bolt shank.

NOTE

- Install nylon washer between damper bearing and bracket.
- If necessary when installing bolt, lightly tap with rawhide mallet.

h. Place damper in damper bracket. Install bolt, countersunk washer, and nylon washer at damper bracket end. Make sure countersunk side of washer is toward bolthead.

NOTE

Restraining cables are not required on dampers, 70106-08100-044, 70106-08100-046, or marked, RS-001C-VI.

i. If restraining cable is required, position cable clip on bolt with bend toward damper assembly.

NOTE

To prevent excessive wear of bolt, install enough washers under nut so that at least half of cotter pin will be inside castellations of nut when cotter pin is installed.

j. Install thick and thin washers, as required, to align cotter pin hole and nut. Install nut.

k. **TORQUE NUT TO 1320 - 1635 INCH-POUNDS.** Install cotter pin.

l. Touch up paint on boltheads, nuts, and exposed threads (PARA 2-6-5).

m. Apply coat of sealing compound, Item 292, Appendix D, to seal gap between damper bracket and main rotor hub.

n. Apply sealing compound, Item 292, Appendix D, to seal areas around boltheads and nuts securing damper bracket to main rotor hub.

o. Make sure area is clean and free of foreign material.

p. Check helicopter logbook to make sure damper attachment bolts torque check is shown due 9 - 11 flight hours after installation (PARA 1-7-5).

q. If required, disengage gust lock (PARA 1-6-18).

5-4-16 DAMPER BEARINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Bearing Installer/Remover, 70700-77306-041
- Freezer (Alternate for Dry Ice)
- Heat Gun
- Inside Micrometer Caliper
- Powertrain Repairer's Toolkit

Materials/Parts

- Abrasive Cloth, Item 13, Appendix D
- Dry Ice, Item 125, Appendix D (Alternate for Freezer)
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 299, Appendix D
- Sealing Compound, Item 301A, Appendix D

References

- Appendix D
- PARA 2-6-5
- PARA 5-4-13
- PARA 5-5-6

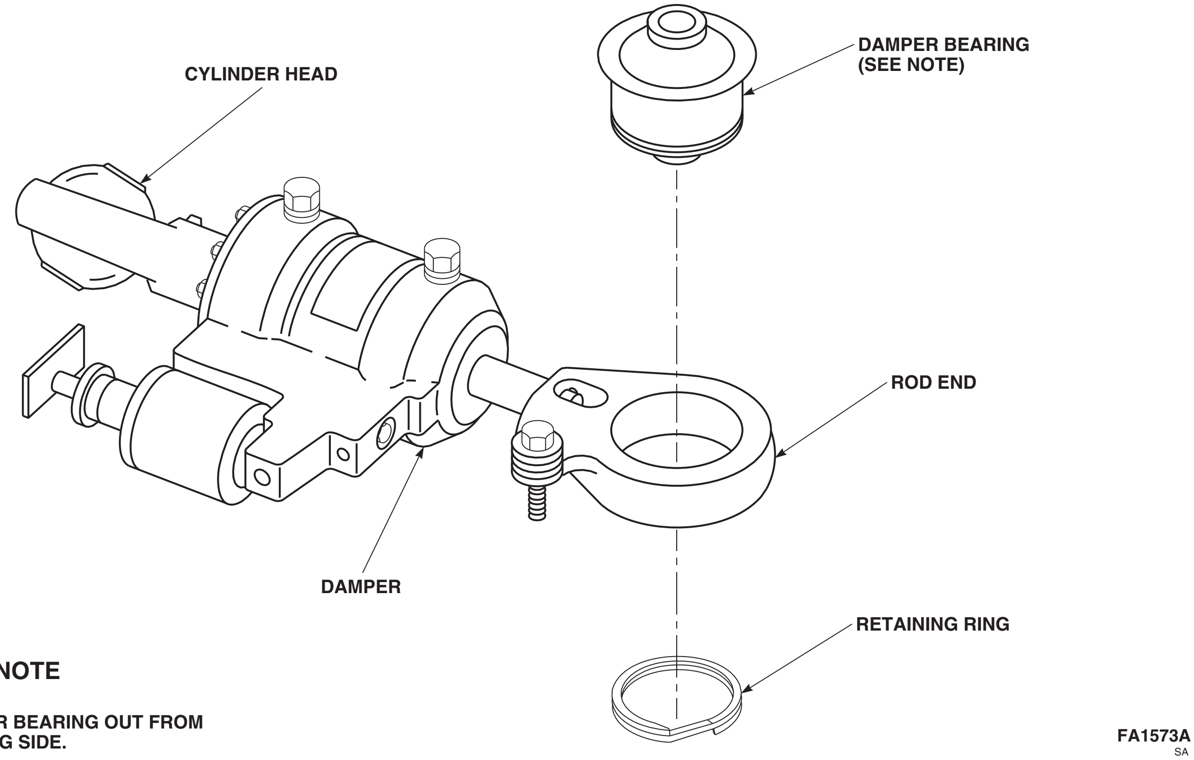


FIGURE 5-4-16-1. REPLACE DAMPER BEARINGS

5-4-16.1 Replace Damper Bearings.

NOTE

- Damper contains two bearings. One installed in the rod end, and the other in the cylinder head.
- Damper does not have to be removed if only one damper bearing is being replaced.
- Damper bearing located in the rod end is shown.

5-4-16.1.1. Remove.

a. Disconnect applicable damper end or remove damper (PARA 5-4-13).

b. Remove retaining ring (Figure 5-4-16-1).

c. Using bearing installer/remover, remove damper bearing from rod end or cylinder head by pressing from retaining ring side of damper bearing.

d. Using methyl ethyl ketone, Item 217, Appendix D, clean primer off rod end bore or cylinder head liner. Do not allow primer to dry.

5-4-16.1.2. Inspect/Repair.

a. Inspect rod end or cylinder head bore for metal pickup as follows:

(1) Using abrasive cloth, Item 13, Appendix D, remove metal pickup by blending out bore.

NOTE

Bore dimensions are different at each end.

(2) Using inside micrometer caliper, measure rod end bore inside diameter (ID) in two places 90° apart. **ID OF ROD END BORE MUST NOT AVERAGE MORE THAN 2.749-INCHES AFTER REWORK.** If ID is beyond limits, replace rod end (PARA 5-5-6).

(3) Using inside micrometer caliper, measure cylinder head inside diameter (ID) in two places 90° apart. **ID OF CYLINDER HEAD BORE (LINERS INSTALLED) MUST NOT AVERAGE MORE THAN 2.748-INCHES AFTER REWORK.** If ID is beyond limits, replace cylinder head (PARA 5-5-6).

b. Visually inspect for security of the bonded liner in the cylinder head. **NO DISBONDING ALLOWED.**

5-4-16.1.3. Install.

a. Install outboard rod end bearing as follows:

(1) If inside diameter (ID) of outboard rod end bore measured less than 2.749-inches (maximum allowable diameter), perform the following:

(a) If available, using heat gun, heat outboard rod end to 250°F (121°C) for 30 minutes.

NOTE

Damper bearings to remain packaged during chilling process.

(b) If available, using dry ice, Item 125, Appendix D, chill replacement damper bearing for a minimum of 30 minutes, or for a minimum of 1 hour in a freezer before installation.

(c) Apply a coat of epoxy primer to rod end bore using epoxy primer coating kit, Item 135, Appendix D. Do not allow primer to dry. Proceed to step a. (3).

CAUTION

Damage to outboard rod end will result if rod end is heated, damper bearing is chilled, or primer is applied when using sealing compound to install damper bearing. Sealing compound will not adhere properly. When using sealing compound to install damper bearing, do not heat rod end, chill damper bearing, or apply primer.

(2) If inside diameter (ID) of outboard rod end bore measured 2.749-inches (maximum allowable diameter), apply sealing compound, Item 299, Appendix D, to damper bearing outside diameter and chamfered shoulder of damper bearing prior to installation.

NOTE

When installing bearing into rod end, make sure bearing is pressed into bore with rod end counterweight bracket facing up.

(3) Using bearing installer/remover, install damper bearing into damper.

(4) Using machinery towel, Item 211, Appendix D, wipe off extra primer or sealing compound.

NOTE

Make sure damper bearing is fully seated in rod end bore.

(5) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to retaining ring and install on the side of bearing marked retaining ring side. If bearing is not marked, install retaining ring on opposite side of bearing chamfer.

(6) Paint around exposed interfacing surface of rod end and outer race of bearing at least half way inbound on surface area of bearing (PARA 2-6-5).

b. Install cylinder head bearing as follows:

CAUTION

Damage to cylinder head will result if cylinder head is heated. Do not heat cylinder head.

NOTE

Damper bearings to remain packaged during chilling process.

(1) If available, using dry ice, Item 125, Appendix D, chill replacement bearing for a minimum of 30 minutes, or for a minimum of 1 hour in a freezer before installation.

(2) Apply a coat of sealing compound, Item 301A, Appendix D, to bearing outside diameter and inside diameter of cylinder head liner.

NOTE

When installing bearing into cylinder head, make sure bearing is pressed into bore from side with chamfered edge.

(3) Using bearing installer/remover, install bearing into cylinder head.

(4) Using machinery towel, Item 211, Appendix D, wipe off excess sealing compound.

(5) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to retaining ring and install on the side of bearing marked retaining ring side. If bearing is not marked, install retaining ring on opposite side of bearing chamfer.

(6) Paint around exposed interfacing surface of cylinder head and outer race of bearing at least half way inbound on surface area of bearing (PARA 2-6-5).

c. Connect damper end or install damper (PARA 5-4-13).

5-4-17 BLEED DAMPER (DAMPER REMOVED).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Bleed Unit, Locally-Made
- Container, 2-Gallon
- Hydraulic Fluid Dispenser, AF-5
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Globe Valve, S-138-4BP
- Hose Clamp
- Hose, MS8005E-204A
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 171, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Reducer, AN919-2D
- Reducer, AN919-6D
- Selector Valve, AN6293-6
- Tiedown Straps
- Urethane Tubing, 70652-02251-174

References

- Appendix D
- PARA 1-3-15
- PARA 5-4-13

5-4-17.1 Bleed Damper (Damper Removed).

NOTE

Bleeding of all dampers is the same.

a. Remove damper (PARA 5-4-13).

b. Locally make bleed unit as follows (Figure 5-4-17-1):

Item	Part Number	Quantity
Container, 2-gallon	-	1
Globe Valve	S-138-4BP	1
Hose	MS8005E-204A	3
Hydraulic Fluid Dispenser	AF-5	1
Reducer	AN919-6D	4
Reducer	AN919-2D	2
Selector Valve	AN6293-6	1
Urethane Tubing	70652-02251-174	2 ft

(1) Connect urethane tubing, fittings, hydraulic fluid dispenser, and valves.

(2) Using tiedown straps, secure urethane tubing to fittings.

CAUTION

At temperatures below -40°F (-40°C), hydraulic fluid, Item 154, Appendix D, must be used.

NOTE

- Do not connect damper until urethane tubing is bled.
- Make sure port numbers on selector valve faceplate are lined up with port numbers stamped on selector valve housing.

(3) Prepare hydraulic fluid dispenser for bleeding. Using at least 4 gallons of hydraulic fluid, Item 154 or Item 155, Appendix D, service hydraulic fluid dispenser.

(4) Place 2-gallon container under drain line (urethane tubing).

c. Bleed lines and fittings of air as follows:

(1) Open bleed valve.

(2) Place selector valve to cylinder 1.

(3) Open hydraulic fluid dispenser shutoff valve and purge line No. 2 of air.

(4) Close hydraulic fluid dispenser shutoff valve and connect line No. 2 to damper.

(5) Place selector valve to cylinder 2.

(6) Open hydraulic fluid dispenser shutoff valve and purge line No. 4 of air.

(7) Close hydraulic fluid dispenser shutoff valve and connect line No. 4 to damper.

(8) Make sure damper is positioned with bleed ports up.

d. Make sure damper is properly installed in bleed unit with bleed ports up, all lines bled, and hydraulic fluid dispenser serviced.

e. Bleed damper of air as follows:

(1) Open bleed valve.

(2) Place selector valve to cylinder 2 (Figure 5-4-17-1, Detail A).

(3) Open hydraulic fluid dispenser shutoff valve until damper piston is fully extended.

(4) Check for air in line No. 2. Leave selector valve in cylinder 2 position until line No. 2 is purged of all air bubbles (Figure 5-4-17-1).

(5) Place selector valve to cylinder 1. Damper piston should fully retract (Figure 5-4-17-1, Detail B).

(6) Check for air in line No. 4. Leave selector valve in cylinder 1 position until line No. 4 is purged of all air bubbles (Figure 5-4-17-1).

f. Repeat step e. until no air bubbles appear in drain line. Leave damper piston in retracted position (selector valve at cylinder 1). Close hydraulic fluid dispenser shutoff valve.

g. Bleed damper indicator of air as follows:

(1) Stand damper on inboard end.

(2) Make sure selector valve is in cylinder 1 position (Figure 5-4-17-1, Detail B).

(3) Open hydraulic fluid dispenser shutoff valve and close bleed valve. Allow damper indicator to fill with hydraulic fluid (damper indicator piston fully extended) (Figure 5-4-17-1).

(4) Close hydraulic fluid dispenser shutoff valve and place selector valve to cylinder 2 (Figure 5-4-17-1, Detail A). Allow damper indicator to bleed down (damper indicator piston will retract), and check for air bubbles in line No. 2 (Figure 5-4-17-1). Press in on damper indicator to aid in bleeding.

(5) If air bubbles are seen, purge line No. 2 of air by opening bleed valve and hydraulic fluid dispenser shutoff valve.

h. Bleed damper indicator of air again as follows:

(1) Close bleed valve and place selector valve to cylinder 1 (Figure 5-4-17-1, Detail B). Allow damper indicator to fill with hydraulic fluid (damper indicator piston fully extended).

(2) Close hydraulic fluid dispenser shutoff valve and place selector valve to cylinder 2 (Figure 5-4-17-1, Detail A).

(3) As damper indicator bleeds down, tilt damper about 45° in direction that will allow air to flow to outboard bleed port (Figure 5-4-17-2).

(4) Check for air bubbles in line No. 2. Press in on damper indicator to aid in bleeding.

(5) If air bubbles are seen, purge line No. 2 of air by opening bleed valve and hydraulic fluid dispenser shutoff valve (Figure 5-4-17-1).

(6) When air bubbles are gone, close hydraulic fluid dispenser shutoff valve and place selector valve to port 2.

i. Repeat step h. until no air bubbles appear.

j. Bleed damper indicator again as follows:

(1) Open hydraulic fluid dispenser shutoff valve and close bleed valve. Allow damper indicator to fill with hydraulic fluid (damper indicator piston fully extended).

(2) Close hydraulic fluid dispenser shutoff valve.

(3) Hold damper in upright position with inboard bleed port up (Figure 5-4-17-2).

(4) Open bleed valve and allow damper indicator to bleed down (damper indicator piston will retract) (Figure 5-4-17-1).

(5) As damper indicator bleeds down, tilt damper from side-to-side and watch for air bubbles in line No. 4 (Figure 5-4-17-2).

(6) If air bubbles are seen, purge line No. 4 of air by opening hydraulic fluid dispenser shutoff valve (selector valve still at cylinder 1 and bleed valve still open) (Figure 5-4-17-1).

(7) When all air bubbles are gone, close hydraulic fluid dispenser shutoff valve.

k. Turn damper 180° so it rests on inboard end. Repeat step j. until no air bubbles appear.

l. Bleed damper of air as follows:

(1) Place selector valve to cylinder 2 (Figure 5-4-17-1, Detail A).

(2) Open hydraulic fluid dispenser shutoff valve until damper piston is fully extended (Figure 5-4-17-1).

(3) Check for air in line No. 2. Leave selector valve in cylinder 2 position until line No. 2 is purged of all air bubbles.

(4) Place selector valve to cylinder 1 (Figure 5-4-17-1, Detail B). Damper piston should fully retract.

(5) Check for air in line No. 4 (Figure 5-4-17-1). Leave selector valve in cylinder 1 position until line No. 4 is purged of all air bubbles.

m. Repeat step l. until air bubbles no longer appear in lines. Leave damper piston in retracted position (selector valve at cylinder 1).

NOTE

When installing hose clamp on damper indicator, be careful not to score or scratch gold band on damper indicator. If necessary, pad hose clamp.

n. With selector valve still at cylinder 1, close bleed valve. Allow damper indicator to fill with hydraulic fluid (damper indicator piston extended). Install hose clamp on damper indicator piston to prevent it from retracting. Close hydraulic fluid dispenser shutoff valve. Perform the following:

(1) Remove urethane tubing from damper.

(2) Remove fittings from bleed ports.

(3) Install packings lubricated with hydraulic fluid, Item 154, Appendix D, on bleed plugs.

(4) Install bleed plugs on damper.

(5) **TIGHTEN BLEED PLUGS HANDTIGHT.**

(6) Remove hose clamp.

o. Crack bleed plug on outboard bleed port, and perform the following:

(1) If air is present, it can be heard as a hissing sound. **WHEN HISSING STOPS, TIGHTEN BLEED PLUG.**

(2) Check gold band on damper indicator. If full gold band is not visible, repeat entire bleeding procedure.

(3) **IF GOLD BAND IS VISIBLE, TIGHTEN BLEED PLUG.**

p. Repeat step o. for inboard bleed port.

q. **TORQUE BOTH BLEED PLUGS TO 65 - 75 INCH-POUNDS.**

r. Using lockwire, Item 197, Appendix D, lockwire bleed plugs and damper indicator.

s. Using heat-shrinkable insulation sleeving, Item 171, Appendix D, cover lockwire.

t. Using machinery towel, Item 211, Appendix D, clean damper of any spilled hydraulic fluid.

u. Install damper (PARA 5-4-13).

v. Check damper as follows:

(1) Position main rotor blade over nose of helicopter.

(2) Grasp main rotor blade at tip and move toward lead position. Rapidly change main rotor blade to lag position. Main rotor blade tip should move 3 - 9-inches. Do not pull main rotor blade down more than 6-inches.

(3) Have assistant observe damper indicator for the following as direction of main rotor blade is changed:

(a) If movement of damper is ⅛-inch or less, damper has been properly bled and serviced. No further action is required.

(b) If damper indicator moves more than ⅛-inch, service damper on helicopter (PARA 1-3-15).

(c) After servicing, repeat damper check (step v.).

(d) If damper indicator still moves more than ⅛-inch, remove damper and repeat entire bleeding procedure.

w. Make sure to note type of hydraulic fluid used during bleeding. This information will be recorded in helicopter logbook after installation.

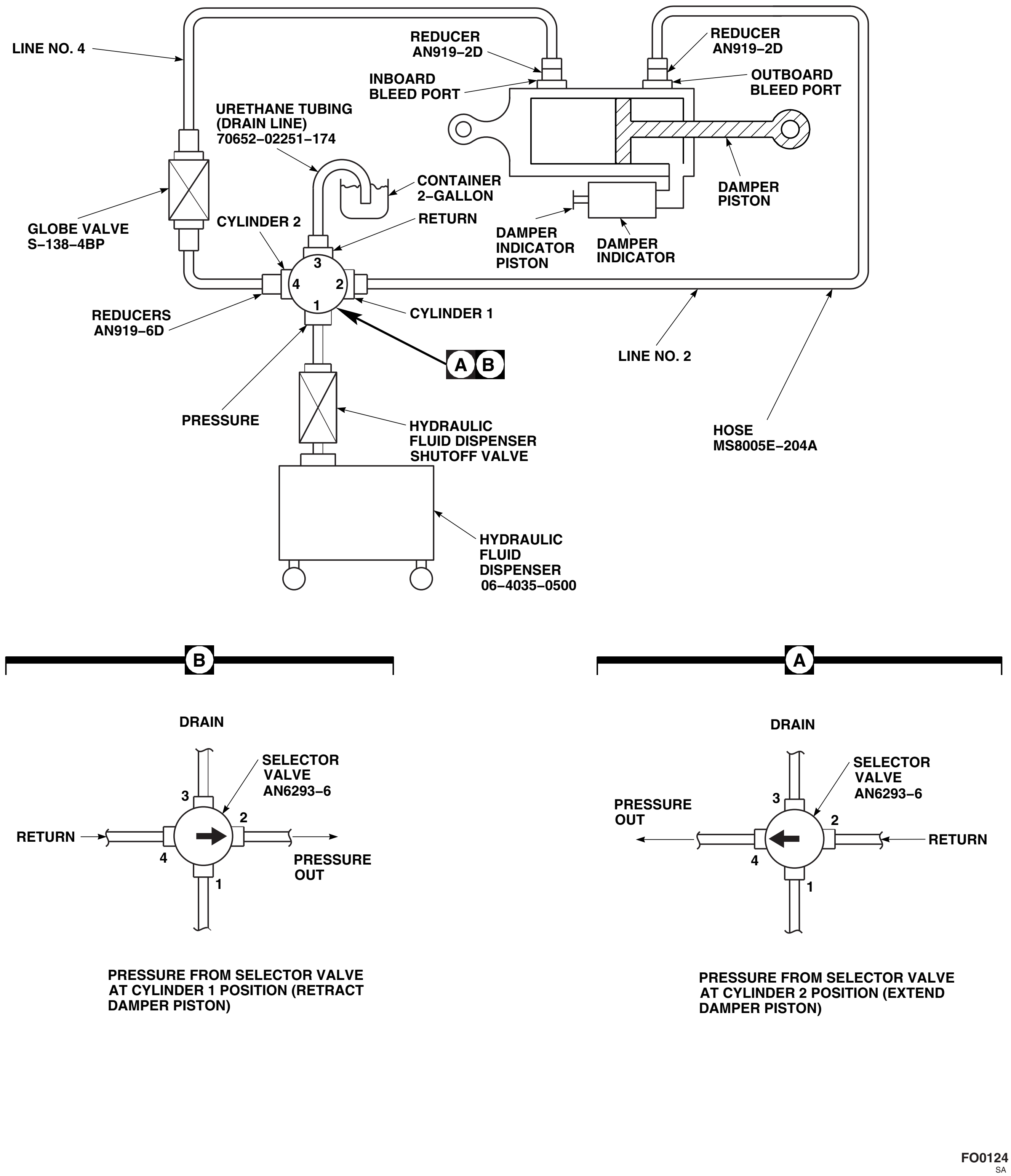


FIGURE 5-4-17-1. LOCALLY MAKE/SETUP BLEED UNIT

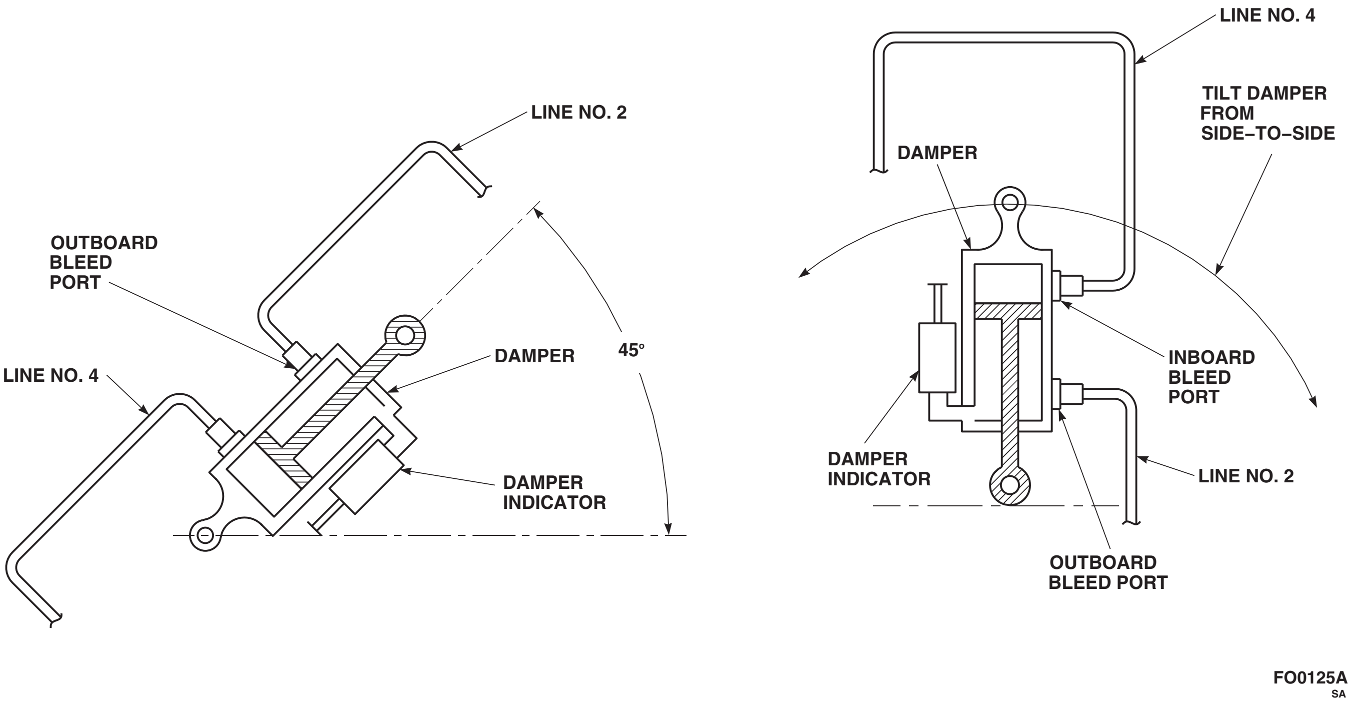


FIGURE 5-4-17-2. BLEED POSITIONS

5-4-18 PITCH CONTROL RODS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit
- Main Rotor Head Wrench
- Medium Cut File
- Pitch Control Link Tool, Locally-Made, Figure H-256, Appendix H
- Powertrain Repairer's Toolkit
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 500 - 2500 in. lbs

Materials/Parts

- Abrasive Cloth, Item 12, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Lockwire, Item 197, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- Appropriate MTF
- ASTM E1417
- PARA 1-6-18
- PARA 1-7-5
- PARA 2-6-5
- PARA 5-3-5
- PARA 5-4-4
- PARA 5-4-7
- PARA 5-4-20
- PARA 5-4-22
- TM 1-70-VIB

5-4-18.1 Replace Pitch Control Rods.

NOTE

Replacement of all pitch control rods is the same.

5-4-18.1.1 Remove.

a. If not already done, engage gust lock (PARA 1-6-18).

b. Locally make pitch control link tool (Figure H-256, Appendix H).

CAUTION

Damage to swashplate and spindle horn lug areas will result if extreme caution is not used when disconnecting pitch control rod. Use extreme caution when disconnecting pitch control rod to prevent damage to swashplate and spindle horn lug areas.

c. Remove bolts from upper and lower pitch control rod end and swashplate as follows (Figure 5-4-18-1, Detail A):

NOTE

- Use the following steps to remove both upper and lower bolts.
- It may be necessary to gently rock main rotor blade to remove binding bolt securing pitch control rod.

(1) Loosen nut on bolt.

(2) Depress retaining pawl on bolt with thumb, and continue backing off nut until castellations clear retaining pawl. Release pawl and remove nut from bolt.

(3) Depress retaining pawl on bolt with thumb, and remove washer from bolt.

NOTE

Retaining pawl on bolt will make bolt hard to remove once inside rod end. Use force required to pull bolt through rod end.

(4) Depress retaining pawl on bolt with thumb, and start removing bolt from rod end. Use force required to pull bolt from rod end.

(5) Depress retaining pawl on bolt with thumb, and remove countersunk washer from upper bolt.

(6) Depress retaining pawl on bolt with thumb, and remove spacer and countersunk washer from lower bolt.

d. Remove pitch control rod.

e. Measure dimension X on removed pitch control rod. Note number of exposed threads on lower rod end (Figure 5-4-18-1, Detail B).

5-4-18.1.2 Inspect/Repair.

a. Inspect bolts (PARA 5-4-4).

b. Inspect washers for cracks. **NONE ALLOWED.** If crack is found, replace washers.

c. Inspect bushings in swashplate lugs for corrosion pitting. **PITTING MUST NOT BE DEEPER THAN 0.005-INCH IN SHOULDER AREA OR 0.040-INCH IN CHAMFER AREA.** If pitting exceeds limits, replace spacer (PARA 5-4-22).

d. Inspect bushings on spindle horn (PARA 5-4-7).

e. Inspect spacer as follows:

(1) Inspect spacer for corrosion pitting. **PITTING MUST NOT BE DEEPER THAN 0.003-INCH ON INSIDE DIAMETER OR 0.005-INCH ON OUTSIDE DIAMETER.** If pitting exceeds limits, replace spacer (PARA 5-4-22).

(2) Inspect spacer for wear. **MAXIMUM WEAR IS 0.020-INCH.** If wear exceeds limit, replace spacer.

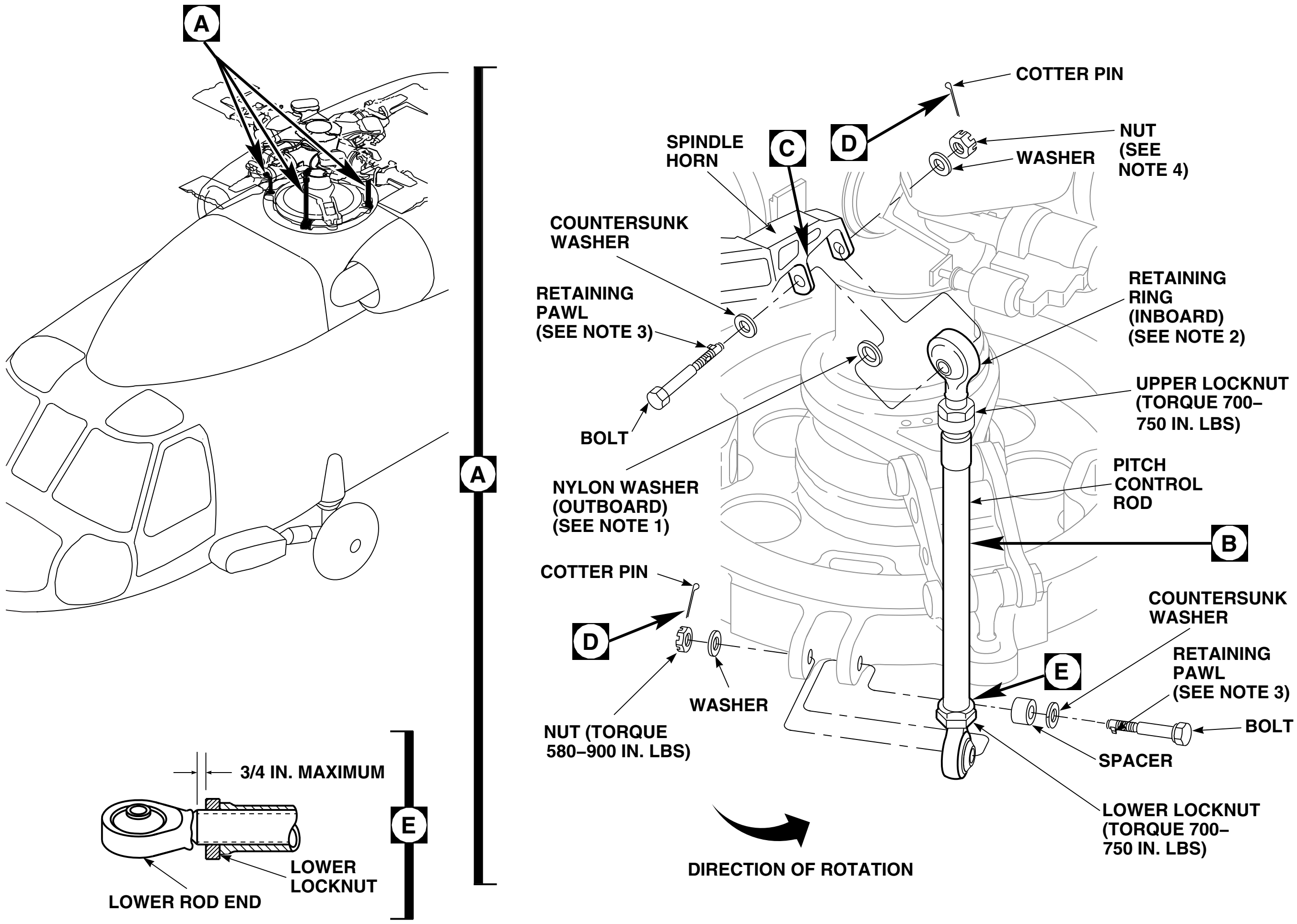
(3) Inspect spacer for nicks, scratches, or gouges. **NO DAMAGE PENETRATING THROUGH SILVER-PLATING ALLOWED.** If damage has penetrated through silver plating, replace spacer.

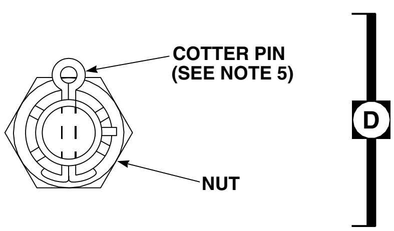
f. Inspect rod end bearings (PARA 5-3-5).

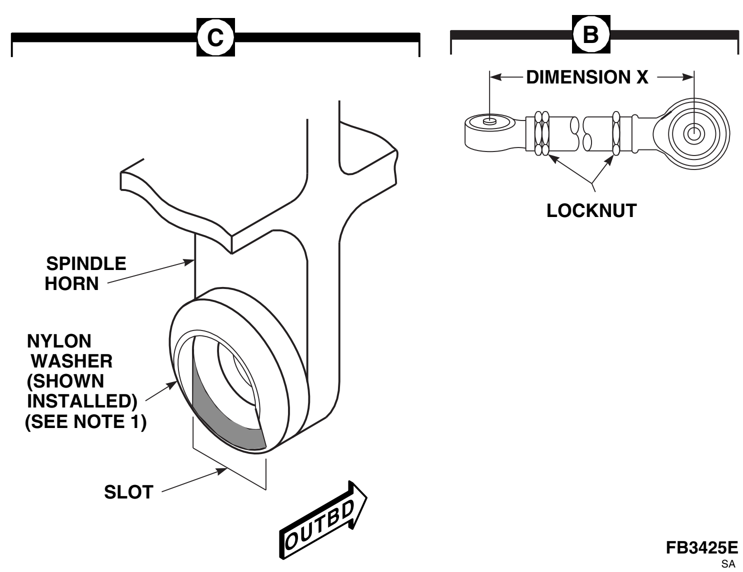
NOTE

On pitch control rods with Teflon-lined bearings, make sure movement is not between spherical bearing and outer race before replacing components.

g. Inspect control rod for movement (looseness) between link, locknuts, and rod ends. **NONE**







NOTES

1. ON PITCH CONTROL RODS WITH ROD END SPHERICAL BEARINGS, NYLON WASHER IS REQUIRED.
2. ON PITCH CONTROL RODS WITH ROD END SPHERICAL BEARINGS ONLY.
3. INSTALL BOLT WITH RETAINING PAWL IN 6 O'CLOCK POSITION.
4. ON PITCH CONTROL RODS WITH ROD END SPHERICAL BEARINGS, TORQUE NUT 1050–1500 IN. LBS. ON PITCH CONTROL RODS WITH ROD END ELASTOMERIC BEARINGS, TORQUE NUT 1350–1800 IN. LBS.
5. INSTALL COTTER PIN HEAD PERPENDICULAR TO NUT SLOT–BEND PINS OVER NUT FLAT.

FIGURE 5-4-18-1. REPLACE PITCH CONTROL RODS

ALLOWED. If movement is detected, inspect upper and/or lower rod end, link, and locknut(s) (PARA 5-4-20).

h. Inspect control rod locknut part numbers. **UPPER LOCKNUT MUST BE, 70101-08004-107. LOWER LOCKNUT MUST BE, 70101-08004-108.** If part numbers are incorrect, replace locknuts (PARA 5-4-20).

i. Inspect pitch control rod for nicks and gouges, and repair as follows:

(1) **CORNER DAMAGE UP TO 0.050-INCH DEEP MAY BE BLENDED OUT TO ½-INCH RADIUS. MULTIPLE CORNER DAMAGE MUST BE AT LEAST 1-INCH APART.**

(2) **SURFACE DAMAGE ON LINK UP TO 0.060-INCH DEEP MAY BE BLENDED OUT TO ½-INCH RADIUS. DAMAGED AREA MUST NOT EXCEED 25% OF LINK CIRCUMFERENCE.**

(3) **SURFACE DAMAGE UP TO 0.030-INCH DEEP, IN ROUND PORTION OF ROD END BETWEEN THREADED AREA AND LUG, MAY BE BLENDED OUT TO ½-INCH RADIUS. DAMAGED AREA MUST NOT EXCEED 25% OF LINK CIRCUMFERENCE.**

(4) **SURFACE DAMAGE UP TO 0.040-INCH DEEP, IN LUG PORTION OF ROD END ACROSS ANY SINGLE SURFACE, MAY BE BLENDED OUT TO ½-INCH RADIUS. MULTIPLE DAMAGE MUST BE AT LEAST 1-INCH APART.**

(5) If nicks and gouges exceed limits, replace pitch control rod.

(6) Remove damage using abrasive cloth, Item 12, Appendix D, abrasive wheel, Item 22, Appendix D, and/or medium cut file, as required.

(7) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect repaired areas (ASTM E1417).

(8) Touch up paint on reworked areas (PARA 2-6-5).

5-4-18.1.3 Install.

NOTE

If recorded dimensions and angle are not available, set length of rod to 25.22-inches, with 0.43-inch of thread exposure from face of lower locknut to end of threads on lower rod end only.

a. Adjust replacement or repaired pitch control rod to length of dimension X measured on removed pitch control rod (Figure 5-4-18-1, Detail B). Make sure lower rod end has same amount of exposed threads as noted on removal.

b. **TIGHTEN UPPER AND LOWER LOCKNUTS HANDTIGHT.**

NOTE

Nylon washer is required on helicopters with rod end spherical bearings installed only. If rod end elastomeric bearings are installed, do not install nylon washers.

c. On helicopters with rod end spherical bearings installed, perform the following:

CAUTION

Damage to pitch control rod and spindle horn will result if nylon washer is installed incorrectly. Make sure nylon washer is installed on side opposite bearing retaining ring.

(1) Install nylon washer on outboard lug of spindle horn. Position slot on washer toward swashplate with slot in line with centerline of pitch control rod (Figure 5-4-18-1, Detail C). Apply sealing compound, Item 292, Appendix D, to bond washer in place.

(2) Apply bead of sealing compound, Item 292, Appendix D, around edge of nylon washer and lug of spindle horn.

d. Install pitch control rod upper end on spindle horn with lockring key outboard. Install bolt as follows (Figure 5-4-18-1, Detail A):

WARNING

FSCAP

Verification of proper hardware installation, torque and safety installation of bolts are critical characteristics.

CAUTION

Damage to bolthead will result if countersunk washer is not installed correctly. When installing countersunk washers, make sure to install the washer with the countersunk side toward bolthead.

NOTE

Make sure bolthead faces outboard.

(1) Depress retaining pawl on bolt with thumb, and install countersunk washer. Make sure countersunk side of washer is toward bolthead.

(2) Depress retaining pawl on bolt with thumb, and install bolt and countersunk washer through pitch control rod and spindle horn, with pawl in 6 o'clock position.

(3) Depress retaining pawl on bolt with thumb, and install washer and nut. Do not torque upper nut now.

CAUTION

Damage to swashplate and spindle horn lug areas will result if extreme caution is not used during installation of pitch control rod. Use extreme caution when installing pitch control rod to prevent damage to swashplate and spindle horn lug areas.

NOTE

- It may be necessary to gently rock or pitch main rotor blade or spindle up to allow lower rod end to swing over swashplate lug from side-to-side and be inserted into clevis from the top, and to line up pitch control rod for bolt insertion.
- On pitch control rods which have the lower rod end bearing retained by a retaining ring, make sure the retaining ring is installed in direction of rotation (counterclockwise) toward leading edge of main rotor blade.

e. Install pitch control rod lower rod end in swashplate. Install bolt as follows:

CAUTION

Damage to bolthead will result if countersunk washer is not installed correctly. When installing countersunk washers, make sure to install the washer with countersunk side toward bolthead.

(1) Depress retaining pawl on bolt with thumb, and install countersunk washer on bolt. Make sure countersunk side of washer is toward bolthead. Install spacer on bolt.

(2) Depress retaining pawl on bolt with thumb, and install bolt and washer through swashplate lugs and pitch control rod end bearing, with pawl in 6 o'clock position.

(3) Depress retaining pawl on bolt with thumb, and install washer and nut. Do not torque lower nut now.

f. Use main rotor head wrench on upper nut. **ON ROD END SPHERICAL BEARINGS, TORQUE UPPER NUT TO 1050 - 1500 INCH-POUNDS. ON ROD END ELASTOMERIC BEARINGS, TORQUE UPPER NUT TO 1350 - 1800 INCH-POUNDS.** Install cotter pin in castellated nut with head perpendicular to nut slot, and bend pins over hex nut flats (Figure 5-4-18-1, Detail D).

g. **TORQUE LOWER NUT TO 580 - 900 INCH-POUNDS** (Figure 5-4-18-1, Detail A). Install cotter pin in castellated nut with head perpendicular to nut slot, and bend pins over hex nut flats (Figure 5-4-18-1, Detail D).

h. Check to make sure both upper and lower rod ends are parallel to their respective attachment lugs at same time. Rotate pitch control rod by hand to make sure there is no metal-to-metal contact on lower rod end attachment lugs. Make small adjustments as required. **EXPOSED THREADS ON LOWER PITCH CONTROL RODS MUST NOT EXCEED ¾-INCH** (Figure 5-4-18-1, Detail E). Tighten upper and lower locknuts as follows:

NOTE

To prevent improper torque of upper locknut on pitch control rod, make sure torque wrench is not in contact with lock key when torquing.

(1) **SECURE LOCK KEY IN NOTCH BY TIGHTENING UPPER LOCKNUT.**

(2) **TORQUE UPPER LOCKNUT TO 700 - 750 INCH-POUNDS** (Figure 5-4-18-1, Detail A).

(3) **TORQUE LOWER LOCKNUT TO 700 - 750 INCH-POUNDS.**

i. Using lockwire, Item 197, Appendix D, lockwire lower locknut to link, and then back to lower locknut (double-safety).

j. Using lockwire, Item 197, Appendix D, lockwire from upper locknut to lock key, and then back to opposite side of upper locknut (double-safety).

CAUTION

Damage to bolt may result if paint or sealing compound is allowed to contact retaining pawl area of bolt. Do not allow paint or sealing compound to come in contact with retaining pawl area of bolt.

k. Touch up paint on nut and bolthead (PARA 2-6-5).

l. Perform main rotor blade track and balance (TM 1-70-VIB).

NOTE

A maintenance test flight is not required if only one end of pitch control rod was disconnected at a time and no adjustments were made.

m. Perform maintenance test flight (Appropriate MTF).

NOTE

Apply sealing compound only to pitch control rod attachment hardware. Do not apply sealing compound to upper and lower locknuts or rod end threads.

n. Apply sealing compound, Item 292, Appendix D, to seal areas around pitch control link adjustment rod ends, nuts, washers, and boltheads on both upper and lower ends.

o. On helicopters with pitch control rods equipped with elastomeric rod ends, check helicopter logbook to make sure pitch control rod upper and lower locknuts retorque is shown due 9 - 11 flight hours after installation (PARA 1-7-5).

p If required, disengage gust lock (PARA 1-6-18).

5-4-19 PITCH CONTROL ROD END SPHERICAL BEARINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Bearing Installer/Remover, 65700-40030-041
- Bearing Installer/Remover, 70700-77306-041
- Freezer
- Powertrain Repairer's Toolkit
- Protective Covering

Materials/Parts

- Abrasive Cloth, Item 13, Appendix D
- Dry Ice, Item 125, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 258, Appendix D

References

- Appendix D
- PARA 5-4-18
- PARA 5-4-20

5-4-19.1 Replace Pitch Control Rod End Spherical Bearings.

5-4-19.1.1 Remove.

WARNING

FSCAP

Upper and lower end of pitch control rods are critical characteristic surfaces which must not be damaged during and after removal. Use extreme caution when removing or disconnecting rod end and provide protective covering after removal.

CAUTION

Damage to swashplate and spindle horn lug areas will result if extreme caution is not used when disconnecting pitch control rod end. Use extreme caution when disconnecting pitch control rod end to prevent damage to swashplate and spindle horn lug areas.

NOTE

- Replacement of all pitch control rod end spherical bearings is the same, except as noted.
- Pitch control rod does not have to be removed if only one pitch control rod end spherical bearing is being replaced. Disconnect pitch control rod end with spherical bearing being replaced only.
- A maintenance test flight is not required if only one end of pitch control rod was disconnected at a time and no adjustments were made.

a. Disconnect applicable pitch control rod end (PARA 5-4-18).

b. Install protective covering to pitch control rod end(s).

c. Remove retaining ring (Figure 5-4-19-1).

d. Using bearing installer/remover, 70700-77306-041, on upper rod end, or bearing installer/remover, 65700-40030-041, on lower rod end, remove pitch control rod end spherical bearing from rod end by pressing from retaining ring side of pitch control rod end spherical bearing.

e. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean primer out of rod end bore. Do not allow primer to dry in bore.

5-4-19.1.2 Inspect/Repair.

a. Inspect upper rod end bore for metal pickup. Using abrasive cloth, Item 13, Appendix D, remove metal pickup by blending out bore.

NOTE

Bore dimensions for upper and lower rod ends are different.

b. Measure upper rod end bore inside diameter (ID) in two places 90° apart and take average of these two measurements. **ID MUST NOT AVERAGE MORE THAN 2.749-INCHES AFTER REWORK.** If ID is beyond limits, replace rod end (PARA 5-4-20).

c. Using abrasive cloth, Item 13, Appendix D, inspect lower rod end bore for metal pickup. Remove metal pickup by blending out bore.

d. Measure lower rod end bore inside diameter (ID) in two places 90° apart and take average of these two measurements. **ID MUST NOT AVERAGE MORE THAN 1.8747-INCHES AFTER REWORK.** If ID is beyond limits, replace rod end (PARA 5-4-20).

5-4-19.1.3 Install.

NOTE

Pitch control rod end spherical bearings are to remain packaged during chilling process.

a. If available, place new pitch control rod end spherical bearing in dry ice, Item 125, Ap-

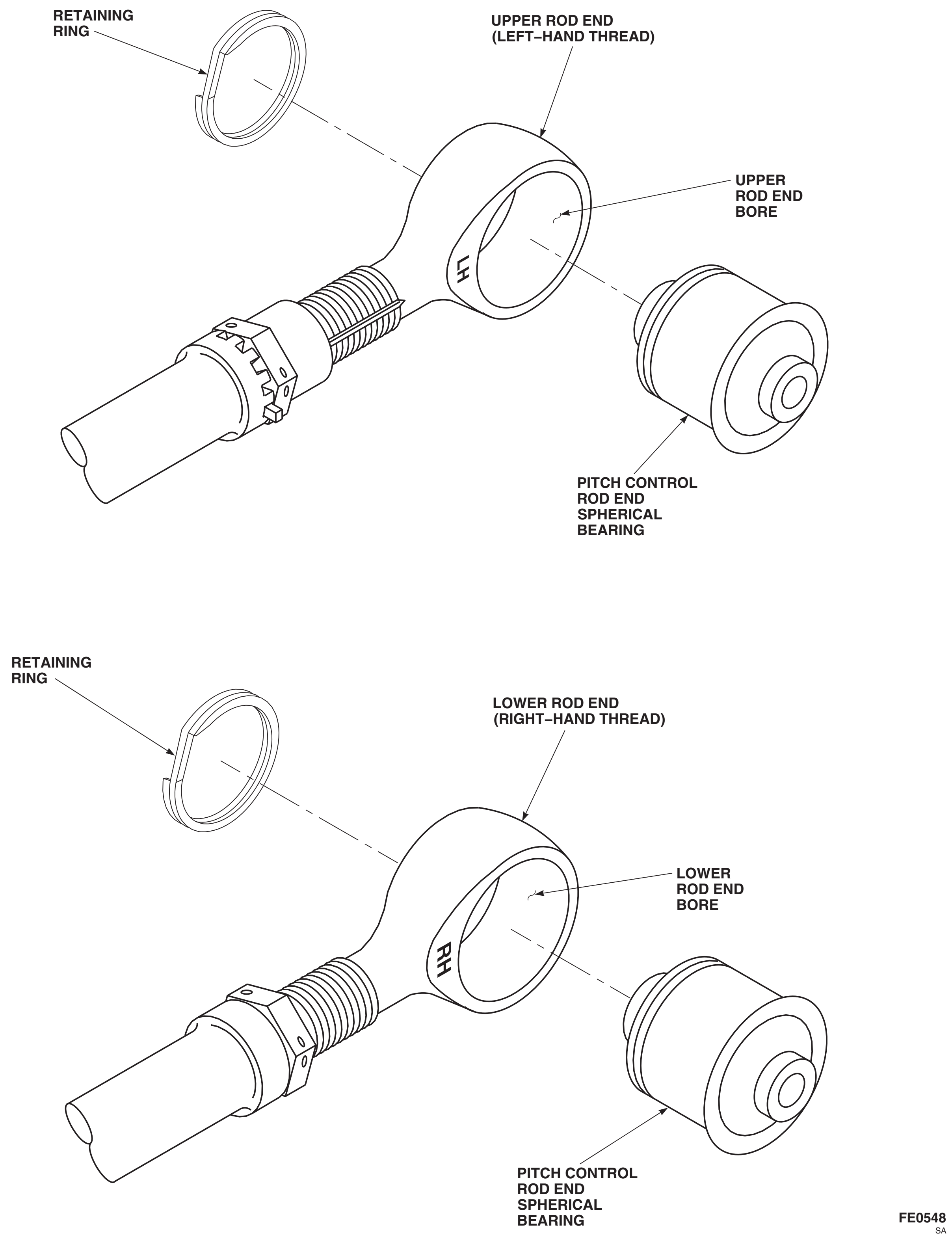


FIGURE 5-4-19-1. REPLACE PITCH CONTROL ROD END SPHERICAL BEARINGS pendix D, for at least 30 minutes or freezer for at least 1 hour before installation.

NOTE

Make sure upper pitch control rod end spherical bearing is pressed into bore so that retaining ring is installed opposite key slot.

b. Apply primer, Item 258, Appendix D, to inside of rod end bore. Do not allow primer to dry.

c. Using bearing remove/installer, 70700-77306-041, on upper rod end, or bearing remover/installer, 65700-40030-041, on lower rod end, install pitch control rod end spherical bearing into rod end (Figure 5-4-19-1).

d. Using machinery towel, Item 211, Appendix D, wipe off extra primer.

NOTE

Make sure pitch control rod end spherical bearing is fully seated in rod end bore.

e. Install retaining ring.

f. Remove protective covering to pitch control rod end(s).

g. Connect applicable pitch control rod end (PARA 5-4-18).

5-4-20 PITCH CONTROL ROD UPPER AND LOWER ROD ENDS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Cotton Gloves
- Fluorescent Penetrant Inspection Kit
- Medium Cut File
- Powertrain Repairer's Toolkit

Materials/Parts

- Abrasive Cloth, Item 12, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Acetone, Item 25, Appendix D
- Lubricant, Solid Film, Item 201, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 301A, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5
- PARA 5-4-18

5-4-20.1 Replace Pitch Control Rod Upper and Lower Rod Ends.

NOTE

Replacement of all pitch control rod upper and lower rod ends is the same.

5-4-20.1.1 Remove.

NOTE

If rod end spherical bearing is replaced with rod end elastomeric bearing, both upper and lower rod end bearings on that pitch control rod must be replaced as well as both upper and lower ends of opposing pitch control rod.

a. Remove pitch control rod (PARA 5-4-18).

b. Loosen upper or lower locknut and remove upper or lower rod end (Figure 5-4-20-1).

c. Remove lock key from upper rod end.

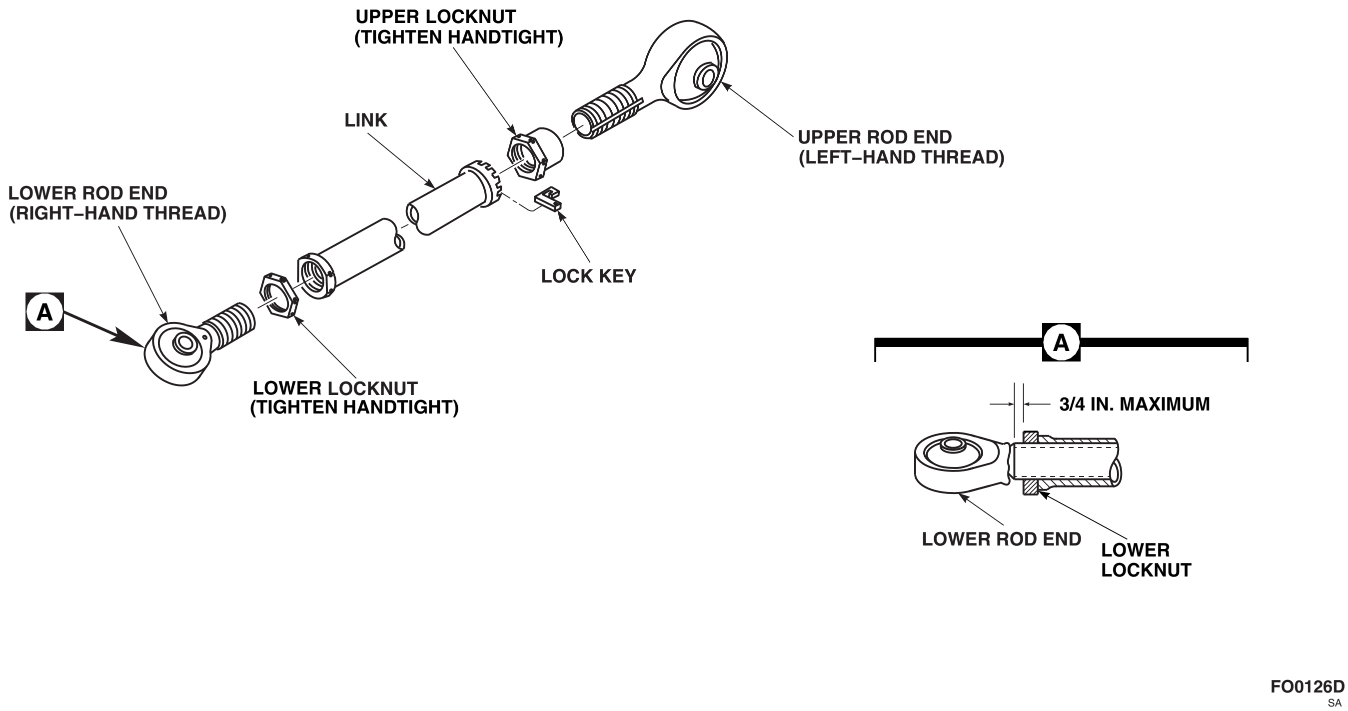


FIGURE 5-4-20-1. REPLACE PITCH CONTROL ROD UPPER AND LOWER ROD ENDS

5-4-20.1.2 Inspect.

NOTE

- Upper locknut, 70101-08004-101, must be replaced with locknut, 70101-08004-107, regardless of condition. Do not reuse locknut, 70101-08004-101.
- Lower locknut, 70101-08004-102, must be replaced with locknut, 70101-08004-108, regardless of condition. Do not reuse locknut, 70101-08004-102.

a. Inspect thread condition of link, locknut, and rod ends for damage. **CORRODED, FLATTENED, STRIPPED, OR CROSSED THREADS NOT ALLOWED.** If damage is found, replace components as required.

b. Inspect link for nicks and gouges, and repair as follows:

(1) **CORNER DAMAGE, UP TO 0.050-INCH DEEP, MAY BE BLENDED OUT TO ½-INCH RADIUS. MULTIPLE CORNER DAMAGE MUST BE AT LEAST 1-INCH APART.**

(2) **SURFACE DAMAGE, UP TO 0.060-INCH DEEP, MAY BE BLENDED OUT TO ½-INCH RADIUS. DAMAGED AREA MUST NOT EXCEED 25% OF LINK CIRCUMFERENCE.**

(3) If nicks and gouges exceed limits, replace link.

(4) Remove damage using abrasive cloth, Item 12, Appendix D, abrasive wheel, Item 22, Appendix D, and/or medium cut file, as required.

(5) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect repaired areas (ASTM E1417).

(6) Touch up paint on reworked areas (PARA 2-6-5).

c. Inspect rod ends for nicks and gouges, and repair as follows:

(1) **CORNER DAMAGE, UP TO 0.050-INCH DEEP, MAY BE BLENDED OUT TO ½-INCH RADIUS. MULTIPLE CORNER DAMAGE MUST BE AT LEAST 1-INCH APART.**

(2) **SURFACE DAMAGE, UP TO 0.030-INCH DEEP, IN ROUND PORTION OF ROD END BETWEEN THREADED AREA AND LUG, MAY BE BLENDED OUT TO ½-INCH RADIUS. DAMAGED AREA MUST NOT EXCEED 25% OF LINK CIRCUMFERENCE.**

(3) **SURFACE DAMAGE, UP TO 0.040-INCH DEEP, IN LUG PORTION OF ROD END ACROSS ANY SINGLE SURFACE, MAY BE BLENDED OUT TO ½-INCH RADIUS. MULTIPLE DAMAGE MUST BE AT LEAST 1-INCH APART.**

(4) If nicks and gouges exceed limits, replace rod end.

(5) Remove damage using abrasive cloth, Item 12, Appendix D, abrasive wheel, Item 22, Appendix D, and/or medium cut file, as required.

(6) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect repaired areas (ASTM E1417).

(7) Touch up paint on reworked areas (PARA 2-6-5).

d. Inspect threads of rod ends for missing solid film lubricant. **NONE ALLOWED.** If lubricant is missing, repair as follows:

(1) Using clean, dry machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, clean areas of bare metal. Allow to dry thoroughly.

NOTE

Solid film lubricant will not adhere to contaminated surfaces. Do not touch cleaned areas with bare hands. Make sure to wear clean cotton gloves when handling rod end.

(2) At room temperature, spray exposed area with solid film lubricant, Item 201, Appendix D. Apply multiple thin coats, allowing 15 - 20 minutes drying time between coats.

(3) Allow solid film lubricant to cure minimum of 12 hours before using rod end.

5-4-20.1.3 Install.

NOTE

- On helicopters with rod end elastomeric bearings, if rod end(s) will be replaced, make sure shelf life for replacement rod end(s) has not exceeded 60 months. If shelf life has exceeded 60 months, do not use.
- Upper locknut, 70101-08004-101, must be replaced with locknut, 70101-08004-107, regardless of condition. Do not reuse locknut, 70101-08004-101.
- Lower locknut, 70101-08004-102, must be replaced with locknut, 70101-08004-108, regardless of condition. Do not reuse locknut, 70101-08004-102.

a. Apply coat of sealing compound, Item 301A, Appendix D, to threads of upper and lower locknuts.

b. Assemble pitch control rod as follows (Figure 5-4-20-1):

(1) Install locknut(s) on rod end(s).

(2) Install lower rod end in link.

(3) Install upper rod end and lock key in link.

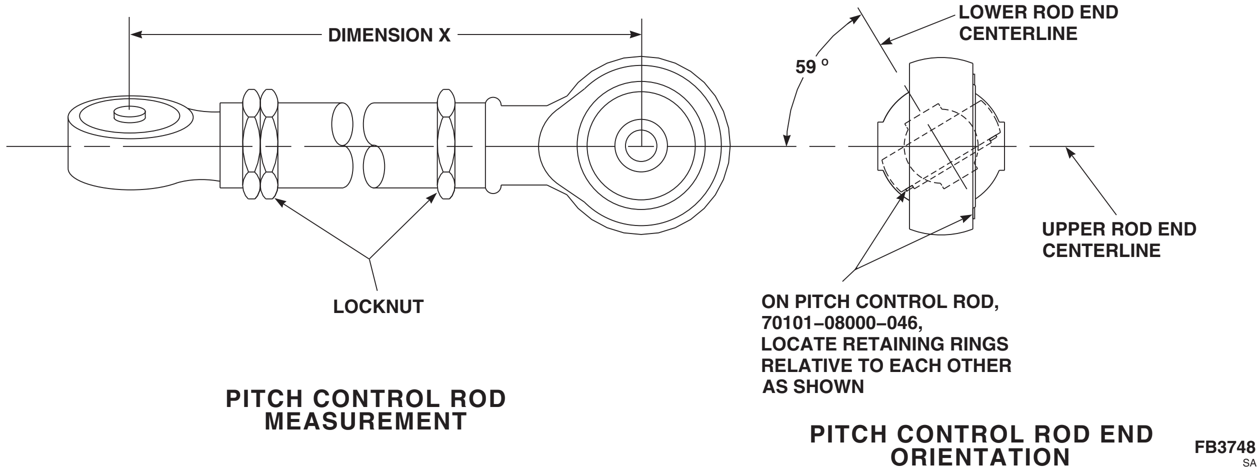


FIGURE 5-4-20-2. PITCH CONTROL ROD ORIENTATION

NOTE

- If dimension X is not available, set control rod length to 25.22-inches. When setting control rod length to 25.22-inches, length of exposd lower rod end threads must not exceed 0.43-inch.
- On pitch control rod, 70101-08000-046, make sure spherical bearing retaining rings face the same direction.

c. Adjust pitch control rod length to length of dimension X as noted during disassembly, and set rod end angle to 59° (Figure 5-4-20-2). Make sure length of exposed lower rod end threads does not exceed ¾-inch (Figure 5-4-20-1, Detail A). Engage lock key on upper rod end.

d. **TIGHTEN UPPER OR LOWER LOCKNUT HANDTIGHT.**

NOTE

Upper or lower locknut are torqued and lockwired during installation of pitch control rod (PARA 5-4-18).

e. Install pitch control rod (PARA 5-4-18).

5-4-21 PITCH CONTROL ROD ADJUSTMENT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Powertrain Repairer's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-7-5
- TM 1-70-VIB

5-4-21.1 Adjust Pitch Control Rod.

NOTE

- For every three notches on pitch control rod, there will be a 1% rotor speed (Nr) change.
- For purpose of adjusting autorotation rpm, all four pitch control rods must be adjusted in equal amounts.
- A one notch change to a pitch control rod will affect track by approximately ¼-inch.

a. Adjust pitch control rods as follows:

(1) Loosen upper and lower locknuts and slide lock key upward enough to allow link to be free to rotate (Figure 5-4-21-1, Detail A).

(2) To decrease rpm, turn link in +2 MINUTES/NOTCH (counterclockwise) direction (Figure 5-4-21-1, Detail B).

(3) To increase rpm, turn link in -2 MINUTES/NOTCH (clockwise) direction.

(4) Check for no more than ¾-inch of threads showing on lower end of pitch control rod (Figure 5-4-21-1, Detail A). If exposed thread is more than ¾-inch, pitch control rod has been improperly adjusted. Recheck adjustment.

b. Tighten upper and lower locknuts as follows:

NOTE

To prevent improper torque of upper locknut on pitch control rod, make sure torque wrench is not in contact with lock key when torquing.

(1) **SECURE LOCK KEY IN NOTCH BY TIGHTENING UPPER LOCKNUT.**

(2) **TORQUE UPPER LOCKNUT TO 700 - 750 INCH-POUNDS.**

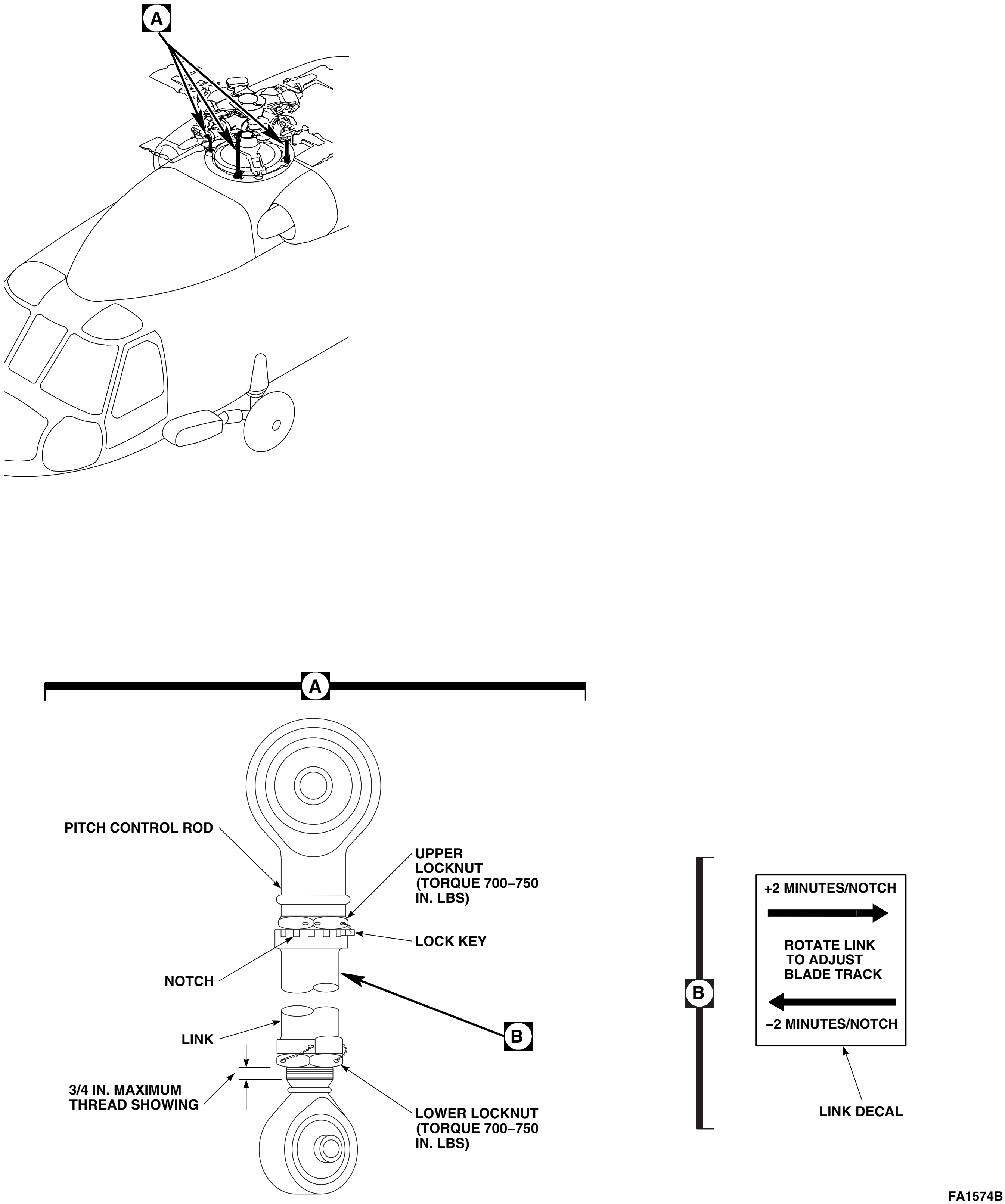


FIGURE 5-4-21-1. ADJUST PITCH CONTROL ROD

(3) **TORQUE LOWER LOCKNUT TO 700 - 750 INCH-POUNDS.**

c. Using lockwire, Item 197, Appendix D, lockwire lower locknut to link, and then back to lower locknut (double-safety).

d. Using lockwire, Item 197, Appendix D, lockwire from upper locknut to lock key, and then back to opposite side of upper locknut (double-safety).

e. Make sure area is clean and free of foreign material.

f. Perform main rotor blade track and balance (TM 1-70-VIB).

g. Perform maintenance test flight (Appropriate MTF).

h. On helicopters with pitch control rods equipped with elastomeric rod ends, check helicopter logbook to make sure pitch control rod upper and lower locknuts retorque is shown due 9 to 11 flight hours after installation (PARA 1-7-5).

5-4-22 SWASHPLATE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, ¼-inch
- Clamps
- Drill
- Fluorescent Penetrant Inspection Kit
- Grease Pencil
- Hoist, Capacity to Support Swashplate (Alternate for Cabin Maintenance Crane)
- Hoisting Sling, 70700-20320-042
- Medium Cut File
- Powertrain Repairer's Toolkit
- Protective Covering
- Shipping Container (Alternate for Work Platform)
- Syringe
- Torque Adapter Set
- Torque Wrench, 150 - 700 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Work Platform (Alternate for Shipping Container)

Materials/Parts

- Abrasive Cloth, Item 5, Appendix D
- Abrasive Cloth, Item 12, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 32, Appendix D
- Alcohol, Denatured, Item 70, Appendix D
- Brush, Artist, Item 85, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Masking Tape, 2-inches, Item 214, Appendix D
- Paint, Polyurethane, Item 223, Appendix D
- Paint, Polyurethane, Item 226, Appendix D
- Paint, Polyurethane, Item 227A, Appendix D

References

- Appendix D
- Appropriate MTF
- ASTM E1417
- PARA 1-5-3
- PARA 1-6-18
- PARA 2-6-5
- PARA 5-3-6

- PARA 5-4-1
- PARA 5-4-24
- PARA 5-4-27
- PARA 11-4-4
- PARA 18-4-1

5-4-22.1 Replace Swashplate.

NOTE

Swashplate weighs about 180 pounds.

5-4-22.1.1 Remove.

a. If cabin maintenance crane will be used, assemble and install cabin maintenance crane (PARA 18-4-1).

b. Remove main rotor head (PARA 5-4-1).

c. Reinstall bolts, washers, spacers, and nuts securing pitch control rods to swashplate, back in swashplate.

CAUTION

Damage to main rotor shaft can result when removing lower pressure plate. Use care when removing lower pressure plate.

d. Place straps of hoisting sling around bolts in pitch control rod attachment points on swashplate (Figure 5-4-22-1, Sheet 1, Detail A).

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

e. Attach hoisting sling to hoist or cabin maintenance crane.

f. Adjust straps before lifting for even distribution of weight, and so swashplate can be lifted straight up.

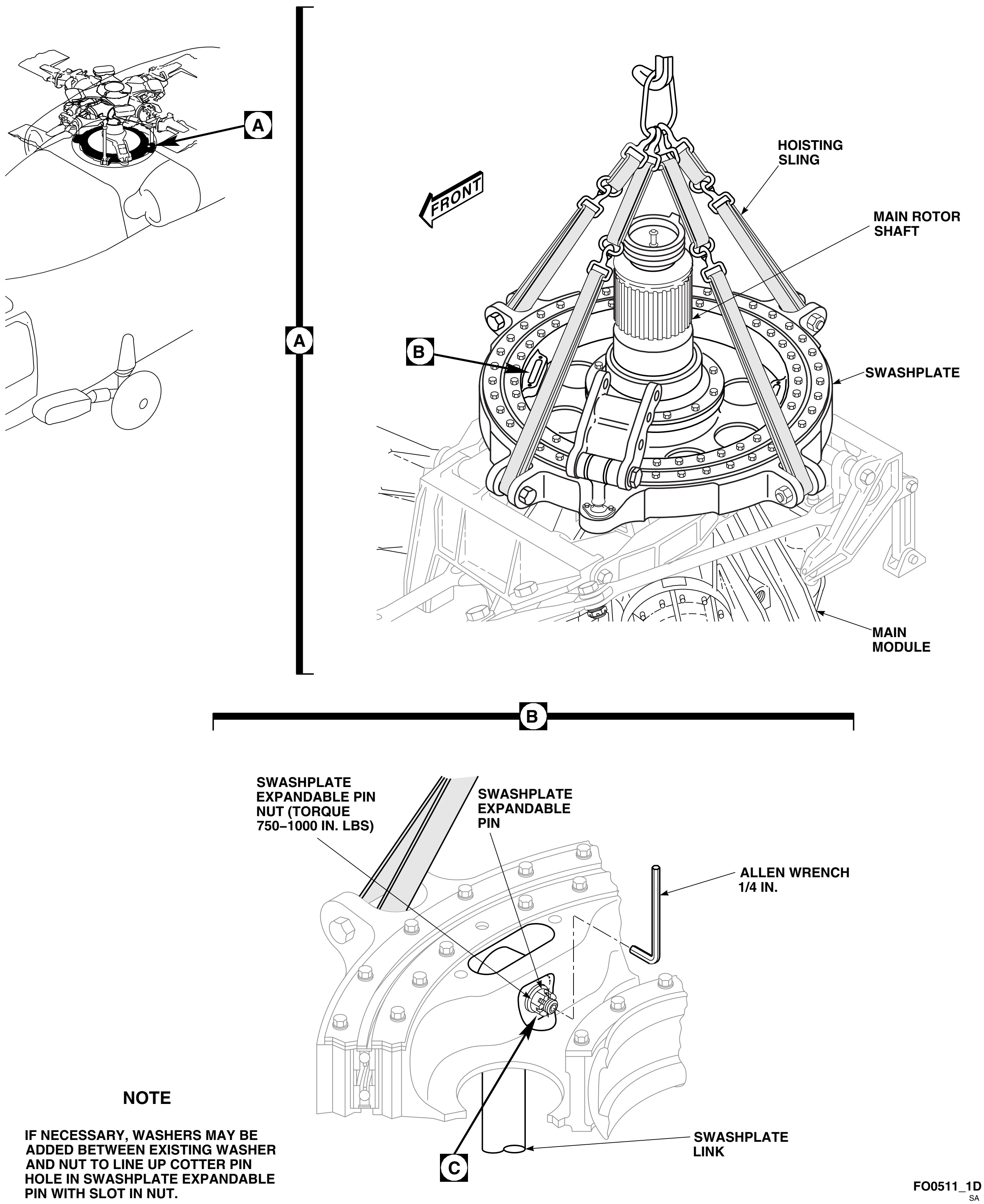
g. Insert 1/4-inch Allen wrench into center pin of swashplate expandable pin securing swashplate links (Figure 5-4-22-1, Sheet 1, Detail B). While holding 1/4-inch Allen wrench, remove swashplate expandable pin nut and washers (Figure 5-4-22-1, Sheet 2, Detail C). Do not allow swashplate expandable pin to turn.

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting swashplate. Keep area clear when hoisting swashplate.

h. Lift hoist enough to carry weight of swashplate.

i. Remove swashplate expandable pins.



NOTE

IF NECESSARY, WASHERS MAY BE ADDED BETWEEN EXISTING WASHER AND NUT TO LINE UP COTTER PIN HOLE IN SWASHPLATE EXPANDABLE PIN WITH SLOT IN NUT.

FIGURE 5-4-22-1. REPLACE SWASHPLATE (SHEET 1 OF 2)

WARNING

FSCAP

Exposed surfaces of main rotor shaft and swashplate link lug areas are critical surfaces which must not be damaged during and after removal of swashplate. Use extreme caution when removing and provide protective covering over critical surfaces after removal of swashplate.

j. Remove swashplate from main module, taking care not to damage main rotor shaft and swashplate links (Figure 5-4-22-1, Sheet 1, Detail A).

k. Install protective covering to exposed surfaces of main rotor shaft and swashplate link lug areas.

l. Inspect swashplate expandable pins and swashplate expandable pin nuts (PARA 5-4-24).

m. Install swashplate expandable pins in swashplate. Use only enough torque to hold swashplate expandable pin.

n. Place swashplate on work platform or in shipping container.

o. Remove rotating scissors (PARA 5-4-27).

p. Remove hoisting sling.

q. If used, disassemble and remove cabin maintenance crane (PARA 18-4-1).

5-4-22.1.2 Inspect.

Inspect swashplate (PARA 5-3-6 and PARA 5-4-22.2).

5-4-22.1.3 Install.

NOTE

If new or replacement swashplate was built-up more than 12 months prior to installation date, new or replacement swashplate must be lubricated prior to installation.

a. Check new or replacement swashplate build up date. If new or replacement swashplate was built-up more than 12 months prior to installation, lubricate swashplate (PARA 1-5-3).

b. Connect straps of hoisting sling around bolts in pitch control rod attachment points on swashplate. Adjust length of straps to evenly distribute weight when hoisting (Figure 5-4-22-1, Sheet 1, Detail A).

c. Attach hoisting sling to hoist or cabin maintenance crane.

d. Install rotating scissors (PARA 5-4-27).

WARNING

FSCAP

Exposed surfaces of main rotor shaft and swashplate link lug areas are critical surfaces which must not be damaged during installation of swashplate. After removal of protective covering from main rotor shaft and swashplate links, use extreme caution when installing swashplate.

e. Remove protective covering from exposed surfaces of main rotor shaft and swashplate link lug areas.

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting swashplate. Keep area clear when hoisting swashplate.

f. Lift and place swashplate onto main rotor shaft, taking care not to damage main rotor shaft.

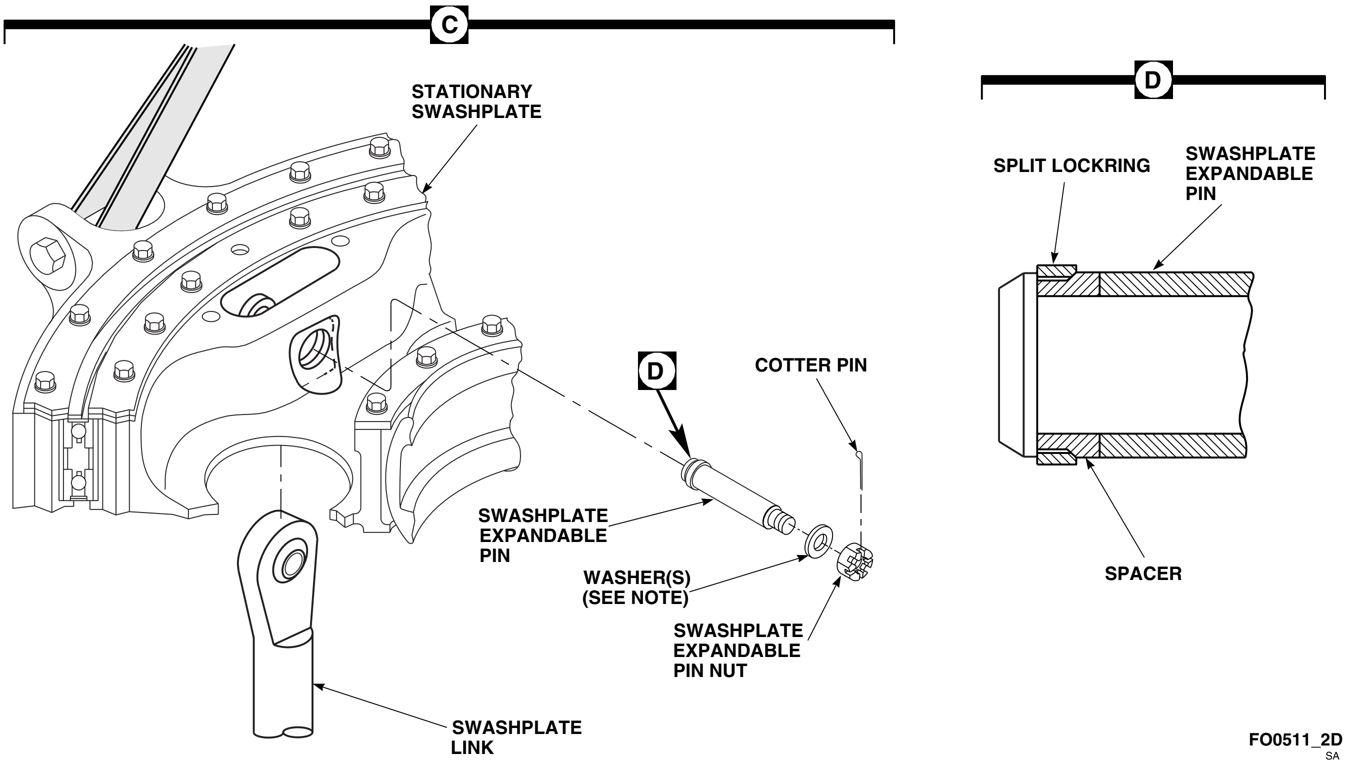


FIGURE 5-4-22-1. REPLACE SWASHPLATE (SHEET 2 OF 2)

g. Lower swashplate onto main module and guide swashplate links into position (Figure 5-4-22-1, Sheet 2, Detail C).

h. Back off swashplate expandable pin nut until split lockring segments are loose on swashplate expandable pin (Figure 5-4-22-1, Sheet 2, Detail C and Detail D).

WARNING

FSCAP

- Verification of proper hardware installation and torque of swashplate expandable pin nut is a critical characteristic.
- Verification of correct installation of swashplate expandable pin split lockring and male spacer is a critical characteristic.

NOTE

If necessary, add washer between existing washer and swashplate expandable pin nut to line up cotter pin hole in swashplate expandable pin with slot in nut.

i. Insert swashplate expandable pin through stationary swashplate and swashplate link of control rod end. Make sure washer under swashplate expandable pin nut is flush with inner side of swashplate (Figure 5-4-22-1, Sheet 2, Detail C).

j. **TIGHTEN SWASHPLATE EXPANDABLE NUT HANDTIGHT.**

k. Insert 1/4-inch Allen wrench into center on swashplate expandable pin. **USING TORQUE ADAPTER, HOLD ALLEN WRENCH AND TORQUE SWASHPLATE EXPANDABLE NUT TO 750 - 1000 INCH-POUNDS** (Figure 5-4-22-1, Sheet 1, Detail B). **DO NOT ALLOW SWASHPLATE EXPANDABLE PIN TO TURN.**

l. **VERIFY CLEARANCE UNDER WASHER DOES NOT EXCEED 0.002-INCH.** This indicates that bolt is locked in place.

m. Install cotter pin.

n. Remove hoisting sling.

NOTE

If cabin maintenance crane will be used to install maintenance rotor head, do not disassemble crane.

o. If used, disassemble and remove cabin maintenance crane (PARA 18-4-1).

p. Install main rotor head (PARA 5-4-1).

q. Perform main rotor system rig check (PARA 11-4-4).

r. Make sure area is clean and free of foreign material.

NOTE

After installation of new or relubricated swashplate, considerable grease may be displaced from duplex ball bearing between stationary and rotating swashplate during normal helicopter operation.

s. Perform maintenance test flight (Appropriate MTF).

5-4-22.2 Inspect/Repair Swashplate.

NOTE

In the following steps, damage should be blended out using abrasive wheel, Item 22, Appendix D, abrasive cloth, Item 12, Appendix D, and/or medium cut file, as required.

a. Visually inspect swashplate grease shield for cracks. **NO MORE THAN 15 CRACKS ALLOWED IN GREASE SHIELD. CRACKS MUST NOT BE LESS THAN 1-INCH APART.** If crack is found in grease shield, perform the following:

(1) Turn main rotor head until red index marks on rotating and stationary swashplates line up.

(2) If not already done, engage gust lock or rotor brake (PARA 1-6-18).

(3) Remove bolts and washers securing stationary swashplate bearing retaining ring to stationary swashplate. Slide bearing retaining ring to one side (Figure 5-4-22-2, Detail B).

(4) Using drill, stop-drill cracks.

(5) Install bearing retaining ring with bolts and washers, making sure red index marks line up. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**

b. If crack is suspected in any other part of swashplate, perform the following:

(1) Clean and strip suspected area. If necessary, using 2-inch masking tape, Item 214, Appendix D, protect bearings.

(2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace swashplate.

(3) If crack is not found, using primer and paint, touch up stripped area (PARA 2-6-5).

c. Inspect corners of swashplate for nicks and gouges. **DAMAGE TO CORNERS MUST NOT EXCEED 0.10-INCH DEEP.** If necessary, repair as follows:

(1) Blend damage to ½-inch radius.

(2) Remove any raised material around damaged area.

(3) Touch up paint repaired area (PARA 2-6-5).

d. Inspect nonmating surfaces of swashplate for nicks and gouges. **DAMAGE TO ANOINTING SURFACES, EXCEPT PITCH CONTROL ROD LUGS SURFACES, MUST NOT EXCEED 0.050-INCH DEEP. DAM-**

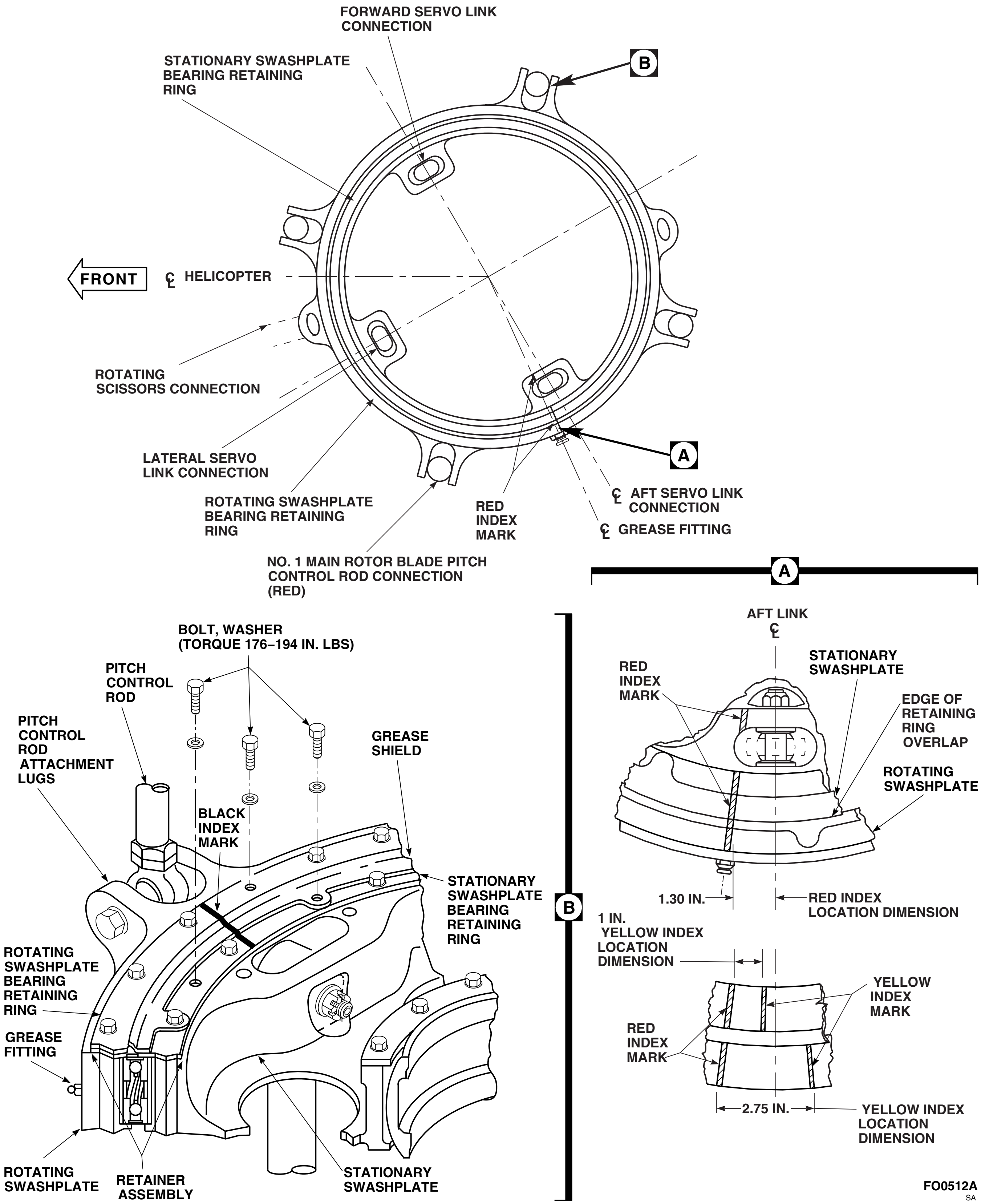


FIGURE 5-4-22-2. INSPECT/REPAIR SWASHPLATE

AGE TO PITCH CONTROL ROD SURFACES MUST NOT EXCEED 0.030-INCH DEEP. If necessary, repair as follows:

(1) Blend damage to ½-inch radius.

(2) Remove any raised material around damaged area.

(3) Touch up paint repaired area (PARA 2-6-5).

e. Inspect mating surface of swashplate for nicks and gouges. **DAMAGE MUST NOT EXCEED 0.020-INCH DEEP ON MORE THAN 10% OF MATING AREA.** If necessary, repair as follows:

(1) Blend damage out to ½-inch radius.

(2) Remove any raised material around damaged area.

(3) Touch up paint repaired area (PARA 2-6-5).

f. Inspect bushings in pitch control rod attachment lugs for corrosion pitting. **IF PITTING IS DEEPER THAN 0.005-INCH IN SHOULDER AREA, OR 0.040-INCH IN CHAMFER AREA, REPLACE SWASHPLATE.**

g. Visually inspect swashplate for voids or disbonding of grease shield on retainer assembly. If disbond or void is over 5-inches, perform the following:

(1) Turn main rotor head until red index marks on rotating and stationary swashplates line up (Figure 5-4-22-2, Detail A).

(2) Remove bolts and washers securing retainer assembly to stationary swashplate (Figure 5-4-22-2, Detail B).

(3) Raise retainer assembly and cover swashplate.

CAUTION

Damage to swashplate will result if solvents or adhesives enter grease. Provide adequate protection to swashplate while using solvents and adhesives.

(4) Using syringe, clean disbonded area by injecting denatured alcohol, Item 70, Appendix D, into area.

(5) If possible, using abrasive cloth, Item 5, Appendix D, sand area. Using denatured alcohol, Item 70, Appendix D, clean again. Allow to dry.

(6) Prepare adhesive, Item 32, Appendix D, per manufacturer's directions.

(7) Using syringe, inject mixed adhesive into void areas.

(8) Using clamps, apply pressure on rebonded area.

(9) Allow to cure for 24 hours at room temperature.

(10) Install retainer assembly by performing the following:

(a) Align red index marks (Figure 5-4-22-2, Detail A).

(b) Secure with bolts and washers (Figure 5-4-22-2, Detail B).

(c) Make sure all nine tabs of grease shield receive a bolt.

(d) **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**

h. If either or both red blade fold index marks on bearing retaining rings are missing, or lines cannot be matched, perform the following:

(1) Turn rotating swashplate so that rotating scissors connection is approximately between helicopter centerline and lateral servo link on stationary swashplate (main rotor blade No. 1 on left front side).

(2) Using clean machinery towel, Item 211, Appendix D, dampened with acetone,

Item 25, Appendix D, thoroughly clean areas on retaining rings and aft servo link boss near grease fitting. Wipe area again; do not allow acetone to dry on parts.

(3) Locate and mark centerline of aft servo link on stationary swashplate bearing retaining ring using grease pencil (Figure 5-4-22-2).

(4) Locate and mark centerline of grease fitting on rotating swashplate bearing retaining ring using grease pencil.

(5) Turn rotating swashplate until measured distance between aft servo link centerline and grease fitting centerline equal 1.30-inches on overlap edge of bearing retaining ring.

(6) If not already done, engage gust lock or rotor brake (PARA 1-6-18).

(7) Using 1-inch masking tape, Item 212, Appendix D, make a guide on both sides of grease pencil centerline to form line approximately 0.10-inch wide from outer bearing retainer ring edge to inner servo link boss. Remove grease pencil marks, using cleaning instructions in step h. (2).

(8) Using polyurethane paint, Item 226, Appendix D, and artist brush, Item 85, Appendix D, paint red index mark between guides.

(9) Remove masking tape when paint has dried.

(10) Make sure area is clean and free of foreign material.

(11) Disengage gust lock or rotor brake (PARA 1-6-18).

i. If either or both yellow blade fold index marks on both rotating swashplate bearing retaining ring and stationary swashplate bearing retaining ring are missing, or lines cannot be matched, perform the following:

(1) Align red index lines on both rotating swashplate bearing retaining ring and stationary swashplate bearing retaining ring.

(2) If not already done, engage gust lock or rotor brake (PARA 1-6-18).

(3) Using clean machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, thoroughly clean area between red index mark and aft servo link on both bearing retaining rings. Wipe area again; do not allow acetone to dry on parts.

(4) On inside diameter of stationary swashplate bearing retaining ring, measure 1-inch from center of red index line, and using grease pencil, mark yellow index centerline.

(5) On outside diameter of rotating swashplate bearing retaining ring, measure 2 ¾-inches from center of red index line, and using grease pencil, mark yellow index centerline.

(6) Using 1-inch masking tape, Item 212, Appendix D, make a guide on both sides of both grease pencil centerlines to form a line approximately 0.10-inch wide across each bearing retainer ring. Remove grease pencil marks, using cleaning instructions in step i. (2).

(7) Using polyurethane paint, Item 227A, Appendix D, and artist brush, Item 85, Appendix D, paint both yellow index lines between guides.

(8) Remove masking tape when paint has dried.

(9) Make sure area is clean and free of foreign material.

(10) Disengage gust lock or rotor brake (PARA 1-6-18).

j. If any black rigging index marks on both rotating swashplate bearing retaining ring and stationary swashplate bearing retaining ring are missing, or lines cannot be matched (total 4 each retaining ring), perform the following:

(1) Turn rotating swashplate to align pitch control rod centerlines with servo link centerlines.

(2) If not already done, engage gust lock or rotor brake (PARA 1-6-18).

(3) Using clean machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, thoroughly clean area to be painted on both retaining rings behind pitch control rods (4 places). Wipe area again; do not allow acetone to dry on parts.

(4) Using grease pencil, carefully mark centerlines of each pitch control rod across both retaining rings (Figure 5-4-22-2, Detail B).

(5) Using 1-inch masking tape, Item 212, Appendix D, make a guide on both sides of grease pencil centerlines to form a line approximately 0.12-inch wide across both retaining rings. Remove grease pencil marks, using cleaning instructions in step j. (2).

(6) Using polyurethane paint, Item 223, Appendix D, and artist brush, paint black index lines between guides.

(7) Remove masking tape when paint has dried.

(8) Make sure area is clean and free of foreign material.

(9) Disengage gust lock or rotor brake (PARA 1-6-18).

k. On aluminum areas that were reworked, perform the following:

(1) Using corrosion resistant coating, Item 118, Appendix D, touch up areas that were reworked.

(2) Using primer and paint, touch up areas (PARA 2-6-5).

l. On nonaluminum areas that were reworked, using primer and paint, touch up areas (PARA 2-6-5).

m. Inspect swashplate for corrosion.

n. Measure thickness of part being inspected at annealed surface where corrosion is found.

o. Inspect swashplate for corrosion pitting as follows:

(1) **ON OPEN AREAS (NONMATING SURFACES), REMOVE PITTING UP TO 0.050-INCH DEEP. NO TWO REWORKED AREAS MAY BE CLOSER TOGETHER THAN THICKNESS OF PART.** If pitting exceeds limits, replace swashplate.

(2) **ON MATING SURFACES, REMOVE PITTING UP TO 0.010-INCH DEEP ON NO MORE THAN 10% OF SURFACE BY BLENDING OUT.** If pitting exceeds limits, replace swashplate.

p. On aluminum areas that were reworked, perform the following:

(1) Using corrosion resistant coating, Item 118, Appendix D, touch up areas that were reworked.

(2) Using primer and paint, touch up areas (PARA 2-6-5).

q. On nonaluminum areas that were reworked, using primer and paint, touch up areas (PARA 2-6-5).

5-4-23 SWASHPLATE SCISSORS ATTACHMENT SPHERICAL BEARING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Powertrain Repairer's Toolkit
- Spherical Bearing Remover/Installer, Locally-Made, Figure H-195, Appendix H
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry Ice, Item 125, Appendix D
- Primer, Item 258, Appendix D

References

- Appendix D
- Appendix H
- PARA 5-4-26
- PARA 5-4-27

5-4-23.1 Replace Swashplate Scissors Attachment Spherical Bearing.

NOTE

Replacement of all swashplate scissors attachment spherical bearings is the same.

5-4-23.1.1. Remove.

a. Locally make spherical bearing remover/installer, tools A through E (Figure H-195, Appendix H).

b. Disconnect rotating scissors lower link from spherical bearing (PARA 5-4-27).

c. Remove bolts and washers securing retainer to rotating swashplate (Figure 5-4-23-1).

d. Using tools A, B, and C of spherical bearing remover/installer, remove swashplate scissors attachment spherical bearing as follows (Figure 5-4-23-1, Detail A):

(1) Position tool A above rotating swashplate to receive swashplate scissors attachment spherical bearing.

(2) Position tool B below rotating swashplate to push swashplate scissors attachment spherical bearing.

(3) Install tool C through tools A, B, and swashplate scissors attachment spherical bearing.

(4) **TIGHTEN TOOL C UNTIL SWASHPLATE SCISSORS ATTACHMENT SPHERICAL BEARING IS PUSHED**

OUT OF SWASHPLATE. Remove tools A, B, C, and swashplate scissors attachment spherical bearing.

5-4-23.1.2. Install.

a. Using tools C, D, and E of spherical bearing remover/installer, install swashplate scissors attachment spherical bearing as follows (Figure 5-4-23-1, Detail B):

(1) Using dry ice, Item 125, Appendix D, chill replacement swashplate scissors attachment spherical bearing for a minimum of 30 minutes.

(2) Remove swashplate scissors attachment spherical bearing from dry ice.

(3) Apply a thin coat of primer, Item 258, Appendix D, to swashplate scissors attachment spherical bearing and rotation swashplate bearing bore mating surfaces.

(4) Position swashplate scissors attachment spherical bearing above swashplate bearing bore.

(5) Position tool E above swashplate and swashplate scissors attachment spherical bearing to push swashplate scissors attachment spherical bearing into swashplate bearing bore.

(6) Position tool D below swashplate to provide surface for tool C to pull against.

(7) Install tool C through tools E, D, and swashplate scissors attachment spherical bearing.

(8) **TIGHTEN TOOL C UNTIL SWASHPLATE SCISSORS ATTACHMENT SPHERICAL BEARING IS BOTTOMED INTO SWASHPLATE BEARING BORE.** Remove tools C, D, and E.

b. Check inserts as follows:

(1) Install bolts in insert until bottom of bolthead is 0.150-inch from rotating swashplate. Do not install retainer or washers.

(2) Check breakaway torque for each bolt. **BREAKAWAY TORQUE SHALL BE 3.5 - 30 INCH-POUNDS.**

(3) If breakaway torque is beyond limits, replace bolt and clean insert. Recheck breakaway torque. Repeat steps b. (1) and b. (2).

(4) If breakaway torque is still beyond limits, replace insert (PARA 5-4-26).

c. Install retainer with bolts and washers (Figure 5-4-23-1).

d. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

e. Connect rotating scissors lower link to spherical bearing (PARA 5-4-27).

f. Make sure area is clean and free of foreign material.

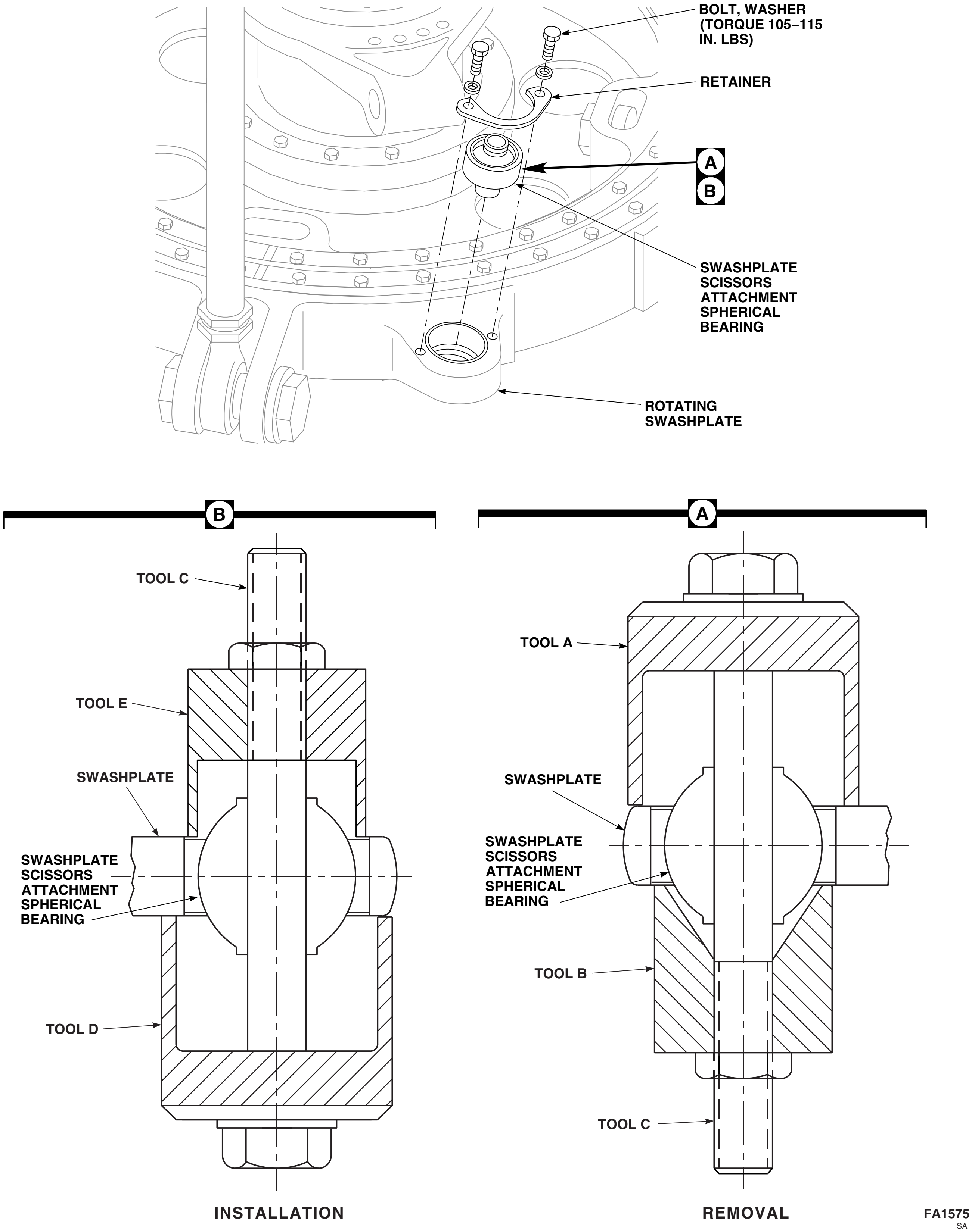


FIGURE 5-4-23-1. REPLACE SWASHPLATE SCISSORS ATTACHMENT SPHERICAL BEARING

5-4-24 SWASHPLATE EXPANDABLE PIN REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Powertrain Repairer's Toolkit

Materials/Parts

- Abrasive Mat, Item 14, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

References

- Appendix D

5-4-24.1 Inspect/Repair Swashplate Expandable Pins.

NOTE

Inspection/repair of all swashplate expandable pins is the same.

a. Using dry-cleaning solvent, Item 124, Appendix D, clean swashplate expandable pin.

b. Using abrasive mat, Item 14, Appendix D, remove surface corrosion.

c. Visually inspect swashplate expandable pin collar and threads for corrosion, fretting, and scoring damage. **NONE ALLOWED.** If corrosion, fretting, and scoring damage are found, replace swashplate expandable pin.

d. Visually inspect swashplate expandable pin bushing outside diameter (OD). If OD is missing more than 25% of dry film lubricant, replace swashplate expandable pin.

5-4-24.2 Inspect Swashplate Expandable Pin Nuts.

NOTE

Inspection of all swashplate expandable pin nuts is the same.

Visually inspect swashplate expandable pin nut for cracks or fractures. **NONE ALLOWED.** If cracks or fractures are found, replace swashplate expandable pin.

5-4-25 SWASHPLATE SPHERICAL BEARING RETAINER INSERTS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit
- Insert Kit, K4-314-314L
- Powertrain Repairer's Toolkit
- Thread Gage, 5/16-24UNF-3B

Materials/Parts

- Primer, Item 258, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 5-4-22
- PARA 5-5-7

5-4-25.1 Replace Swashplate Spherical Bearing Retainer Inserts.

NOTE

Replacement of all swashplate spherical bearing retainer inserts is the same.

5-4-25.1.1 Inspect.

a. Inspect condition of threads of swashplate spherical bearing retainer inserts as follows:

(1) Visually inspect for looseness. **NONE ALLOWED.** If looseness is found, replace inserts.

(2) Using thread gage, check for crossed, stripped, or flattened threads. **NONE ALLOWED.** If damage is found, replace inserts.

b. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect holes of insert for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace swashplate (PARA 5-4-22).

5-4-25.1.2 Remove.

a. If required, remove retainer and shim (PARA 5-5-7).

b. Using insert kit, remove inserts as follows:

(1) Using removal tool, counterbore 0.110-inch deep × 0.374 - 0.379-inch diameter.

(2) Using drive wrench, back out and remove insert (Figure 5-4-25-1).

c. If insert cannot be removed, send swashplate to Depot.

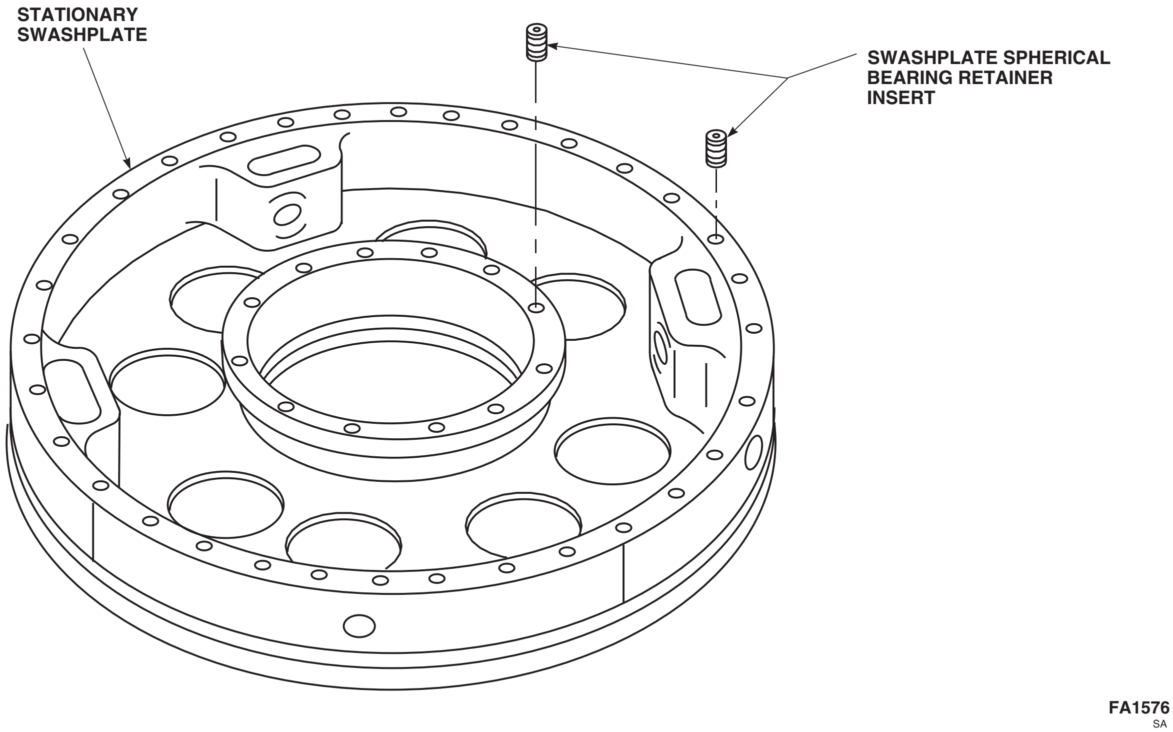


FIGURE 5-4-25-1. REPLACE SWASHPLATE SPHERICAL BEARING RETAINER INSERTS

5-4-25.1.3 Install.

a. Apply coat of primer, Item 258, Appendix D, to insert.

b. Using insert kit, install insert as follows:

(1) Using drive wrench, thread insert 0.015 - 0.025-inch below surface of swashplate (Figure 5-4-25-1).

(2) Place swage tool in insert and apply enough downward force to bottom nylon protective shoulder on surface of swashplate.

c. If required, install retainer and shim (PARA 5-5-7).

5-4-26 SWASHPLATE SCISSORS ATTACHMENT RETAINER INSERTS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Insert Kit, K4-258-258L
- Powertrain Repairer's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Primer, Item 258, Appendix D

References

- Appendix D

5-4-26.1 Replace Swashplate Scissors Attachment Retainer Inserts.

NOTE

Replacement of all swashplate scissors attachment retainer inserts is the same.

5-4-26.1.1. Remove.

a. Remove bolts and washers securing retainer to rotating swashplate.

b. Using insert kit, remove insert as follows:

(1) Using removal tool, counterbore 0.095-inch deep by 0.311 - 0.316-inch diameter.

(2) Back out and remove insert from swashplate (Figure 5-4-26-1).

5-4-26.1.2. Install.

a. Apply a coat of primer, Item 258, Appendix D, to insert.

b. Using insert kit, install insert as follows:

(1) Thread insert 0.015 - 0.025-inch below surface of rotating swashplate (Figure 5-4-26-1).

(2) Place swage tool in insert and apply enough downward force to bottom nylon protective shoulders on surface of swashplate.

c. Check inserts as follows:

(1) Install bolts in inserts until bottom of bolthead is 0.150-inch from rotating swashplate. Do not install retainer or washers.

(2) Check breakaway torque for each bolt. **BREAKAWAY TORQUE SHALL BE 3.5 - 30 INCH-POUNDS.**

(3) If breakaway torque is beyond limits, replace bolt and clean insert. Repeat steps c. (1) and c. (2).

(4) If breakaway torque is still beyond limits, replace insert.

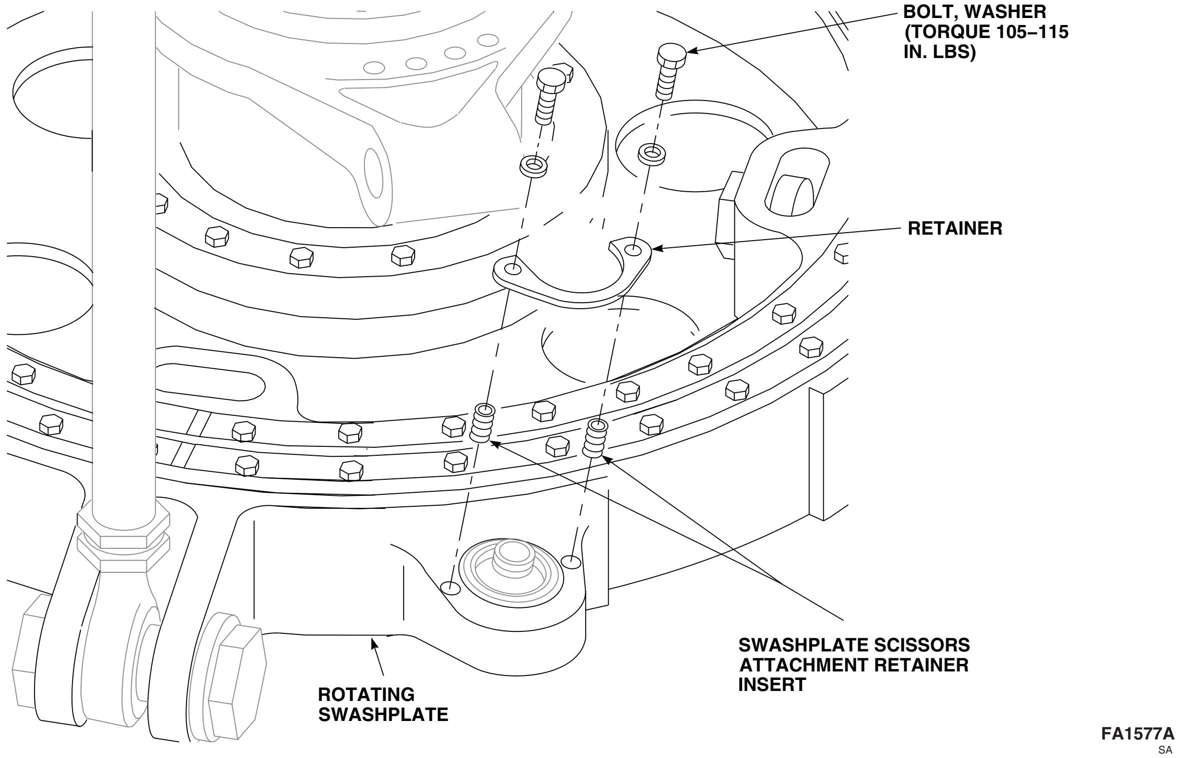


FIGURE 5-4-26-1. REPLACE SWASHPLATE SCISSORS ATTACHMENT RETAINER INSERTS

d. Install retainer on rotating swashplate with bolts and washers.

e. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

5-4-27 ROTATING SCISSORS.

PARAGRAPH BREAKDOWN

PAGE

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Medium Cut File
- Powertrain Repairer's Toolkit
- Torque Wrench, 500 - 2500 in. lbs

Materials/Parts

- Abrasive Cloth, Item 12, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Lubricant, Solid Film, Item 201, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Paint, Polyurethane, Item 222A, Appendix D
- Paint, Polyurethane, Item 223, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 258, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 1-6-18
- PARA 2-6-5
- PARA 5-3-7
- PARA 5-4-4
- PARA 5-5-8
- PARA 5-5-9

5-4-27.1 Replace Rotating Scissors.

NOTE

Replacement of all rotating scissors is the same.

5-4-27.1.1 Remove.

a. Inspect rotating scissors upper and lower link bearings for play (PARA 5-3-7).

b. Turn main rotor head so rotating scissors are easy to reach.

c. If not already done, engage gust lock or rotor brake (PARA 1-6-18).

d. Remove bolt, countersunk washer, washer, shim, and nut securing upper link to lower link (Figure 5-4-27-1, Detail B).

e. Remove bolt, countersunk washer, washer, shims, and nut securing upper link to lower

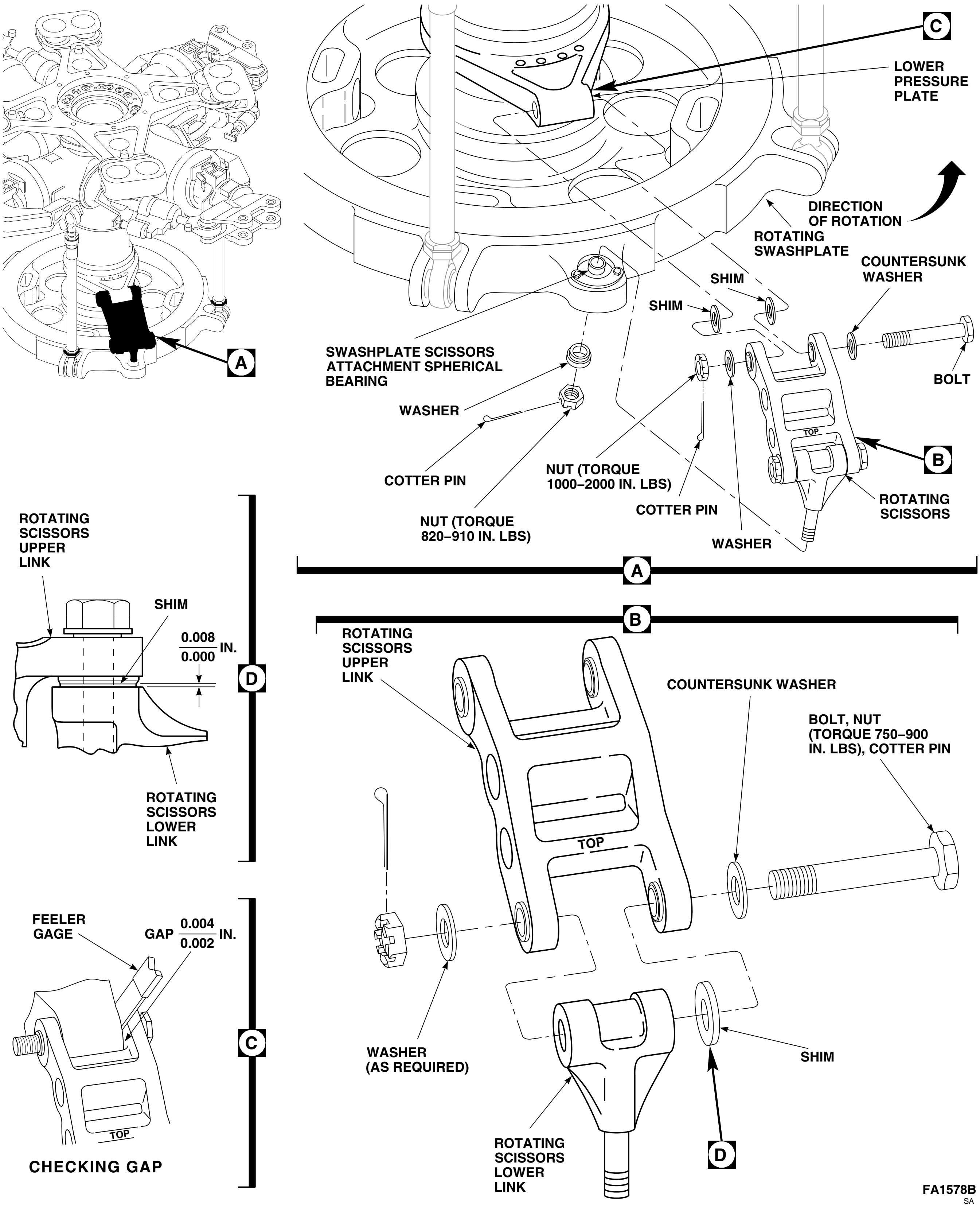


FIGURE 5-4-27-1. REPLACE ROTATING SCISSORS pressure plate (Figure 5-4-27-1, Detail A). Remove upper link.

f. Remove washer and nut securing lower link to spherical bearing. Remove lower link.

5-4-27.1.2 Inspect/Repair.

NOTE

In the following steps, damage should be blended out using abrasive wheel, Item 22, Appendix D, abrasive cloth, Item 12, Appendix D, and/or medium cut file, as required.

a. Inspect bolt securing upper link to lower pressure plate (PARA 5-4-4).

b. Visually inspect rotating scissors for cracks. **NONE ALLOWED.** If crack is suspected, perform fluorescent penetrant inspection of rotating scissors as follows:

(1) Using paint remover, Item 230, Appendix D, clean and strip suspected area.

(2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect for cracks (ASTM E1417). **NONE ALLOWED.** Perform either of the following as required:

(a) If crack is found, replace rotating scissors.

(b) If no cracks are found, using primer and paint, touch up stripped area (PARA 2-6-5).

c. Inspect rotating scissors for nicks and gouges as follows:

(1) **CORNER DAMAGE UP TO 0.10-INCH DEEP, EXCEPT IN LUG AREA WHERE LIMIT IS 0.050-INCH, MAY BE BLENDED OUT TO A ½-INCH RADIUS. NO TWO REWORKED AREAS MAY BE CLOSER TOGETHER THAN ONE-INCH.** If nicks and gouges exceed limits, replace rotating scissors.

(2) **ON OPEN AREAS (NONMATING SURFACES), SURFACE DAMAGE UP TO 0.050-INCH DEEP, EXCEPT LUG AREA WHERE LIMIT IS 0.030-INCH, MAY BE BLENDED OUT TO A ½-INCH RADIUS.** If surface damage exceeds limits, replace rotating scissors.

(3) **ON MATING SURFACES, REMOVE DAMAGE UP TO 0.020-INCH ON NO MORE THAN 10% OF MATING AREA. REMOVE ANY RAISED MATERIAL AROUND DAMAGED AREA AND BLEND OUT TO A ½-INCH RADIUS.** If damage exceeds limits, replace rotating scissors.

(4) On aluminum areas that were reworked, perform the following:

(a) Using corrosion resistant coating, Item 118, Appendix D, touch up areas that were reworked.

(b) Using primer and paint, touch up areas (PARA 2-6-5).

d. Inspect rotating scissors for corrosion pitting as follows:

(1) **ON OPEN AREAS (NONMATING SURFACES), REMOVE PITTING UP TO 0.050-INCH DEEP. NO TWO REWORKED AREAS MAY BE CLOSER TOGETHER THAN 1-INCH.** If pitting exceeds limits, replace rotating scissors.

(2) **ON MATING SURFACES, REMOVE PITTING UP TO 0.010-INCH DEEP ON NO MORE THAN 10% OF SURFACES, BY BLENDING OUT.** If pitting exceeds limits, replace rotating scissors.

(3) On aluminum areas that were reworked, perform the following:

(a) Using corrosion resistant coating, Item 118, Appendix D, touch up areas that were reworked.

(b) Using primer and paint, touch up areas (PARA 2-6-5).

e. Paint both upper and lower links as follows (PARA 2-6-5):

(1) Using 1-inch masking tape, Item 212, Appendix D, mask bearing bores.

(2) Apply coat of primer, Item 258, Appendix D, to surface.

(3) Using polyurethane paint, Item 222A, Appendix D, apply two coats.

(4) Using polyurethane paint, Item 223, Appendix D, stencil "TOP" and "BOTTOM" on top and bottom surface of upper link.

(5) Remove masking tape from bearing bores.

f. Measure outside diameter (OD) of lower link assembly shaft. **OD MUST NOT BE LESS THAN 0.6235-INCH.** If OD is below limit, replace lower link.

g. Measure inside diameter (ID) of lower link assembly bushing. **ID MUST NOT EXCEED 0.7510-INCH.** If ID exceeds limit, replace lower link bushing (PARA 5-5-9).

5-4-27.1.3 Install.

a. Loosely assemble upper link to lower link with bolt (Figure 5-4-27-1, Detail B).

b. Using feeler gage, measure gap between upper and lower links. **PEEL LAMINATED SHIM TO MAKE 0.000 - 0.008-INCH TOTAL CLEARANCE BETWEEN UPPER AND LOWER LINKS** (Figure 5-4-27-1, Detail D).

WARNING

FSCAP

Verification of proper hardware installation, torque, and safety installation is a critical characteristic.

c. Assemble upper link to lower link with bolt, countersunk washer, shim, washer (as required), and nut. Make sure countersunk side of washer is toward bolthead (Figure 5-4-27-1, Detail B).

d. Using solid film lubricant, Item 201, Appendix D, lubricate threads of bolt.

e. **TORQUE NUT TO 750 - 900 INCH-POUNDS.** Install cotter pin.

f. Apply torque stripe (PARA 2-6-5).

WARNING

FSCAP

Verification of proper hardware installation, torque, and safety installation are critical characteristics.

g. Install lower link in spherical bearing and secure with washer and nut. Place shoulder of washer facing swashplate scissors attachment spherical bearing as shown (Figure 5-4-27-1, Detail A).

h. Using solid film lubricant, Item 201, Appendix D, lubricate threads on lower link.

i. **TORQUE NUT TO 820 - 910 INCH-POUNDS.** Install cotter pin.

j. Apply torque stripe (PARA 2-6-5).

k. If previously removed rotating scissors will be installed and lower pressure plate was not replaced, perform the following (Figure 5-4-27-1, Detail A and Detail C):

(1) Loosely connect rotating scissors upper links to lower pressure plate with bolts and previously installed shims.

(2) Using feeler gage, measure gap between on each side of lower pressure plate lugs. **OVERALL GAP ON EACH SIDE MUST BE 0.002 - 0.004-INCH.**

(3) If gap is within limits, remove bolts and shims. Proceed to step m.

(4) If gap is not within limits, perform the following:

(a) Remove bolts and shims.

(b) Inspect upper link bearing flange height (PARA 5-5-8).

(c) Proceed to step l. (1) through step l. (5).

l. If new rotating scissors are being installed or if lower pressure plate has been replaced, perform the following:

(1) Loosely connect upper links to lower pressure plate with bolts. Do not install nuts at this time.

(2) Using feeler gage, measure gap between upper link on each side of lower pressure plate lugs. **OVERALL GAP ON EACH SIDE MUST BE 0.002 - 0.004-INCH.**

(3) Remove bolt.

(4) Peel laminated shims equally on both sides of upper link to allow for 0.002 - 0.004-inch gap on each side of lug.

(5) Repeat step l. until 0.002 - 0.004-inch gap on each side of lugs is obtained.

WARNING

FSCAP

Verification of proper hardware installation, torque, and safety installation are critical characteristics.

NOTE

Install bolt with bolthead in direction of rotation (counterclockwise).

m. Connect upper link to lower pressure plate with bolt, countersunk washer, shims, washer, and nut (Figure 5-4-27-1, Detail C). Make sure countersunk side of washer is toward bolthead.

n. Using solid film lubricant, Item 201, Appendix D, lubricate bolt threads.

o. **TORQUE NUT TO 1000 - 2000 INCH-POUNDS.** Install cotter pin.

p. If required, disengage gust lock or rotor brake (PARA 1-6-18).

q. Make sure area is clean and free of foreign material.

5-4-28 BIFILAR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Drive Tool
- Driver Wrench
- Extension, 3-inch
- Handle Socket Wrench
- Hex Wrench
- Hoist, Capacity to Support Bifilar (Alternate for Cabin Maintenance Crane)
- Hoisting Sling, 70700-20320-042
- Lockring Removal Tool
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer's Toolkit
- Shipping Container (Alternate for suitable work area)
- Socket, 13⁄16-inch
- Swage Tool
- Torque Wrench, Dial, 0 - 75 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Torque Wrench, 0 - 300 ft lbs

Materials/Parts

- Abrasive Cloth, Item 5, Appendix D
- Abrasive Paper, Item 19, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D

References

- Appendix D
- PARA 1-5-2
- PARA 1-6-18
- PARA 1-7-5
- PARA 2-6-5

- PARA 5-4-4
- PARA 5-4-29
- PARA 5-4-30
- PARA 12-4-26
- PARA 18-4-1

5-4-28.1 Replace Bifilar.

5-4-28.1.1 Remove.

NOTE

Bifilar weighs about 165 pounds.

a. If cabin maintenance crane will be used, assemble and install cabin maintenance crane (PARA 18-4-1).

b. If not already done, engage gust lock (PARA 1-6-18).

c. If blade deice kit is installed, remove main rotor blade deice slipring assembly (PARA 12-4-26).

d. If blade deice kit is not installed, remove bifilar cover (PARA 5-4-30).

e. Inspect bifilar cover gasket for damage. **NONE ALLOWED.** If damage is found, replace gasket (PARA 5-4-30).

f. If required, remove bifilar weights (PARA 5-4-29).

g. Install straps of hoisting sling on bifilar arms (Figure 5-4-28-1, Sheet 1, Detail A).

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing component from helicopter. Do not allow hook assembly to contact pulley.

h. Attach hoisting sling to hoist or cabin maintenance crane. Adjust lengths of straps to evenly distribute weight when hoisting.

i. Remove bolts, washers, and lockring from bifilar (Figure 5-4-28-1, Sheet 1, Detail B and Sheet 2, Detail C).

j. Loosen expandable pins, but do not remove nuts. Use hex wrench to prevent expandable pin from turning while loosening nut.

k. Remove bifilar bolts and washers (4 places) from bifilar.

WARNING

To avoid damage to equipment and personnel injury, do not remove hub-to-shaft extension bolts (NAS bolts) that have spacers under bolthead (4 places).

l. Using handle socket wrench, 3-inch extension, and 13/16-inch socket, remove hub-to-shaft extension bolts and washers (8 places) securing bifilar to main rotor head.

m. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean primer off bolts.

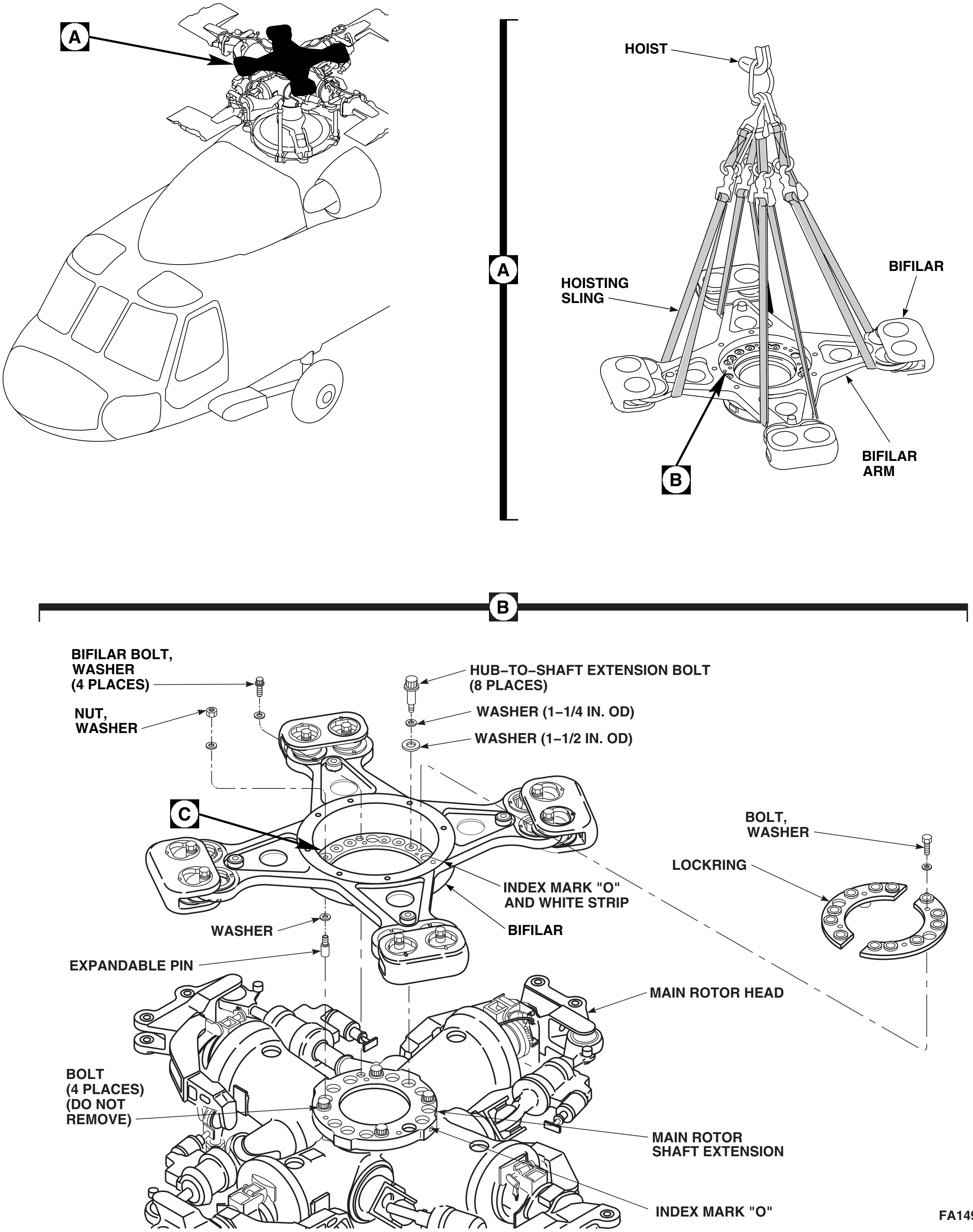
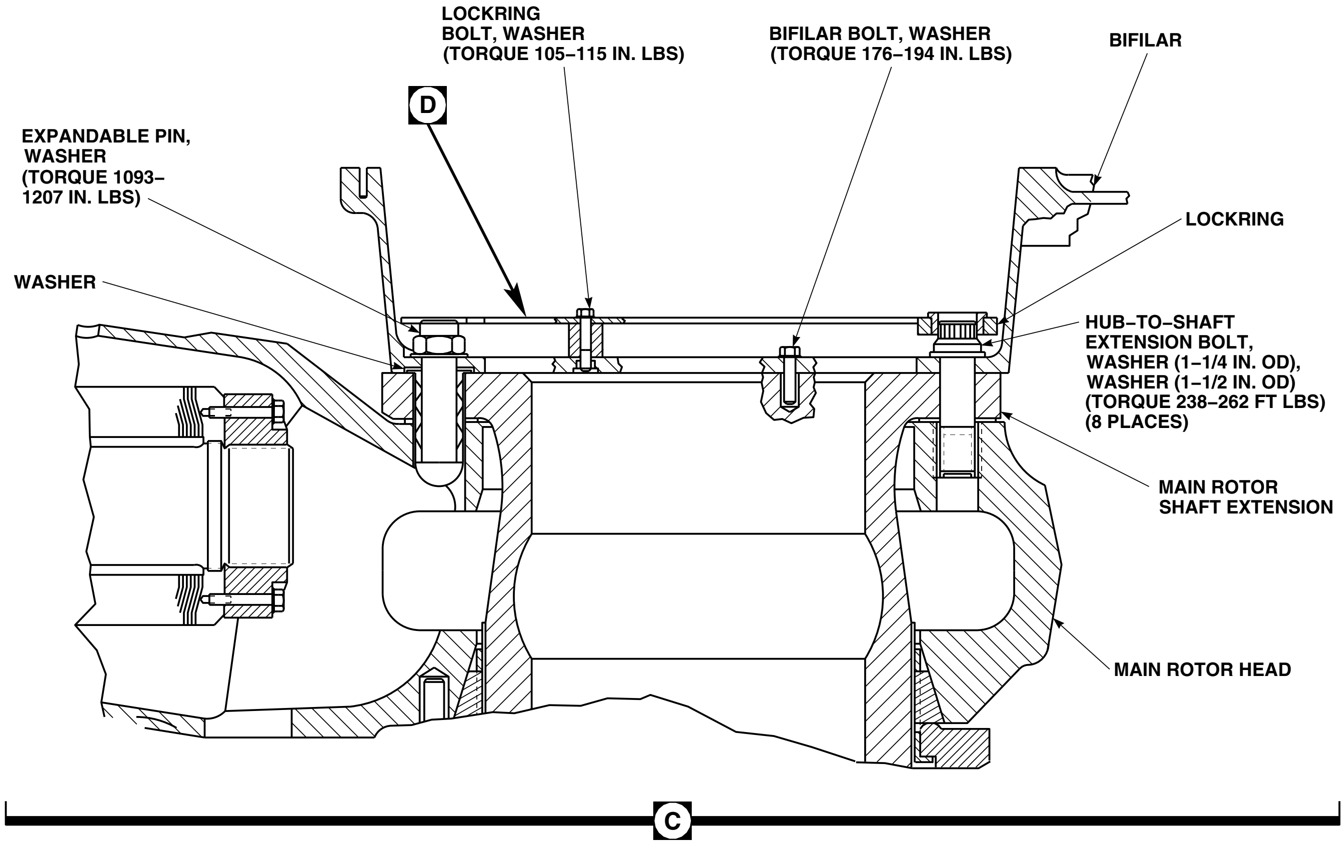


FIGURE 5-4-28-1. REPLACE BIFILAR (SHEET 1 OF 2)



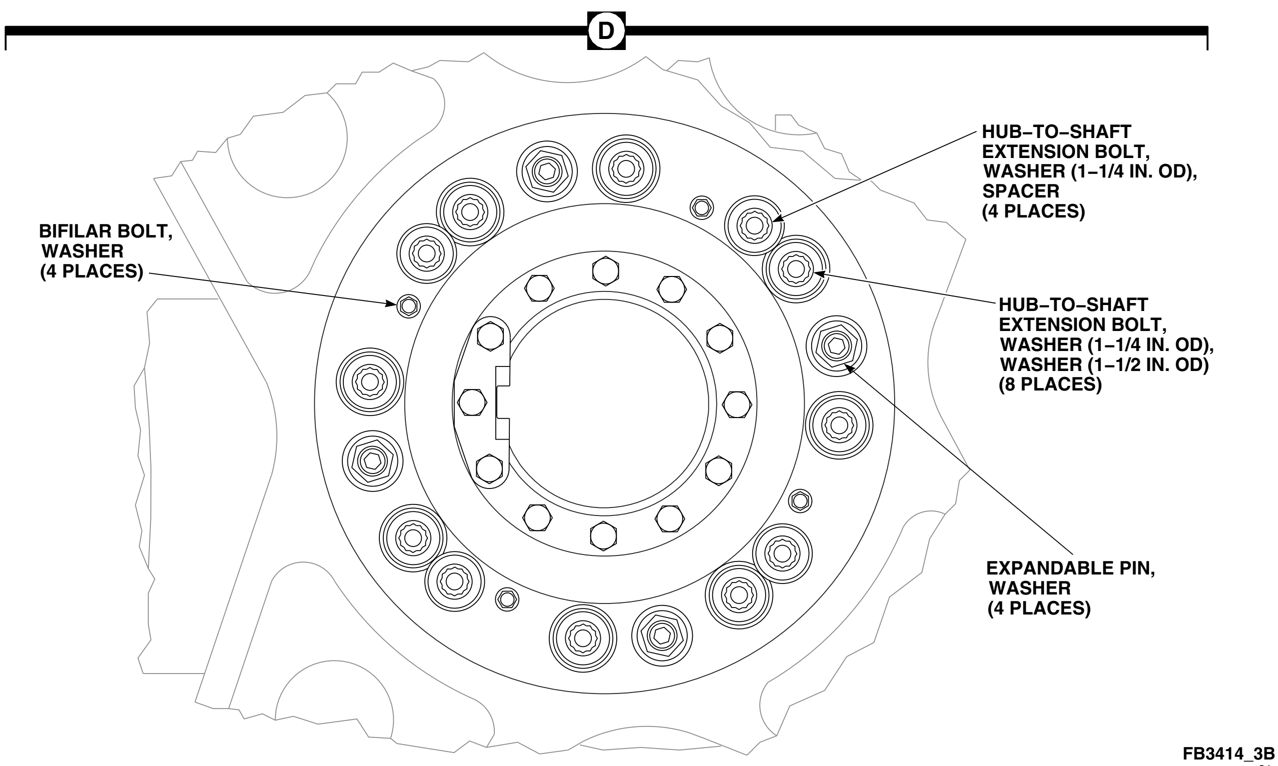


FIGURE 5-4-28-1. REPLACE BIFILAR (SHEET 2 OF 2)

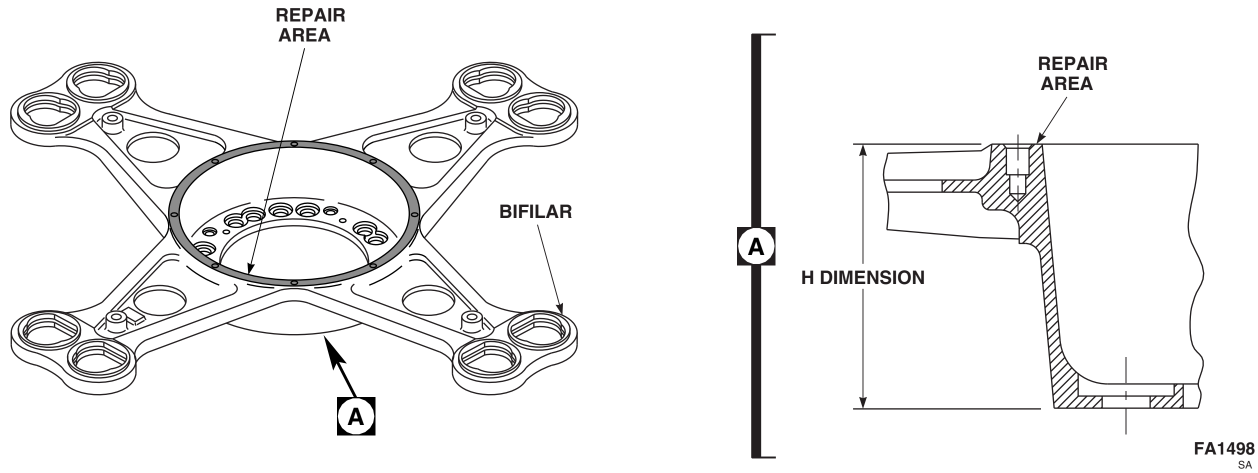


FIGURE 5-4-28-2. INSPECT/REPAIR BIFILAR

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting bifilar. Keep area clear when hoisting bifilar.

n. Lift bifilar from main rotor head with expandable pins still in, and place on suitable work area or in shipping container.

CAUTION

Damage to main rotor head and bifilar will result if metallic scrapers are used when cleaning rotor head components. Do not use metallic items to remove sealing compound.

o. Using nonmetallic scraper, remove sealing compound from bifilar and main rotor head.

p. Remove expandable pins (Figure 5-4-28-1, Sheet 1, Detail B).

q. Remove hoisting sling from bifilar arms (Figure 5-4-28-1, Sheet 1, Detail A).

r. Inspect bolts securing bifilar to main rotor head (PARA 5-4-4).

5-4-28.1.2 Inspect/Repair.

a. Inspect bifilar cover/blade deice distributor (if installed) mating surface of bifilar for corrosion or pitting and repair as follows:

(1) **USING ABRASIVE CLOTH, ITEM 5, APPENDIX D, REMOVE PITTING UP TO 0.010-INCH DEEP. UP TO 100% OF REPAIR AREA MAY BE BLENDED, PROVIDED A MAXIMUM ″H″ DIMENSION OF 4.40-INCHES IS MAINTAINED** (Figure 5-4-28-2, Detail A).

(2) **USING ABRASIVE PAPER, ITEM 19, APPENDIX D, REMOVE PITTING UP TO 0.020-INCH DEEP. MAINTAIN 10:1 LENGTH-TO-DEPTH RATIO. UP TO 25% OF REPAIR AREA MAY BE BLENDED, PROVIDED A MAXIMUM ″H″ DIMENSION OF 4.390-INCHES IS MAINTAINED. MAXIMUM REPAIR OF 0.015-INCH DEEP IS ALLOWED WITHIN ½-INCH RADIUS OF ATTACHMENT HOLES.**

(3) **USING ABRASIVE PAPER, ITEM 19, APPENDIX D, REMOVE PITTING UP TO 0.030-INCH DEEP. MAINTAIN 10:1 LENGTH-TO-DEPTH RATIO. UP TO 5% OF REPAIR AREA MAY BE BLENDED, PROVIDED A MAXIMUM ″H″ DIMENSION OF 4.380-INCHES IS**

MAINTAINED. MAXIMUM REPAIR OF 0.015-INCH DEEP IS ALLOWED WITHIN ½-INCH RADIUS OF ATTACHMENT HOLES.

(4) **REPAIR STEP A. (1) MAY BE COMBINED WITH REPAIR STEP A. (2) OR STEP A. (3) UP TO MAXIMUM LIMITS FOR EACH TYPE OF REPAIR.**

(5) If repairs exceed limits, replace bifilar.

b. Using corrosion resistant coating, Item 118, Appendix D, touch up repaired areas.

5-4-28.1.3 Install.

WARNING

Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing component from helicopter. Do not allow hook assembly to contact pulley.

a. Install straps of hoisting sling on bifilar arms. Adjust length of straps to evenly distribute weight when hoisting (Figure 5-4-28-1, Sheet 1, Detail A).

b. Attach hoisting sling to hoist or cabin maintenance crane.

c. Remove nuts from expandable pins and place washers that go between main rotor shaft extension and bifilar on expandable pins. Insert top of expandable pin up through holes in main rotor shaft extension. Install second washers on expandable pins. Loosely install nuts (Figure 5-4-28-1, Sheet 1, Detail B and Sheet 2, Detail C).

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting bifilar. Keep area clear when hoisting bifilar.

d. Lift bifilar and position over main rotor head (Figure 5-4-28-1, Sheet 1, Detail A and Detail B).

e. Line up ″O″ index mark (white stripe) on bifilar to ″O″ index mark on main rotor shaft extension. Lower bifilar and guide expandable pins into holes (Figure 5-4-28-1, Sheet 1, Detail B).

NOTE

Do not use hex wrench to torque expandable pins.

f. **TIGHTEN EXPANDABLE PINS. TORQUE NUTS TO 1093 - 1207INCH-POUNDS.** Use hex wrench to stop expandable pin from turning (Figure 5-4-28-1, Sheet 2, Detail C and Detail D).

g. Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer or sealing compound to shank, threads, and area under hub-to-shaft extension boltheads.

h. Install hub-to-shaft extension bolts, washers (1 ¼-inch outside diameter (OD)), and washers (1 ½-inch OD) (8 places) securing bifilar to main rotor shaft extension while primer is still wet. Install hub-to-shaft extension bolts with extension and socket.

i. **TORQUE HUB-TO-SHAFT EXTENSION BOLTS TO 238 - 262 INCH-POUNDS. STABILIZE TORQUE AT 238 - 262 INCH-POUNDS BY TURNING EACH HUB-TO-SHAFT EXTENSION BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT.**

j. Break torque on expandable pins.

k. **TORQUE EIGHT HUB-TO-SHAFT EXTENSION BOLTS TO 238 - 262 FOOT-POUNDS. STABILIZE TORQUE AT 238 - 262 FOOT-POUNDS BY TURNING EACH HUB-TO-**

SHAFT EXTENSION BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT (Figure 5-4-28-1, Sheet 2, Detail C).

l. Using machinery towel, Item 211, Appendix D, wipe off any primer that has seeped from under boltheads.

m. **TORQUE NUTS ON FOUR EXPANDABLE PINS TO 1093 - 1207 INCH-POUNDS. STABILIZE TORQUE AT 1093 - 1207 INCH-POUNDS BY TURNING EACH BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT.**

n. Inspect bifilar threaded inserts as follows:

(1) Thread in bifilar bolts until boltheads are 0.030-inch away from surface of bifilar.

(2) Using dial torque wrench, back off each bolt individually. **MINIMUM BREAKAWAY TORQUE IS 6.5 INCH-POUNDS. MAXIMUM REMOVAL TORQUE IS 60 INCH-POUNDS.**

o. If breakaway or removal torque exceeds limits, perform the following:

(1) Inspect bolts for evidence of corrosion or wear. **NONE ALLOWED.** If corrosion or wear is found, replace bolts.

(2) Inspect threaded insert for cleanness. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean threaded insert.

(3) Repeat step n. If breakaway or removal torque still exceeds limits, replace threaded insert (PARA 5-4-4).

p. Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer or sealing compound to shank, threads, and area under bifilar boltheads.

q. Install bifilar bolts and washers while primer or sealing compound is still wet.

r. **TORQUE BIFILAR BOLTS TO 176 - 194 INCH-POUNDS.**

s. Using machinery towel, Item 211, Appendix D, wipe off any primer that has seeped from under boltheads.

t. Apply sealing compound, Item 292, Appendix D, to seal around all boltheads.

WARNING

FSCAP

Verification of proper installation and torque of bolt, and presence and proper installation of lockring is a critical characteristic.

u. Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply coat of primer or sealing compound to shank, threads, and area under lockring boltheads.

v. Make sure lockring is properly seated over boltheads. Install upper lockring over hub-to-shaft extension bolts, and secure with lockring bolts and washers while primer is still wet.

CAUTION

Damage to lockring will result if it is not seated properly prior to torquing bolts. Make sure lockring is properly seated over boltheads. If necessary, turn bushings to allow lockring to seat over boltheads. Lockring must sit flat before torquing bolts.

w. **TORQUE LOCKRING BOLTS TO 105 - 115 INCH-POUNDS.**

x. Using machinery towel, Item 211, Appendix D, wipe off any primer that has seeped from under boltheads.

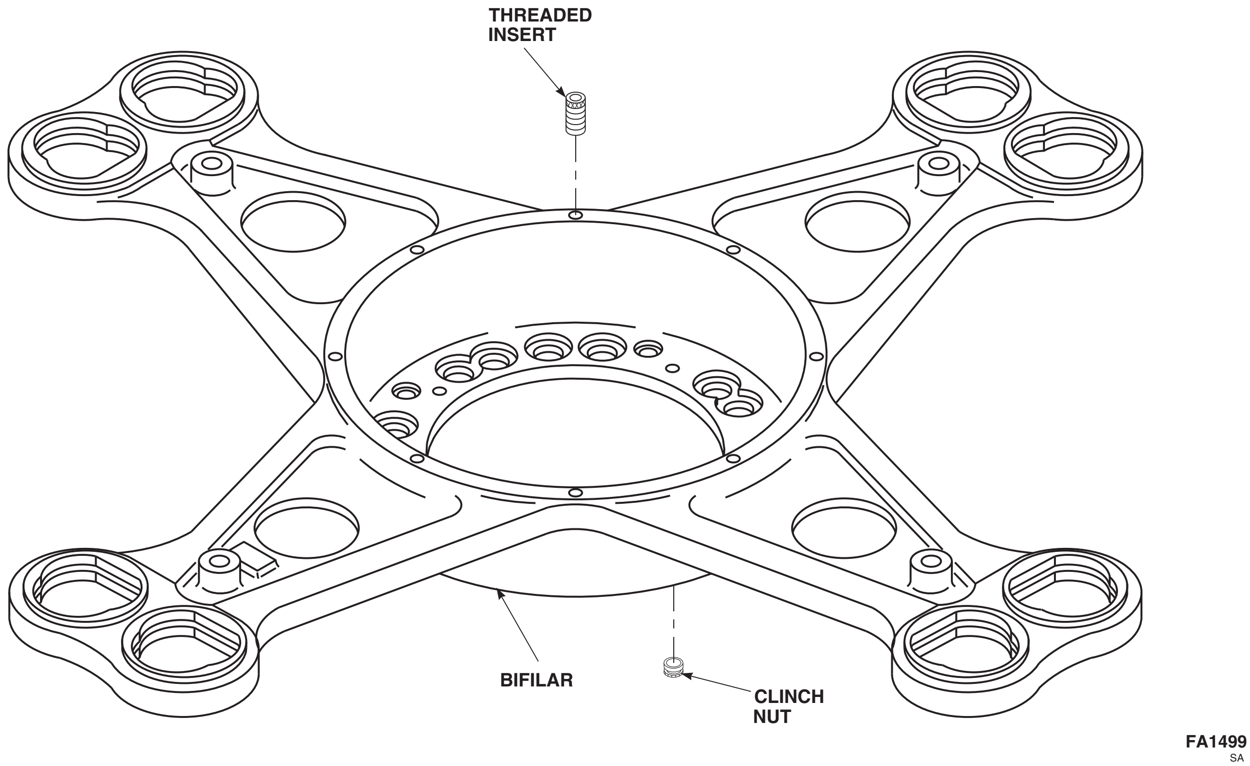


FIGURE 5-4-28-3. REPLACE THREADED INSERT AND CLINCH NUT

y. Touch up paint on boltheads (PARA 2-6-5).

z. Apply sealing compound, Item 292, Appendix D, to seal areas around bifilar boltheads, and gap between bifilar and main rotor shaft extension.

aa. Spray corrosion preventive compound, Item 113, Appendix D, inside main rotor shaft extension and on main rotor shaft nut before installing bifilar cover or main rotor blade deice slipring assembly (if installed).

ab. If blade deice kit is installed, install main rotor blade deice slipring assembly (PARA 12-4-26).

ac. If blade deice kit is not installed, install bifilar cover (PARA 5-4-30).

ad. If required, install bifilar weights (PARA 5-4-29).

ae. Lubricate bifilar (PARA 1-5-2).

af. Remove hoisting sling from bifilar arms (Figure 5-4-28-1, Sheet 1, Detail A).

ag. If required, disengage gust lock (PARA 1-6-18).

ah. If used, disassemble and remove cabin maintenance crane (PARA 18-4-1).

ai. Make sure area is clean and free of foreign material.

aj. Check helicopter logbook to make sure hub-to-shaft extension bolts and expandable pins torque check is shown due 9 - 11 flight hours after installation (PARA 1-7-5).

5-4-28.2 Replace Threaded Insert.

5-4-28.2.1 Remove.

a. Using lockring removal tool, drill out threaded insert lockring to 0.082-inch (Figure 5-4-28-3).

b. Using drive tool, remove remainder of threaded insert.

c. Clear threaded insert hole and threads of any foreign material and chips.

5-4-28.2.2 Install.

a. Using driver wrench, install threaded insert in hole, until top of insert is 0.015 - 0.025-inch below surface (Figure 5-4-28-3).

b. Using swage tool, swage threaded insert until shoulder of tool touches surface of bifilar.

5-4-28.3 Replace Clinch Nut.

5-4-28.3.1 Remove.

Using drive tool, remove clinch nut from bifilar (Figure 5-4-28-3).

5-4-28.3.2 Install.

a. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to mating surface of clinch nut and bore of bifilar.

b. Using drive tool, install clinch nut into bifilar while epoxy primer is still wet (Figure 5-4-28-3).

5-4-29 BIFILAR WEIGHT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 700 - 1600 in. lbs

References

- PARA 1-5-2
- PARA 1-6-18
- PARA 5-3-10

5-4-29.1 Replace Bifilar Weight.

NOTE

Replacement of all bifilar weights is the same.

5-4-29.1.1 Remove.

a. Turn main rotor head so bifilar weight is easy to reach.

b. If not already done, engage gust lock (PARA 1-6-18).

c. Remove bolt, countersunk washers, tapered washers, pins, and nuts securing bifilar weight to bifilar (Figure 5-4-29-1, Detail A).

d. Remove bifilar weight.

5-4-29.1.2 Inspect/Repair.

Inspect/repair bifilar weights (PARA 5-3-10).

5-4-29.1.3 Install.

NOTE

Install thin washers (maximum of two) as required to prevent nut from bottoming on bolt.

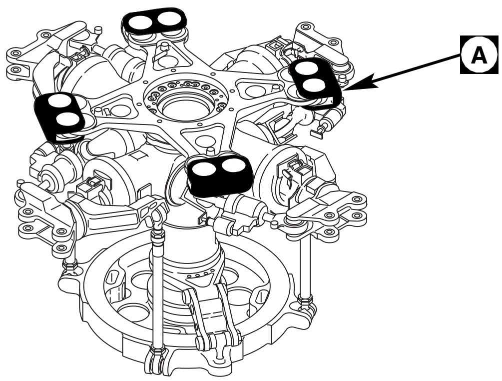
a. Place bifilar weight, long pins, and tapered washers on bifilar. Install bolts, countersunk washers, short pins, washers, thin washers (as required), and nuts. Make sure countersunk side of washer is toward bolthead (Figure 5-4-29-1, Detail A).

b. **TORQUE NUTS TO 1333 - 1367 INCH-POUNDS.**

c. Lubricate bifilar (PARA 1-5-2).

d. Make sure area is clean and free of foreign material.

e. If required, disengage gust lock (PARA 1-6-18).



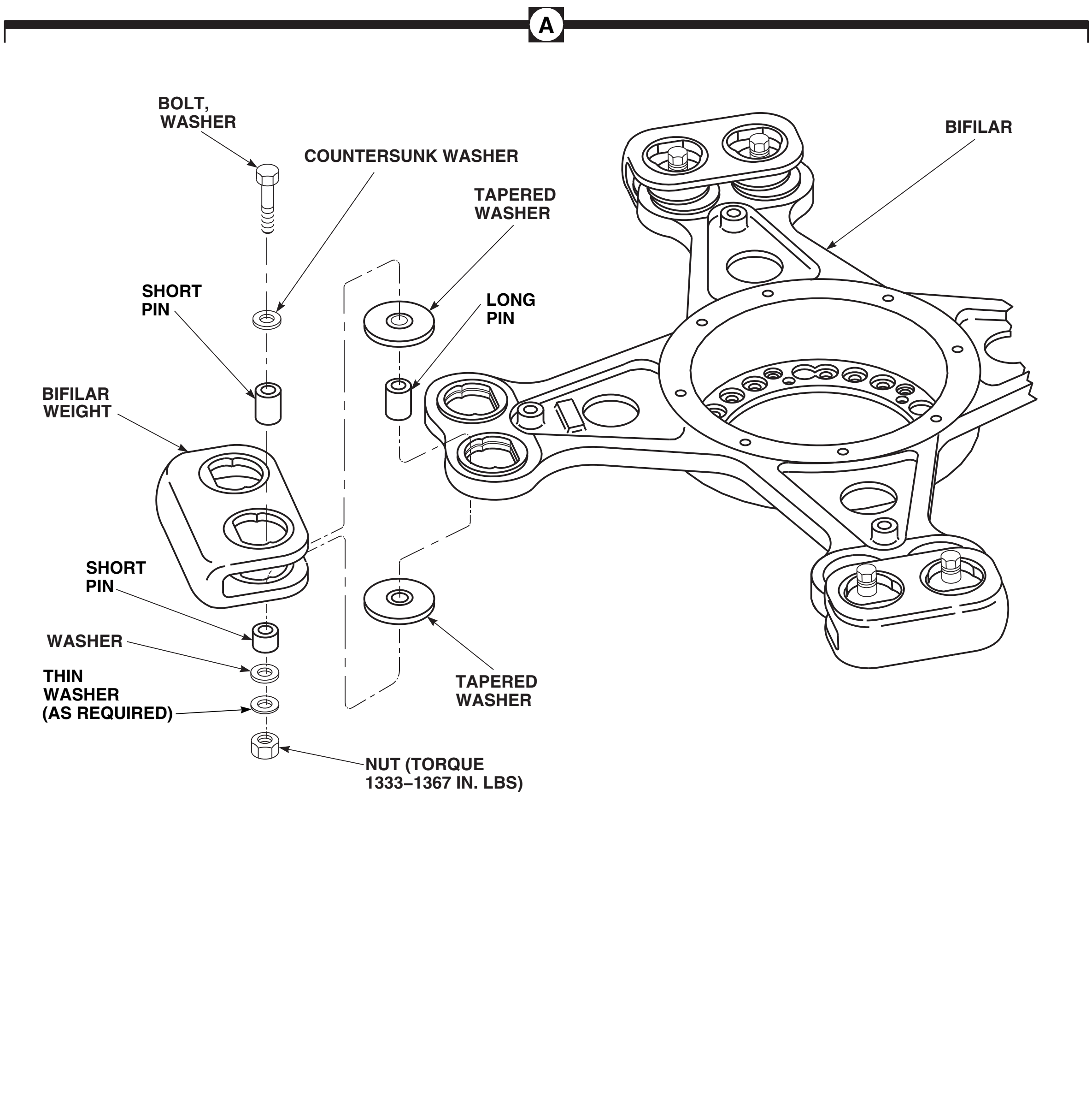


FIGURE 5-4-29-1. REPLACE BIFILAR WEIGHT

5-4-30 BIFILAR COVER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Paper, Item 18, Appendix D
- Adhesive, Item 37, Appendix D
- Cheesecloth, Item 90, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D

References

- Appendix D
- PARA 2-6-5

5-4-30.1 Replace Bifilar Cover.

5-4-30.1.1 Remove.

a. Using nonmetallic scraper, remove sealing compound from edge of cover, bifilar, and area around boltheads.

b. Remove bolts and washers securing bifilar cover to bifilar. Remove bifilar cover (Figure 5-4-30-1, Detail A).

5-4-30.1.2 Inspect.

Inspect bifilar cover gasket for damage. **NONE ALLOWED.** If damage is found, replace gasket (PARA 5-4-30.2).

5-4-30.1.3 Install.

a. Make sure area is clean and free of foreign material.

b. Apply coat of primer or sealing compound to shank, threads, and area under bifilar boltheads using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D.

c. Position bifilar cover over bifilar and install bolts and washers while primer is still wet (Figure 5-4-30-1, Detail A).

d. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

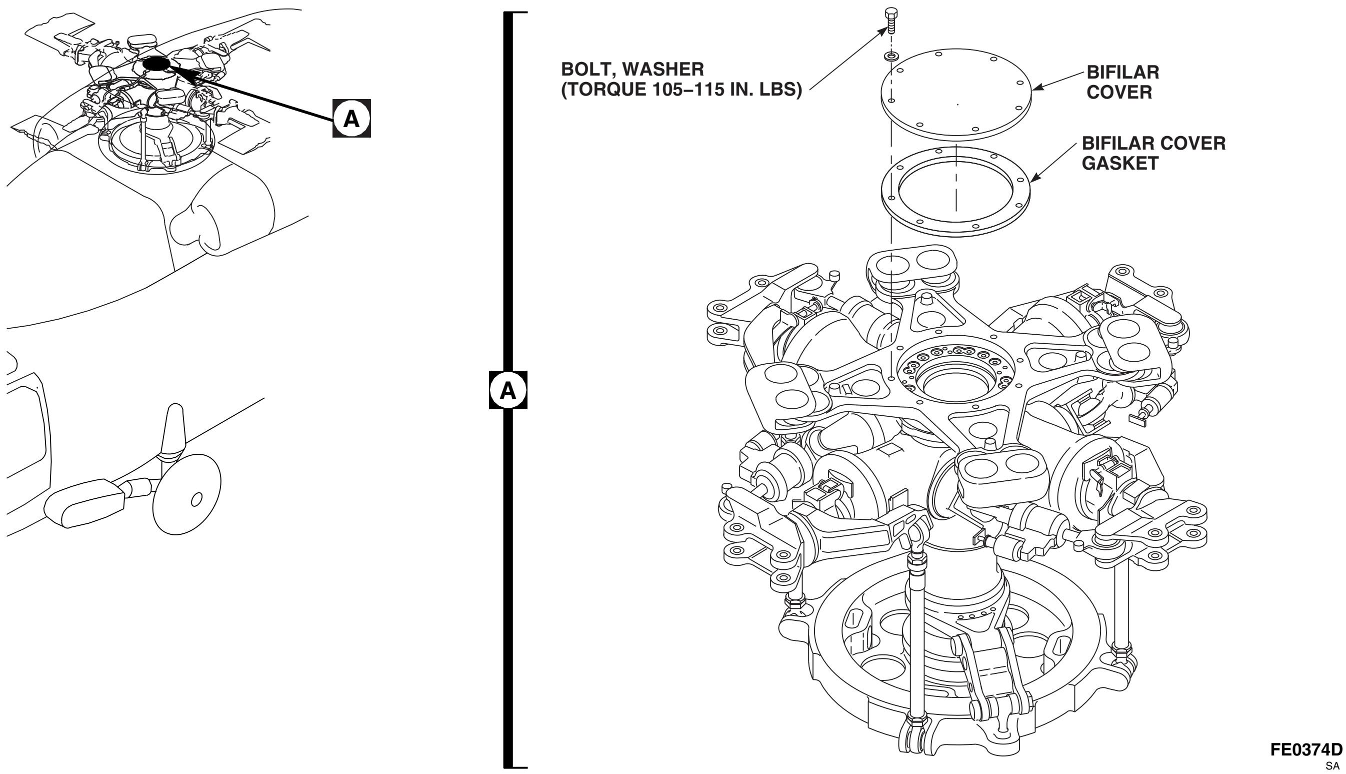


FIGURE 5-4-30-1. REPLACE BIFILAR COVER

e. Using machinery towel, Item 211, Appendix D, wipe off any primer that has seeped from under boltheads.

f. Using sealing compound, Item 292, Appendix D, seal edge of bifilar cover and areas around bifilar boltheads.

g. Touch up paint on boltheads as necessary (PARA 2-6-5).

5-4-30.2 Replace Bifilar Cover Gasket.

5-4-30.2.1 Remove.

a. Remove bifilar cover (PARA 5-4-30.1.1).

b. Using nonmetallic scraper, remove bifilar cover gasket (Figure 5-4-30-1, Detail A).

c. Using cheesecloth, Item 90, Appendix D moistened with methyl ethyl ketone, Item 217, Appendix D, remove remaining residue from bifilar cover.

5-4-30.2.2 Install.

a. Using abrasive paper, Item 18, Appendix D, thoroughly scuff up surface of bifilar cover gasket and bifilar cover to be mated (Figure 5-4-30-1, Detail A).

b. Using cheesecloth, Item 90, Appendix D moistened with methyl ethyl ketone, Item 217, Appendix D, clean mating surface of cover and gasket.

c. Apply adhesive, Item 37, Appendix D, to mating surface of bifilar cover and bifilar cover gasket.

NOTE

Allow adhesive to air-dry for 10 minutes until most of the solvent has evaporated and adhesive is very tacky.

d. Install bifilar cover gasket on bifilar cover. Align holes in gasket with holes in bifilar cover.

e. Lay cover assembly on flat surface with bifilar cover gasket side down.

f. Press down firmly on bifilar cover to make sure bifilar cover and bifilar cover gasket mate.

g. Allow adhesive to cure for 24 hours.

h. Install bifilar cover (PARA 5-4-30.1.2).

5-4-31 MAIN ROTOR BLADES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Blade Clamp Assembly, 70700-20324-041
- Box Wrench, 7/8-inch, Closed End
- Bungee (Elastic) Cord
- C-Clamps
- Center Punch
- Clamping Plates (or equivalent)
- Composites Repair Kit, consisting of:
    - Router
    - Router Bit
    - Spanner Bar
- Container
- Cup (large enough to prepare adhesive)
- Cutter
- DC Ammeter
- DC Voltmeter
- Drill
- Drill Bit, No. 40, 0.098-inch diameter
- Driving Tool
- Electrical Power Source, 110 VAC
- Electrical Power Source, 115 VAC, 60 Hz
- End Mill, 1/2-inch diameter
- Felt
- Fine Cut File
- Flexible Metal Spatula (or equivalent)
- Guide Rope
- Heat Gun
- Heat Lamps
- Heater Mat Resistance Test Set, Locally-Made, Figure H-258, Appendix H
- Hoist, Capacity to Support Main Rotor Blade (Alternate for Cabin Maintenance Crane)
- Holding Fixture or Padded Area (Alternate for Main and Tail Rotor Blade Adapter)
- Main and Tail Rotor Blade Adapter, 70700-20426-041 (Alternate for Holding Fixture or Padded Area)
- Main Rotor Blade Solid Pin Removal Tool, Locally-Made, Figure H-266, Appendix H
- Maintenance Sling, 70700-20323-043
- Maintenance Trailer
- Medium Cut File
- NO SMOKING Signs
- Nonmetallic Scraper, Locally-Made
- Pencil, Soft
- Plastic Tool, Thin (or equivalent)
- Pressure Plates (or equivalent)
- Pressure Strips, Wood
- Pressure/Heat Curing Fixture
- Protection, Eye, Skin, and Breathing
- Protective Covering
- Rotary Wing Blade Repair Kit, 70072-15001-012 through -021
- Rubber Mat

- Safety Strap
- Screw Extractor Set (or equivalent)
- Screwdriver, Flat-Bladed (or equivalent)
- Shot Bags (or equivalent)
- Small Brush
- Squeegee
- Straightedge, Plastic or Wood
- Torque Wrench, 0 - 75 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Vacuum
- Weights (or equivalent)
- Wire Brush
- Workstand
- Wrench

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Cloth, Item 3, Appendix D
- Abrasive Cloth, Item 4, Appendix D
- Abrasive Cloth, Item 5, Appendix D
- Abrasive Cloth, Item 7, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Abrasive Cloth, Medium-to-Fine Grit
- Abrasive Paper, Item 18, Appendix D
- Abrasive Paper, Item 19, Appendix D
- Acetone, Item 25, Appendix D
- Adhesion Promoter, Item 25A, Appendix D
- Adhesive, Item 32, Appendix D
- Adhesive, Item 35, Appendix D
- Adhesive, Item 66F, Appendix D
- Alcohol, Denatured, Item 70, Appendix D
- Aluminum Mesh Screen, Item 73A, Appendix D
- Applicator, Item 78, Appendix D (or equivalent)
- Brush, Acid, Item 84, Appendix D
- Brush, Paint, Item 86, Appendix D
- Cheesecloth, Item 90, Appendix D
- Chevron Patch, Locally-Made, Figure H-264, Appendix H
- Corrosion Resistant Coating, Item 118, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Fiberglass Repair Kit, Item 138, Appendix D
- Fuel Tank Parts Kit, Item 141B, Appendix D
- Heavy Paper (or equivalent protective covering)
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Methyl Isobutyl Ketone, Item 218, Appendix D
- Paint, Polyurethane, Item 223, Appendix D
- Paint, Polyurethane, Item 227, Appendix D
- Plastic Film Peel Ply
- Plastic Sheet, Item 235A, Appendix D

- Plastic Strip, Item 237, Appendix D
- Plastic, Polyvinyl Sheet, Item 237B, Appendix D
- Polyurethane Coating, Item 241D, Appendix D
- Polyurethane Patch, 70150-09100-101
- Pressure Sensitive Adhesive Tape, Item 244B, Appendix D
- Primer, Item 258, Appendix D
- Primer Coating, Item 259, Appendix D
- Satin Cloth, Item 269A, Appendix D
- Sealing Compound, Item 290, Appendix D
- Sealing Compound, Item 296, Appendix D
- Sealing Compound, Item 301, Appendix D
- Stirring Stick, Beverage, Item 315C, Appendix D
- Tape
- Tiedown Strap, Item 328, Appendix D
- Toluene, Item 339, Appendix D
- Tongue Depressor (or equivalent)
- Vacuum Bags (or equivalent) (vacuum bag must be large enough to cover repair)

References

- Appendix D
- Appendix H
- Appropriate MTF
- PARA 1-3-29
- PARA 1-4-6
- PARA 1-6-16
- PARA 1-6-18
- PARA 1-7-4
- PARA 1-7-5
- PARA 2-6-5
- PARA 5-3-9
- PARA 5-3-11
- PARA 5-4-8
- PARA 5-5-12
- PARA 12-2-5
- PARA 12-4-26
- PARA 18-4-1
- TM 1-70-VIB

5-4-31.1 Replace Main Rotor Blade.

CAUTION

Damage to main rotor blade will result if blade trim tab is bent or damaged. While working on main rotor blade, be careful not to bend or damage blade trim tab. Bending trim tab will cause blade out-of-track condition. Position workstand under tip of blade for assistant to stand on to help guide blade clear of helicopter.

NOTE

- Cabin maintenance crane may be used instead of hoist. In this procedure, the hoist is illustrated.
- Replacement of all main rotor blades is the same.

5-4-31.1.1 Remove.

a. If cabin maintenance crane will be used, erect cabin maintenance crane (PARA 18-4-1).

b. If necessary, install main and tail rotor blade adapter on maintenance trailer.

c. Turn off all electrical power (PARA 1-6-16).

d. If not already done, engage gust lock (PARA 1-6-18).

NOTE

Main rotor blade's pretrack number will be used to adjust the pitch control rod during installation of replacement main rotor blade.

e. Write down main rotor blade's pretrack number and location.

WARNING

Blade clamp assembly may open during removal, causing injury to personnel and damage to equipment. If maintenance sling is used, one leg of maintenance sling should be fastened loosely around blade clamp assembly, in case blade clamp assembly opens. Install safety strap over blade clamp assembly before removing main rotor blade.

f. Position blade clamp assembly between balance stripes and lock to main rotor blade. Install safety strap over blade clamp assembly and through shackle (Figure 5-4-31-1, Detail B).

g. Attach guide ropes to main rotor blade cuff and tip.

h. Position assistants at main rotor blade tip and main rotor blade cuff, to hold guide rope to help guide main rotor blade.

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

NOTE

To prevent undue binding of hoisting hook and blade clamp assembly hoisting ring, hook must fit ring with sufficient clearance.

i. Attach blade clamp assembly to hoist or cabin maintenance crane. Use maintenance sling to attach blade clamp assembly to any hoist other than cabin maintenance crane.

CAUTION

To prevent damage to antiflap cam and antiflap stop attaching bolts, antiflap cam must be secured in the out or flight position whenever main rotor blades are removed.

j. Pull up on antiflap cam levers (flight position) and using tiedown strap, Item 328, Appendix D, secure to spindle horn.

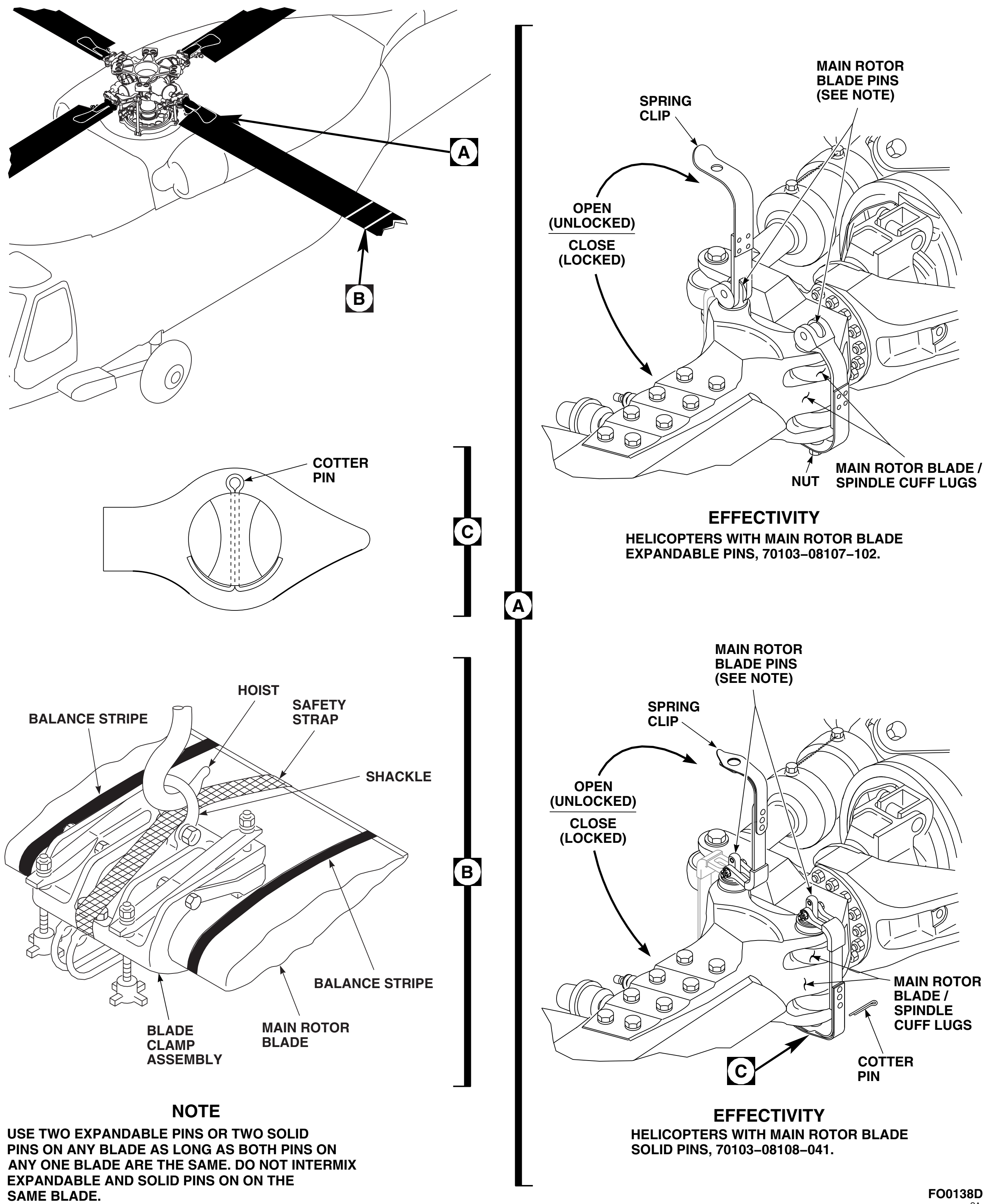


FIGURE 5-4-31-1. REPLACE MAIN ROTOR BLADE

WARNING

FSCAP

Main rotor hub is flight critical. Do not secure any items to main rotor hub.

k. If blade deice kit is installed, disconnect blade deice electrical connector and secure electrical connector away from all moving parts (PARA 12-4-26).

WARNING

FSCAP

Outside lug surface of main rotor blade cuffs are critical surfaces which must not be damaged during and after main rotor blade removal. Use extreme caution when removing and provide protective covering after removal.

NOTE

Main rotor blades may be secured by two different kinds of pins, main rotor blade expandable pins, 70103-08107-102, or main rotor blade solid pins, 70103-08108-041. In the following procedures, both types of pins are referred to as main rotor blade pins, except where noted.

l. Relieve load on main rotor blade pins by slightly lifting main rotor blade.

m. Remove main rotor blade pins as follows (Figure 5-4-31-1, Detail A):

WARNING

FSCAP

Main rotor blade pins contain critical surfaces which must not be damaged during and after removal from main rotor blade. Use extreme caution when removing and provide protective covering for pins after removal, if not immediately installed in spindle.

(1) On helicopters with main rotor blade expandable pins, perform the following:

(a) Release spring clip portion of handle from nut and open handle on expandable pin.

(b) Raise handle to fully unlocked (open) position.

CAUTION

Damage to expandable pin and main rotor blade cuff will result if pin is removed before lockring is fully compressed. Pin will jam in main rotor blade cuff. Make sure lockring is fully compressed before attempting to remove expandable pin.

(c) Make sure lockring on each pin is fully compressed by rotating it around pin core. If lockring is not compressed, or shows evidence of dirt or contaminants, clean out lockring before removing pin (PARA 1-4-6).

CAUTION

Expandable pins must be installed in same hole from which they were removed. Note and record pin hole and blade location of all pins removed.

(d) Remove expandable pins from main rotor blade/spindle cuff lugs. If expandable pin is stuck, remove stuck pin (PARA 5-4-31.24).

(e) Repeat for other expandable pin.

(2) On helicopters with main rotor blade solid pins, perform the following:

(a) Remove cotter pin securing solid pin.

(b) Raise handle to fully unlocked (open) position.

(c) Remove solid pins from main rotor blade/spindle cuff lugs. If solid pin is stuck, remove stuck pin (PARA 5-4-31.24).

(d) Repeat for other solid pin.

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting main rotor blade. Keep area clear when hoisting main rotor blade.

n. Remove main rotor blade.

CAUTION

Damage to main rotor pins may result if not immediately installed in spindle. If main rotor blade pins are not immediately installed, install protective covering on pins.

o. Install main rotor blade pins in spindle.

p. Lower main rotor blade and place on main and tail rotor blade adapter, holding fixture, or padded area.

q. If used, remove maintenance sling from main rotor blade. Remove safety strap; unlock and remove blade clamp assembly from main rotor blade.

r. Install protective covering on outside surfaces of main rotor blade cuff lugs.

s. If installed, remove guide ropes.

t. If more than one main rotor blade will be removed, disengage gust lock (PARA 1-6-18), and repeat entire procedure.

u. Inspect spindle lugs for security of bonded washers. If bonded washers are loose, rebond washers (PARA 5-4-8).

5-4-31.1.2 Deice Heater Mat Failure Test.

NOTE

- The procedures contained in this deice heater mat failure test are intended for checkout of main rotor blades, 70150-09100, that have been removed from helicopter for suspected heater mat resistance faults.
- Main rotor blades that are installed on helicopter can be checked with digital low-resistance ohmmeter (PARA 12-2-5). For resistance checks in PARA 12-2-5 to be accurate, power must have been applied to main rotor blade under test during system operational check a maximum of one hour prior to taking resistance readings.
- Resistance test for main rotor blade deice heater mats using heater mat resistance test set requires use of an external DC voltmeter and DC ammeter.
- Test points TP1, TP2, and TP3 on the heater mat resistance test set are not used during the deice heater mat failure test.

a. Locally make heater mat resistance test set (Figure H-258, Appendix H).

b. Make sure controls on heater mat resistance test set are in the following positions:

CONTROL	POSITION
CB1 PWR SUPPLY circuit breaker	Disengaged

CONTROL	POSITION
CB2 115 VAC circuit breaker	Disengaged
HEATING ELEMENTS (S1) switch	Set to 1-2 position
S2 switch	OPEN
S3 switch	OPEN
POWER SUPPLY ADJUSTMENT	Fully counterclockwise position

c. Connect heater mat resistance test set to 115 VAC, 60 Hz electrical power source.

d. Connect electrical connector, M83723/86W2212N, on heater mat resistance test set to main rotor blade to be tested.

e. Connect DC voltmeter, set to 20 VDC range, to VOLTMETER jacks TP4 (+) and TP5 (-) on heater mat resistance test set.

f. Connect DC ammeter, set to 2 amp DC range, to AMMETER jacks TP6 (-) and TP7 (+) on heater mat resistance test set.

g. Engage CB2 115 VAC circuit breaker on heater mat resistance test set. Verify that AC POWER ON IND lamp illuminates.

h. Engage CB1 PWR SUPPLY circuit breaker on heater mat resistance test set. Verify that DC voltmeter displays between 11.9 - 12.1 VDC reading. Allow heater mat resistance test set to warm up for at least 15 minutes before continuing.

i. Set switches, S2 and S3, to CLOSE position on heater mat resistance test set.

CAUTION

If less than 6 VDC reading is displayed on the DC voltmeter, do not proceed further to prevent possible damage to the main rotor blade and/or heater mat resistance test set.

j. Verify that DC voltmeter displays greater than 6 VDC reading. If less than 6 VDC reading is displayed on DC voltmeter, do not proceed further to prevent possible damage to main rotor blade and/or heater mat resistance test set.

k. Verify that HEATING ELEMENTS (S1) switch is set to 1-2 position.

l. Adjust POWER SUPPLY ADJUSTMENT on heater mat resistance test set slowly clockwise until 1.0 AMP DC reading is displayed on DC ammeter.

NOTE

The DC voltmeter will now correspond directly to a resistance reading.

m. Verify that DC voltmeter displays a reading between 7 - 10 VDC.

n. Return POWER SUPPLY ADJUSTMENT on heater mat resistance test set fully to clockwise position.

o. Return switches, S2 and S3, to OPEN position on heater mat resistance test set.

p. Repeat step i. through step n. with HEATING ELEMENTS (S1) switch in each of the following positions:

(1) 1-3

(2) 2-3

(3) 4-5

(4) 4-6

(5) 5-6

(6) 7-8

(7) 7-9

(8) 8-9

(9) 10-11

(10) 10-12

(11) 11-12

q. Disengage CB2 115 VAC circuit breaker on heater mat resistance test set. Verify that AC POWER ON IND. lamp is not illuminated.

r. Disengage CB1 PWR SUPPLY circuit breaker on heater mat resistance test set.

s. Disconnect DC voltmeter from VOLTMETER jacks TP4 (+) and TP5 (-) on heater mat resistance test set.

t. Disconnect DC ammeter from AMMETER jacks TP6 (-) and TP7 (+) on heater mat resistance test set.

u. Disconnect heater mat resistance test set from main rotor blade.

5-4-31.1.3 Inspect.

a. Inspect or service main rotor blade as follows:

(1) If removed main rotor blade will be installed, inspect blade (PARA 5-3-11).

(2) If new, overhauled, or stored main rotor blade from supply system will be installed, check and service main rotor blade (PARA 1-3-29).

b. Inspect main rotor blade cuff for cracks between each bolt and leading and trailing edges of main rotor blade cuff on upper and lower surfaces. **NONE ALLOWED.** If crack is found, replace main rotor blade.

c. Inspect main rotor blade cuff liners for damage or disbonding. If damage or disbonding is found, replace main rotor blade cuff liners (PARA 5-5-12).

d. Inspect main rotor blade pins (PARA 5-3-9).

e. Test operation of BIM® pressure indicator (PARA 1-3-29).

f. Inspect trim tab for possible damage. Do not fold, bend, or adjust trim tab in any way.

5-4-31.1.4 Install.

a. If not already done, engage gust lock (PARA 1-6-18).

WARNING

Injury to personnel and damage to equipment will result if blade clamp assembly opens during main rotor blade installation. Make sure safety strap is installed over blade clamp assembly and through shackle prior to installing main rotor blade.

b. Position blade clamp assembly between balance stripes and lock to main rotor blade. Install safety strap over blade clamp assembly and through shackle (Figure 5-4-31-1, Detail B).

NOTE

Make sure leading edge of main rotor blade faces direction of rotation.

c. Attach blade clamp assembly to hoist or cabin maintenance crane. Use maintenance sling to attach blade clamp assembly to any hoist other than cabin maintenance crane.

d. Attach guide ropes to main rotor blade cuff and tip.

e. Position assistants at main rotor blade tip and main rotor blade cuff, to hold guide rope to help guide main rotor blade.

WARNING

FSCAP

Main rotor blade pins contain critical surfaces which must not be damaged during removal from main rotor blade. Use extreme caution when removing pins.

CAUTION

On helicopters with main rotor blade expandable pins, expandable pins must be installed in the same hole from which they were removed. Note and record pin hole and blade location of all pins removed.

NOTE

Main rotor blades may be secured by two different kinds of pins, main rotor blade expandable pins, 70103-08107-102, or main rotor blade solid pins, 70103-08108-041. In the following procedures, both types of pins are referred to as main rotor blade pins, except where noted.

f. If installed in spindles, remove main rotor blade pins. Install protective covering on pins.

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting main rotor blade. Keep area clear when hoisting main rotor blade.

g. Lift main rotor blade into place.

h. Line up main rotor blade/spindle cuff lugs.

WARNING

FSCAP

Outside lug surface of main rotor blade cuffs are critical surfaces which must not be damaged during main rotor blade installation. After removal of protective covering, use extreme caution when installing.

i. Remove protective covering from outside surfaces of main rotor blade cuff lugs and main rotor blade pins.

j. Install main rotor blade pins as follows:

WARNING

FSCAP

- Main rotor blade pins contain critical surfaces which must not be damaged during installation in main rotor blade. Use extreme caution when installing pins.
- Do not intermix expandable pins with solid pins on same main rotor blade. Intermixing expandable and solid pins on same blade may cause fatigue damage to spindle, cuff, and main rotor blade pins.
- It is permissible to use expandable pins or solid pins on any main rotor blade, as long as both pins on any one blade are the same.
- Main rotor blade solid pins can not be used on spindle cuffs with eight lugs. If spindle cuff is configured with eight lugs, main rotor blade expandable pins must be used.

(1) On helicopters with main rotor blade expandable pins, perform the following:

CAUTION

Make sure pins are installed in same hole noted and recorded during remove procedure.

NOTE

- Expandable pin, 70103-08107-102, is manufactured by different suppliers and slight variations in the design are evident; however, their function and operation are the same.
- Expandable pins manufactured by Shur-Lok Corporation weigh slightly less than those supplied by other suppliers. Expandable pins manufactured by Shur-Lok Corporation can be identified by CAGE code: 97393.
- Intermixing expandable pins manufactured by Shur-Lok Corporation with heavier expandable pins is permitted; however, this may cause helicopter main rotor vibrations to increase.
- Where possible, expandable pins of the same manufacturer must be used for all eight positions. If expandable pins of the same manufacturer are not available, install available expandable pins and check main rotor balance (TM 1-70-VIB).
- Unseating of expandable pins up to a maximum of 0.125-inch is acceptable, provided lockring does not contact main rotor blade cuff.

(a) Insert expandable pins through main rotor blade/spindle cuff lugs (Figure 5-4-31-1, Detail A).

NOTE

Do not hit or force expandable pin into position.

(b) Align handle with main rotor blade cuff decal. If necessary, gently rock main rotor blade to get proper alignment.

NOTE

Make sure blade tension is off main rotor blade expandable pins during tension check. Lower or raise blade as necessary to relieve tension.

(c) Perform spring clip/handle tension check (PARA 5-3-9).

(d) Close handle, making sure spring clip of pin snaps over and fully engages hex on nut. Visually check for expansion of lockring on end of pin. Make sure slots in spring clip line up with points of nut.

(2) On helicopters with main rotor blade solid pins, perform the following:

(a) Insert solid pin through main rotor blade/spindle cuff lugs.

NOTE

Do not hit or force solid pin into position.

(b) Align handle with main rotor blade cuff decal. If necessary, gently rock main rotor blade to get proper alignment.

(c) Close handle, making sure retaining clip fork engages groove in pin and that bottom of pin protrudes through hole in spring clip.

(d) Install cotter pin from side of pin closest to main rotor hub (Figure 5-4-31-1, Detail C).

k. Cut and remove tiedown strap securing antiflap cam in flight position.

l. If used, remove maintenance sling from main rotor blade. Remove safety strap; unlock and remove blade clamp assembly from main rotor blade.

m. If used, disassemble and remove cabin maintenance crane (PARA 18-4-1).

n. If blade deice kit is installed, connect blade deice electrical connector (PARA 12-4-26).

o. Make entry in helicopter logbook showing tip cap screw torque stabilization check is shown due 9 - 11 flight hours after installation (PARA 1-7-4).

5-4-31.1.5 Adjust.

NOTE

- Adjust procedure, including the maintenance test flight, is not required when in-flight characteristics are normal and main rotor blades have been removed for inspection or shipment, have not been abused or damaged, require no adjustments or replacement, and are reinstalled in their original positions and locations.
- A one notch change to a pitch control rod will affect track by approximately ¼-inch.

a. Adjust pitch control rod as follows:

(1) Compare pretrack number of removed (old) main rotor blade with number on main rotor blade now installed.

NOTE

Pretrack numbers are stenciled on bottom of main rotor blade. Adjustment is not necessary if numbers are alike. Proceed to step c.

(2) Subtract pretrack number of installed (new) main rotor blade from number of removed (old) main rotor blade.

(3) If difference between removed (old) main rotor blade and installed (new) main rotor blade is a plus (+) number, loosen upper and lower locknuts on pitch control rod and turn in -2 MINUTES/NOTCH direction (Figure 5-4-31-2, Detail B) (Table 5-4-31-1 and Table 5-4-31-2).

(4) If difference between old main rotor blade and new main rotor blade is a minus (-) number, loosen upper and lower locknuts on pitch control rod and turn in +2 MINUTES/NOTCH direction.

(5) Check for no more than ¾-inch of threads showing on lower end of pitch control rod (Figure 5-4-31-2, Detail A). If exposed thread exceeds ¾-inch, pitch control rod has been improperly adjusted. Recheck adjustment.

NOTE

To prevent improper torque of upper locknut on pitch control rod, make sure torque wrench is not in contact with lock key when torquing.

(6) **SECURE LOCK KEY IN NOTCH BY TIGHTENING UPPER LOCKNUT.**

(7) **TORQUE UPPER LOCKNUT TO 700 - 750 INCH-POUNDS.**

(8) **TORQUE LOWER LOCKNUT TO 700 - 750 INCH-POUNDS.**

(9) Using lockwire, Item 197, Appendix D, lockwire lower locknut to link, and then back to lower locknut (double-safety).

(10) Using lockwire, Item 197, Appendix D, lockwire from upper locknut to lock key, and then back to opposite side of upper locknut (double-safety).

b. If more than one main rotor blade will be installed, disengage gust lock (PARA 1-6-18), before turning rotor head, and repeat entire procedure.

c. Make sure area is clean and free of foreign material.

d. If different main rotor blade expandable pins were installed, check main rotor balance (TM 1-70-VIB).

e. If different main rotor blade was installed, check main rotor balance (TM 1-70-VIB).

NOTE

Maintenance test flight is not required when in-flight characteristics are normal and main rotor blades have been removed for inspection or shipment, have not been abused or damaged, require no adjustments or replacement, and are reinstalled in their original positions and locations.

f. Perform maintenance test flight (Appropriate MTF) if:

(1) Main rotor blades or components have been abused or damaged in operational or nonoperational conditions.

(2) In-flight characteristics are abnormal, requiring adjustments to main rotor blades or components to correct.

(3) Different main rotor blade or components are installed.

g. Check helicopter logbook to make sure pitch control rod upper and lower locknuts retorque is shown due 9 - 11 flight hours after installation (PARA 1-7-5).

Table 5-4-31-1. Examples

Example 1 (See Note 2).	+8 MINUTES	removed OB
subtract	+4 MINUTES	installed NB
Total	+4 MINUTES	
Example 2 (See Note 3).	+8 MINUTES	removed OB
subtract	-4 MINUTES	installed NB
Total	+12 MINUTES	
Example 3 (See Note 4).	-2 MINUTES	removed OB
subtract	+10 MINUTES	installed NB
Total	-12 MINUTES	

NOTES

1. OB = old blade, NB = new blade.
2. Example 1 shows old blade at +8 and new blade at +4; therefore, adjust pitch control rod 2 notches in -2 MINUTES/NOTCH direction.
3. Example 2 shows old blade at +8 and new blade at -4; therefore, adjust pitch control rod 6 notches in -2 MINUTES/NOTCH direction.
4. Example 3 shows old blade at -2 and new blade at +10; therefore, adjust pitch control rod 6 notches in +2 MINUTES/NOTCH direction.

Table 5-4-31-2. Examples

Minutes Shown on Main Rotor Blade	Example 1	Example 2	Example 3
+12			
+10			***NB
+8	*OB	**OB	
+6			
+4	*NB		
+2			
0			
-2			***OB
-4		**NB	
-6			
-8			
-10			
-12			

NOTES

1. OB = old blade, NB = new blade.

2. * Example 1 shows old blade at +8 and new blade at +4; therefore, adjust pitch control rod 2 notches in -2 MINUTES/NOTCH direction.

3. ** Example 2 shows old blade at +8 and new blade at -4; therefore, adjust pitch control rod 6 notches in -2 MINUTES/NOTCH direction.

4. *** Example 3 shows old blade at -2 and new blade at +10; therefore, adjust pitch control rod 6 notches in +2 MINUTES/NOTCH direction.

5-4-31.2 Adhesive Filler Repair.

WARNING

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion-proof electrical equipment in areas where combustible vapors or materials may be present.
- Wash hands and face before eating or smoking.

NOTE

- Post NO SMOKING signs in area and keep materials away from spark or flame.
- This repair applies to repairing damage in the main rotor blade spar and trailing edge regions as shown in Figure 5-4-31-3, Sheet 1, Region A and Region C. For damage limits and repair procedures on main rotor blade, refer to Table 5-3-11-1 (PARA 5-3-11), and Figure 5-4-31-3, Sheet 1 and Sheet 2.
- Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.

a. Using suitable masking tape, mask repair area around damage ½-inch larger than damage outline (Figure 5-4-31-4, Sheet 1, Step A).

CAUTION

Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints or laminate fibers. Do not allow solvent to seep into bonded joints or laminate fibers. Machinery towels wet with methyl ethyl ketone should not be left lying on main rotor blade.

b. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, remove paint inside repair area (Figure 5-4-31-4, Sheet 1, Step B).

c. On repairs to leading edge fairing, slightly undercut foam, leaving cavity slightly larger than hole in fairing. Remove all loose foam from hole.

CAUTION

Damage to skin of main rotor blade will result if main rotor blade is sanded excessively. Sand only until yellow primer is removed.

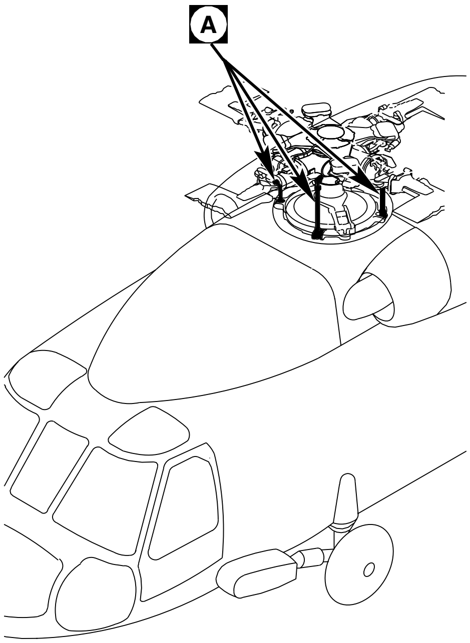
d. Using medium-to-fine-grit abrasive cloth, sand yellow primer from repair area (Figure 5-4-31-4, Sheet 1, Step C).

e. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean sanded area (Figure 5-4-31-4, Sheet 1, Step D). Wipe dry before methyl ethyl ketone evaporates.

NOTE

- Adhesive, Item 32, Appendix D, is available in 32-gram, pre-measured packages or in bulk form. Either may be used. Prepare accordingly.
- Pot life of mixture is 25 minutes at 75°F (24°C).

f. Prepare adhesive, Item 32, Appendix D, either step f. (1) or step f. (2) as follows:



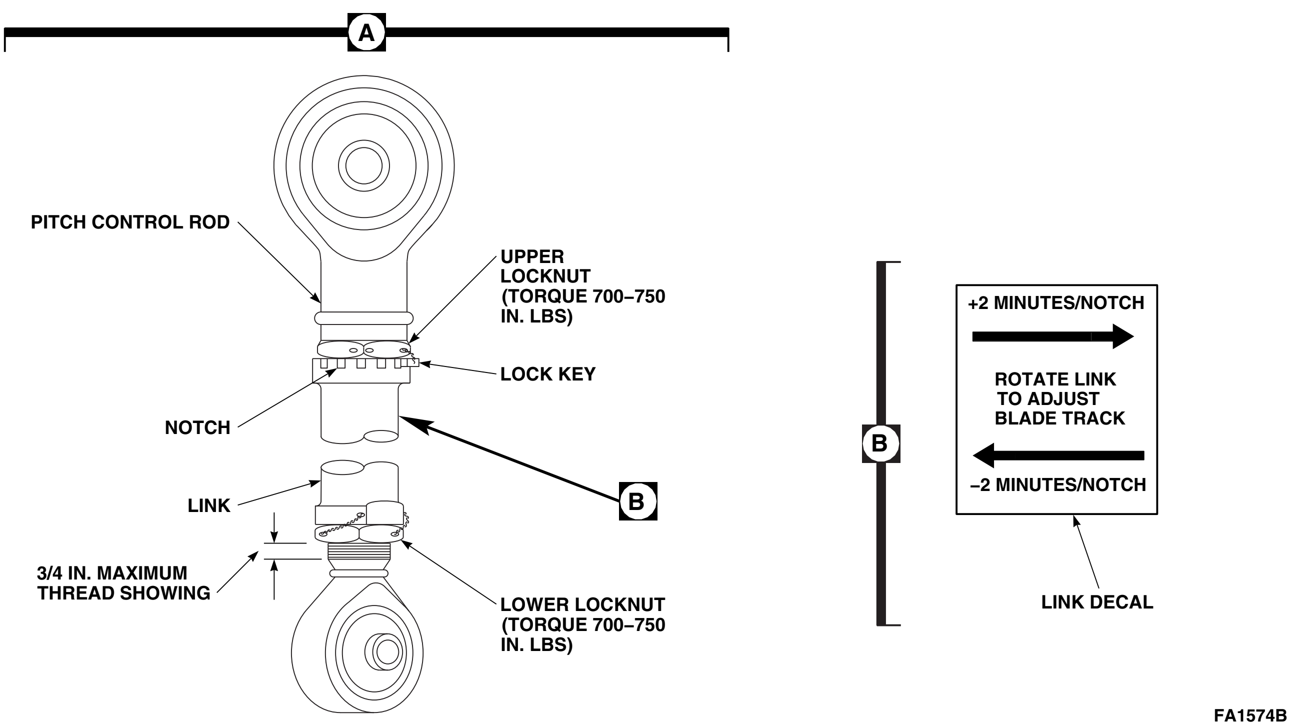
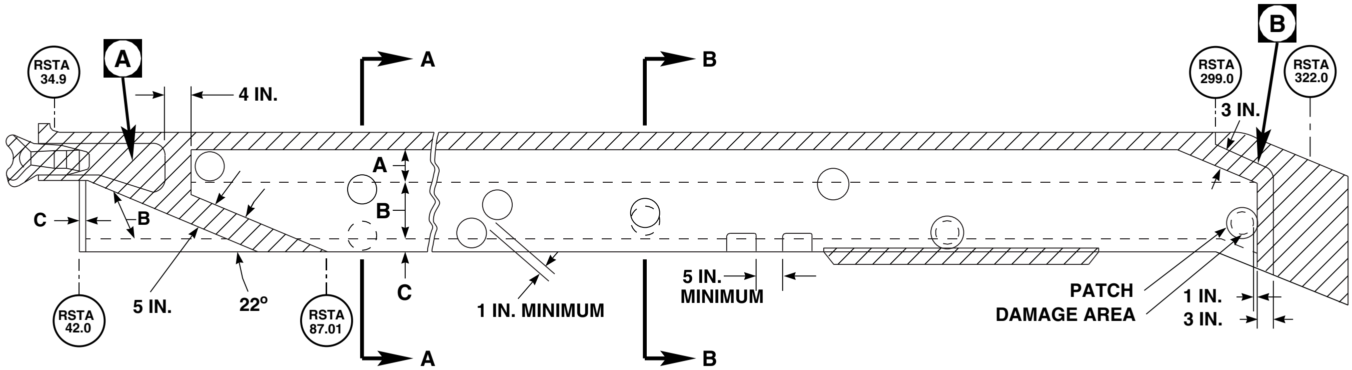
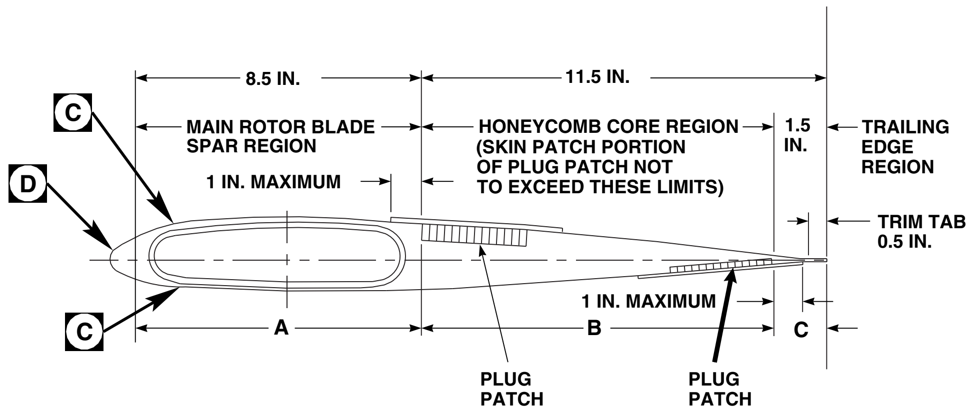


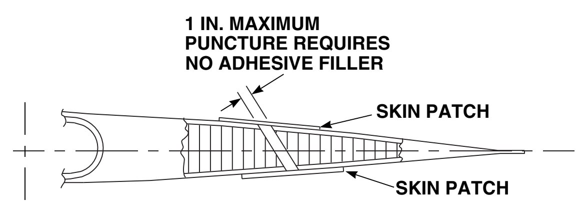
FIGURE 5-4-31-2. ADJUST PITCH CONTROL ROD



MAIN ROTOR BLADE



TYPICAL REPAIR LIMITS
SECTION A–A



TYPICAL PUNCTURE REPAIR
SECTION B–B

LEGEND

A – MAIN ROTOR BLADE SPAR REGION – SURFACE REPAIR OF FIBERGLASS BY SKIN PATCH OR ADHESIVE FILLER.

B – HONEYCOMB CORE REGION–SURFACE, SUBSURFACE REPAIR BY SKIN AND PLUG PATCHES (ONE OR BOTH SIDES), AND ADHESIVE INJECTION.

C – TRAILING EDGE REGION–SURFACE, SUBSURFACE REPAIR BY SKIN PATCH, ADHESIVE FILLER, OR ADHESIVE INJECTION. CRACKS IN ADHESIVE, EA9309, ONLY FILL WITH SEALING COMPOUND, ITEM 296, APPENDIX D.

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FIGURE 5-4-31-3. MAIN ROTOR BLADE GENERAL REPAIR LIMITS (SHEET 1 OF 2)

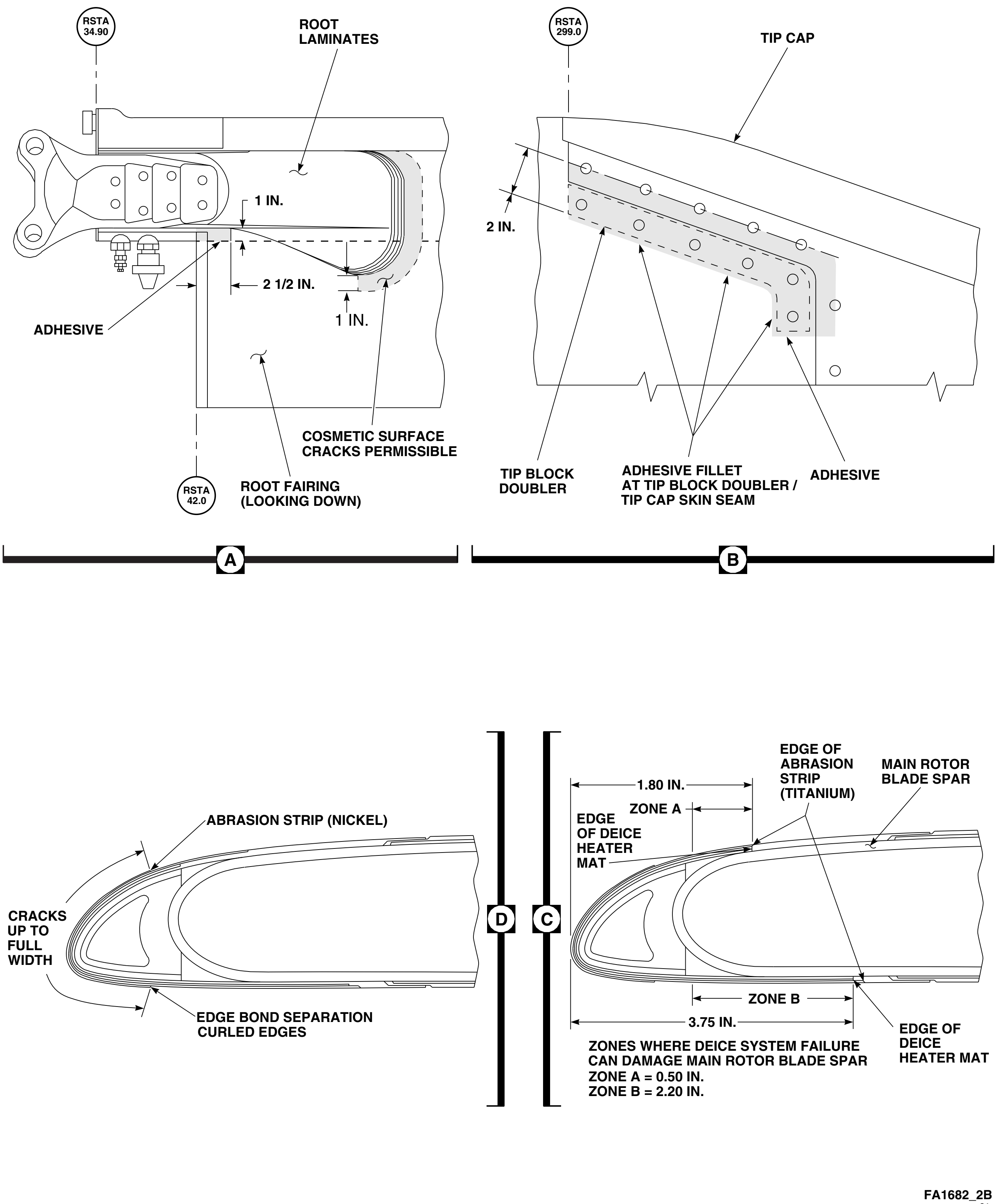


FIGURE 5-4-31-3. MAIN ROTOR BLADE GENERAL REPAIR LIMITS (SHEET 2 OF 2)

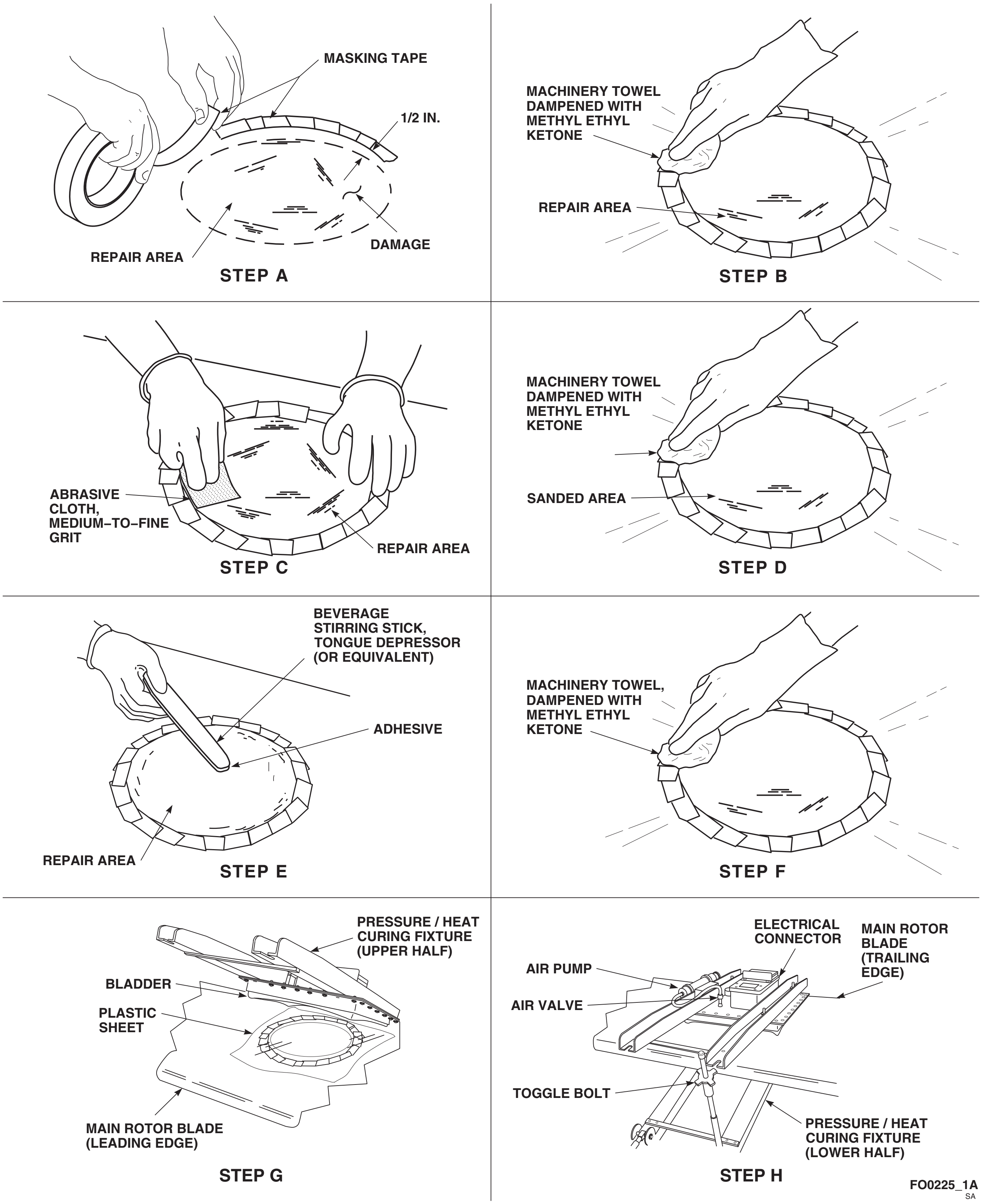


FIGURE 5-4-31-4. ADHESIVE FILLER REPAIR (SHEET 1 OF 2)

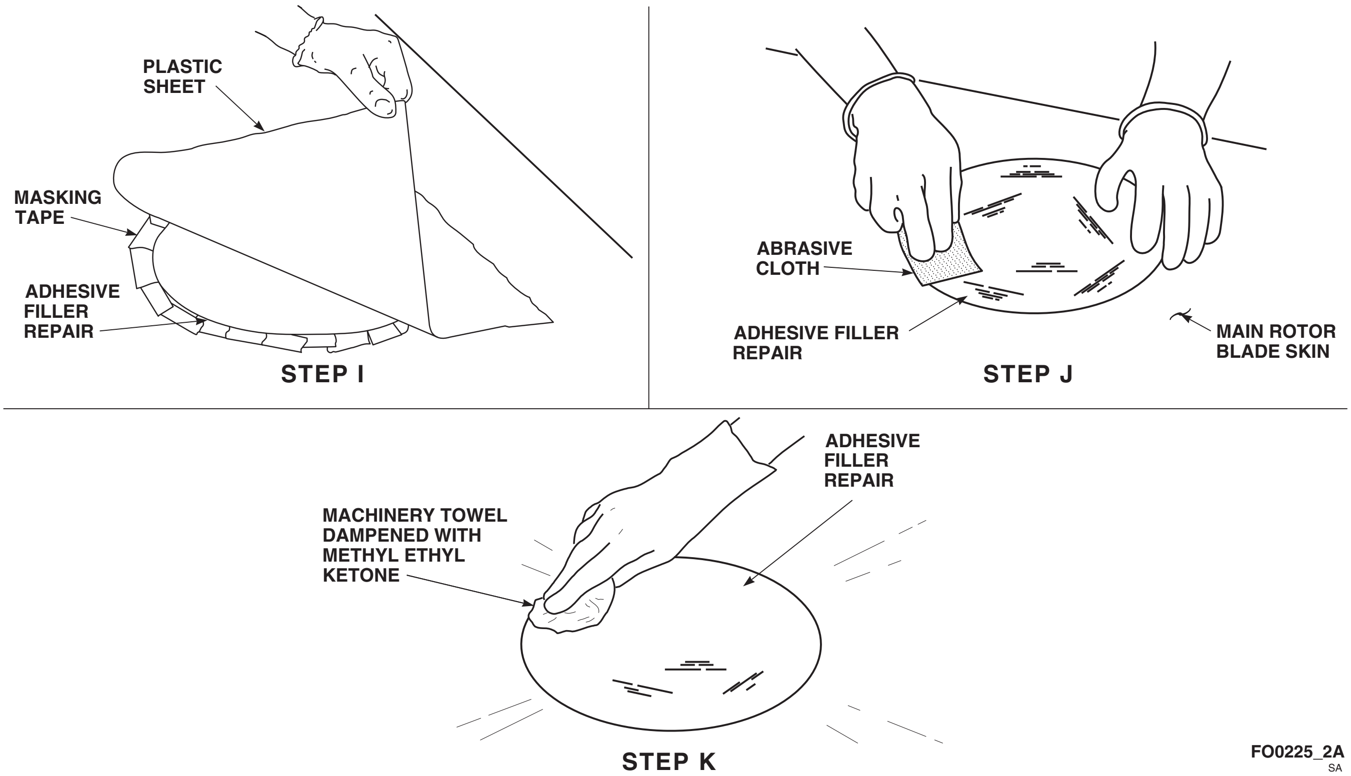


FIGURE 5-4-31-4. ADHESIVE FILLER REPAIR (SHEET 2 OF 2)

NOTE

Each 32-gram package is pre-measured with curing agent and resin. Mix full package and use as required. Discard leftover mixture.

(1) **32-Gram, Pre-Measured Package.**

(a) Empty package of adhesive resin into cup.

(b) Empty tube of curing agent into cup of adhesive resin.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

(2) **Bulk Adhesive.**

NOTE

Do not try to use all adhesive; use as required. Discard leftover mixture.

(a) Mix 100 parts of ″A″ part to 22 parts of ″B″ part by weight.

(b) Weigh each part and place both parts in cup.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

g. Using beverage stirring stick, Item 315C, Appendix D, or tongue depressor (or equivalent), fill in repair area with adhesive (Figure 5-4-31-4, Sheet 1, Step E).

NOTE

To reduce sanding, remove extra adhesive from repair area.

h. Using wood or plastic straightedge, bridge repair spanwise and scrape off extra adhesive in chordwise direction. Make sure level of adhesive is even with contour of skin.

i. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean area outside repair (Figure 5-4-31-4, Sheet 1, Step F).

j. Using plastic sheet, Item 235A, Appendix D, large enough to cover bladder area of pressure/heat curing fixture, cover repair area (Figure 5-4-31-4, Sheet 1, Step G).

NOTE

- Do not use pressure/heat curing fixture on leading edge fairing repair.
- Alternate curing time without pressure/heat curing fixture is 24 hours at 75°F (24°C).

k. Place pressure/heat curing fixture on main rotor blade so that bladder is over repair and hinges are toward trailing edge. Close pressure/heat curing fixture and engage toggle bolts. **TIGHTEN TOGGLE BOLTS UNTIL STOPS ARE CONTACTED** (Figure 5-4-31-4, Sheet 1, Step H).

NOTE

Relief valve on pump prevents pressure over 5 psi.

l. Actuate air pump until gage reads 4 psi.

m. Disconnect air pump hose from air valve.

n. Connect pressure/heat curing fixture to 110 VAC electrical power source.

o. Maintain heat and pressure for 1 hour.

p. Disconnect electrical power and depress plunger in valve stem, on top of pack assembly, to release air pressure.

q. Remove pressure/heat curing fixture from main rotor blade.

r. Remove plastic sheet and masking tape from adhesive filler repair (Figure 5-4-31-4, Sheet 2, Step I).

CAUTION

Damage to main rotor blade will result if main rotor blade is sanded excessively. Heavy sanding will damage skin of main rotor blade. Use only enough force to blend repair into main rotor blade.

s. Using abrasive cloth, Item 5, Appendix D, lightly sand edges of adhesive filler repair to blend repair to surrounding main rotor blade skin (Figure 5-4-31-4, Sheet 2, Step J). Leave 0.062-inch bead of adhesive over damaged area when repairing leading edges fairing.

t. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean adhesive filler repair (Figure 5-4-31-4, Sheet 2, Step K). Using clean machinery towel, Item 211, Appendix D, wipe dry.

u. Touch up paint (PARA 5-4-31.20).

5-4-31.3 Adhesive Injection Repair.

WARNING

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion-proof electrical equipment in areas where combustible vapors or materials may be present.
- Wash hands and face before eating or smoking.

NOTE

- Post NO SMOKING signs in area and keep materials away from spark or flame.
- This repair applies to repairing damage in honeycomb core and trailing edge regions as shown in Figure 5-4-31-3, Sheet 1, Regions B and C. For damage limits and repair procedures on main rotor blade, refer to Table 5-3-11-1 (PARA 5-3-11), and Figure 5-4-31-3, Sheet 1 and Sheet 2.
- Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.
- Refer to Figure 5-4-31-3, Sheet 1 and Sheet 2 for adhesive injection repair limitations.

a. Position main rotor blade horizontally, with leading edge 2 - 3-inches higher than trailing edge.

b. Using denatured alcohol, Item 70, Appendix D, clean out separation.

NOTE

- Adhesive, Item 32, Appendix D, is available in 32-gram, pre-measured packages or in bulk form. Either may be used. Prepare accordingly.
- Pot life of mixture is 25 minutes at 75°F (24°C).

c. Prepare adhesive, Item 32, Appendix D, either step c. (1) or step c. (2) as follows:

NOTE

Each 32-gram package is pre-measured with curing agent and resin. Mix full package and use as required. Discard leftover mixture.

(1) **32-Gram, Pre-Measured Package.**

(a) Empty package of adhesive resin into cup.

(b) Empty tube of curing agent into cup of adhesive resin.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

(2) **Bulk Adhesive.**

NOTE

Do not try to use all adhesive; use as required. Discard leftover mixture.

(a) Mix 100 parts of ″A″ part to 22 parts of ″B″ part by weight.

(b) Weigh each part and place both parts in cup.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

CAUTION

Damage to main rotor blade skin will result if excessive force is used to open up separation in main rotor blade. Do not apply too much pressure with plastic tool.

d. Using clean, thin plastic tool (or equivalent), carefully open up separation and apply adhesive to slot. Using applicator, Item 78, Appendix D (or equivalent), force adhesive into separation (Figure 5-4-31-5, Step A).

e. Remove tool and finish filling slot with adhesive.

f. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean excess adhesive

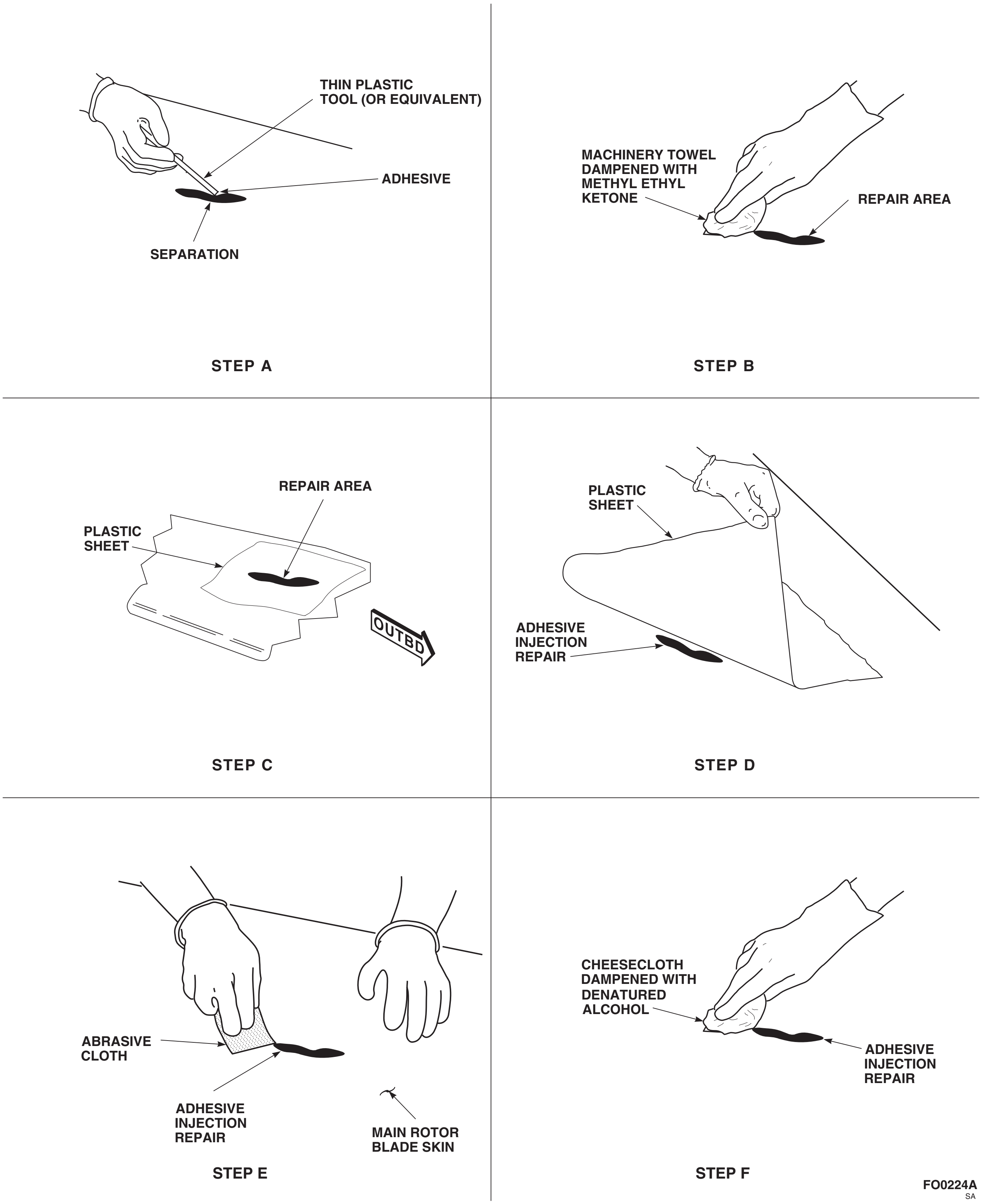


FIGURE 5-4-31-5. ADHESIVE INJECTION REPAIR

from repair area (Figure 5-4-31-5, Step B). Wipe dry before methyl ethyl ketone evaporates.

g. Using plastic sheet, Item 235A, Appendix D, cover repair area (Figure 5-4-31-5, Step C).

h. Using C-clamps, clamp flat wood pressure strips to sides of tabs with light but firm pressure.

i. Allow adhesive to cure for 8 hours at 75°F (24°C), or using heat lamps, 1 hour at 160°F (71°C).

j. Remove C-clamps, wood pressure strips, and plastic sheet from adhesive injection repair (Figure 5-4-31-5, Step D).

k. Using abrasive cloth, lightly sand edges of adhesive injection repair to blend repair to surrounding main rotor blade skin (Figure 5-4-31-5, Step E).

l. Using cheesecloth, Item 90, Appendix D, dampened with denatured alcohol, Item 70, Appendix D, clean adhesive injection repair (Figure 5-4-31-5, Step F).

m. Touch up paint (PARA 5-4-31.20).

5-4-31.4 Skin Patch Repair.

WARNING

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion-proof electrical equipment in areas where combustible vapors or materials may be present.
- Wash hands and face before eating or smoking.

NOTE

- Post NO SMOKING signs in area and keep materials away from spark or flame.
- This repair applies to repairing damage in the main rotor blade spar, honeycomb core, and trailing edge regions as shown in Figure 5-4-31-3, Sheet 1, Region A, Region B, and Region C. For damage limits and repair procedures on main rotor blade, refer to Table 5-3-11-1 (PARA 5-3-11), and Figure 5-4-31-3, Sheet 1 and Sheet 2.
- This repair applies to 3, 5, and 8-inch skin patches (rotary wing blade repair kits, 70072-15001-012 through 70072-15001-014, in that order). Use smallest skin patch size that will allow 1-inch overlap of cleaned up damage.
- Edge of new skin patch repair must not be closer than 1-inch from edge of previously installed skin and plug patch repairs on same side of main rotor blade.
- Use skin patch when damage does not extend through skin to main rotor blade spar or when core damage is less than 1-inch diameter (honeycomb core region). Refer to Table 5-3-11-1 (PARA 5-3-11).
- Use skin patch on two sides if damage is through both skins but core damage is less than 1-inch diameter (honeycomb core region).
- If core damage is greater than 1-inch diameter, use plug patch repair (PARA 5-4-31.5).
- Skin patch may straddle area between main rotor blade spar and honeycomb core regions.
- Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.
- Damaged main rotor blades exceeding repairable limits will be sent to manufacturer.

a. Using fiberglass repair kit, Item 138, Appendix D, select correct template. Use 5, 7, or 10-inch template for 3, 5, or 8-inch skin patch, respectively (Figure 5-4-31-6, Sheet 1).

b. Center template over damage, and using soft pencil, draw circle around damage. Circle should be 1-inch larger than damage on all sides (Figure 5-4-31-6, Sheet 2, Step A).

CAUTION

Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints, laminate fibers, or core. Do not allow solvent to seep into bonded joints, laminate fibers, or core. Machinery towels wet with methyl ethyl ketone should not be left lying on main rotor blade.

c. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, remove paint inside circle/repair area (Figure 5-4-31-6, Sheet 2, Step B).

CAUTION

Damage to main rotor blade skin will result if main rotor blade is sanded excessively. Sand only until yellow primer is removed.

d. Using abrasive cloth, Item 7, Appendix D, sand yellow primer from repair area (Figure 5-4-31-6, Sheet 2, Step C).

NOTE

When damage results from rips, tears, or holes less than 1-inch diameter, remove debris using vacuum.

e. Using fiberglass repair kit, Item 138, Appendix D, select correct template, and using soft pencil, draw circle ½-inch larger than patch to be applied (Figure 5-4-31-6, Sheet 2, Step D).

f. Using 1-inch masking tape, Item 212, Appendix D, mask area outside circle/repair area (Figure 5-4-31-6, Sheet 2, Step E).

g. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean repair area inside masked circle (Figure 5-4-31-6, Sheet 2, Step F). Wipe dry before methyl ethyl ketone evaporates.

h. Protect area to keep surface clean.

NOTE

- Adhesive, Item 32, Appendix D, is available in 32-gram, pre-measured packages or in bulk form. Either may be used. Prepare accordingly.
- Pot life of mixture is 25 minutes at 75°F (24°C).

i. Prepare adhesive, Item 32, Appendix D, either step i. (1) or step i. (2) as follows:

NOTE

Each 32-gram package is pre-measured with curing agent and resin. Mix full package and use as required. Discard leftover mixture.

(1) **32-Gram, Pre-Measured Package.**

(a) Empty package of adhesive resin into cup.

(b) Empty tube of curing agent into cup of adhesive resin.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

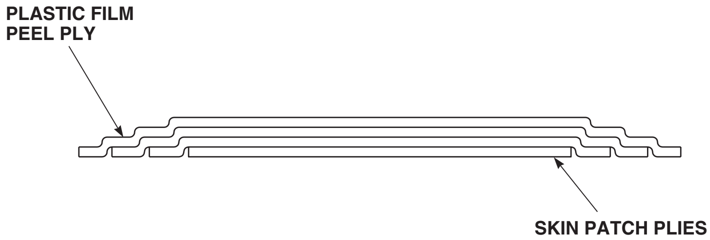
(2) **Bulk Adhesive.**

NOTE

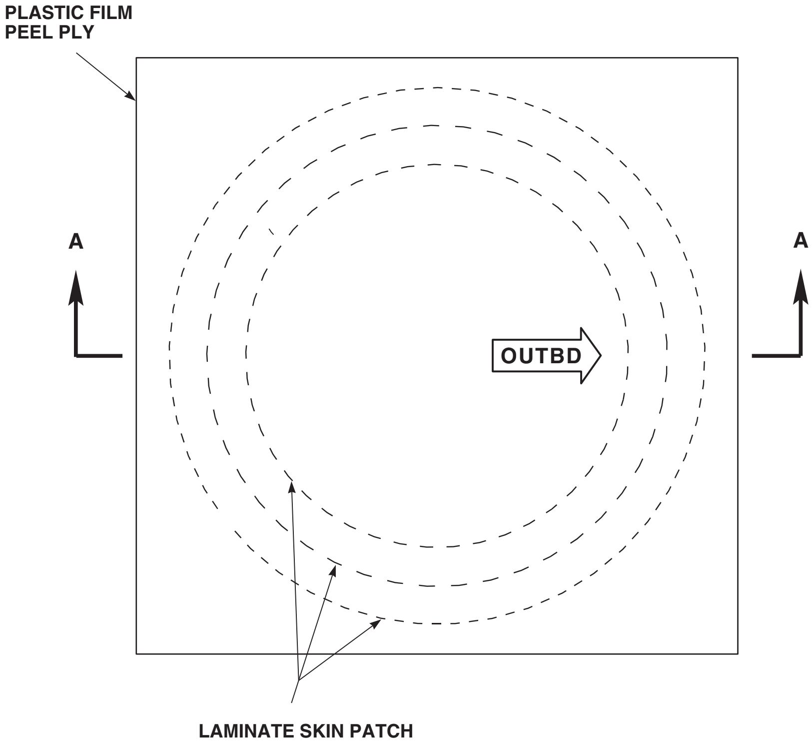
Do not try to use all adhesive; use as required. Discard leftover mixture.

(a) Mix 100 parts of ″A″ part to 22 parts of ″B″ part by weight.

(b) Weigh each part then place both parts in cup.



SECTION A–A



TYPICAL SKIN PATCH

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FIGURE 5-4-31-6. SKIN PATCH REPAIR (SHEET 1 OF 3)

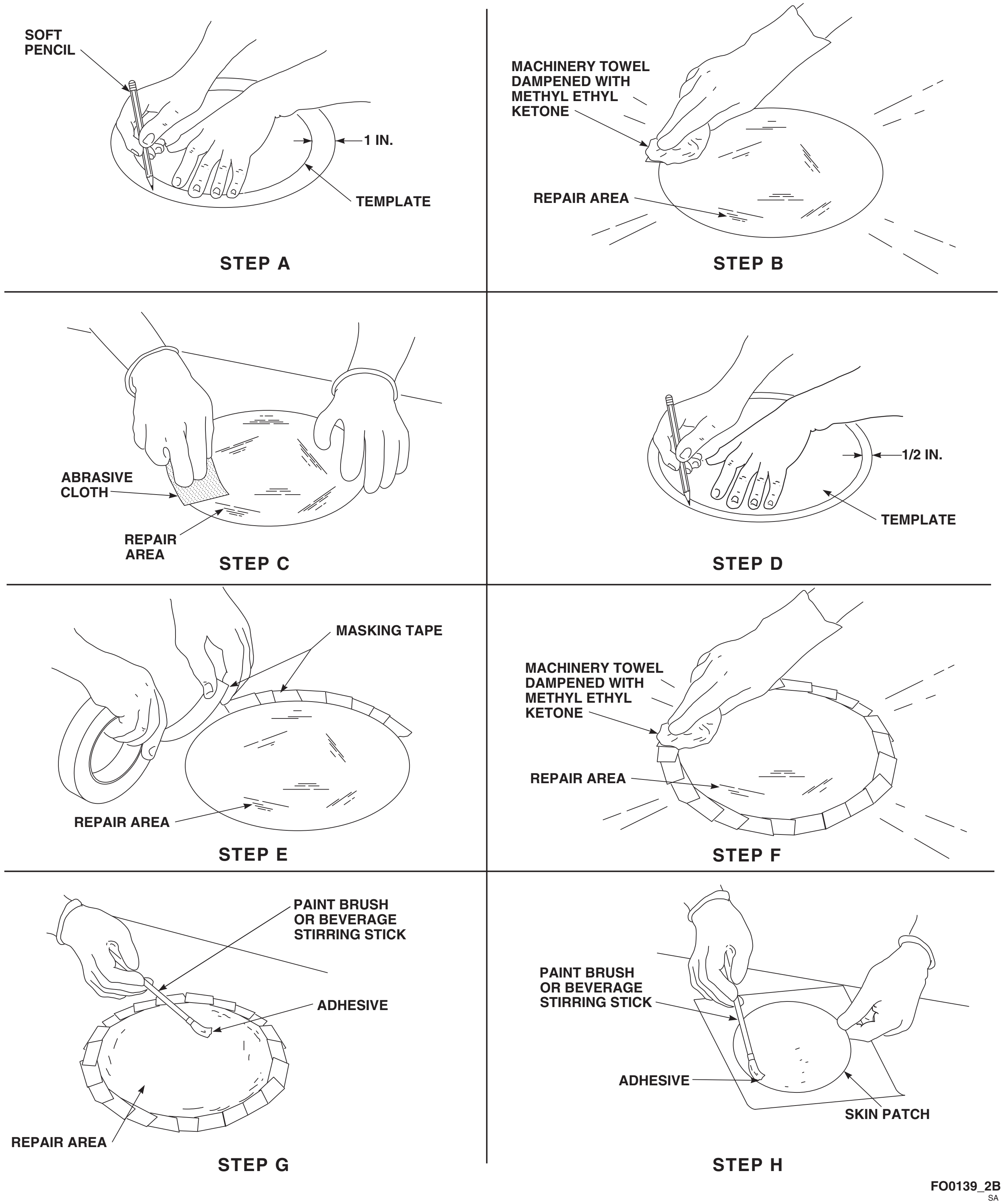


FIGURE 5-4-31-6. SKIN PATCH REPAIR (SHEET 2 OF 3)

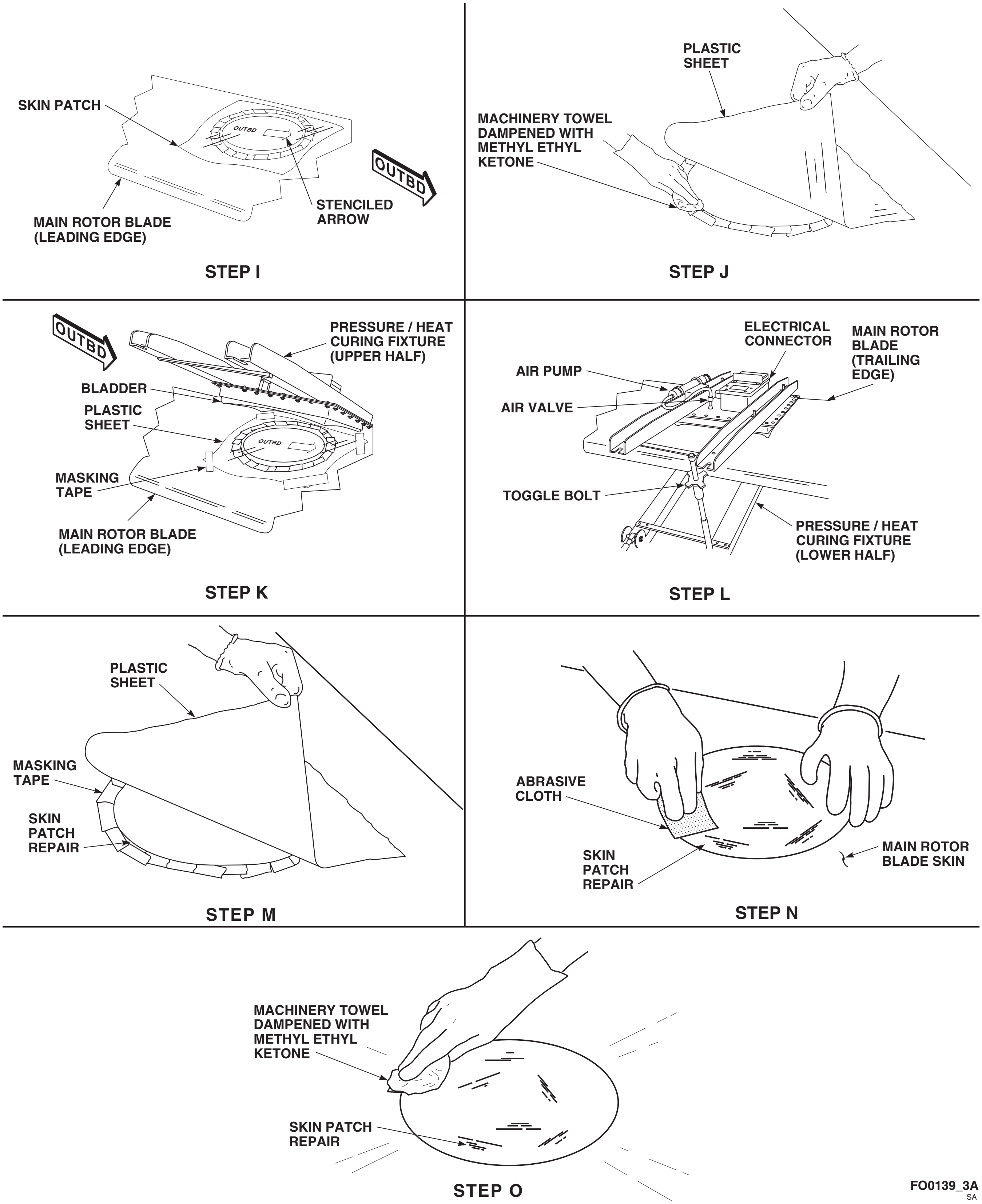


FIGURE 5-4-31-6. SKIN PATCH REPAIR (SHEET 3 OF 3)

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

NOTE

If core is damaged less than 1-inch diameter, smear adhesive on frayed ends of honeycomb material. Routing out core is not necessary for this repair (PARA 5-3-11). Do not pack adhesive into cells of honeycomb. Added adhesive weight will affect main rotor blade balance.

j. Using clean, 1-inch paint brush, Item 86, Appendix D, or beverage stirring stick, Item 315C, Appendix D, apply light, even coat of adhesive to repair area inside masked circle (Figure 5-4-31-6, Sheet 2, Step G).

NOTE

Skin patches are made from several plies of material (PARA 5-3-11). Plastic sheet on the outer surface keeps adhesive from contacting pressure/heat curing fixture during curing. An arrow is stenciled on plastic sheet for correct positioning of skin patch (Figure 5-4-31-6, Sheet 1). Skin patches are packaged in plastic to protect bonding surface. Do not remove plastic sheet from skin patch until repair is cured.

k. Carefully remove skin patch from plastic wrapping. Do not handle inner (bonding) surface.

l. Using same 1-inch paint brush, Item 86, Appendix D, or beverage stirring stick, Item 315C, Appendix D, apply thin, even coat of adhesive to skin patch bonding surface (Figure 5-4-31-6, Sheet 2, Step H).

NOTE

Skin patch must overlap damage by 1-inch on all sides.

m. Position skin patch on main rotor blade skin with stenciled arrow pointing outboard (Figure 5-4-31-6, Sheet 3, Step I). Move skin patch back-and-forth with light hand pressure to seat skin patch and work out air bubbles in adhesive.

n. Carefully lift edges of plastic sheet, and using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone Item 217, Appendix D, clean excess adhesive from around skin patch (Figure 5-4-31-6, Sheet 3, Step J).

o. Check skin patch for correct position, and using 1-inch masking tape, Item 212, Appendix D, tape down four corners to prevent movement (Figure 5-4-31-6, Sheet 3, Step K).

p. Place pressure/heat curing fixture on main rotor blade so that bladder is over repair and hinges are toward trailing edge. Close pressure/heat curing fixture and engage toggle bolts. **TIGHTEN BOLTS UNTIL METAL SKIRT AROUND BLADDER IS 1/8-INCH FROM SKIN OF MAIN ROTOR BLADE** (Figure 5-4-31-6, Sheet 3, Step K and Step L).

q. Actuate air pump until gage reads 4 psi.

r. Disconnect air pump hose from air valve.

s. Connect pressure/heat curing fixture to 110 VAC electrical power source.

t. Maintain heat and pressure for one hour.

u. Disconnect electrical source and press plunger in valve stem on top of pressure/heat curing fixture to release air pressure.

v. Remove pressure/heat curing fixture from main rotor blade.

w. Allow main rotor blade to cool to room temperature.

x. Remove plastic sheet and masking tape from skin patch repair (Figure 5-4-31-6, Sheet 3, Step M).

CAUTION

Damage to main rotor blade skin will result if main rotor blade is sanded excessively. Use only enough force to blend edges of skin patch into main rotor blade.

y. Using abrasive cloth, Item 7, Appendix D, lightly sand edges of skin patch repair to blend repair to surrounding main rotor blade skin (Figure 5-4-31-6, Sheet 3, Step N).

z. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean skin patch repair (Figure 5-4-31-6, Sheet 3, Step O). Using clean machinery towel, Item 211, Appendix D, wipe dry.

aa. Touch up paint (PARA 5-4-31.20).

NOTE

If required, make sure all other skin patch, plug patch, and trailing edge patch repairs are performed prior to calculating and adjusting tip cap weight chordwise and spanwise unbalance.

ab. Calculate and adjust tip block weight chordwise and spanwise unbalance (PARA 5-4-31.14).

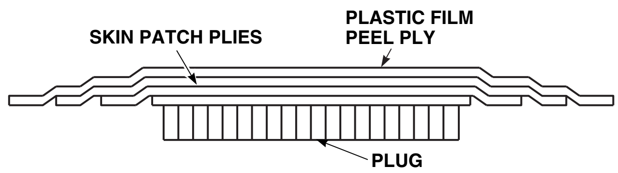
5-4-31.5 Plug Patch Repair.

WARNING

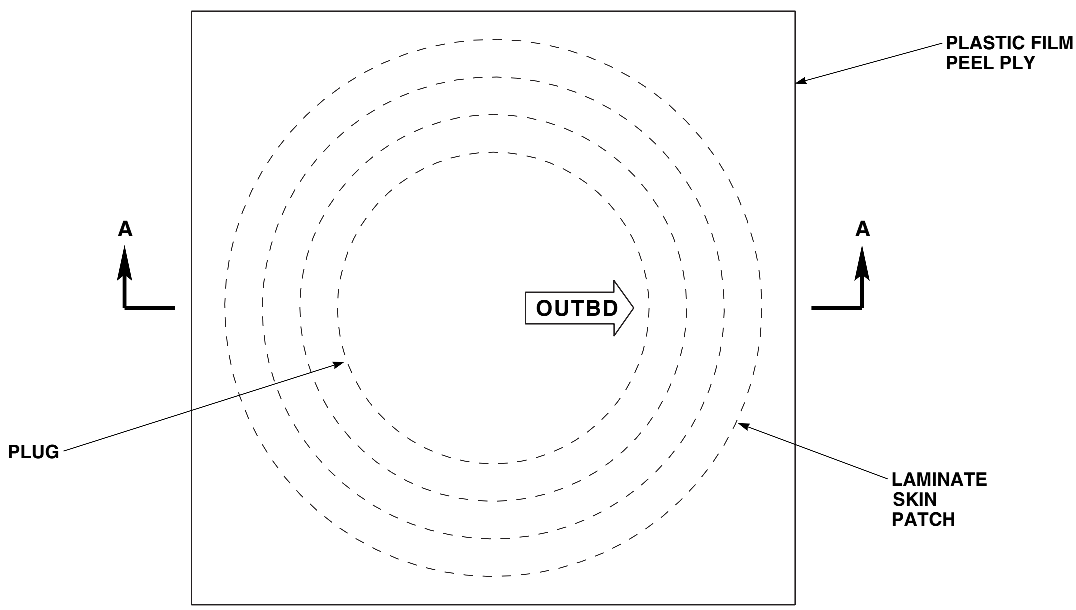
- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion-proof electrical equipment in areas where combustible vapors or materials may be present.
- Wash hands and face before eating or smoking.

NOTE

- Post NO SMOKING signs in area and keep materials away from spark or flame.
- This repair applies to repairing damage in the honeycomb core region as shown in Figure 5-4-31-3, Sheet 1, Region B. For damage limits and repair procedures on main rotor blade, refer to Table 5-3-11-1 (PARA 5-3-11), and Figure 5-4-31-3, Sheet 1 and Sheet 2.
- This repair applies to 3-inch and 6-inch plug patches (rotary wing blade repair kits, 70072-15001-015 through 70072-15001-020, in that order) (Table 5-4-31-3). Use size that allows 1-inch skin overlap of cleaned up damage.
- The skin patch portion of plug patch must overlap a minimum of 1-inch beyond damage, and may extend 1-inch outside honeycomb core region as shown in Figure 5-4-31-3, Sheet 1, Section A-A.
- Edge of new plug patch repair must not be closer than 1-inch from edge of previously installed skin and plug patch repairs on same side of main rotor blade.
- Holes routed for plug patch must be within dimensions allowed in PARA 5-3-11.
- Overlapping plug patch skin may extend 1-inch into main rotor blade spar or trailing edge region.
- For damage through both skins and core damage greater than 1-inch diameter, use plug patch on both sides (Figure 5-4-31-7, Sheet 2). Complete repair on one side of main rotor blade before starting repair on reverse side.
- Plug patches 3 and 6-inches in diameter are made from prefabricated skin patch



SECTION A–A



TYPICAL PLUG PATCH

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FIGURE 5-4-31-7. PLUG PATCH REPAIR (SHEET 1 OF 5)

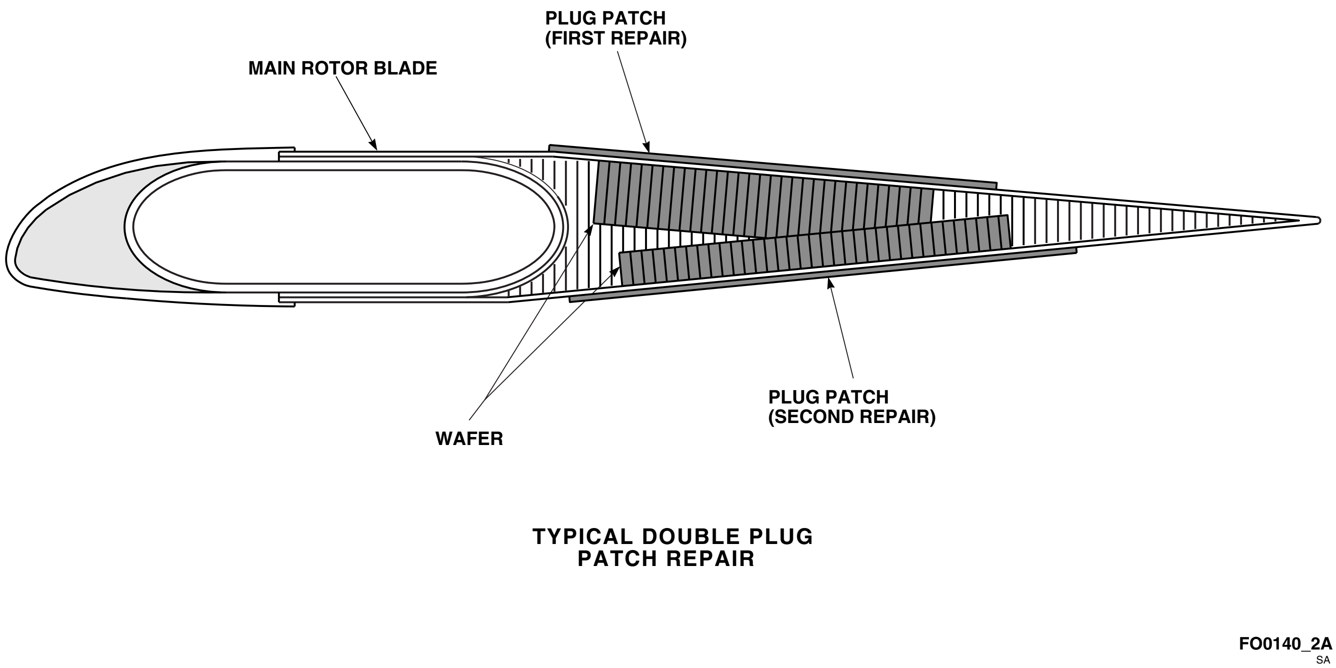


FIGURE 5-4-31-7. PLUG PATCH REPAIR (SHEET 2 OF 5)

and Nomex honeycomb plug. Refer to Table 5-4-31-3 for plug patch size and plug patch repair kit part number. Plastic sheet on the outer surface keeps adhesive from contacting pressure/heat curing fixture during curing. An arrow is stenciled on plastic sheet for correct positioning of plug (Figure 5-4-31-7, Sheet 1). A reinforced plastic wafer provides bonding surface for honeycomb core and plug core.

- Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.

- Damaged main rotor blades exceeding repairable limits will be sent to manufacturer.

a. Select correct template from rotary wing blade repair kit. Use 3 - 7-inch template for 3-inch plug, and 6 - 10-inch template for 6-inch plug.

Table 5-4-31-3. Plug Patch Repair

Rotary Wing Blade Repair Kit	Plug Patch Diameter	Plug Patch Thickness
70072-15001-015	3-inch	1/4-inch
70072-15001-016	3-inch	1/2-inch
70072-15001-017	3-inch	7/8-inch
70072-15001-018	6-inch	1/4-inch
70072-15001-019	6-inch	1/2-inch
70072-15001-020	6-inch	7/8-inch

b. Center template over damage, and using soft pencil, draw two circles around damage. Outer circle will be 2-inches in diameter larger than inner circle (Figure 5-4-31-7, Sheet 3, Step A).

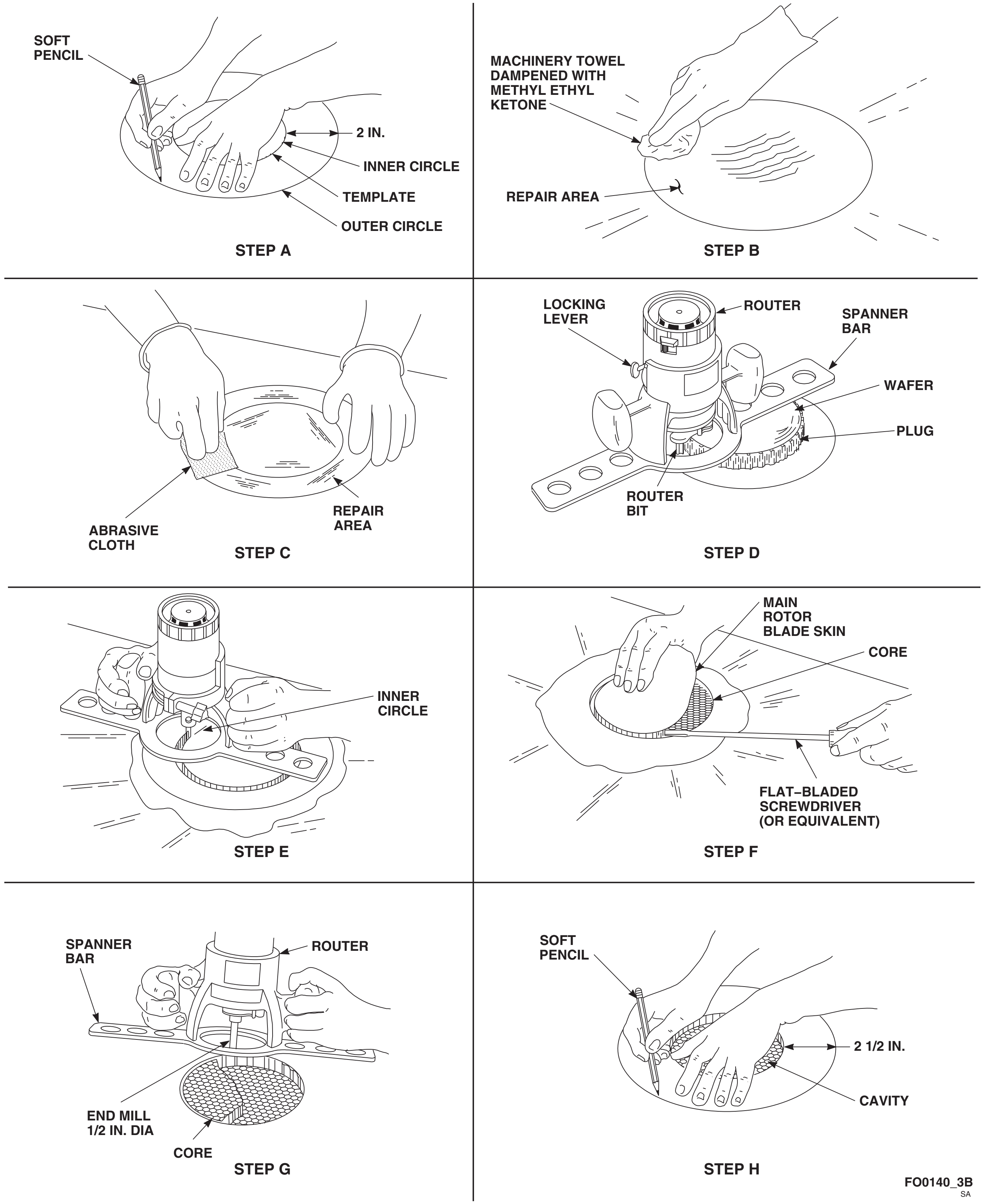


FIGURE 5-4-31-7. PLUG PATCH REPAIR (SHEET 3 OF 5)

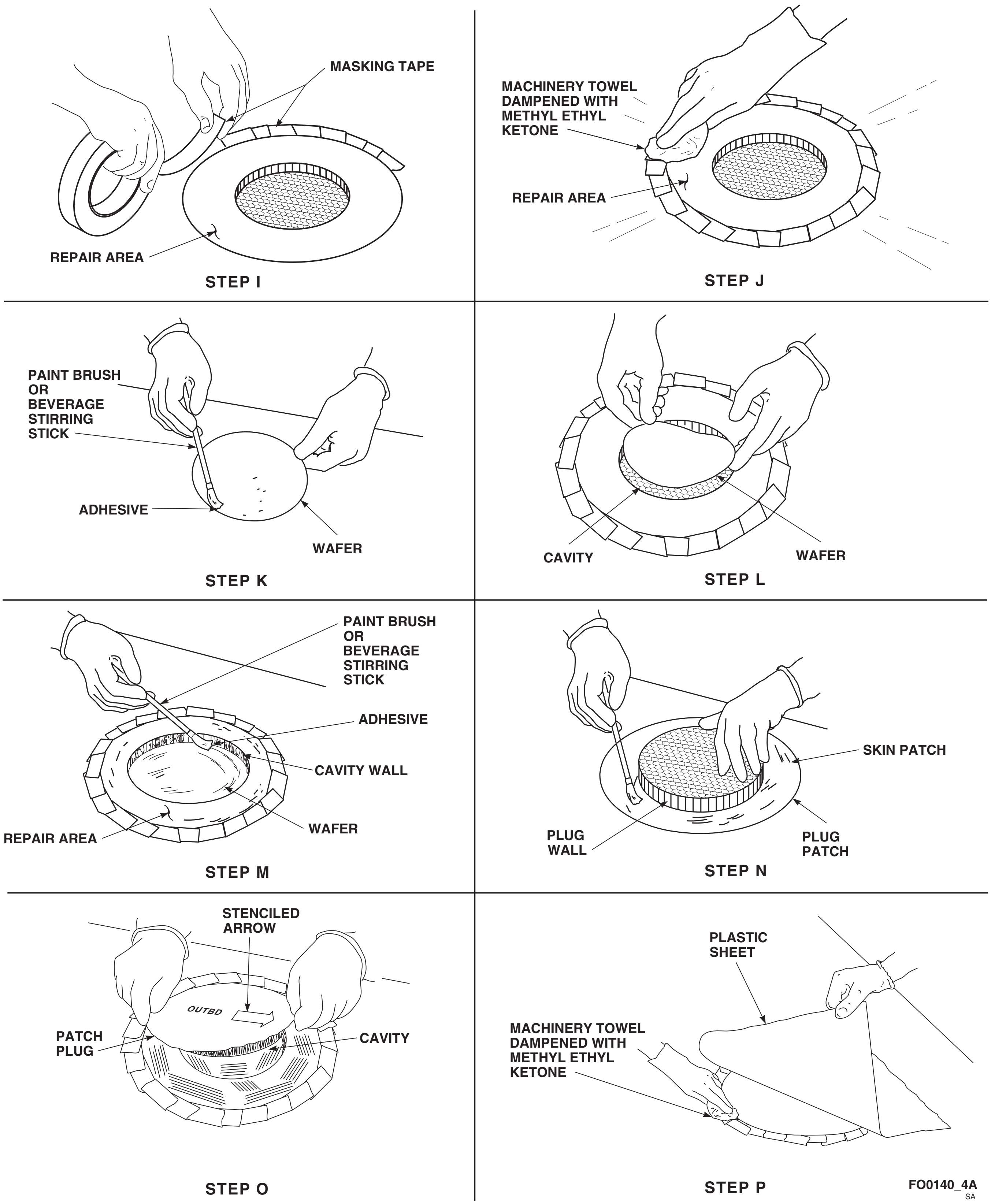


FIGURE 5-4-31-7. PLUG PATCH REPAIR (SHEET 4 OF 5)

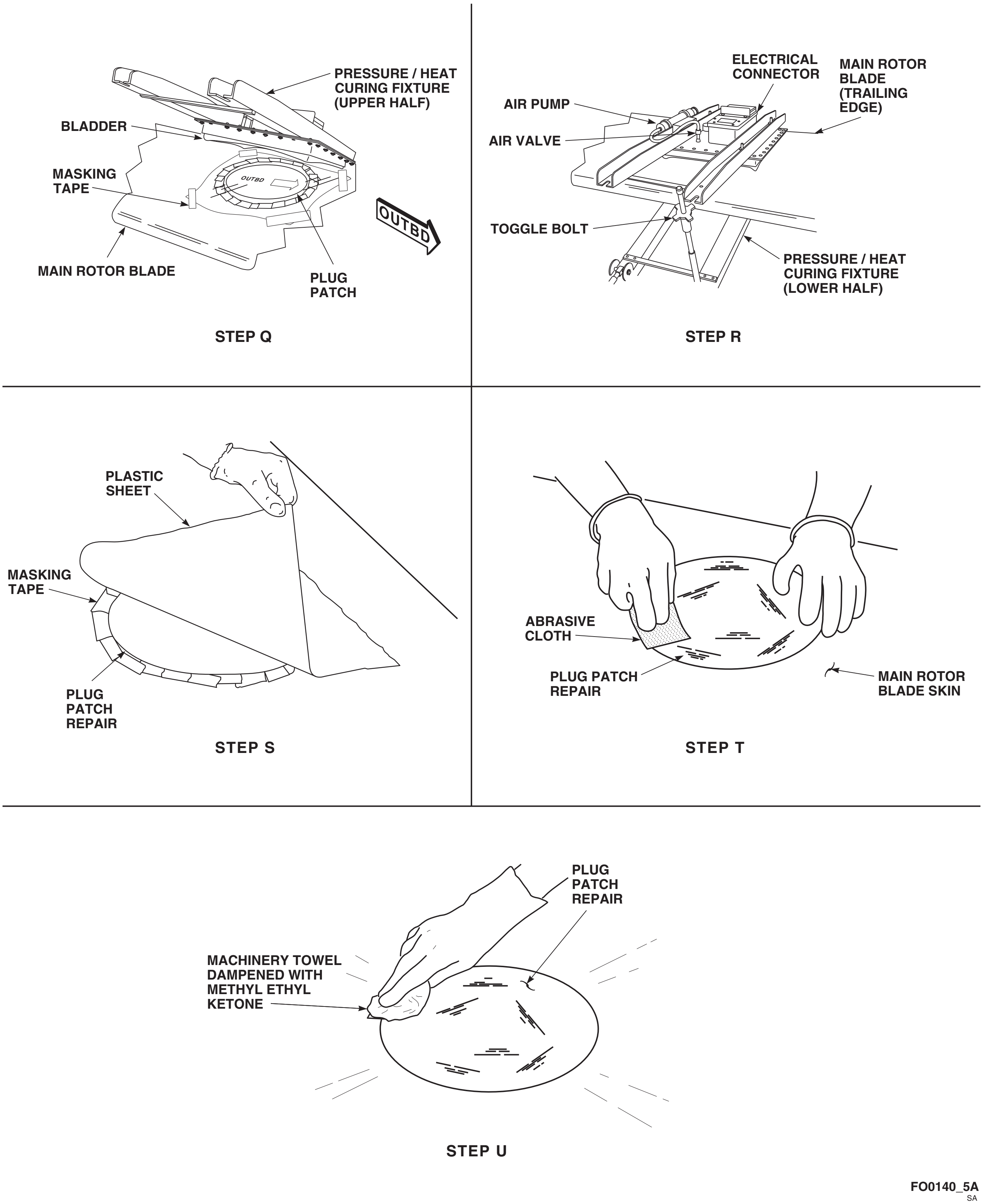


FIGURE 5-4-31-7. PLUG PATCH REPAIR (SHEET 5 OF 5)

c. Inner circle will be diameter of routed hole. Depth will be ¼-, ½-, or ⅞-inch, depending upon core thickness at point of damage. Limits in Table 5-4-31-4 show how close cavity depth can come to trailing edge without breaking through opposite skin.

Table 5-4-31-4. Plug Patch Repair

Plug Depth	Closest Distance to Trailing Edge
¼-inch	1 ½-inches
½-inch	3-inches
⅞-inch	5 ½-inches

CAUTION

Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints, laminate fibers, or core. Do not allow solvent to seep into bonded joints, laminate fibers, or core. Machinery towels wet with methyl ethyl ketone should not be left lying on main rotor blade.

d. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, remove paint from inside both circles/repair area (Figure 5-4-31-7, Sheet 3, Step B).

e. If necessary, using soft pencil, redraw inner circle (Figure 5-4-31-7, Sheet 3, Step A).

CAUTION

Damage to main rotor blade skin will result if main rotor blade is sanded excessively. Sand only until yellow primer is removed.

f. Using abrasive cloth, Item 7, Appendix D, sand yellow primer from repair area between both circles (Figure 5-4-31-7, Sheet 3, Step C).

NOTE

When patching both sides of main rotor blade, start with larger diameter repair.

g. To use plug and wafer as gage for setting router bit depth, perform the following (Figure 5-4-31-7, Sheet 3, Step D):

(1) Place plug patch on flat surface with plug facing up.

(2) Place wafer on plug.

(3) Set base of router on wafer with router bit along side of plug.

(4) Adjust spanner bar (router base) until router bit touches inner surface of plug.

(5) **TIGHTEN LOCKING LEVER TO LOCK SPANNER BAR (ROUTER BASE).**

h. If necessary, using soft pencil, redraw inner circle for routing operation (Figure 5-4-31-7, Sheet 3, Step A).

i. Insert router bit in router collet and set depth of cut to match depth of required plug patch (including wafer).

j. Using router, cut around damage as follows:

(1) Start router and hold with both hands.

(2) Rest edge of spanner bar (router base) on main rotor blade, with router bit parallel to main rotor blade skin.

(3) Tilt router forward towards vertical, bringing router bit in contact with main rotor blade skin near center of inner circle. Keep spanner bar (router base) in contact with main rotor blade skin.

(4) Move router slowly towards pencil line of inner circle, then carefully follow line

until complete circle is cut in main rotor blade skin (Figure 5-4-31-7, Sheet 3, Step E).

(5) Shut off router and remove from cut.

k. Using flat-bladed screwdriver (or equivalent), lift edge of cut main rotor blade skin, inside circle, and peel from core (Figure 5-4-31-7, Sheet 3, Step F).

l. Using same depth of cut, insert ½-inch diameter end mill in router, and route out damage as follows:

(1) Start router and hold with both hands.

(2) Reinsert router and with spanner bar (router base) spanning cut, route out core material using back-and-forth motion (Figure 5-4-31-7, Sheet 3, Step G).

(3) Shut off router and remove from cavity.

NOTE

To make sure plug patch will fit routed hole, check diameters of plug and hole.

m. Clean up debris in cavity. Using vacuum, remove dust and small particles.

NOTE

Template for 3-inch plug has 5 ½-inch outside diameter (OD), and template for 6-inch plug has 8 ½-inch OD.

n. Using soft pencil, draw circle on main rotor blade skin 2 ½-inches larger than cavity (Figure 5-4-31-7, Sheet 3, Step H).

o. Using 1-inch masking tape, Item 212, Appendix D, mask outside this circle/repair area (Figure 5-4-31-7, Sheet 4, Step I).

p. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean repair area to be bonded (Figure 5-4-31-7, Sheet 4, Step J). Wipe dry before methyl ethyl ketone evaporates.

q. Protect area to keep surface clean.

NOTE

- Adhesive, Item 32, Appendix D, is available in 32-gram, pre-measured packages or in bulk form. Either may be used. Prepare accordingly.
- Pot life of mixture is 25 minutes at 75°F (24°C).

r. Prepare adhesive, Item 32, Appendix D, either step r. (1) or step r. (2) as follows:

NOTE

Each 32-gram package is pre-measured with curing agent and resin. Mix full package and use as required. Discard leftover mixture.

(1) **32-Gram, Pre-Measured Package.**

(a) Empty package of adhesive resin into cup.

(b) Empty tube of curing agent into cup of adhesive resin.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

(2) **Bulk Adhesive.**

NOTE

Do not try to use all adhesive; use as required. Discard leftover mixture.

(a) Mix 100 parts of ″A″ part to 22 parts of ″B″ part by weight.

(b) Weigh each part and place both parts in cup.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

Table 5-4-31-5. Plug Patch Repair

Plug Diameter	Plug Depth	Quantity
3-inch	¼-inch	⅓ Pkg
	½-inch	⅓ Pkg
	⅞-inch	⅔ Pkg
6-inch	¼-inch	1 Pkg
	½-inch	1 ¼ Pkgs
	⅞-inch	2 Pkgs

s. Using clean, 1-inch paint brush, Item 86, Appendix D, or beverage stirring stick, Item 315C, Appendix D, apply liberal coat of mixed adhesive to one side of wafer (Figure 5-4-31-7, Sheet 4, Step K).

t. Place wafer in cavity with adhesive side down (Figure 5-4-31-7, Sheet 4, Step L), when repairing top side of main rotor blade, or place wafer on plug side of plug patch when repairing bottom side of main rotor blade.

CAUTION

Damage to main rotor blade will result if honeycombs are packed with adhesive. Added adhesive weight will affect main rotor blade balance. Do not pack honeycomb cells with adhesive.

u. Using same 1-inch paint brush, Item 86, Appendix D, or beverage stirring stick, Item 315C, Appendix D, apply heavy coat of adhesive, Item 32, Appendix D, to exposed side of wafer and cavity wall (Figure 5-4-31-7, Sheet 4, Step M). Apply thin, even coat of adhesive, Item 32, Appendix D, to repair area to be bonded.

v. Remove plug patch from plastic bag, and using same 1-inch paint brush, Item 86, Appendix D, or beverage stirring stick, Item 315C, Appendix D, apply light coat of adhesive, Item 32, Appendix D, to mating surfaces of skin patch and plug wall (Figure 5-4-31-7, Sheet 4, Step N). Do not apply adhesive to plug end.

w. Position plug patch in cavity with stenciled arrow pointing outboard (Figure 5-4-31-7, Sheet 4, Step O) and press firmly in place. With firm hand-pressure, work out air bubbles and extra adhesive from skin patch overlapping main rotor blade skin.

x. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, carefully lift edges of plastic sheet and wipe squeezed-out adhesive from around skin patch (Figure 5-4-31-7, Sheet 4, Step P).

y. Check plug patch for correct position, and using 1-inch masking tape, Item 212, Appendix D, tape down four corners to prevent movement (Figure 5-4-31-7, Sheet 5, Step Q).

z. Place pressure/heat curing fixture on main rotor blade so that bladder is over repair and hinges are toward trailing edge. Close pressure/heat curing fixture and engage toggle bolts. **TIGHTEN BOLTS UNTIL METAL SKIRT AROUND BLADDER IS ⅛-INCH FROM MAIN ROTOR BLADE SKIN** (Figure 5-4-31-7, Sheet 5, Step Q and Step R).

aa. Actuate air pump until gage reads 4 psi.

ab. Disconnect air pump hose from air valve.

ac. Connect pressure/heat curing fixture to 110 VAC electrical power source.

ad. For 3-inch plug, maintain heat and pressure for one hour.

ae. For 6-inch plug, maintain heat and pressure for one hour and 15 minutes.

af. Disconnect electrical source and press plunger in valve stem on top of pressure/heat curing fixture to release air pressure.

ag. Remove pressure/heat curing fixture from main rotor blade.

ah. Allow main rotor blade to cool to room temperature.

ai. Remove plastic sheet and masking tape from plug patch repair (Figure 5-4-31-7, Sheet 5, Step S).

CAUTION

Damage to main rotor blade skin will occur if main rotor blade is sanded excessively. Use only enough force to blend edge of plug patch into main rotor blade.

aj. Using abrasive cloth, Item 7, Appendix D, lightly sand edges of plug patch repair to blend repair to surrounding main rotor blade skin (Figure 5-4-31-7, Sheet 5, Step T).

ak. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean plug patch repair (Figure 5-4-31-7, Sheet 5, Step U). Wipe dry before methyl ethyl ketone evaporates.

NOTE

If lower side of main rotor blade requires plug patch (Figure 5-4-31-7, Sheet 2), repeat this procedure for lower side.

al. Touch up paint (PARA 5-4-31.20).

NOTE

If required, make sure all other skin patch, plug patch, and trailing edge patch repairs are performed prior to calculating and adjusting tip cap weight chordwise and spanwise unbalance.

am. Calculate and adjust tip block weight chordwise and spanwise unbalance (PARA 5-4-31.14).

5-4-31.6 Trailing Edge Patch Repair.

WARNING

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion-proof electrical equipment in areas where combustible vapors or materials may be present.
- Wash hands and face before eating or smoking.

NOTE

- Post NO SMOKING signs in area and keep materials away from spark or flame.
- This repair applies to repairing damage in the trailing edge region as shown in Figure 5-4-31-3, Sheet 1, Region C. For damage limits and repair procedures on main rotor blade, refer to Table 5-3-11-1 (PARA 5-3-11), and Figure 5-4-31-3, Sheet 1 and Sheet 2.
- This repair applies to the 3 × 5-inch V-shaped doubler patch (rotary wing blade repair kit, 70072-15001-021) bonded to both sides of trailing edge. Trailing edge patch must overlap damage, on all sides, by at least 1-inch (Figure 5-4-31-3, Sheet 1).
- Trailing edge patch repairs must not be closer than 5-inches from previously installed trailing edge patch repairs.
- Tab groove damage greater than 0.060-inch deep and up to 3-inches long spanwise can be repaired.
- Trailing edge gashes up to ½-inch chordwise (from trailing edge) and 1 ½-inches spanwise can be repaired.

- Skin tears no greater than 2-inches chordwise and 3-inches spanwise can be repaired.
- Trailing edge patch is made from piles of impregnated glass cloth fabric. Plastic sheet on the outer surface keeps adhesive from contacting pressure/heat curing fixture during curing. Patch is packaged in plastic to protect bonding surface.
- Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.
- Damaged main rotor blades exceeding repairable limits will be sent to manufacturer.

a. Turn main rotor blade to safe, comfortable working position, and support it to prevent movement or droop.

b. Position workstand and adjust to safe, comfortable working height.

c. Center kit template spanwise over damaged area, and using soft pencil, draw line around template on upper and lower surface of main rotor blade (Figure 5-4-31-8, Sheet 1, Step A).

CAUTION

Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints. Machinery towels wet with methyl ethyl ketone should not be left lying on main rotor blade.

d. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, remove paint from inside template guide lines/repair area (Figure 5-4-31-8, Sheet 1, Step B).

CAUTION

Damage to main rotor blade will result if main rotor blade is sanded excessively. Sand only until yellow primer is removed.

e. Using abrasive cloth, Item 7, Appendix D, sand yellow primer from inside repair area. Sand off damaged material that rises above main rotor blade contour. Do not sand undamaged main rotor blade skin fibers (Figure 5-4-31-8, Sheet 1, Step C).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

f. Using vacuum or low-pressure air (5 - 10 psi), clean up sanding dust and small particles.

g. Using same template, redraw guide lines if necessary (Figure 5-4-31-8, Sheet 1, Step A).

h. Using 1-inch masking tape, Item 212, Appendix D, mask around guide lines (Figure 5-4-31-8, Sheet 1, Step D).

i. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean repair area to be bonded (Figure 5-4-31-8, Sheet 1, Step E). Wipe dry before methyl ethyl ketone evaporates.

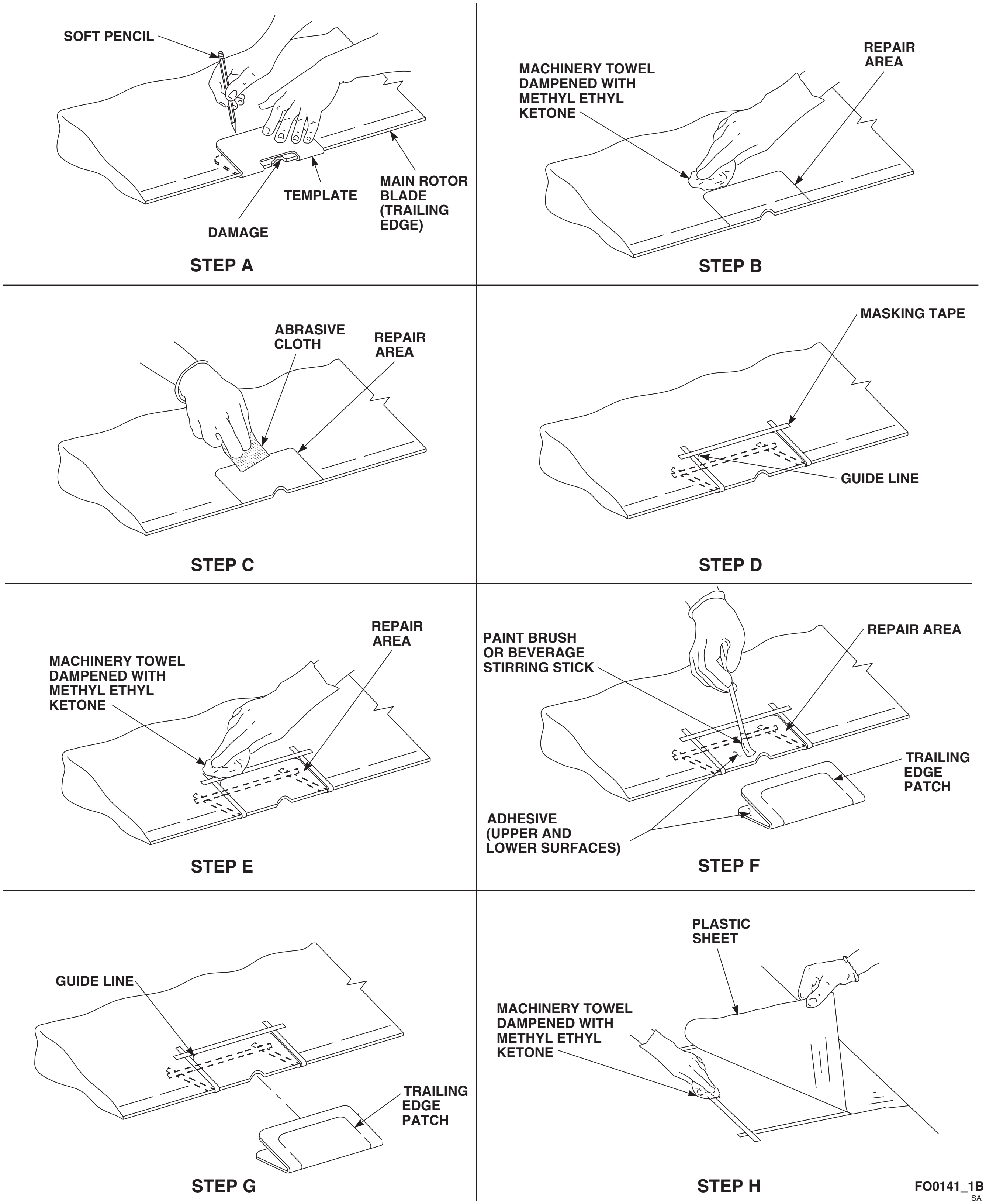


FIGURE 5-4-31-8. TRAILING EDGE PATCH REPAIR (SHEET 1 OF 2)

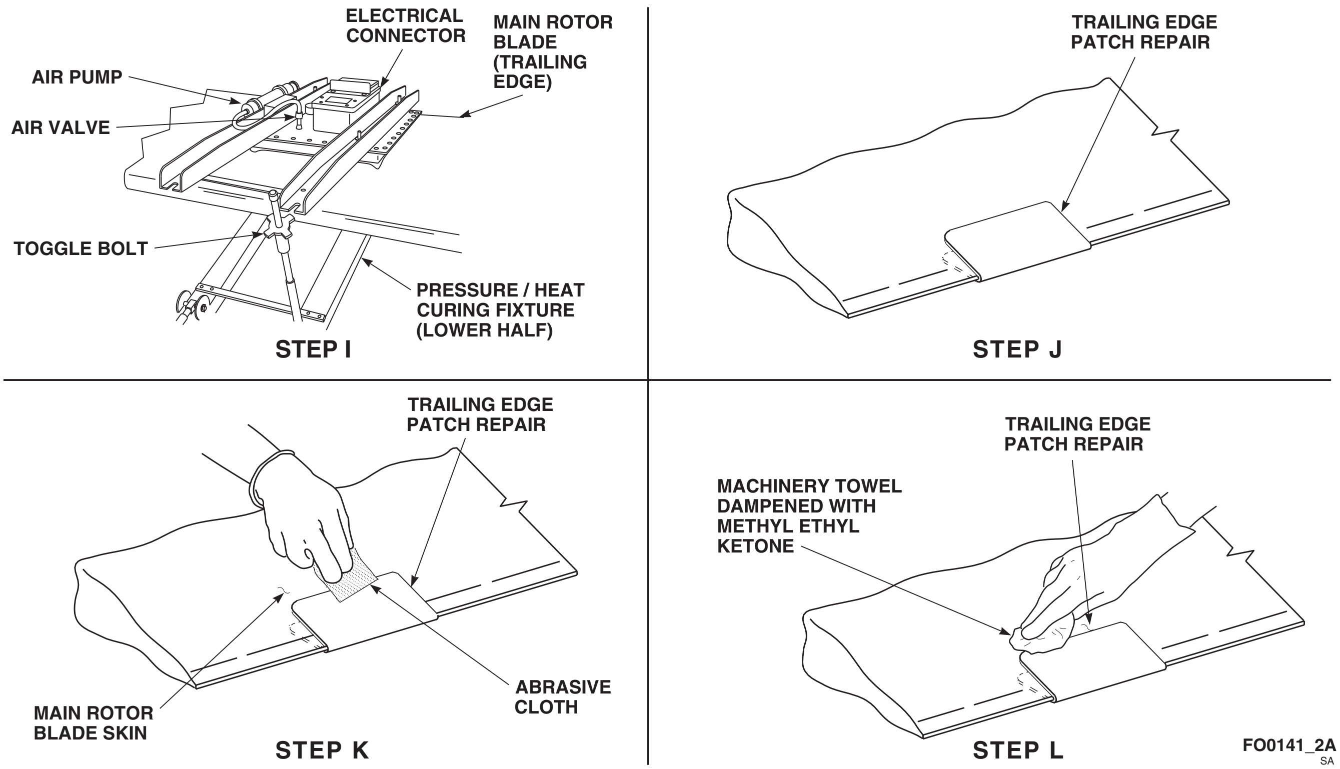


FIGURE 5-4-31-8. TRAILING EDGE PATCH REPAIR (SHEET 2 OF 2)

NOTE

- Adhesive, Item 32, Appendix D, is available in 32-gram, pre-measured packages or in bulk form. Either may be used. Prepare accordingly.
- Pot life of mixture is 25 minutes at 75°F (24°C).

j. Prepare adhesive, Item 32, Appendix D, either step j. (1) or step j. (2) as follows:

NOTE

Each 32-gram package is pre-measured with curing agent and resin. Mix full package and use as required. Discard leftover mixture.

(1) **32-Gram, Pre-Measured Package.**

(a) Empty package of adhesive resin into cup.

(b) Empty tube of curing agent into cup of adhesive resin.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

(2) **Bulk Adhesive.**

NOTE

Do not try to use all adhesive; use as required. Discard leftover mixture.

(a) Mix 100 parts of ″A″ part to 22 parts of ″B″ part by weight.

(b) Weigh each part and place both parts in cup.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

k. Using clean, 1-inch paint brush, Item 86, Appendix D, or beverage stirring stick, Item 315C, Appendix D, apply light, even coat of adhesive to area of trailing edge inside repair area and to inside of trailing edge patch (Figure 5-4-31-8, Sheet 1, Step F).

l. Center trailing edge patch within guide lines and press in place (Figure 5-4-31-8, Sheet 1, Step G). To seat trailing edge patch, move it back-and-forth with light hand pressure.

m. Push trailing edge patch firmly against trailing edge and work out air bubbles and extra adhesive, working from center to edge of trailing edge patch.

n. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, carefully lift edges of plastic sheet and wipe squeezed-out adhesive from around trailing edge patch (Figure 5-4-31-8, Sheet 1, Step H).

o. Check trailing edge patch for correct position, and using 1-inch masking tape, Item 212, Appendix D, tape down four corners to prevent movement.

p. Place pressure/heat curing fixture on main rotor blade so that bladder is over repair and hinges are toward trailing edge. Close pressure/heat curing fixture and engage toggle bolts. **TIGHTEN BOLTS UNTIL METAL SKIRT AROUND BLADDER IS 1/8-INCH FROM MAIN ROTOR BLADE SKIN** (Figure 5-4-31-8, Sheet 2, Step I).

q. Actuate air pump until gage reads 4 psi.

r. Disconnect air pump hose from air valve.

s. Connect pressure/heat curing fixture to 110 VAC electrical power source (thermostat will keep curing temperature of 160°F (71°C)).

t. Maintain heat and pressure for 1 hour.

u. Disconnect electrical source and press plunger in valve stem on top of pressure/heat curing fixture to release air pressure.

v. Remove pressure/heat curing fixture from main rotor blade.

w. Allow main rotor blade to cool to room temperature.

x. Remove plastic sheet and masking tape from trailing edge patch (Figure 5-4-31-8, Sheet 2, Step J).

CAUTION

Damage to main rotor blade will result if main rotor blade is sanded excessively. Sand only until yellow primer is removed.

y. Using abrasive cloth, Item 7, Appendix D, lightly sand edges of trailing edge patch repair to blend repair to surrounding main rotor blade skin (Figure 5-4-31-8, Sheet 2, Step K).

z. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean trailing edge patch repair (Figure 5-4-31-8, Sheet 2, Step L). Wipe dry before methyl ethyl ketone evaporates.

aa. Touch up paint (PARA 5-4-31.20).

NOTE

If required, make sure all other skin patch, plug patch, and trailing edge patch repairs are performed prior to calculating and adjusting tip cap weight chordwise and spanwise unbalance.

ab. Calculate and adjust tip block weight chordwise and spanwise unbalance (PARA 5-4-31.14).

5-4-31.7 Replace Tip Cap.

NOTE

Replacement of all main rotor blade tip caps is the same.

5-4-31.7.1 Remove.

a. Turn main rotor blade to safe, comfortable working position and support to prevent movement or droop.

b. Locate workstand and adjust to safe, comfortable, working height.

c. Remove screws from holes 1 through 6 (from leading edge), upper and lower, and store separately (Figure 5-4-31-9, Sheet 1, Sheet 2, and Sheet 3).

d. Remove remaining screws from holes 7 through 12 (rear), upper and lower, and store separately.

e. Using sharp nonmetallic scraper, scrape out old sealing compound from joint of tip cap and main rotor blade.

CAUTION

Damage to main rotor blade and tip cap will result if tip cap is pried off without removing sealing compound. Use care when using flexible metal spatula (or equivalent) to cut through sealing compound that bonds tip cap to tip block.

f. Remove tip cap. If necessary, using flexible metal spatula (or equivalent), cut through sealing compound that bonds tip cap to tip block.

5-4-31.7.2 Inspect.

a. Inspect interior lower airfoil skin of tip cap in area where spanwise weights are located for missing paint or indication of contact by spanwise weights. If evidence of contact is found, repair spanwise weights (PARA 5-4-31.15).

b. Inspect tip cap for damage (PARA 5-3-11).

c. Using screw extractor set (or equivalent), remove broken screws. Center punch broken screw before drilling screw extractor hole.

d. Inspect tip block inserts in first six holes (from leading edge) for damage (stripped, corroded, loose, or broken). If damaged, repair tip block inserts, MS124735 (PARA 5-4-31.17).

e. Inspect/repair tip block weights (PARA 5-4-31.15).

5-4-31.7.3 Install.

NOTE

Composite tip caps, 70150-09207-041, with radius at outboard trailing edge corner are acceptable. Sikorsky Aircraft Corporation incorporated an in-line change to the outboard trailing edge radius to prevent cracking and for ease of manufacturing. When installing composite tip caps, it is permissible for trailing edge of tip cap to be shorter than, or extend beyond, trailing edge of main rotor blade. **NO TRIMMING ALLOWED.**

a. Using nonmetallic scraper, scrape/clean old sealing compound from main rotor blade tip end and tip cap (if tip cap will be reused).

b. Using nonmetallic scraper, scrape/clean old adhesive from screws and attachment holes.

CAUTION

Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow methyl ethyl ketone to seep into bonded joints. Machinery towels wet with methyl ethyl ketone should not be left lying on main rotor blade.

c. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean area.

NOTE

If tip cap needs to be filed for proper fit, use fine cut file parallel to edge to prevent bending and cracking of abrasion strip (nickel).

d. Trial fit replacement tip cap on main rotor blade. If abrasion strip binds or does not fit, using fine cut file, file binding end so that tip cap will fit without putting pressure on abrasion strip. **EDGE OF ABRASION STRIP MUST BE WITHIN 0.005 - 0.090-INCH OF MAIN ROTOR BLADE AS LONG AS SEAL CAN BE MAINTAINED WITH SEALING COMPOUND.**

e. Using sealing compound, Item 290, Appendix D, apply bead to gap where tip cap butts to main rotor blade.

CAUTION

Different screws are used to secure tip cap to main rotor blade, 70150-09100-041 and 70150-09100-043. Refer to Figure 5-4-31-9, Sheet 2 and Sheet 3, before installing screws.

NOTE

Tip cap, 70150-09107-055, uses pan head screws, NAS7803P5, at holes 4 and 5 on lower surface of tip cap (Figure 5-4-31-9, Sheet 3, Detail C).

f. Position tip cap on main rotor blade and install screws, wet with sealing compound, Item 301, Appendix D, or primer, Item 258, Appendix D (Figure 5-4-31-9, Sheet 1, Detail A). Refer to Figure 5-4-31-9, Sheet 2, Detail B and Sheet 3, Detail B and Detail C, for part number of screws.

WARNING

FSCAP

Verification of tip cap screw torque is a critical characteristic.

g. **TORQUE SCREWS TO 29 - 31 INCH-POUNDS.**

h. Using nonmetallic scraper, fair sealing compound even with tip cap and main rotor blade surface to form a smooth transition. Wipe off excess sealing compound from around main rotor blade/tip cap joint.

i. Make entry in helicopter logbook showing tip cap screw torque stabilization check is shown due 9 - 11 flight hours after installation (PARA 1-7-4).

j. Perform main rotor blade track and balance (TM 1-70-VIB).

k. If tip cap abrasion strip was filed, perform the following:

(1) Using sealing compound, Item 296, Appendix D, apply to fill area to form flush contour.

(2) Place some sealing compound on a test specimen.

(3) Finish sealing tip cap-to-main rotor blade joint (PARA 5-4-31.8).

(4) Place test specimen of sealing compound next to repair. Repair sealing compound is cured when test specimen is cured.

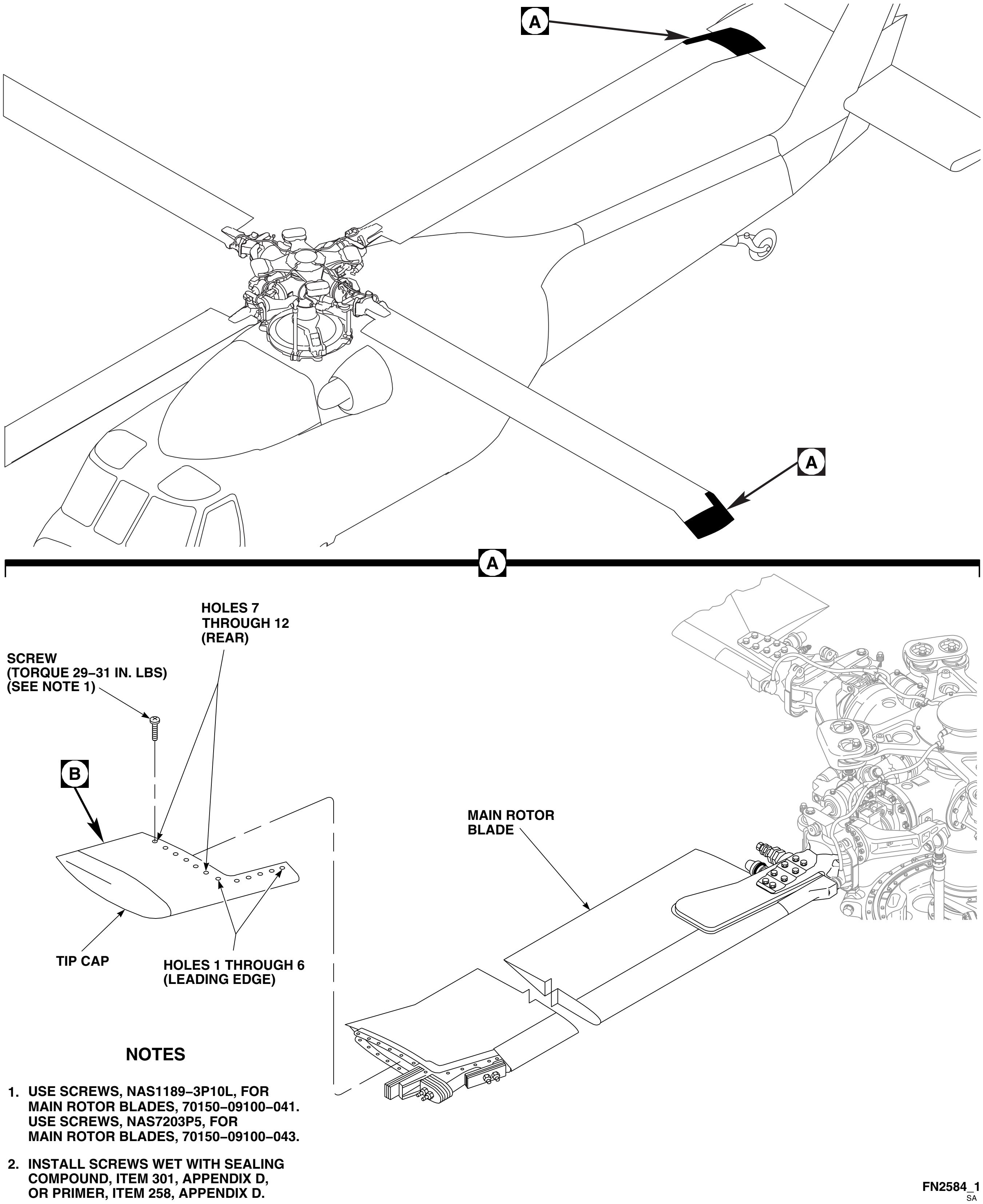
(5) After sealing compound has cured, sand filled area flush with contour.

(6) Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean repair area.

(7) Using primer coating, Item 259, Appendix D, spray area.

(8) Apply 10-inch long × 2-inch wide plastic strip, Item 237, Appendix D, centered over joint between tip cap and main rotor blade.

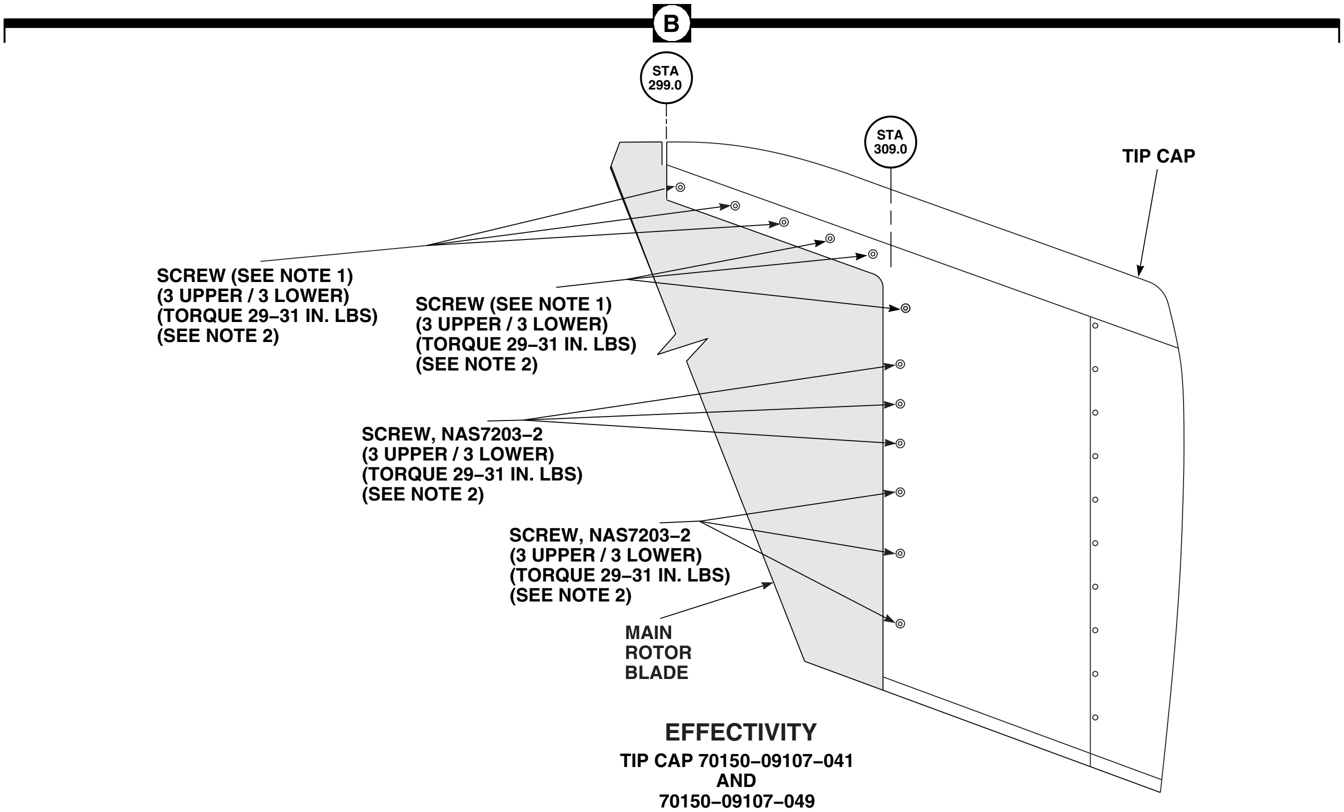
l. Touch up paint (PARA 5-4-31.20).



NOTES

1. USE SCREWS, NAS1189–3P10L, FOR MAIN ROTOR BLADES, 70150–09100–041. USE SCREWS, NAS7203P5, FOR MAIN ROTOR BLADES, 70150–09100–043.
2. INSTALL SCREWS WET WITH SEALING COMPOUND, ITEM 301, APPENDIX D, OR PRIMER, ITEM 258, APPENDIX D.

FIGURE 5-4-31-9. REPLACE TIP CAP (SHEET 1 OF 3)



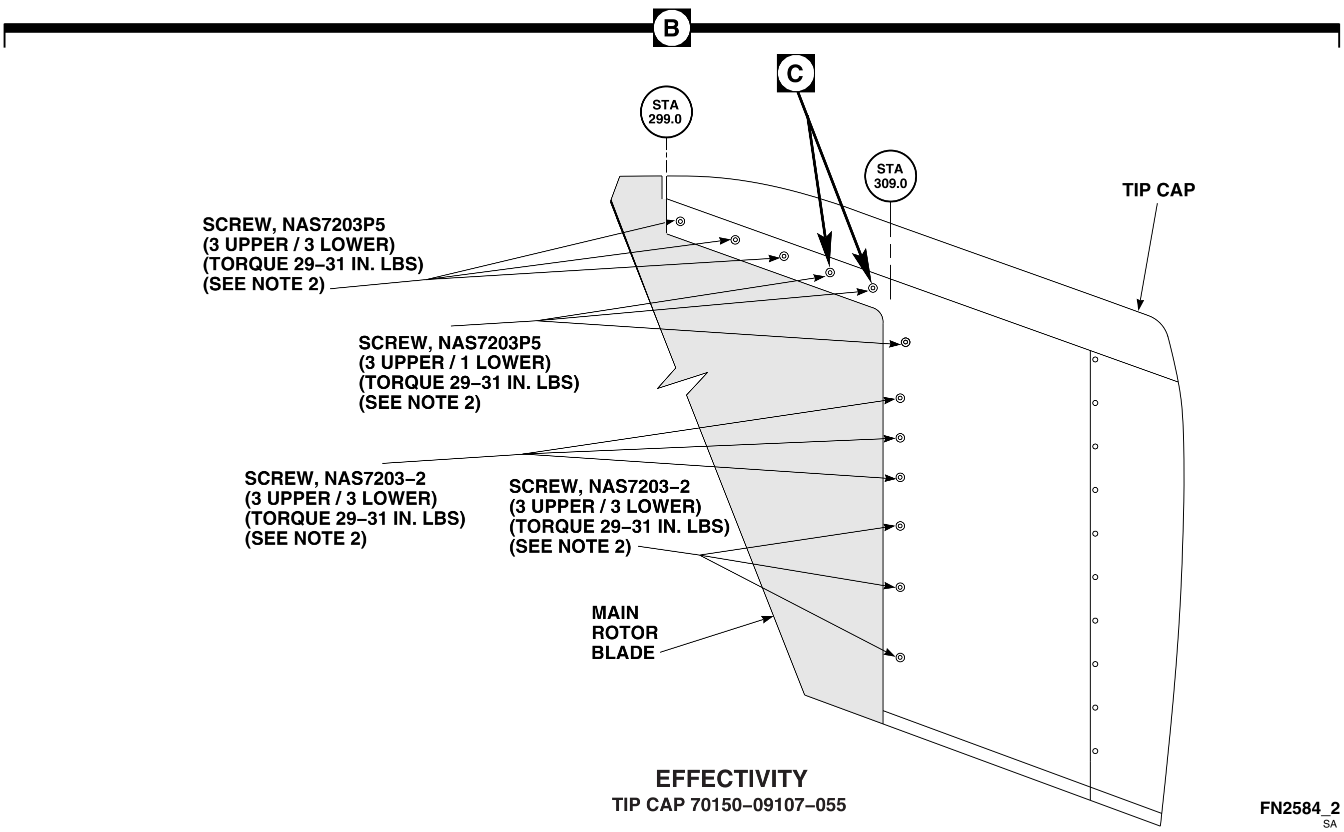
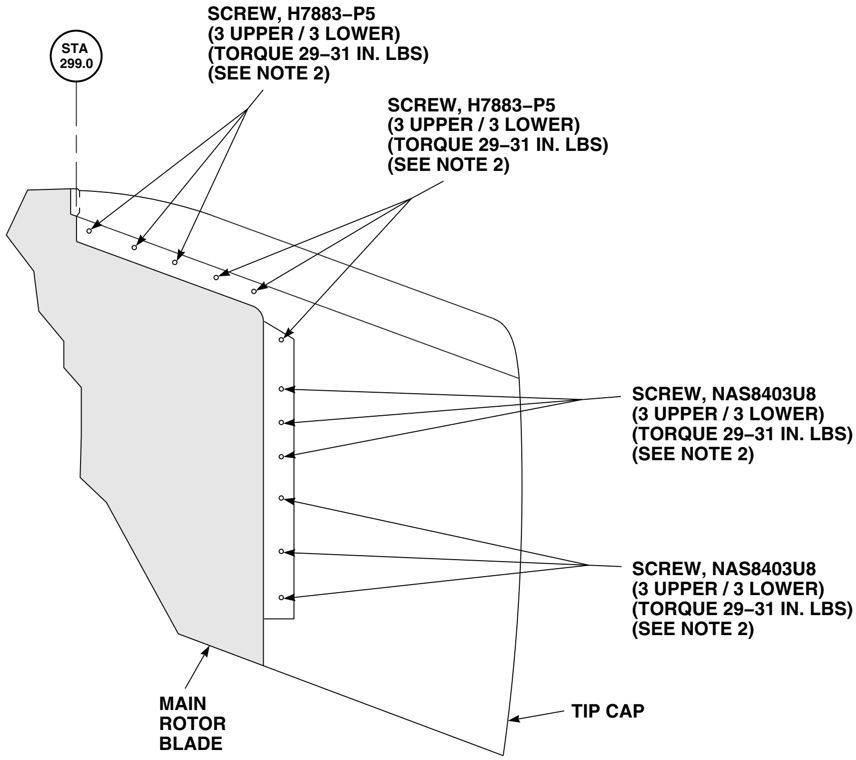


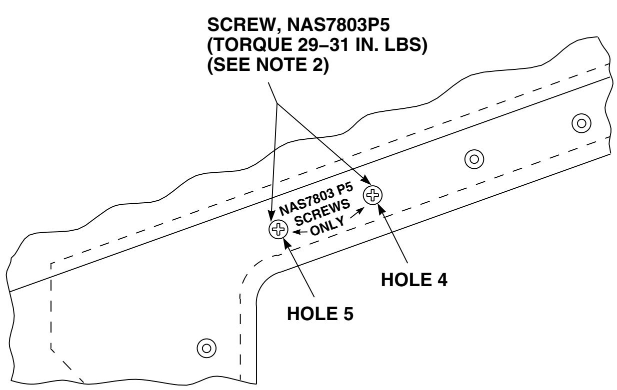
FIGURE 5-4-31-9. REPLACE TIP CAP (SHEET 2 OF 3)



EFFECTIVITY
TIP CAP 70150–09207–041
AND
70150–09207–045

B

C



LOWER SURFACE

FN2584_3
SA

FIGURE 5-4-31-9. REPLACE TIP CAP (SHEET 3 OF 3)

5-4-31.8 Tip Cap-to-Main Rotor Blade Sealing.

CAUTION

Main rotor blade will be damaged if solvent seeps into bonded joints or contact with reinforced plastic is prolonged. Bonded joints will separate or plastic will dissolve. Do not allow solvent to seep into bonded joints or remain in contact with reinforced plastic.

a. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean metal and reinforced plastic surfaces thoroughly.

b. Using sealing compound, Item 296, Appendix D, apply to main rotor blade/tip cap joint (Figure 5-4-31-10, Detail A, Detail B, and Detail C).

c. Using nonmetallic scraper, fair sealing compound even with tip cap and main rotor blade surface to form a smooth transition. Wipe off excess sealing compound from around main rotor blade/tip cap joint.

d. Place test specimen of sealing compound next to repair. Repair sealing compound is cured when test specimen is cured.

e. Touch up paint (PARA 5-4-31.20).

f. Perform main rotor blade track and balance (TM 1-70-VIB).

5-4-31.9 Bond Repair.

WARNING

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion-proof electrical equipment in areas where combustible vapors or materials may be present.
- Wash hands and face before eating or smoking.
- Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

CAUTION

Main rotor blade will be damaged if solvent seeps into bonded joints or contact with reinforced plastic is prolonged. Bonded joints will separate or plastic will dissolve. Do not allow solvent to seep into bonded joints or remain in contact with reinforced plastic.

NOTE

- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.

a. Clean bond separation by first wetting separation with methyl ethyl ketone, Item 217, Appendix D. While wet, blow out with clean, filtered, forced air (5 - 10 psi).

b. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, carefully insert

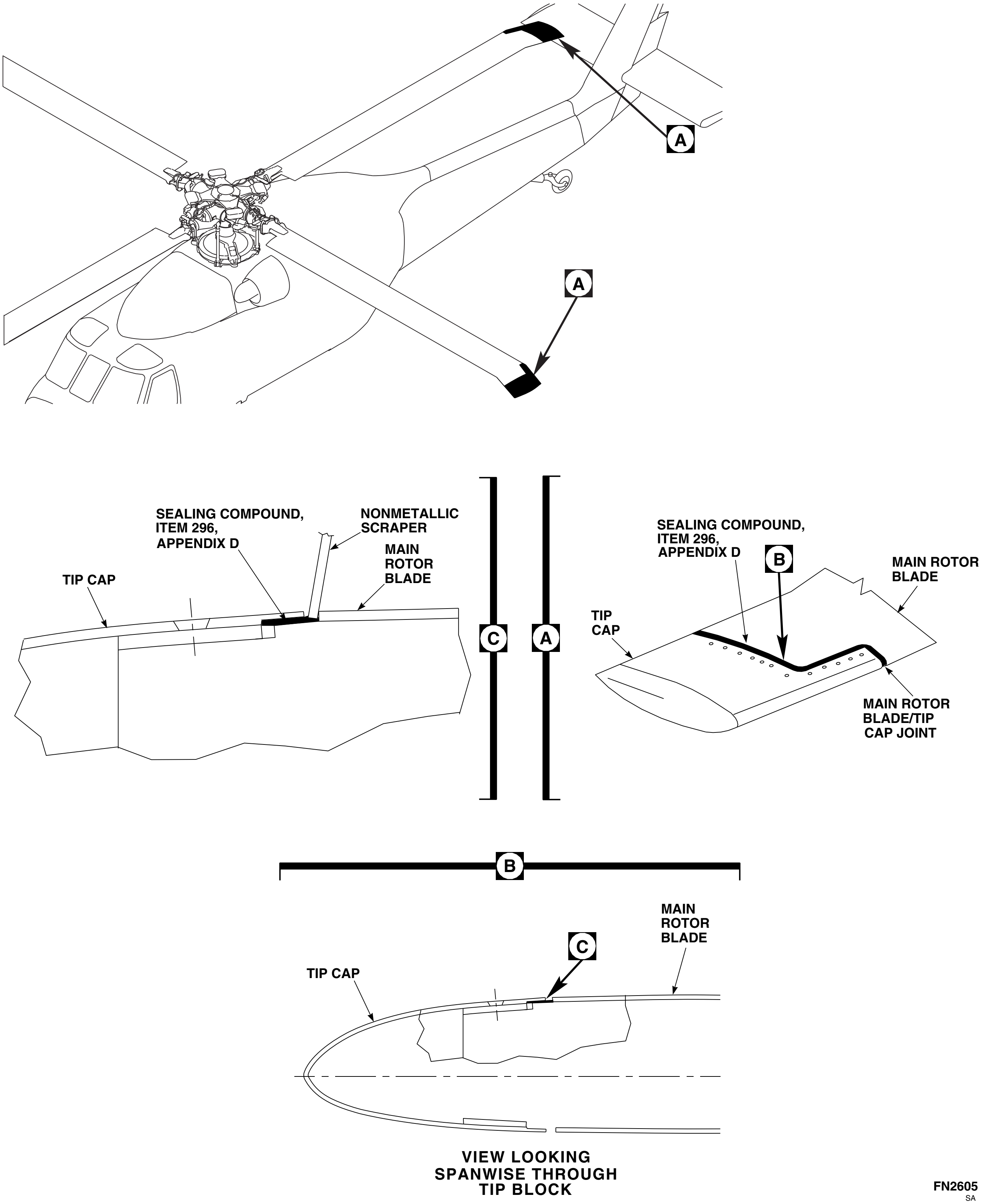


FIGURE 5-4-31-10. TIP CAP-TO-MAIN ROTOR BLADE SEALING between separation. Clean with back-and-forth motion. Change machinery towel, as required, until machinery towel remains clean.

NOTE

- Adhesive, Item 32, Appendix D, is available in 32-gram, pre-measured packages or in bulk form. Either may be used. Prepare accordingly.
- Pot life of mixture is 25 minutes at 75°F (24°C).

c. Prepare adhesive, Item 32, Appendix D, either step c. (1) or step c. (2) as follows:

NOTE

Each 32-gram package is pre-measured with curing agent and resin. Mix full package and use as required. Discard leftover mixture.

(1) **32-Gram, Pre-Measured Package.**

(a) Empty package of adhesive resin into cup.

(b) Empty tube of curing agent into cup of adhesive resin.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

(2) **Bulk Adhesive.**

NOTE

Do not try to use all adhesive; use as required. Discard leftover mixture.

(a) Mix 100 parts of ″A″ part to 22 parts of ″B″ part by weight.

(b) Weigh each part then place both parts in cup.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

d. Using applicator, Item 78, Appendix D (or equivalent), apply mixed adhesive inside separation. Make sure both surfaces are coated.

e. Using C-clamps, bungee (elastic) cord, or tape, apply enough pressure to keep surfaces mated.

f. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean up squeezed-out adhesive.

g. Allow adhesive to cure for 24 hours at 75°F (24°C), or using heat lamps, 1 hour at 160°F (71°C).

h. Remove C-clamps, bungee (elastic) cord, or tape from repair.

i. Touch up paint (PARA 5-4-31.20).

5-4-31.10 Corrosion Removal.

WARNING

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion-proof electrical equipment in areas where combustible vapors or materials may be present.
- Wash hands and face before eating or smoking.

CAUTION

Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints. Machinery towels wet with methyl ethyl ketone should not be left lying on main rotor blade.

NOTE

Post NO SMOKING signs in area and keep materials away from spark or flame.

a. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, remove paint from corroded area.

b. Using abrasive cloth, Item 3 and Item 4, Appendix D, lightly sand aluminum surface until corrosion is removed.

c. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean repair area. Using clean machinery towel, Item 211, Appendix D, wipe dry.

d. Using corrosion resistant coating, Item 118, Appendix D, apply to reworked surface (PARA 2-6-5).

e. Allow chemical conversion material to air-dry.

f. Prime and paint reworked surface (PARA 5-4-31.20).

5-4-31.11 Tip Cap Holes Repair.

WARNING

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion-proof electrical equipment in areas where combustible vapors or materials may be present.
- Wash hands and face before eating or smoking.

NOTE

- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.

a. Using fine cut file or abrasive paper, Item 18, Appendix D, smooth edges of hole.

b. Using abrasive cloth, Item 5, Appendix D, polish edges.

NOTE

Holes with up to ½-inch diameter or length in tip cap, with no internal damage, may be patched using 2-inch diameter reinforced glass fabric patch.

c. Using 1-inch masking tape, Item 212, Appendix D, outline area around damage ¼-inch larger than patch size.

d. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean off paint inside masked area. Wipe dry before methyl ethyl ketone evaporates.

NOTE

- Adhesive, Item 32, Appendix D, is available in 32-gram, pre-measured packages or in bulk form. Either may be used. Prepare accordingly.
- Pot life of mixture is 25 minutes at 75°F (24°C).

e. Prepare adhesive, Item 32, Appendix D, either step f. (1) or step f. (2) as follows:

NOTE

Each 32-gram package is pre-measured with curing agent and resin. Mix full package and use as required. Discard leftover mixture.

(1) **32-Gram, Pre-Measured Package.**

(a) Empty package of adhesive resin into cup.

(b) Empty tube of curing agent into cup of adhesive resin.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

(2) **Bulk Adhesive.**

NOTE

Do not try to use all adhesive; use as required. Discard leftover mixture.

(a) Mix 100 parts of ″A″ part to 22 parts of ″B″ part by weight.

(b) Weigh each part then place both parts in cup.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

f. On clean smooth surface, lay up three 2-inch diameter plies of glass fabric saturated with mixed adhesive. Using squeegee, squeegee out extra adhesive.

g. Using light, even coat of adhesive, Item 32, Appendix D, apply on repair area inside masking tape.

h. Center reinforced glass fabric patch (prepared in step g.) over hole and press firmly into place.

i. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean up squeezed-out adhesive.

j. Using plastic sheet, Item 235A, Appendix D, cover reinforced glass fabric patch and place felt or rubber mat over plastic sheet.

k. Place shot bag or weights (or equivalent) on felt or rubber mat to keep reinforced glass fabric patch in good contact while curing.

l. Allow adhesive to cure for 24 hours at 75°F (24°C), or using heat lamps, 1 hour at 160°F (71°C).

m. Remove shot bag or weights (or equivalent) from felt or rubber mat and peel off plastic sheet.

n. Lightly sand reinforced glass fabric patch edges to blend with surrounding main rotor blade skin.

o. Remove masking tape.

p. Touch up paint (PARA 5-4-31.20).

5-4-31.12 Tip Cap Skin and Fairing Crack Repair.

NOTE

- Cracks up to 2-inches long in Kevlar fairing are patched.
- Cracks in aluminum skin up to 1-inch long are stop-drilled and patched.
- Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.

a. Using drill with No. 40 drill bit, stop-drill cracks in aluminum skin only.

NOTE

Repair procedure for tip cap skin and fairing cracks is the same as for the tip cap holes.

b. Prepare surface and patch over cracks with reinforced glass fabric patches (PARA 5-4-31.11).

5-4-31.13 Abrasion Strip Repair.

NOTE

Cracks or negligible voids in abrasion strips on main rotor blade and tip cap, within limits described in PARA 5-3-11, are sealed to prevent moisture from entering bond line and causing further damage.

a. Using adhesive, Item 32, Appendix D, apply smooth, even coat over crack or void (PARA 5-4-31.9).

b. Bond separations listed as repairable in PARA 5-3-11 are bonded using adhesive, Item 35, Appendix D, per PARA 5-4-31.9.

5-4-31.14 Tip Block Weight Chordwise and Spanwise Unbalance Calculation and Adjustment.

NOTE

- If required, make sure all other skin patch, plug patch, and trailing edge patch repairs are done prior to performing this procedure.
- Repairs inboard of zone 1 (RSTA 60.0) do not need to be recorded. Repairs in this area will not affect chordwise or spanwise balance.

a. After repairing main rotor blade, record repair in helicopter logbook on component log card (SA Form 7343-21) as follows:

(1) Refer to Figure 5-4-31-11 to find zone of repair.

(2) Refer to Table 5-4-31-6 to calculate chordwise weight unbalance, and Table 5-4-31-7 to calculate spanwise weight unbalance.

(3) If determined that no adjustment to tip block weights is required, unbalance calculation and adjustment is complete.

(4) If determined that adjustment to tip block weights is required, perform the following:

(a) Record date, description of damage, zone, part number of rotary wing blade repair kit, and chordwise and spanwise weight unbalances as shown in Table 5-4-31-8.

(b) Proceed to step b.

b. Add chordwise weight adjustment numbers for all repairs on main rotor blade (if tip block weights have been adjusted previously, add all numbers from last new chordwise weight unbalance). Round this number to nearest whole number. Record remaining fraction under new chordwise weight unbalance (Table 5-4-31-6).

c. Remove tip cap (PARA 5-4-31.7).

d. Remove nuts and washers securing front and rear chordwise weights from tip block studs (Figure 5-4-31-12, Detail A).

NOTE

If condition exists where there are not enough rear chordwise weights to be moved, the tracking and balancing step at the end of procedure will be the determining factor.

e. Move whole number (obtained from above) of tip block weights from rear chordwise studs (bolts), and install on front chordwise studs.

f. Add spanwise weight unbalances for all repairs on main rotor blade (if tip block weights have been adjusted previously, add all numbers from last new spanwise weight unbalance) (Table 5-4-31-7).

g. Remove nuts and washers securing tip block weights from spanwise studs.

NOTE

If condition exists where there are not enough spanwise weights to be removed, the tracking and balancing step at the end of procedure will be the determining factor.

h. Remove combination of weights from spanwise studs whose weight comes closest to, but is not greater than total spanwise weight unbalances.

i. Record difference between weight removed and total spanwise weight unbalances as new spanwise weight unbalance.

j. To complete adjustment, inspect, and if required, repair tip block weights (PARA 5-4-31.15).

NOTE

New unbalance numbers must be added to future repairs and balances.

k. Install tip cap (PARA 5-4-31.7).

l. Perform main rotor blade track and balance (TM 1-70-VIB).

Table 5-4-31-6. Calculate Chordwise Weight Unbalance (Number of Rear Chordwise Weights to be Moved from Rear Chordwise Studs to Front Chordwise Studs)

Spanwise Chordwise	0 - 6-inch A	6 - 9-inch B	9 - 12-inch C	12 - 15-inch D	Rotary Wing Blade Repair Kit, 70072-15001
Zone 1 RSTA 60.0 - 140.0 (See Note)	0.4	0.3	0.2	0.1	-014, -020
	0.3	0.2	0.2	0.1	-018, -019
	0.2	0.1	0.1	0.0	-013, -015, -016, -017
	0.2	-	-	-	-021
	0.1	0.0	0.0	0.0	-012
Zone 2 RSTA 140.0 - 200.0	0.7	0.6	0.4	0.2	-014, -020
	0.5	0.4	0.3	0.1	-018, -019
	0.3	0.2	0.1	0.1	-013, -015, -016, -017
	0.3	-	-	-	-021

Table 5-4-31-6. Calculate Chordwise Weight Unbalance (Number of Rear Chordwise Weights to be Moved from Rear Chordwise Studs to Front Chordwise Studs) (Cont)

Spanwise Chordwise	0 - 6-inch A	6 - 9-inch B	9 - 12-inch C	12 - 15-inch D	Rotary Wing Blade Repair Kit, 70072-15001
	0.1	0.1	0.0	0.0	-012
Zone 3 RSTA 200.0 - 260.0	0.9	0.8	0.5	0.2	-014, -020
	0.7	0.6	0.4	0.1	-018, -019
	0.4	0.3	0.2	0.1	-013, -015, -016, -017
	0.4	-	-	-	-021
	0.1	0.1	0.1	0.0	-012
Zone 4 RSTA 260.0 - 300.0	1.2	0.9	0.6	0.3	-014, -020
	0.8	0.7	0.4	0.2	-018, -019
	0.5	0.3	0.2	0.1	-013, -015, -016, -017
	0.5	-	-	-	-021
	0.2	0.1	0.1	0.0	-012

Example 1. Chordwise Weight Calculation (with 1 Repair)

Given Repair: Zone 3C and Kit No. -018

Find: Chordwise Weight Unbalance = 0.4 weight

Round (to nearest whole number): In this case, round down 0.4 weight = 0.0 weight

Action: No action necessary.

Example 2. Chordwise Weight Calculation (with Multiple Repairs)

Table 5-4-31-6. Calculate Chordwise Weight Unbalance (Number of Rear Chordwise Weights to be Moved from Rear Chordwise Studs to Front Chordwise Studs) (Cont)

Spanwise Chordwise	0 - 6-inch A	6 - 9-inch B	9 - 12-inch C	12 - 15-inch D	Rotary Wing Blade Repair Kit, 70072-15001

Given Repairs: Zone 1A and Kit No. -020
Zone 2B and Kit No. -013
Zone 3A and Kit No. -014

Find: Chordwise Weight Unbalance = 0.4 + 0.2 + 0.9 weight = 1.5 weights

Round (to nearest whole number): In this case, round up 1.5 weights = 2.0 weights

Action: Remove 2.0 rear chordwise weights from rear chordwise studs and add to front chordwise studs.

NOTE

Repairs inboard of zone 1 (RSTA 60.0) do not need to be recorded. Repairs in this area will not affect chordwise or spanwise balance.

Table 5-4-31-7. Calculate Spanwise Weight Unbalance (Amount of Spanwise Weight to be Removed from Spanwise Studs due to Skin, Plug, or Trailing Edge Patch Repairs)

Zone Kit No.	-014, -020	-018, -019	-013, -015, -016, -017, -021	-012
Zone 1 RSTA 60.0 - 140.0 (See Note)	0.036 lb	0.026 lb	0.013 lb	0.004 lb
Zone 2 RSTA 140.0 - 200.0	0.061 lb	0.044 lb	0.022 lb	0.007 lb
Zone 3 RSTA 200.0 - 260.0	0.083 lb	0.060 lb	0.030 lb	0.010 lb
Zone 4 RSTA 260.0 - 300.0	0.101 lb	0.073 lb	0.036 lb	0.012 lb

Example. Spanwise Weight Determination

Given Repair: Zone 3 and Kit No. -019

Find: Spanwise Weight Unbalance = 0.060 lb

Action: Remove 0.060 lb spanwise weights from spanwise studs.

NOTE

Repairs inboard of zone 1 (RSTA 60.0) do not need to be recorded. Repairs in this area will not affect chordwise or spanwise balance.

Table 5-4-31-8. Examples of Main Rotor Blade Damage Repair Historical Record Entries

Date	Description of Damage	Zone	Rotary Wing Blade Repair Kit, 70072-15001	Chordwise Weight Unbalance	Spanwise Weight Unbalance
4/19/91	Damage 1-inch × 5-inch × ½-inch deep (bottom)	3C	-019	+ 4 weights	+ 0.060 lb
8/21/91	Skin tear 5-inch long (top)	3A	-014	+ 9 weights	+ 0.083 lb
1/20/92	Skin disbond 1-inch × 4-inch (top)	4A	-014	+ 1.2 weights	+ 0.101 lb
1/20/92	Blade Tip Wgt. Adjustment Chordwise - 2 weight adjusted Spanwise -0.224 lb removed			New + 0.5 weights	New + 0.020 lb

5-4-31.15 Tip Block Weight Inspection/ Repair.

a. Remove tip cap (PARA 5-4-31.7).

b. Visually inspect tip block weights for discoloration (slight corrosion). Using abrasive cloth, Item 10, Appendix D, remove any discoloration.

c. Inspect tip block weights for corrosion pitting. **NONE ALLOWED.** If pitting is found, repair as follows:

NOTE

Under each nut securing tip block weights is one or more washers used for fine-tuning main rotor blade balance. Record exact number and location of each tip block weight, and exact number and location of washers under each nut, for reinstallation.

(1) Remove nuts, washers, and tip block weights from tip block (Figure 5-4-31-12, Detail A).

(2) Repair tip block weights for pitting as follows:

(a) Using abrasive cloth, Item 5, Appendix D, **REMOVE PITTING FROM TIP BLOCK WEIGHTS UP TO 0.005-INCH.**

(b) If pitting exceeds limits, replace tip block weights.

(c) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to tip block weights (PARA 5-4-31.20).

(3) Inspect tip block studs for pitting and repair using wire brush as follows:

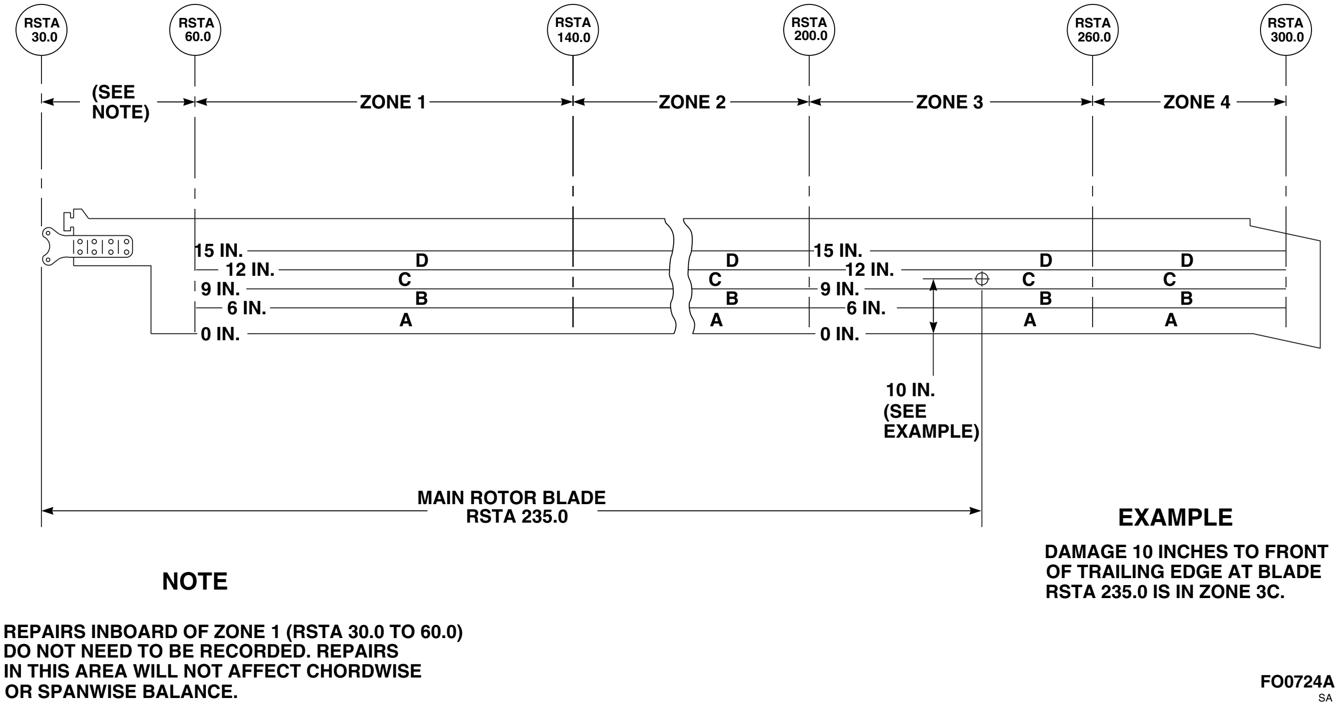


FIGURE 5-4-31-11. MAIN ROTOR BLADE REPAIR ZONE DETERMINATION

(a) **REMOVE PITTING FROM MINOR DIAMETER OF THREADS INBOARD OF RETENTION NUTS UP TO 0.005-INCH DEEP.**

(b) **REMOVE PITTING FROM MAJOR DIAMETER OF THREADS UP TO 0.010-INCH DEEP.**

(c) If pitting exceeds limits, replace main rotor blade.

(d) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to repaired threads (PARA 5-4-31.20).

(4) Using wire brush, remove light corrosion from washers and nuts by brushing lightly.

(5) Using primer, Item 258, Appendix D, reprime tip block studs, bolts, nuts, washers, and tip block weights. While primer is still wet, install tip block weights, washers, and nuts in exact location and stack up as recorded when removed.

(6) **TORQUE NUTS ON FRONT AND REAR CHORDWISE WEIGHTS TO 156 - 173 INCH-POUNDS.**

(7) **WHILE PUSHING SPANWISE WEIGHTS TOWARD TOP OF MAIN ROTOR BLADE, TORQUE NUTS TO 262 - 288 INCH-POUNDS.**

d. Using straightedge, check spanwise weights for protrusion below bottom edge of tip block (Figure 5-4-31-12, Detail B and Detail C). If spanwise weights protrude below bottom edge of tip block, perform the following:

(1) Loosen nuts securing spanwise weights.

(2) Using straightedge, push spanwise weights toward top of main rotor blade and check for protrusion below bottom edge of tip block.

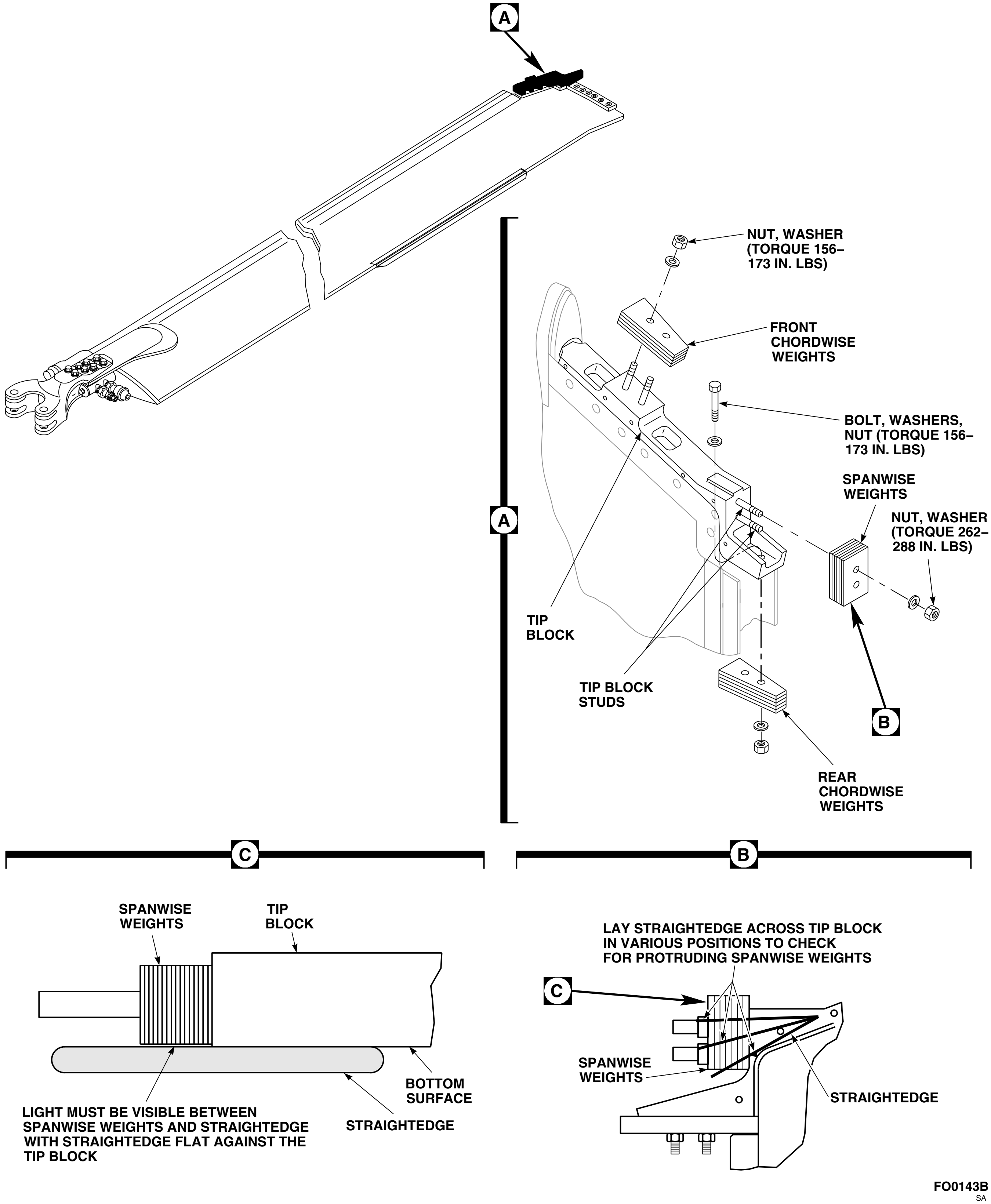


FIGURE 5-4-31-12. INSPECT/REPAIR TIP BLOCK WEIGHT

(3) If spanwise weights cannot be positioned to eliminate protrusion below bottom edge of tip block, perform the following:

(a) Mark spanwise weights which protrude below bottom edge of tip block.

(b) Remove nuts, washers, and spanwise weights from main rotor blade.

(c) Using medium cut file, remove material from protruding spanwise weights.

(d) Install spanwise weights on main rotor blade, push weights toward top of main rotor blade, and using straightedge, check for protrusion below bottom edge of tip block.

(e) Repeat step d. (3) (c) and step d. (3) (d) until all protruding spanwise weights are eliminated.

(4) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to reworked areas of spanwise weights.

(5) Install spanwise weights, washers, and nuts in exact location and stackup as when removed. **WHILE PUSHING SPANWISE WEIGHTS TOWARD TOP OF MAIN ROTOR BLADE, TORQUE NUTS TO 262 - 288 INCH-POUNDS.**

e. Install tip cap (PARA 5-4-31.7).

5-4-31.16 Tip Cap Rib Cracks Repair.

NOTE

Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.

a. Remove tip cap (PARA 5-4-31.7).

NOTE

Repairs on tip cap ribs will not affect weight or balance of tip cap.

b. Clip tip cap ribs as shown in Figure 5-4-31-13.

c. Using primer, Item 258, Appendix D, touch up exposed clipped edges.

d. Install tip cap (PARA 5-4-31.7).

5-4-31.17 Tip Block Insert, RZA12487, Repair.

a. Remove tip cap (PARA 5-4-31.7).

b. Thread pilot from cutter into tip block insert and slide cutter over pilot (Figure 5-4-31-14, Step A). Cut serrations from top of tip block insert and lockring. Do not allow cutter to remove metal from bottom of lockring counterbore.

c. Remove cutter and pilot. Lightly tap screw extractor into tip block insert and unscrew tip block insert from threaded hole. Remainder of lockring will lift out as tip block insert is removed (Figure 5-4-31-14, Step B).

d. If lockring does not come out with tip block insert, remove by carefully prying out, or using center punch, collapsing it.

e. Clear threaded hole and threads of any foreign material and chips.

f. Using fingers only, start new tip block insert into threaded hole to make sure proper engagement of threads (Figure 5-4-31-14, Step C).

g. Using wrench, screw tip block insert into threaded hole until top is 0.005 - 0.010-inch below surface of tip block (Figure 5-4-31-14, Step D).

h. Place lockring over top of tip block insert. Line up lockring outer serrations with original serrations in sides of counterbore (Figure 5-4-31-14, Step E).

i. Using driving tool, drive lockring down flush with surface of tip block, or not exceeding 0.010-inch below surface (Figure 5-4-31-14, Step E and Step F).

j. Using primer, Item 258, Appendix D, apply coat to seal lockring, insert, and counterbore.

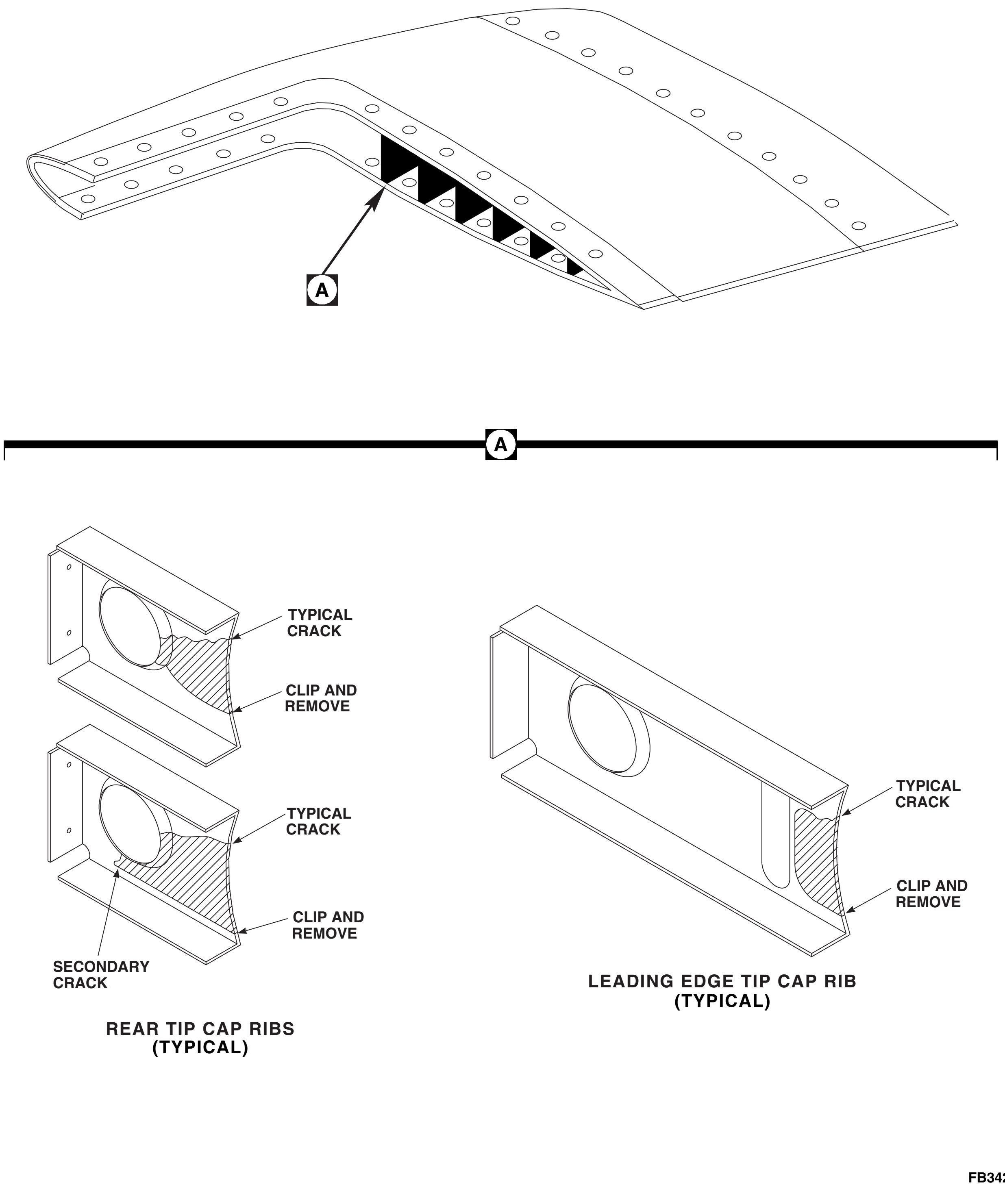


FIGURE 5-4-31-13. REPAIR TIP CAP RIB CRACKS

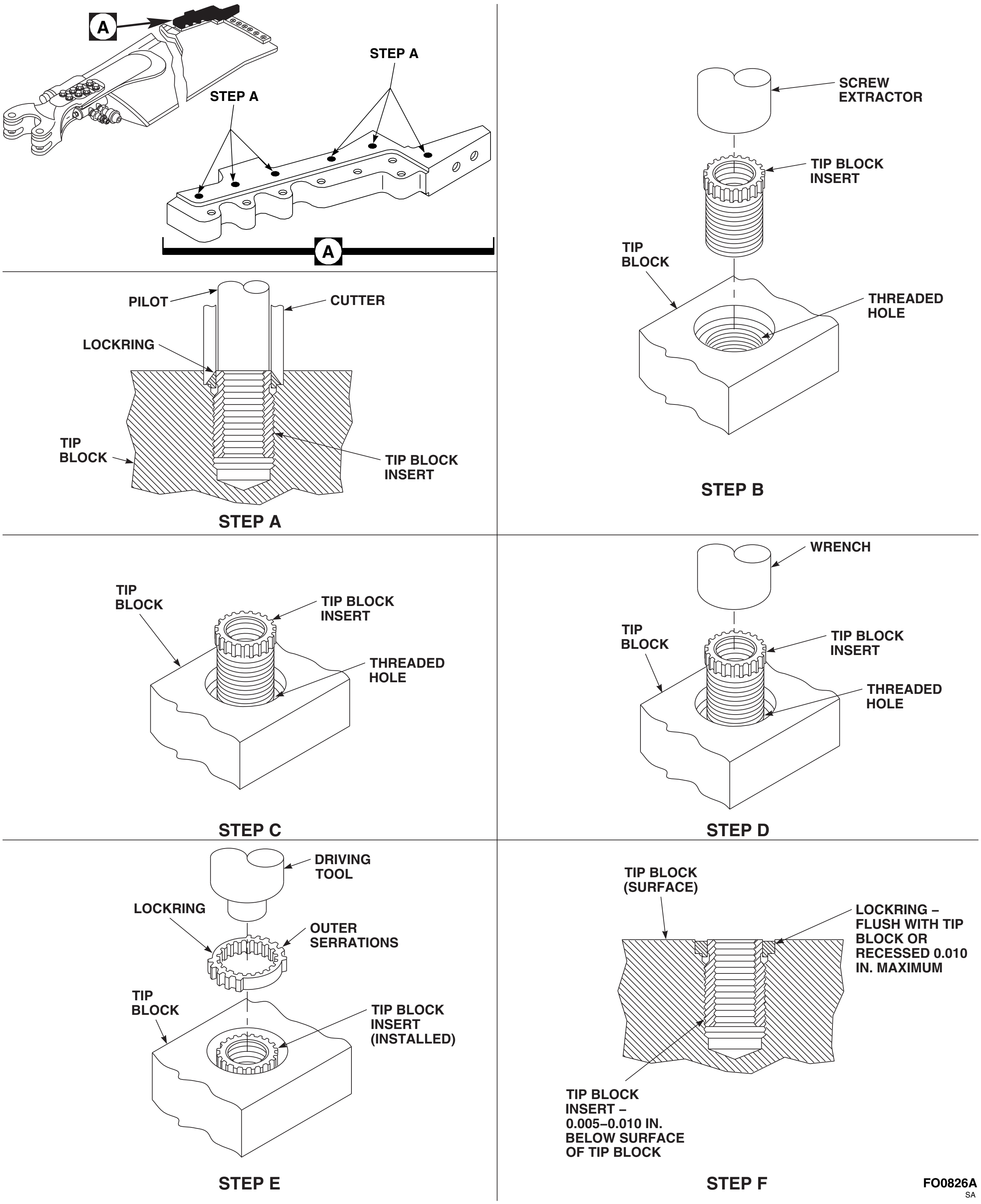


FIGURE 5-4-31-14. REPAIR TIP BLOCK INSERT, RZA12487

k. Install tip cap (PARA 5-4-31.7).

5-4-31.18 Aluminum Wire Mesh Lightning Protection Repair.

WARNING

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion-proof electrical equipment in areas where combustible vapors or materials may be present.
- Wash hands and face before eating or smoking.

NOTE

- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.

a. Using suitable masking tape, mask area around corrosion ½-inch larger than corrosion outline.

CAUTION

Damage to main rotor blade will result if main rotor blade is sanded excessively. Heavy sanding will damage main rotor blade. Do not sand through underlying fiberglass skin plies. Use only enough force to remove paint, primer and corrosion.

NOTE

- Sand only in immediate areas of pin holes.
- Up to 20% of chordwise width of aluminum wire mesh lightning protection may be removed without affecting lightning protection.

b. Sand away paint, primer, and corrosion.

CAUTION

Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints or inside of main rotor blade. Do not allow solvent to seep into bonded joints or inside of main rotor blade. Machinery towels wet with methyl ethyl ketone should not be left lying on main rotor blade.

c. Using clean cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean sanded area. Wipe dry before methyl ethyl ketone evaporates.

d. Using corrosion resistant coating, Item 118, Appendix D, treat exposed aluminum wire mesh lightning protection.

e. Using epoxy primer coating kit, Item 135, Appendix D, apply to repair area (PARA 5-4-31.20).

f. Using epoxy coating kit, Item 126A, Appendix D, paint repair area.

g. Using epoxy primer coating kit, Item 135, Appendix D, apply another light coat of epoxy primer (PARA 5-4-31.20).

h. Apply top coat of touchup paint (PARA 5-4-31.20).

5-4-31.19 Replace Polyurethane Patch.

NOTE

Replacement of all polyurethane patches is the same.

5-4-31.19.1 Remove.

a. Peel off old polyurethane patch from upper and lower leading edge surfaces of main rotor blade. If necessary, using nonmetallic scraper, remove any remaining polyurethane patch (Figure 5-4-31-15, Detail A).

b. Using cheesecloth, Item 90, Appendix D, dampened with acetone, Item 25, Appendix D, or methyl ethyl ketone, Item 217, Appendix D, clean remaining adhesive from upper and lower leading edge surfaces of main rotor blade.

5-4-31.19.2 Install.

CAUTION

Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints or laminate fibers. Cheesecloth wet with methyl ethyl ketone should not be left lying on main rotor blade.

a. If not previously performed, using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean upper and lower leading edge surfaces of main rotor blade.

b. Brush or spray adhesion promoter, Item 25A, Appendix D, onto upper and lower leading edge surfaces of main rotor blade.

c. Remove backing material from polyurethane patch.

d. Position polyurethane patch on root end of deice heater mat (Figure 5-4-31-15, Detail B).

e. With firm hand-pressure, work out all wrinkles and air bubbles from polyurethane patch to obtain good adhesion to main rotor blade surface.

5-4-31.20 Paint Touchup.

WARNING

FSCAP

- When performing paint touchup of main rotor blades, only apply maximum of one light coat to repaired area, or touchup paint with small brush. Repainting of the entire main rotor blade (even by dust-coating, and even if main rotor blade had large areas of erosion) will cause static imbalance causing one-per-revolution vibration, and is therefore not allowed. Following paint touchup, annotate on main rotor blade log card the requirement for a main rotor track and balance (TM 1-70-VIB).

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.

- Use properly grounded and explosion-proof electrical equipment in areas where combustible vapors or materials may be present.

- Wash hands and face before eating or smoking.

NOTE

Post NO SMOKING signs in area and keep materials away from spark or flame.

a. Refer to Table 5-4-31-9 for touchup paint requirements.

Table 5-4-31-9. Paint Touchup

Area or Details	Paint Color, Type and Specification	Number of Coats
Main Rotor Blade, overall	Epoxy Primer Coating Kit, Item 135, Appendix D	One
	Polyurethane Paint, Item 223, Appendix D	Two
Data Markings and Balance Stripes	Polyurethane Paint, Item 227, Appendix D	One
Blade Cuff Serial No. Etched Markings	Polyurethane Coating, Item 241D, Appendix D	Two

CAUTION

- Main rotor blade imbalance will result if excessive paint is applied to main rotor blade. Weight of paint, although slight, has large effects on main rotor blade balance. Restrict application of paint to avoid inducing balance problems.
- Removal of paint to allow for cosmetic repainting of main rotor blades is not allowed.
- Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints. Machinery towels wet with methyl ethyl ketone should not be left lying on main rotor blade.

NOTE

- Paint may be missing from main rotor blades due to erosion. There is no limit to the areas of erosion that can be reprimed and repainted if paint is missing.
- When repainting eroded or repaired areas of a main rotor blade, apply epoxy primer and paint to those areas only. Spot-spray or touch up with small brush. Apply just enough paint for good coverage.

b. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean surface to be painted.

CAUTION

Damage to main rotor blade will occur if main rotor blade is sanded excessively. Use only enough force to remove resin gloss.

c. Using abrasive paper, Item 19, Appendix D (or equivalent), lightly sand surface to be painted to remove resin gloss.

d. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, rewipe surface. Wipe dry before methyl ethyl ketone evaporates.

e. Using 1-inch masking tape, Item 212, Appendix D, and heavy paper (or equivalent protective covering), mask around touchup area to reduce cleanup.

NOTE

- Do not apply epoxy primer when temperature is below 50°F (10°C). Recommended temperature is 75° - 86°F (24° - 30°C). Lower temperature will cause longer drying time. Apply epoxy primer by brush or spray directly over cleaned reinforced plastic surfaces.

- Epoxy primer coating kit, Item 135, Appendix D, consists of two components packaged separately. Add component II to component I. Do not reverse mixing, or mix components from different manufacturers.

- Pot life of mixture is 8 hours at 75°F (24°C).

f. Using epoxy primer coating kit, Item 135, Appendix D, prepare epoxy primer as follows:

(1) Pour required amount of component I (base) into clean container.

(2) Add equal amount, by volume, of component II (converter).

(3) Stir mixture slowly and thoroughly for at least 10 minutes.

NOTE

Thinner is mixture of equal amounts of methyl isobutyl ketone, Item 218, Appendix D, and toluene, Item 339, Appendix D.

(4) Thin mixture for spraying by adding one part by volume of thinner to two parts by volume of mixed epoxy primer.

(5) Mix thoroughly, strain mixture, and allow to stand 1 hour before using. Stir again just before using.

g. Spray light even coat of epoxy primer on prepared surface.

h. Allow epoxy primer to cure for 1 hour at 75°F (24°C).

NOTE

- Minor feathering during repair painting into existing painted areas is acceptable.

- When repainting erosion damage on the forward edge of the main rotor blade, a 3-inch wide feathering into painted areas is acceptable on both upper and lower main rotor blade surfaces along entire length of main rotor blade. Maintain even coverage along main rotor blade to evenly distribute weight of paint.

i. Apply top coat of proper paint over epoxy primer. Refer to Table 5-4-31-9 for paint requirements.

j. Remove masking tape and heavy paper (or equivalent protective covering) from repair area.

k. Using abrasive cloth, Item 1A, Appendix D, lightly sand repair area to produce a uniform surface finish.

l. On main rotor blade log card, annotate requirement for main rotor track and balance (TM 1-70-VIB).

5-4-31.21 Cosmetic Adhesive Crack Repair.

CAUTION

Damage to main rotor blade will result if main rotor blade is sanded excessively. Heavy sanding will damage main rotor blade. Do not sand through underlying fiberglass skin plies. Use only enough force to remove paint, primer, and wire mesh.

NOTE

- Cosmetic cracking is limited to cracking of the adhesive and/or wire mesh only.

Cracking of skin substrate is not cosmetic and is not an acceptable condition for this repair. Any cracking of skin substrate must be reported to Sikorsky Engineering.

- If possible, minimize removal of wire mesh. Up to 20% of chordwise width of aluminum wire mesh may be removed without affecting lightning protection.

a. Using abrasive cloth, Item 4, Appendix D, sand away paint, primer, and aluminum wire mesh from repair area (Figure 5-4-31-3, Sheet 2, Detail A). Sand in a 45° direction to spar.

CAUTION

Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints. Do not leave cheesecloth, wet with methyl ethyl ketone, lying on main rotor blade.

b. Using clean cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean sanded area. Wipe dry before methyl ethyl ketone evaporates.

NOTE

Cut satin cloth so that direction of woven cloth strands will be at 45° angle to crack when applied.

c. Cut two pieces of satin cloth, Item 269A, Appendix D, as follows:

(1) Cut first piece to overlap adhesive crack by minimum of 1-inch in both spanwise and chordwise direction.

(2) Cut second piece 0.250 - 0.350-inch larger than first piece in both spanwise and chordwise directions.

d. Soak previously cut pieces of satin cloth in adhesive, Item 32 or Item 35, Appendix D.

NOTE

Make sure satin cloth overlaps adhesive crack minimum of 1-inch in both spanwise and chordwise direction.

e. Apply first (smaller) piece of adhesive soaked satin cloth over crack. Satin cloth must be applied with woven cloth (strands) positioned at 45° angle to crack line maintaining 1-inch overlap of crack.

NOTE

Make sure satin cloth is 0.250 - 0.350-inch larger than first piece.

f. Apply second (larger) piece of adhesive soaked satin cloth over first piece. Satin cloth must be applied with woven cloth (strands) positioned at a 45° angle to crack.

g. Allow adhesive to cure for minimum of 24 hours at 70° - 85°F (21° - 29°C). Cure time may be accelerated as follows:

(1) Using heat gun, cure adhesive, Item 32, Appendix D, for minimum of 1 hour at 140° - 160°F (60° - 71°C).

(2) Using heat gun, cure adhesive, Item 35, Appendix D, for minimum of 1 hour at 180° - 200°F (82° - 93°C).

NOTE

Do not excessively sand area. Sand only to remove excessive adhesive squeeze out.

h. Using abrasive cloth, Item 5, Appendix D, remove excessive adhesive squeezeout by lightly sanding area to smooth transition.

i. If more than 20% of chordwise width of aluminum wire mesh was removed, apply wire mesh patch as follows:

(1) Using abrasive cloth, Item 5, Appendix D, sand 0.500-inch minimum overlap area just enough to expose surrounding aluminum wire mesh.

CAUTION

Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints. Do not leave cheesecloth, wet with methyl ethyl ketone, lying on main rotor blade.

(2) Using clean cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean sanded area. Wipe dry before methyl ethyl ketone evaporates.

(3) Using corrosion resistant coating, Item 118, Appendix D, treat exposed aluminum wire mesh.

(4) Cut aluminum mesh screen, Item 73A, Appendix D, 0.500-inch larger than repair area.

(5) Place pressure sensitive adhesive tape, Item 244B, Appendix D, around repair area to limit excess adhesive squeezeout.

(6) Using adhesive, Item 32, Appendix D, apply to repair area and embed aluminum mesh screen in adhesive. If required, reapply adhesive to any aluminum mesh screen areas that are not fully saturated.

(7) Cover repair with plastic film peel ply.

(8) Using vacuum bag from fuel tank parts kit, Item 141B, Appendix D, tape, or mechanical means, apply pressure.

(9) Allow adhesive to cure for 24 hours at 70° - 85°F (21° - 29°C), or using heat gun, for 1 hour minimum at 140° - 160°F (60° - 71°C).

(10) Remove pressure (vacuum bag, tape, or mechanical means) and plastic film peel ply.

(11) Using abrasive cloth, Item 4, Appendix D, lightly scuff repair.

CAUTION

Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints. Do not leave cheesecloth, wet with methyl ethyl ketone, lying on main rotor blade.

(12) Using clean cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean repair area. Wipe dry before methyl ethyl ketone evaporates.

(13) Remove pressure sensitive adhesive tape from main rotor blade.

j. Using epoxy primer coating kit, Item 135, Appendix D, apply to repair area (PARA 5-4-31.20).

k. Using epoxy coating kit, Item 126A, Appendix D, paint repair area.

l. Using epoxy primer coating kit, Item 135, Appendix D, apply another light coat of epoxy primer (PARA 5-4-31.20).

m. Apply top coat of touchup paint (PARA 5-4-31.20).

5-4-31.22 Discrepant Root End Skin Plies Repair.

a. Procure or locally make chevron patch (Figure H-264, Appendix H).

CAUTION

Damage to main rotor blade will result if main rotor blade is sanded excessively. Heavy sanding will damage main rotor blade. Do not sand through underlying fiberglass skin plies. Use only enough force to remove paint, primer, and wire mesh.

b. Using abrasive cloth, Item 4, Appendix D, sand away paint, primer, and aluminum wire mesh lightning protection from repair area.

CAUTION

Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints. Do not leave cheesecloth, wet with methyl ethyl ketone, lying on main rotor blade.

c. Using clean cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean sanded area. Wipe dry before methyl ethyl ketone evaporates.

d. Prepare adhesive, Item 66F, Appendix D, per manufacturer's instructions.

NOTE

Use of scrim cloth in the bondline is optional.

e. Bond repair patch to blade using adhesive, Item 66F, Appendix D (Figure 5-4-31-16, Detail A).

f. Apply pressure to repaired plies by vacuum bag, clamping plates, or system of weights and pressure plates.

g. Allow adhesive to cure for 5 days at 60° - 80°F (16° - 27°C), or for 1 hour at 140° - 250°F (60° - 121°C) using heat gun or heat lamps.

h. Using coin-tapping method, inspect repair area for disbonds (PARA 5-3-11).

i. Install aluminum wire mesh lightning protection (PARA 5-4-31.21, step i. (1) through step i (13)).

j. Touch up paint, as required (PARA 5-4-31.20).

k. Mark main rotor blades ″RS-100-1″. Make sure blades are marked so that this number will not be added to part number by mistake.

l. Perform main rotor blade track and balance (TM 1-70-VIB).

m. Record repair on component log card (SA Form 7343-21).

5-4-31.23 Trailing Edge Root Skin Crack (RSTA 42.0 - 50.0) Repair.

CAUTION

Damage to main rotor blade will result if main rotor blade is sanded excessively. Heavy sanding will damage main rotor blade. Do not sand through underlying fiberglass skin plies. Use only enough force to remove paint, primer, and wire mesh.

a. Remove any existing repair patches and sealant from repair area.

b. Using abrasive cloth, Item 4, Appendix D, sand away paint, primer, and aluminum wire mesh lightning protection from repair area. Sand in 45° direction to spar.

c. Check cracked skin area for presence of water entrapped in underlying core. **NONE ALLOWED.** If water is present, remove by vacuum bagging and applying heat (up to 160°F (71°C)) until water is removed.

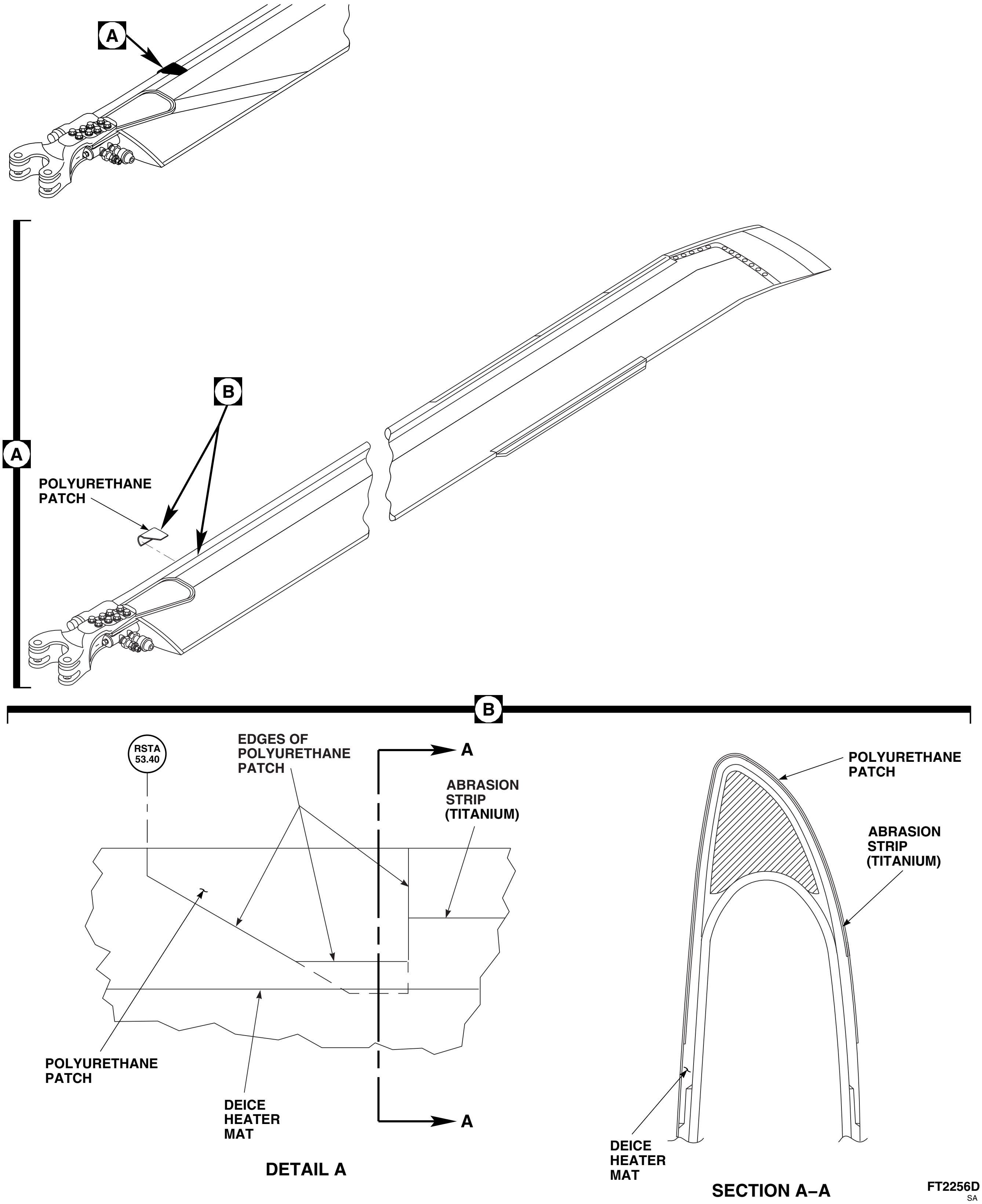
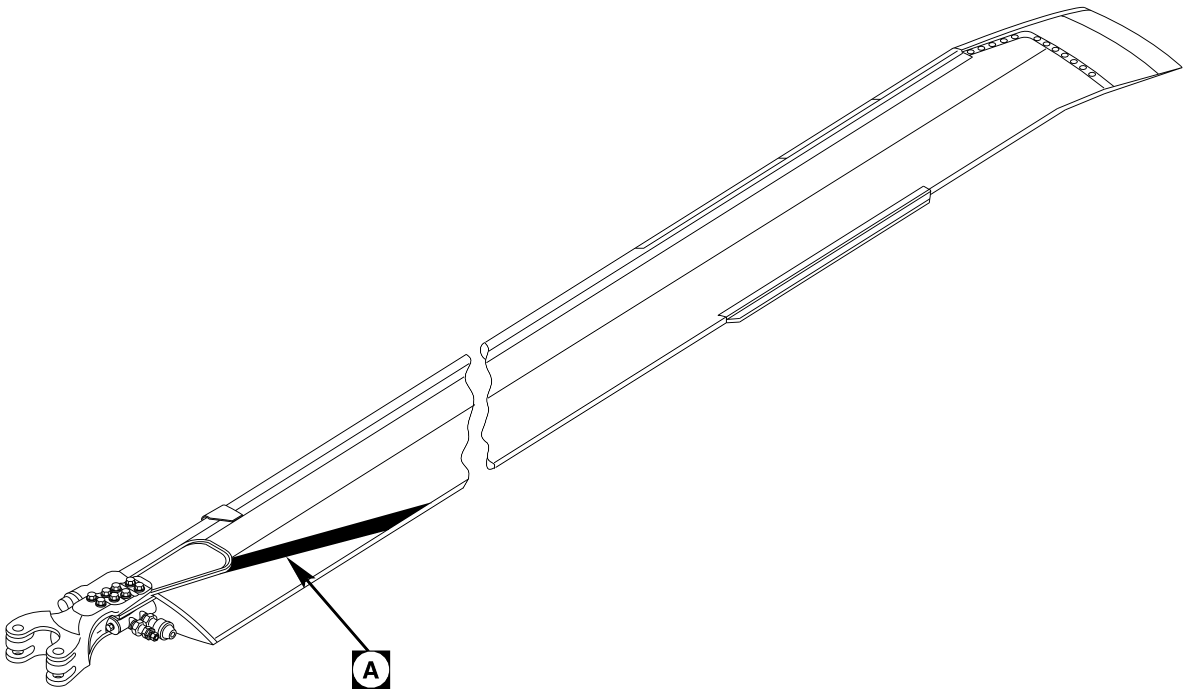


FIGURE 5-4-31-15. REPLACE POLYURETHANE PATCH



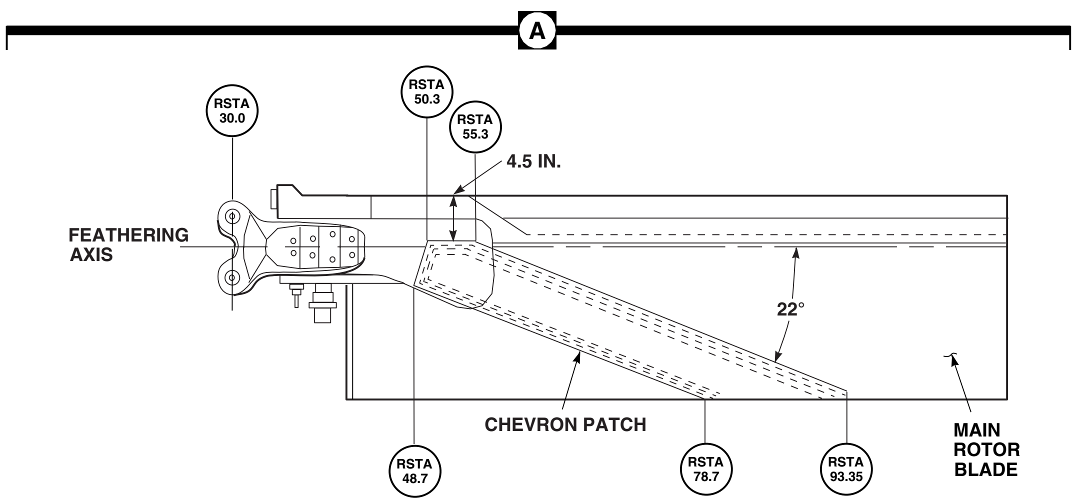


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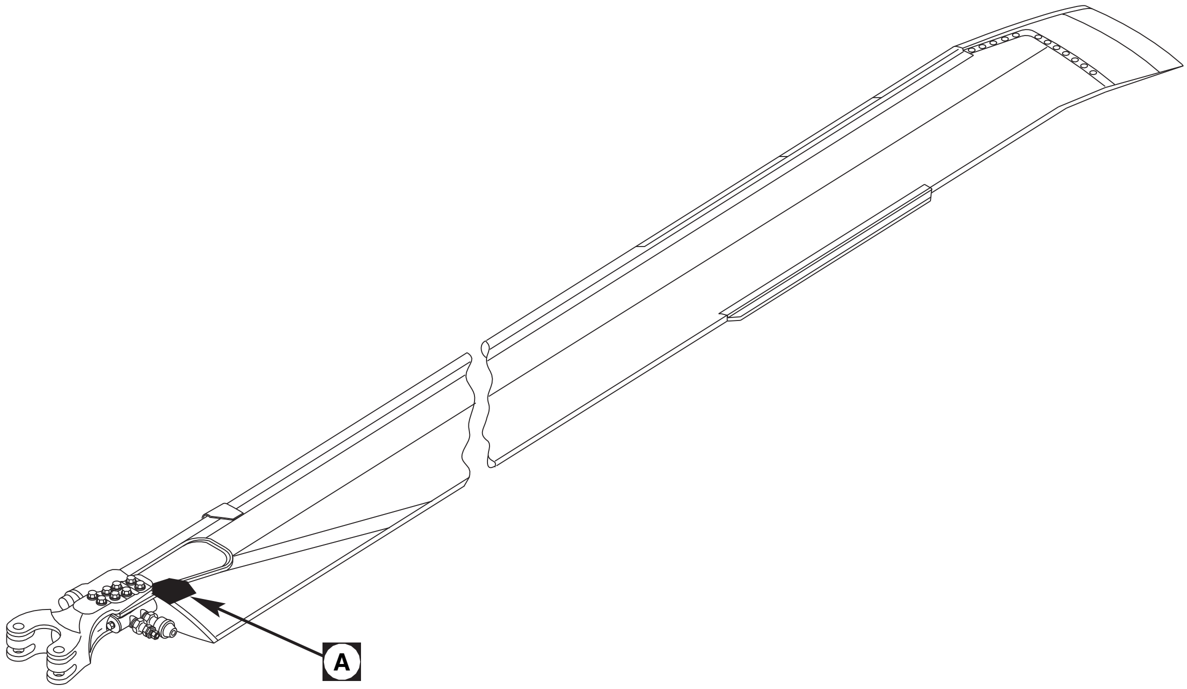
FIGURE 5-4-31-16. REPAIR DISCREPANT ROOT END SKIN PLIES



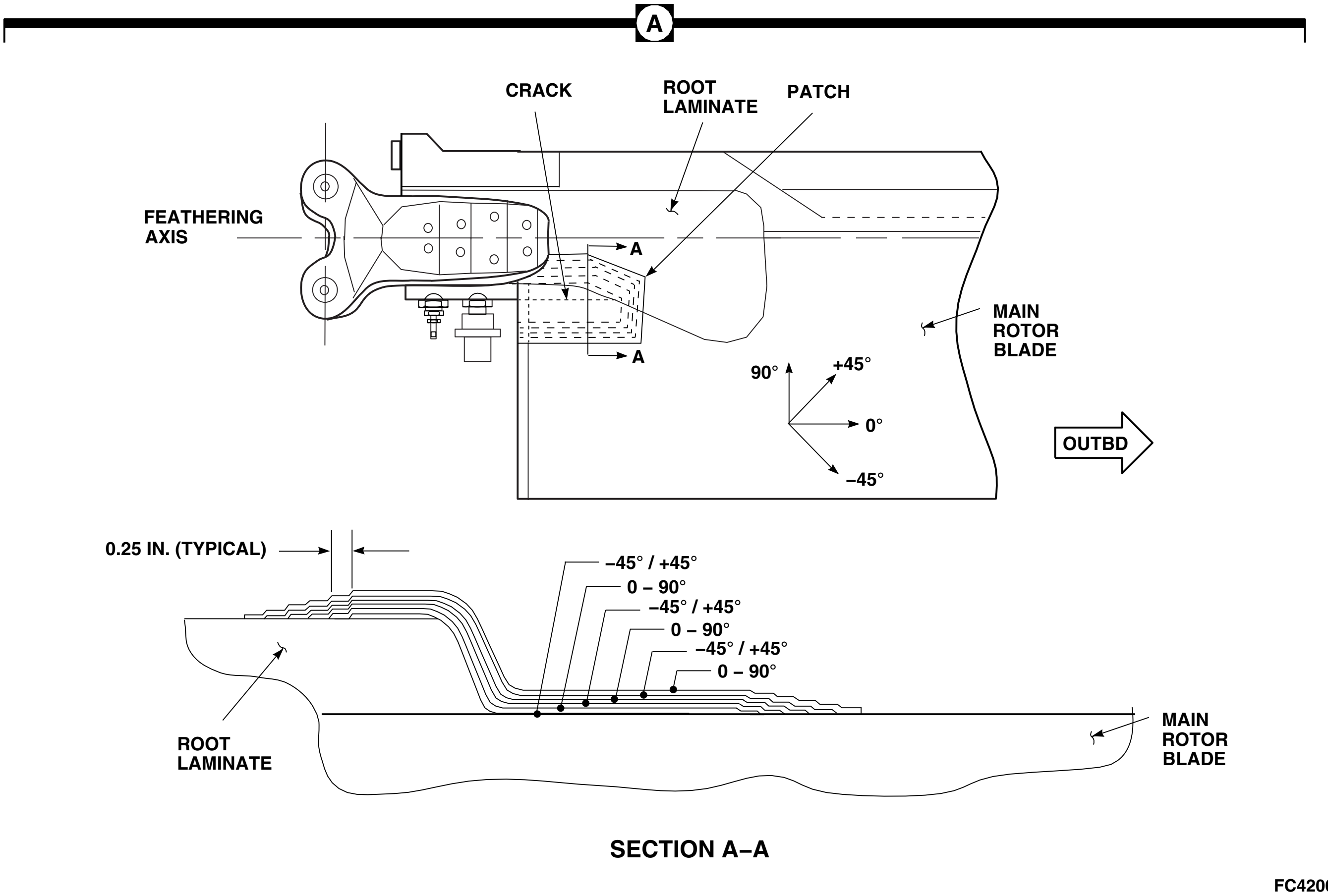


FIGURE 5-4-31-17. REPAIR TRAILING EDGE ROOT SKIN CRACK (RSTA 42.0 - 50.0)

CAUTION

Damage to main rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints. Do not leave cheesecloth, wet with methyl ethyl ketone, lying on main rotor blade.

d. Using clean cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean sanded area. Wipe dry before methyl ethyl ketone evaporates.

NOTE

- Cut plies so that they overlap onto root laminate and past outboard end of crack by approximately 1 ½-inches. Repair plies must cover damage or previous repair by 1 ½-inches in all directions.
- When cutting satin cloth, cut satin cloth plies in order of lay up. Cut each piece so that direction of woven cloth strands will be at 45° angle to that of previous cut piece (Figure 5-4-31-17, Detail A).

e. Cut six repair plies of satin cloth, Item 269A, Appendix D.

f. Prepare adhesive, Item 35, Appendix D, per manufacturer's instructions.

g. Spread repair plies on flat surface (in order of lay up) covered with polyvinyl plastic sheet, Item 237B, Appendix D.

h. Saturate repair plies with mixed adhesive using acid brush, Item 84, Appendix D, or beverage stirring stick, Item 315C, Appendix D.

i. Using acid brush, Item 84, Appendix D, apply coat of mixed adhesive to repair area on main rotor blade.

j. Position repair plies in place on blade with each ply overlapping undamaged skin or previous repair ply at least 1 ½-inches in all directions. Orientate each repair ply 45° (Figure 5-4-31-17, Detail A).

k. Cover repair with polyvinyl plastic sheet, Item 237B, Appendix D.

l. Squeegee trapped air and excess resin out of repair patch.

m. Use vacuum bags or clamping plates to provide pressure during cure.

n. Allow adhesive to cure for minimum of 24 hours at 70° - 85°F (21° - 29°C), or for minimum of 1 hour at 180° - 200°F (82° - 93°C) using heat gun.

o. Using coin-tapping method, inspect repair area for disbonds (PARA 5-3-11).

p. Install aluminum wire mesh lightning protection (PARA 5-4-31.21, step i. (1) through step i. (13)).

q. Touch up paint, as required (PARA 5-4-31.20).

r. Mark main rotor blades ″RS-100-2″. Make sure blades are marked so that this number will not be added to part number by mistake.

s. Perform main rotor blade track and balance (TM 1-70-VIB).

t. Record repair on component log card (SA Form 7343-21).

5-4-31.24 Remove Stuck Main Rotor Blade Pin.

NOTE

To remove stuck main rotor blade expandable pin, proceed to step a. To remove stuck main rotor blade solid pin, proceed to step b.

a. To remove stuck main rotor blade expandable pin, perform the following:

(1) If not already done, relieve load on expandable pin by slightly lifting main rotor blade.

(2) With main rotor blade lifted to relieve load on pin, apply upward force on pin,

and gently lead and lag main rotor blade as necessary until pin begins to move up (Figure 5-4-31-18, Detail A).

(3) Stop lead and lag of main rotor blade once up-and-down movement of pin is achieved.

(4) While applying upward force on pin, gently lower or raise main rotor blade as necessary until pressure is relieved and pin can be removed.

(5) Repeat step a. (2) and step a. (3) until pin can be removed.

(6) If main rotor blade pin still cannot be removed, loosen nut and repeat step a. (2) through step a. (4).

WARNING

FSCAP

Main rotor blade pin is a critical component. **NO DISASSEMBLY OF THE PIN IS ALLOWED, EXCEPT AS A LAST MEANS TO REMOVE A STUCK PIN. IF DISASSEMBLY IS REQUIRED TO REMOVE A STUCK PIN, THE PIN MUST BE REPLACED.**

(7) If pin still cannot be removed, disassemble pin and remove from main rotor blade. Replace pin.

b. Locally make main rotor blade solid pin removal tool (Figure H-266, Appendix H).

c. Back off nut and extend rod to bottom of removal tool.

d. Remove T-pin from rod.

e. If not already done, relieve load on pin by slightly lifting main rotor blade.

f. Lower handle on pin enough to position removal tool over main rotor blade pin.

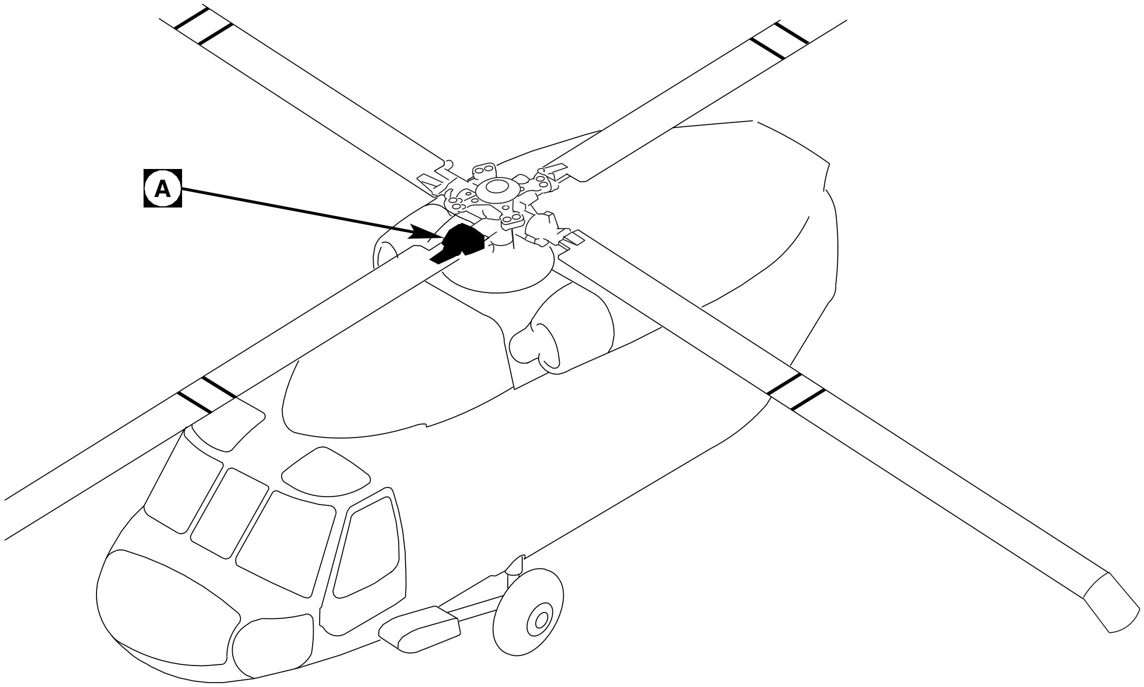
g. Position removal tool over main rotor blade pin and onto main rotor blade cuff (Figure 5-4-31-19, Detail A).

h. Align hole in top lug of pin with slot in removal tool. Insert T-pin fully through slot and pin (Figure 5-4-31-19, View A-A).

i. Handtighten nut until bottom of removal tool sits flush on main rotor blade cuff (Figure 5-4-31-19, Detail A).

j. Support removal tool by handle, and using ⅞-inch closed-end box wrench, extract pin from main rotor blade/spindle cuff lugs.

k. Disengage T-pin and remove pin from tool.



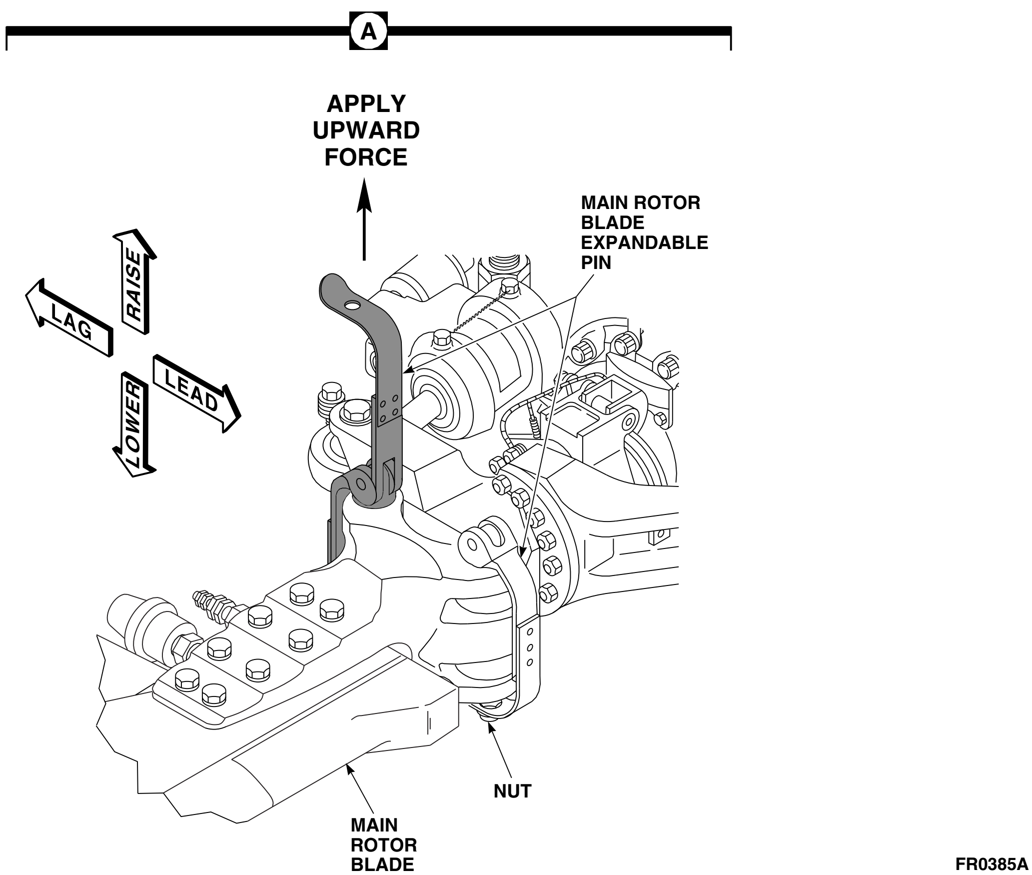
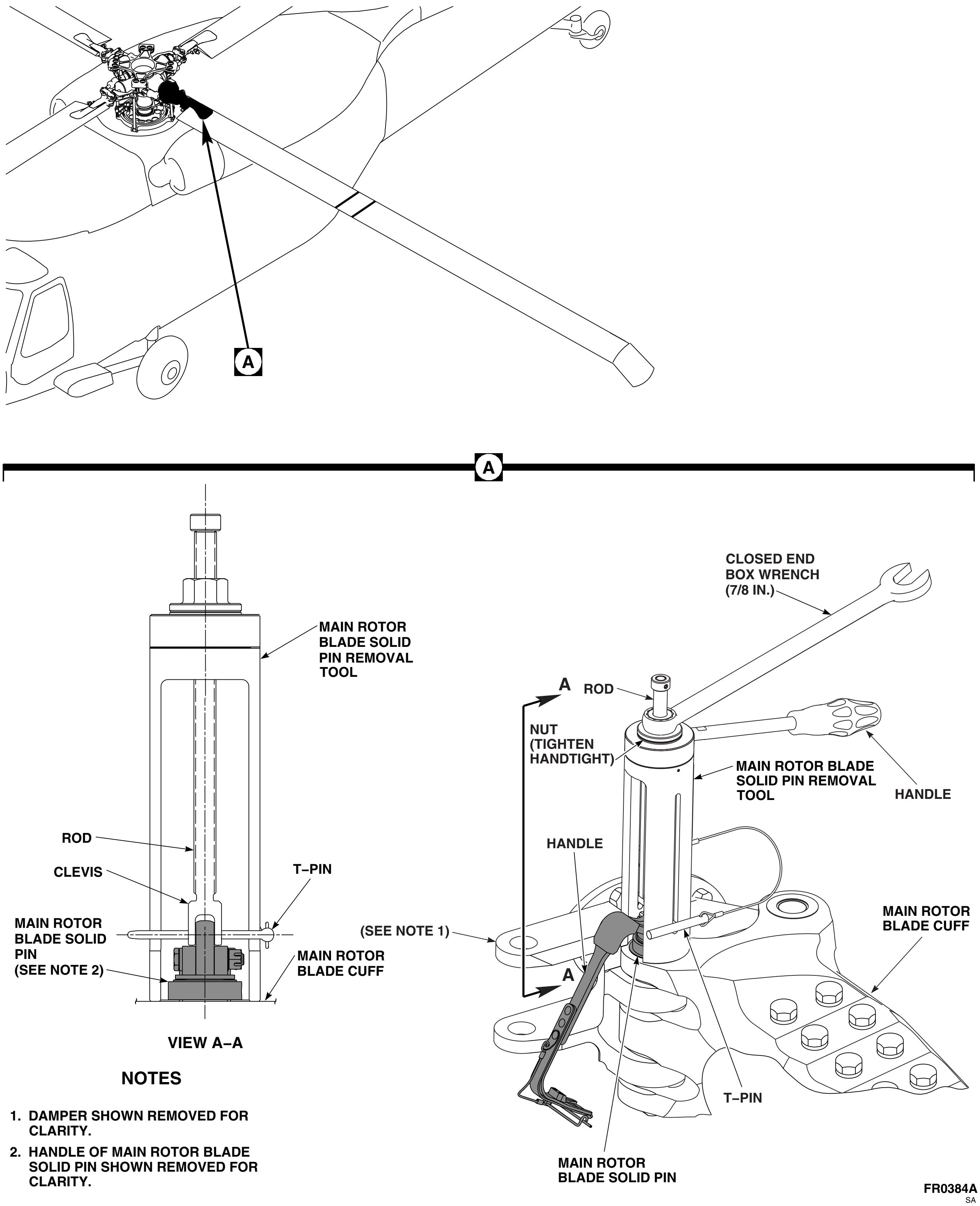


FIGURE 5-4-31-18. REMOVE STUCK MAIN ROTOR BLADE EXPANDABLE PIN



NOTES

1. DAMPER SHOWN REMOVED FOR CLARITY.
2. HANDLE OF MAIN ROTOR BLADE SOLID PIN SHOWN REMOVED FOR CLARITY.

FIGURE 5-4-31-19. REMOVE STUCK MAIN ROTOR BLADE SOLID PIN

5-4-32 MAIN ROTOR BLADE CUFF DECAL.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Grease Pencil
- Heat Gun
- Nonmetallic Wedge, Locally-Made
- Rawhide Mallet
- Roller

Materials/Parts

- Alcohol, Denatured, Item 70, Appendix D
- Cheesecloth, Item 90, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 231, Appendix D
- Sealing Compound, Item 271A, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D

5-4-32.1 Replace Main Rotor Blade Cuff Decal.

NOTE

Replacement of all main rotor blade cuff decals is the same.

5-4-32.1.1 Remove.

a. Turn main rotor head until main rotor blade cuff is easy to access.

b. Using grease pencil, mark position of main rotor blade cuff decal on cuff (Figure 5-4-32-1, Detail A).

c. Using nonmetallic wedge and rawhide mallet, remove old decal from cuff.

5-4-32.1.2 Install.

a. Using paint remover, Item 231, Appendix D, remove old sealing compound from main rotor blade cuff.

b. Using denatured alcohol, Item 70, Appendix D, and cheesecloth, Item 90, Appendix D, clean cuff and back of new main rotor blade cuff decal.

c. Using machinery towel, Item 211, Appendix D, dry cleaned areas.

d. Using sealing compound, Item 271A, Appendix D, apply (sticky side) to decal. Smooth out tape to remove any trapped air.

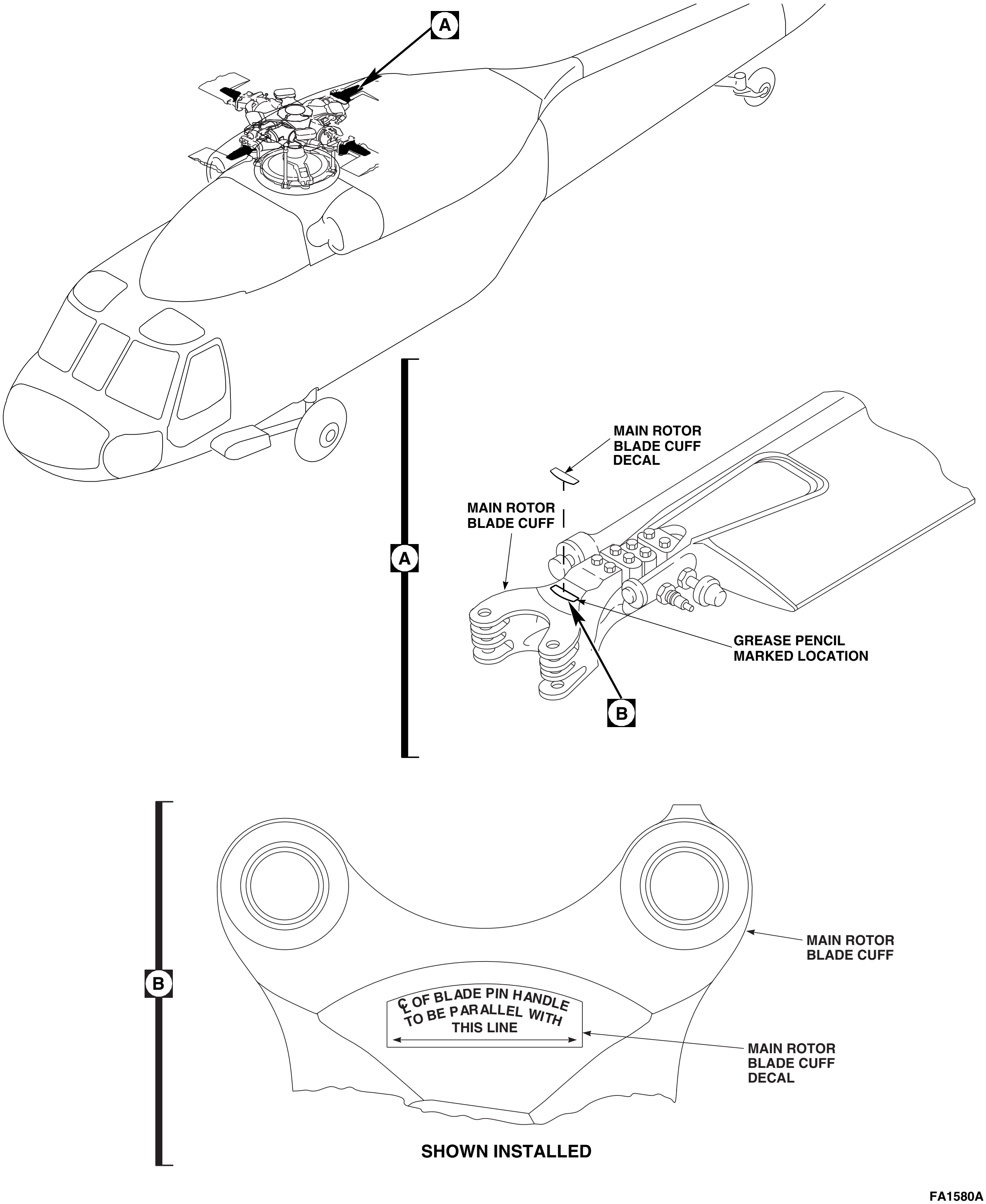


FIGURE 5-4-32-1. REPLACE MAIN ROTOR BLADE CUFF DECAL

e. Remove backing paper from tape and install decal on cuff in same position as decal just removed (location marked with grease pencil) (Figure 5-4-32-1, Detail B).

f. Using roller, apply firm pressure to main rotor blade cuff decal to obtain a strong bond.

NOTE

Bond strength may be improved by applying heat and pressure.

g. For maximum contact, using heat gun, apply heat, 150°F (66°C), and pressure to main rotor blade cuff decal.

h. Using sealing compound, Item 292, Appendix D, seal all edges of main rotor blade cuff decal.

i. Allow main rotor blade cuff decal to cure for 24 hours at room temperature.

j. Make sure area is clean and free of foreign material.

5-4-33 MAIN ROTOR BLADE RECEIVER ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Installation Tool, 70150-09148-041T046
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 5-4-31

5-4-33.1 Replace Main Rotor Blade Receiver Assembly.

5-4-33.1.1. Remove.

a. Using installation tool, remove main rotor blade receiver assembly from main rotor blade (Figure 5-4-33-1, Detail A).

b. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean area.

5-4-33.1.2. Inspect.

a. Inspect receiver block on main rotor blade for cracks. **NONE ALLOWED.** If cracks are found, replace main rotor blade (PARA 5-4-31).

b. Inspect receiver block on main rotor blade for disbonding from main rotor blade. **NONE ALLOWED.** If disbonding is found, replace main rotor blade (PARA 5-4-31).

c. Visually inspect threaded insert for damage or missing threads. **NONE ALLOWED.** If damaged or missing threads are found, perform the following:

(1) Remove threaded insert (Figure 5-4-33-1, Detail A).

(2) Inspect receiver block threads for damage. **NONE ALLOWED.** If damage is found, replace main rotor blade (PARA 5-4-31).

(3) Install threaded insert into receiver block.

5-4-33.1.3. Install.

a. Using installation tool, install main rotor blade receiver assembly into main rotor blade (Figure 5-4-33-1, Detail A). **MINIMUM RUNNING TORQUE IS 70 INCH-POUNDS.** If minimum running torque is not met, inspect threaded insert (PARA 5-4-33.1.2).

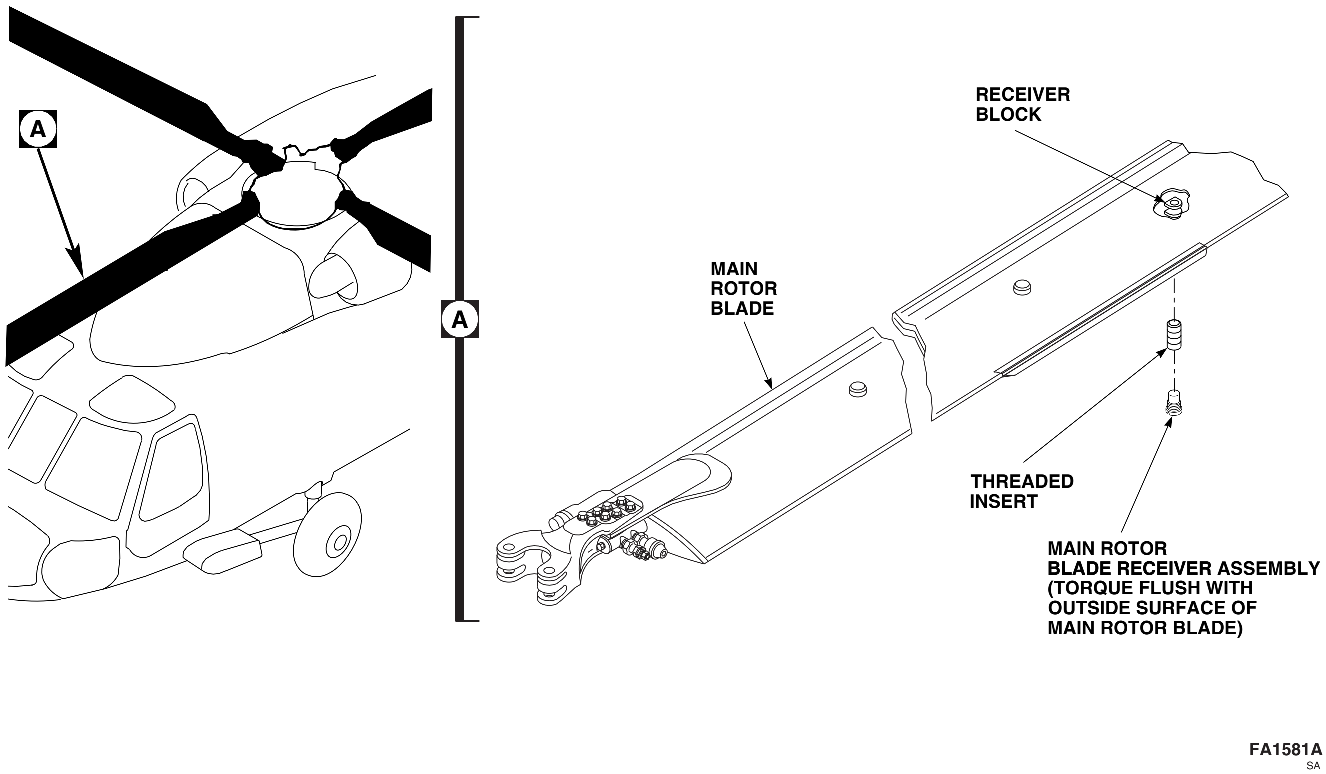


FIGURE 5-4-33-1. REPLACE MAIN ROTOR BLADE RECEIVER ASSEMBLY

b. **FINAL TORQUE MAIN ROTOR BLADE RECEIVER ASSEMBLY UNTIL FLUSH WITH MAIN ROTOR BLADE OUTSIDE SURFACE.**

c. Apply sealing compound, Item 292, Appendix D, to outside diameter of main rotor blade receiver assembly.

5-4-34 BLADE INSPECTION METHOD (BIM®) PRESSURE INDICATOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Spanner Wrench Set, AN8514-1
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Vacuum

Materials/Parts

- Dishwashing Compound, Item 122, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 296, Appendix D

References

- Appendix D
- PARA 1-3-29
- PARA 5-4-36

5-4-34.1 Replace Blade Inspection Method (BIM®) Pressure Indicator.

NOTE

Replacement of all BIM® pressure indicators is the same.

NOTE

If any repair is performed because main rotor blade spar pressure is below limits, one hour whirl test is required (PARA 5-4-36).

5-4-34.1.1 Remove.

a. Release pressure in main rotor blade spar by removing cap from main rotor blade air valve and loosening outer hexnut 2 ½ turns while keeping inner hexnut and main rotor blade air valve adapter from turning (Figure 5-4-34-1, Sheet 1, Detail B and Sheet 2, Detail C).

b. Using spanner wrench with pin arm, AN8514-4, installed, install wrench on body only and remove BIM® pressure indicator and packing (Figure 5-4-34-1, Sheet 1, Detail B). Do not remove BIM® pressure indicator adapter in which pressure indicator was installed. Discard packing.

c. Immediately install plug in opening in adapter to prevent air from entering main rotor blade spar.

d. Install cap over opening in BIM® pressure indicator.

5-4-34.1.2 Inspect.

Perform torque check on BIM® pressure indicator adapter. **TORQUE SHALL BE 176 - 194 INCH-POUNDS.**

5-4-34.1.3 Install.

a. Using petrolatum, Item 234, Appendix D, lubricate new packing. Install packing on BIM® pressure indicator (Figure 5-4-34-1, Sheet 1, Detail B).

b. Remove cap from replacement pressure indicator and remove plug from opening in BIM® pressure indicator adapter.

c. Immediately install pressure indicator and packing in adapter.

d. Using spanner wrench with pin arm, AN8514-4, installed, install spanner wrench only on body of pressure indicator.

e. **TORQUE PRESSURE INDICATOR TO 105 - 115 INCH-POUNDS.**

f. Service main rotor blade spar to normal pressure (PARA 1-3-29).

g. Using test liquid, such as leak detection fluid, Item 189, Appendix D, or dishwashing compound, Item 122, Appendix D, check packing for leakage by applying over and around base of pressure indicator and watching for bubbles to form.

h. When check is completed and before test liquid dries, using clean, fresh water, clean area thoroughly to remove all traces of test liquid.

i. Using machinery towel, Item 211, Appendix D, dry main rotor blade.

j. Test BIM® pressure indicator for proper operation (PARA 1-3-29).

k. Using lockwire, Item 197, Appendix D, lockwire pressure indicator to BIM® pressure indicator adapter, and then pressure indicator adapter to main rotor blade air valve adapter (Figure 5-4-34-1, Sheet 2, Detail C).

l. If required, perform main rotor blade whirl test (one hour) (PARA 5-4-36).

5-4-34.2 Replace Blade Inspection Method (BIM®) Pressure Indicator Manual Test Lever.

NOTE

Replacement of all BIM® pressure indicator manual test levers is the same.

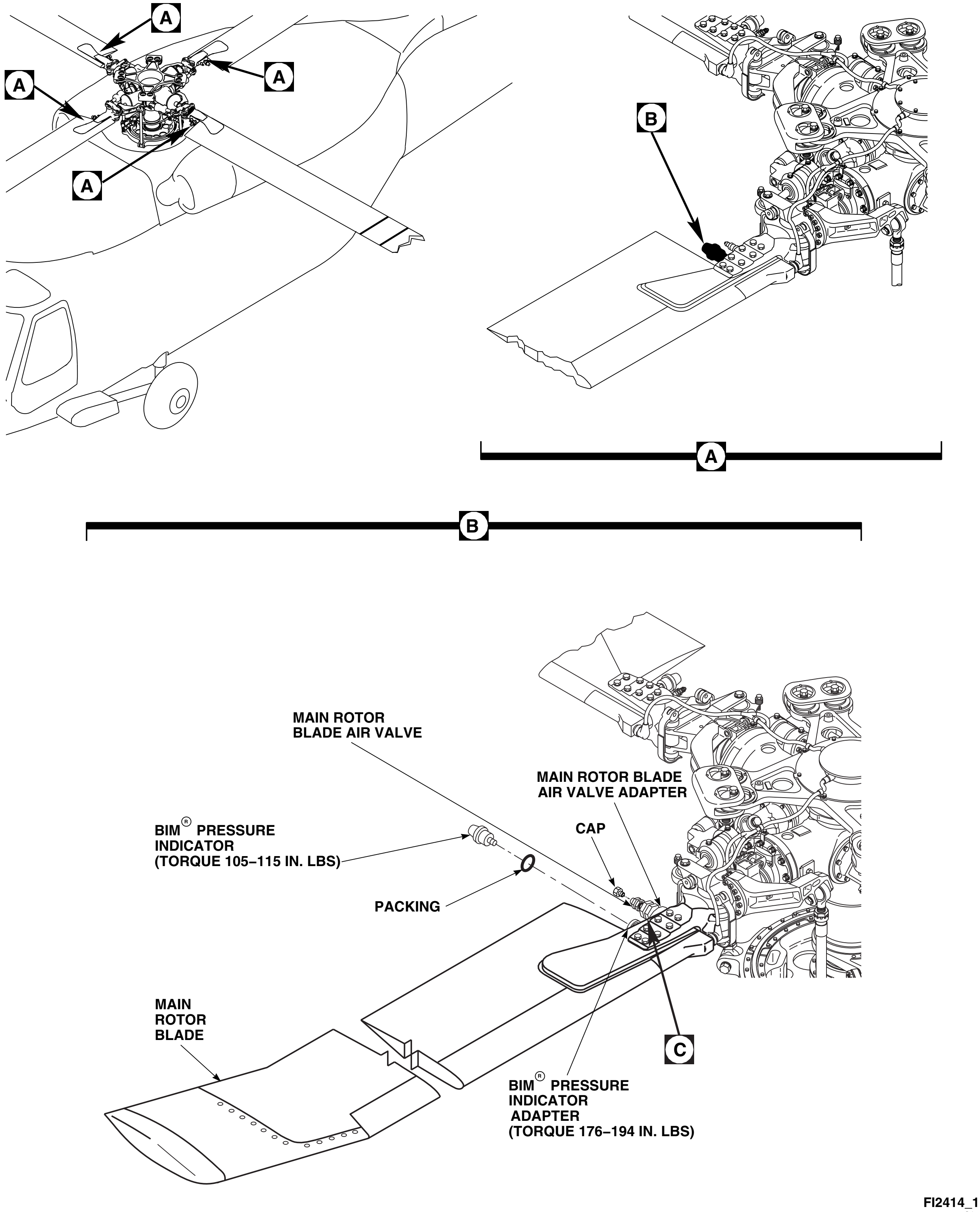


FIGURE 5-4-34-1. REPLACE BLADE INSPECTION METHOD (BIM®) PRESSURE INDICATOR (SHEET 1 OF 2)

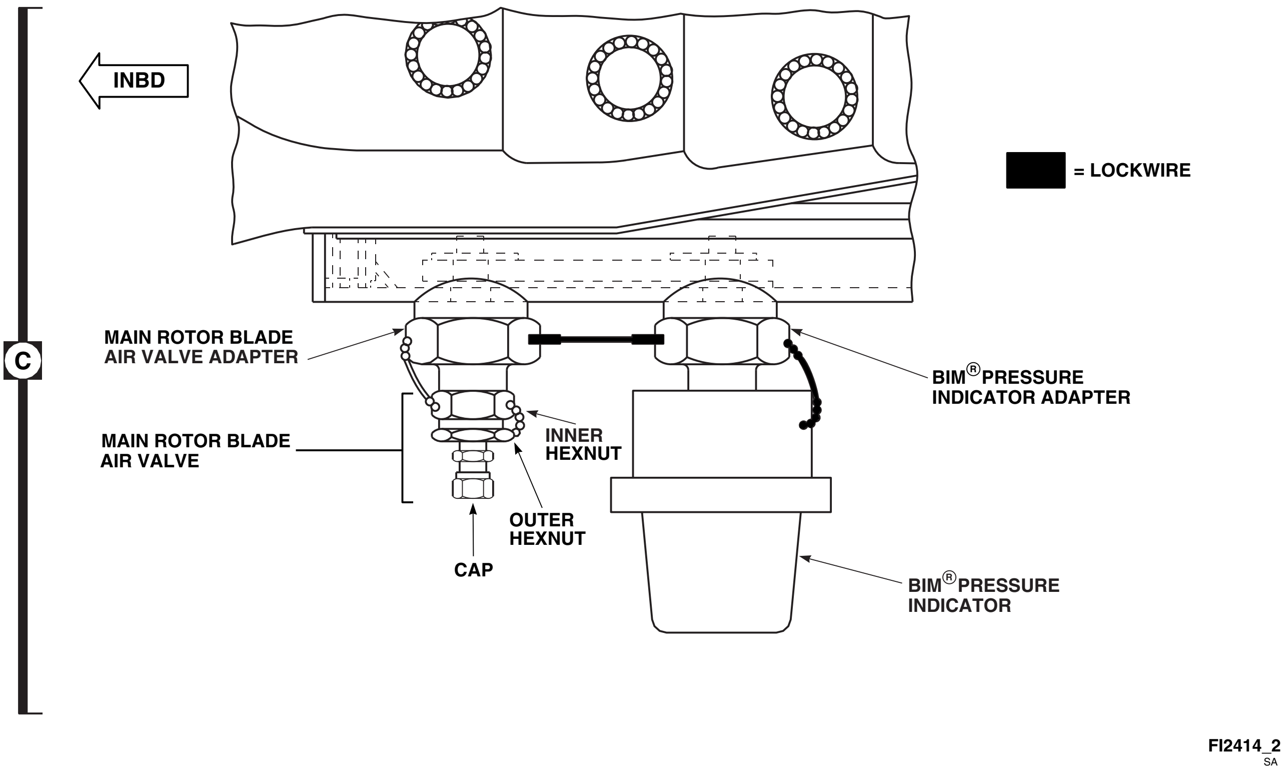


FIGURE 5-4-34-1. REPLACE BLADE INSPECTION METHOD (BIM®) PRESSURE INDICATOR (SHEET 2 OF 2)

5-4-34.2.1 Remove.

a. Remove screw securing BIM® pressure indicator manual test lever to base of BIM® pressure indicator (Figure 5-4-34-2, Detail B).

b. Remove manual test lever.

5-4-34.2.2 Install.

a. Place BIM® pressure indicator manual test lever on base of BIM® pressure indicator and install screw (Figure 5-4-34-2, Detail B).

b. Test BIM® pressure indicator for proper operation (PARA 1-3-29).

5-4-34.3 Replace Blade Inspection Method (BIM®) Pressure Indicator Adapter.

NOTE

- Replacement of all BIM® pressure indicators is the same.
- If any repair is done because main rotor blade spar pressure is below limits, perform one hour whirl test (PARA 5-4-36).

5-4-34.3.1 Remove.

a. Remove BIM® pressure indicator (PARA 5-4-34.1.1).

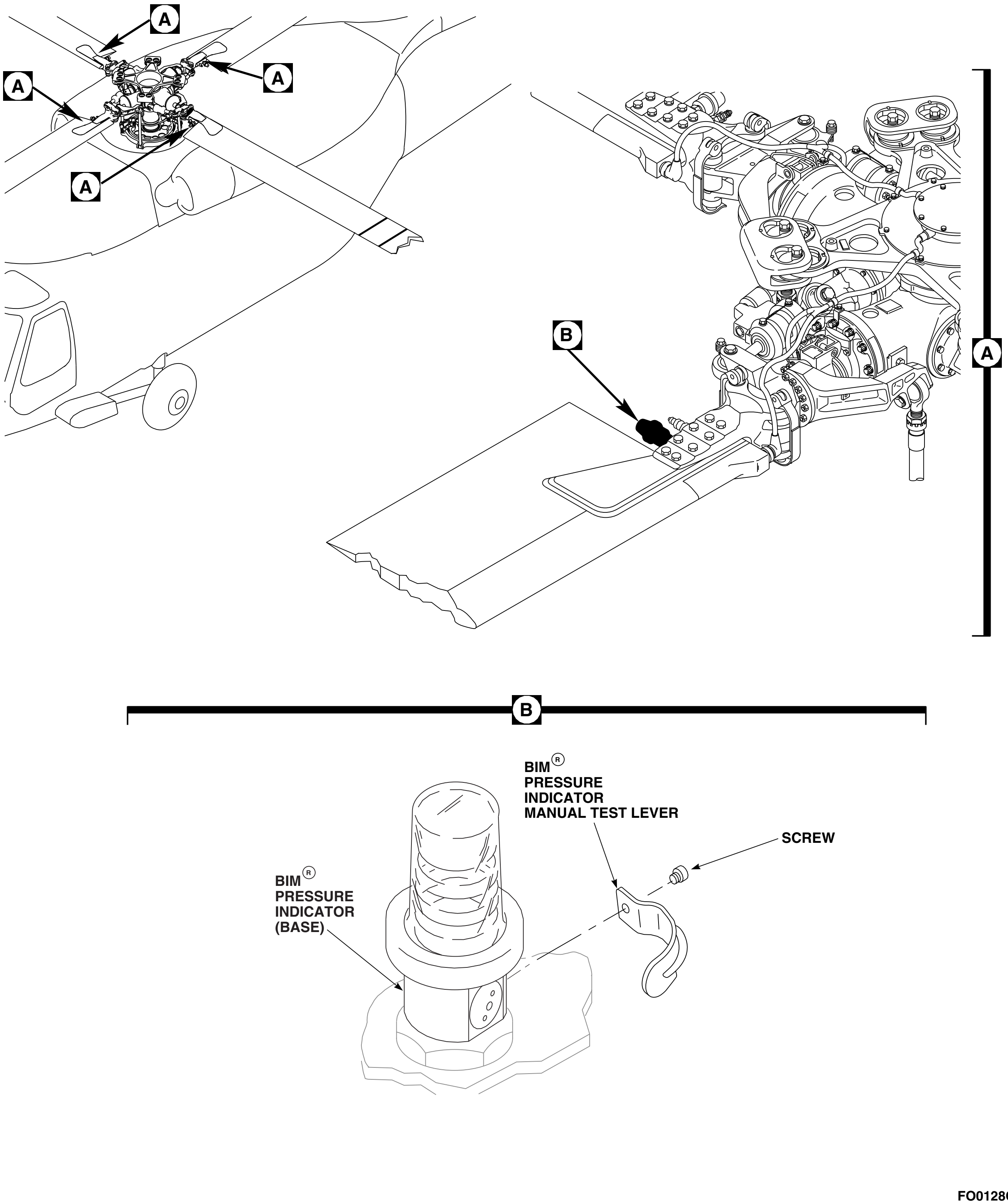


FIGURE 5-4-34-2. REPLACE BLADE INSPECTION METHOD (BIM®) PRESSURE INDICATOR MANUAL TEST LEVER

b. Remove BIM® pressure indicator adapter from housing (Figure 5-4-34-3, Sheet 1, Detail B).

c. Remove and discard packing.

5-4-34.3.2 Clean/Inspect.

CAUTION

Damage to main rotor blade spar will result if metallic tools are used to remove sealing compound from adapter cavity. Use only nonmetallic tools to remove sealing compound.

NOTE

All traces of sealing compound do not need to be removed from cavity. Only enough sealing compound should be removed to provide good seal.

a. Using nonmetallic scraper, remove enough sealing compound from adapter cavity to provide good seal (Figure 5-4-34-1, Sheet 1, Detail C).

CAUTION

Damage to main rotor blade will result if debris is not removed from housing prior to installing new adapter. Make sure all debris is removed from housing.

b. Using vacuum, remove debris from housing.

5-4-34.3.3 Install.

a. Apply sealing compound, Item 296, Appendix D, to fill cavity of housing (Figure 5-4-34-3, Sheet 1, Detail C).

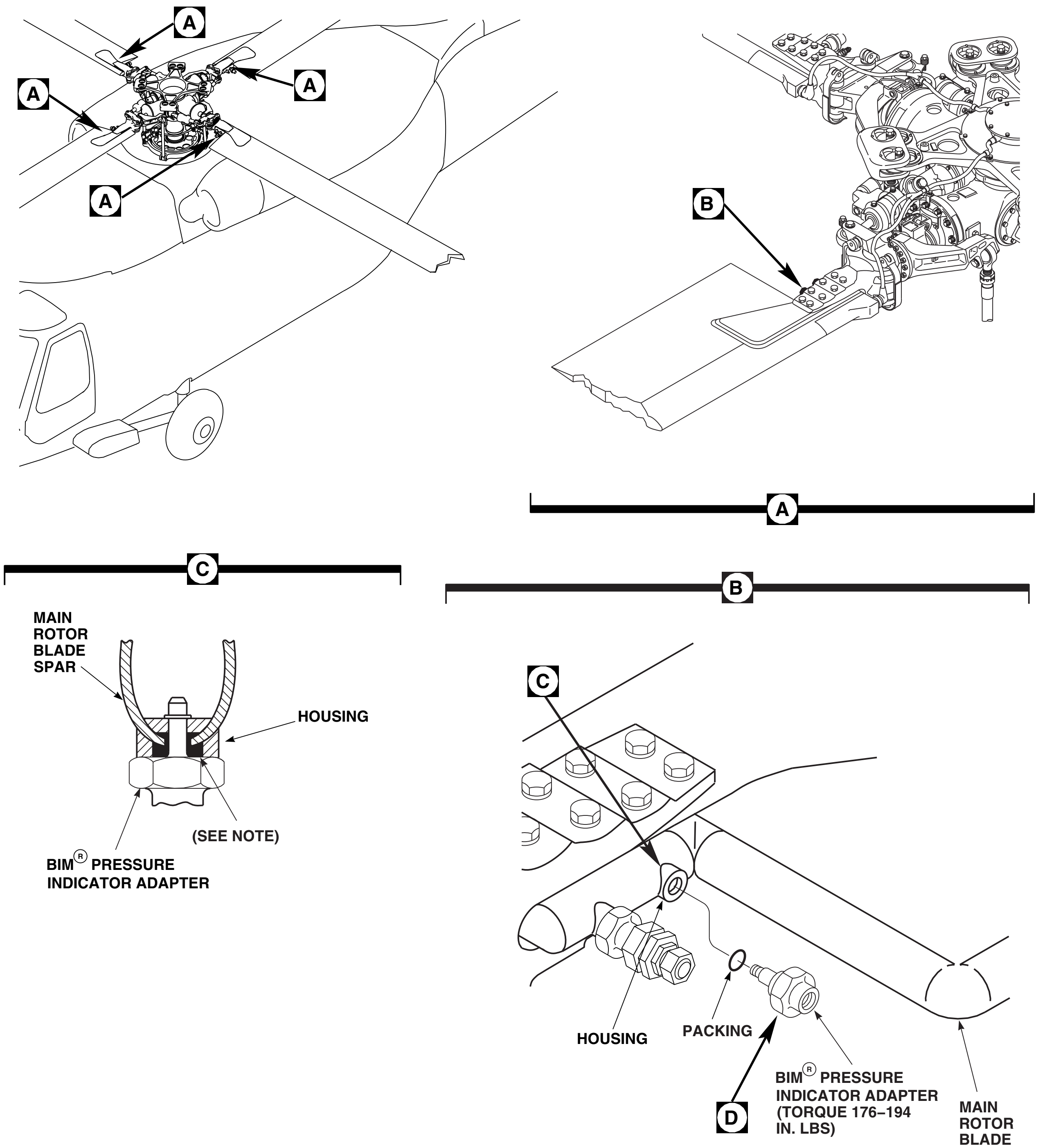
b. Install BIM® pressure indicator adapter in housing (Figure 5-4-34-3, Sheet 1, Detail B).

c. **TORQUE ADAPTER TO 176 - 194 INCH-POUNDS.**

d. Using lockwire, Item 197, Appendix D, lockwire BIM® pressure indicator adapter to main rotor blade air valve adapter (Figure 5-4-34-3, Sheet 2, Detail D).

e. Inspect BIM® pressure indicator (PARA 5-4-34.1.2).

f. Install BIM® pressure indicator (PARA 5-4-34.1.3).



NOTE

FILL CAVITY WITH SEALING COMPOUND UPON INSTALLATION OF ADAPTER.

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FIGURE 5-4-34-3. REPLACE BLADE INSPECTION METHOD (BIM®) PRESSURE INDICATOR ADAPTER (SHEET 1 OF 2)

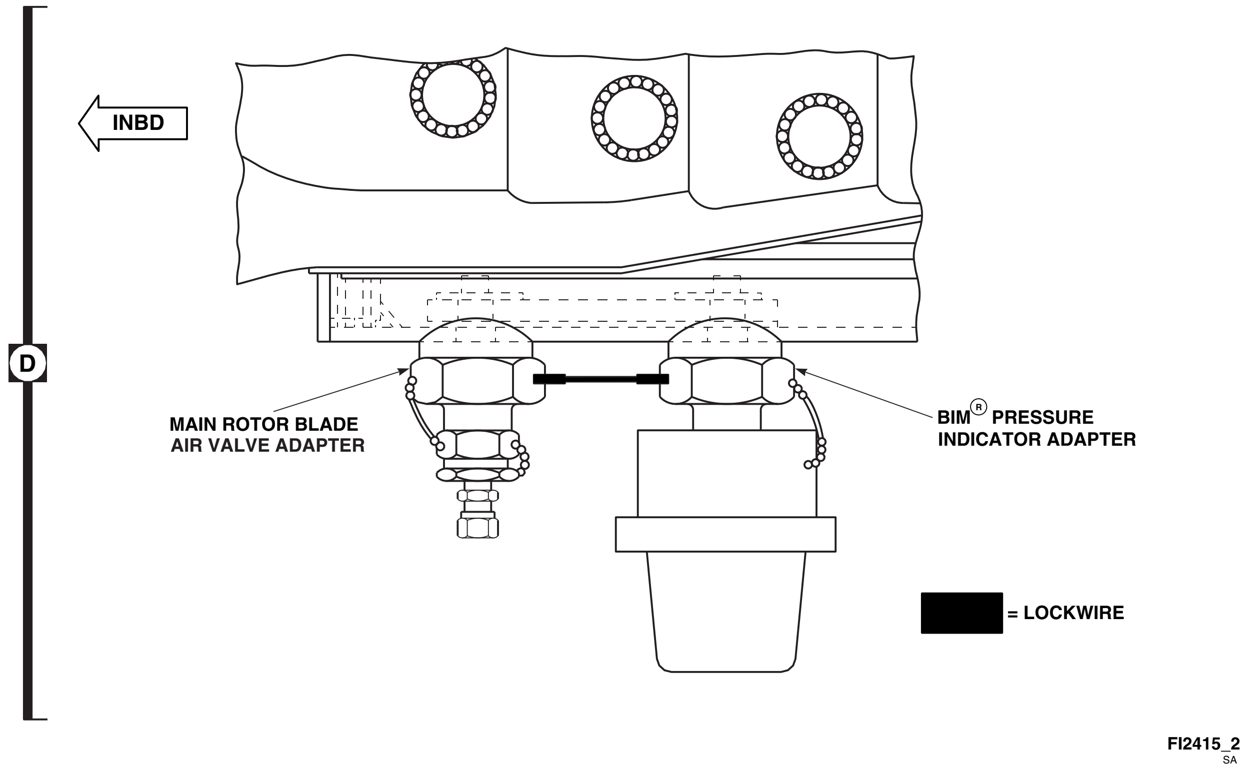


FIGURE 5-4-34-3. REPLACE BLADE INSPECTION METHOD (BIM®) PRESSURE INDICATOR ADAPTER (SHEET 2 OF 2)

5-4-35 MAIN ROTOR BLADE AIR VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Vacuum

Materials/Parts

- Dishwashing Compound, Item 122, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Packing, MS29512-07
- Protective Caps and Plugs
- Sealing Compound, Item 296, Appendix D

References

- Appendix D
- PARA 1-3-29
- PARA 5-4-36
- PARA 5-4-37

5-4-35.1 Replace Main Rotor Blade Air Valve.

NOTE

Replacement of all main rotor blade air valves is the same.

NOTE

If any repair is performed because main rotor blade spar pressure is below limits, main rotor blade whirl test (one hour) is required (PARA 5-4-36).

5-4-35.1.1 Remove.

a. Release pressure in main rotor blade spar by removing cap from main rotor blade air valve and loosening outer hexnut 2 ½ turns while keeping inner hexnut and main rotor blade air valve adapter from turning (Figure 5-4-35-1, Sheet 1, Detail B and Sheet 2, Detail C).

b. Remove air valve and packing keeping air valve adapter from turning.

c. Immediately install plug in opening in adapter to prevent air from entering main rotor blade spar.

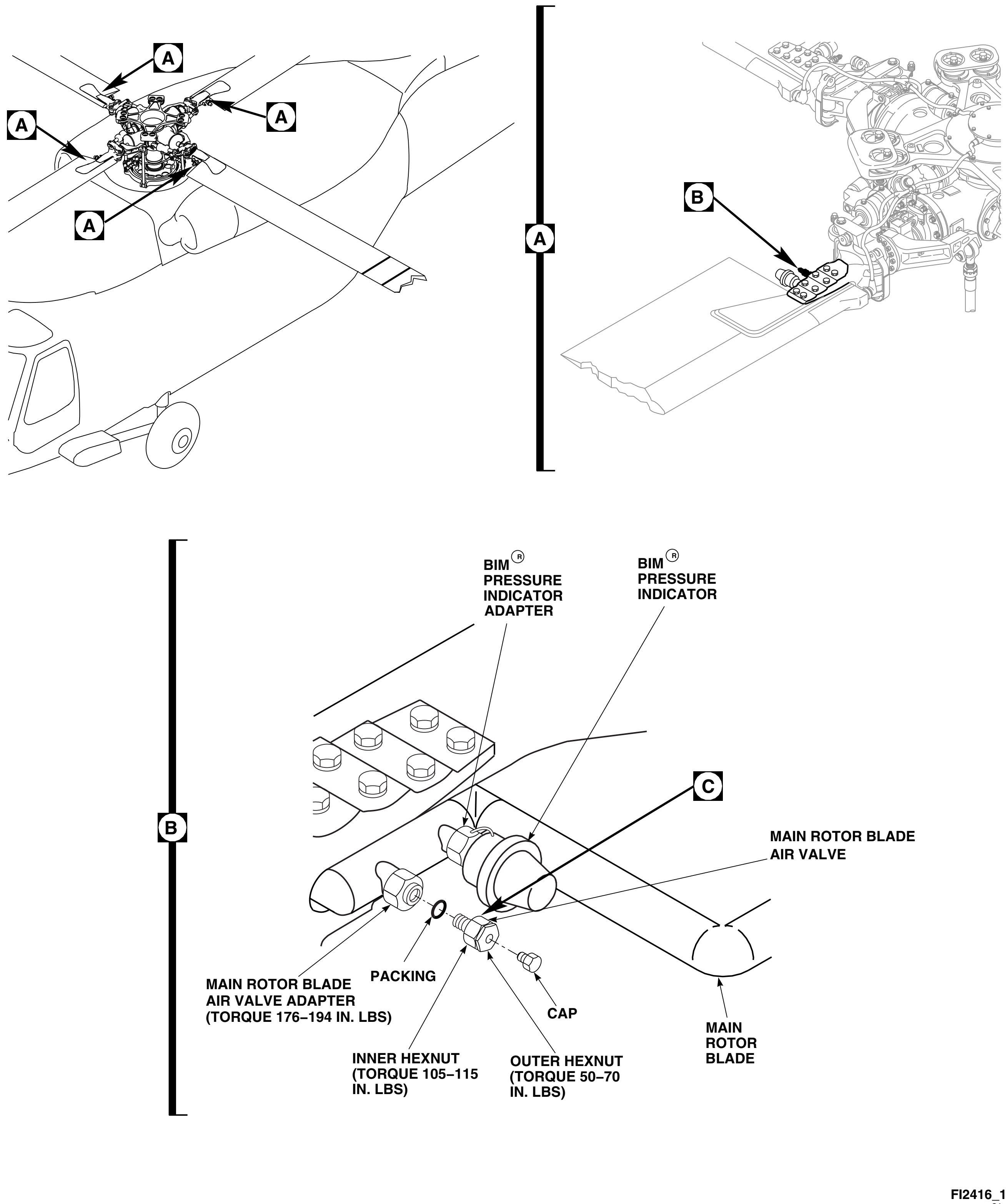


FIGURE 5-4-35-1. REPLACE MAIN ROTOR BLADE AIR VALVE (SHEET 1 OF 2)

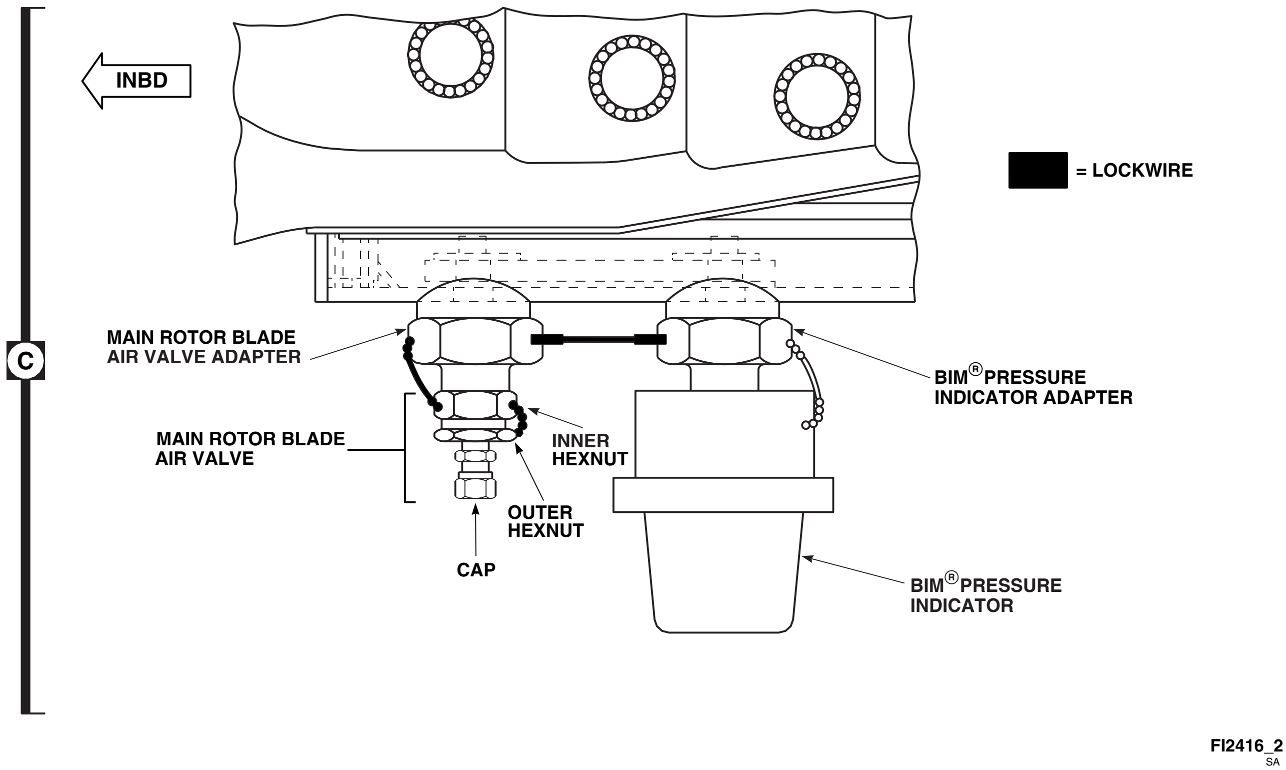


FIGURE 5-4-35-1. REPLACE MAIN ROTOR BLADE AIR VALVE (SHEET 2 OF 2)

5-4-35.1.2 Inspect.

Perform torque check on air valve adapter. **TORQUE SHALL BE 176 - 194 INCH-POUNDS** (Figure 5-4-35-1, Sheet 1, Detail B).

5-4-35.1.3 Install.

a. Remove and discard packing that is provided with new main rotor blade air valve. Install packing, MS29512-07, on air valve (Figure 5-4-35-1, Sheet 1, Detail B).

b. Remove plug from opening in main rotor blade air valve adapter.

c. Immediately install air valve and packing in adapter.

d. **TORQUE AIR VALVE TO 105 - 115 INCH-POUNDS.**

e. Using lockwire, Item 197, Appendix D, lockwire air valve to air valve adapter, and then air valve adapter to BIM® pressure indicator adapter (Figure 5-4-35-1, Sheet 2, Detail C).

f. **IF MAIN ROTOR BLADE SPAR WILL NOT BE SERVICED IMMEDIATELY, TORQUE OUTER HEXNUT TO 50 - 70 INCH-POUNDS TO PREVENT AIR AND/OR MOISTURE FROM ENTERING MAIN ROTOR BLADE SPAR** (Figure 5-4-35-1, Sheet 1, Detail B).

g. If air valve was replaced because of an unsafe indication on BIM® pressure indicator, perform the following:

   (1) Check air valve for leakage (PARA 5-4-37).

   (2) Perform spar leakage test (PARA 5-4-37).

NOTE

Make sure air valve outer hexnut is lockwired to inner hexnut of air valve when servicing is finished.

h. Service main rotor blade spar to normal pressure (PARA 1-3-29).

i. If main rotor blade spar pressure was within acceptable limits when air valve was replaced, check air valve for leakage by applying leak detection fluid, Item 189, Appendix D, or dishwashing compound, Item 122, Appendix D, over and around entire air valve and watching for bubbles to form.

j. When check is completed and before test liquid dries, using clean, fresh water, clean area thoroughly to remove all traces of test liquid.

k. Using machinery towel, Item 211, Appendix D, dry main rotor blade.

l. Using lockwire, Item 197, Appendix D, lockwire outer hexnut to inner hexnut.

m. If required, perform main rotor blade whirl test (one hour) (PARA 5-4-36).

5-4-35.2 Replace Main Rotor Blade Air Valve Adapter.

NOTE

- Replacement of all main rotor blade air valve adapters is the same.
- If any repair is performed because main rotor blade spar pressure is below limits, one hour whirl test is required (PARA 5-4-36).

5-4-35.2.1 Remove.

a. Remove main rotor blade air valve (PARA 5-4-35.1.1).

b. Remove main rotor blade air valve adapter from housing (Figure 5-4-35-2, Sheet 1, Detail B).

c. Remove and discard packing.

5-4-35.2.2 Clean/Inspect.

CAUTION

Damage to main rotor blade spar will result if metallic tools are used to remove sealing compound from adapter cavity. Use only nonmetallic tools to remove sealing compound.

NOTE

All traces of sealing compound do not need to be removed from cavity. Only enough sealing compound should be removed to provide good seal.

a. Using nonmetallic scraper, remove enough sealing compound from adapter cavity to provide good seal (Figure 5-4-35-1, Sheet 1, Detail C).

CAUTION

Damage to main rotor blade will result if debris is not removed from housing prior to installing new adapter. Make sure all debris is removed from housing.

b. Using vacuum, remove debris from housing.

5-4-35.2.3 Install.

a. Remove and discard packing that is provided with new main rotor blade air valve. Install packing, MS29512-07, on air valve adapter.

b. Apply sealing compound, Item 296, Appendix D, to fill cavity of housing (Figure 5-4-35-2, Sheet 1, Detail C).

c. Install air valve adapter in housing (Figure 5-4-35-2, Sheet 1, Detail B).

d. **TORQUE ADAPTER TO 176 - 194 INCH-POUNDS.**

e. Using lockwire, Item 197, Appendix D, lockwire main rotor blade air valve adapter to BIM® pressure indicator adapter (Figure 5-4-35-2, Sheet 2, Detail D).

f. Inspect air valve (PARA 5-4-35.1.2).

g. Install air valve (PARA 5-4-35.1.3).

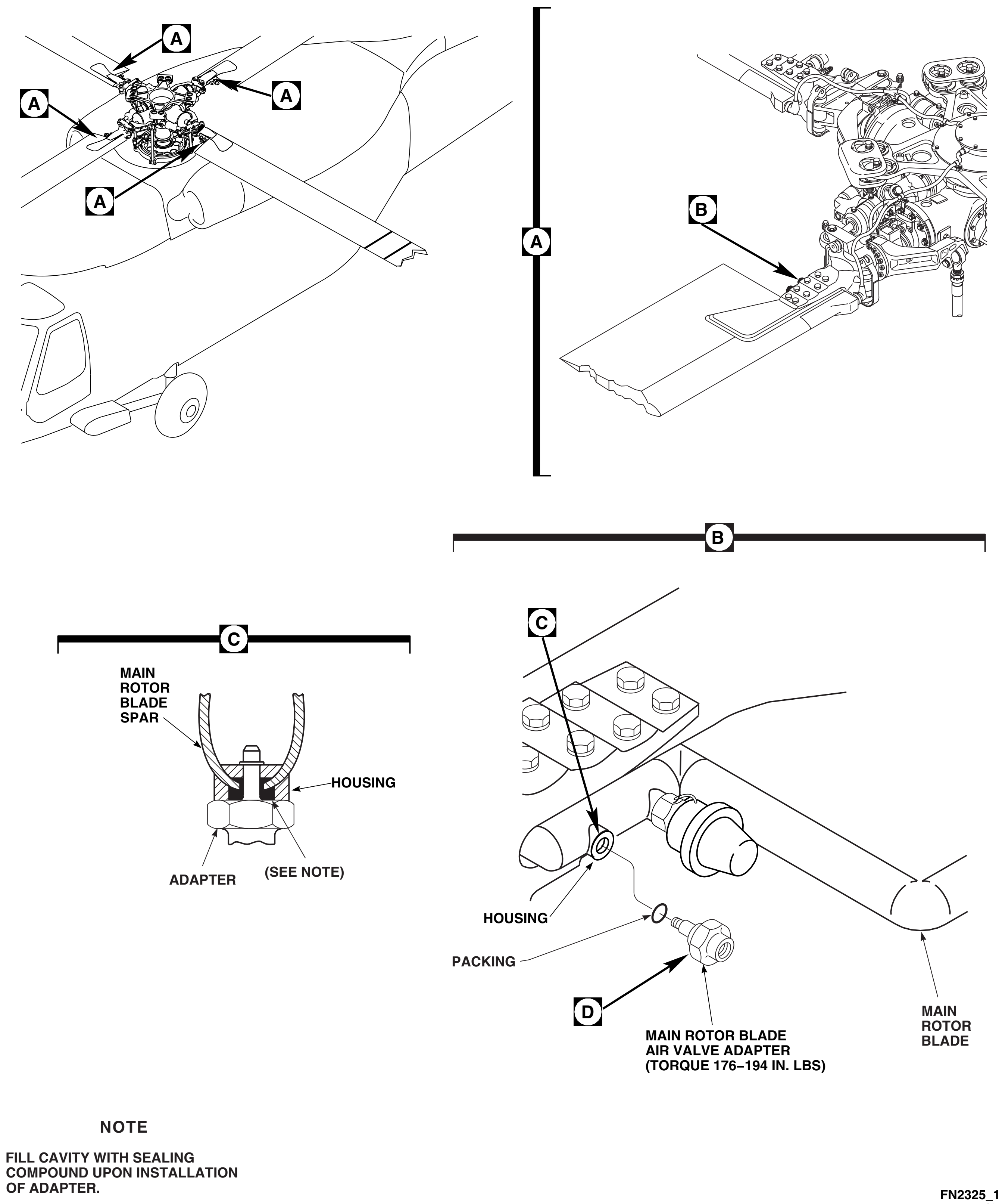


FIGURE 5-4-35-2. REPLACE MAIN ROTOR BLADE AIR VALVE ADAPTER (SHEET 1 OF 2)

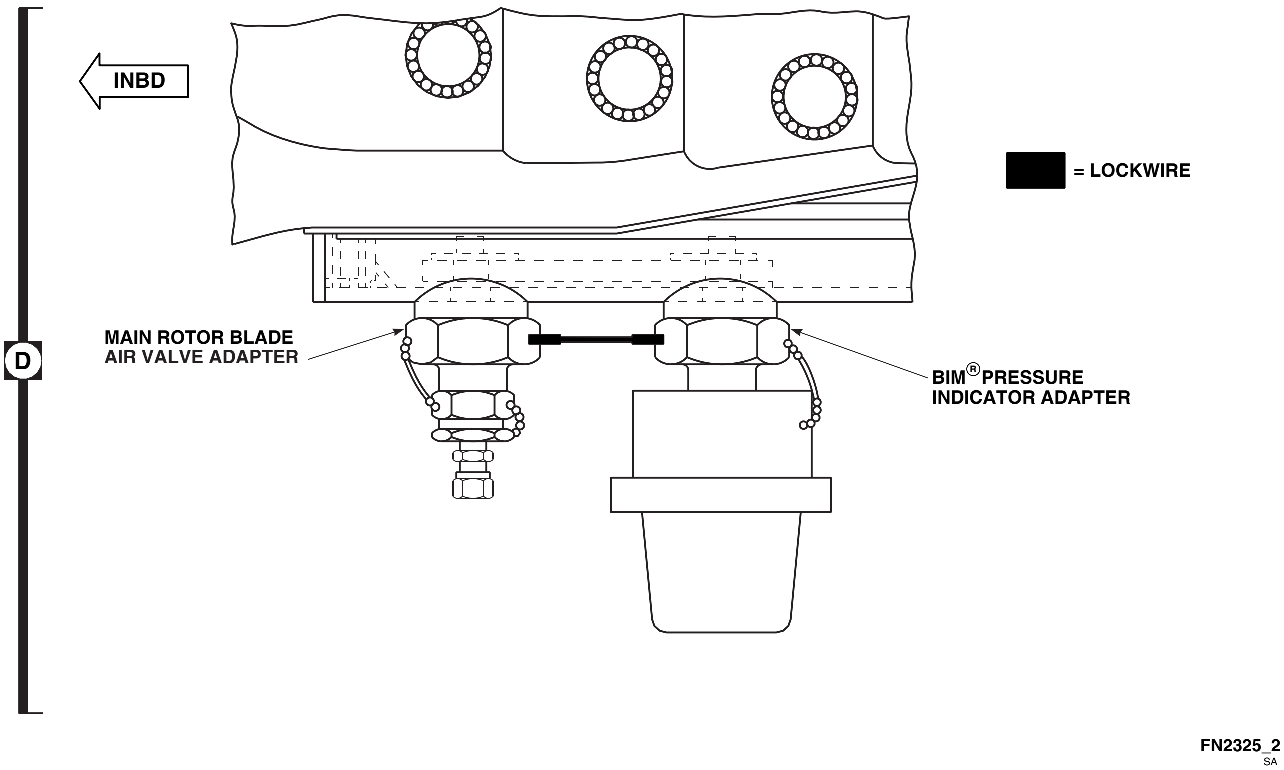


FIGURE 5-4-35-2. REPLACE MAIN ROTOR BLADE AIR VALVE ADAPTER (SHEET 2 OF 2)

5-4-36 MAIN ROTOR BLADE WHIRL TEST (ONE HOUR).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- Appropriate Flight Manual
- PARA 1-3-29

5-4-36.1 Main Rotor Blade Whirl Test (One Hour).

NOTE

Whirl test is a final check on the integrity of a main rotor blade spar that has had a loss of spar pressure. This test is made after all other corrective actions have been taken.

a. Test BIM® pressure indicator for proper operation (PARA 1-3-29).

b. Check spar pressure to make sure it is within limits shown in Table 1-3-29-1 (PARA 1-3-29). Pressure must be at ″Maximum When Servicing Blade″ pressure for actual measured spar temperature. If required, service to that pressure. If pressure is within limits, record pressure to nearest 0.1 psi, record temperature within 1°F (-17°C), and proceed with one hour whirl test.

c. If pressure exceeds limits, service main rotor blade to ″Maximum When Servicing Blade″ pressure for actual measured spar temperature. Record pressure to nearest 0.1 psi, record temperature within 1°F (-17°C), and proceed with one hour whirl test.

d. With qualified personnel at controls, ground run helicopter for ½ hour period at 100% rpm, low collective stick, and neutral cyclic stick and directional control (Appropriate Flight Manual).

e. At end of ½ hour period, check BIM® pressure indicator. **BIM® PRESSURE INDICATOR STRIPES SHALL BE YELLOW.** If stripes are not yellow, send main rotor blade to manufacturer.

f. With qualified personnel at controls, ground run helicopter for ½ hour period at 100% rpm, low collective stick, and neutral cyclic stick and directional control (Appropriate Flight Manual).

g. At end of ½ hour period, measure spar pressure and temperature. Corrected spar pressure is ″Maximum When Servicing Blade″ pressure obtained in Table 5-4-36-1 at the actual measured spar temperature at the end of test.

Table 5-4-36-1. Example

	Pressure	Temperature	
	(psi)	°F	°C
Initial Reading:	18	73	22.8
Reading after 24 hours:	17	63	17.2

Temperature change is 10°F (5.6°C) decrease

correction to be added to reading:

For °F: $10 \times 0.07 = 0.70$ psi

or

For °C: $5.6 \times 0.126 = 0.70$ psi

Therefore the actual corrected pressure in the main rotor blade spar is:

$17 + 0.70 = 17.7$ psi

h. Subtract measured pressure from corrected pressure. **PRESSURE DIFFERENCE SHALL NOT BE GREATER THAN 0.25 PSI.** If pressure exceeds limits, send main rotor blade to manufacturer for repair.

5-4-37 MAIN ROTOR BLADE SPAR LEAKAGE TEST.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Dishwashing Compound, Item 122, Appendix D
- Leak Detection Fluid, Item 189, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitrogen, Item 221, Appendix D

References

- Appendix D
- PARA 1-3-29
- PARA 5-2-2
- PARA 5-4-36

5-4-37.1 Test Components for Leakage.

NOTE

- Perform these procedures following an unsafe BIM® indication.
- The procedures outlined in this task are to be used with and in the order of, the instructions in the Bench procedures in PARA 5-2-2.
- This procedure contains instructions for testing for leakage at the BIM® pressure indicator valve and packing, main rotor blade air valve, root end plate seal, and main rotor blade cuff/spar bolts.
- Maximum pressure in spar not to exceed 18 psig.

a. Using nitrogen, Item 221, Appendix D, pressurize main rotor blade spar to 18 psig (PARA 1-3-29).

b. Make sure that outer hexnut on main rotor blade air valve is torqued and lockwired correctly (PARA 1-3-29).

NOTE

- A significant leak should be readily determined by this test. However, a slow leak may be more difficult to find. Should there be any doubt that the source of leak has been found, send main rotor blade to manufacturer for further tests.
- To prevent false indications of leakage, make certain that bubbles formed during application of liquid are allowed to settle out.

c. Using test liquid, such as leak detection fluid, Item 189, Appendix D, or dishwashing compound, Item 122, Appendix D, check the following areas for leakage:

(1) Spread test liquid over and around base of BIM® pressure indicator and around test valve under handle.

(2) Spread test liquid over and around main rotor blade air valve, especially stem area.

(3) Spread test liquid over and around entire edge of root endplate.

(4) Spread test liquid over and around each bolthead and nut that secures main rotor blade cuff to main rotor blade spar.

d. Watch carefully at all of these areas for bubbles to form.

e. When check is completed and before test liquid dries, using clean, fresh water, clean area thoroughly to remove all traces of test liquid.

f. Using machinery towel, Item 211, Appendix D, dry main rotor blade.

g. If no leakage is found, perform main rotor blade one hour whirl test (PARA 5-4-36).

h. Service main rotor blade to "Maximum When Servicing Blade" pressure for actual measured spar temperature (PARA 1-3-29).

i. If repairable leakage is found, repeat step a. through step g. after completion of repair.

j. If nonrepairable leakage is found, forward the damaged main rotor blade to manufacturer. A main rotor blade component record card entry should be made whenever a main rotor blade is removed from service stating reason for removal.

k. If a main rotor blade is to be forwarded to manufacturer it must have with it a statement that describes accurately the specific maintenance action, or actions that were performed. Statement should include:

- Spar pressure that existed when main rotor-blade was inducted and when it left.
- Specific reason for forwarding main rotor-blade to manufacturer.
- All data, including identification of original-problem, should also be with main rotor-blade.

5-4-38 TAIL ROTOR BLADES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- C-Clamps (4-inch)
- Caliper Set
- Clecos, ⅛-inch diameter
- Countersink, 100°
- Cup, Large Enough to Prepare Adhesive
- Drill
- Drill Bit, ⅛-inch diameter
- Drill Bit, No. 30, 0.1285-inch diameter
- Drill Bit, No. 50, 0.070-inch diameter
- Drill Bit, No. 60
- Flashlight
- Fluorescent Penetrant Inspection Kit
- Grease Pencil
- Heat Gun
- Heat Lamps
- Hoist, Capacity to Support Tail Rotor Blade (Alternate for Tail Rotor Maintenance Crane)
- Hoisting Sling, 70700-20320-042
- Inspection Mirror
- Magnifying Glass, 10X
- Main and Tail Rotor Blade Adapter, 70700-20426-041
- Maintenance Trailer
- NO SMOKING Signs
- Outside Micrometer
- Pin Punch, 3⁄32-inch diameter
- Plastic Tool, Thin (or equivalent)

- Pneumatic Blind Riveter
- Pressure Strips, Metal or Wood
- Protection, Eye, Skin, and Breathing
- Single Cut Mill File
- Small Brush
- Socket, Deep Well
- Spray Applicator
- Straight Edge, Wood or Plastic
- Tail Rotor Blade Installation Checking Shim ″A″, Locally-Made, Figure H-208, Appendix H
- Tail Rotor Blade Installation Checking Shim ″B″, Locally-Made, Figure H-112, Appendix H
- Tool, Small, Thin
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Wooden Wedges, Locally-Made

Materials/Parts

- Abrasion Resistant Tape, Item 1, Appendix D
- Abrasive Cloth, Item 1A, Appendix D (or equivalent)
- Abrasive Cloth, Item 4, Appendix D
- Abrasive Cloth, Item 6, Appendix D
- Abrasive Cloth, Item 7, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Abrasive Paper, Item 16, Appendix D
- Abrasive Paper, Item 18, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Acetone, Item 25, Appendix D
- Adhesion Promoter, Item 25A, Appendix D
- Adhesive, Item 32, Appendix D
- Adhesive, Item 42, Appendix D
- Alcohol, Denatured, Item 70, Appendix D
- Alcohol, Isopropyl, Item 72, Appendix D
- Applicator, Item 78, Appendix D (or equivalent)
- Brush, Acid, Item 84, Appendix D
- Cheesecloth, Item 90, Appendix D
- Cloth, Satin, Item 105, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dishwashing Compound, Item 122, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Heavy Paper (or equivalent protective covering)
- Lacquer, Item 183, Appendix D
- Lacquer, Item 184, Appendix D
- Lubricant, Solid Film, Item 201, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Methyl Isobutyl Ketone, Item 218, Appendix D
- Paint, Polyurethane, Item 223, Appendix D
- Paint, Polyurethane, Item 227, Appendix D
- Paint Remover, Item 230, Appendix D
- Plastic Sheet, Item 235, Appendix D

- Plastic Sheet, Item 235A, Appendix D
- Polyurethane Tape, Abrasion Resistant, Item 241A, Appendix D
- Polyurethane Tape, Abrasion Resistant, Item 241B, Appendix D
- Polyurethane Tape, Item 241C, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D
- Stirring Stick, Beverage, Item 315C, Appendix D
- Tags
- Toluene, Item 339, Appendix D
- Tongue Depressor (or equivalent)
- Toothpick

References

- Appendix D
- Appendix H
- Appropriate MTF
- ASTM E1417
- MIL-C-46168
- MIL-L-19538
- PARA 1-7-7
- PARA 2-6-2
- PARA 2-6-5
- PARA 5-3-12
- PARA 5-3-13
- PARA 5-4-39
- PARA 5-4-40
- PARA 5-4-41
- PARA 5-4-43
- PARA 5-5-16
- PARA 5-5-17
- PARA 5-5-18
- PARA 5-5-19
- PARA 11-2-3
- PARA 11-4-113
- PARA 12-4-52
- PARA 18-4-2
- TM 1-70-VIB

5-4-38.1 Replace Tail Rotor Blades.

5-4-38.1.1 Remove.

a. If tail rotor maintenance crane will be used, assemble and install tail rotor maintenance crane (PARA 18-4-2).

b. If necessary, install main and tail rotor blade adapter on maintenance trailer.

c. Remove tail rotor pitch control rods (PARA 5-4-39).

d. Remove tail rotor pitch beam (PARA 5-4-40).

e. Using 1-inch masking tape, Item 212, Appendix D, protect threads of pitch control shaft.

f. If blade deice kit is installed, disconnect blade deice electrical connectors (PARA 12-4-52).

NOTE

If tail rotor blades being removed will be reinstalled on same tail gear box, and color bands have worn off, index marks must be applied on tail rotor blades, retention plates, tail rotor pitch control rods, and tail rotor pitch beam so they are reinstalled in the same clockwise and inboard/outboard position, to minimize rebalancing of tail rotor.

g. If necessary, using grease pencil, apply index marks on tail rotor blades, retention plates, tail rotor pitch control rods, and tail rotor pitch beam to facilitate proper installation.

CAUTION

Damage to outboard retention plate will result if wrenches are used to remove bolts. Use deep well socket when removing balance weight bolts.

NOTE

- If balance weights are installed with bolt through both retention plates, note location of hole on outboard retention plate. Keep all washers, bolt, and self-locking nut together for that specific hole. Do not remove balance weights installed with screw and self-locking nut in one retention plate, or on tail rotor blade.
- Keep bolts, washers, and self-locking nuts for reinstallation.

h. If installed, remove bolts, washers, and self-locking nuts securing balance weights to retention plates (Figure 5-4-38-1, Detail A). Tag hardware and note location of hole from which hardware was removed.

CAUTION

Damage to outboard retention plate will result if wrenches are used to remove bolts. Use deep well socket when removing outboard retention plate bolts.

NOTE

Keep bolts, countersunk washers, washers, shims and spacers (if installed), and self-locking nuts for reinstallation.

i. Remove bolts, countersunk washers, washers, shims, spacers (if installed), and self-locking nuts securing outboard retention plate to inboard retention plate and tail rotor blade deice slipring rotor (if installed). Remove outboard retention plate.

j. Using hoisting sling, remove outboard and inboard tail rotor blades individually as follows:

NOTE

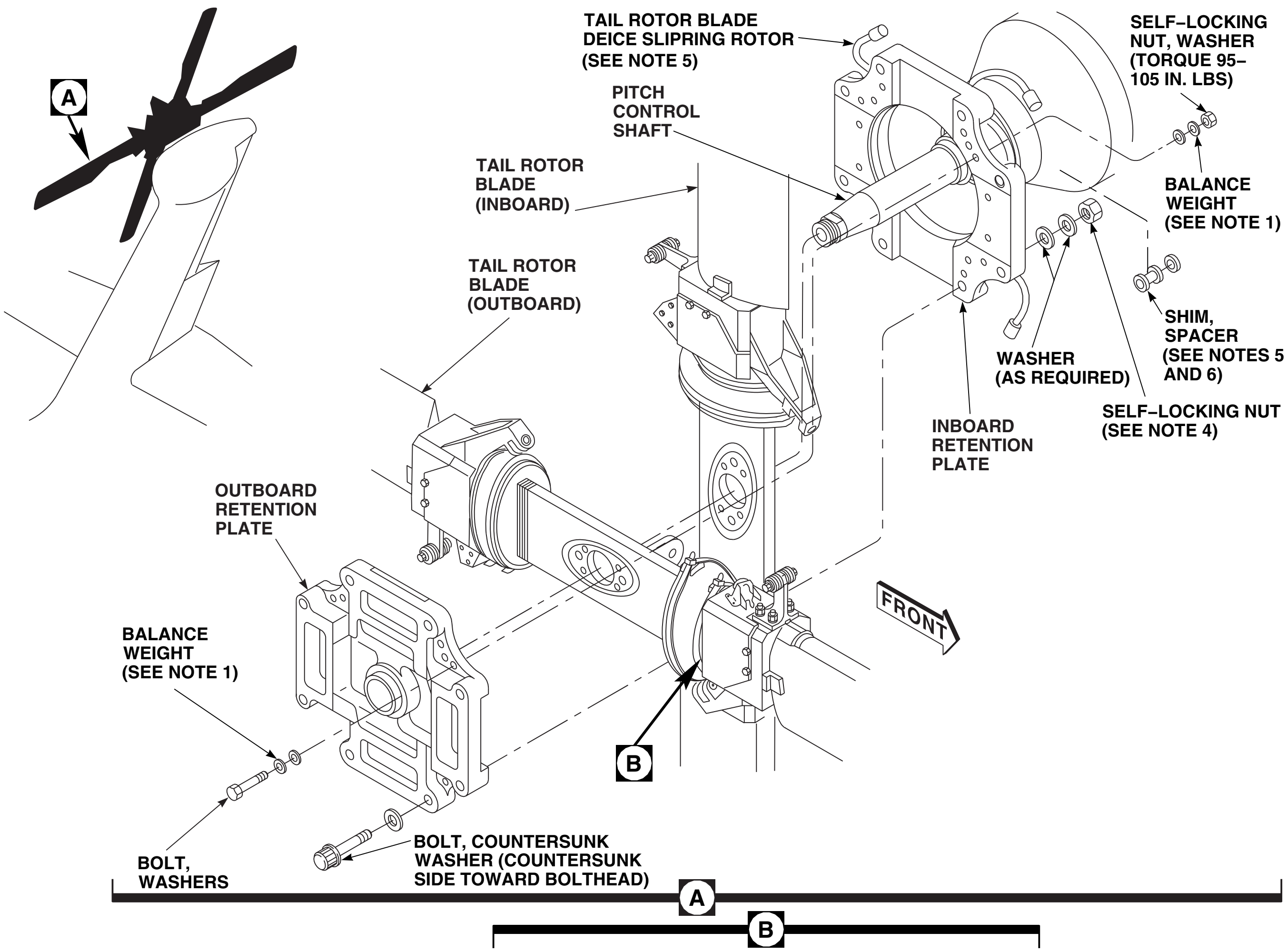
Outboard tail rotor blade must be removed first.

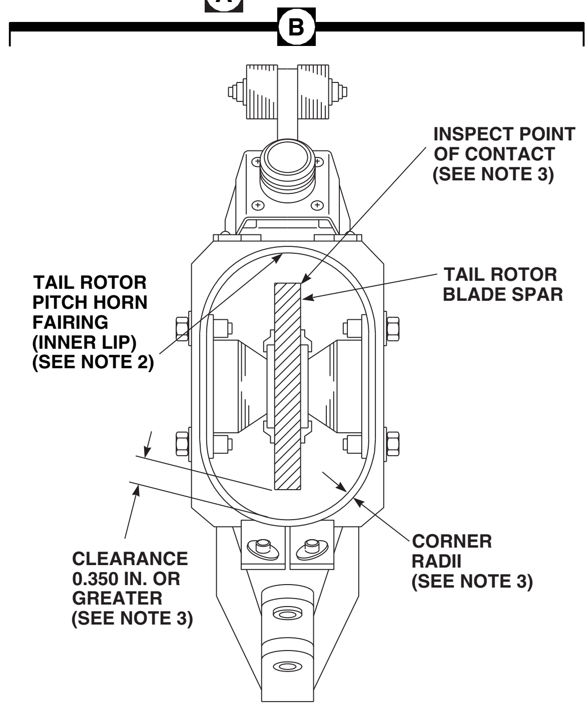
(1) Place straps around tail rotor blade.

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

(2) Attach hoisting sling to hoist or tail rotor maintenance crane, and adjust straps.





NOTES

1. IF INSTALLED, BALANCE WEIGHTS MAY BE AT ANY OF 4 LOCATIONS ON RETENTION PLATES.
2. TAIL ROTOR BOOT NOT SHOWN FOR CLARITY.
3. CHECK ALL FOUR CORNERS.
4. TORQUE IN CRISSCROSS PATTERN TO 209–231 IN. LBS, THEN 418–462 IN. LBS, THEN TO 627–693 IN. LBS.
5. ON HELICOPTERS WITH BLADE DEICE KIT INSTALLED.
6. SHIM AND SPACER INSTALLED AT LOCATIONS WHERE TAIL ROTOR BLADE DEICE SLIPRING ROTOR IS CONNECTED.

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FIGURE 5-4-38-1. REPLACE TAIL ROTOR BLADES

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting tail rotor blade. Keep area clear when hoisting tail rotor blade.

CAUTION

Damage to tail rotor blade and pitch control shaft will result if excessive force is used to remove tail rotor blades. Guide tail rotor blade carefully over pitch control shaft.

(3) Remove and lower tail rotor blade from pitch control shaft and lower tail rotor blade on main and tail rotor blade adapter.

(4) Remove hoisting sling.

(5) Repeat step j. (1) through step j. (4) for inboard tail rotor blade.

5-4-38.1.2 Inspect Tail Rotor Boot.

Inspect tail rotor boot for cuts, tears, breaks, and dry rot. **NONE ALLOWED.** If cuts, tears, breaks, or dry rot are found, replace tail rotor boot (PARA 5-4-41).

5-4-38.1.3 Inspect Tail Rotor Blade Deice Bracket.

Inspect tail rotor blade deice bracket for cracks. **NONE ALLOWED.** If cracks are found, replace tail rotor blade deice bracket (PARA 5-4-43).

5-4-38.1.4 Inspect Tail Rotor Blade Spar.

a. Remove one of the tail rotor boot ties and roll back tail rotor boot (PARA 5-4-41).

b. Using caliper set, measure clearance between inner lip of tail rotor pitch horn fairing and tail rotor blade spar (Figure 5-4-38-1, Detail B).

c. Record minimum clearance at leading and trailing edges. **MINIMUM CLEARANCE MUST NOT BE LESS THAN 0.350-INCH.**

d. If clearance between inner lip of tail rotor pitch horn fairing and tail rotor blade spar is less than minimum acceptable limits, rework tail rotor pitch horn fairing as follows:

(1) Using single cut mill file, remove material from inner lip of tail rotor pitch horn fairing at all four corner radii, and leading and trailing edges. **INNER LIP SHOULD BE 0.060 - 0.100-INCH BELOW INSIDE TAIL ROTOR PITCH HORN FAIRING SURFACE.**

(2) Using straight edge, measure along entire perimeter of corner radii.

e. Remove tail rotor pitch horn fairing from tail rotor blade assembly (PARA 5-5-17). Inspect for delaminations and splinters on tail rotor blade spar in areas near inner lip of tail rotor pitch horn fairing. **NONE ALLOWED.** If any signs of delaminations or splinters are found, replace tail rotor blade assembly.

f. Prior to sealing installation hardware, recheck clearance between inner lip of tail rotor pitch horn fairing and tail rotor blade spar. If clearance is below minimum acceptable limit, remove tail rotor pitch horn fairing and rework (PARA 5-5-17).

g. If no sign of delaminations or splinters are found, measure depth of contact damage. **MAXIMUM CONTACT DAMAGE DEPTH ALLOWED ON TAIL ROTOR BLADE SPAR IS 0.070-INCH.**

h. Measure depth of damage to tail rotor blade spar as follows:

(1) Place No. 50 (0.070-inch diameter) drill bit into damage, and place straight edge perpendicular to drill bit.

(2) If straight edge touches drill bit, damage is less than 0.070-inch deep. Using next smaller drill bit, repeat measurement.

Continue repeating measurements with a progressively smaller drill bit until straight edge touches both adjacent sides without touching drill bit.

(3) If straight edge touches No. 60 drill bit when placed in damage, discontinue measuring.

5-4-38.1.5 Inspect Tail Rotor Boot Supports.

a. Inspect tail rotor boot supports for displacement from their proper position. **NONE ALLOWED.** If displacement is found, replace tail rotor boot supports (PARA 5-5-19).

b. Inspect tail rotor boot supports for looseness using hand pressure. **NONE ALLOWED.** If looseness is found, replace tail rotor boot supports (PARA 5-5-19).

c. Inspect inboard and outboard tail rotor boot support adhesive fillets adjacent to tail rotor blade spar for cracks. Repair cracks in adhesive per adhesive injection repair instructions (PARA 5-4-38.3).

d. Apply sealing compound, Item 292, Appendix D, to inboard and outboard adhesive fillets as follows:

(1) Using cheesecloth, Item 90, Appendix D, dampened with isopropyl alcohol, Item 72, Appendix D, clean inboard and outboard adhesive fillets.

CAUTION

Damage to repair may result if sealing compound bead comes in contact with inboard and outboard retention plates. When applying sealing compound to 0.10-inch adhesive fillet, make sure sealing compound bead and adhesive fillet combined is ¼-inch maximum.

(2) Apply sealing compound, Item 292, Appendix D, to adhesive fillets and edge area (Figure 5-4-38-2).

(3) Make sure sealing compound beads do not interfere with retention plate during tail rotor blade installation.

5-4-38.1.6 Inspect Tail Rotor Blade Pivot Bearing.

a. Using flashlight and inspection mirror, inspect tail rotor blade pivot bearing for delaminations, and bearing retainer for disbond, as follows:

(1) Inspect tail rotor blade pivot bearings for separations within rubber layers. **IF SEPARATION EXCEEDS ⅛-INCH DEEP AROUND ¼ OF CIRCUMFERENCE, REPLACE TAIL ROTOR BLADE PIVOT BEARING** (PARA 5-5-16).

(2) Inspect tail rotor blade pivot bearings for eraser-type extrusion. **NONE ALLOWED.** If extrusions are found, replace tail rotor blade pivot bearing (PARA 5-5-16).

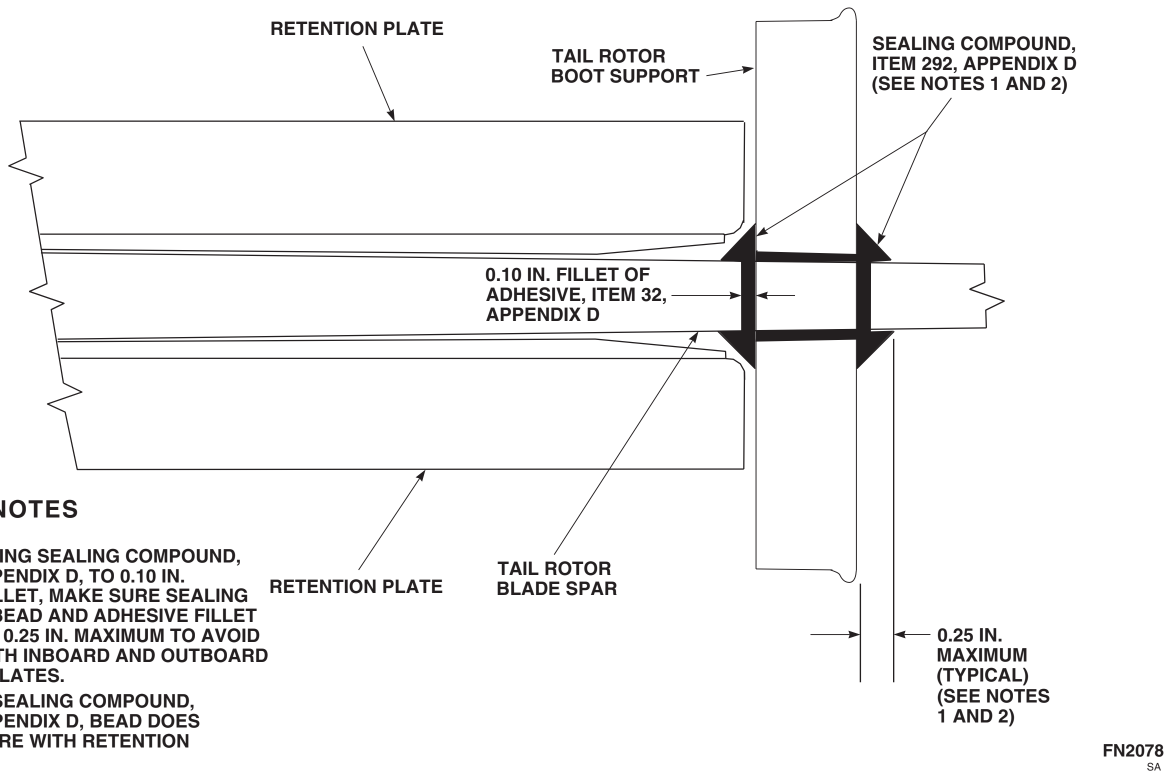
(3) Inspect tail rotor blade pivot bearings for rubber-to-shim separation. **NONE ALLOWED. BENT SHIMS ARE ACCEPTABLE.** If shim separations are found, replace tail rotor blade pivot bearings (PARA 5-5-16).

(4) Inspect tail rotor blade pivot bearing retainers for disbonding. **NONE ALLOWED.** If any tail rotor blade pivot bearing retainer is disbonded (four tail rotor blade pivot bearing retainers per tail rotor blade), replace tail rotor blade pivot bearing retainer (PARA 5-5-18).

(5) If no visible disbond of tail rotor blade pivot bearing retainer or delamination of tail rotor blade pivot bearing is detected, perform the following:

(a) Press directly on tail rotor pitch horn over tail rotor blade pivot bearing attachment bolts on each side of tail rotor blade.

(b) Inspect for looseness of tail rotor pitch horn relative to tail rotor blade



NOTES

1. WHEN APPLYING SEALING COMPOUND, ITEM 292, APPENDIX D, TO 0.10 IN. ADHESIVE FILLET, MAKE SURE SEALING COMPOUND BEAD AND ADHESIVE FILLET COMBINED IS 0.25 IN. MAXIMUM TO AVOID CONTACT WITH INBOARD AND OUTBOARD RETENTION PLATES.
2. MAKE SURE SEALING COMPOUND, ITEM 292, APPENDIX D, BEAD DOES NOT INTERFERE WITH RETENTION PLATES.

FIGURE 5-4-38-2. APPLICATION OF SEALING COMPOUND TO TAIL ROTOR BOOT SUPPORT

spar. **NONE ALLOWED.** If any looseness is found, remove pivot bearings and inspect pivot bearings and retainers (PARA 5-5-16).

b. Using flashlight and inspection mirror, inspect tail rotor blade pivot bearing baseplate for cracks. Typical cracks run spanwise from near edge of baseplate towards tip of tail rotor blade. **NONE ALLOWED.** If crack is suspected, remove and inspect tail rotor blade pivot bearing (PARA 5-5-16).

c. Using flashlight and inspection mirror, inspect inside surfaces of tail rotor pitch horn, tail rotor blade pivot bearing retainer, and tail rotor blade pivot bearing baseplates for missing or damaged primer. **NONE ALLOWED.** If damaged or missing primer is found, remove tail rotor blade pivot bearing and replace primer (PARA 5-5-16).

d. Secure tail rotor boot (PARA 5-4-41).

5-4-38.1.7 Inspect Tail Rotor Pitch Horn.

a. Visually inspect tail rotor pitch horn for cracks, paying particular attention to upper, lower, and trailing edge faces of box section as follows:

(1) Using machinery towel, Item 211, Appendix D, dampened with isopropyl alcohol, Item 72, Appendix D, clean surface of tail rotor pitch horn.

(2) Using 10X magnifying glass, visually inspect painted surfaces for cracks. If crack is suspected, perform the following:

(a) Using paint remover, Item 230, Appendix D, strip paint from suspected area.

(b) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect tail rotor pitch horn for cracks (ASTM E1417). **NONE AL-**

LOWED. If cracks are found, replace tail rotor blade.

(3) Using corrosion resistant coating, Item 118, Appendix D, touch up surface of tail rotor pitch horn. Allow to remain on surface for 3 - 5 minutes.

(4) Using cold water, rinse surface of tail rotor pitch horn. Allow to dry.

(5) Using epoxy primer coating kit, Item 135, Appendix D, apply one coat of epoxy primer to surface of tail rotor pitch horn.

(6) Touch up paint on tail rotor pitch horn (PARA 2-6-5).

b. Inspect tail rotor pitch horn for nicks and gouges. Using abrasive wheel, Item 22, Appendix D, remove raised edges and blend out damage. **REPLACE TAIL ROTOR BLADE IF:**

NOTE

Torque tube attachment area is where tail rotor pitch horn attaches to tail rotor blade.

(1) **CORNER DAMAGE IN TAIL ROTOR PITCH HORN AFTER BLENDING EXCEEDS 0.020-INCH DEEP OR 0.010-INCH DEEP IN LUG AND TORQUE TUBE ATTACHMENT AREAS.**

(2) **SURFACE DAMAGE IN TAIL ROTOR PITCH HORN AFTER BLENDING EXCEEDS 0.010-INCH DEEP.**

(3) **NICKS AND GOUGES IN TAIL ROTOR PITCH HORN MATING SURFACES ARE GREATER THAN 0.010-INCH DEEP, OR COVER MORE THAN 10% OF MATING SURFACE.**

c. Inspect tail rotor pitch horn for corrosion. **REPLACE TAIL ROTOR BLADES IF:**

(1) **TAIL ROTOR PITCH HORN HAS CORROSION EXCEEDING 0.010-INCH DEEP.**

(2) **CORRODED AREAS ARE LESS THAN 1-INCH APART.**

d. Inspect tail rotor pitch horn fairing for edge disbonds. Apply sealing compound, Item 292, Appendix D, to rebond edges.

e. Inspect bushing bore and flange surface for corrosion. Remove corrosion from, inspect, and replace tail rotor pitch horn as required.

f. Refer to PARA 5-3-12 for damage limits for other components of tail rotor system.

g. Inspect tail rotor blade for damage (PARA 5-3-13).

5-4-38.1.8 Inspect/Repair Retention Plate Bolts.

a. Visually inspect solid film lubricant on outboard retention plate bolt. Bolt shank must be completely covered with solid film lubricant. **NO SIGNS OF BASE METAL ALLOWED.** If solid film lubricant is missing, perform the following:

(1) Inspect bolt shank for any material displaced above surface of bolt. Using abrasive cloth, Item 10, Appendix D, remove displaced material.

(2) Using outside micrometer, inspect bolt shank diameter. **MINIMUM BOLT SHANK DIAMETER IS 0.4970-INCH.** If shank diameter is less than limit, replace bolt.

(3) If solid film lubricant is missing and bolt shank diameter is within limits, using fluorescent penetrant inspection kit, fluorescent penetrant inspect bolts for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace bolts.

b. Using abrasive paper, Item 18, Appendix D, blend out nicks and gouges smoothly to ½-inch radius. Using abrasive paper, Item 18, Appendix D, polish out sanding marks.

c. Using abrasive paper, Item 18, Appendix D, remove corrosion by lightly sanding surface until corrosion is removed.

d. If outboard retention plate bolt shank is missing solid film lubricant or has been reworked and is still within acceptable limits, reapply solid film lubricant, Item 201, Appendix D, as follows:

(1) Using dry machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean areas of exposed base metal. Allow to dry thoroughly.

(2) At room temperature, using solid film lubricant, Item 201, Appendix D, spray exposed area. Apply multiple thin coats allowing 15 - 20 minutes drying time between coats.

(3) Allow solid film lubricant to cure a minimum of 12 hours before using bolt. **REPLACE BOLT IF:**

(a) **COATING IS NOT ″SLATE-BLACK″ AND UNIFORM IN COLOR.**

(b) **COATING HAS ANY RUNS, BLISTERS, ROUGHNESS, OR SEPARATION OF INGREDIENTS.**

(c) **COATING HAS ANY OTHER SURFACE IMPERFECTIONS.**

e. After solid film lubricant has cured, check solid film lubricant repair as follows:

(1) Firmly press strip of 1-inch masking tape, Item 212, Appendix D, onto surface of touched-up area.

(2) Remove masking tape with an abrupt motion.

(3) If solid film lubricant leaves a uniform deposit of powdery material clinging to tape, repair is acceptable. Proceed to step f.

(4) **IF SOLID FILM LUBRICANT FLAKES OR LEAVES ANY BARE METAL EXPOSED, REPLACE BOLT.**

f. If solid film lubricant repair is successful, reinstall bolts.

5-4-38.1.9 Install.

NOTE

- If tail rotor blades being installed are the same ones that were removed, make certain they are installed in the same clockwise and inboard/outboard position, to minimize rebalancing of tail rotor.

- To prevent additional tail rotor balancing, match color bands or index marks on tail rotor blades, retention plates, tail rotor pitch control rods, and tail rotor pitch beam during installation.

- A complete tail rotor system rig check is required whenever changes in tail rotor flight controls are affected by removal/installation of ″BLADES″, which requires removal of tail rotor pitch control rods, tail rotor pitch beams, etc.

- Tail rotor blades, 70101-11200-042 through 70101-11200-044, are individually interchangeable with each other.

- Tail rotor blades with erosion protection kits installed must not be mixed with tail rotor blades that do not have erosion protection kits installed. Tail rotor balance and vibration levels will be adversely affected.

- When shim ″A″ is made from 0.001-inch thick stock, use 70102-11109-101 as shim ″B″. When shim ″A″ is made from 0.002-inch thick stock, use 70102-11109-102 as shim ″B″.

a. Locally make tail rotor blade installation checking shims ″A″ (Figure H-208, Appendix H) and ″B″ (Figure H-112, Appendix H).

NOTE

Make sure dust plug is installed on pitch control shaft.

b. Visually inspect exposed portion of pitch control shaft (PARA 11-4-113).

c. Place straps around inboard tail rotor blade. Attach hoisting sling to hoist and adjust straps.

d. Make sure pitch control shaft threads are protected with 1-inch masking tape, Item 212, Appendix D.

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting inboard tail rotor blade. Keep area clear when hoisting inboard tail rotor blade.

CAUTION

Damage to tail rotor blade will result if inboard tail rotor blade is dragged across pitch control shaft. Guide inboard tail rotor blade carefully over pitch control shaft.

NOTE

During assembly, make sure all inboard tail rotor blade leading edges are facing proper direction.

e. Lift inboard tail rotor blade into place. Place inboard tail rotor blade into deep slot of inboard retention plate (Figure 5-4-38-1, Detail A).

f. Remove hoisting sling.

g. Place straps around outboard tail rotor blade. Attach hoisting sling to hoist and adjust straps.

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting outboard tail rotor blade. Keep area clear when hoisting outboard tail rotor blade.

CAUTION

Damage to outboard tail rotor blade will result if outboard tail rotor blade is dragged across pitch control shaft. Guide outboard tail rotor blade carefully over pitch control shaft.

NOTE

During assembly, make sure all outboard tail rotor blade leading edges are facing proper direction.

h. Lift outboard tail rotor blade into place.

i. Remove hoisting sling.

j. Visually inspect outboard retention plate and make sure dust seal is installed. Line up index marks on inboard and outboard retention plates. Outboard tail rotor blade should line up with deep slot in outboard retention plate.

NOTE

If original inboard retention plate, outboard retention plate, tail rotor blades, and tail gear box will be installed, proceed to step l.

k. If a new or different inboard retention plate, outboard retention plate, tail rotor blades, or tail gear box has been installed, perform tail rotor clamp-up check procedure as follows:

(1) Temporarily install shims ″A″ (8 places) between outboard retention plate and tail rotor blade spars (Figure 5-4-38-3, Detail A and Detail B).

(2) Temporarily install shims ″B″ (4 places) between outboard and inboard retention plate flanges. Make sure shims ″B″ do not contact tail rotor blades (Figure 5-4-38-3, Detail B).

CAUTION

Damage to outboard retention plate will result if wrenches are used to install bolts. Use deep well socket when installing outboard retention plate bolts.

(3) With shims ″A″ and ″B″ in place, install bolts, countersunk washers, washers, shims, and spacers (if installed), and self-locking nuts securing retention plate halves and tail rotor blade deice slipring rotor (if installed) together. Make sure countersunk side of washer is toward bolthead. Snug up self-locking nuts evenly (Figure 5-4-38-3, Detail C).

(4) **TORQUE SELF-LOCKING NUTS IN CRISSCROSS PATTERN TO 209 - 231 INCH-POUNDS.**

(5) **TORQUE SELF-LOCKING NUTS IN CRISSCROSS PATTERN TO 418 - 462 INCH-POUNDS.**

(6) **TORQUE SELF-LOCKING NUTS IN CRISSCROSS PATTERN TO A FINAL TORQUE OF 627 - 693 INCH-POUNDS.**

(7) Verify that 0.080 - 0.125-inch of bolt extends beyond self-locking nut. Add or subtract washers under self-locking nut, as required, to correct dimension.

(8) Shims ″A″ should remain in place when normal finger-force is applied to remove them. If shims ″A″ remain in place, loosen retention plate self-locking nuts and remove shims ″A″ and shims ″B″. Proceed to step m.

(9) If shims ″A″ do not remain in place, loosen retention plate bolts and repeat step k. (1) through step k. (8).

(10) If shims ″A″ still do not remain in place, perform the following:

(a) Inspect tail rotor blade thickness in area of nylon wrap. **THICKNESS SHOULD BE 0.719-INCH MINIMUM.**

(b) Inspect thickness of nylon pads of inboard and outboard retention plates. **THICKNESS SHOULD BE 0.123-INCH MINIMUM.**

(c) Inspect spar channels of inboard and outboard retention plates with nylon wrap removed. **MAXIMUM DEPTH MUST NOT EXCEED 0.821-INCH (DEEP CHANNEL), AND MAXIMUM DEPTH MUST NOT EXCEED 0.141-INCH (SHALLOW CHANNEL).**

(d) If above requirements are exceeded, replace necessary component.

(e) Repeat step k. (1) through k. (8).

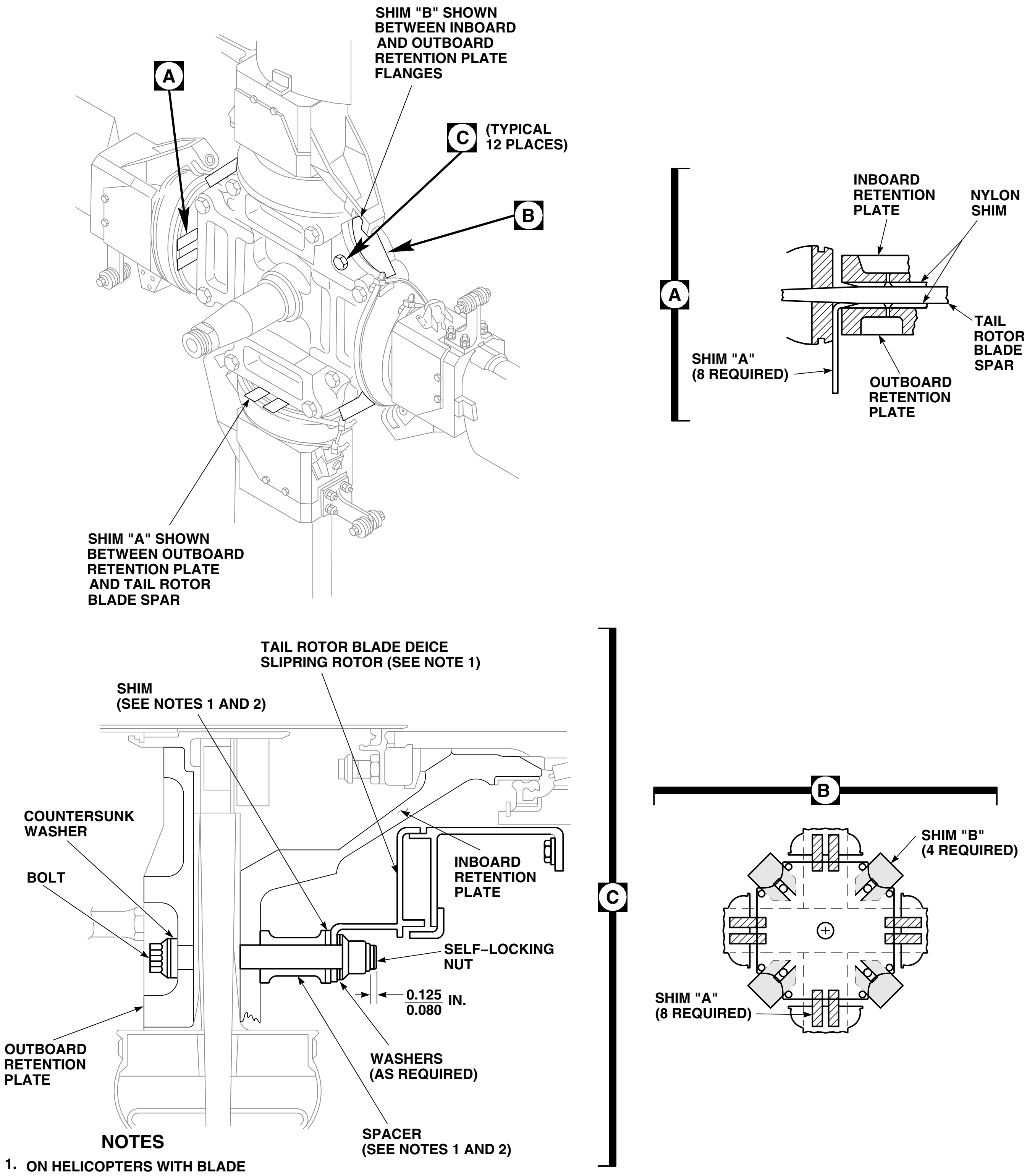
l. If original inboard retention plate, outboard retention plate, tail rotor blades, and tail gear box are being installed, install bolts, countersunk washers, washers, shims, and spacers (if installed) securing retention plate halves and tail rotor blade deice slipring rotor (if installed) together. Make sure countersunk side of washer is toward bolthead.

m. Check friction torque on self-locking nuts as follows:

(1) Install self-locking nuts on bolts.

(2) **TIGHTEN SELF-LOCKING NUTS FINGERTIGHT.**

(3) Using torque wrench, turn self-locking nut in tightening direction. **TORQUE MUST BE GREATER THAN 18 INCH-POUNDS BEFORE MORE THAN 0.080 - 0.125-INCH OF BOLT EXTENDS BEYOND SELF-LOCKING NUT.**



NOTES

1. ON HELICOPTERS WITH BLADE DEICE KIT INSTALLED.
2. SHIM AND SPACER INSTALLED AT LOCATIONS WHERE TAIL ROTOR BLADE DEICE SLIPRING ROTOR IS CONNECTED.

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FIGURE 5-4-38-3. SHIM TAIL ROTOR BLADES

(4) **IF TORQUE IS LESS THAN 18 INCH-POUNDS, REPLACE SELF-LOCKING NUT.**

CAUTION

Damage to outboard retention plate will result if wrenches are used when torquing hardware. Use deep well sockets when torquing hardware.

n. **TORQUE SELF-LOCKING NUTS IN CRISSCROSS PATTERN TO 209 - 231 INCH-POUNDS.**

o. **TORQUE SELF-LOCKING NUTS IN CRISSCROSS PATTERN TO 418 - 462 INCH-POUNDS.**

p. **TORQUE SELF-LOCKING NUTS IN CRISSCROSS PATTERN TO A FINAL TORQUE OF 627 - 693 INCH-POUNDS.**

q. Verify that 0.080 - 0.125-inch of bolt extends beyond self-locking nut. Add or subtract washers under self-locking nut as required to get correct dimension.

r. If blade deice kit is installed, check for proper blade deice slipring brush to slipring rotor contact (PARA 12-4-52).

CAUTION

Damage to outboard retention plate will result if wrenches are used when torquing hardware. Use deep well sockets when torquing hardware.

NOTE

- Balance weights must be stacked on both sides of retention plates.
- Make sure washer, NAS1149C0463R, is between balance weight and retention plate on both sides.

s. If required, remove tags and install balance weights in assigned holes with bolt, washers, and self-locking nut.

t. **TORQUE SELF-LOCKING NUT TO 95 - 105 INCH-POUNDS.**

u. Install tail rotor pitch beam (PARA 5-4-40).

v. Install tail rotor pitch control rods (PARA 5-4-39).

w. If blade deice kit is installed, connect blade deice electrical connectors (PARA 12-4-52).

x. If used, disassemble and remove tail rotor maintenance crane (PARA 18-4-2).

y. Make sure area is clean and free of foreign material.

z. Perform operational check of tail rotor flight control system (PARA 11-2-3).

NOTE

Certain work performed on tail rotor system requires a balance check to be completed prior to flight. This includes the following:

- Removal of outboard retention plate
- Removal of tail rotor blades
- Skin repairs to tail rotor blades
- Any other work performed to tail rotor system that may affect tail rotor balance

aa. Perform tail rotor balance check (TM 1-70-VIB).

ab. Perform maintenance test flight (Appropriate MTF).

NOTE

Inboard retention plate nut torque must be stabilized prior to performing outboard retention plate bolt and tail rotor pitch beam retaining nut torque check.

ac. Check helicopter logbook to make sure outboard retention plate nut and tail rotor pitch beam retaining nut torque checks are shown due 9 - 11 flight hours after installation (PARA 1-7-7).

5-4-38.2 Adhesive Filler Repair.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wash hands and face before eating or smoking.
- Refer to Table 5-3-13-1 for reference to damage limits and repair procedures on main rotor blade.
- Perform all repairs carefully and according to instructions. Even small repairs affect tail rotor blade track and balance.

a. Using masking tape, mask repair area around damage ½-inch larger than damage outline.

CAUTION

Damage to tail rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints or inside of tail rotor blade. Do not allow solvent to seep into bonded joints or inside of tail rotor blade. Machinery towels wet with methyl ethyl ketone should not be left lying on tail rotor blade.

b. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, remove paint inside repair area.

CAUTION

Damage to tail rotor blade will result if tail rotor blade is sanded excessively. Sand only until yellow primer is removed.

c. Using abrasive cloth, Item 7, Appendix D, sand yellow primer from repair area.

d. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean sanded area. Wipe dry before methyl ethyl ketone evaporates.

NOTE

- Adhesive, Item 32, Appendix D, is available in 32-gram, pre-measured packages or in bulk form. Either may be used. Prepare accordingly.
- Pot life of mixture is 25 minutes at 75°F (24°C).

e. Prepare adhesive, Item 32, Appendix D, either step e. (1) or step e. (2) as follows:

NOTE

Each 32-gram package is pre-measured with curing agent and resin. Mix full package and use as required. Discard leftover mixture.

(1) **32-Gram, Pre-Measured Package.**

(a) Empty package of adhesive resin into cup.

(b) Empty tube of curing agent into cup of adhesive resin.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

(2) **Bulk Adhesive.**

NOTE

Do not try to use all adhesive; use as required. Discard leftover mixture.

(a) Mix 100 parts of ″A″ part to 22 parts of ″B″ part by weight.

(b) Weigh each part then place both parts in cup.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

f. Using beverage stirring stick, Item 315C, Appendix D, tongue depressor (or equivalent), fill in repair area with adhesive.

NOTE

To reduce sanding, remove extra adhesive from repair area.

g. Using wood or plastic straight edge, bridge repair spanwise and scrape off extra adhesive in chordwise direction. Make sure level of adhesive is even with contour of skin.

h. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean area outside repair.

i. Allow adhesive to cure for 24 hours at 75°F (24°C).

j. Remove masking tape from repair area.

CAUTION

Damage to tail rotor blade will result if tail rotor blade is sanded excessively. Heavy sanding will damage skin of tail rotor blade. Use only enough force to blend repair into tail rotor blade.

k. Using abrasive cloth, Item 7, Appendix D, lightly sand edges of adhesive filler repair to blend repair to surrounding tail rotor blade skin.

l. Touch up paint (PARA 5-4-38.10).

5-4-38.3 Adhesive Injection Repair.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wash hands and face before eating or smoking.
- Refer to Table 5-3-13-1 for reference to damage limits and repair procedures on main rotor blade.
- Perform all repairs carefully and according to instructions. Even small repairs affect tail rotor blade track and balance.

a. Position tail rotor blade horizontally, with leading edge 2 - 3-inches higher than trailing edge.

b. Using denatured alcohol, Item 70, Appendix D, clean out separation.

NOTE

- Adhesive, Item 32, Appendix D, is available in 32-gram, pre-measured packages or in bulk form. Either may be used. Prepare accordingly.
- Pot life of mixture is 25 minutes at 75°F (24°C).

c. Prepare adhesive, Item 32, Appendix D, either step c. (1) or step c. (2) as follows:

NOTE

Each 32-gram package is pre-measured with curing agent and resin. Mix full package and use as required. Discard leftover mixture.

(1) **32-Gram, Pre-Measured Package.**

(a) Empty package of adhesive resin into cup.

(b) Empty tube of curing agent into cup of adhesive resin.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

(2) **Bulk Adhesive.**

NOTE

Do not try to use all adhesive; use as required. Discard leftover mixture.

(a) Mix 100 parts of ″A″ part to 22 parts of ″B″ part by weight.

(b) Weigh each part then place both parts in cup.

(c) Using beverage stirring stick, Item 315C, Appendix D, stir until mixture is free of streaks and color is uniform.

CAUTION

Damage to tail rotor blade skin will result if excessive force is used to open up separation in tail rotor blade. Do not apply too much pressure with plastic tool.

d. Using clean, thin plastic tool (or equivalent), carefully open up separation and apply adhesive to slot. Using small applicator, Item 78, Appendix D (or equivalent), force adhesive into separation.

e. Remove tool and finish filling slot with adhesive.

f. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean excess adhesive from repair area. Wipe dry before methyl ethyl ketone evaporates.

g. Using plastic sheet, Item 235A, Appendix D, cover repair area.

h. Using C-clamps, clamp flat wood pressure strips to sides of tabs with light but firm pressure.

i. Allow adhesive to cure for 8 hours at 75°F (24°C), or using heat lamps, 1 hour at 160°F (71°C).

j. Remove C-clamps and wood pressure strips and plastic sheet from adhesive injection repair.

k. Using abrasive cloth, Item 7, Appendix D, lightly sand edges of adhesive injection repair to blend repair to surrounding tail rotor blade skin.

l. Using cheesecloth, Item 90, Appendix D, dampened with denatured alcohol, Item 70, Appendix D, clean adhesive injection repair.

m. Touch up paint (PARA 5-4-38.10).

5-4-38.4 Polyurethane Abrasion Strip Repair (Primary Method).

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wash hands and face before eating or smoking.
- Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.
- Any repair in polyurethane abrasion strip on one tail rotor blade must be counterbalanced by same sized patch in same location on opposite tail rotor blade.

a. Remove all loose or torn pieces of polyurethane abrasion strip.

b. Fill in cavities or areas worn below contour of leading edge skin (PARA 5-4-38.2).

c. Using abrasive paper, Item 16, Appendix D, lightly sand edges of polyurethane abrasion strip repair to blend repair to surrounding contour of leading edge skin.

CAUTION

Damage to polyurethane abrasion strip will result if care is not taken when sanding polyurethane abrasion strip. Damage caused by sanding to polyurethane abrasion strip beyond repair area will reduce its life. Do not sand polyurethane abrasion strip beyond repair area.

d. Using abrasive paper, Item 16, Appendix D, lightly sand polyurethane abrasion strip on tail rotor blade where patch will overlap.

e. Using methyl ethyl ketone, Item 217, Appendix D, acetone, Item 25, Appendix D, or isopropyl alcohol, Item 72, Appendix D, clean area. Using dry machinery towel, Item 211, Appendix D, wipe dry before utilized cleaning agent evaporates.

f. Make flat pattern for new strip. **EDGES OF NEW POLYURETHANE ABRASION STRIP MUST BE AT LEAST EVEN WITH, BUT NOT TO EXCEED ⅛-INCH GREATER THAN EXISTING POLYURETHANE ABRASION STRIP. CORNERS MUST BE ROUNDED TO ½-INCH RADIUS.**

g. Install abrasion resistant polyurethane tape, Item 241A, Appendix D, or Item 241B, Appendix D, as follows:

NOTE

If necessary, adhesion promoter and/or wetting solution may be used to aid installation. If either or both will be used, perform step g. (1) and/or step g. (2) as necessary. If neither will be used, proceed to step g. (3).

(1) If necessary, using adhesion promoter, Item 25A, Appendix D, apply to nonmetal surfaces.

NOTE

A wetting solution may be used to aid installation and eliminate air bubbles.

(2) If necessary, apply wetting solution as follows:

(a) Mix wetting solution consisting of 75% water, 25% isopropyl alcohol, Item 72, Appendix D, and 20 drops per liter of dishwashing compound, Item 122, Appendix D.

(b) Using spray applicator, apply wetting solution to tail rotor blade and abrasion resistant polyurethane tape.

(3) Remove backing strip from abrasion resistant polyurethane tape, Item 241A, Appendix D, or Item 241B, Appendix D.

(4) If wetting solution is not used, apply abrasion resistant polyurethane tape as follows:

(a) Apply abrasion resistant polyurethane tape by pressing one spanwise edge of abrasion resistant polyurethane tape to prepared surface, making sure rest of abrasion resistant polyurethane tape does not touch tail rotor blade.

(b) Continue lightly smoothing abrasion resistant polyurethane tape across its width, applying abrasion resistant polyurethane tape to tail rotor blade over leading edge. Make sure abrasion resistant polyurethane tape is applied smoothly, with no air bubbles.

(c) Once abrasion resistant polyurethane tape is lightly applied, with firm hand-pressure, work out all wrinkles and air bubbles from installation, to obtain good adhesion to main rotor blade surface.

(5) If wetting solution was used, apply abrasion resistant polyurethane tape as follows:

(a) Position abrasion resistant polyurethane tape by sliding it, and rub wetting solution and any air bubbles out from under abrasion resistant polyurethane tape.

(b) Allow to cure for 24 hours.

h. Touch up paint (PARA 5-4-38.10).

5-4-38.5 Polyurethane Abrasion Strip Repair (Alternate Method).

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wash hands and face before eating or smoking.
- Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.
- Any repair in polyurethane abrasion strip on one tail rotor blade must be counterbalanced by same sized patch in same location on opposite tail rotor blade.

a. For full length repair, mask tail rotor blades as follows:

(1) Mask to area 1/4-inch beyond outboard end of existing strip.

(2) Mask to raised portion of deice heater mat.

(3) Mask area chordwise 1/4-inch beyond edges of existing strip.

b. For partial repairs, mask tail rotor blades as follows:

(1) Mask area at least 1-inch beyond either side (spanwise) of damage.

(2) Mask area chordwise ¼-inch beyond edges of existing strip.

(3) Mask opposite tail rotor blade in same locations.

c. Remove all loose or torn pieces of polyurethane abrasion strip.

d. Fill in cavities or areas worn below contour of leading edge skin per PARA 5-4-38.2.

e. Using abrasive paper, Item 16, Appendix D, lightly sand edges of polyurethane abrasion strip repair to blend repair to surrounding contour of leading edge skin.

CAUTION

Damage to polyurethane abrasion strip will result if care is not taken when sanding polyurethane abrasion strip. Damage caused by sanding to polyurethane abrasion strip beyond repair area will reduce its life. Do not sand polyurethane abrasion strip beyond repair area.

f. Using abrasive paper, Item 16, Appendix D, lightly sand polyurethane abrasion strip on tail rotor blade where patch will overlap.

g. Using methyl ethyl ketone, Item 217, Appendix D, clean areas. Using dry machinery towel, Item 211, Appendix D, wipe dry before methyl ethyl ketone evaporates.

h. Apply thin coat of adhesive, Item 42, Appendix D, to cover cleaned area.

i. Allow adhesive to air cure for 15 - 20 minutes.

j. Make flat pattern for new strip. **EDGES OF NEW POLYURETHANE ABRASION STRIP MUST BE AT LEAST EVEN WITH, BUT NOT TO EXCEED ⅛-INCH GREATER THAN EXISTING POLYURETHANE ABRASION STRIP. CORNERS MUST BE ROUNDED TO ½-INCH RADIUS.**

k. Install polyurethane tape, Item 241C, Appendix D, as follows:

(1) Cut polyurethane tape, Item 241C, Appendix D, to flat pattern size.

(2) Remove backing strip from polyurethane tape, Item 241C, Appendix D.

(3) Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, activate adhesive under backing strip by wiping surface two times.

(4) Allow adhesive on polyurethane tape air-dry for 1 - 2 minutes until it is tacky.

(5) When adhesive is tacky, press one spanwise edge of polyurethane tape to prepared surface, making sure rest of polyurethane tape does not touch tail rotor blade.

(6) Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, lightly smooth tacked area in place by hand.

(7) Continue lightly smoothing polyurethane tape across its width, applying polyurethane tape to tail rotor blade over leading edge. Make sure polyurethane tape is applied smoothly, with no air bubbles.

(8) Once polyurethane tape is lightly applied, using machinery towel, Item 211, Appendix D, firmly rub it down.

(9) Allow adhesive to cure for 1 hour.

(10) Apply adhesive, Item 42, Appendix D, to seal edges of tape.

(11) Allow adhesive to dry for 20 - 30 minutes, and apply second coat of adhesive, Item 42, Appendix D, over first coat.

(12) Allow repair to cure a minimum of 12 hours at 70°F (21°C).

l. If using abrasion resistant tape, Item 1, Appendix D, perform the following:

(1) Cut abrasion resistant tape, Item 1, Appendix D, to flat pattern size.

(2) Mix wetting solution consisting of 75% water, 25% isopropyl alcohol, Item 72, Appendix D, and 20 drops per liter of dishwashing compound, Item 122, Appendix D.

(3) Using spray applicator, apply wetting solution to tail rotor blade and adhesive layer of abrasion resistant tape, Item 1, Appendix D.

NOTE

Abrasion resistant tape must be applied at 50°F (10°C), or above ambient temperature.

(4) Apply abrasion resistant tape, Item 1, Appendix D, to tail rotor blade and squeegee out wetting solution, bubbles, and wrinkles starting at center of tape and working toward edges.

(5) Allow abrasion resistant tape, Item 1, Appendix D, to cure for 24 hours at room temperature.

m. Touch up paint (PARA 5-4-38.10).

5-4-38.6 Replace Tip Cap.

NOTE

Replacement of all tip caps is the same.

a. Turn tail rotor blade to safe, comfortable working position.

CAUTION

Damage to tail rotor blade will result if tail rotor blade tip attachment rivets are drilled out carelessly. Holes will be enlarged. Use care in drilling out tip cap attachment rivets.

b. Using drill with ⅛-inch diameter drill bit, drill out 12 rivet heads attaching tip cap to tip rib.

c. Using 3/32-inch diameter drive pin punch, carefully remove 12 rivets attaching tip cap to tip rib.

CAUTION

Damage to tail rotor blade will result if counterweights on tail rotor blade tip are disturbed when removing tip cap. Tail rotor blade balance will be lost. Do not disturb counterweights on tip of tail rotor blade.

d. Carefully remove tip cap and clean out rivet debris from tip end of tail rotor blade.

CAUTION

Damage to tip cap will result if pilot holes are opened-up carelessly. Use care when drilling through nickel abrasion cap when opening-up pilot holes in new tip cap.

e. Using drill and No. 30 drill bit, carefully open pilot holes to full size in new tip cap. Use care when drilling through nickel abrasion cap.

f. Using 100° countersink, countersink all holes to 0.225-inch diameter.

g. Deburr holes.

h. Using corrosion resistant coating, Item 118, Appendix D, treat holes and countersunk areas (PARA 5-4-38.10).

i. Position tip cap on tail rotor blade and hold in place using ⅛-inch-diameter clecos in two top and two bottom holes.

NOTE

When installing a new tip cap on tail rotor blade, 70101-11200, position tip cap on tail rotor blade. Match drill second hole from leading edge of tip cap into tip rib, both top and bottom.

j. Apply light coat of primer, Item 258, Appendix D, to holes. Using pneumatic blind riveter, install required rivets while primer is wet.

k. Install twelve blind rivets in holes, top and bottom, from leading edge (Figure 5-4-38-4, Sheet 1, Detail A).

l. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, wipe off excess primer from around rivets.

m. Touch up paint (PARA 5-4-38.10).

5-4-38.7 Abrasion Strip Repair.

NOTE

Cracks or negligible voids in abrasion strips on tail rotor blade and tip cap, within limits described in PARA 5-3-12, are sealed to prevent moisture from entering bond line and causing further damage.

Apply smooth, even coat of adhesive, Item 32, Appendix D, over crack or void (PARA 5-4-38.3).

5-4-38.8 Aluminum Wire Mesh Lightning Protection Repair.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wash hands and face before eating or smoking.
- Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.

a. Using masking tape, mask area around corrosion ½-inch larger than corrosion outline.

CAUTION

Damage to tail rotor blade will result if tail rotor blade is sanded excessively. Heavy sanding will damage tail rotor blade. Do not sand through underlying fiberglass skin plies. Use only enough force to remove paint, primer and corrosion.

NOTE

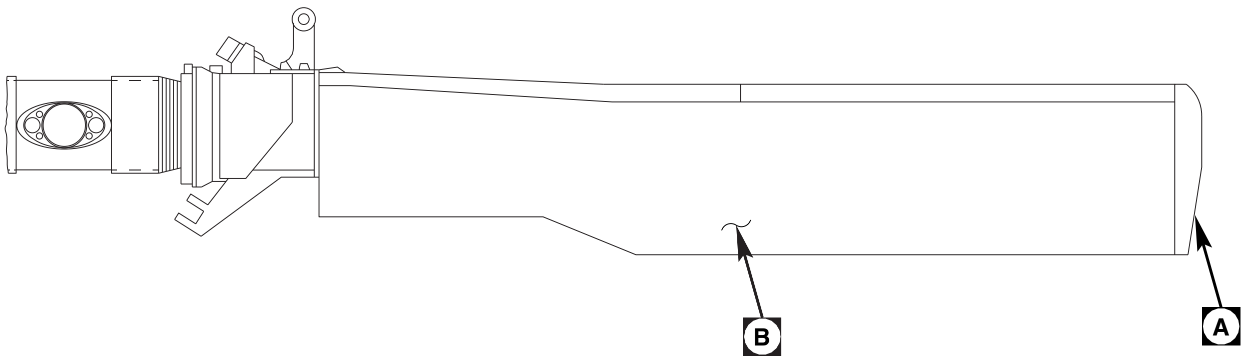
- Sand only in immediate areas of pin holes.
- Up to 20% of chordwise width of aluminum wire mesh lightning protection may be removed without affecting lightning protection.

b. Sand away paint, primer, and corrosion.

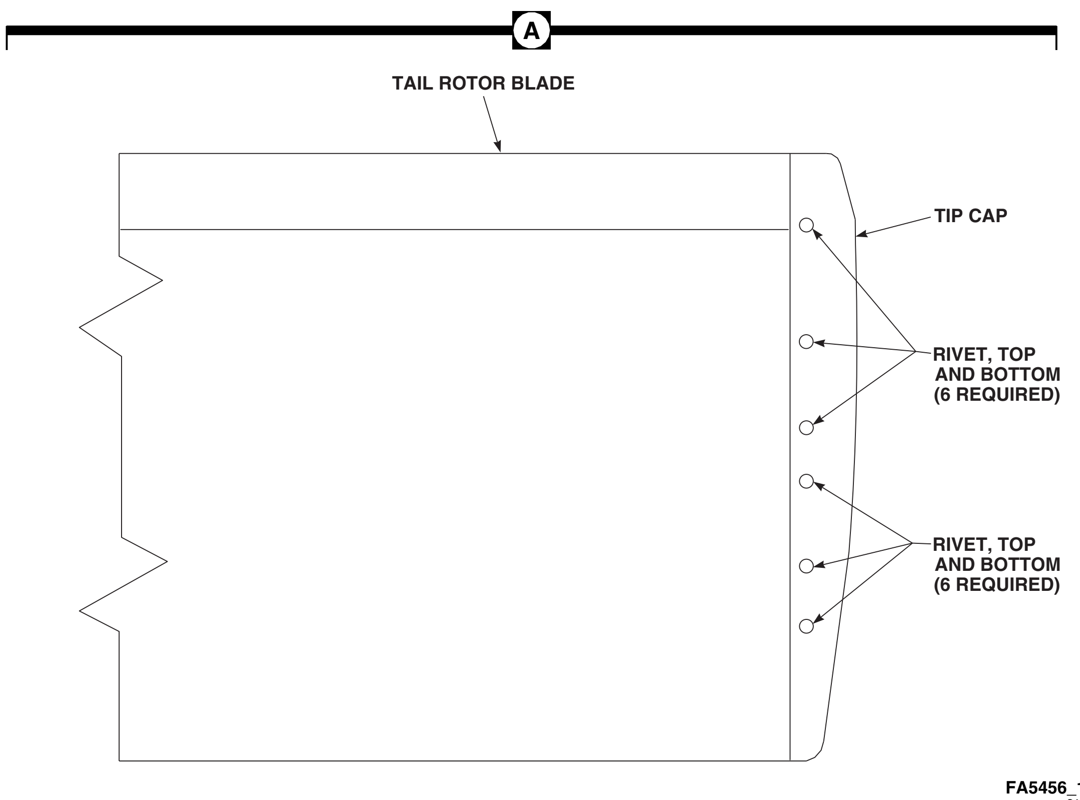
CAUTION

Damage to tail rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints or inside of tail rotor blade. Do not allow solvent to seep into bonded joints or inside of tail rotor blade. Machinery towels wet with methyl ethyl ketone should not be left lying on tail rotor blade.

c. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone,



TAIL ROTOR BLADE



NOTES
1. ALL CORNERS TO HAVE 1/4 IN. RADIUS.
2. SLIT AS NECESSARY TO PERMIT PLY TO LIE FLAT ON REPAIR.

FIGURE 5-4-38-4. REPAIR PROCEDURES (TAIL ROTOR BLADE) (SHEET 1 OF 2)

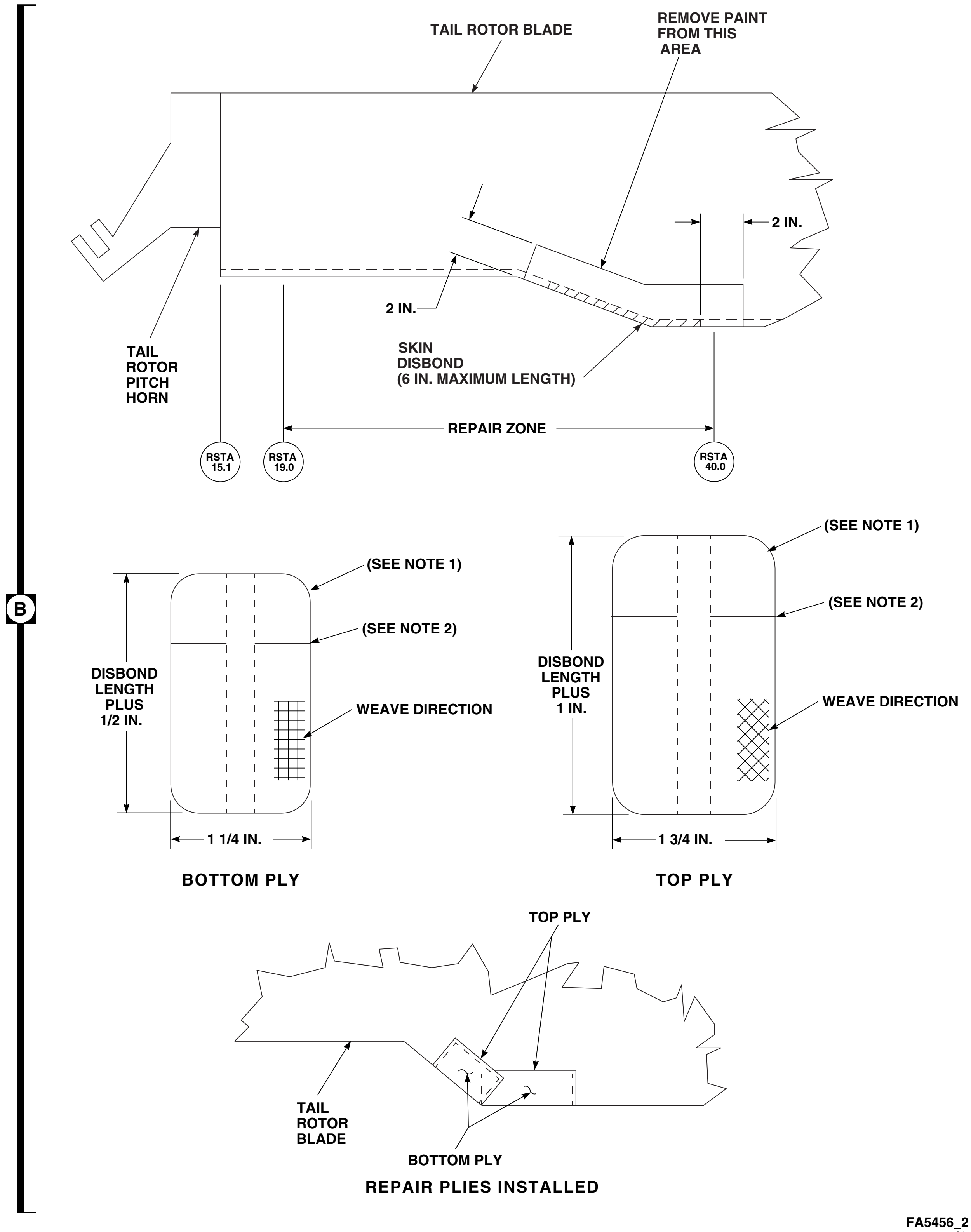


FIGURE 5-4-38-4. REPAIR PROCEDURES (TAIL ROTOR BLADE) (SHEET 2 OF 2)

Item 217, Appendix D, clean sanded area. Wipe dry before methyl ethyl ketone evaporates.

d. Using corrosion resistant coating, Item 118, Appendix D, treat exposed aluminum wire mesh lightning protection.

e. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to repair area (PARA 5-4-38.10).

f. Using epoxy coating kit, Item 126A, Appendix D, paint repair area with epoxy coating.

g. Using epoxy primer coating kit, Item 135, Appendix D, apply another light coat of epoxy primer (PARA 5-4-38.10).

h. Apply top coat of touchup paint (PARA 5-4-38.10).

5-4-38.9 Trailing Edge Disbond Repair.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wash hands and face before eating or smoking.
- Perform all repairs carefully and according to instructions. Even small repairs affect main rotor blade track and balance.
- If damage is found on both tail rotor blade ends, larger damaged area determines size of repairs applied to both ends.
- Repair must be duplicated on opposite end of tail rotor blade to maintain tail rotor blade balance.

a. Using C-clamps, clamp trailing edge just beyond each end of disbond.

CAUTION

Damage to tail rotor blade will result if skins are forced apart during repair. Length of disbond will be extended. Do not force skins apart during repair. Use care when inserting wooden wedges.

b. Insert small wooden wedges to spread skins apart for access.

c. Using abrasive paper, Item 16, Appendix D, wrapped around thin tool to penetrate opening, sand bond surfaces of both skins to remove old adhesive.

d. Sand into end of split so adhesive can form uniform bond line.

e. Using small pieces of cheesecloth, Item 90, Appendix D, wrapped around toothpick or small, thin tool, and dampened with methyl ethyl ketone, Item 217, Appendix D, clean bond surfaces. Repeat until cheesecloth remains clean.

f. Using cheesecloth, Item 90, Appendix D, dampened with isopropyl alcohol, Item 72, Appendix D, clean surfaces.

g. Prepare adhesive, Item 32, Appendix D (PARA 2-6-2).

CAUTION

Damage to tail rotor blade will result if adhesive is allowed to run inside of tail rotor blade. Do not allow adhesive to run inside tail rotor blade.

h. Using toothpick or small, thin tool, apply adhesive to entire length of disbond, moving wooden wedges as required.

i. Remove wooden wedges.

j. Using C-clamps, clamp metal or wood pressure strips to apply pressure enough (10 - 15 psi) for intimate contact at bond line.

k. Allow adhesive to cure for 24 hours at 70°F (22°C), or using heat gun, 2 hours at 140° - 160°F (60° - 71°C).

l. Remove all C-clamps and metal or wood pressure strips.

m. Using abrasive cloth, Item 4, Appendix D, remove excess adhesive squeezeout and paint from repair area (Figure 5-4-38-4, Sheet 2, Detail B).

n. Using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean repair area.

o. Using cheesecloth, Item 90, Appendix D, dampened with isopropyl alcohol, Item 72, Appendix D, clean repair area.

p. Cut repair plies from satin cloth, Item 105, Appendix D.

q. Prepare adhesive, Item 32, Appendix D (PARA 2-6-2).

r. Place repair plies on flat surface covered with plastic sheet, Item 235, Appendix D.

s. Using adhesive, and acid brush, Item 84, or beverage stirring stick, Item 315C, Appendix D, saturate repair plies.

t. Apply adhesive, Item 32, Appendix D, to sanded surface of tail rotor blade in repair area.

u. Apply bottom repair ply, centering it on disbond repair. Slit ply at tail rotor blade transition area if necessary.

v. Apply top repair ply, slitting if necessary, overlapping bottom repair ply.

w. Using plastic sheet, Item 235, Appendix D, cover repair area.

x. Using beverage stirring stick, Item 315C, Appendix D, as a squeegee, work out bubbles and excess adhesive.

y. Allow repair to cure for 24 hours at 70°F (22°C), or using heat gun, 2 hours at 140° - 160°F (60° - 71°C).

z. Remove plastic sheet.

aa. Using abrasive cloth, Item 4 and Item 6, Appendix D, remove excess adhesive and feather edge of repair area.

ab. Touch up paint (PARA 5-4-38.10).

5-4-38.10 Paint Touchup.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wash hands and face before eating or smoking.

a. Refer to Table 5-4-38-1 for touch up paint requirements.

Table 5-4-38-1. Paint Touchup

Area or Details	Tail Rotor Blade Effectivity	Paint Color, Type and Specification	Number of Coats
Tail Rotor Blade, overall	Data Plate Marked Paint, per MIL-L-19538	Epoxy Primer Coating Kit, Item 135, Appendix D	One
		Lacquer, Item 184, Appendix D	Two
Data Markings		Lacquer, Item 183, Appendix D	One
Tail Rotor Blade, overall	Data Plate Marked Paint, per MIL-C-46168	Epoxy Primer Coating Kit, Item 135, Appendix D	One
		Polyurethane Paint, Item 223, Appendix D	Two
Data Markings		Polyurethane Paint, Item 227, Appendix D	One

CAUTION

- Tail rotor blade imbalance will result if excessive paint is applied to tail rotor blade. The weight of the paint, although slight, has large effects on tail rotor blade balance. Restrict application of paint to avoid inducing balance problems. Do not repaint entire tail rotor blade, or large areas of the tail rotor blade. Spot-spray or touch up with small brush all repaired areas. Apply just enough paint for good coverage.
- Damage to tail rotor blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints. Machinery towels wet with methyl ethyl ketone should not be left lying on tail rotor blade.

NOTE

Do not apply any paint on abrasion strip, including tip cap.

b. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean surface to be painted.

CAUTION

Damage to tail rotor blade will occur if tail rotor blade is sanded excessively. Use only enough force to remove resin gloss.

NOTE

Removal of paint to allow cosmetic painting is not allowed.

c. Using abrasive cloth, Item 1A, Appendix D (or equivalent), lightly sand surface to be painted to remove resin gloss.

d. Using clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, rewipe surface. Wipe dry before methyl ethyl ketone evaporates.

e. Using 1-inch masking tape, Item 212, Appendix D, and heavy paper (or equivalent protective covering), mask around touchup area to reduce cleanup.

NOTE

- Do not apply epoxy primer when temperature is below 50°F (10°C). Recommended temperature is 75° - 86°F (24° - 30°C). Lower temperature will cause longer drying time. Apply epoxy primer by brush or spray directly over cleaned reinforced plastic surfaces.
- Aluminum surfaces which have finish removed must be treated with corrosion resistant coating, Item 118, Appendix D, before epoxy primer coating is applied.
- Epoxy primer coating kit, Item 135, Appendix D, consists of two components packaged separately. Add component II to component I. Do not reverse mixing, or mix components from different manufacturers.
- Pot life of mixture is 8 hours at 75°F (24°C).

f. Using epoxy primer coating kit, Item 135, Appendix D, prepare epoxy primer as follows:

(1) Pour required amount of component I (base) into clean container.

(2) Add equal amount, by volume, of component II (converter).

(3) Stir mixture slowly and thoroughly for at least 10 minutes.

NOTE

Thinner is mixture of equal amounts of methyl isobutyl ketone, Item 218, Appendix D, and toluene, Item 339, Appendix D.

(4) Thin mixture for spraying by adding one part, by volume, thinner to two parts, by volume, of mixed epoxy primer.

(5) Mix thoroughly, strain mixture, and allow to stand 1 hour before using. Stir again just before using.

g. Spray light even coat of epoxy primer on prepared surface.

h. Allow epoxy primer to cure for 1 hour at 75°F (24°C).

i. Apply top coat of proper paint over epoxy primer. Refer to Table 5-4-38-1 for paint requirements.

j. Remove masking tape and heavy paper (or equivalent protective covering) from repair area.

5-4-39 TAIL ROTOR PITCH CONTROL RODS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit
- Micrometer
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Antiseize Compound, Item 74, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- Appropriate MTF
- ASTM E1417
- PARA 1-6-18
- PARA 5-3-12
- PARA 5-5-15
- PARA 11-4-2
- PARA 11-4-6

5-4-39.1 Replace Tail Rotor Pitch Control Rods.

NOTE

Replacement of all tail rotor pitch control rods is the same.

5-4-39.1.1 Remove.

a. If not already done, engage gust lock (PARA 1-6-18).

b. Install tail rotor servo rigging pin (PARA 11-4-2).

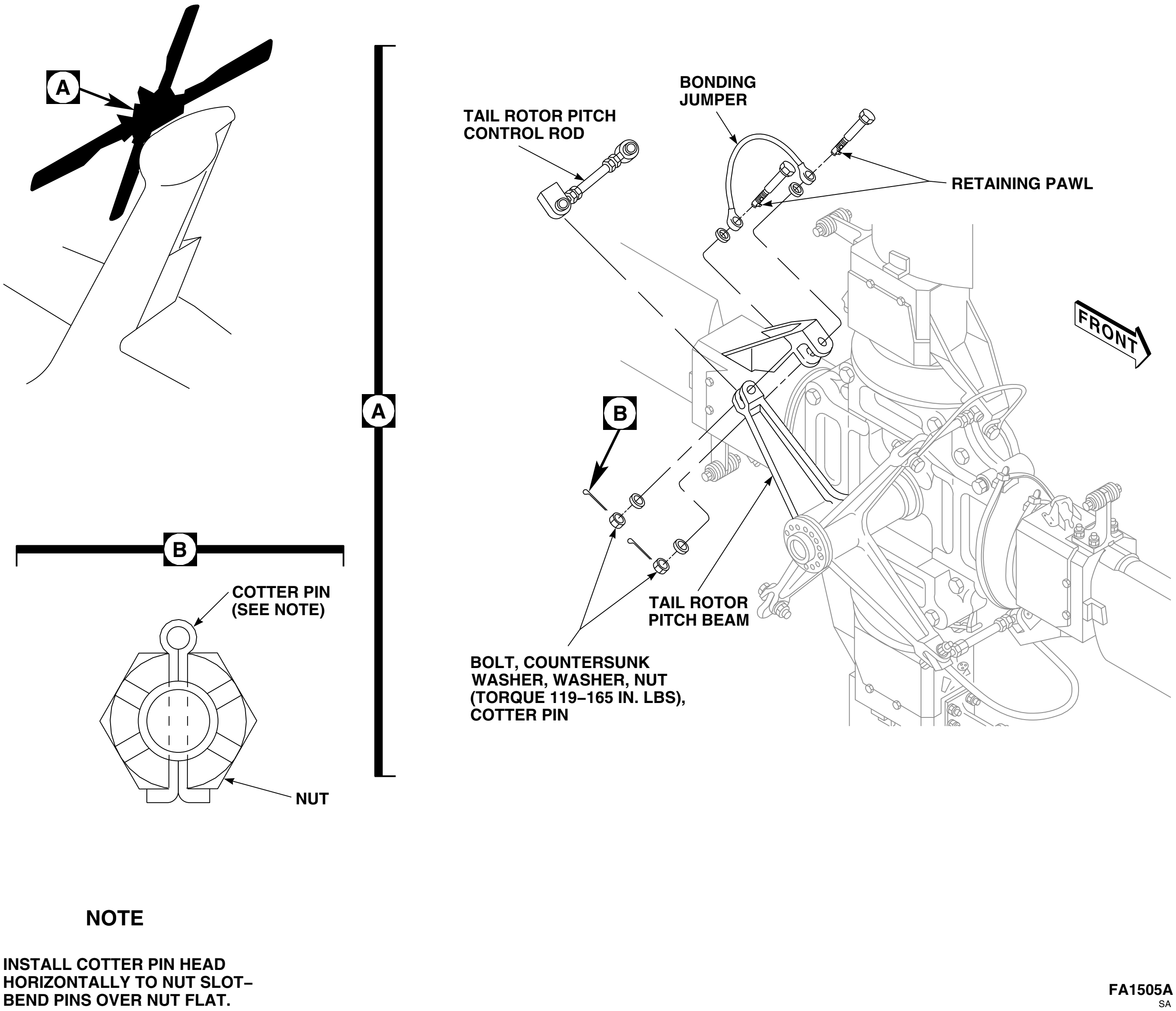


FIGURE 5-4-39-1. REPLACE TAIL ROTOR PITCH CONTROL RODS

c. Remove bolts from both tail rotor pitch control rod ends as follows (Figure 5-4-39-1, Detail A):

(1) Loosen nut on bolt.

(2) Depress retaining pawl on bolt with thumb, and continue backing off nut until castellations clear retaining pawl. Release retaining pawl and remove nut from bolt.

(3) Depress retaining pawl on bolt with thumb, and remove washer from bolt.

NOTE

Retaining pawl on bolt will make bolt hard to remove once inside rod end. Use required force to pull bolt through rod end.

(4) Depress retaining pawl on bolt with thumb, and start removing bolts from rod ends. Use force required to pull bolts from rod ends.

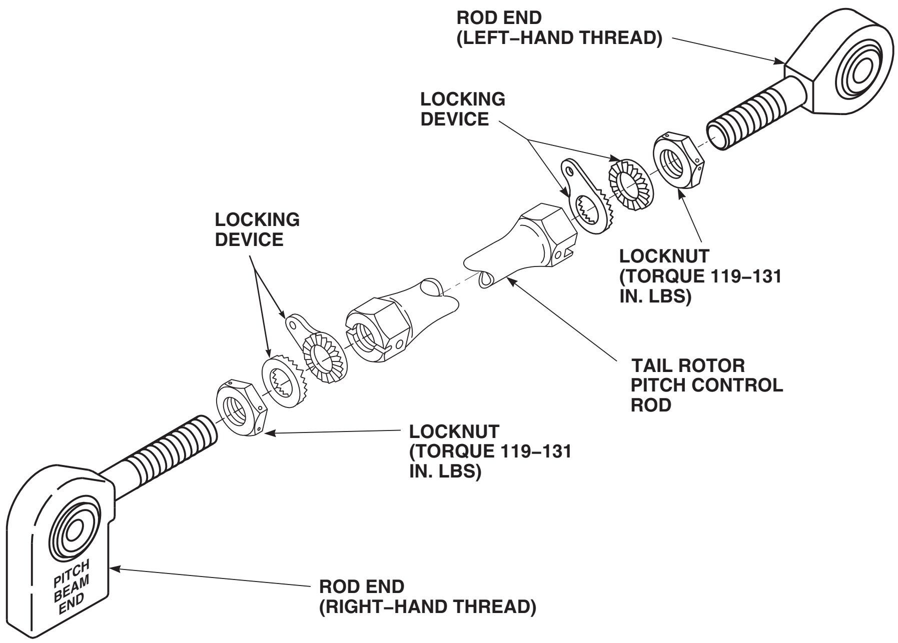


FIGURE 5-4-39-2. REPLACE SPHERICAL AND ELASTOMERIC ROD ENDS

(5) Depress retaining pawl on bolt with thumb, and remove bonding jumper and countersunk washer from bolt.

d. Remove tail rotor pitch control rod.

e. Inspect spherical or elastomeric rod ends (PARA 5-3-12).

5-4-39.1.2 Remove Spherical and Elastomeric Rod End.

a. Loosen locknut securing rod end to tail rotor pitch control rod (Figure 5-4-39-2).

b. Remove rod end from tail rotor pitch control rod.

c. Remove locknut and locking devices from rod end.

d. If required, replace rod end spherical bearing (PARA 5-5-15).

e. Inspect/repair tail rotor pitch control rod, rod ends, and attaching hardware (PARA 5-4-39.1.3).

5-4-39.1.3 Inspect/Repair.

a. Inspect tail rotor pitch control rod attaching washers for cracks. **NONE ALLOWED.** If crack is found, replace washers.

b. Inspect rod end spherical bearings for corrosion by performing the following:

(1) Inspect spherical bearing surface for corrosion pitting. **IF 5 OR MORE PITS ARE VISIBLE IN A 1/2-INCH-DIAMETER CIRCLE OR ANY PIT IS LARGER THAN 0.050-INCH, REPLACE SPHERICAL BEARING.**

CAUTION

Damage to Teflon liner will result if solvent is used to clean bearings. Do not use solvent to clean bearings.

(2) Using scouring pad, Item 270, Appendix D, remove surface corrosion.

c. Inspect spherical and elastomeric rod end threads for damage. **REPLACE TAIL ROTOR PITCH CONTROL ROD OR ROD ENDS IF:**

(1) **NICKS OR GOUGES ARE VISIBLE IN THREAD AREA.**

(2) **ANY DAMAGE TO THREADS THAT PREVENTS FREE MOVEMENT OF ROD ENDS OR LOCKNUTS IS FOUND.**

d. Inspect tail rotor pitch control rods for nicks and gouges. Blend out damage. **REPLACE TAIL ROTOR PITCH CONTROL ROD IF:**

(1) **AFTER BLENDING, CORNER DAMAGE IS GREATER THAN 0.035-INCH DEEP.**

(2) **AFTER BLENDING, SURFACE DAMAGE IS DEEPER THAN 0.020-INCH OR COVERS MORE THAN ¼ OF CIRCUMFERENCE OF TAIL ROTOR PITCH CONTROL ROD BETWEEN LUG AND THREADS.**

(3) **AFTER BLENDING, SURFACE DAMAGE IN LUG PORTION OF ROD ENDS IS GREATER THAN 0.030-INCH DEEP.**

e. Inspect tail rotor pitch control rod for cracks. **NONE ALLOWED.** If crack is found, replace tail rotor pitch control rod.

f. Inspect rod end and locknut for corrosion pitting. If pitting is found, using scouring pad, Item 270, Appendix D, repair as follows:

(1) **REMOVE PITTING FROM ROD END UP TO 0.005-INCH DEEP.**

(2) If pitting exceeds limits, replace rod end.

(3) Remove pitting from locknut. **IF 5 OR MORE PITS ARE FOUND ON ANY SINGLE FLAT, REPLACE LOCKNUT.**

(4) Using 1-inch masking tape, Item 212, Appendix D, mask rod end bearing and threads.

(5) Using epoxy primer coating kit, Item 135, Appendix D, apply two coats of epoxy primer to rod end and locknut.

g. Inspect attachment bolts as follows (Figure 5-4-39-3):

(1) Using dry-cleaning solvent, Item 124, Appendix D, clean any oil or grease from bolts.

(2) Inspect bolts for any material displaced above normal surface of bolt. **NONE ALLOWED.** Using abrasive cloth, Item 10, Appendix D, remove any displaced material.

(3) Inspect bolts for scoring or grooves. **LENGTHWISE SCORE MARKS OR GROOVES MAY BE SMOOTHED OUT, BUT CAN BE LEFT AS IS IF WITHIN LIMITS** (Figure 5-4-39-3). **CIRCUMFERENTIAL SCORE MARKS EXCEEDING MAXIMUM LIMIT MUST BE BLENDED OUT AS FOLLOWS:**

(a) Before reworking bolt, using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, bolt for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, discard bolt.

(b) Using abrasive wheel, Item 22, Appendix D, blend out scratches. Using abrasive cloth, Item 10, Appendix D, blend area smooth.

(4) After rework, using micrometer, measure bolt diameter. **DISCARD BOLT IF:**

(a) **BOLT DIAMETER IS LESS THAN SPECIFIED.**

(b) **AFTER REWORK, REPAIR AREA IS OVER 30% OF SHANK SURFACE AREA.**

(c) **AFTER REWORK, REPAIR AREA IS OVER 30% OF SHANK CIRCUMFERENCE.**

(5) After reworking bolt, using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, bolt for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, discard bolt.

h. Inspect bonding jumper for broken strands. **NONE ALLOWED. CRACKS AND CHIPS IN PLASTIC COATING ARE ACCEPTABLE.** If broken strands are found, replace bonding jumper.

5-4-39.1.4 Install Spherical and Elastomeric Rod End.

NOTE

- If rod end spherical bearing is replaced with rod end elastomeric bearing, both rod end bearings on that tail rotor pitch control rod must be replaced as well as both ends of opposing tail rotor pitch control rod.
- On helicopters with rod end elastomeric bearings, if rod end(s) will be replaced, make sure the shelf life for the replacement rod end(s), SB7112-101 and SB7112-102, has not exceeded 60 months. If shelf life has exceeded 60 months, do not use.

a. Using antiseize compound, Item 74, Appendix D, apply light coat to threads of rod end.

NOTE

Locking devices' serrated sides face each other.

b. Install locknuts and locking devices onto rod end (Figure 5-4-39-2).

c. Thread rod ends into tail rotor pitch control rod. Make sure each rod end is threaded into tail rotor pitch control rod in the same amount until a measurement of 5.564 - 6.804-inches, measured from center of each rod end is obtained.

d. Adjust tail rotor pitch control rod (PARA 5-4-39.1.5).

5-4-39.1.5 Adjust.

a. Adjust replacement tail rotor pitch control rod (PARA 11-4-6, step o. through step z.).

b. Make sure locking device is in proper position. **TORQUE LOCKNUTS TO 119 - 131 INCH-POUNDS** (Figure 5-4-39-2).

c. Using lockwire, Item 197, Appendix D, lockwire from locknut to locking device then back to opposite side of locknut (double-safety).

5-4-39.1.6 Install.

a. Make sure replacement tail rotor pitch control rod has been adjusted (PARA 5-4-39.1.5).

WARNING

FSCAP

Verification of proper hardware, bonding jumper, cotter pin installation, and torque are critical characteristics.

b. Install tail rotor pitch control rod in tail rotor pitch beam and tail rotor pitch horn. Install bolts as follows (Figure 5-4-39-1, Detail A):

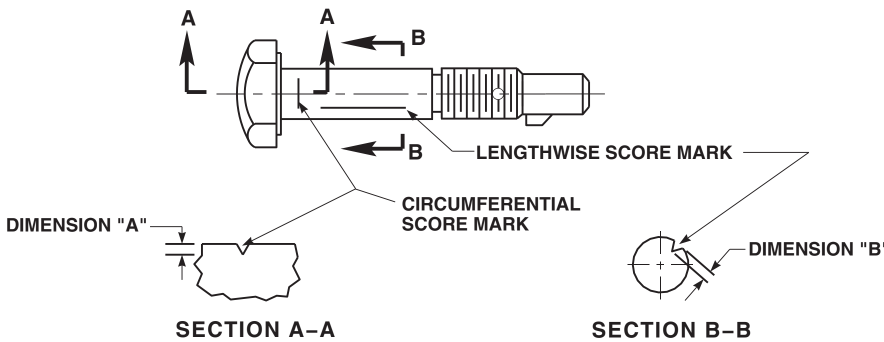
CAUTION

Damage to bolthead will result if countersunk washer is not installed correctly. When installing countersunk washers, make sure to install the washer with the countersunk side toward bolthead.

NOTE

Bending of bonding jumper terminal up to 30° is allowed to prevent interference between bonding jumper and tail rotor pitch beam or tail rotor pitch horn.

(1) Depress retaining pawl on bolt with thumb, and install countersunk washer



BOLT PART NUMBER	LOCATION	DIMENSION "A" MAXIMUM	DIMENSION "B" MAXIMUM	MINIMUM ALLOWABLE DIAMETER AFTER REWORK
70103–08801–101	TAIL ROTOR PITCH CONTROL ROD (BOTH ENDS)	0.004 IN.	0.004 IN.	0.3655 IN.

FA1503
SA

FIGURE 5-4-39-3. INSPECT/REPAIR TAIL ROTOR PITCH CONTROL ROD BOLT

and bonding jumper on bolt. Make sure countersunk side of washer is toward bolthead.

CAUTION

Loss of torque on bolts will result if bonding jumper is not bent up to 30° before installation. Make sure bonding jumper is bent up to 30° before installation.

NOTE

Paint or dirt under bonding jumper washer will prevent proper bonding of parts. Make sure area of tail rotor pitch beam and tail rotor pitch horn where bonding jumper is connected is completely clean and free of paint or contaminants.

(2) If not already done, bend bonding jumper up to 30° to eliminate chafing against tail rotor pitch beam and tail rotor pitch horn.

(3) Using corrosion preventive compound, Item 114, Appendix D, apply coat on inside diameter of tail rotor pitch beam bushings, and on tail rotor horn bushings at tail rotor pitch control rod connection.

NOTE

If installing tail rotor pitch control rod with rod end elastomeric bearings, install with rod end bearing marked with yellow dot at tail rotor pitch beam end.

(4) Line up rod end bearing with tail rotor pitch horn and install bolt with head facing away from direction of rotation.

(5) Before installing nut, using machinery towel, Item 211, Appendix D, wipe off excess corrosion preventive compound from bolt threads.

(6) Depress retaining pawl on bolt with thumb, and install washer on bolt and install nut. Do not torque nut now.

(7) Repeat step b. (1) through step b. (6) to install tail rotor pitch beam end of tail rotor pitch control rod. Center rod ends within ears of tail rotor pitch beam.

(8) Make sure bolt retaining pawl is facing outward towards tail rotor blade tip cap. **TORQUE NUT TO 119 - 165 INCH-POUNDS.** Install cotter pins in castellated nut, with head horizontal to nut slot and bend pins over hex flats (Figure 5-4-39-1, Detail B).

c. Remove tail rotor servo rigging pin (PARA 11-4-2).

d. Make sure area is clean and free of foreign material.

e. Disengage gust lock (PARA 1-6-18).

f. Perform maintenance test flight (Appropriate MTF)

5-4-40 TAIL ROTOR PITCH BEAM.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit
- Pitch Beam Puller, 70700-77389-041
- Rawhide Mallet
- Spanner Wrench
- Thread Protection
- Torque Wrench, 700 - 1600 in. lbs

Materials/Parts

- Abrasive Wheel, Item 22, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 38, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Corrosion Preventive Compound, Item 115, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 172, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 1-7-7
- PARA 2-6-5
- PARA 5-4-39
- PARA 11-4-6
- PARA 11-4-113

5-4-40.1 Replace Tail Rotor Pitch Beam.

5-4-40.1.1 Remove.

a. Disconnect tail rotor pitch control rods at tail rotor pitch beam only (PARA 5-4-39).

b. Remove screws and washers securing retaining nut to tail rotor pitch beam (Figure 5-4-40-1, Sheet 1, Detail A).

c. Using spanner wrench, remove retaining nut. Remove washer (Figure 5-4-40-1, Sheet 1, Detail B and Sheet 2, Detail C).

CAUTION

Damage to dust plug on pitch control shaft will result if excessive force is used to remove retaining nut and tail rotor pitch beam. Use care when removing retaining nut and tail rotor pitch beam.

d. Using pitch beam puller, remove tail rotor pitch beam. If necessary, using rawhide mallet, tap on inboard side.

e. Install thread protection on pitch control shaft.

5-4-40.1.2 Inspect/Repair.

a. Visually inspect washer for cracks. **NONE ALLOWED.** If cracks are found, replace washer.

b. Inspect surfaces of washer which come in contact with retaining nut for nicks, gouges, corrosion, or fretting. Using abrasive wheel, Item 22, Appendix D, remove damage. **IF DAMAGE AFTER BLENDING EXCEEDS 0.010-INCH DEEP OR EXCEEDS 15% OF SURFACE, REPLACE TAIL ROTOR PITCH BEAM WASHER.**

c. Visually inspect retaining nut for cracks. **NONE ALLOWED.** If cracks are found, replace retaining nut.

d. Visually inspect retaining nut for thread face wear or chafing. **NONE ALLOWED.** If threads are damaged, replace retaining nut.

e. Inspect surfaces of retaining nut that come in contact with washer for nicks, gouges, corrosion, or fretting. Using abrasive wheel, Item 22, Appendix D, remove damage. **IF DAMAGE AFTER BLENDING EXCEEDS 0.010-INCH DEEP OR EXCEEDS 15% OF SURFACE, REPLACE RETAINING NUT.**

f. Visually inspect tail rotor pitch beam for cracks. If cracks are suspected, using fluorescent penetrant inspection kit, fluorescent penetrant inspect for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace tail rotor pitch beam.

g. Inspect for nicks, gouges, and fretting. Using abrasive wheel, Item 22, Appendix D, remove raised edges and blend out damage. **REPLACE TAIL ROTOR PITCH BEAM IF:**

(1) **CORNER DAMAGE AFTER BLENDING IS GREATER THAN 0.100-INCH DEEP, OR 0.020-INCH DEEP IN LUG AREA. MULTIPLE DAMAGE MUST BE ½-INCH APART.**

(2) **SURFACE DAMAGE AFTER BLENDING IS GREATER THAN 0.050-INCH DEEP, OR 0.020-INCH DEEP IN LUG AREA. MULTIPLE DAMAGE MUST BE AT LEAST 1-INCH APART.**

(3) **RADIAL OR AXIAL PLAY IS FOUND ON TAIL ROTOR PITCH BEAM TO PITCH CONTROL SHAFT INSTALLATION.**

h. Inspect for corrosion. **REPLACE TAIL ROTOR PITCH BEAM IF CORROSION IS:**

NOTE

Removal of corrosion on mating surface (conical area) of pitch control shaft is not allowed. If corrosion is found, replace tail rotor pitch beam.

(1) **FOUND ON PITCH CONTROL SHAFT MATING SURFACE.**

(2) **DEEPER THAN 0.010-INCH OR COVERS MORE THAN 10% OF SURFACE AREA IN BUSHING BORES.**

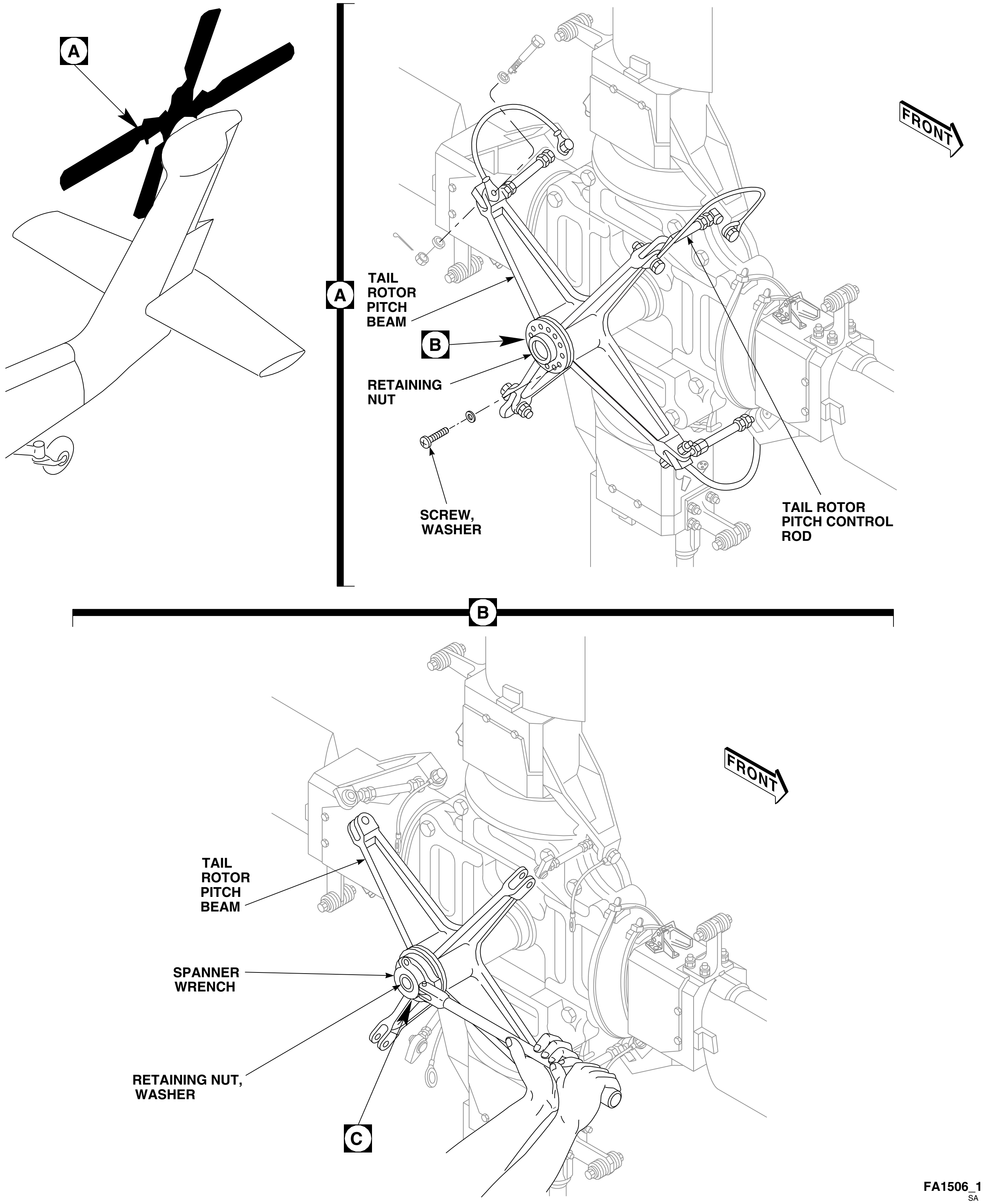


FIGURE 5-4-40-1. REPLACE TAIL ROTOR PITCH BEAM (SHEET 1 OF 2)

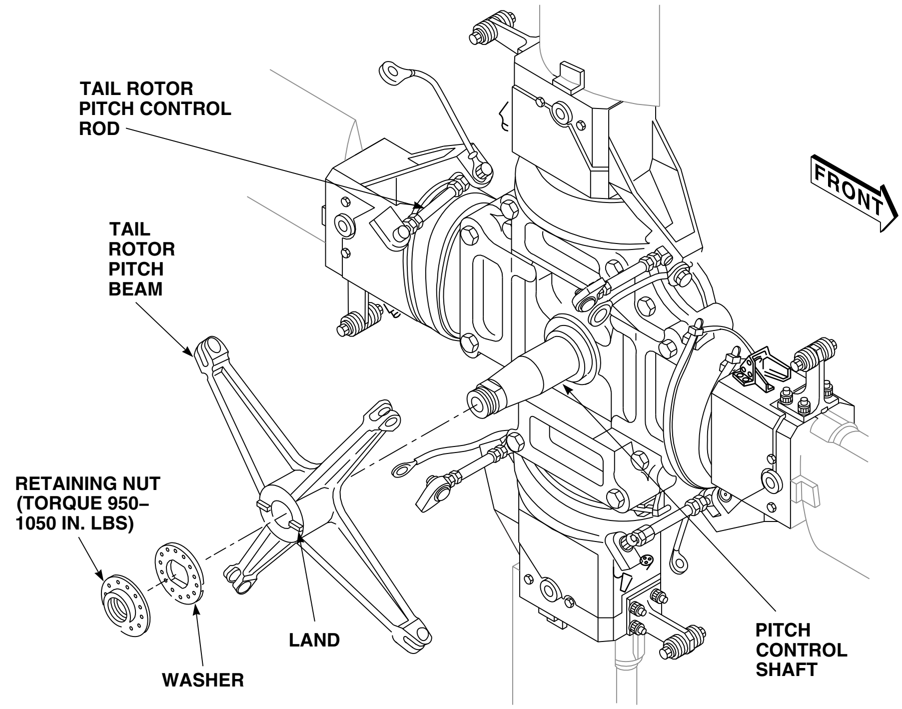


FIGURE 5-4-40-1. REPLACE TAIL ROTOR PITCH BEAM (SHEET 2 OF 2)

(3) **DEEPER THAN 0.030-INCH IN OPEN SECTIONS.**

(4) **DEEPER THAN 0.020-INCH IN LUG AREAS.**

NOTE

Do not use any type of Scotch-Brite pad.

(5) Using abrasive wheel, Item 22, Appendix D, remove corrosion and/or any discoloration that is less than limits.

i. Inspect flat surfaces of tail rotor pitch beam which come in contact with washer for nicks, gouges, corrosion, or fretting. Using abrasive wheel, Item 22, Appendix D, remove damage. **IF DAMAGE AFTER BLENDING EXCEEDS 0.010-INCH DEEP OR EXCEEDS 15% OF SURFACE, REPLACE TAIL ROTOR PITCH BEAM.**

j. If repair has been made, or if crack is suspected, using fluorescent penetrant inspection kit, fluorescent penetrant inspect tail rotor pitch beam for cracks (ASTM E1417). **NONE ALLOWED.** If cracked, replace tail rotor pitch beam.

k. Using corrosion resistant coating, Item 118, Appendix D, apply to repaired areas.

l. Touch up paint on tail rotor pitch beam (PARA 2-6-5).

m. Inspect exposed portion of pitch control shaft (PARA 11-4-113).

n. If necessary, remove thread protection, and visually inspect pitch control shaft and make sure dust plug is installed. If dust plug is not installed, perform the following:

(1) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean dust plug opening.

(2) Using acid brush, Item 84, Appendix D, apply coat of adhesive, Item 38, Appendix D, around dust plug.

(3) Install dust plug in opening on pitch control shaft. Make sure dust plug is seated firmly against pitch control shaft.

(4) Apply bead of sealing compound, Item 292, Appendix D, around edges of dust plug. Allow to dry.

(5) If necessary, install thread protection on pitch control shaft.

5-4-40.1.3 Install.

a. Using machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, or dry-cleaning solvent, Item 124, Appendix D, clean outside diameter of pitch control shaft, conical seat of tail rotor pitch beam, washer, and retaining nut.

b. Remove thread protection from pitch control shaft.

WARNING

Improper torque of retaining nut will result if corrosion preventive compound is applied to retaining nut or washer mating surface. Do not apply corrosion preventive compound to retaining nut or washer mating surfaces.

NOTE

Corrosion preventive compound, Item 115, Appendix D, is preferred for use on pitch control shaft mating surface. If corrosion preventive compound, Item 115, Appendix D, is not available, corrosion preventive compound, Item 114, Appendix D, may be used as an alternate.

c. Using corrosion preventive compound, Item 115, or alternate, Item 114, Appendix D, lightly coat tapered surface and threads of pitch control shaft.

CAUTION

Damage to dust plug on pitch control shaft will result if excessive force is used to remove retaining nut and tail rotor pitch beam. Use care when removing retaining nut and tail rotor pitch beam.

d. Slide tail rotor pitch beam on pitch control shaft (Figure 5-4-40-1, Sheet 2, Detail C).

e. Using machinery towel, Item 211, Appendix D, wipe off any excess corrosion preventive compound on pitch control shaft and tail rotor pitch beam.

f. Remove thread protection from pitch control shaft.

g. Install washer. Line up lands on tail rotor pitch beam with slots in washer. Turn tail rotor pitch beam as required. Install retaining nut.

h. **USING SPANNER WRENCH, TORQUE RETAINING NUT TO 950 - 1050 INCH-POUNDS. DO NOT EXCEED 1050 INCH-POUNDS TORQUE LIMIT WHEN ATTEMPTING TO ALIGN HOLES IN RETAINING NUT AND WASHER** (Figure 5-4-40-1, Sheet 1, Detail B and Sheet 2, Detail C).

i. If holes in retaining nut do not align with holes in washer when 1050 inch-pounds is reached, perform the following:

(1) Remove retaining nut, washer, and tail rotor pitch beam.

(2) Inspect tail rotor pitch beam conical seat and washer mating surface, retaining nut, washer, and pitch control shaft tapered surface for damage or debris which may prevent alignment of holes. Replace components as required.

(3) Reinstall tail rotor pitch beam.

WARNING

Make sure steel washer, 70103-11015-106, is used. All aluminum washers, 70103-11015-101/-104, must not be used and must be purged from the supply system.

(4) Install washer 180° from original position.

(5) Install retaining nut. **USING SPANNER WRENCH, TORQUE RETAINING NUT TO 950 - 1050 INCH-POUNDS.**

j. Install screws and washers securing retaining nut (Figure 5-4-40-1, Sheet 1, Detail A).

k. **TIGHTEN SCREWS SNUG; DO NOT TORQUE.**

l. Using lockwire, Item 197, Appendix D, lockwire screws.

m. Install heat-shrinkable insulation sleeving, Item 172, Appendix D, to lockwire to prevent chafing.

NOTE

Corrosion preventive compound, Item 115, Appendix D, is preferred for use on exterior surfaces of retaining nut and washer. If corrosion preventive compound, Item 115, Appendix D, is not available, corrosion preventive compound, Item 114, Appendix D, may be used as an alternate.

n. Using corrosion preventive compound, Item 115, or alternate, Item 114, Appendix D, apply to exterior surfaces of retaining nut and washer.

o. Connect tail rotor pitch control rods to tail rotor pitch beam (PARA 5-4-39).

p. Perform tail rotor system rig check (PARA 11-4-6).

q. Clean corrosion preventive compound from exposed portion of pitch control shaft tapered surface and exterior surfaces of retaining nut and washer.

r. After torque has stabilized, apply sealing compound, Item 292, Appendix D, to inboard tail rotor pitch beam to pitch control shaft surface and to retaining nut, washer, and pitch change shaft thread surfaces.

s. Make sure area is clean and free of foreign material.

t. Check helicopter logbook to make sure retaining nut torque check is shown due 9 - 11 flight hours after installation (PARA 1-7-7).

5-4-41 TAIL ROTOR BOOT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Spring Scale
- Tail Rotor Blade Boot Installation Tool, Locally-Made, Figure H-207, Appendix H

Materials/Parts

- Insulation Tape, Item 181, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D
- Tiedown Strap, Item 336, Appendix D

References

- Appendix D
- Appendix H
- PARA 5-4-39

5-4-41.1 Replace Tail Rotor Boot.

NOTE

Replacement of all tail rotor boots is the same.

5-4-41.1.1 Remove.

a. Remove tiedown straps and clips from tail rotor boot (Figure 5-4-41-1, Detail A).

b. Disconnect tail rotor pitch control rod at tail rotor pitch horn only (PARA 5-4-39).

c. Cut off old tail rotor boot.

d. Inspect insulation tape on tail rotor boot support and tail rotor pitch horn fairing for condition. If unserviceable, remove tape as follows (Figure 5-4-41-1, Detail B):

(1) Remove tape from boot support and/or pitch horn fairing.

(2) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean boot support and pitch horn fairing where insulation tape was removed.

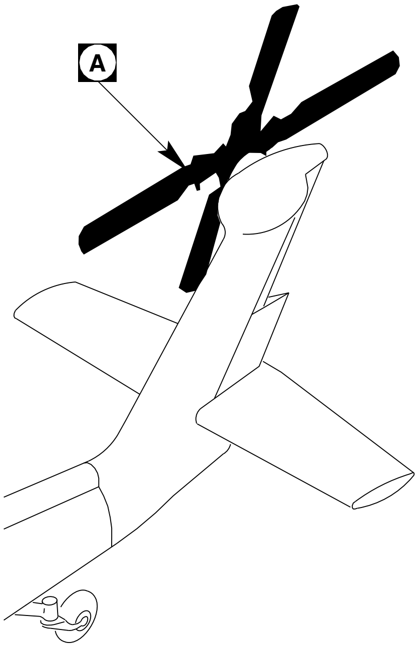
5-4-41.1.2 Inspect.

Inspect tail rotor boot for cuts, tears, breaks, and dry rot. **NONE ALLOWED.** If cuts, tears, breaks, or dry rot are found, replace tail rotor boot.

5-4-41.1.3 Install.

a. Locally make tail rotor blade boot installation tool (Figure H-207, Appendix H).

b. If insulation tape was removed, apply insulation tape, Item 181, Appendix D, to ends of tail rotor boot support and tail rotor pitch horn fairing (Figure 5-4-41-1, Detail B).



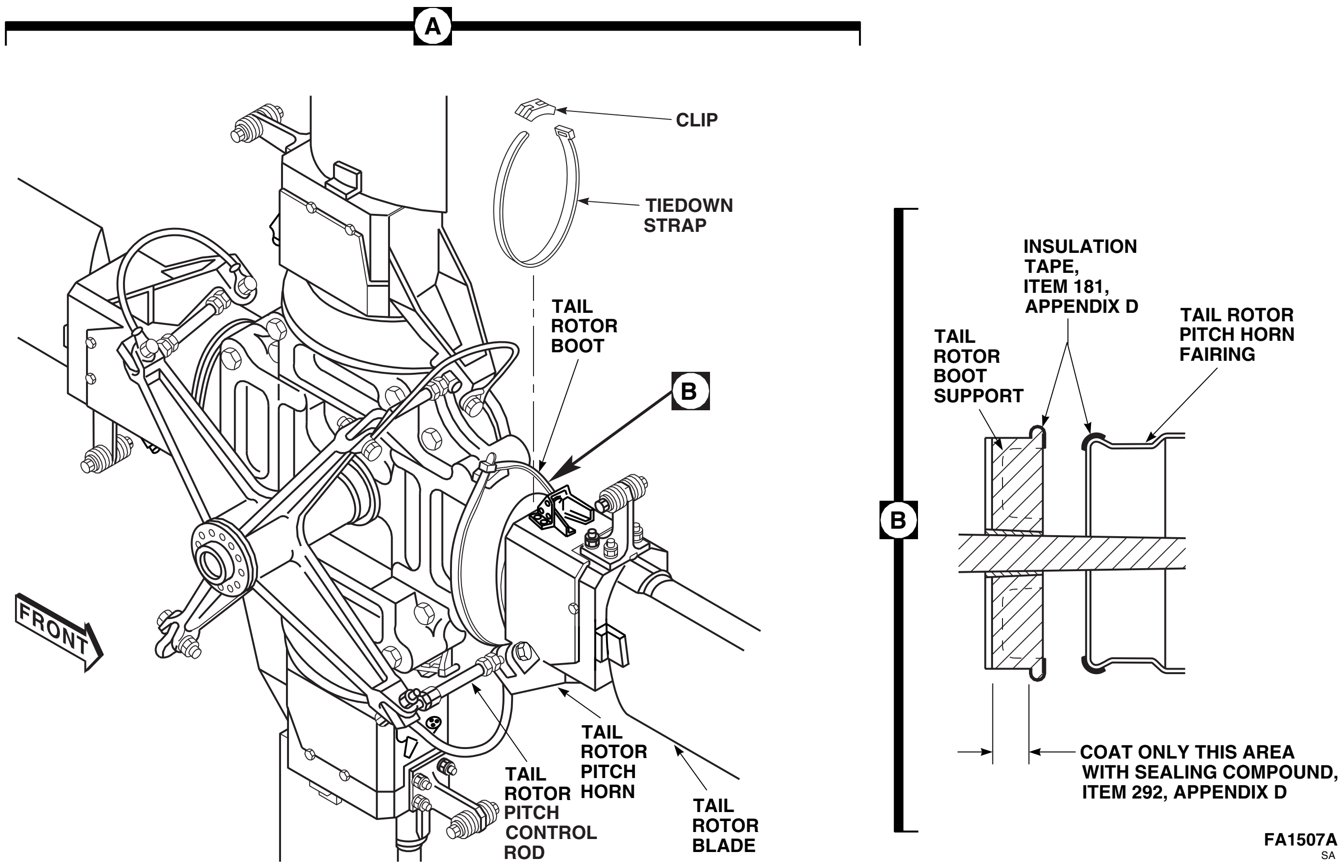


FIGURE 5-4-41-1. REPLACE TAIL ROTOR BOOT

c. Apply coat of sealing compound, Item 292, Appendix D, to mating surface on tail rotor boot support.

d. Using water, lubricate leading edge and trailing edge of tail rotor blade.

e. Stretch tail rotor boot over tail rotor blade. Make sure egg-shaped end of tail rotor boot goes on first (Figure 5-4-41-1, Detail A).

f. Using tail rotor blade boot installation tool, slide tail rotor boot along tail rotor blade spar to tail rotor pitch horn. Stretch tail rotor boot over tail rotor pitch horn. Make sure both ends of tail rotor boot are seated in grooves.

g. Place clip on tail rotor boot at leading edge, and using tiedown strap, Item 336, Appendix D, secure tail rotor boot and clip.

h. Using spring scale, apply 10 - 20 pounds tension to end of tiedown straps.

i. Connect tail rotor pitch control rod to tail rotor pitch horn (PARA 5-4-39).

5-4-42 OUTBOARD RETENTION PLATE REPAIR (PLATE REMOVED).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, 5⁄32-inch
- Micrometer Set
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Alcohol, Denatured, Item 70, Appendix D
- Pressure Sensitive Adhesive Tape, Item 247, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 280, Appendix D
- Sealing Primer, Item 306, Appendix D

References

- Appendix D
- PARA 2-6-5

5-4-42.1 Inspect/Repair Outboard Retention Plate.

a. Inspect outboard retention plate for cracks. **NONE ALLOWED.** If crack is found, replace outboard retention plate.

b. Visually inspect outboard retention plate for fretting, gouges, nicks, and scoring.

c. Using abrasive wheel, Item 22, Appendix D, remove raised material and blend out damage. **REPLACE OUTBOARD RETENTION PLATE IF:**

(1) **AFTER BLENDING, CORNER DAMAGE IS GREATER THAN 0.100-INCH DEEP, OR 0.040-INCH DEEP IN BOLTHOLE AREAS. MULTIPLE DAMAGE MUST BE AT LEAST 1-INCH APART.**

(2) **AFTER BLENDING, SURFACE DAMAGE IS GREATER THAN 0.040-INCH DEEP IN OPEN AREAS, OR GREATER THAN 0.020-INCH DEEP IN BOLTHOLE AREAS. MULTIPLE DAMAGE MUST BE AT LEAST 1-INCH APART.**

(3) **AFTER BLENDING, SURFACE DAMAGE IN MATING AREAS IS GREATER THAN 0.020-INCH DEEP, OR COVERS MORE THAN 10% OF MATING AREA.**

(4) **FRETTING ON AREAS THAT MATE WITH INBOARD RETENTION PLATE AFTER BLENDING IS GREATER THAN 0.010-INCH DEEP, OR COVERS MORE THAN 10% OF MATING AREA.**

d. Touch up paint on repaired areas (PARA 2-6-5).

e. Remove screws securing nylon shims to outboard retention plate (Figure 5-4-42-1).

f. Using micrometer, inspect nylon shims 1-inch in from edge of each corner. **NYLON SHIM THICKNESS MUST BE 0.123 - 0.126-INCH.** If thickness exceeds limits, or if surface is not flat, replace nylon shim.

g. If new nylon shims are being installed, inspect per step f. Do not use nylon shim that exceeds limits.

h. Inspect threaded inserts as follows:

(1) Install screws in threaded inserts and check drag torque required to install screws. **DRAG TORQUE MUST BE 2 - 18 INCH-POUNDS.**

(2) If drag torque exceeds limits, replace screw and clean threaded insert.

(3) Recheck drag torque. If drag torque still exceeds limits, replace threaded insert as follows:

(a) Using 5/32 -inch Allen wrench, remove threaded insert.

(b) Using denatured alcohol, Item 70, Appendix D, clean tapped hole in outboard retention plate.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

(c) Using clean, dry compressed air, blow dry cleaned area.

(d) Inspect tapped hole in outboard retention plate for thread yielding. If more than 10% of thread has yielded, replace outboard retention plate.

(e) Using denatured alcohol, Item 70, Appendix D, clean outer threads of threaded insert.

(f) Using clean, dry compressed air, blow dry cleaned area.

(g) Using sealing primer, Item 306, Appendix D, apply thin coat to threads of threaded insert and threads of outboard retention plate.

(h) Apply thin coat of sealing compound, Item 280, Appendix D, to first two threads of threaded insert.

(i) **USING 5/32-INCH ALLEN WRENCH, INSTALL THREADED INSERT FLUSH TO 0.025-INCH BELOW SURFACE WITHIN TAPPED HOLE, TO DRIVING TORQUE OF 25 - 300 INCH-POUNDS.**

i. Install nylon shims and screws.

j. **HANDTIGHTEN SCREWS.**

k. Remove seal from outboard retention plate.

l. Visually inspect retaining ring and end of bushing for corrosion pitting. **IF PITTING IS DEEPER THAN 0.003-INCH ON RETAINING RING, OR 0.020-INCH ON END OF BUSHING, REPLACE RETAINING RING AND/OR BUSHING.**

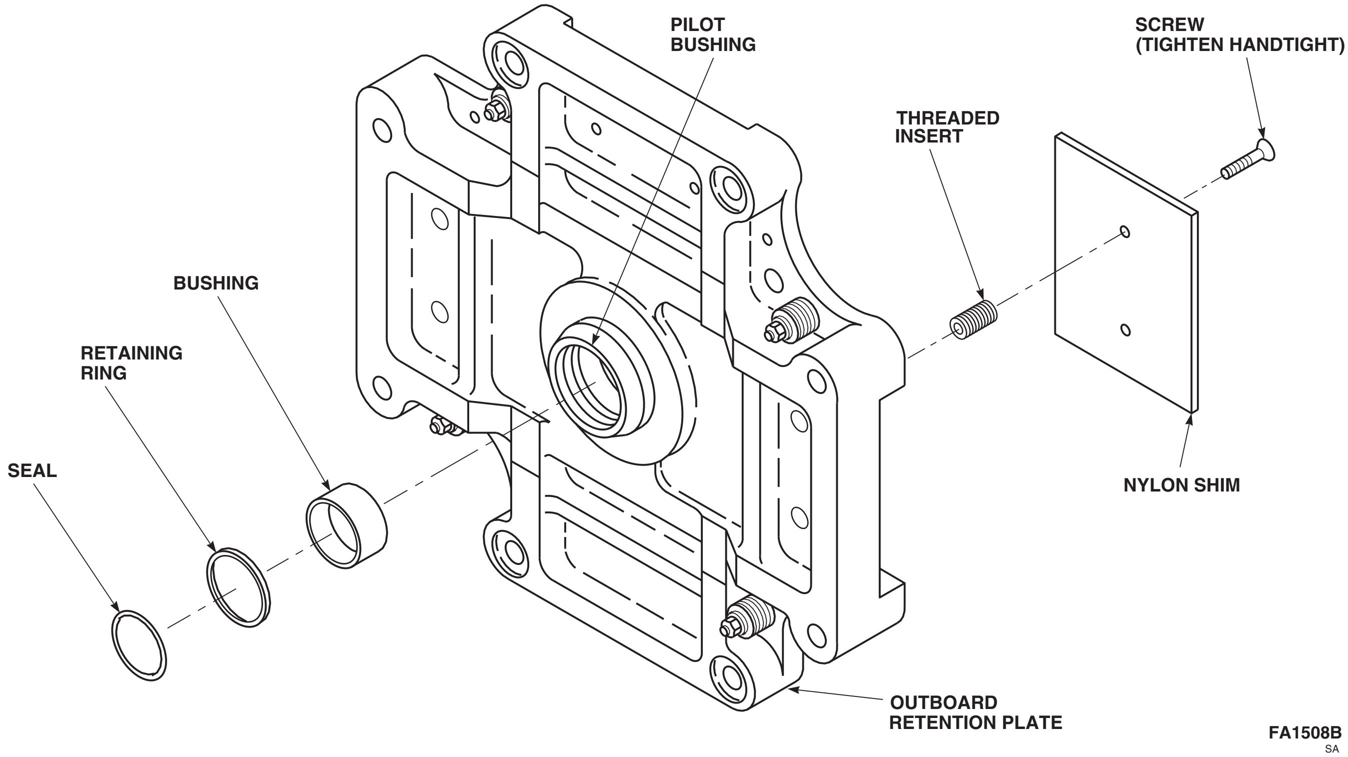


FIGURE 5-4-42-1. INSPECT/REPAIR OUTBOARD RETENTION PLATE

m. Visually inspect inside diameter (ID) of bushing. Surface will appear bronze speckled under normal wear. **REPLACE BUSHING IF:**

(1) **70% OF ID APPEARS BRONZE.**

(2) **BRONZE APPEARS SMEARED.**

(3) **BRONZE LAYER IS CHIPPED AND STEEL BACKING IS EXPOSED.**

n. If corrosion pitting is less than above limits, perform the following:

(1) Remove retaining ring from pilot bushing.

(2) Using abrasive cloth, Item 10, Appendix D, remove corrosion from retaining ring using limits in Table 5-4-42-1.

(3) Remove corrosion from bushing as follows:

(a) Insert clean, dry cloth inside bushing.

CAUTION

Do not use crocus cloth on ID of bushing.

(b) Using abrasive cloth, Item 10, Appendix D, remove corrosion from top of bushing only.

CAUTION

When removing cloth, be careful not to imbed any grit on ID of Teflon-impregnated bushing.

(c) Remove cloth from inside bushing.

(d) Using clean, dry cloth, wipe bushing clean.

o. Using pressure sensitive adhesive tape, Item 247, Appendix D, mask ID of bushing.

p. Apply coat of primer, Item 258, Appendix D, to outside diameter of bushing. Press in bushing.

q. Apply coat of primer, Item 258, Appendix D, to top of bushing, retaining ring groove in pilot bushing, and retaining ring.

r. While primer is still wet, remove pressure sensitive tape from bushing, and install retaining ring.

s. With flange of seal facing outboard retention plate, install seal.

Table 5-4-42-1. Retaining Ring Rework Limits

Total Thickness - 0.049-inch	Width = 0.148-inch
After Cleaning - 0,040-inch	Width = 0.142-inch
Total Diameter of Retaining Ring	Diameter = not less than 2.670-inches

5-4-43 TAIL ROTOR BLADE DEICE BRACKET.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 197, Appendix D
- Protective Cap
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-7-20
- PARA 5-4-41
- PARA 12-4-52

5-4-43.1 Replace Tail Rotor Blade Deice Bracket.

NOTE

Replacement of all tail rotor blade deice brackets is the same.

5-4-43.1.1 Remove.

a. Using nonmetallic scraper, remove sealing compound from applicable electrical connector and mounting surfaces of tail rotor blade deice bracket.

b. If blade deice kit is installed, disconnect applicable blade deice electrical connector (PARA 12-4-52).

c. If blade deice kit is not installed, remove metal cap from electrical connector.

d. Remove screws, washers, and nuts securing electrical connector, J611, J612, J613, or J614, to tail rotor blade deice bracket (Figure 5-4-43-1, Detail A).

e. Remove electrical connector from tail rotor blade deice bracket.

f. If blade deice kit is not installed, install metal cap on electrical connector. If blade deice kit is installed, install protective cap on electrical connector

g. Remove screws securing tail rotor blade deice bracket to tail rotor pitch horn fairing.

h. Remove tail rotor blade deice bracket.

5-4-43.1.2 Inspect.

Inspect nutplates on tail rotor pitch horn fairing for damage and security. **NO DAMAGED OR LOOSE NUTPLATES ALLOWED.** If a damaged or loose nutplate is found, replace as follows:

(1) Remove tail rotor boot (PARA 5-4-41).

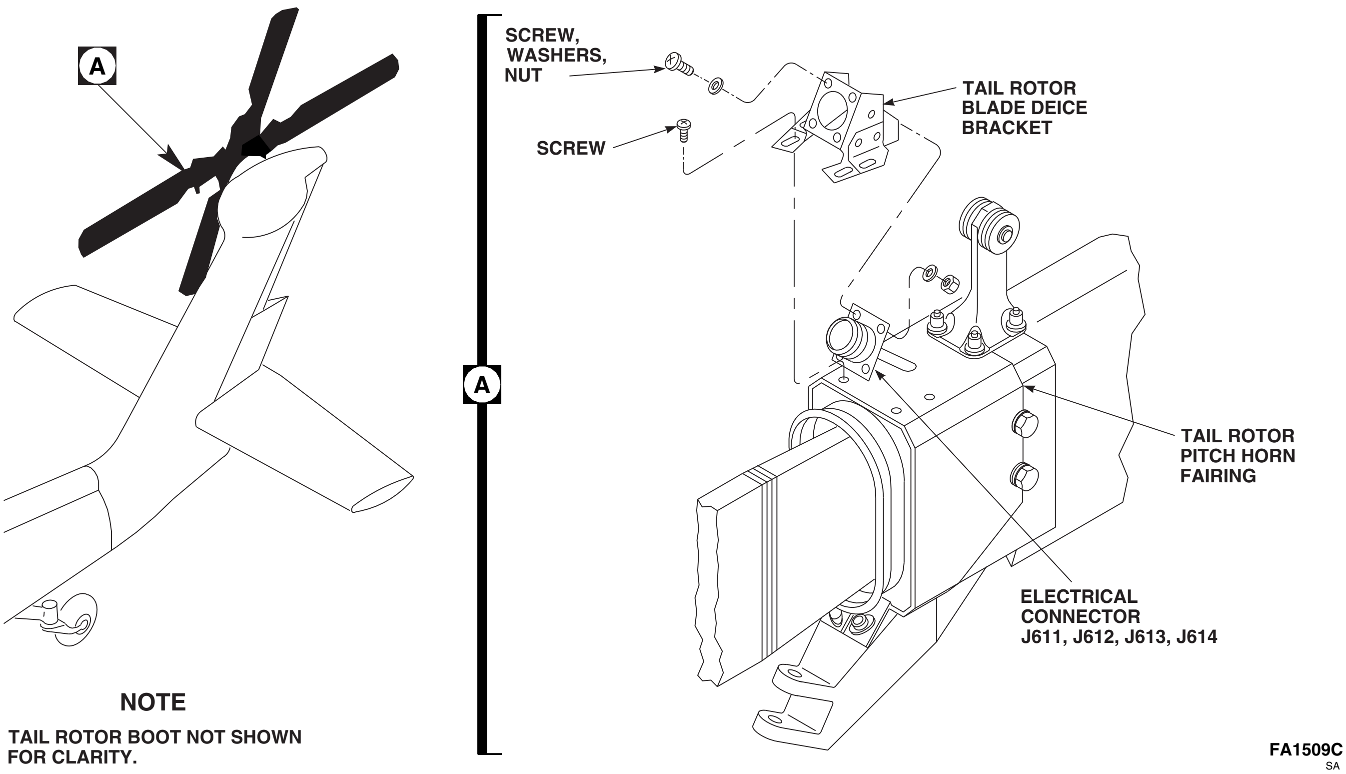


FIGURE 5-4-43-1. REPLACE TAIL ROTOR BLADE DEICE BRACKET

(2) Replace nutplates.

(3) Install tail rotor boot (PARA 5-4-41).

5-4-43.1.3 Install.

a. Install tail rotor blade deice bracket to tail rotor pitch horn fairing with screws (Figure 5-4-43-1, Detail A).

b. If blade deice kit is not installed, remove metal cap on electrical connector. If blade deice kit is installed, remove protective cap on electrical connector

c. Inspect and treat electrical connector (PARA 1-7-20).

d. Connect electrical connector, J611, J612, J613, or J614, in tail rotor blade deice bracket.

e. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to screws, washers, and nuts.

f. While epoxy primer is still wet, secure electrical connector to tail rotor blade deice bracket with screws, washers, and nuts.

g. Apply sealing compound, Item 292, Appendix D, to edges of tail rotor blade deice bracket, tail rotor pitch horn fairing mounting surfaces, electrical connector, and tail rotor blade deice bracket mating surfaces.

h. If blade deice kit is not installed, perform the following:

(1) Install metal cap on electrical connector.

(2) Using lockwire, Item 197, Appendix D, lockwire metal cap.

i. If blade deice kit is installed, connect blade deice electrical connector (PARA 12-4-52).

SECTION 5-5

MAINTENANCE PROCEDURES (BENCH)

SECTION OVERVIEW

5-5-1 MAIN ROTOR SHAFT NUT (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Drift, 7⁄16-inch diameter
- Inside Micrometer Caliper
- Powertrain Repairer's Toolkit

Materials/Parts

- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lacquer, Item 182B, Appendix D
- Lacquer, Item 187, Appendix D
- Masking Tape, Item 212, Appendix D
- Sealing Compound, Item 302, Appendix D

References

- Appendix D
- ASTM E1444

5-5-1.1 Replace Insert (BENCH).

NOTE

Replacement of all inserts is the same.

5-5-1.1.1 Remove.

Using arbor press and 7⁄16-inch diameter drift, press insert out from top of main rotor shaft nut (Figure 5-5-1-1, Detail A).

5-5-1.1.2 Inspect.

a. Magnetic particle inspect main rotor shaft nut (ASTM E1444).

b. Using inside micrometer caliper, measure inside diameter (ID) of hole where insert is to be installed. **ID MUST NOT BE GREATER THAN 0.501-INCH.** If ID is beyond limits, replace main rotor shaft nut.

5-5-1.1.3 Install.

a. Apply sealing compound, Item 302, Appendix D, per manufacturer's instructions.

NOTE

Before installing insert, make sure chamfer on insert is facing bottom of main rotor shaft nut.

b. Using arbor press and 7⁄16-inch diameter drift, press insert in from bottom of main rotor shaft nut (Figure 5-5-1-1, Detail A).

c. Using masking tape, Item 212, Appendix D, mask threads and inserts.

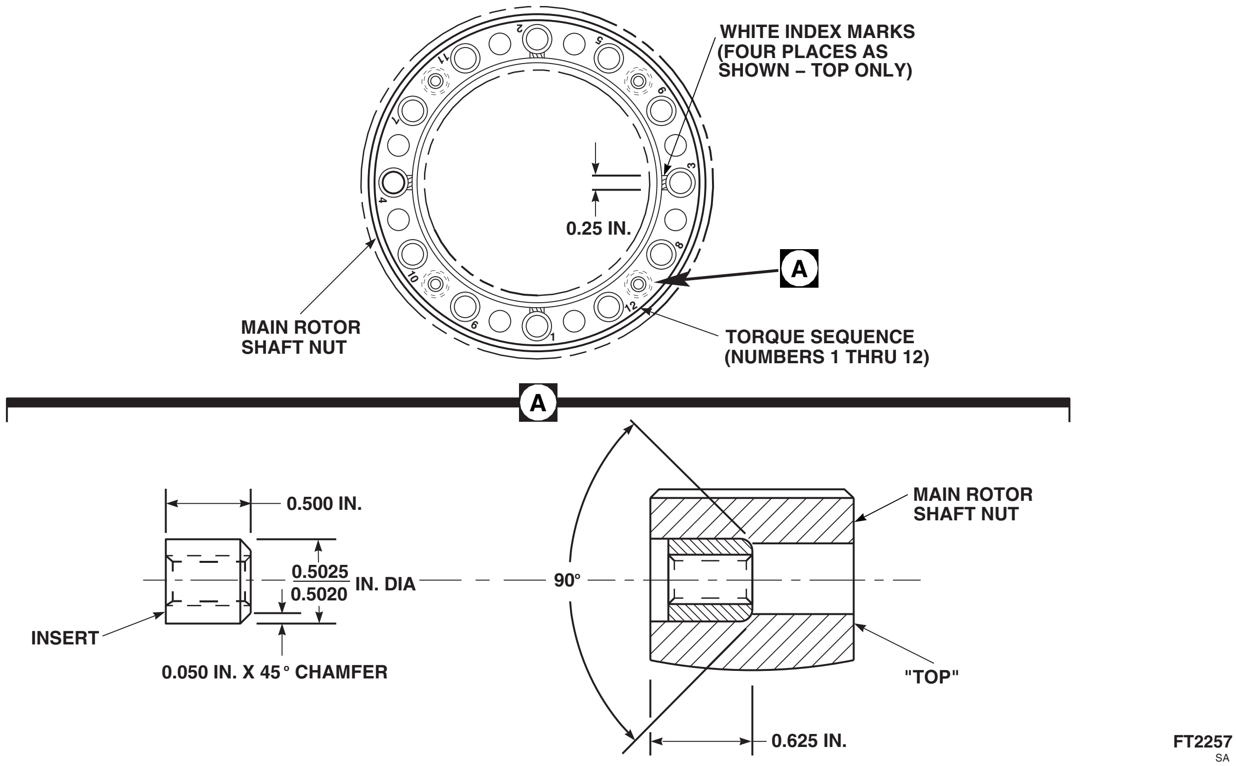


FIGURE 5-5-1-1. REPLACE INSERT (BENCH)

d. Using epoxy primer coating kit, Item 135, Appendix D, apply one coat of epoxy primer to main rotor shaft nut.

e. Using lacquer, Item 187, Appendix D, apply two coats to main rotor shaft nut.

f. Using lacquer, Item 182B, Appendix D, paint index marks on main rotor shaft nut (Figure 5-5-1-1).

g. Using lacquer, Item 182B, Appendix D, paint numbers indicating torque sequence over existing stencil marks.

5-5-2 SPINDLE SLEEVE BEARING OUTER SLEEVE (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Arbor Press
- Oven
- Powertrain Repairer's Toolkit

5-5-2.1 Replace Spindle Sleeve Bearing Outer Sleeve (BENCH).

NOTE

The following procedure applies to droop stop support ring, 70105-08056-101, only.

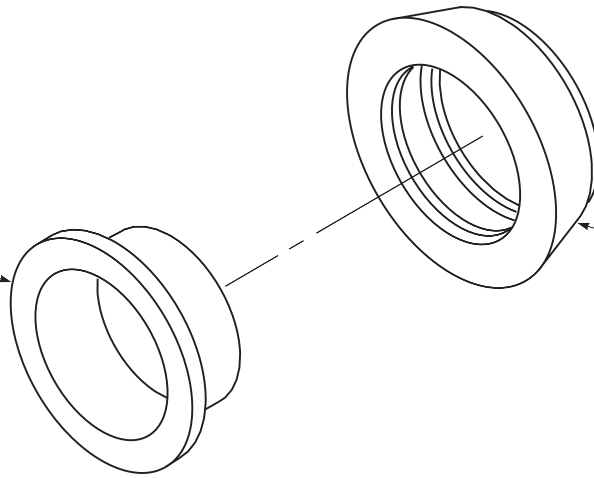
5-5-2.1.1. Remove.

Place droop stop support ring and sleeve bearing outer sleeve in arbor press and carefully remove sleeve bearing outer sleeve (Figure 5-5-2-1).

5-5-2.1.2. Install.

a. Heat droop stop support ring in an oven not over 200°F (93°C) for 30 minutes.

b. Using old sleeve bearing outer sleeve as a drift, carefully press sleeve bearing outer sleeve into droop stop support ring (Figure 5-5-2-1).



EFFECTIVITY

ON DROOP STOP SUPPORT RING,
70105–08056–101.

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FIGURE 5-5-2-1. REPLACE SPINDLE SLEEVE BEARING OUTER SLEEVE (BENCH)

5-5-3 SPINDLE CUFF LUG BUSHINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Arbor Press
- Blade/Spindle Cuff Bushing Installation Tools, Locally-Made, Figure H-161, Appendix H
- Inside Micrometer Caliper
- Powertrain Repairer's Toolkit

Materials/Parts

- Dry Ice, Item 125, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- PARA 5-4-5

5-5-3.1 Replace Spindle Cuff Lug Bushings (BENCH).

NOTE

Replacement of all spindle cuff lug bushings is the same.

5-5-3.1.1 Remove.

a. Locally make blade/spindle cuff bushing installation tools, A, B, and C (Figure H-161, Appendix H).

b. Using tool A and arbor press, press out spindle cuff lug bushing (Figure 5-5-3-1).

5-5-3.1.2 Inspect.

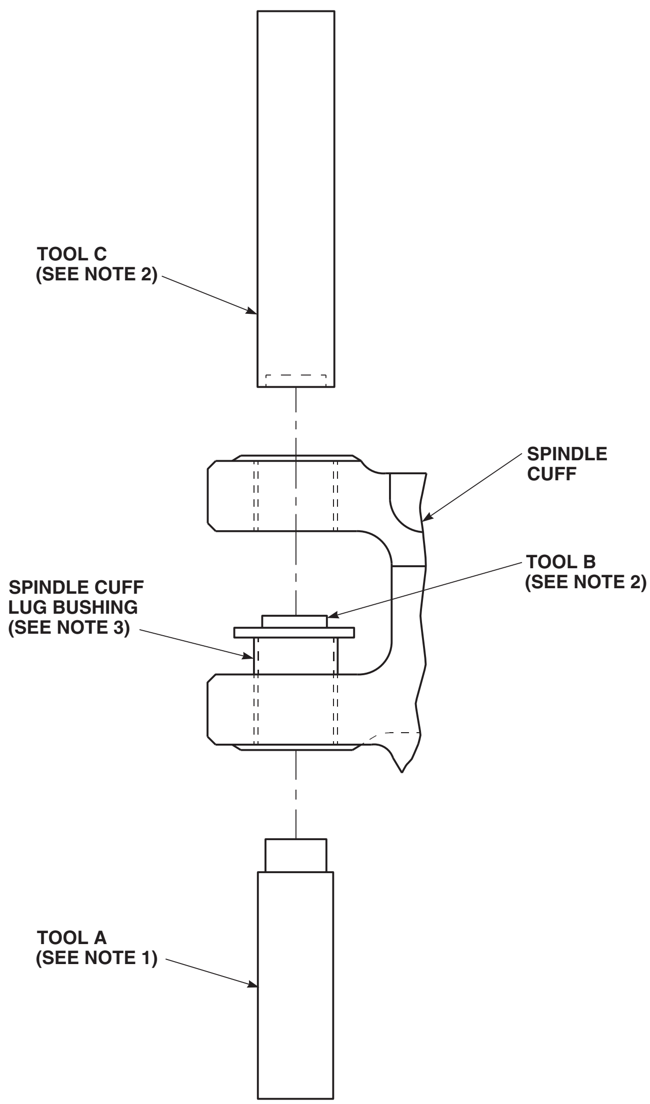
a. Inspect bores for scoring or corrosion. **NONE ALLOWED.** If damage is found, replace spindle (PARA 5-4-5).

b. Using inside micrometer caliper, measure inside diameter (ID) of spindle cuff lug bushing bore. **MAXIMUM ID IS 1.125-INCHES.** If ID is beyond limits, replace spindle (PARA 5-4-5).

5-5-3.1.3 Install.

a. Using dry ice, Item 125, Appendix D, chill replacement spindle cuff lug bushings for a minimum of 3 hours before installation.

b. Remove spindle cuff lug bushings from dry ice.



NOTES

1. TOOL A IS USED WITH ARBOR PRESS TO REMOVE SPINDLE CUFF LUG BUSHING.
2. TOOL B IS USED WITH TOOL C TO INSTALL SPINDLE CUFF LUG BUSHING.
3. INSTALL FLUSH TO 0.003 IN. BELOW SPINDLE CUFF SURFACE.

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FIGURE 5-5-3-1. REPLACE SPINDLE CUFF LUG BUSHINGS (BENCH)

c. Apply coat of primer, Item 258, Appendix D, to spindle cuff lug bushing outer diameter and bushing bore.

d. Using tools B, C, and arbor press, press in spindle cuff lug bushings while primer is still wet flush to 0.003-inch below surface (Figure 5-5-3-1).

e. Apply bead of sealing compound, Item 292, Appendix D, to edges of spindle cuff lug bushing.

5-5-4 SPINDLE ELASTOMERIC BEARING ASSEMBLY (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Feeler Gage, 0.002-inch
- Inside Micrometer Caliper
- Nonmetallic Scraper, Locally-Made
- Oven
- Powertrain Repairer's Toolkit
- Protective Gloves
- Rawhide Mallet
- Sleeve Bearing Removal/Installation Tools, Locally-Made, Figure H-198, Appendix H
- Soft-Bristle Brush
- Spherical Bearing Support Tool, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 9A, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dishwashing Compound, Item 122, Appendix D
- Dry Ice, Item 125, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Mild Detergent
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D

References

- Appendix D
- Appendix H
- PARA 5-3-1
- PARA 5-3-3
- PARA 5-4-9
- PARA 5-4-12

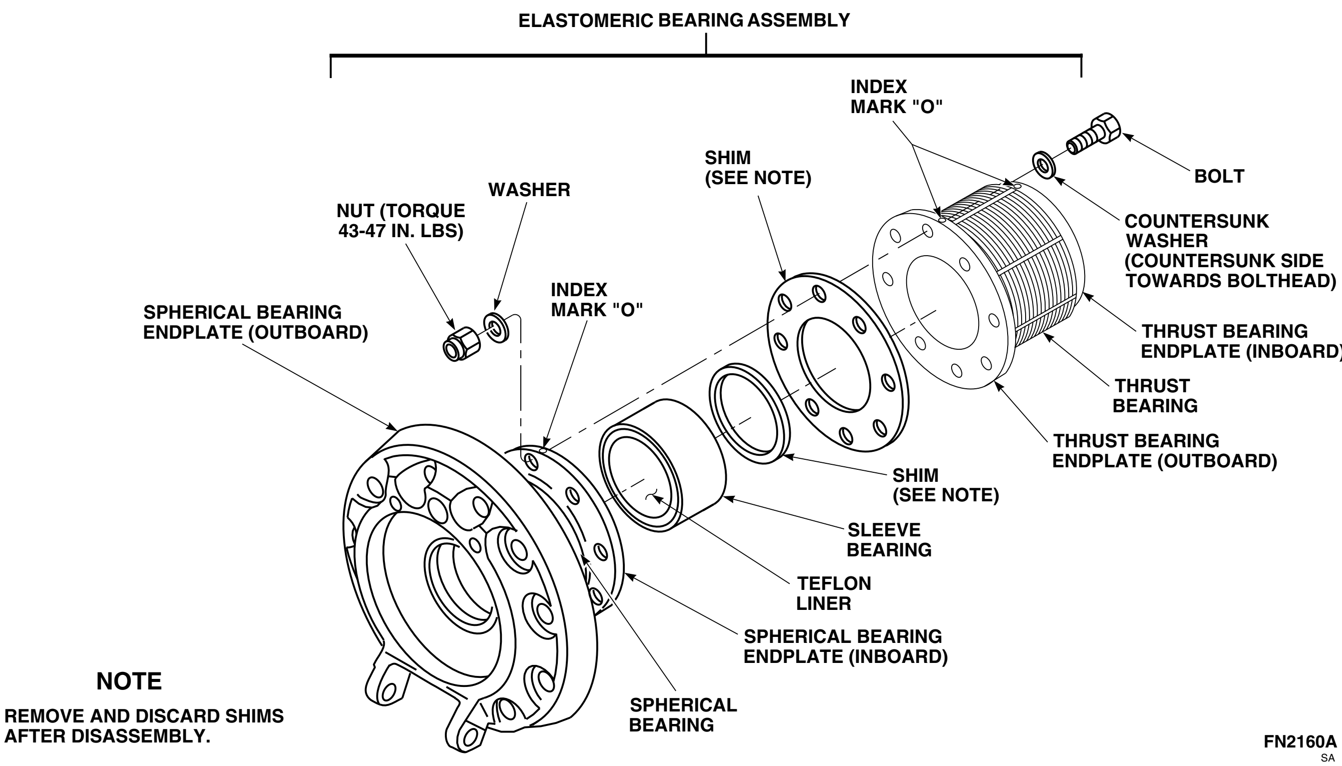


FIGURE 5-5-4-1. REPAIR SPINDLE ELASTOMERIC BEARING ASSEMBLY (BENCH)

5-5-4.1 Repair Spindle Elastomeric Bearing Assembly (BENCH).

5-5-4.1.1 Disassemble.

a. Locally make sleeve bearing removal/ installation tools (Figure H-198, Appendix H).

b. Using nonmetallic scraper, remove sealing compound from boltheads.

c. Remove bolts, countersunk washers, washers, and nuts securing spherical and thrust bearings together (Figure 5-5-4-1). Discard nuts.

d. Separate spherical bearing from thrust bearing.

NOTE

- Shims, SS52C3000Z3237, between sleeve bearing and thrust bearing, or shims, 70102-08119-101, between spherical and thrust bearing endplates will no longer be required.

- If shims, 70102-08119-101, are found between spherical bearing and thrust bearing endplates, bolts securing bearings must be replaced with shorter bolts.

e. If shims are found between spherical bearing, sleeve bearing, and thrust bearing, discard shims.

f. If shims are found between spherical bearing and thrust bearing endplates, discard shims and bolts.

5-5-4.1.2 Clean.

CAUTION

If any oil, grease, or other fluids are spilled or accidentally get on spherical and thrust bearings, do not use solvent to remove them. Use only warm water, dishwashing compound, or mild detergent, and soft-bristle brush to clean elastomeric bearings. After cleaning, rinse with clear water and then dry.

a. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean spherical and thrust bearing titanium and aluminum endplates, and bolts and washers.

b. Using machinery towel, Item 211, Appendix D, dishwashing compound, Item 122, Appendix D, or mild detergent, and soft-bristle brush, clean spherical and thrust bearings.

5-5-4.1.3 Inspect/Repair.

a. Inspect spherical and thrust bearings (PARA 5-3-1).

b. Inspect droop stop lug bushings (PARA 5-3-3).

c. Inspect droop stop pad (PARA 5-4-12).

d. Inspect antiflap stop threaded inserts (PARA 5-4-9).

e. Inspect spherical bearing endplates as follows:

(1) Inspect spherical bearing endplates for cracks. **NONE ALLOWED.** If crack is found, replace spherical bearing.

(2) Inspect spherical bearing endplate (outboard) for corner and surface damage, corrosion pitting, nicks, and scratches. Replace spherical bearing if:

(a) **MAXIMUM DAMAGE EXCEEDS 0.050-INCH DEEP.**

(b) **MINIMUM DISTANCE BETWEEN MULTIPLE DAMAGE EXCEEDS DEPTH OF DAMAGE.**

(c) **DAMAGE EXCEEDS 20% OF TOTAL SURFACE AREA.**

(3) Inspect spherical bearing endplate (inboard) and bearing seat area (3.2465 - 3.2485-inch diameter) for corner and surface damage, corrosion pitting, nicks, and scratches. Replace spherical bearing if:

(a) **MAXIMUM DAMAGE EXCEEDS 0.050-INCH DEEP.**

(b) **MINIMUM DISTANCE BETWEEN MULTIPLE DAMAGE EXCEEDS DEPTH OF DAMAGE.**

(c) **DAMAGE EXCEEDS 20% OF TOTAL SURFACE AREA.**

(4) **NO REPAIR ALLOWED IN 0.255 - 0.260-INCH HOLES.**

f. Repair spherical bearing endplates as follows:

(1) Using abrasive cloth, Item 9A, Appendix D, blend out repairable areas.

(2) Using abrasive cloth, Item 10, Appendix D, polish reworked area to 0.001 - 0.002-inch below damage depth. Blend to 0.005-inch radius.

(3) Using corrosion resistant coating, Item 118, Appendix D, apply to repaired inboard endplate surfaces.

g. Inspect thrust bearing endplates as follows:

(1) Inspect thrust bearing endplates for cracks. **NONE ALLOWED.** If crack is found, replace thrust bearing.

(2) Inspect thrust bearing endplates for corner and surface damage, corrosion pitting,

nicks, and scratches. Replace thrust bearing if:

(a) **MAXIMUM DAMAGE EXCEEDS 0.050-INCH DEEP.**

(b) **MINIMUM DISTANCE BETWEEN MULTIPLE DAMAGE EXCEEDS DEPTH OF DAMAGE.**

(c) **DAMAGE EXCEEDS 20% OF TOTAL SURFACE AREA.**

(3) **NO REPAIR ALLOWED IN 0.255 - 0.260-INCH HOLES OR IN SPLINE AREAS.**

h. Repair thrust bearing endplates as follows:

(1) Using abrasive cloth, Item 9A, Appendix D, blend out repairable areas.

(2) Using abrasive cloth, Item 10, Appendix D, polish reworked area, to 0.001 - 0.002-inch below damage depth. Blend to 0.005-inch radius.

(3) Using corrosion resistant coating, Item 118, Appendix D, apply to repaired outboard aluminum endplate surfaces.

i. Using inside micrometer caliper, measure journal bearing inside diameter (ID) 0.050-inch from each end of ID in four places, at 12/6 o'clock position and at 3/9 o'clock position. **AVERAGE MAXIMUM ID IS 2.922-INCHES. OUT-OF-ROUND MUST NOT EXCEED 0.014-INCH.** If out-of-round exceeds limits, replace sleeve bearing.

j. Inspect journal bearing Teflon surface for disbonding. **EDGE DISBOND MUST NOT EXCEED 1/4-INCH. MULTIPLE EDGE DISBONDS MUST NOT BE CLOSER THAN 1-INCH APART.** If disbonding exceeds limits, replace journal bearing.

k. Inspect journal bearing Teflon surface for scoring (either circumferential or longitudinal), nicks, or gouges. Measure damage and outer race ID. Chart results on graph (Figure 5-5-4-2). If depth of damage extends into NOT ACCEPTABLE area of graph, replace journal bearing.

l. To remove journal bearing from spherical bearing, perform the following:

(1) Using spherical bearing support tool, support spherical bearing in arbor press with droop stop lugs facing up.

(2) Using journal bearing removal/installation tool, remove sleeve bearing from spherical bearing. Discard sleeve bearing.

(3) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean sleeve bearing aluminum seat in spherical bearing.

5-5-4.1.4 Assemble.

WARNING

Injury to personnel and damage to equipment will result if spherical and thrust bearings are assembled without installing sleeve bearing outer race into spherical bearing. Make sure sleeve bearing outer race is installed into spherical bearing prior to assembling spherical and thrust bearings.

NOTE

- If spherical and/or thrust bearing of elastomeric bearing assembly will be replaced, make sure shelf life of replacement bearing(s), SB7001-048 and SB7002-048, has not exceeded 60 months. If shelf life has exceeded 60 months, do not use.
- Sleeve bearing outer race is 2 1/4-inches long with Teflon fabric lining on inside diameter.

a. Check spherical bearing for journal bearing installation. If bearing is not installed, install bearing into spherical bearing as follows:

WARNING

Wear protective gloves when handling dry ice.

(1) Using dry ice, Item 125, Appendix D, chill replacement sleeve bearing for minimum of one hour.

(2) Heat spherical bearing in oven at 140° - 160°F (60° - 71°C) for maximum of 20 minutes.

(3) Position spherical bearing support tool in arbor press.

(4) Have rawhide mallet and sleeve bearing installation tool ready at arbor press to install bearing.

(5) Remove spherical bearing from oven, and using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, coat spherical bearing bore.

(6) Position spherical bearing into spherical bearing support tool with droop stop lug down.

(7) Remove journal bearing from dry ice and carefully position into spherical bearing bore. Tap bearing evenly to align bearing and prevent cocking.

(8) Using sleeve bearing installation tool and arbor press, press bearing into spherical bearing 1 3/4 - 2-inches. Remove sleeve bearing installation tool and spherical bearing from support tool and check leading edge of sleeve bearing for burrs and debris pickup. Clean burrs and pickup and continue pressing sleeve bearing into spherical bearing until bottomed.

(9) Using 0.002-inch feeler gage, make sure bearing is fully seated in spherical bearing bore.

(10) Allow assembly to return to room temperature.

(11) Using inside micrometer caliper, measure bearing inside diameter (ID) 0.050-inch from each end of ID in four places, at 12/6 o'clock position and at 3/9 o'clock position. **AVERAGE MINIMUM ID IS 2.9145-INCHES.** If ID exceeds limit, replace bearing.

b. Using epoxy primer coating kit, Item 135, Appendix D, primer, Item 258, Appendix D, or sealing compound, Item 301A, Appendix D, apply thin coat to mating surfaces of spherical and thrust bearings, and bolts.

c. Line up three index marks "O" on spherical and thrust bearings. Install bolts and countersunk washers. Make sure countersunk side of washer is toward bolthead, with bolthead facing thrust bearing side (Figure 5-5-4-1).

d. Install washers and nuts on bolts.

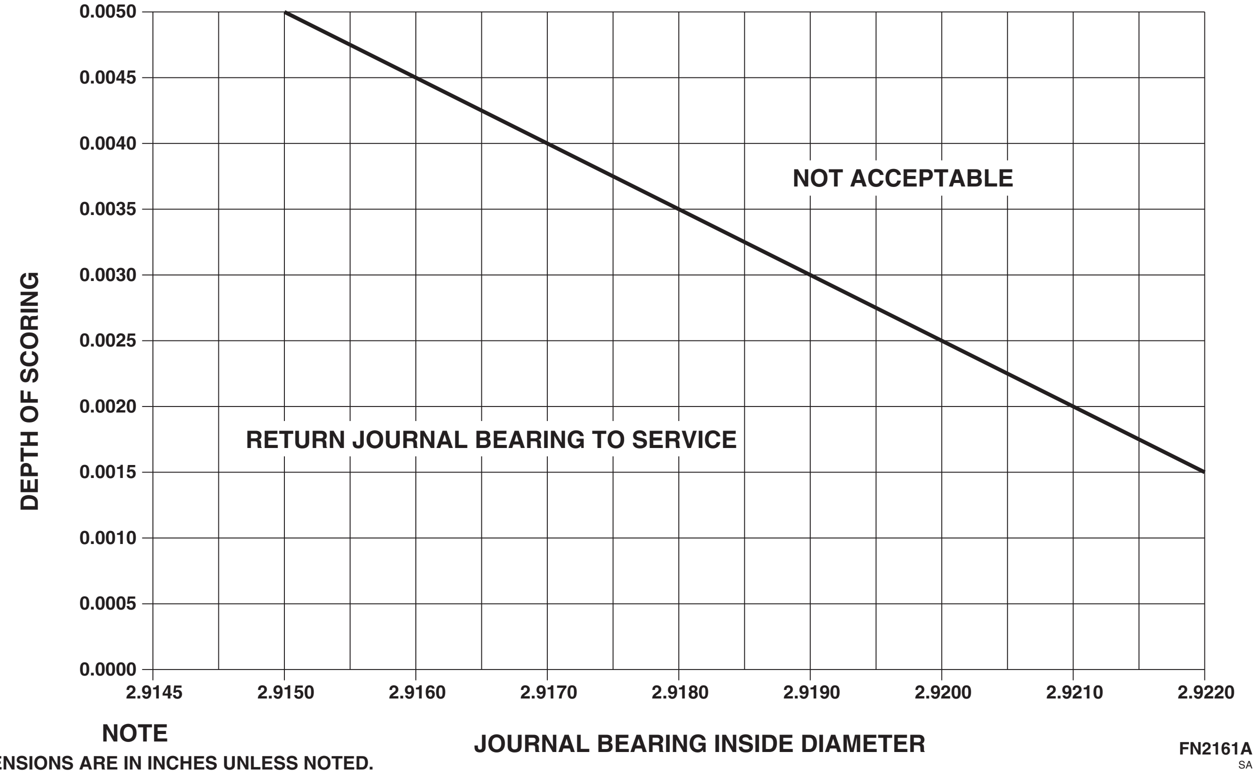
e. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

f. Using machinery towel, Item 211, Appendix D, wipe excess primer from bolts and bearing mating surfaces.

g. Apply bead of sealing compound, Item 292, Appendix D, to boltheads, nuts, and joint between spherical and thrust bearings.

h. Using epoxy primer coating kit, Item 135, Appendix D, touch up exposed surfaces of bolts, washers, and nuts with epoxy primer.

i. If shim, 70102-08119-101, was removed, reidentify elastomeric bearing assembly to part number: 70102-08100-044.



NOTE
ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

FIGURE 5-5-4-2. INSPECT JOURNAL BEARING (BENCH)

5-5-5 DAMPER INDICATOR (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench
- Damper Indicator Cradle, Locally-Made, Figure H-164, Appendix H
- Damper Indicator Insertion Tool, Locally-Made, Figure H-165, Appendix H
- Damper Indicator Installation Tool, Locally-Made, Figure H-166, Appendix H
- Drill
- Drill Bit, No. 52
- Hydraulic Repairer's Toolkit
- Hydraulic Servicing Unit
- Pin Punch
- Protection, Eye
- Spanner Wrench, Adjustable

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Grease, Item 148, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Sealing Compound, Item 271A, Appendix D

References

- Appendix D
- Appendix H

5-5-5.1 Repair Damper Indicator (BENCH).

5-5-5.1.1. Disassemble.

a. Locally make the following tools:

(1) Damper indicator cradle (Figure H-164, Appendix H).

(2) Damper indicator insertion tool (Figure H-165, Appendix H).

(3) Damper indicator installation tool (Figure H-166, Appendix H).

b. Manually pull handle to extend piston and position in damper indicator cradle.

c. Using pin punch, tap spring pin into handle (Figure 5-5-5-1, Sheet 1 or Sheet 2).

d. Remove damper indicator from damper indicator cradle.

e. Mark housing in line with vent on the locking ring.

f. Using Allen wrench, remove two setscrews from locking ring.

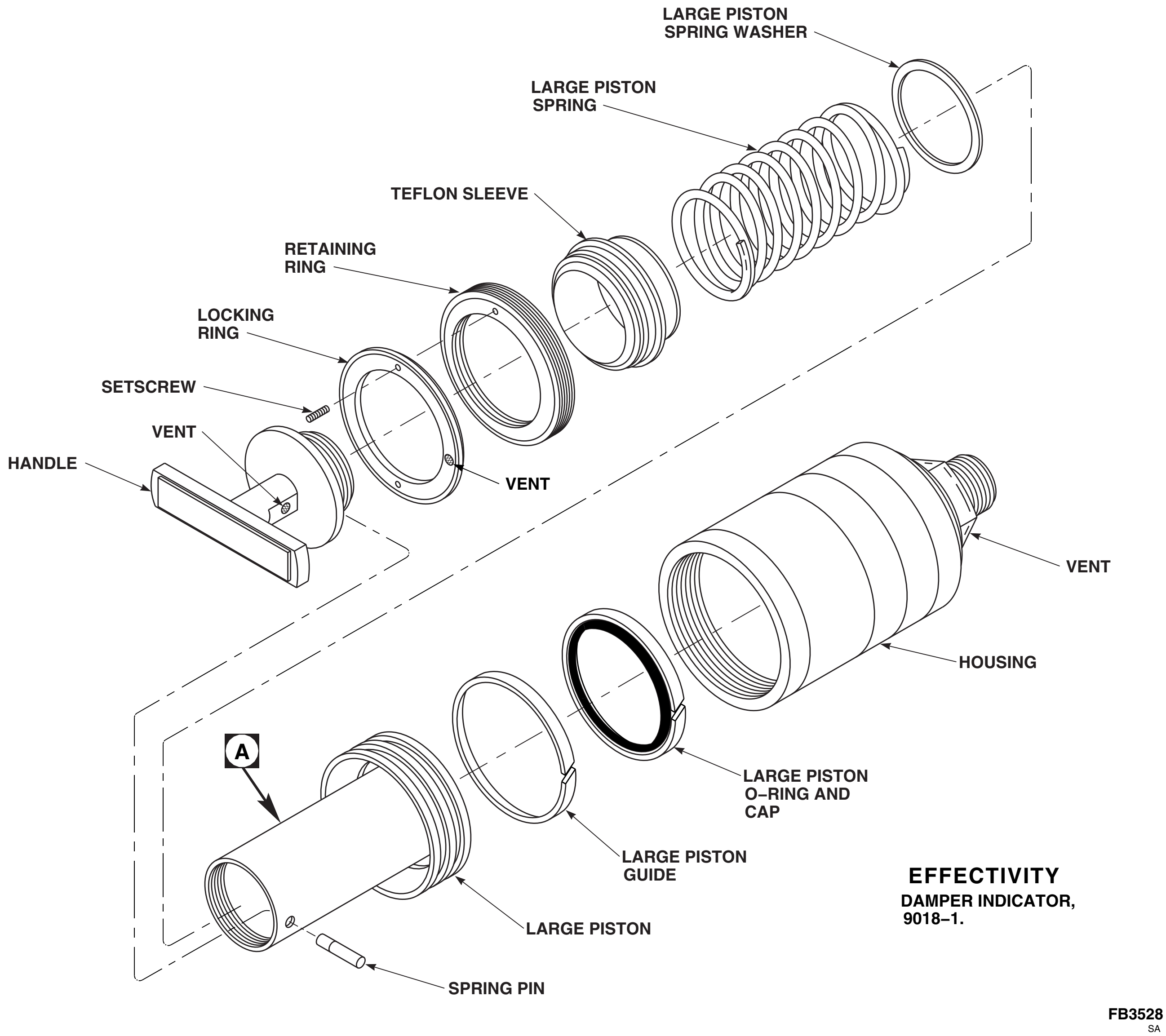


FIGURE 5-5-5-1. REPAIR DAMPER INDICATOR (BENCH) (SHEET 1 OF 3)

g. On damper indicator, 9018-1, disassemble as follows (Figure 5-5-5-1, Sheet 1):

(1) Using adjustable spanner wrench, unscrew locking ring from housing and slide it over the handle.

(2) Using same adjustable spanner wrench, unscrew retaining ring and slide it over handle.

(3) Grasping handle, slowly remove large piston from housing.

WARNING

Injury to personnel and damage to equipment will result if care is not taken when removing handle. Piston springs are compressed. Take care when removing handle.

(4) Slowly unscrew handle from large piston.

(5) Remove spring pin from inside handle.

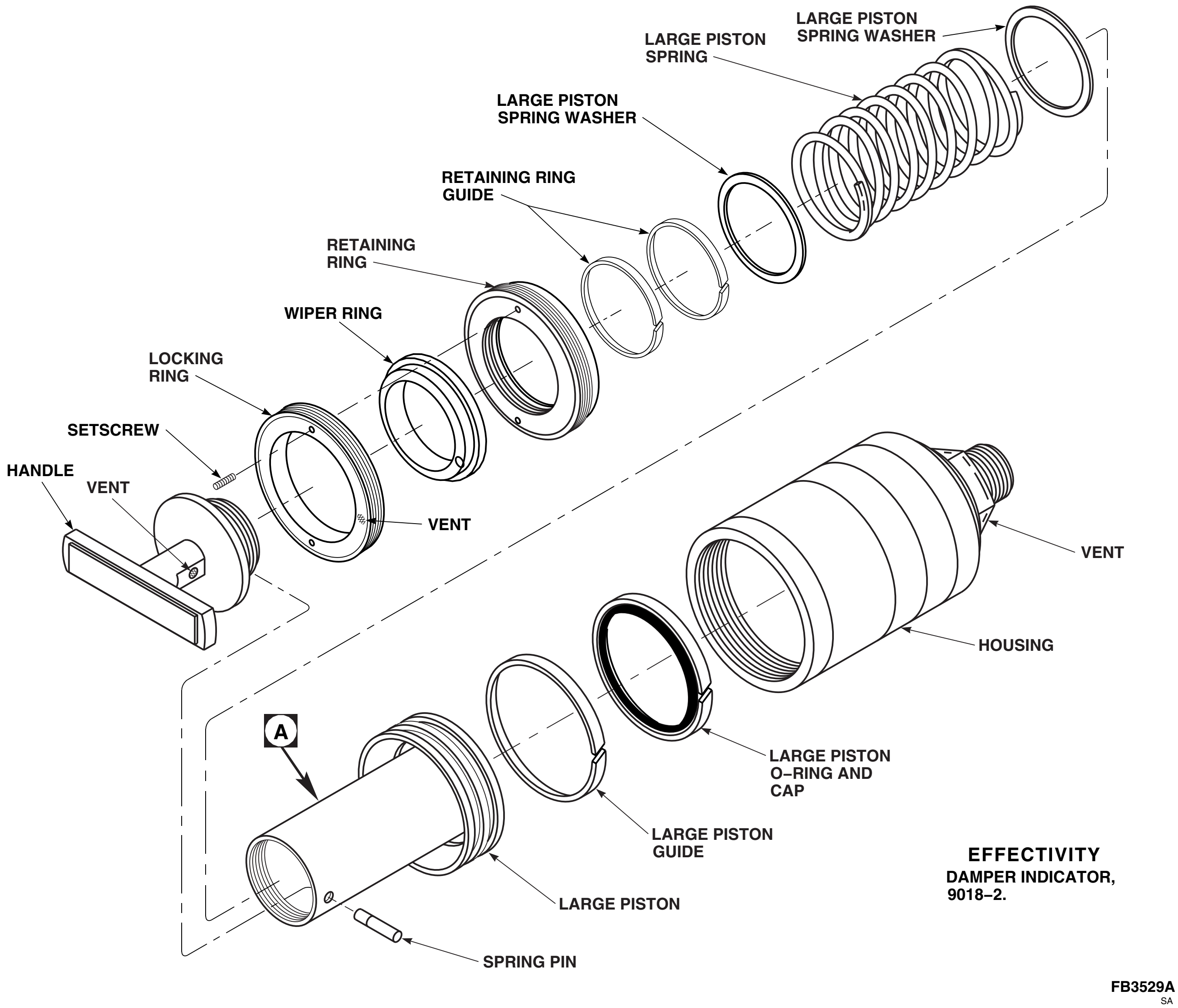


FIGURE 5-5-5-1. REPAIR DAMPER INDICATOR (BENCH) (SHEET 2 OF 3)

(6) Remove large piston spring and Teflon sleeve.

(7) Remove small piston spring from inside large piston (Figure 5-5-5-1, Sheet 3, Detail A).

(8) Push small piston out of large piston.

(9) Remove small piston O-ring, large piston O-ring and cap, small piston guides and large piston guide from large and small pistons (Figure 5-5-5-1, Sheet 1 and Sheet 3, Detail A). Discard O-rings.

(10) Remove spring washers from large and small pistons, including small spring washer from underside of the handle.

h. On damper indicator, 9018-2, disassemble as follows (Figure 5-5-5-1, Sheet 2):

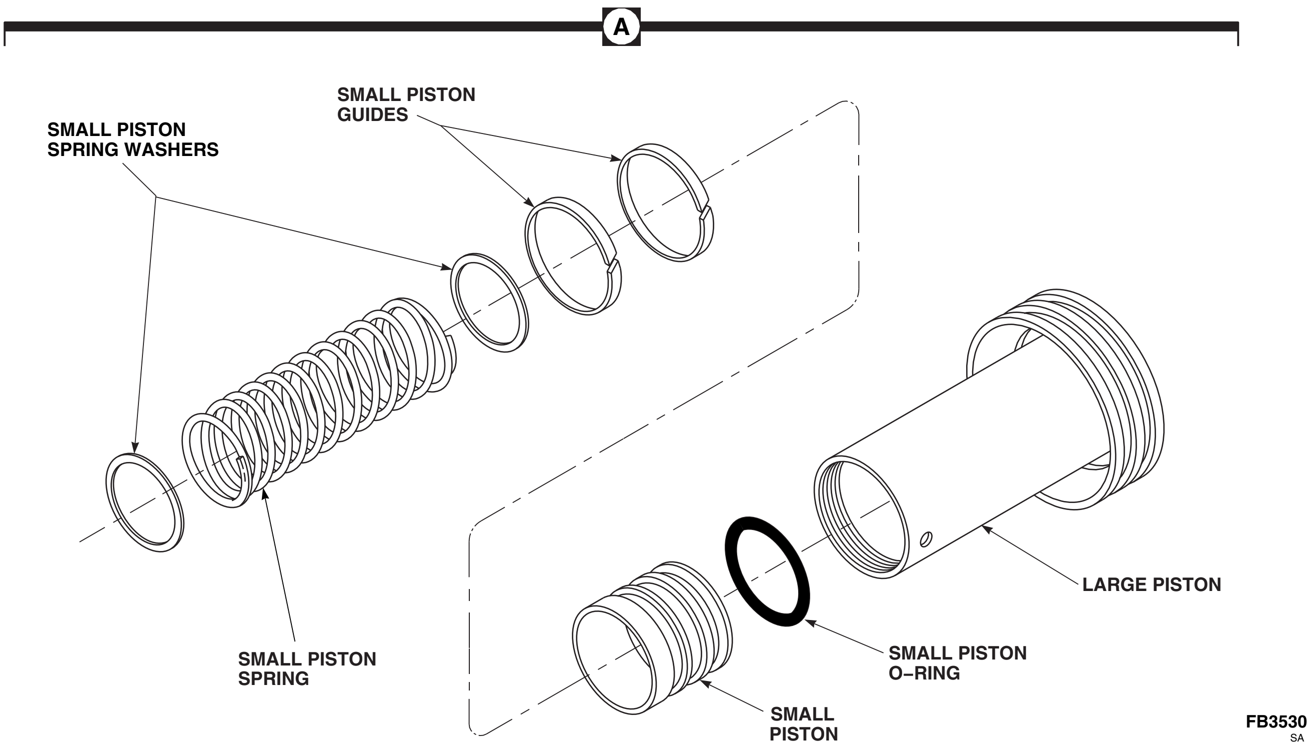


FIGURE 5-5-5-1. REPAIR DAMPER INDICATOR (BENCH) (SHEET 3 OF 3)

WARNING

Injury to personnel and damage to equipment will result if care is not taken when removing handle. Piston springs are compressed. Take care when removing handle.

(1) Slowly unscrew handle from large piston.

(2) Remove spring pin from inside handle.

(3) Remove small piston spring and small piston spring washers from inside large piston (Figure 5-5-5-1, Sheet 3, Detail A).

(4) Using adjustable spanner wrench, unscrew locking ring and wiper ring from housing (Figure 5-5-5-1, Sheet 2). Separate wiper ring and locking ring.

(5) Using same adjustable spanner wrench, unscrew retaining ring from housing. Remove retaining ring guides from retaining ring.

(6) Remove large piston spring and large piston spring washers.

(7) Slowly remove large piston from housing.

(8) Push small piston out of large piston (Figure 5-5-5-1, Sheet 3, Detail A).

(9) Remove small piston O-ring, large piston O-ring and cap, small piston guides and large piston guide from large and small pistons (Figure 5-5-5-1, Sheet 2 and Sheet 3, Detail A). Discard O-rings.

i. Using dry-cleaning solvent, Item 124, Appendix D, clean all parts.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

j. Using clean, dry, compressed air, blow dry all parts.

5-5-5.1.2. Inspect.

a. Inspect exterior and interior of large and small pistons for nicks, scratches, gouges, and general condition. Pay particular attention to piston guide grooves and lands. **NONE ALLOWED.** If damaged, replace piston.

b. Inspect servicing marks on exterior of large piston. If black or gold band anodizing is worn through to aluminum, replace large piston.

c. Inspect exterior and interior of housing for damage. **NONE ALLOWED.** If damaged, replace housing.

d. Inspect handle, locking ring, and retaining ring for damage. Pay particular attention for cracks in the locking ring. **NONE ALLOWED.** If damaged, replace handle, locking ring, and/or retaining ring.

e. Inspect large and small piston springs for corrosion and cracks. **NONE ALLOWED.** If corroded or cracked, replace large and/or small piston spring.

f. Inspect stainless steel mesh in the three vent holes for integrity and damage. **NO DAMAGE ALLOWED.** If stainless steel mesh integrity is questionable or damaged, replace as necessary.

g. Inspect for any damaged, loose, or missing labels and decals on exterior of housing. **NONE ALLOWED.** If damaged, loose, or missing, replace as necessary.

5-5-5.1.3. Assemble.

a. If new large piston or new handle is being used, perform the following:

(1) Thread handle into large piston (Figure 5-5-5-1, Sheet 1 and Sheet 2).

(2) If spring pin holes do not line up, using drill with No. 52 drill bit, drill hole in handle in line with existing hole in large piston. Deburr hole.

(3) Unthread handle from large piston. Using dry-cleaning solvent, Item 124, Appendix D, clean any metal particles from threads.

(4) Thread retaining ring and locking ring to housing. Remove retaining and locking rings. Using dry-cleaning solvent, Item 124, Appendix D, clean any metal particles from threads.

b. Using hydraulic fluid, Item 154, Appendix D, coat all internal parts before assembly.

c. Install new small piston guides and small piston O-ring on small piston (Figure 5-5-5-1, Sheet 3, Detail A).

d. Using damper indicator installation tool, install new large piston guide, and O-ring and cap on large piston (Figure 5-5-5-1, Sheet 1, Sheet 2, and Sheet 3, Detail A).

e. Install new large and small piston spring washers in large and small pistons, including small spring washer in underside of handle.

f. Install small piston into bore of large piston until it bottoms. Make sure small piston O-ring and small piston guides remain in their proper position (Figure 5-5-5-1, Sheet 3, Detail A).

g. On damper indicator, 9018-1, assemble as follows (Figure 5-5-5-1, Sheet 1):

(1) Using grease, Item 148, Appendix D, apply a thin coat to threads of handle and large piston.

(2) Place small piston spring into bore of large piston. Make sure spring seats

properly inside small piston (Figure 5-5-5-1, Sheet 3, Detail A).

(3) Place large piston spring over large piston. Place new Teflon sleeve in top of large spring.

(4) Place handle into Teflon sleeve. Push down on handle, compressing both springs, and screw into large piston.

(5) Using damper indicator insertion tool, install large piston into housing. Make sure large piston O-ring and cap and large piston guide remain in their proper position.

(6) Using grease, Item 148, Appendix D, apply a thin coat to threads of housing, retaining ring, and locking ring.

(7) Screw retaining ring into housing until large piston no longer feels loose in housing (approximately three full turns).

(8) Slip locking ring over handle. Screw locking ring into housing until it contacts retaining ring.

(9) **CONTINUE TO TIGHTEN LOCKING RING UNTIL VENT ALIGNS WITH MARK ON HOUSING.**

h. On damper indicator, 9018-2, assemble as follows (Figure 5-5-5-1, Sheet 2):

(1) Using damper indicator insertion tool, install large piston into housing. Make sure large piston O-ring and cap and large piston guide remain in their proper position.

(2) Using grease, Item 148, Appendix D, apply a thin coat to threads of housing, retaining ring, and locking ring.

(3) Place large piston spring over large piston.

(4) Install retaining ring guides into retaining ring. Place large piston spring washer and retaining ring on top of large spring.

(5) Push down on retaining ring and screw into housing until top of retaining ring is approximately ¼-inch below top of housing. Make sure retaining ring guides remain in their proper position.

(6) Install wiper ring into locking ring.

(7) Screw locking ring and wiper ring assembly into housing until it contacts retaining ring.

(8) **CONTINUE TO TIGHTEN LOCKING RING UNTIL VENT ALIGNS WITH MARKS ON HOUSING.**

(9) Using grease, Item 148, Appendix D, apply a thin coat to threads of handle and large piston.

(10) Place small piston spring into bore of large piston (Figure 5-5-5-1, Sheet 3, Detail A). Place washer and handle on top of spring (Figure 5-5-5-1, Sheet 1 or Sheet 2).

(11) Push down on handle, compressing spring, and screw into large piston (Figure 5-5-5-1, Sheet 2).

i. Manually pull handle to extend piston and position in damper indicator cradle.

j. Using pin punch, install spring pin into large piston and handle. Remove damper indicator from damper indicator cradle.

k. Apply sealing compound, Item 271A, Appendix D, to setscrews.

l. **USING ALLEN WRENCH, TIGHTEN SETSCREWS INTO LOCKING RING. DO NOT OVERTIGHTEN.**

5-5-5.1.4. Test.

a. Connect hydraulic servicing unit to damper indicator.

CAUTION

Damage to piston may result if hydraulic pressure exceeds 100 psi. Do not exceed 100 psi hydraulic pressure.

b. Slowly apply hydraulic pressure, and fully extend piston. Check valve in bottom of housing should release fluid before 100 psi.

c. Allow damper indicator to sit under pressure for 30 minutes. Check for leakage. **NONE ALLOWED.** If damper indicator leaks, repeat repair procedures. Pay particular attention to installation of new packings.

5-5-6 DAMPER ROD END, CYLINDER HEAD, AND SEALS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Container
- Fluorescent Penetrant Inspection Kit
- Hydraulic Repairer's Toolkit
- Inside Micrometer Caliper
- Main Rotor Damper Holding Fixture, Locally-Made, Figure H-223, Appendix H
- Main Rotor Damper Piston Rod Insertion Tool, Locally-Made, Figure H-225, Appendix H
- Main Rotor Damper Piston Rod Wrench, Locally-Made, Figure H-226, Appendix H
- Main Rotor Damper Seal Expansion Tool, Locally-Made, Figure H-224, Appendix H
- Main Rotor Damper Sleeve Bearing Installation/Removal Tool, Locally-Made, Figure H-222, Appendix H
- Outside Micrometer Caliper
- Packing, MS28778-6
- Plug, AN814-6D
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

Materials/Parts

- Abrasive Cloth, Item 4, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Abrasive Cloth, Item 13, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Compound, Item 108, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint Remover, Item 230, Appendix D
- Tags

References

- Appendix D
- Appendix H
- ASTM E1417

- PARA 2-6-5
- PARA 5-3-4
- PARA 5-4-14
- PARA 5-4-17

5-5-6.1 Replace Damper Rod End, Cylinder Head, and Seals (BENCH).

NOTE

If rod end and/or cylinder head will be replaced, both are procured with rod end bearing installed.

5-5-6.1.1 Disassemble.

a. Locally make the following tools:

(1) Main rotor damper sleeve bearing installation/removal tool (Figure H-222, Appendix H).

(2) Main rotor damper holding fixture (Figure H-223, Appendix H).

(3) Main rotor damper seal expansion tool (Figure H-224, Appendix H).

(4) Main rotor damper piston rod insertion tool (Figure H-225, Appendix H).

(5) Main rotor damper piston rod wrench (Figure H-226, Appendix H).

b. Remove damper indicator (PARA 5-4-14). Install packing, MS28778-6, on plug, AN814-6D. Plug port.

WARNING

Injury to personnel will result if pressure is not released from damper prior to disassembly. Damper may have residual pressure. Relieve pressure from damper prior to disassembly.

CAUTION

Damper assembly/disassembly can only be done in clean-room environment. Care must be taken during disassembly to avoid introducing dirt or other debris which will contaminate damper.

NOTE

Any damper removed from helicopter due to vibration problems must be forwarded to Depot level facility for repair. Tag damper with tag stating ″DAMPER REQUIRES FORCED VELOCITY TESTING″.

c. Slowly remove bleed plugs from damper, and remove packings. Discard packings. Drain hydraulic fluid from damper into suitable container.

NOTE

Note relationship between cylinder head and cylinder assembly. Components must be reassembled in the same position.

d. Remove nuts, washers, and cylinder head (Figure 5-5-6-1).

e. Remove packing and packing retainer from cylinder head. Discard packing.

f. Remove packing and ring from bushing in cylinder head. Discard packing.

g. Remove retaining ring and bushing from cylinder head. Remove packing from cylinder head (Figure 5-5-6-1, Detail A). Discard packing.

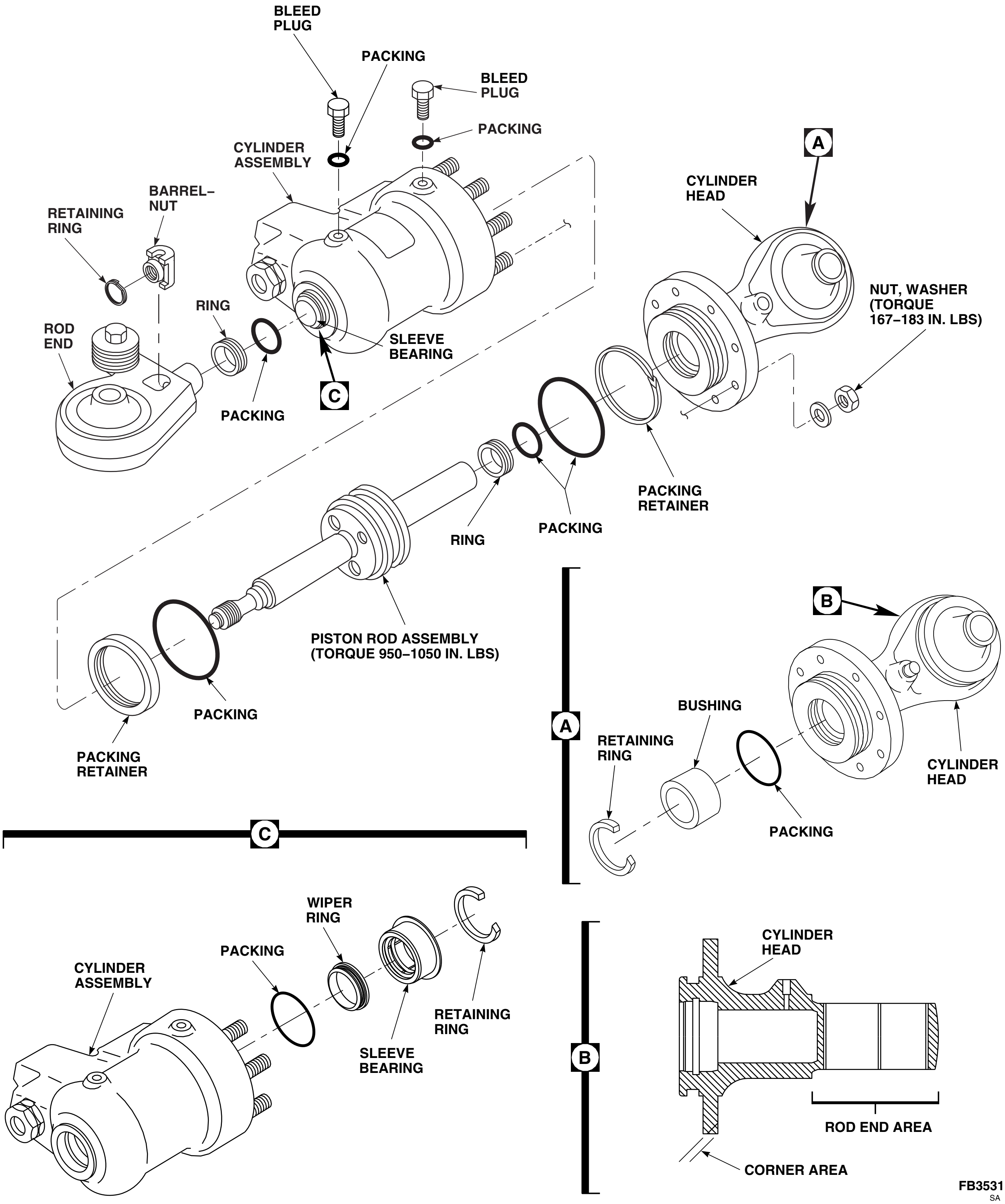


FIGURE 5-5-6-1. REPLACE DAMPER ROD END, CYLINDER HEAD, AND SEALS (BENCH)

h. Remove retaining ring from rod end on piston rod assembly (Figure 5-5-6-1).

CAUTION

Damage to cylinder assembly will result if main rotor damper piston rod wrench comes in contact with interior of cylinder. Use extreme caution when inserting main rotor damper piston rod wrench into cylinder assembly.

i. Install main rotor damper piston rod wrench into cylinder assembly until it engages piston rod assembly. Place cylinder assembly into main rotor damper holding fixture.

j. Using main rotor damper piston rod wrench, break torque to loosen piston rod assembly from rod end.

k. Remove barrel-nut and rod end from piston rod assembly.

l. Remove cylinder assembly from main rotor damper holding fixture and remove main rotor damper piston rod wrench.

CAUTION

Damage to cylinder assembly will result if piston rod assembly comes in contact with interior of cylinder. Use extreme caution when removing piston rod assembly from cylinder assembly.

m. Remove piston rod assembly from cylinder assembly.

n. Remove packing and packing retainer from piston rod assembly. Discard packing and packing retainer.

o. Remove packing and ring from sleeve bearing in cylinder assembly. Discard packing.

p. Remove wiper ring from sleeve bearing (Figure 5-5-6-1, Detail C). Discard wiper ring.

q. Remove retaining ring from cylinder assembly.

r. Using main rotor damper sleeve bearing installation/removal tool and arbor press, press out sleeve bearing from cylinder assembly.

s. Remove packing from cylinder assembly. Discard packing.

5-5-6.1.2 Clean.

a. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean all parts.

b. Using machinery towel, Item 211, Appendix D, wipe all parts dry.

5-5-6.1.3 Inspect/Repair.

a. Inspect bushing on cylinder head for burrs, scratches, nicks, or corrosion (Figure 5-5-6-1, Detail A). **NONE ALLOWED.** If damage is found, replace bushing.

b. Using inside micrometer caliper, measure inside diameter (ID) of bushing on cylinder head. **MAXIMUM ID IS 1-INCH.** If ID exceeds limit, replace bushing.

c. Using outside micrometer caliper, measure outside diameter (OD) of bushing on cylinder head. **MAXIMUM OD IS 1.438-INCHES.** If OD exceeds limit, replace bushing.

d. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect bushing on cylinder head for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace bushing.

e. Inspect damper bearing on cylinder head (PARA 5-3-4).

f. Inspect open areas on cylinder head for nicks, gouges, and corrosion. Replace cylinder head if:

(1) **DAMAGE EXCEEDS 0.040-INCH DEEP OR 15% OF CIRCUMFERENCE ON CIRCULAR FEATURES.**

(2) **DAMAGE EXCEEDS 1-INCH AXIAL LENGTH (EACH REWORK).**

(3) **DAMAGE EXCEEDS 1-INCH DIAMETER.**

(4) **DAMAGE IS CLOSER THAN 1-INCH TO OTHER REWORK.**

g. Inspect rod end area on cylinder head for nicks, gouges, burrs, or corrosion (Figure 5-5-6-1, Detail B). **NONE ALLOWED.** If damage exceeds limits, replace cylinder head.

h. Inspect corner areas of cylinder head for nicks, gouges, and corrosion. Replace cylinder head if:

(1) **CORNER DAMAGE EXCEEDS 0.060-INCH DEEP OR 30% OF CIRCUMFERENCE ON CIRCULAR FEATURES TOTAL.**

(2) **CORNER DAMAGE EXCEEDS 1-INCH IN LENGTH ON LINEAR FEATURES EACH REWORK.**

(3) **CORNER DAMAGE IS CLOSER THAN 1-INCH TO OTHER REWORK.**

i. Inspect mating surface of cylinder head for nicks, gouges, and corrosion. **DAMAGE MUST NOT EXCEED 0.010-INCH DEEP OR 10% OF SURFACE AREA. MATING PART MUST NOT BE REWORKED IN SAME AREA.** If damage exceeds limits, replace cylinder head.

j. Inspect bushing bore diameter on cylinder head. **MAXIMUM DIAMETER MUST NOT EXCEED 1.438-INCHES. OUT-OF-ROUND MUST NOT EXCEED 0.001-INCH.** If diameter or out-of-round exceed limits, replace cylinder head.

k. Inspect piston rod bore diameter on cylinder head. **MAXIMUM DIAMETER IS 1.070-INCH.** If diameter exceeds limit, replace cylinder head.

l. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect cylinder head for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace cylinder head.

m. Inspect cylinder head for missing or damaged protective coating. If necessary, repair cylinder head as follows:

(1) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean damaged areas.

(2) Using paint remover, Item 230, Appendix D, remove paint from corroded area. Rinse part thoroughly with water to remove paint remover.

(3) Using acid brush, Item 84, Appendix D, apply corrosion compound, Item 108, Appendix D, to corroded area. On heavy corrosion, rub area using abrasive cloth, Item 4, Appendix D, while corrosion compound is in corroded area. Do not allow corrosion compound to remain on area for more than 10 minutes.

(4) Using machinery towel, Item 211, Appendix D, remove corrosion compound from area while rinsing with clean water. Repeat step m. (3) and step m. (4) until all corrosion is removed.

(5) Using abrasive cloth, Item 4, Appendix D, blend out area. Do not exceed repair limits.

(6) Using abrasive wheel, Item 22, Appendix D, remove gouges and dents on external surfaces. Blend to 0.001 - 0.002-inch deeper than original damage.

(7) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect repaired area for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace cylinder head.

(8) Using corrosion resistant coating, Item 118, Appendix D, apply to exposed metal.

(9) Touch up paint (PARA 2-6-5).

n. Inspect damper bearing in rod end (PARA 5-3-4).

o. Inspect outer lug areas on rod end for nicks, gouges, and corrosion. **CORROSION MUST NOT EXCEED 0.002-INCH DEEP OR 10% OF TOTAL SURFACE AREA.** If damage exceeds limits, replace rod end.

p. Inspect lug faces on both sides of rod end for nicks, gouges, and corrosion. **NONE ALLOWED.** If damage exceeds limits, replace rod end.

q. Inspect remaining areas of rod end for nicks, gouges, and corrosion. **DAMAGE MUST NOT EXCEED 0.010-INCH DEEP OR 10% OF TOTAL SURFACE.** If damage exceeds limits, replace rod end.

r. Inspect plated areas of piston rod assembly for scratches, nicks, burrs, or corrosion. **NONE ALLOWED.** If damage exceeds limits, replace piston rod assembly.

s. Inspect shaft diameter of piston rod assembly. **DIAMETER MUST NOT BE LESS THAN 0.996-INCH.** If diameter is less than limits, replace piston rod assembly.

t. Inspect piston diameter of piston rod assembly. **DIAMETER MUST NOT BE LESS THAN 2.970-INCHES.** If diameter is less than limits, replace piston rod assembly.

u. Inspect threads of piston rod assembly for crossed, stripped, or flattened threads. **NONE ALLOWED.** If crossed, stripped, or flattened threads are found, replace piston rod assembly.

v. Inspect piston rod assembly for missing or damaged protective coating (chrome-plating). **NONE ALLOWED.** If protective coating is missing or damaged, replace piston rod assembly.

w. Inspect packing groove diameter on piston rod assembly. **DIAMETER MUST NOT BE LESS THAN 2.621-INCHES.** If diameter is less than limits, replace piston rod assembly.

x. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect piston rod assembly for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace piston rod assembly.

y. If necessary, repair piston rod assembly as follows:

(1) Using abrasive cloth, Item 13, Appendix D, and abrasive cloth, Item 10, Appendix D, polish minor axial scoring on 0.996 - 0.998-inch chrome-plated diameter not deeper than 0.0002-inch.

(2) **IF SCORING IS DEEPER THAN 0.0002-INCH, REPLACE PISTON ROD ASSEMBLY.**

z. Inspect sleeve bearing of cylinder assembly for scratches, nicks, burrs, or corrosion. **NONE ALLOWED.** If scratches, nicks, burrs, or corrosion is found, replace sleeve bearing.

aa. Inspect OD of sleeve bearing on cylinder assembly. **OD MUST NOT BE LESS THAN 1.438-INCH.** If OD exceeds limit, replace cylinder assembly.

ab. Inspect ID of sleeve bearing on cylinder assembly. **ID MUST NOT EXCEED 1-INCH.** If ID exceeds limit, replace cylinder assembly.

ac. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect sleeve bearing on cylinder assembly for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace sleeve bearing.

ad. Inspect open areas of cylinder assembly for nicks, gouges, and corrosion. Replace cylinder assembly if:

(1) **DAMAGE EXCEEDS 0.040-INCH DEEP.**

(2) **WALL THICKNESS IS LESS THAN 0.275-INCH IN ALL OPEN AREAS, EXCEPT CENTER OF FORGING.**

(3) **DAMAGE IN CENTER OF FORGING EXCEEDS 1-INCH WIDE, OR EXCEEDS 0.020-INCH DEEP.**

(4) **DAMAGE EXCEEDS 15% OF CIRCUMFERENCE ON CIRCULAR FEATURES, AND EXCEEDS 1-INCH AXIAL LENGTH.**

(5) **EACH REWORK IS CLOSER THAN 1-INCH TO OTHER REWORK.**

(6) **DAMAGE EXCEEDS 1-INCH DIAMETER.**

(7) **DAMAGE IS CLOSER THAN 1-INCH TO OTHER DAMAGE.**

ae. Inspect corner areas of cylinder assembly for nicks, gouges, and corrosion. Replace cylinder assembly if:

(1) **CORNER DAMAGE EXCEEDS 0.060-INCH DEEP.**

(2) **CORNER DAMAGE ON CORNER OF CYLINDER ADJACENT TO CYLINDER HEAD STUDS EXCEEDS 0.040-INCH DEEP.**

(3) **DAMAGE EXCEEDS 30% OF CIRCUMFERENCE ON CIRCULAR FEATURES, AND EXCEEDS 1-INCH LENGTH ON LINEAR FEATURES.**

(4) **EACH REWORK IS CLOSER THAN 1-INCH TO OTHER REWORK.**

af. Inspect mating surfaces of cylinder assembly for damage. Replace cylinder assembly if:

(1) **DAMAGE EXCEEDS 0.010-INCH DEEP.**

(2) **DAMAGE EXCEEDS 10% OF SURFACE AREAS.**

(3) **MATING PART IS REWORKED IN SAME AREA.**

ag. Inspect ID of cylinder assembly. **ID MUST NOT EXCEED 2.997-INCHES. NO MISSING OR DAMAGED CHROME-PLATING OR PITTING ALLOWED.** If ID or damage exceeds limits, replace cylinder assembly.

ah. Inspect sleeve bearing bore diameter on cylinder assembly. **DIAMETER MUST NOT EXCEED 1.438-INCHES.** If diameter exceeds limit, replace cylinder assembly.

ai. Inspect exposed threads of cylinder assembly for crossed, stripped, or flattened threads. **NONE ALLOWED.** If crossed, stripped, or flattened threads are found, replace cylinder assembly.

aj. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect cylinder assembly for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace cylinder assembly.

ak. Inspect exterior of cylinder assembly for missing or damaged protective coating. If necessary, repair cylinder assembly as follows:

(1) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean damaged areas.

(2) Using paint remover, Item 230, Appendix D, remove paint from corroded area. Rinse part thoroughly with water to remove paint remover.

(3) Using acid brush, Item 84, Appendix D, apply corrosion compound, Item 108, Appendix D, to corroded area. On heavy corrosion, rub area using abrasive cloth, Item 4, Appendix D, while corrosion compound is in corroded area. Do not allow corrosion compound to remain on area for more than 10 minutes.

(4) Using machinery towel, Item 211, Appendix D, remove corrosion compound from area while rinsing with clean water. Repeat step ak. (3) and step ak. (4) until all corrosion is removed.

(5) Using abrasive cloth, Item 4, Appendix D, blend out area. Do not exceed repair limits.

(6) Using abrasive wheel, Item 22, Appendix D, remove gouges and dents on external

surfaces. Blend to 0.001 - 0.002-inch deeper than original damage.

(7) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect repaired area for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace cylinder assembly.

(8) Using corrosion resistant coating, Item 118, Appendix D, apply to exposed metal.

(9) Touch up paint (PARA 2-6-5).

5-5-6.1.4 Assemble.

a. Using hydraulic fluid, Item 154, Appendix D, coat all internal parts before assembly.

NOTE

After installation, make sure seals and packings are fully seated in packing grooves and are not twisted.

b. Install packing into groove in cylinder assembly (Figure 5-5-6-1, Detail C).

CAUTION

Damage to packing will result if sleeve bearing is forced into cylinder assembly. Use extreme caution when installing sleeve bearing.

c. Using main rotor damper sleeve bearing installation/removal tool and arbor press, install sleeve bearing into cylinder assembly. Secure with retaining ring.

d. Install packing and ring into sleeve bearing (Figure 5-5-6-1).

e. Using clean hydraulic fluid, Item 154, Appendix D, coat outside diameter (OD) of main rotor damper seal expansion tool and inside diameter (ID) of sleeve bearing.

f. Insert main rotor damper seal expansion tool into bore of sleeve bearing with twisting motion until ring and packing come in contact with largest OD of main rotor damper seal expansion tool. Remove main rotor damper seal expansion tool.

g. Install wiper ring into sleeve bearing (Figure 5-5-6-1, Detail C).

h. Using clean hydraulic fluid, Item 154, Appendix D, coat OD of main rotor damper seal expansion tool and ID of sleeve bearing.

i. Insert main rotor damper seal expansion tool into bore of sleeve bearing with twisting motion until wiper ring comes in contact with largest OD of main rotor damper seal expansion tool. Remove main rotor damper seal expansion tool.

j. Install packing and packing retainer on piston rod assembly (Figure 5-5-6-1).

k. Attach main rotor damper piston rod insertion tool to cylinder assembly. Using clean hydraulic fluid, Item 154, Appendix D, coat ID of main rotor damper piston rod insertion tool.

CAUTION

- Damage to cylinder assembly, ring and packing, or wiper ring will result if piston rod is forced during installation. Use extreme caution when installing piston rod assembly.
- Damage to cylinder assembly will result if piston rod assembly comes in contact with interior of cylinder assembly. Use extreme caution when installing piston rod assembly into cylinder assembly.

l. Using clean hydraulic fluid, Item 154, Appendix D, coat shank (threaded side) of piston rod assembly and ID of cylinder assembly. Carefully insert piston rod assembly (threaded end) into cylinder assembly. Remove main rotor damper piston rod insertion tool.

m. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217,

Appendix D, thoroughly clean and dry threaded section of piston rod assembly. Remove all traces of hydraulic fluid.

n. Install rod end and barrel-nut to piston rod assembly.

CAUTION

Damage to cylinder assembly will result if main rotor damper piston rod wrench comes in contact with interior of cylinder. Use extreme caution when inserting main rotor damper piston rod wrench into cylinder assembly.

o. Install main rotor damper piston rod wrench into cylinder assembly until it engages piston rod assembly.

p. Place cylinder assembly with main rotor damper piston rod wrench into main rotor damper holding fixture.

q. **TORQUE PISTON ROD ASSEMBLY TO 950 - 1050 INCH-POUNDS.**

r. Remove cylinder assembly from main rotor damper holding fixture.

s. Remove main rotor damper piston rod wrench. Install retaining ring of barrel-nut on end of piston rod assembly.

t. Install packing into cylinder head (Figure 5-5-6-1, Detail A).

CAUTION

Damage to packing will result if sleeve bearing is forced into cylinder head. Use extreme caution when installing sleeve bearing.

u. Using main rotor damper sleeve bearing installation/removal tool and arbor press, install bushing into cylinder head. Secure with retaining ring.

v. Insert packing and ring into bushing (Figure 5-5-6-1).

w. Using clean hydraulic fluid, Item 154, Appendix D, coat OD of main rotor damper seal expansion tool and ID of bushing.

x. Insert main rotor damper seal expansion tool into sleeve bearing bore with twisting motion until ring and packing come in contact with largest OD of main rotor damper seal expansion tool. Remove main rotor damper seal expansion tool.

y. Install packing and packing retainer in OD groove of cylinder head.

NOTE

Make sure cylinder head and cylinder assembly are aligned as shown in Figure 5-5-6-2.

z. Install cylinder head and secure with nuts and washers (Figure 5-5-6-1).

aa. **TORQUE NUTS TO 167 - 183 INCH-POUNDS.**

ab. Remove plug and packing from port. Install damper indicator (PARA 5-4-14).

5-5-6.1.5 Test.

Bleed damper (PARA 5-4-17) and check for leaks.

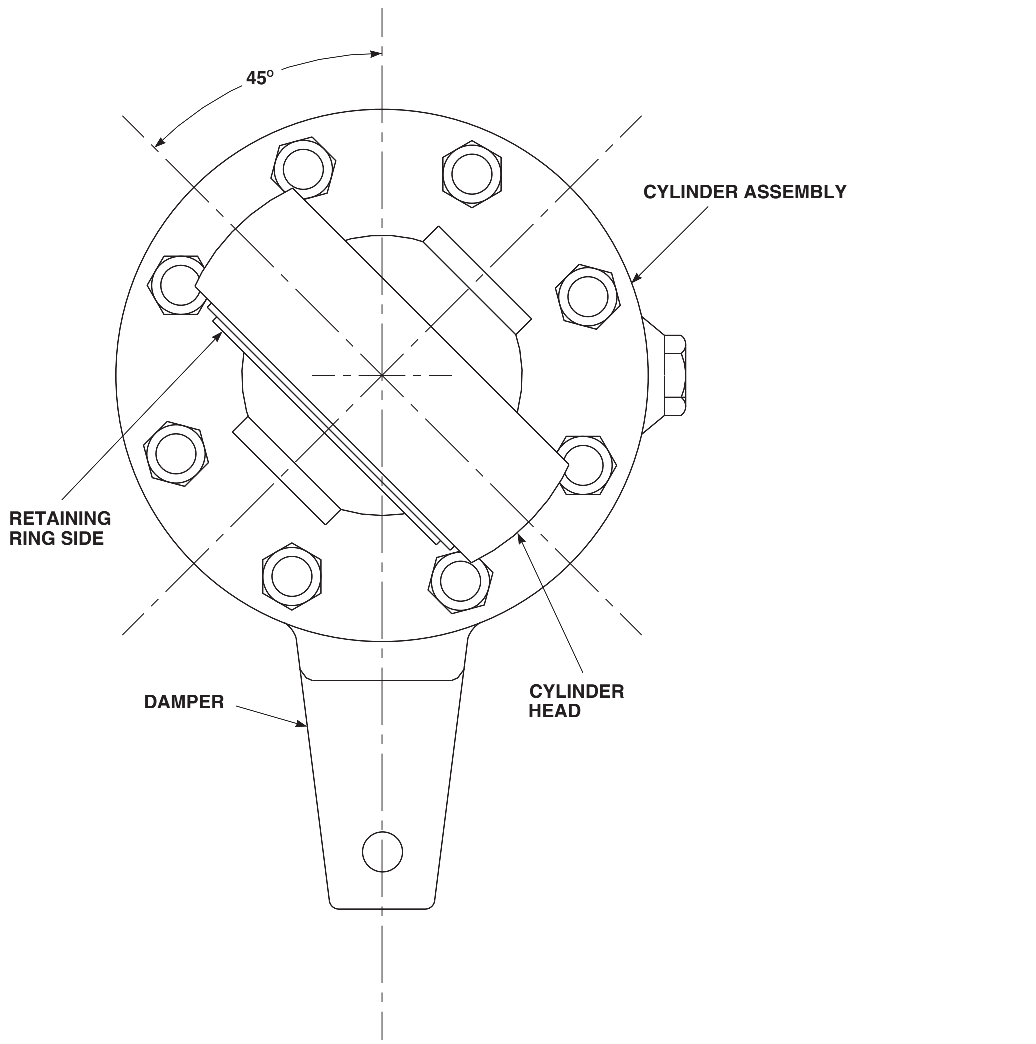


FIGURE 5-5-6-2. ALIGN CYLINDER HEAD AND CYLINDER ASSEMBLY (BENCH)

5-5-7 SWASHPLATE LARGE SPHERICAL BEARING (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage
- Rubber Mallet
- Torque Wrench, 150 - 700 in. lbs

Materials/Parts

- Abrasive Cloth, Item 12, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Primer, Item 258, Appendix D

References

- Appendix D

5-5-7.1 Replace Swashplate Large Spherical Bearing (BENCH).

5-5-7.1.1. Remove.

a. Remove bolts and washers attaching retainer and shim to stationary swashplate (Figure 5-5-7-1).

b. Remove retainer and shim from stationary swashplate.

c. Remove swashplate large spherical bearing from stationary swashplate.

5-5-7.1.2. Inspect/Repair.

NOTE

Repair to inner bore of stationary swashplate is limited to a maximum of 10 repairs with any repair no closer than ½-inch.

a. Inspect inner bore of stationary swashplate for corrosion or damage. Using abrasive cloth, Item 12, Appendix D, blend corrosion or damage. **MAXIMUM ALLOWABLE DAMAGE AFTER BLENDING IS 0.010-INCH IN DEPTH AND 1-INCH IN DIAMETER.**

b. Using corrosion resistant coating, Item 118, Appendix D, touch up repaired area.

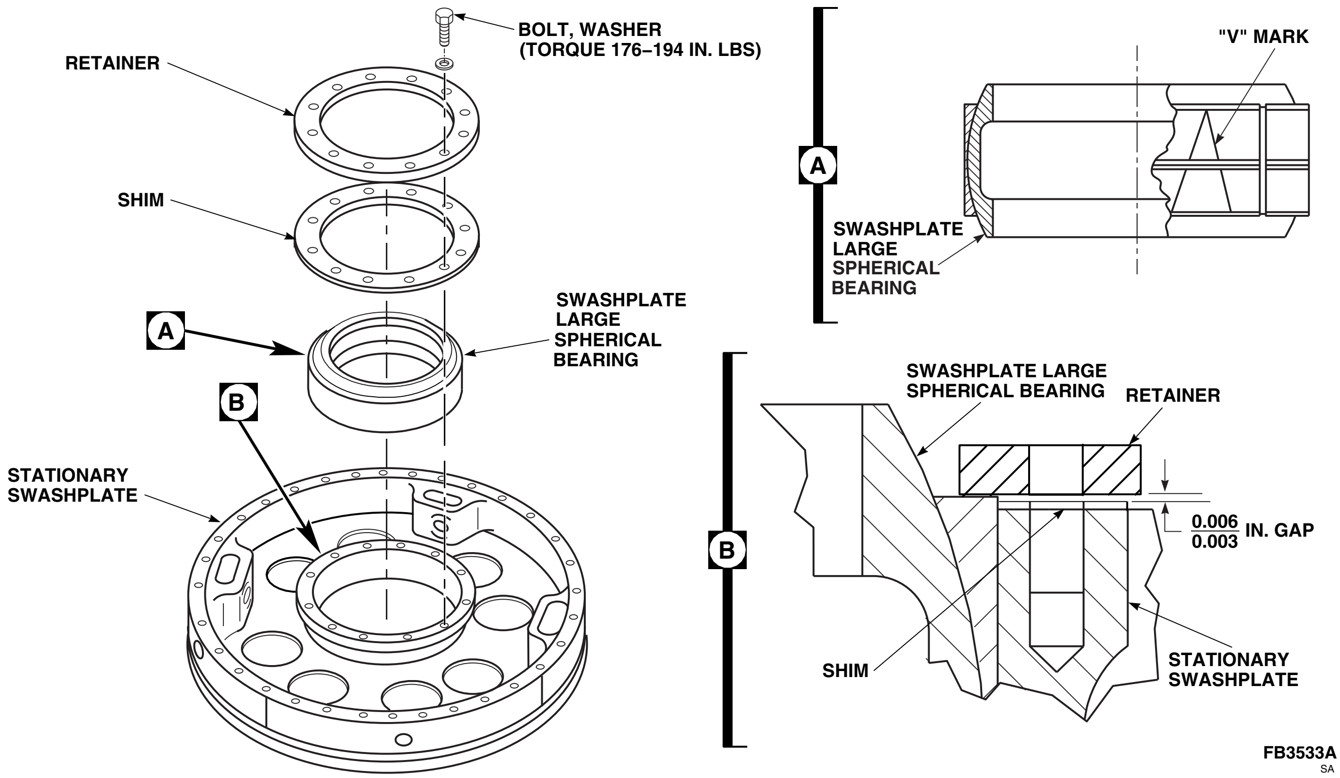


FIGURE 5-5-7-1. REPLACE SWASHPLATE LARGE SPHERICAL BEARING (BENCH)

5-5-7.1.3. Install.

NOTE

- Do not lubricate swashplate large spherical bearing. Swashplate large spherical bearing is installed dry.
- Swashplate large spherical bearing and outer races are a matched set. Make sure all parts have same serial number.

a. Line up etched "V" mark on outer races of swashplate large spherical bearing (Figure 5-5-7-1, Detail A).

CAUTION

Stationary swashplate and/or swashplate large spherical bearing will be damaged if swashplate large spherical bearing is cocked during installation. Make sure swashplate large spherical bearing is correctly aligned during installation.

b. Using hand pressure, or tapping lightly with rubber mallet, install swashplate large spherical bearing in stationary swashplate (Figure 5-5-7-1).

c. Install shim and retainer on stationary swashplate and secure with bolts and washers. Do not torque bolts now.

d. Using feeler gage, measure gap between retainer and shim. **GAP IS TO BE 0.003 - 0.006-INCH** (Figure 5-5-7-1, Detail B). If gap is beyond limits, perform the following:

(1) Remove bolts, washers, retainer, and shim (Figure 5-5-7-1).

(2) Peel shim as necessary to allow for 0.003 - 0.006-inch gap.

(3) Repeat step c. and step d. until 0.003 - 0.006-inch gap is obtained.

e. Remove bolts, washers, retainer, and shim from stationary swashplate.

f. Apply a coat of primer, Item 258, Appendix D, to shank, threads, and under heads of bolts.

g. Install shim and retainer on stationary swashplate with bolts and washers while primer is still wet.

h. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**

i. Operate controls and check installed swashplate large spherical bearing for freedom of movement. **NO RATCHETING OR BINDING ALLOWED.**

5-5-8 ROTATING SCISSORS UPPER LINK BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Fluorescent Penetrant Inspection Kit
- Freezer (Alternate for Dry Ice)
- Powertrain Repairer's Toolkit

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Dry Ice, Item 125, Appendix D (Alternate for Freezer)
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Paint Remover, Item 230, Appendix D
- Sealing Compound, Item 292, Appendix D
- Tags

References

- Appendix D
- ASTM E1417
- PARA 2-6-5

5-5-8.1 Replace Rotating Scissors Upper Link Bearings (BENCH).

NOTE

- Each rotating scissors upper link bearing is composed of an outer bearing and an inner sleeve bearing.
- Bearings are a matched set and cannot be replaced individually. If same bearings will be reinstalled, tag for correct reassembly. If either inner sleeve bearing or outer bearing must be replaced, both must be replaced with a new matched set.
- Replacement of all bearings is the same.

5-5-8.1.1 Remove.

a. Remove inner sleeve bearing from rotating scissors upper link. If same bearings will be reinstalled, tag bearings for correct reassembly (Figure 5-5-8-1).

b. Measure flange height of outer bearing (Figure 5-5-8-1, Detail A). **MINIMUM HEIGHT IS 0.086-INCH.** If height is less than 0.086-inch, replace outer bearing.

c. If necessary, using arbor press, press out outer bearing from rotating scissors upper link.

5-5-8.1.2 Inspect/Repair.

a. Visually inspect upper link for cracks. If crack is suspected, perform the following:

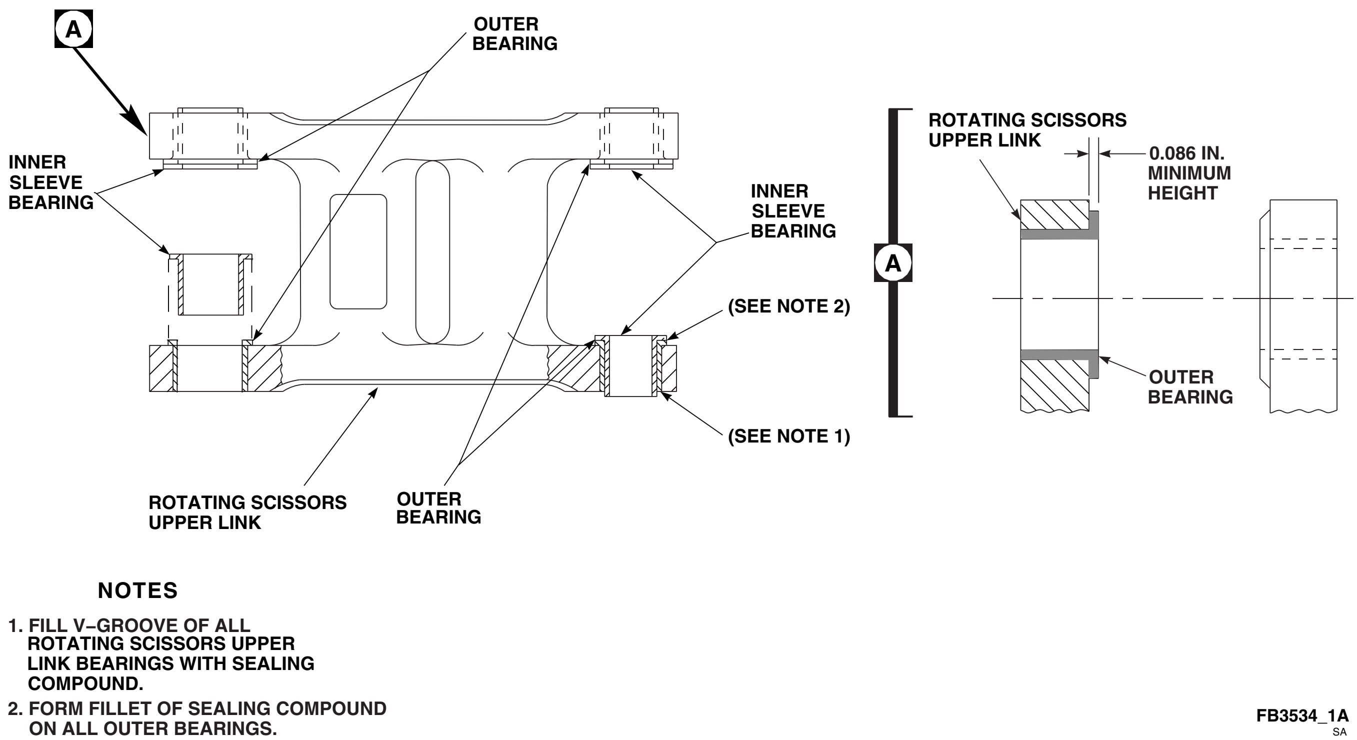


FIGURE 5-5-8-1. REPLACE ROTATING SCISSORS UPPER LINK BEARINGS (BENCH)

(1) Using paint remover, Item 230, Appendix D, clean and strip suspected area.

(2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect for cracks (ASTM E1417). **NONE ALLOWED.** Evaluate as follows:

(a) If crack is found, replace upper link.

(b) If no crack is found, using epoxy primer coating kit, Item 135, Appendix D, and paint, touch up stripped area (PARA 2-6-5).

b. Using abrasive cloth, Item 10, Appendix D, blend out any wear or damage in link bore. **IF DAMAGE IS DEEPER THAN 0.010-INCH AFTER BLENDING, OR BLENDED AREA EXCEEDS 25% OF BORE AREA, REPLACE UPPER LINK.**

5-5-8.1.3 Install.

a. Using dry ice, Item 125, Appendix D, or a freezer, chill replacement bearing for a minimum of 30 minutes in at least -65°F (-54°C).

b. After a minimum of 30 minutes, remove bearing from dry ice (or freezer).

NOTE

- Make sure flange of bearing is seated against flange of upper link.
- Bearings are matched sets. Make sure serial numbers on inner sleeve bearing and outer bearing are the same.

c. Using arbor press, press bearing into bore of upper link (Figure 5-5-8-1).

NOTE

Make sure sealing compound does not get in bearing. Bearing should rotate freely after installation.

d. Apply sealing compound, Item 292, Appendix D, to seal exposed edges of outer bearing, and fill V-groove of bearing.

5-5-9 ROTATING SCISSORS LOWER LINK BUSHING (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Fluorescent Penetrant Inspection Kit
- Hand Reamer Set, Adjustable

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Dry Ice, Item 125, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- ASTM E1417

5-5-9.1 Replace Rotating Scissors Lower Link Bushing (BENCH).

5-5-9.1.1 Remove.

a. Using arbor press, press out bushing from lower link (Figure 5-5-9-1).

b. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean bore.

5-5-9.1.2 Inspect/Repair.

a. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect lower link for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace lower link.

b. Inspect bore of lower link for scoring and corrosion. Using abrasive cloth, Item 10, Appendix D, blend out scoring and corrosion.

5-5-9.1.3 Install.

a. Using dry ice, Item 125, Appendix D, chill replacement bushing for minimum of 30 minutes.

b. Remove lower link bushing from dry ice.

c. Apply coat of primer, Item 258, Appendix D, to outside of bushing.

d. Using arbor press, press bushing into bore of lower link while primer is still wet (Figure 5-5-9-1).

e. Allow parts to return to room temperature.

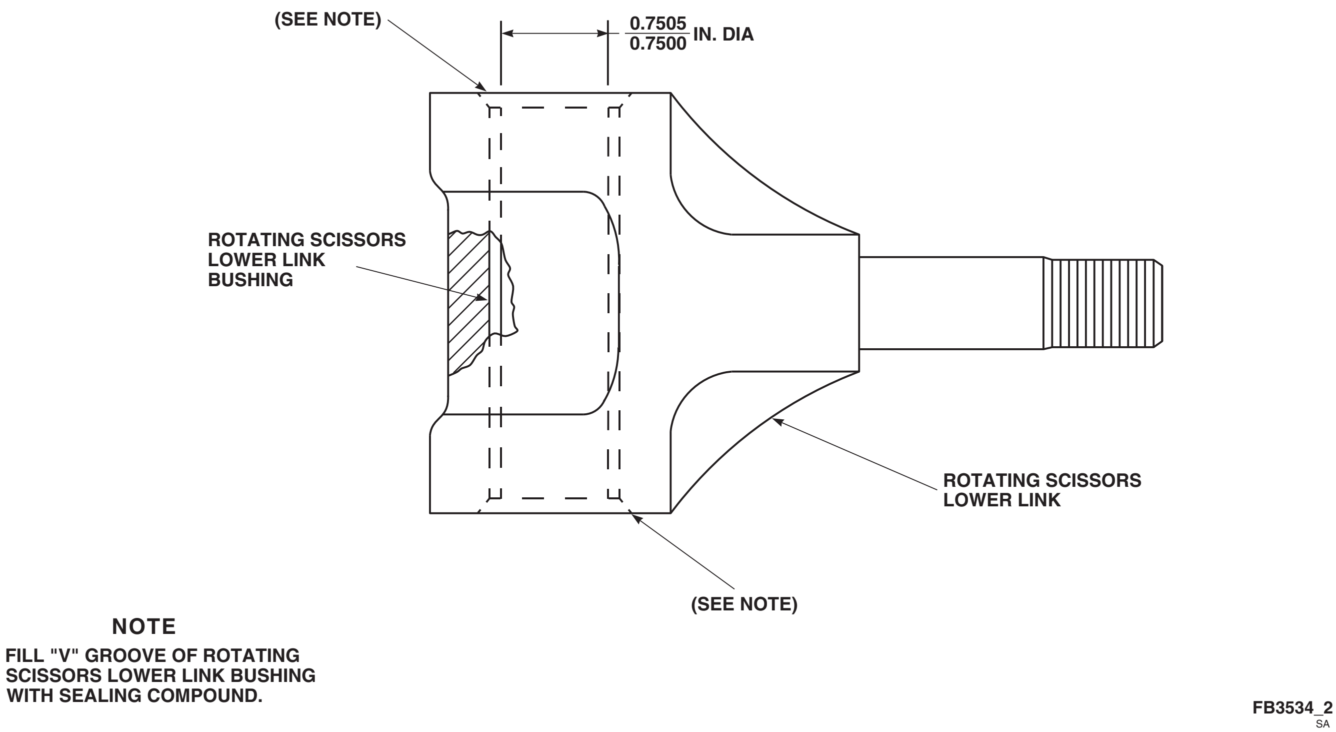


FIGURE 5-5-9-1. REPLACE ROTATING SCISSORS LOWER LINK BUSHING (BENCH)

f. Using adjustable hand reamer set, ream bushing to 0.7500 - 0.7505-inch.

g. Apply sealing compound, Item 292, Appendix D, to fill V-groove of bushing.

5-5-10 BIFILAR ARM BUSHINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Bifilar Bushings Replacement Tools, Locally-Made, Figure H-213, Appendix H, consisting of:
  - Alignment Bar, 2763-7
  - Installer Washer, 2763-6
  - Puller Disk, 2763-5
  - Pulling Tabs, 2763-2
  - Pulling Tabs, 2763-3
  - Pulling Washer, 2763-4
  - Threaded Shaft, 2763-1
- Flat Washer, MS15795-830
- Inside Micrometer Caliper
- Nonmetallic Scraper, Locally-Made
- Nut, AN320-20

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Dry Ice, Item 125, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 288, Appendix D

References

- Appendix D
- Appendix H
- PARA 5-4-29

5-5-10.1 Replace Bifilar Arm Bushings (BENCH).

NOTE

Replacement of all bifilar arm bushings is the same.

5-5-10.1.1 Remove.

a. Locally make bifilar bushings replacement tools (Figure H-213, Appendix H).

b. Remove bifilar weights (PARA 5-4-29).

NOTE

- Note part number on top bifilar arm bushing to make sure of correct bifilar arm bushing replacement.
- To make sure alignment of bifilar arm bushing is maintained, only replace one set of bifilar arm bushings at a time.

c. Select set of pulling tabs, 2763-2 or 2763-3, that best fits groove inside bifilar arm bore between the two bifilar arm bushings (Figure 5-5-10-1, Detail A).

d. Install threaded portion of threaded shaft through bifilar arm bushings from bottom side of bifilar arm with flange of threaded shaft pressed against bottom side of pulling tabs.

e. Place pulling washer on threaded shaft along with one flat washer and nut.

f. **TIGHTEN NUT TO SANDWICH PULLING TABS BETWEEN FLANGE OF THREADED SHAFT AND PULLING WASHER.**

g. Place puller disk down over threaded shaft along with one flat washer and nut.

h. **TIGHTEN NUT DOWN IN PULLER DISK UNTIL TOP BIFILAR ARM BUSHING IS REMOVED.**

i. Remove nut, flat washer, and puller disk from threaded shaft. Leave pulling tabs for lower bifilar arm bushing removal.

j. Place threaded shaft with pulling tabs still secured to threaded shaft down through bifilar arm bore from top, resting pulling tabs on lower bifilar arm bushing.

k. Install puller disk up through threaded shaft over flange of lower bifilar arm bushing.

l. Align cut out on puller disk with bifilar arm bore. Secure with one flat washer and nut.

m. **TIGHTEN NUT AGAINST PULLER DISK UNTIL LOWER BIFILAR ARM BUSHING IS REMOVED.**

n. Remove bifilar bushings replacement tools from bifilar arm.

5-5-10.1.2 Inspect/Repair.

a. Using inside micrometer caliper, inspect inside diameter (ID) of bifilar arm bore for wear. **ID TO BE NO GREATER THAN 3.326-INCHES** (Figure 5-5-10-1, Detail B). If diameter is beyond limits, replace bifilar.

NOTE

Contact area is length × width. Length is circumference of bifilar arm bore, and width is depth of bifilar arm bore.

b. Using abrasive cloth, Item 10, Appendix D, repair bifilar arm bore as follows:

(1) **BLEND NICKS, GOUGES, OR SCORING ON INSIDE DIAMETER OF BIFILAR ARM BORE UP TO 0.010-INCH DEEP. TOTAL BLENDED AREA SHALL NOT EXCEED 25% OF TOTAL CONTACT AREA.**

(2) If repairs are beyond limits, replace bifilar.

5-5-10.1.3 Install.

a. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper, clean all sealing compound from bifilar arm.

b. Using dry ice, Item 125, Appendix D, chill replacement bifilar arm bushings.

c. Remove bifilar arm bushings from dry ice.

d. Using epoxy primer coating kit, Item 135, Appendix D, coat bifilar arm bore with epoxy primer.

e. Line up flange flats of top bifilar arm bushing with alignment bar on top of bifilar arm while epoxy primer is still wet (Figure 5-5-10-1, Detail A).

f. Using threaded shaft, install one installer washer and one flat washer and secure with nut.

g. Insert threaded end of threaded shaft up through bifilar arm, lining up installer washer flat with alignment bar.

h. Place other installer washer down over threaded shaft and on top of new bifilar arm bushing, lining up flats of installer washer and bifilar arm bushing with alignment bar on top. Secure in place with one flat washer and nut.

i. **TIGHTEN NUT AGAINST INSTALLER WASHER UNTIL TOP BIFILAR ARM BUSHING HAS BEEN INSTALLED IN BIFILAR ARM.**

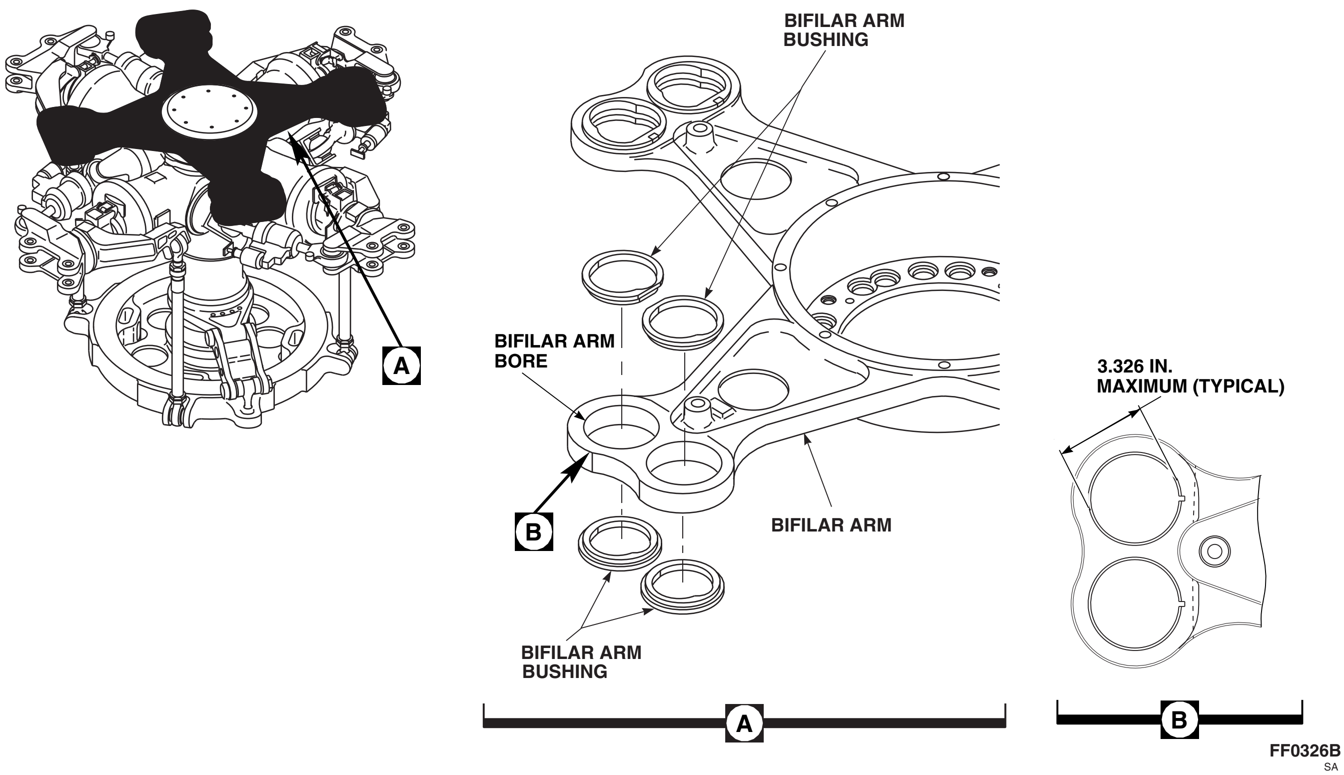


FIGURE 5-5-10-1. REPLACE BIFILAR ARM BUSHINGS (BENCH)

j. Remove nut, flat washer, and installer washer from threaded shaft and remove threaded shaft from bifilar arm. Leave other installing washer secured on threaded shaft.

k. With alignment bar still in place on bifilar arm and installer washer still secured on threaded shaft place bifilar arm bushing over threaded end of threaded shaft with flange against installer washer.

l. Insert threaded end of threaded shaft up through bifilar arm lining up flats on installer washer and bifilar arm bushing with alignment bar.

m. Install other installer washer in threaded shaft from top of bifilar arm, lining up flat of installing washer with alignment bar.

n. Install flat washer and nut.

o. **TIGHTEN NUT UNTIL LOWER BIFILAR ARM BUSHING HAS BEEN INSTALLED IN BIFILAR ARM.**

p. Allow bifilar arm bushing to return to room temperature.

q. Remove bifilar bushings replacement tools from bifilar arm.

r. Apply sealing compound, Item 288, Appendix D, to seal mating surfaces between flanges of bifilar arm bushing and bifilar arm.

s. Install bifilar weights (PARA 5-4-29).

5-5-11 BIFILAR WEIGHT BUSHINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Bifilar Bushings Replacement Tools, Locally-Made, Figure H-213, Appendix H, consisting of:
  - Alignment Bar, 2803-1
  - Press Plate, 2803-2
- Drift
- Mallet
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Dry Ice, Item 125, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- PARA 5-4-29

5-5-11.1 Replace Bifilar Weight Bushings (BENCH).

NOTE

Replacement of all bifilar weight bushings is the same.

5-5-11.1.1 Remove.

a. Locally make bifilar bushings replacement tools (Figure H-213, Appendix H).

NOTE

- Note part number on top bifilar weight bushing to make sure of correct bushing replacement.
- To make sure bifilar weight bushing alignment is maintained, only replace one set of bifilar weight bushings at a time.

b. Remove bifilar weights (PARA 5-4-29).

c. Using suitable drift and mallet, remove one upper and one lower bifilar weight bushing as a set (Figure 5-5-11-1).

5-5-11.1.2 Install.

a. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper, clean all sealing compound from bifilar weight.

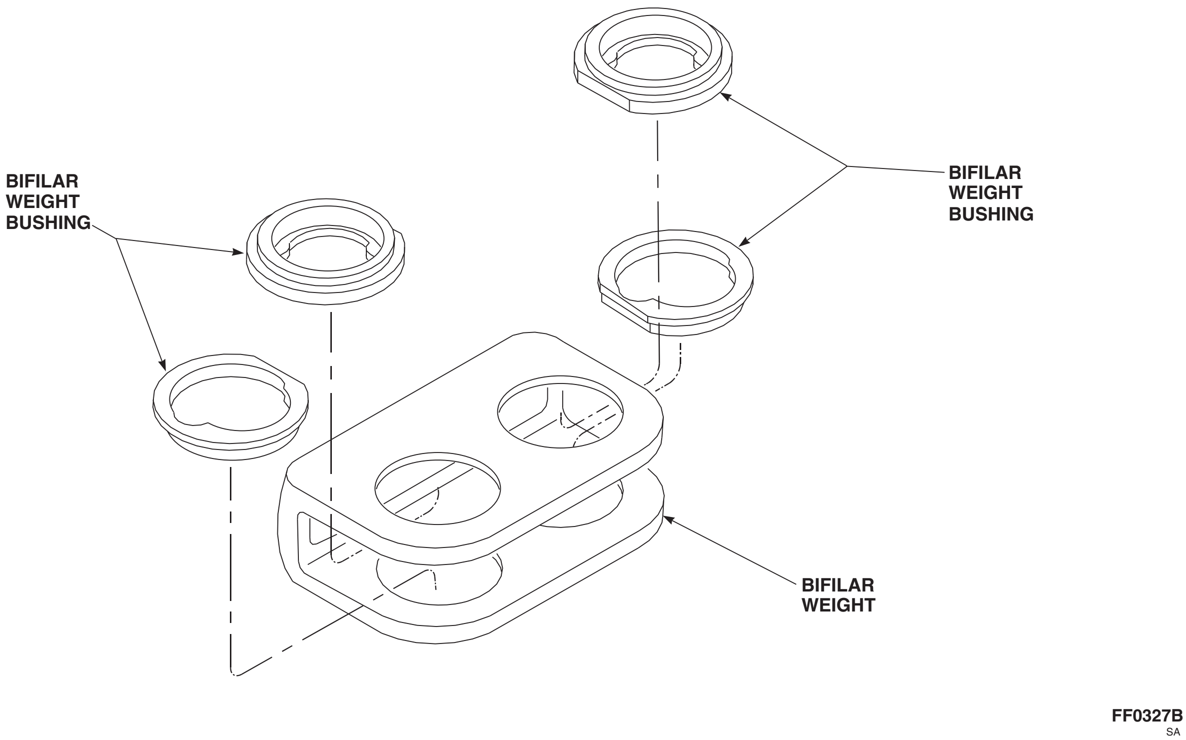


FIGURE 5-5-11-1. REPLACE BIFILAR WEIGHT BUSHINGS (BENCH)

b. Using dry ice, Item 125, Appendix D, chill replacement bifilar weight bushings.

c. Remove bifilar weight bushings from dry ice.

d. Using epoxy primer coating kit, Item 135, Appendix D, coat bore of bifilar weight bushing with epoxy primer.

e. Place alignment bar inside bifilar weight and against flat portion of upper and lower flanges on existing bifilar weight bushings in bifilar weight (Figure 5-5-11-1).

f. Line up top flange flats of bifilar weight bushing with alignment bar on top of bifilar weight while epoxy primer is still wet.

g. Using arbor press, place press plate on top of flange and press bifilar weight bushing into bifilar weight using alignment bar as guide.

h. Repeat step d. through step g. for installation of bottom bifilar weight bushing.

i. Allow bifilar weight bushings to return to room temperature.

j. Using sealing compound, Item 292, Appendix D, seal mating surfaces between flanges of bifilar weight bushings and bifilar weight.

5-5-12 MAIN ROTOR BLADE CUFF BUSHINGS AND LINERS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Blade/Spindle Cuff Bushing Installation Tools, Locally-Made, Figure H-161, Appendix H
- C-Clamp
- Heat Lamp
- Inside Micrometer Caliper
- Magnifying Glass, 10X
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer's Toolkit
- Protection, Eye, Skin, and Breathing

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Adhesive, Item 32, Appendix D
- Adhesive, Item 66, Appendix D
- Cheesecloth, Item 90, Appendix D
- Dry Ice, Item 125, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 258, Appendix D

References

- Appendix D
- Appendix H

5-5-12.1 Replace Main Rotor Blade Cuff Bushings (BENCH).

NOTE

Replacement of all main rotor blade cuff bushings is the same.

5-5-12.1.1 Inspect/Repair.

a. Using inside micrometer caliper, measure inside diameter (ID) of main rotor blade cuff bushings for wear. **MAXIMUM ID IS 1.0025-INCHES.** If ID exceeds limit, replace main rotor blade cuff bushing.

b. Inspect face of main rotor blade cuff bushings for damage. **DAMAGE MUST NOT EXCEED 0.010-INCH.** If damage exceeds limit, replace main rotor blade cuff bushing.

WARNING

Main rotor blade cuff bushings are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds and eyes. To prevent injury, personnel must wear eye protection. Clean up visible dust piles and wash hands and arms after maintenance.

c. Using abrasive cloth, Item 10, Appendix D, repair main rotor blade cuff bushings as follows:

(1) **BLEND NICKS, GOUGES, OR SCORING ON ID OF MAIN ROTOR BLADE CUFF BUSHING UP TO 0.005-INCH DEEP. TOTAL BLENDED AREA MUST NOT EXCEED 25% OF BORE AREA.** If repairs exceed limits, replace main rotor blade cuff bushing.

(2) **BLEND CORROSION ON ID OF MAIN ROTOR BLADE CUFF BUSHING NO MORE THAN MAXIMUM MAIN ROTOR BLADE CUFF BUSHING ID OF 1.0025-INCHES.** If repairs exceed limit, replace main rotor blade cuff bushing.

(3) **BLEND NICKS AND GOUGES ON FLANGE OF MAIN ROTOR BLADE CUFF BUSHING UP TO 0.010-INCH DEEP. CORNER DAMAGE MUST NOT EXCEED 0.030-INCH.** If repairs exceed limits, replace main rotor blade cuff bushing.

5-5-12.1.2 Remove.

a. Locally make blade/spindle cuff bushing installation tools A, B, and C (Figure H-161, Appendix H).

b. Using tool A, press out main rotor blade cuff bushings (Figure 5-5-12-1).

5-5-12.1.3 Inspect.

a. Inspect bores for scoring and corrosion. If necessary, using abrasive cloth, Item 10, Appendix D, remove corrosion and blend out scoring. **IF DAMAGE IS DEEPER THAN 0.010-INCH AFTER BLENDING, OR IF BLENDED AREA IS GREATER THAN 25% OF BORE AREA, REPLACE MAIN ROTOR BLADE.**

b. Using inside micrometer caliper, measure bore diameter of main rotor blade cuff bushing. **IF BORE IS GREATER THAN 1.125-INCHES AFTER BLENDING, REPLACE MAIN ROTOR BLADE.**

c. Inspect main rotor blade cuff for cracks between attachment boltholes and leading and trailing edges of main rotor blade cuff on upper and lower surfaces. **NONE ALLOWED.** If main rotor blade cuff is cracked, replace main rotor blade.

5-5-12.1.4 Install.

a. Using dry ice, Item 125, Appendix D, chill replacement main rotor blade cuff bushings for a minimum of 3 hours before installation.

b. After a minimum of 3 hours, remove main rotor blade cuff bushings from dry ice.

c. Apply coat of primer, Item 258, Appendix D, to main rotor blade cuff bushing outside diameter and bore.

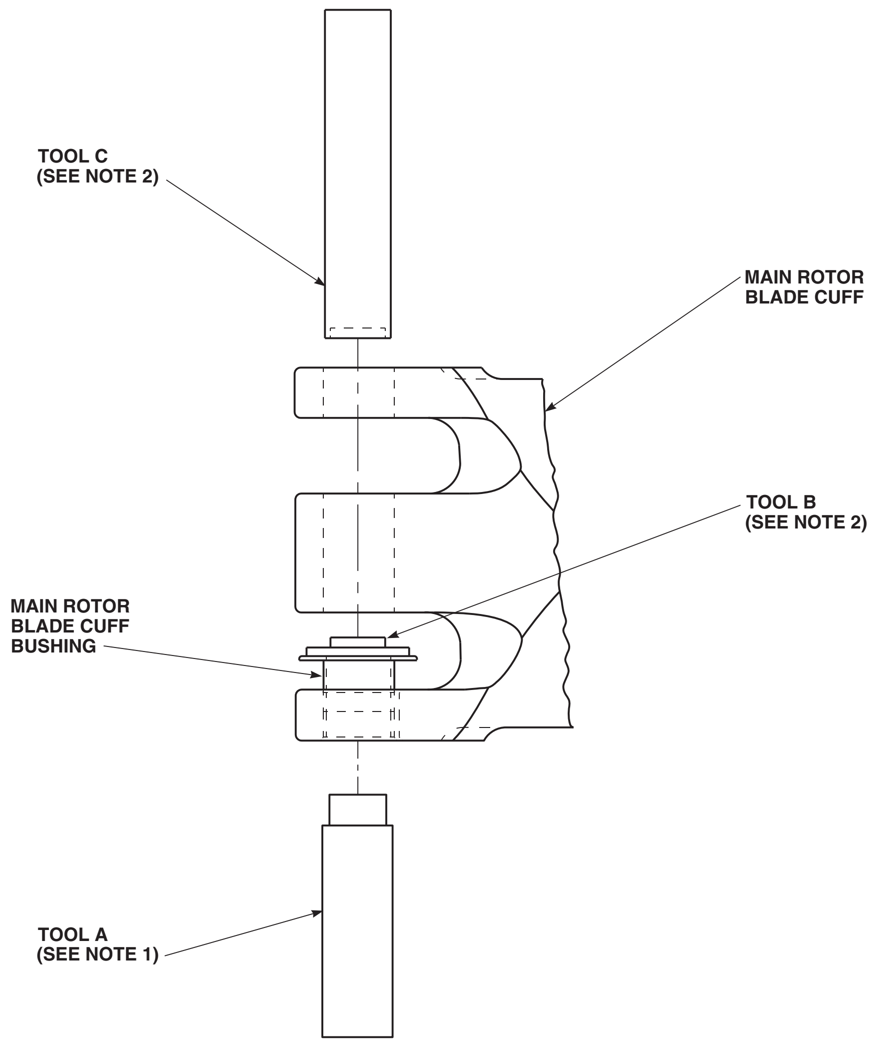
d. Press in main rotor blade cuff bushings while primer is still wet. Using tools B and C, and arbor press, press in main rotor blade cuff bushings (Figure 5-5-12-1). Make sure flange of main rotor blade cuff bushing is firmly seated against lug of main rotor blade.

5-5-12.2 Replace Main Rotor Blade Cuff Liners (BENCH).

NOTE

Replacement of all main rotor blade cuff liners is the same.

a. If required, remove partially disbonded or damaged liners from cuff as follows:



NOTES

1. TOOL A IS USED WITH ARBOR PRESS TO REMOVE MAIN ROTOR BLADE CUFF BUSHING.
2. TOOL B IS USED WITH TOOL C TO INSTALL MAIN ROTOR BLADE CUFF BUSHING.

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FIGURE 5-5-12-1. REPLACE MAIN ROTOR BLADE CUFF BUSHINGS (BENCH)

(1) Using dry ice, Item 125, Appendix D, chill bond line of liner for 20 - 30 minutes.

(2) Using nonmetallic scraper, remove liner from cuff.

b. Using nonmetallic scraper and methyl ethyl ketone, Item 217, Appendix D, remove any remaining adhesive from bonding surface of cuff.

c. Using 10X magnifying glass, inspect faying surfaces of cuff to verify shotpeen characteristics. Absence of shotpeen dimples is cause for blade to be returned to depot.

d. Using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean bonding surface of cuff and liner. Wipe dry using clean, dry cheesecloth, Item 90, Appendix D.

e. Apply thin coat of adhesive, Item 66, Appendix D, to bonding surfaces of cuff and liner. Allow adhesive to cure for 1 hour at room temperature.

NOTE

Mix an excess amount of adhesive for evaluation of cure after total bonding is complete.

f. Prepare adhesive, Item 32, Appendix D, using manufacturer's instructions.

g. Apply thin coat of adhesive, Item 32, Appendix D, to area of cuff where liner will be bonded. Make sure to cover entire area with adhesive.

h. Apply thin coat of adhesive, Item 32, Appendix D, to bottom of liner.

i. Position liner on cuff so that hole in liner is centered with bore in cuff.

j. Clamp liner to cuff using C-clamp, allowing excess adhesive to squeeze out. Remove excess adhesive.

k. Allow adhesive to cure for 24 hours at room temperature, or using heat lamp, cure adhesive at 160°F (71°C) for 1 hour.

l. Inspect bond for proper cure and edge voids.

m. Inspect remaining adhesive in mixing container for proper cure.

5-5-13 TAIL ROTOR PITCH BEAM BUSHINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Fluorescent Penetrant Inspection Kit
- Pitch Beam/Horn Bushing Sizing Block, Locally-Made, Figure H-228, Appendix H
- Powertrain Repairer's Toolkit
- Reamer Set
- Vernier Caliper

Materials/Parts

- Coating Compound, Item 106, Appendix D
- Lacquer, Item 186, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- ASTM E1417
- PARA 2-6-5

5-5-13.1 Replace Tail Rotor Pitch Beam Bushings (BENCH).

NOTE

Replacement of all tail rotor pitch beam bushings is the same.

5-5-13.1.1 Remove.

a. Using arbor press, press out tail rotor pitch beam bushings from tail rotor pitch beam (Figure 5-5-13-1).

b. Using paint remover, Item 230, Appendix D, remove paint from boss area of tail rotor pitch beam bushing. Allow to dry.

5-5-13.1.2 Inspect.

a. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect boss area of tail rotor pitch beam bushing for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace bushing.

b. Visually inspect bore in tail rotor pitch beam for pits and corrosion. **NONE ALLOWED.** If corrosion or pits are found, replace tail rotor pitch beam.

c. Using vernier caliper, measure diameter of bore in tail rotor pitch beam (Figure 5-5-13-1). **MAXIMUM DIAMETER ALLOWED IS**

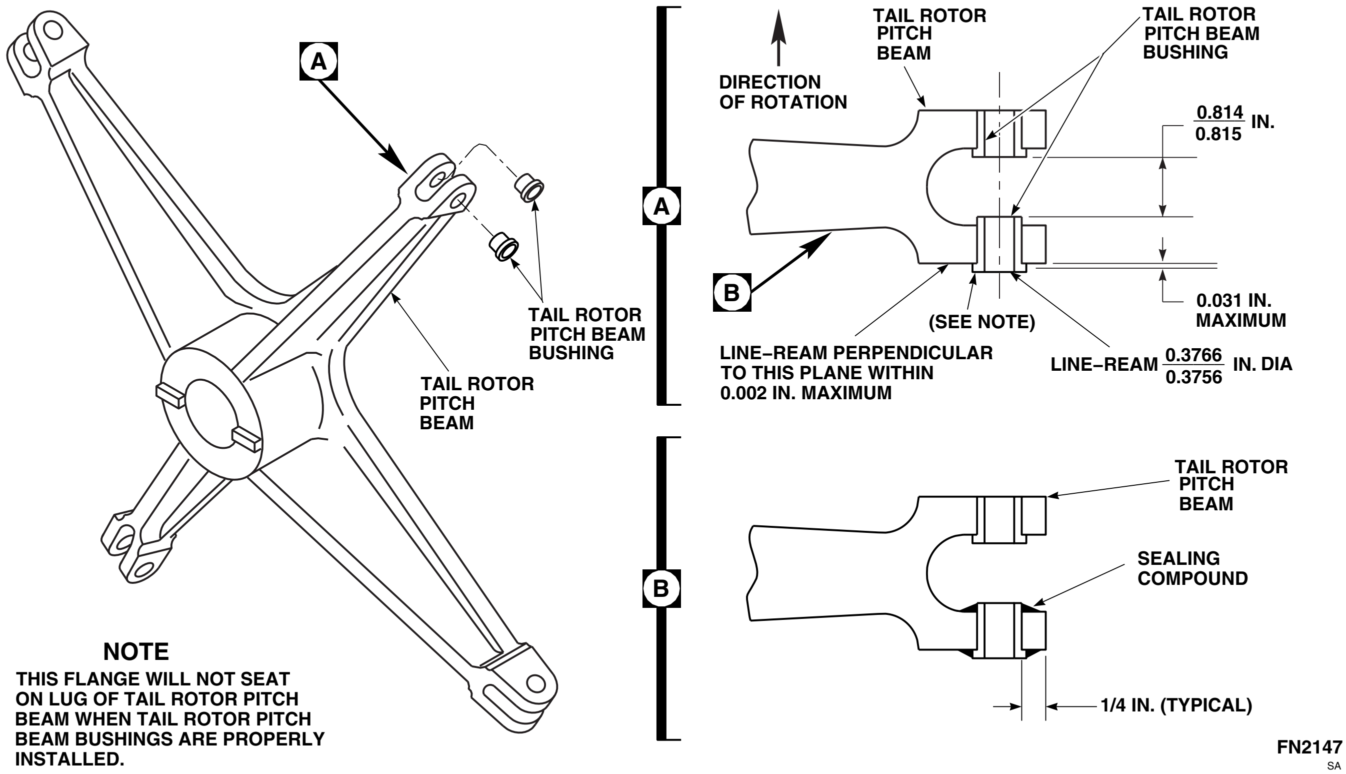


FIGURE 5-5-13-1. REPLACE TAIL ROTOR PITCH BEAM BUSHINGS (BENCH)

0.5001-INCH. If diameter exceeds limits, replace tail rotor pitch beam.

d. **IF TAIL ROTOR PITCH BEAM BUSHING IS MARKED RS-028-I, NEXT TO PART NUMBER, REPLACE TAIL ROTOR PITCH BEAM.**

5-5-13.1.3 Install.

a. Locally make pitch beam/horn bushing sizing block, 0.814 - 0.815-inch (Figure H-228, Appendix H).

b. Using coating compound, Item 106, Appendix D, apply to bare metal area of pitch beam/horn bushing sizing block.

c. Using primer, Item 258, Appendix D, prime tail rotor pitch beam bushings and tail rotor pitch beam bushing holes.

NOTE

Flange of tail rotor pitch beam bushings on outside of tail rotor pitch beam will not seat on lug surface when distance between tail rotor pitch beam bushings is correct.

d. Using arbor press, install tail rotor pitch beam bushings while primer is still wet, using pitch beam/horn bushing sizing block to make sure distance between tail rotor pitch beam bushings is 0.814 - 0.815-inch (Figure 5-5-13-1, Detail A).

e. Using reamer set, line-ream tail rotor pitch beam bushings to 0.3756 - 0.3766-inch.

f. Touch up with lacquer, Item 186, Appendix D (PARA 2-6-5).

g. Apply sealing compound, Item 292, Appendix D, around tail rotor pitch beam bushings (Figure 5-5-13-1, Detail B).

5-5-14 TAIL ROTOR PITCH HORN BUSHINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Ink Marker
- Inside Micrometer Caliper
- Pitch Beam/Horn Bushing Sizing Block, Locally-Made, Figure H-228, Appendix H
- Powertrain Repairer's Toolkit
- Pusher
- Reamer Set

Materials/Parts

- Abrasive Paper, Item 18, Appendix D
- Dry Ice, Item 125, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 258, Appendix D
- Scouring Pad, Item 270, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- PARA 2-6-5
- PARA 5-4-38

5-5-14.1 Replace Tail Rotor Pitch Horn Bushings (BENCH).

NOTE

Replacement of all tail rotor pitch horn bushings is the same.

5-5-14.1.1 Inspect/Repair.

a. Inspect for corrosion of flange surfaces. **CORROSION UP TO 0.005-INCH DEEP AND NOT COVERING MORE THAN 20% OF AREA MAY BE HAND-POLISHED USING ABRASIVE PAPER, ITEM 18, APPENDIX D, OR SCOURING PAD, ITEM 270, APPENDIX D.**

b. Inspect corrosion of bushing inside diameter (ID) (Figure 5-5-14-1, Detail A). Corrosion of ID may be blended using abrasive paper, Item 18, Appendix D, or scouring pad, Item 270, Appendix D. **ID MUST NOT EXCEED 0.3766-INCH.**

c. Using inside micrometer caliper, measure tail rotor pitch horn bushing ID. **ID MUST NOT EXCEED 0.3766-INCH.** If ID exceeds limit, replace tail rotor pitch horn bushing.

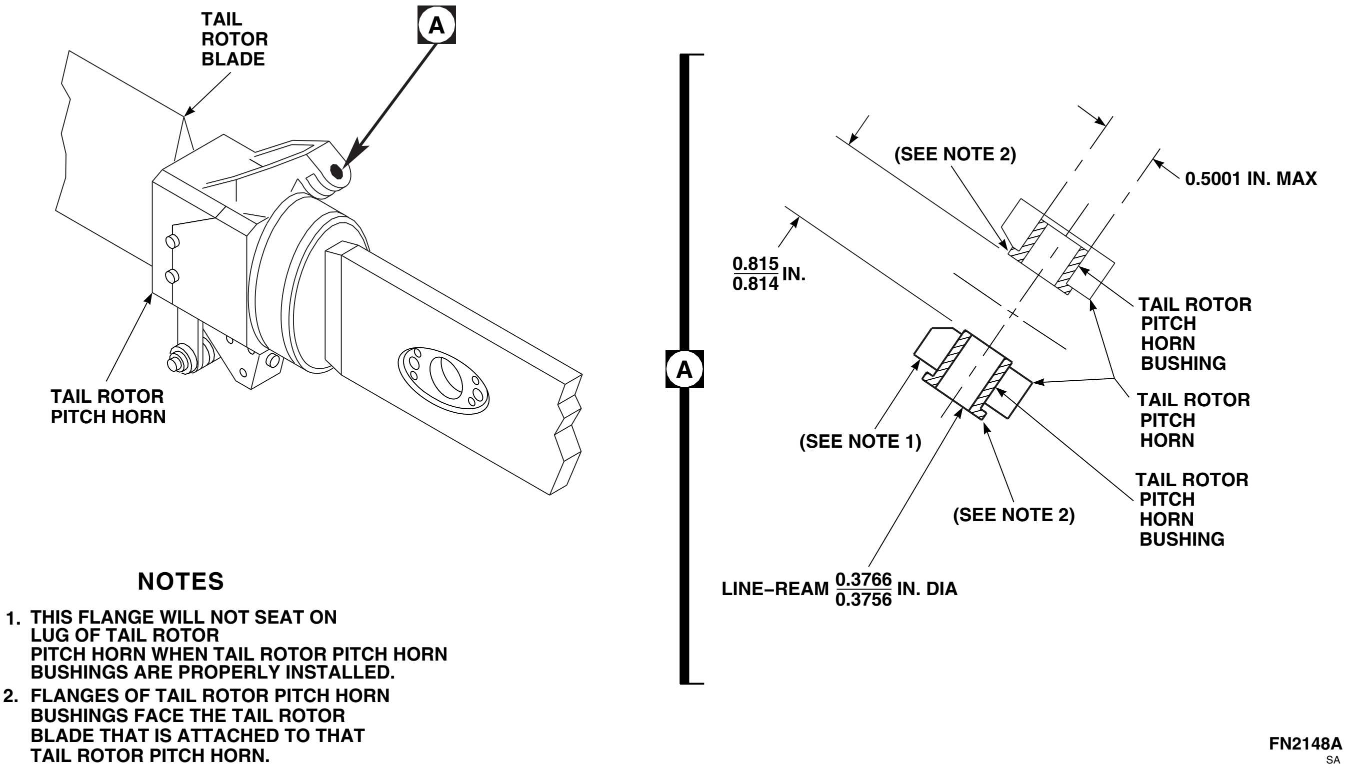


FIGURE 5-5-14-1. REPLACE TAIL ROTOR PITCH HORN BUSHINGS (BENCH)

d. **IF CORROSION PITTING IS STILL FOUND IN BORE OR ON FLANGE, REPLACE TAIL ROTOR PITCH HORN BUSHINGS.**

e. If pits are found, remove tail rotor pitch horn bushings and inspect bores in tail rotor pitch horn.

5-5-14.1.2 Remove.

Using suitable pusher and arbor press, press out tail rotor pitch horn bushings from tail rotor pitch horn.

5-5-14.1.3 Inspect.

Inspect bores in tail rotor pitch horn. **NO CORROSION DAMAGE ALLOWED. BORE MUST NOT EXCEED 0.5001-INCH.** If damage is found or limit exceeded, replace tail rotor blade (PARA 5-4-38).

5-5-14.1.4 Install.

a. Locally make pitch beam/horn bushing sizing block to approximately 0.814 - 0.815-inch (Figure H-228, Appendix H).

b. Using dry ice, Item 125, Appendix D, chill replacement tail rotor pitch horn bushings for at least 30 minutes.

c. Using methyl ethyl ketone, Item 217, Appendix D, clean bores in tail rotor pitch horn.

d. Remove tail rotor pitch horn bushings from dry ice.

e. Using primer, Item 258, Appendix D, prime tail rotor pitch horn bushings.

NOTE

Flange of tail rotor pitch horn bushing on outside of tail rotor pitch horn lug will not seat on lug surface when distance between tail rotor pitch horn bushings is correct.

f. Install tail rotor pitch horn bushings while primer is still wet, using pitch beam/horn bushing sizing block to make sure distance between tail rotor pitch horn bushings is 0.814 - 0.815-inch (Figure 5-5-14-1, Detail A).

g. Using reamer set, line-ream tail rotor pitch horn bushings to 0.3756 - 0.3766-inch.

h. Apply sealing compound, Item 292, Appendix D, to seal edges of tail rotor pitch horn bushings.

i. Using ink marker, mark tail rotor data plate ″RS-006A-II″. Make sure data plate is marked so that this number will not be added to part number by mistake.

j. Touch up paint area (PARA 2-6-5).

5-5-15 TAIL ROTOR PITCH CONTROL ROD END SPHERICAL BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Arbor Press
- Bearing Pusher/Puller, 70700-77454-041
- Bearing Staking Tool Set, 70700-77353-041
- Drill Press
- Feeler Gage, 0.005-inch
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer's Toolkit
- Spinning Tool
- Torque Wrench, 0 - 30 in. lbs
- Vernier Caliper

Materials/Parts

- Abrasive Cloth, Item 5, Appendix D
- Abrasive Cloth, Item 8, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Petrolatum, Item 234, Appendix D
- Primer, Item 258, Appendix D

References

- Appendix D
- PARA 5-4-39

5-5-15.1 Replace Tail Rotor Pitch Control Rod End Spherical Bearings (BENCH).

NOTE

Replacement of both tail rotor pitch control rod end spherical bearings is the same.

5-5-15.1.1. Remove.

a. Using arbor press and bearing pusher/puller, remove tail rotor pitch control rod end spherical bearing (Figure 5-5-15-1).

b. Using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217,

Appendix D, and nonmetallic scraper, clean primer residue from rod end bore and chamfer.

c. Make sure that rod end bore, chamfers, and adjacent surfaces are clean and dry.

5-5-15.1.2. Inspect/Repair.

a. Using abrasive cloth, Item 5, Appendix D, remove minor scoring or metal pickup from rod end bore. Using abrasive cloth, Item 8, Appendix D, blend rod end bore.

b. Using vernier caliper, measure rod end bore. **MAXIMUM INSIDE DIAMETER (ID) SHALL NOT EXCEED 0.9994-INCH.** If ID after repair is beyond limits, replace rod end (PARA 5-4-39).

5-5-15.1.3. Install.

a. Apply a light coat of primer, Item 258, Appendix D, to inside of rod end bore and to outside surface of outer race of tail rotor pitch control rod end spherical bearing.

b. Using arbor press and bearing pusher/puller, install tail rotor pitch control rod end spherical bearing (Figure 5-5-15-1).

c. Using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean primer, if any, from V-grooves. Wipe dry.

d. Using bearing staking tool set, position rod end with tail rotor pitch control rod end spherical bearing on drill press baseplate, with bearing staking tool.

CAUTION

Damage to tail rotor pitch control rod end spherical bearing will result if petrolatum is allowed to come in contact with tail rotor pitch control rod end spherical bearing or exposed edge of fabric liner. If petrolatum gets on these areas, remove petrolatum immediately.

e. Using petrolatum, Item 234, Appendix D, lubricate V-groove of tail rotor pitch control rod end spherical bearing and spinning tool (Figure 5-5-15-1, View A-A).

CAUTION

Damage to tail rotor pitch control rod end spherical bearing will result if bearing staking tool is used improperly. Do not allow bearing staking tool to make contact with spherical ball side of groove as binding may result.

f. Using bearing staking tool, swage V-groove lip over chamfer in rod end bore.

g. Using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean petrolatum from V-groove.

h. Inspect swage by inserting 0.005-inch feeler gage under staked lip of tail rotor pitch control rod end spherical bearing. **NO INSERTION ALLOWED.** If swage is beyond limits, reswage tail rotor pitch control rod end spherical bearing.

i. Proof test both sides of tail rotor pitch control rod end spherical bearing using 1000 pounds.

j. Perform breakaway torque check of spherical ball as follows:

(1) Rotate spherical ball 360° clockwise and 360° counterclockwise.

(2) Misalign spherical ball in at least two directions to full degree permitted by clearance.

(3) **NO-LOAD ROTATIONAL BREAKAWAY TORQUE SHALL NOT EXCEED 10 INCH-POUNDS.**

(4) If beyond limits, replace tail rotor pitch control rod end spherical bearing.

k. Using corrosion preventive compound, Item 113, Appendix D, touch up surfaces of tail rotor pitch control rod end spherical bearing.

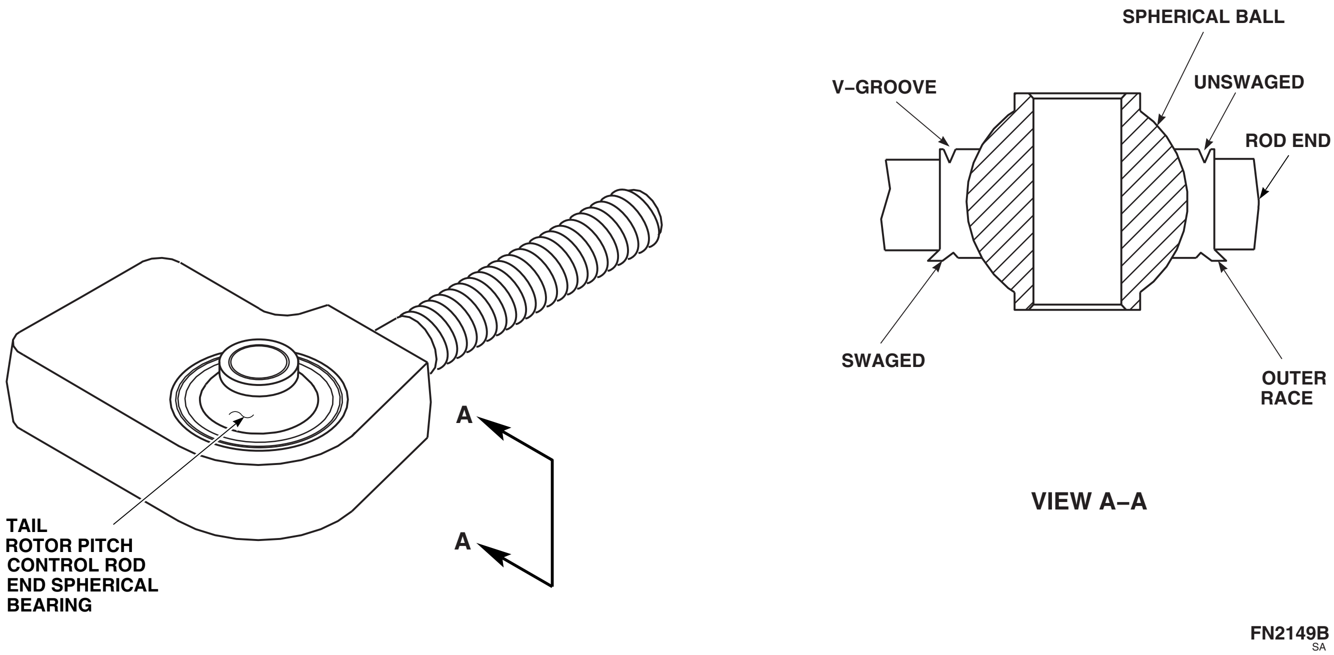


FIGURE 5-5-15-1. REPLACE TAIL ROTOR PITCH CONTROL ROD END SPHERICAL BEARINGS (BENCH)

l. Using epoxy primer coating kit, Item 135, Appendix D, touch up rod end surfaces with epoxy primer.

5-5-16 TAIL ROTOR BLADE PIVOT BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench
- Compression Screw, Y-28446-1-4
- Flashlight
- Inspection Mirror
- Nonmetallic Scraper, Locally-Made
- Pivot Bearing Replacement Wrench, Locally-Made, Figure H-51, Appendix H
- Spring Scale
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Adhesive, Item 41, Appendix D
- Alcohol, Isopropyl, Item 72, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D
- Tiedown Strap, Item 336, Appendix D
- Washers, NAS1149D0416K
- Washers, NAS1149D0463K

References

- Appendix D
- Appendix H
- PARA 5-5-18

5-5-16.1 Replace Tail Rotor Blade Pivot Bearings (BENCH).

NOTE

Replacement of all tail rotor blade pivot bearings is the same.

5-5-16.1.1 Remove.

NOTE

If any one tail rotor blade pivot bearing is removed or requires replacement, its mating tail rotor blade pivot bearing must be removed in order to take measurements for calculating shim thickness.

a. Locally make pivot bearing replacement wrench (Figure H-51, Appendix H).

NOTE

It is not necessary to remove both tiedown straps and clips securing tail rotor boot. Only removal of outboard tiedown strap and clip is necessary.

b. Remove only outboard tiedown strap and clip from tail rotor boot. Pull tail rotor boot back over opposite end (Figure 5-5-16-1, Sheet 1, Detail A).

c. On tail rotor blades with tail rotor pitch horn fairing, 70104-11104-043, installed, remove cover from tail rotor blade pivot bearing hole.

d. Remove screws and washers securing tail rotor pitch horn fairing to tail rotor pitch horn.

e. Remove screws securing tail rotor deice bracket to tail rotor pitch horn fairing.

f. Using nonmetallic scraper, remove bead of sealing compound from around edge of tail rotor pitch horn fairing.

g. Using pivot bearing replacement wrench to hold each nut, loosen bolts securing tail rotor pitch horn fairing and tail rotor blade pivot bearing to tail rotor blade.

h. Carefully remove tail rotor pitch horn fairing from tail rotor blade.

i. Using pivot bearing replacement wrench, remove both bolts and washers securing tail rotor blade pivot bearing to each side of tail rotor blade.

j. Using nonmetallic scraper, remove sealing compound from boltheads.

k. Remove tail rotor blade pivot bearing as follows:

(1) Using Allen wrench, remove compression screw from new tail rotor blade pivot bearing (Figure 5-5-16-1, Sheet 2, Detail B).

(2) Install screw in old tail rotor blade pivot bearing.

(3) Turn compression screw clockwise and compress tail rotor blade pivot bearing enough to remove it.

(4) Carefully remove tail rotor blade pivot bearing by pulling it straight out (Figure 5-5-16-1, Sheet 1, Detail A).

(5) Remove shim and retain.

l. Repeat step f. through step j. to remove mating tail rotor blade pivot bearing.

m. Using nonmetallic scraper, remove sealing compound from tail rotor pitch horn fairing and tail rotor pitch horn.

5-5-16.1.2 Inspect.

a. Inspect all four tail rotor blade pivot bearing retainers for disbonding from tail rotor blade spar. **NONE ALLOWED.** If disbonding is found, replace tail rotor blade pivot bearing retainers (PARA 5-5-18).

NOTE

- A white, wax-like substance on rubber surface is normal and should not be removed.
- Elastomeric pivot bearings consist of laminated elastomer and steel shells. Wear of elastomer in bearing is a gradual process that will take several hundreds of hours from first detection of elastomer damage to point where tail rotor blade pivot bearing must be replaced.

b. Compare tail rotor blade pivot bearings with tail rotor blade pivot bearings in Figure 5-5-16-2 to determine if replacement is required.

c. Tail rotor blade pivot bearings may fail by developing separations between rubber layers and metal shims or endplates, or within rubber layers themselves. **SEPARATION WITHIN RUBBER LAYERS MUST NOT BE GREATER THAN 1/8-INCH DEEP AROUND 1/4 OF CIRCUMFERENCE OF TAIL ROTOR BLADE PIVOT BEARING.** If limit is exceeded, replace tail rotor blade pivot bearing.

d. Inspect tail rotor blade pivot bearings for eraser-type extrusions. **NONE ALLOWED.** If extrusions are found, replace tail rotor blade pivot bearing.

e. Inspect tail rotor blade pivot bearings for rubber-to-shim separations. **NONE ALLOWED. BENT SHIMS ARE ACCEPTABLE.** If separations are found, replace tail rotor blade pivot bearing.

f. Inspect tail rotor blade pivot bearing baseplate for cracks. **NONE ALLOWED.** If cracks are found, replace tail rotor blade pivot bearings.

g. Using flashlight and inspection mirror, inspect inside surfaces of tail rotor pitch horn, tail rotor blade pivot bearing retainer, and tail rotor blade pivot bearing baseplates for missing or damaged primer. **NONE ALLOWED.** If primer is missing or damaged, repair primer (PARA 5-5-16.1.3).

5-5-16.1.3 Repair.

a. Using abrasive cloth, Item 6, Appendix D, lightly sand surfaces of tail rotor pitch horn that mate with shims.

b. Using clean cheesecloth, Item 90, Appendix D, dampened with isopropyl alcohol, Item 72, Appendix D, clean sanded surface.

c. Treat bare aluminum using corrosion resistant coating, Item 118, Appendix D.

d. Using epoxy primer coating kit, Item 135, Appendix D, prime untreated surfaces of tail rotor pitch horn.

5-5-16.1.4 Install.

NOTE

- If tail rotor blade pivot bearing(s) will be replaced, make sure shelf life for tail rotor blade pivot bearing(s), SB7301-103, has not exceeded 60 months. If shelf life has exceeded 60 months, do not use.
- The following procedure applies to all mating tail rotor blade pivot bearings.

a. Calculate required thickness of new shim as follows (Figure 5-5-16-1, Sheet 2, Detail C):

(1) Measure and record inside dimension of tail rotor pitch horn (dimension A).

(2) Measure and record thickness of tail rotor blade spar, including both tail rotor blade pivot bearing retainer seats (dimension B).

(3) Measure and record free height (compression screw removed) of tail rotor blade pivot bearings (dimensions C^1 and C^2). Take measurements of both sides of tail rotor blade pivot bearing centerhole. Average of the two measurements will determine dimensions.

(4) Calculate dimension D using the following equation:

$$B + C^1 + C^2 - 0.066 = D$$

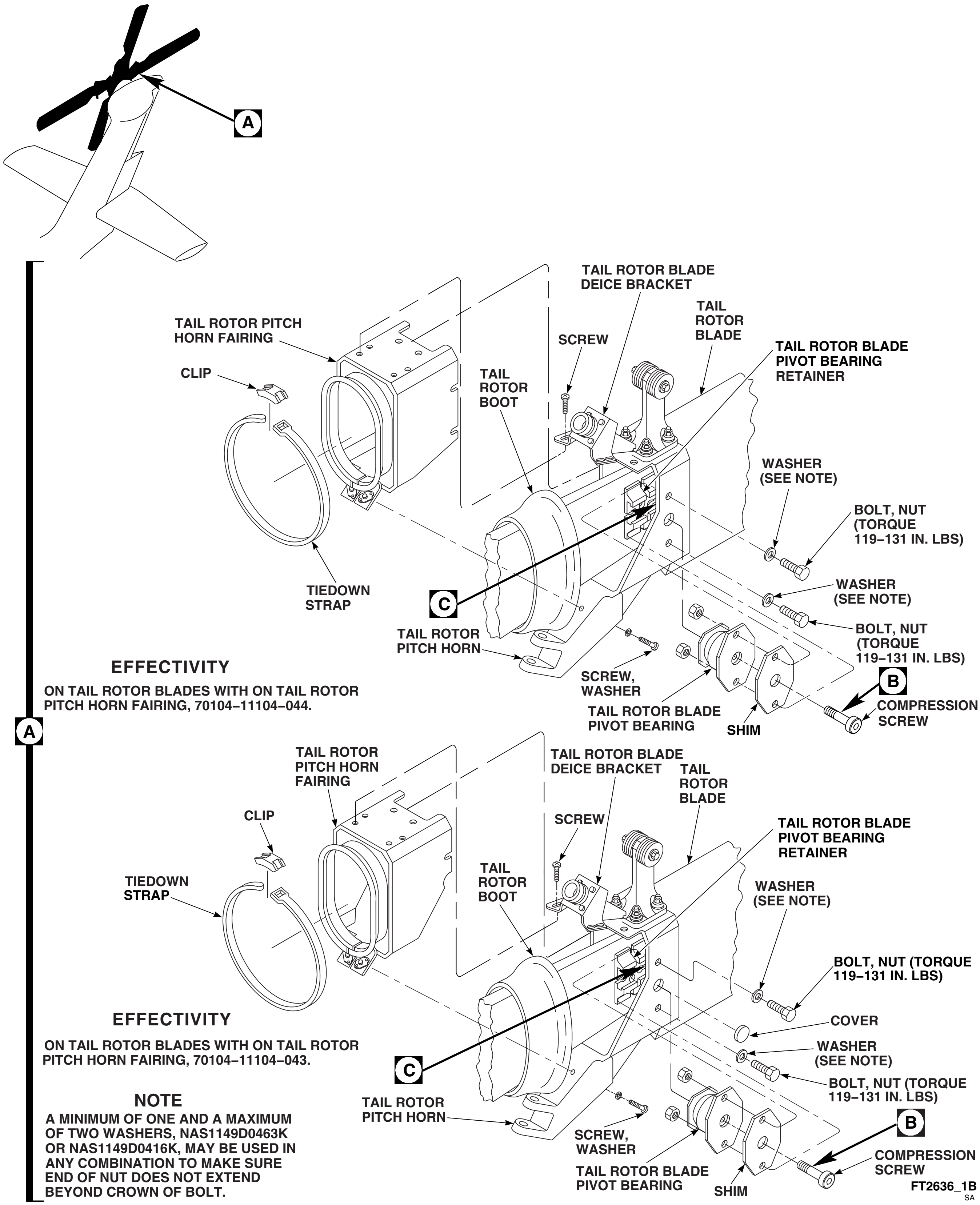
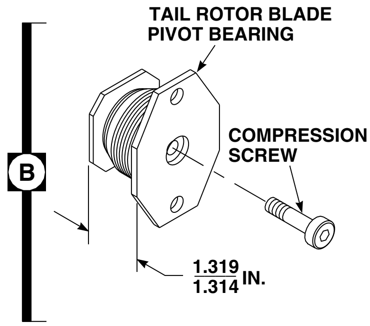
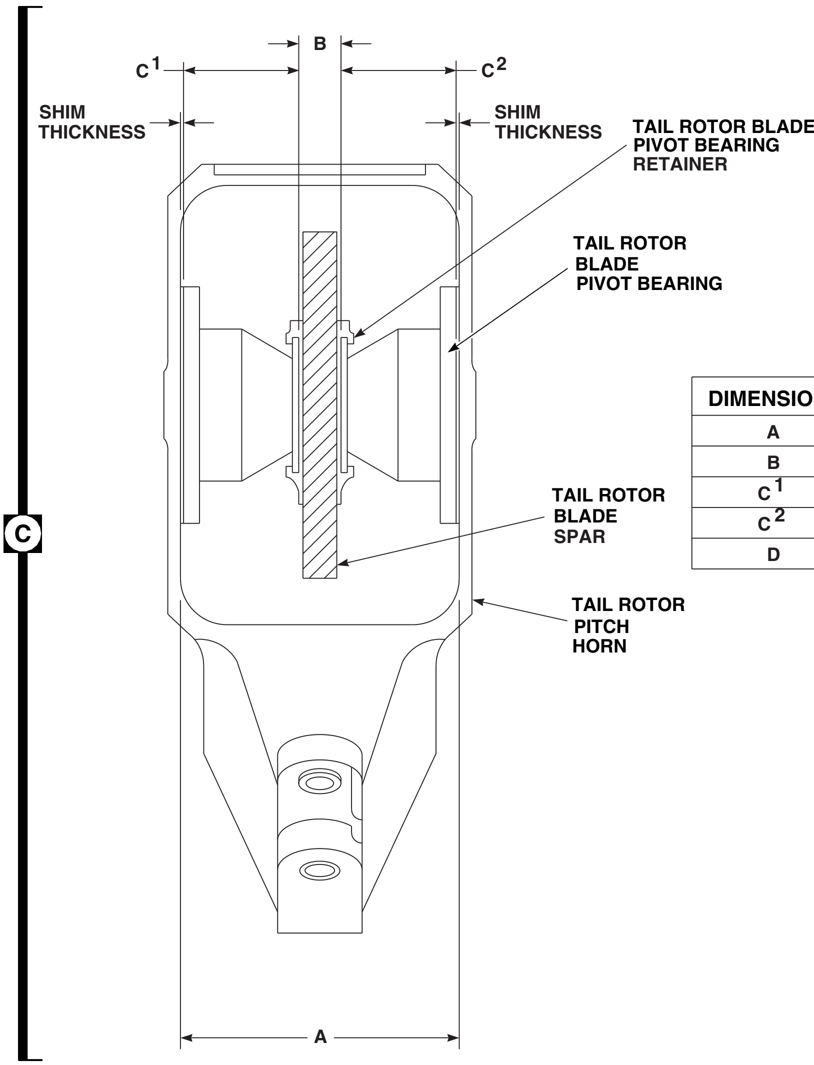


FIGURE 5-5-16-1. REPLACE TAIL ROTOR BLADE PIVOT BEARINGS (BENCH)
(SHEET 1 OF 2)





DIMENSION	MEASUREMENT	EQUATIONS
A		$B + C^1 + C^2 - 0.066 = D$ $(A - D) \div 2$ = SHIM THICKNESS
B		
C^1		
C^2		
D		

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FIGURE 5-5-16-1. REPLACE TAIL ROTOR BLADE PIVOT BEARINGS (BENCH)
(SHEET 2 OF 2)

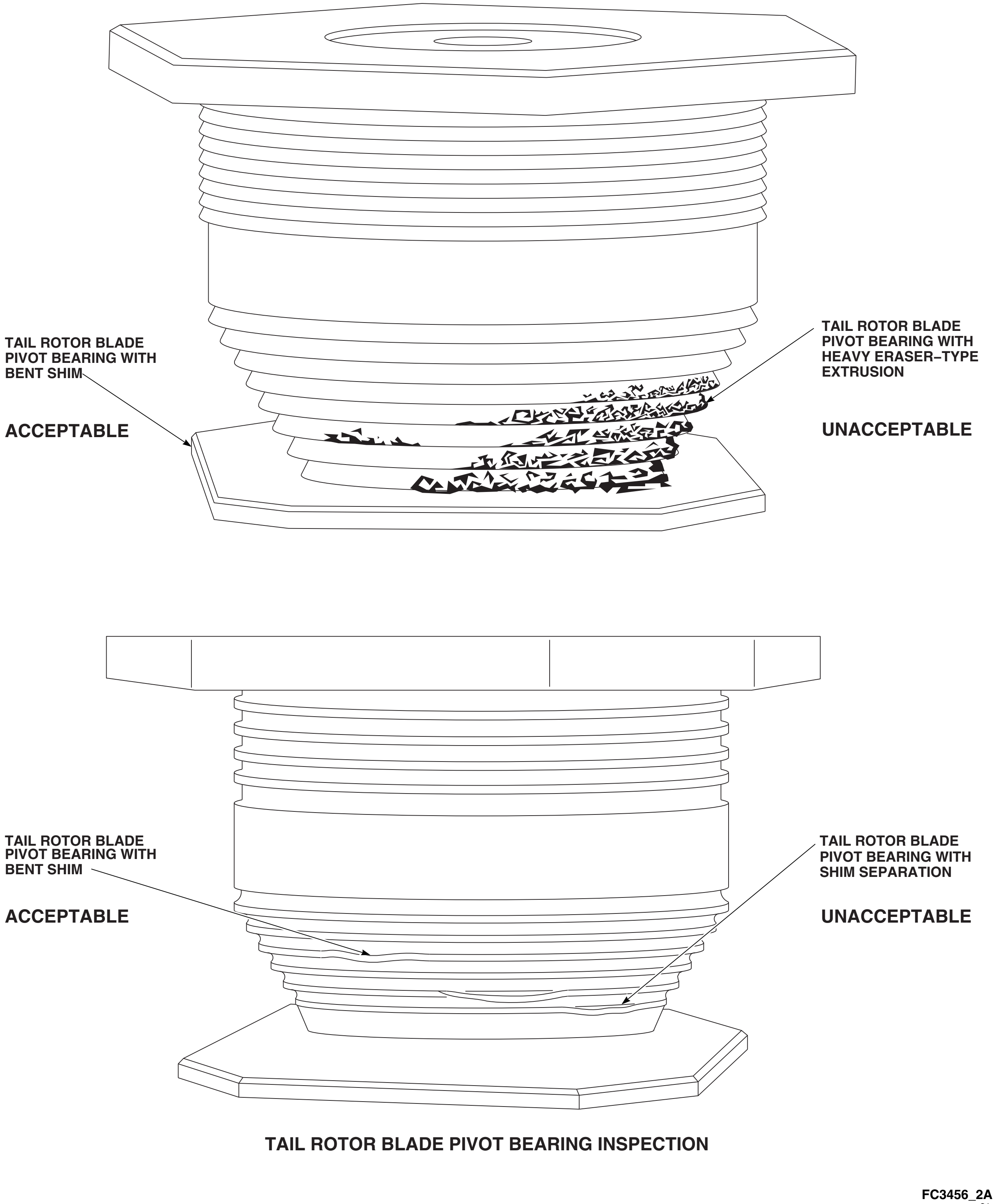


FIGURE 5-5-16-2. INSPECT TAIL ROTOR BLADE PIVOT BEARINGS (BENCH)

(5) Calculate shim thickness using the following equation:

$$(A - D) \div 2 = \text{Shim Thickness}$$

(6) Peel each shim to thickness calculated in step a. (5).

b. Install compression screw into new tail rotor blade pivot bearing (Figure 5-5-16-1, Sheet 2, Detail B). **COMPRESS TAIL ROTOR BLADE PIVOT BEARING TO HEIGHT OF 1.314 - 1.319-INCHES.**

CAUTION

Damage to tail rotor blade pivot bearing will result if primer is allowed to seep into tail rotor blade pivot bearing. Do not allow primer to come in contact with tail rotor blade pivot bearing.

NOTE

All aluminum surfaces of tail rotor blade pivot bearing, shims, and retainer must be primed prior to installation.

c. Apply coat of primer, Item 258, Appendix D, to aluminum surfaces of tail rotor blade pivot bearing, shims, and retainer.

d. Place tail rotor blade pivot bearing and shim in tail rotor pitch horn while epoxy primer is still wet (Figure 5-5-16-1, Sheet 1, Detail A).

e. Line up centerhole and boltholes in tail rotor blade pivot bearing and shim with holes in tail rotor pitch horn. Make sure tail rotor blade pivot bearing is properly engaged with tail rotor blade pivot bearing retainer.

f. Loosely install bolts, washers, and nuts through tail rotor blade and tail rotor blade pivot bearing.

g. Expand tail rotor blade pivot bearing by turning compression screw counterclockwise until screw is removed (Figure 5-5-16-1, Sheet 2, Detail B).

h. Place tail rotor pitch horn fairing on tail rotor pitch horn, and slide cutouts in tail rotor pitch horn fairing under washers and boltheads (Figure 5-5-16-1, Sheet 1, Detail A). Line up screwholes and install both screws and washers securing tail rotor pitch horn fairing to tail rotor pitch horn.

i. To form watertight seal, apply bead of sealing compound, Item 292, Appendix D, around edge of tail rotor pitch horn fairing and tail rotor pitch horn mating surfaces.

j. Torque bolts securing tail rotor blade pivot bearing as follows:

(1) Using pivot bearing replacement wrench, hold nuts.

(2) **TORQUE BOLTS TO 119 - 131 INCH-POUNDS.**

NOTE

- A minimum of one and a maximum of two washers, NAS1149D0463K or NAS1149D0416K, may be used in any combination.
- At least one washer must be installed under bolthead.
- Thread protrusion beyond end of nut is not required.

(3) Make sure end of nut does not extend beyond crown of bolt. Remove or install washers as required. Repeat step h.

k. Install tail rotor deice bracket on tail rotor pitch horn fairing with screws.

l. Using sealing compound, Item 292, Appendix D, completely cover boltheads securing tail rotor blade pivot bearing to tail rotor pitch horn.

m. To form watertight seal, apply bead of sealing compound, Item 292, Appendix D, around edges of tail rotor deice bracket and boltheads.

n. On tail rotor blades with tail rotor pitch horn fairing, 70104-11104-043, installed, using adhesive, Item 41, Appendix D, bond cover to tail rotor blade pivot bearing hole.

o. Pull tail rotor boot forward on tail rotor blade.

p. Place clip on tail rotor boot at leading edge, and using tiedown strap, Item 336, Appendix D, secure tail rotor boot and clip.

q. Using spring scale, apply 10 - 20 pounds tension to end of tiedown straps.

5-5-17 TAIL ROTOR BLADE SPAR AND TAIL ROTOR PITCH HORN FAIRING (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Drill
- Drill Bit, No. 50, 0.070-inch diameter
- Nonmetallic Scraper, Locally-Made
- Sanding Block (Alternate for Single Cut Mill File)
- Single Cut Mill File (Alternate for Sanding Block)
- Straight Edge

References

- PARA 5-5-16

5-5-17.1 Inspect Tail Rotor Blade Spar and Rework Tail Rotor Pitch Horn Fairing (BENCH).

NOTE

- Inspection of both tail rotor blade spars is the same.
- Rework of all tail rotor pitch horn fairings is the same.

a. Inspect for equal spacing on each side of tail rotor pitch horn fairing. If clearance between the tail rotor pitch horn fairing and tail rotor blade spar is not equal, repositioning of the tail rotor pitch horn fairing should be completed before any measurements or rework of the tail rotor pitch horn fairing.

b. Remove tail rotor pitch horn fairing (PARA 5-5-16).

c. Inspect for delaminations and/or splinters on tail rotor blade spar in area near inner lip of tail rotor pitch horn fairing. If any evidence of delamination and/or splintering is found, tail rotor blade should be rejected and tail rotor pitch horn fairing should not be reworked.

d. Inspect for grooves at corners of tail rotor blade spar in area near inner lip of tail rotor pitch horn fairing. If no grooves are found, rework tail rotor pitch horn fairing. If a groove is found, measure its depth as follows:

(1) Using drill with No. 50 drill bit, place into damage notch and place a straight edge in contact notch with surfaces directly adjacent to damage, holding straight edge perpendicular to drill centerline.

(2) **IF STRAIGHT EDGE TOUCHES ADJACENT SURFACE BUT NOT DRILL BIT, DAMAGE EXCEEDS**

0.070-INCH AND TAIL ROTOR BLADE SHOULD BE REJECTED AND TAIL ROTOR PITCH HORN FAIRING SHOULD NOT BE REWORKED.

(3) **IF DAMAGE DOES NOT EXCEED 0.070-INCH, TAIL ROTOR BLADE IS ACCEPTABLE AND TAIL ROTOR PITCH HORN FAIRING SHOULD BE REWORKED.**

e. Using single cut mill file or sanding block, remove material from inner lip of tail rotor pitch horn fairing at all four corner radii. Continue until inner lip is 0.060 - 0.100-inch outboard of the inside tail rotor pitch horn fairing surface under the tail rotor boot. Tiedown strap area as measured with straight edge.

f. If tail rotor blade spar inspection has revealed no delaminations and/or splinters and no grooves deeper than 0.070-inch, using nonmetallic scraper, clean old sealing compound from tail rotor pitch horn fairing and tail rotor pitch horn.

g. Install tail rotor pitch horn fairing (PARA 5-5-16).

5-5-18 TAIL ROTOR BLADE PIVOT BEARING RETAINERS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Compression Screw, Y-28446-1-4
- Heat Gun
- Pivot Bearing Replacement Wrench, Locally-Made, Figure H-51, Appendix H
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 4, Appendix D
- Abrasive Paper, Item 18, Appendix D
- Adhesive, Item 41, Appendix D
- Alcohol, Denatured, Item 70, Appendix D
- Brush, Acid, Item 84, Appendix D
- Cheesecloth, Item 90, Appendix D
- Mat, Reinforcing, Fibrous, Item 216, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 258, Appendix D
- Stirring Stick, Beverage, Item 315C, Appendix D

References

- Appendix D
- Appendix H
- PARA 2-6-3
- PARA 5-4-38
- PARA 5-5-16

5-5-18.1 Replace Tail Rotor Blade Pivot Bearing Retainers (BENCH).

NOTE

Replacement of all tail rotor blade pivot bearing retainers is the same.

5-5-18.1.1 Inspect.

a. Remove tiedown straps and boots from both ends of tail rotor blade.

b. Inspect all four tail rotor blade pivot bearing retainers for disbonding from tail rotor blade. **NONE ALLOWED.** If disbonding is found, replace tail rotor blade pivot bearing retainer.

c. Tail rotor blade pivot bearing retainer disbond can cause trailing edge disbond. Prior to repair of tail rotor blade pivot bearing retainers, inspect trailing edge for damage (PARA 5-4-38).

5-5-18.1.2 Remove.

a. Remove tail rotor blade pivot bearings (PARA 5-5-16).

b. Remove disbonded tail rotor blade pivot bearing retainer.

5-5-18.1.3 Install.

NOTE

If tail rotor blade pivot bearing(s) will be replaced, make sure shelf life for replacement tail rotor blade pivot bearing(s), SB7301-103, has not exceeded 60 months. If shelf life has exceeded 60 months, do not use.

a. Locally make pivot bearing replacement wrench (Figure H-51, Appendix H).

CAUTION

Damage to graphite plies in tail rotor blade spar will result if care is not taken when removing residual adhesive from tail rotor blade. Use care in smoothing and removing residual adhesive from tail rotor blade.

b. Using abrasive cloth, Item 4, Appendix D, sand adhesive remaining on tail rotor blade. Continue to sand only until smooth bond surface is attained.

c. Using clean cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, wipe bond surface.

d. Using clean cheesecloth, Item 90, Appendix D, dampened with denatured alcohol, Item 70, Appendix D, wipe bond surface.

e. Prepare bond surface on tail rotor blade pivot bearing retainer as follows:

(1) Using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean rubber surface.

(2) Using abrasive paper, Item 18, Appendix D, scuff rubber surface.

(3) Using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, wipe rubber.

(4) Using clean cheesecloth, Item 90, Appendix D, dampened with denatured alcohol, Item 70, Appendix D, wipe rubber again.

f. Calculate required thickness of new shim as follows (Figure 5-5-18-1):

(1) Measure and record inside dimension of tail rotor pitch horn (dimension A).

(2) Measure and record thickness of tail rotor blade spar, including both tail rotor blade pivot bearing retainer seats (dimension B).

(3) Measure and record free height (compression screw removed) of tail rotor blade pivot bearings (dimensions C^1 and C^2). Take measurements of both sides of tail rotor blade pivot bearing centerhole. Average of the two measurements will determine dimensions.

(4) Calculate dimension D using the following equation:

$$B + C^1 + C^2 - 0.066 = D$$

(5) Calculate shim thickness using the following equation:

$$(A - D) \div 2 = \text{Shim Thickness}$$

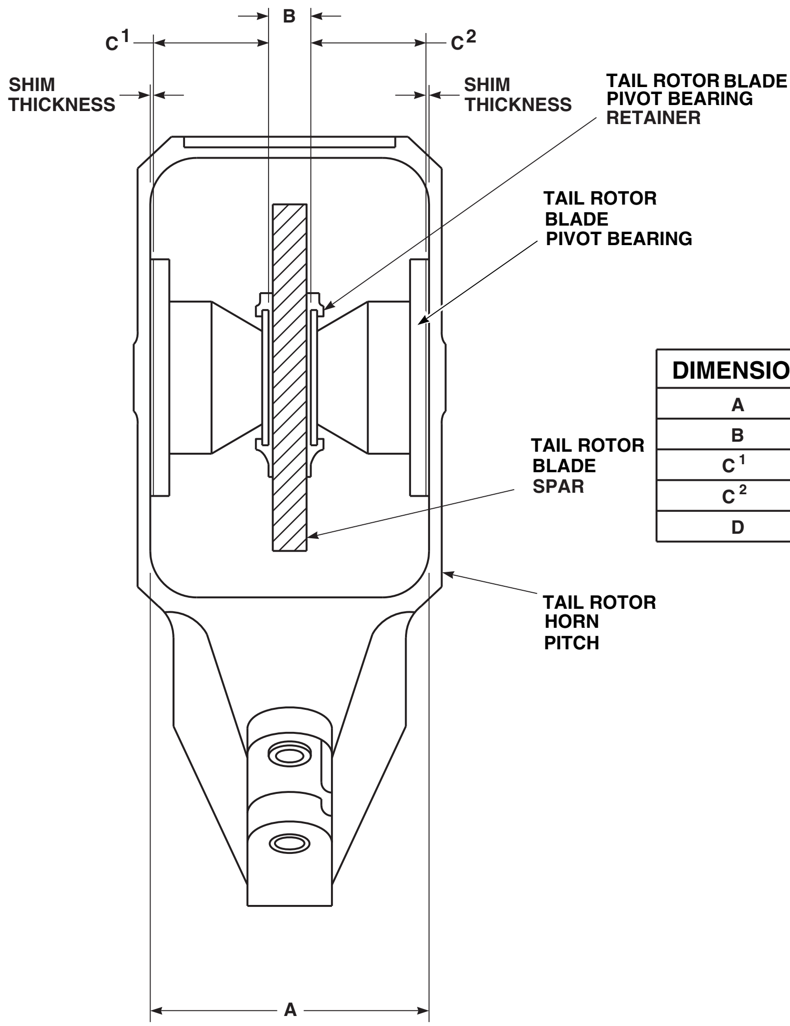
(6) Peel each shim to thickness calculated in step f. (5).

g. Install compression screw in tail rotor blade pivot bearing.

NOTE

Keep tail rotor blade pivot bearing from twisting during compression, and align top and bottom plates when compressed.

h. Compress tail rotor blade pivot bearing as required to allow installation of tail rotor blade



DIMENSION	MEASUREMENT	EQUATIONS
A		$B + C^1 + C^2 - 0.066 = D$ $(A - D) \div 2 =$ SHIM THICKNESS
B		
C^1		
C^2		
D		

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FIGURE 5-5-18-1. REPLACE TAIL ROTOR BLADE PIVOT BEARING RETAINERS (BENCH)

pivot bearing and shim into tail rotor blade (PARA 5-5-16).

i. Install tail rotor blade pivot bearing retainer on tail rotor blade pivot bearing. Make sure retainer is fully seated.

j. Prepare adhesive, Item 41, Appendix D (PARA 2-6-3).

NOTE

If both tail rotor blade pivot bearing retainers are disbonded, completely bond only one side at a time. Bond opposite side only after first side is cured.

k. Using acid brush, Item 84, Appendix D, or beverage stirring stick, Item 315C, Appendix D, apply even coat of adhesive, Item 41, Appendix D, to bond surface of tail rotor blade spar and rubber bond surface of tail rotor blade pivot bearing retainer.

l. Embed fibrous reinforcing mat, Item 216, Appendix D, in wet adhesive.

m. Push tail rotor pitch horn to deflect it away from tail rotor blade spar.

CAUTION

Poor bonding will result if adhesive is smeared during installation of tail rotor blade pivot bearing retainer. Do not smear adhesive during installation of tail rotor blade pivot bearing retainer.

n. Loosely install bolts, washers, and nuts through tail rotor blade, tail rotor blade pivot bearing, tail rotor blade pivot bearing retainer, and shim.

o. Using pivot bearing replacement wrench, hold nuts. **TORQUE BOLTS TO 119 - 131 INCH-POUNDS.**

p. Compress opposite tail rotor blade pivot bearing as required to allow installation of tail rotor blade pivot bearing and shim into tail rotor blade. Install tail rotor blade pivot bearing with shim.

q. If opposite tail rotor blade pivot bearing retainer is also disbonded, install tail rotor blade pivot bearing retainer on tail rotor blade pivot bearing and install assembly with bolts, washers, and nuts. **TORQUE BOLTS TO 119 - 131 INCH-POUNDS.**

r. Release and remove compression screw on lower side (not being bonded) to provide bonding pressure against tail rotor blade.

s. Allow adhesive to cure adhesive for 24 hours at 70°F (22°C), or using heat gun, for 2 hours at 140° - 160°F (60° - 71°C).

t. Repeat procedures for each disbonded tail rotor blade pivot bearing retainer on tail rotor blade.

CAUTION

Damage to rubber will result if contacted by primer. Prevent primer from contacting rubber parts of tail rotor blade pivot bearing.

u. Apply coat of primer, Item 258, Appendix D, to tail rotor blade pivot bearing endplates, tail rotor blade pivot bearing retainers, shims, and all attaching hardware.

v. Install shims, tail rotor blade pivot bearings, and tail rotor pitch horn fairing (PARA 5-5-16).

5-5-19 TAIL ROTOR BOOT SUPPORTS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Boot Support Locator, Locally-Made, Figure H-253, Appendix H
- Heat Gun
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D
- Abrasive Cloth, Item 4, Appendix D
- Adhesive, Item 32, Appendix D
- Alcohol, Ethyl, Item 71, Appendix D
- Alcohol, Isopropyl, Item 72, Appendix D
- Aluminum Mesh Screen, Item 73A, Appendix D
- Cheesecloth, Item 90, Appendix D
- Fuel Tank Parts Kit, Item 141B, Appendix D
- Liner, 70104-11110-103, Procured or Locally-Made
- Mat, Reinforcing, Fibrous, Item 216, Appendix D
- Peel Ply, Plastic Film
- Pressure Sensitive Adhesive Tape, Item 244B, Appendix D
- Rubber Sheet, 0.050-inch thick, Item 267B, Appendix D (for Liner, 70104-11110-103, Locally-Made)
- Sealing Compound, Item 292, Appendix D
- Tail Rotor Boot Supports, 70104-11110-041/-042
- Tape

References

- Appendix D
- Appendix H
- PARA 5-3-12
- PARA 5-4-38

5-5-19.1 Replace Tail Rotor Boot Supports (BENCH).

NOTE

Replacement of all tail rotor boot supports is the same.

5-5-19.1.1. Remove.

CAUTION

Damage to tail rotor boot supports or tail rotor blade spar may result if care is not taken when removing sealing compound. Use care when removing sealing compound.

a. Using nonmetallic scraper, carefully remove sealing compound coating screws.

b. Remove screws securing tail rotor boot supports to tail rotor blade spar and remove tail rotor boot supports (Figure 5-5-19-1).

c. Using nonmetallic scraper, carefully remove adhesive from tail rotor blade spar.

d. Inspect tail rotor blade spar for damage (PARA 5-3-12 and PARA 5-4-38).

5-5-19.1.2. Install.

NOTE

- As an alternate to procuring new tail rotor boot supports, 70104-11110-041/-042, if required, existing tail rotor boot supports, 70104-11110-041/-042, may be refurbished, if not significantly damaged.
- As an alternate to procuring tail rotor boot supports, 70104-11110-041/-042, existing tail rotor boot supports, having other part numbers, may be reused and modified to the 70104-11110-041/-042 configuration, if not significantly damaged.
- If new tail rotor boot supports, 70104-11110-041/-042, are to be procured and used, perform step b. If existing tail rotor boot supports are to be refurbished or modified to the 70104-11110-041/-042 configuration, perform step a.

a. If existing tail rotor boot supports are to be refurbished or modified to the 70104-11110-041/-042 configuration, perform the following (Figure 5-5-19-1, Sheet 1, Detail A):

(1) Remove sealing compound and rubber remnants from bond area, and inspect threaded inserts for good condition. If tail rotor boot supports fail inspection, replace tail rotor boot support. If tail rotor boot supports pass inspection, perform step a. (2).

(2) Procure or locally make liners, 70104-11110-103. To locally make liners, cut rubber sheet, 0.050-inch thick, Item 267B, Appendix D, to the following dimensions: 4.7-inches long × 0.77-inch wide (Figure 5-5-19-1, Sheet 1, Detail B).

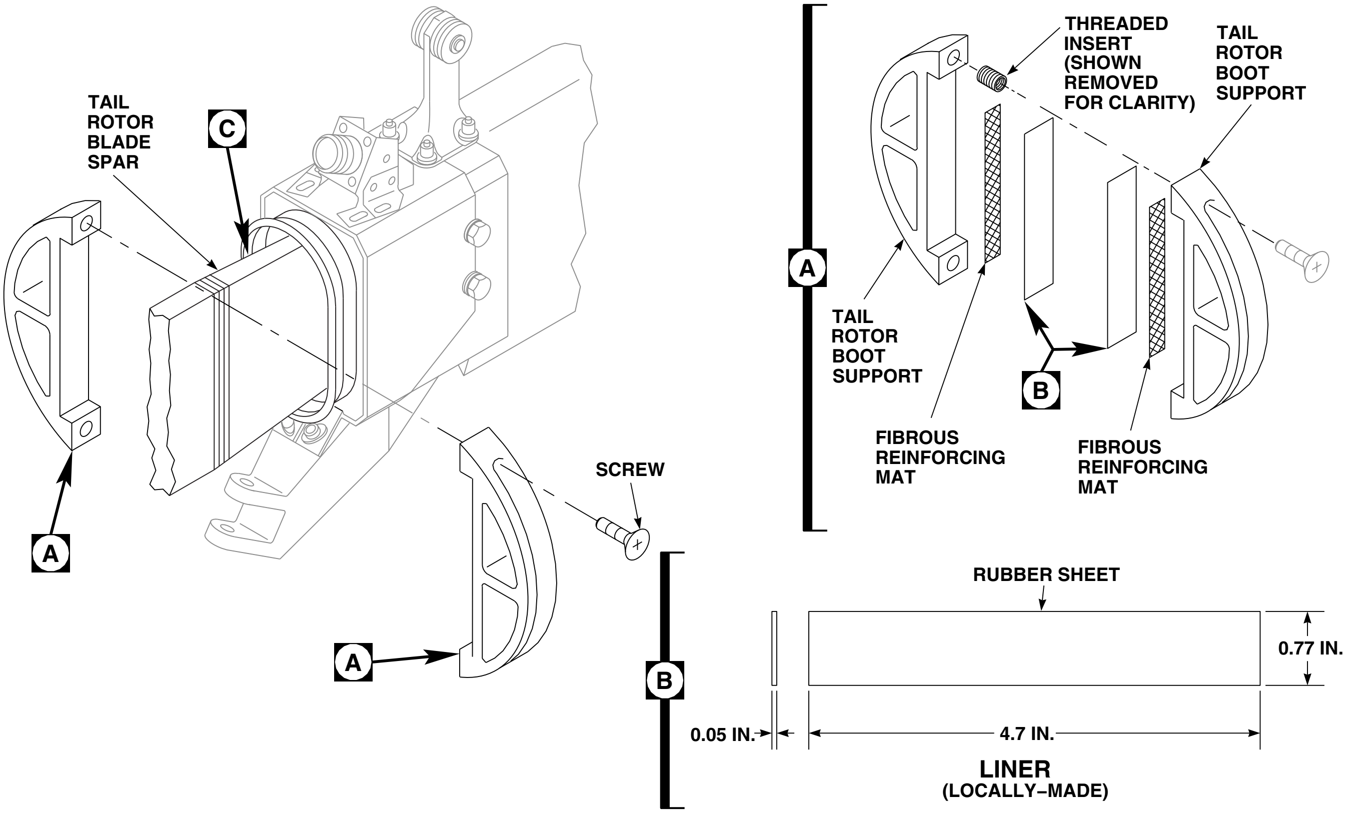
(3) Using cheesecloth, Item 90, Appendix D, dampened with ethyl alcohol, Item 71 Appendix D, or isopropyl alcohol, Item 72, Appendix D, wipe liner.

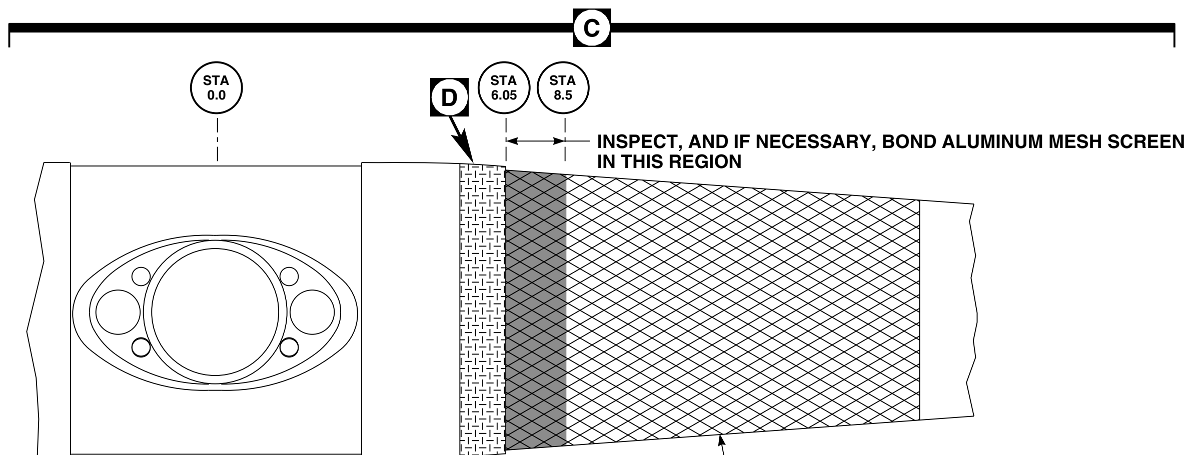
(4) Using abrasive cloth, Item 2, Appendix D, abrade liner.

(5) Using cheesecloth, Item 90, Appendix D, dampened with ethyl alcohol, Item 71 Appendix D, or isopropyl alcohol, Item 72, Appendix D, wipe liner again and repeat until cheesecloth remains clean.

(6) Using cheesecloth, Item 90, Appendix D, dampened with ethyl alcohol, Item 71 Appendix D, or isopropyl alcohol, Item 72, Appendix D, wipe tail rotor boot supports bonding surfaces (Figure 5-5-19-1, Sheet 1, Detail A).

(7) Cut fibrous reinforcing mat, Item 216, Appendix D, 0.125-inch smaller than bond area all around to avoid wicking water into the bondline during service.





NOTES

1. ON INITIAL TORQUE OF SCREWS (3 IN. LBS), MAKE SURE GAPS ARE APPROXIMATELY EQUAL.
2. FILL SCREW COUNTERBORES AND GAPS WITH SEALING COMPOUND.
3. WHEN APPLYING SEALING COMPOUND TO 0.10 IN. ADHESIVE FILLET, MAKE SURE SEALING COMPOUND BEAD AND ADHESIVE FILLET COMBINED IS 0.25 IN. MAXIMUM TO AVOID CONTACT WITH INBOARD AND OUTBOARD RETENTION PLATES.

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FIGURE 5-5-19-1. REPLACE TAIL ROTOR BOOT SUPPORTS (BENCH) (SHEET 1 OF 2)

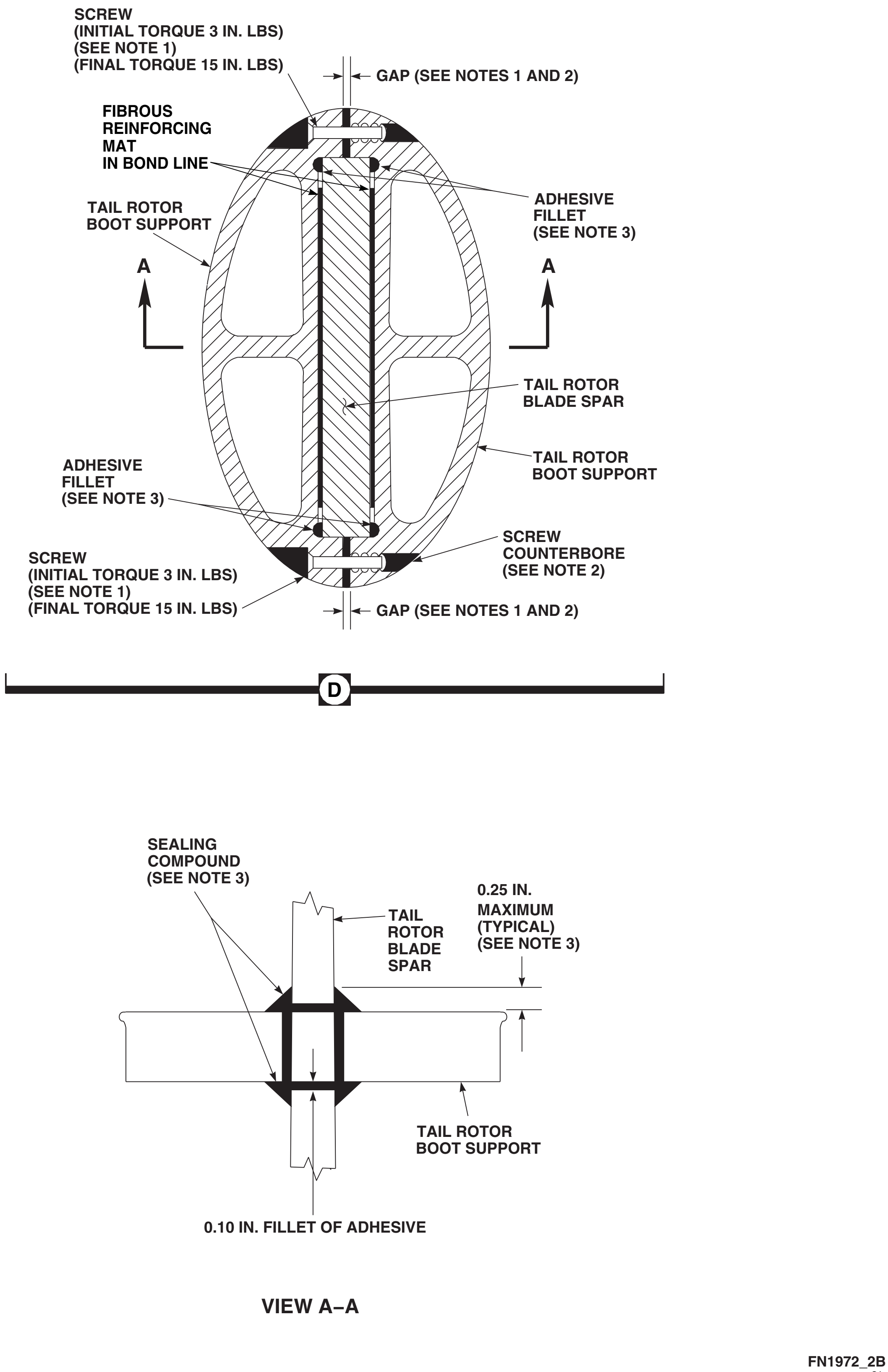

FIGURE 5-5-19-1. REPLACE TAIL ROTOR BOOT SUPPORTS (BENCH) (SHEET 2 OF 2)

(8) Using adhesive, Item 32, Appendix D, bond liner to tail rotor boot support with fibrous reinforcing mat, Item 216, Appendix D, in the bondline.

b. Inspect tail rotor blade spar for presence of aluminum mesh screen from STA 6.05 (outboard edge of nylon wrap) to STA 8.5 (tail rotor boot support bond area). If aluminum mesh screen is not present on tail rotor blade spar, from STA 6.05 (outboard edge of nylon wrap) to STA 8.5, bond aluminum mesh screen as follows (Figure 5-5-19-1, Sheet 1, Detail C):

CAUTION

Damage to tail rotor blade spar will result if sanding penetrates graphite plies on surface of tail rotor blade spar. Do not sand through aluminum mesh screen to graphite plies on surface of tail rotor blade spar.

(1) Using abrasive cloth, Item 4, Appendix D, lightly abrade tail rotor blade spar in spanwise direction only, to remove remaining adhesive.

(2) Using cheesecloth, Item 90, Appendix D, dampened with ethyl alcohol, Item 71, Appendix D, or isopropyl alcohol, Item 72, Appendix D, wipe tail rotor blade spar.

(3) Inspect tail rotor blade spar for damaged, delaminated, or splintered graphite plies. If damaged, delaminated, or splintered graphite plies exist, replace tail rotor blade.

(4) Cut aluminum mesh screen, Item 73A, Appendix D, to surface shape of tail rotor blade spar from STA 6.05 (outboard edge of nylon wrap) to STA 8.5.

(5) Place pressure sensitive adhesive tape, Item 244B, Appendix D, on tail rotor blade spar just inboard and outboard of tail rotor boot support bonding area to limit excess adhesive squeeze-out.

(6) Apply adhesive, Item 32, Appendix D, to surface of tail rotor blade spar and embed aluminum mesh screen in adhesive. If required, reapply adhesive to any aluminum mesh screen areas that are not fully saturated.

(7) Cover repair with plastic film peel ply.

(8) Using vacuum bag from fuel tank parts kit, Item 141B, Appendix D, tape, or mechanical means, apply pressure.

(9) Allow adhesive to cure for 24 hours at 70° - 85°F (21° - 29°C), or using heat gun, for 1 hour minimum at 140° - 160°F (60° - 71°C).

(10) Remove pressure (vacuum bag, tape or mechanical means) and plastic film peel ply.

(11) Using abrasive cloth, Item 4, Appendix D, lightly scuff repair.

(12) Using cheesecloth, Item 90, Appendix D, dampened with ethyl alcohol, Item 71 Appendix D, or isopropyl alcohol, Item 72, Appendix D, wipe repair.

(13) Remove pressure sensitive adhesive tape from tail rotor blade spar.

c. If not already done, prepare exposed rubber surface on tail rotor boot supports. Refer to step a. (3) through step a. (5).

d. Locally make boot support locator (Figure H-253, Appendix H).

e. Using locally made boot support locator, position tail rotor boot supports on tail rotor blade spar with their inboard face 7.18-inches from center of rotation.

f. Cut fibrous reinforcing mat, Item 216, Appendix D, 0.125-inch smaller than bond area all around to avoid wicking water into the bondline during service.

NOTE

When bonding tail rotor boot supports, it may be necessary to wait until the adhesive begins to thicken in order to keep fibrous reinforcing mat in the bondline.

g. Using adhesive, Item 32, Appendix D, bond tail rotor boot supports to tail rotor blade spar with fibrous reinforcing mat, Item 216, Appendix D, in the bondline. Form a 0.10-inch maximum adhesive fillet (Figure 5-5-19-1, Sheet 2, Detail D).

h. Fill upper and lower gaps between tail rotor boot supports with adhesive, Item 32, Appendix D.

NOTE

As screws are being torqued, make sure upper and lower gaps between tail rotor boot supports is approximately equal at leading and trailing edge of tail rotor blade spar.

i. Install screws securing tail rotor boot supports to tail rotor blade spar. **TORQUE SCREWS TO 3 INCH-POUNDS** (Figure 5-5-19-1, Sheet 2, Detail D).

CAUTION

Damage to repair will result if adhesive is not allowed to cure at room temperature for at least 2 hours prior to curing for 1 hour minimum at 140° - 160°F (60° - 71°C) using heat gun. Adhesive may liquefy and run out of bondline at elevated temperature. If accelerated curing method is used (1 hour minimum at 140° - 160°F (60° - 71°C) using heat gun), allow adhesive to cure at room temperature for at least 2 hours prior to accelerated curing.

j. Allow adhesive to cure for 24 hours at 70° - 85°F (21° - 29°C), or using heat gun, for 1 hour minimum at 140° - 160°F (60° - 71°C).

CAUTION

Damage to tail rotor boot supports or tail rotor blade spar may result if care is not taken when removing excess adhesive. Use care when removing excess adhesive.

k. After cure, using nonmetallic scraper, remove any excess adhesive.

l. **RETORQUE SCREWS TO 15 INCH-POUNDS.**

CAUTION

Damage to repair may result if sealing compound bead comes in contact with inboard and outboard retention plates. When applying sealing compound to 0.10-inch adhesive fillet, make sure sealing compound bead and adhesive fillet combined is ¼-inch maximum.

m. Apply sealing compound, Item 292, Appendix D, to adhesive fillets and edge area, screw counterbores, and gaps between tail rotor boot supports (Figure 5-5-19-1, Sheet 2, View A-A).

5-5-20 OUTBOARD RETENTION PLATE (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer's Toolkit

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Lacquer, Item 186, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Pressure Sensitive Adhesive Tape, Item 247, Appendix D
- Primer, Item 258, Appendix D

References

- Appendix D
- ASTM E1417

5-5-20.1 Repair Outboard Retention Plate (BENCH).

NOTE

Do not remove balance weights (washers) from outboard retention plate.

5-5-20.1.1 Disassemble.

Using arbor press, press out pilot bushing from outboard retention plate; then press out bushing from pilot bushing (Figure 5-5-20-1).

5-5-20.1.2 Inspect.

a. Visually inspect outboard retention plate for cracks, nicks, wear, and corrosion.

b. Inspect pilot bushing for nicks, gouges, and scratches on surface (Figure 5-5-20-1, Detail A).

c. **USING ABRASIVE CLOTH, ITEM 6, APPENDIX D, BLEND OUT DAMAGE UP TO 0.010-INCH DEEP COVERING LESS THAN 20% OF SURFACE AND OUTSIDE OF 0.10 RADIUS AREA.** If damage exceeds limits, replace pilot bushing.

d. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, remove all paint on inspection surface.

NOTE

Do not repaint pilot bushing.

e. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace pilot bushing.

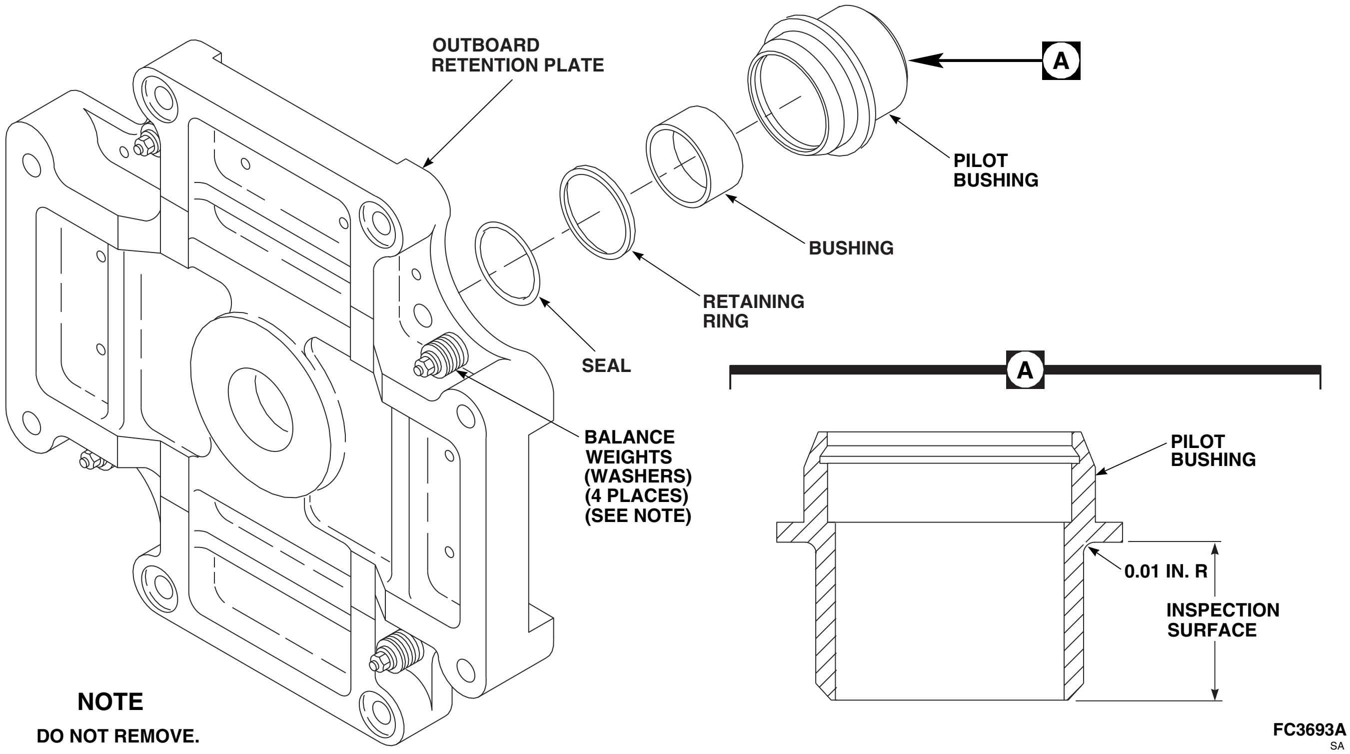

FIGURE 5-5-20-1. REPAIR OUTBOARD RETENTION PLATE (BENCH)

5-5-20.1.3 Assemble.

a. Apply coat of primer, Item 258, Appendix D, to outboard surface of pilot bushing outside diameter.

b. Using arbor press, press pilot bushing in outboard retention plate while primer is still wet. Be careful not to cock pilot bushing (Figure 5-5-20-1).

c. Make sure shoulder on pilot bushing is fully seated all around on outboard retention plate.

d. Using pressure sensitive adhesive tape, Item 247, Appendix D, mask inside diameter of pilot bushing.

e. Apply coat of primer, Item 258, Appendix D, to outside diameter of pilot bushing.

f. Apply coat of primer, Item 258, to top of bushing, retaining ring groove in pilot bushing, and retaining ring.

g. Remove pressure sensitive tape and install retaining ring while primer is still wet.

h. Install seal.

i. If necessary, using lacquer, Item 186, Appendix D, touch up outboard retention plate.